Mathematical Handholds

One-page explorations of number, shape, chance and change
Geometric, Probabilistic and Networked–Lattice Number Theory

from foothills to handholds

Geometric | Probabilistic | Networked

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How to read this book

Every sheet is a single page you can work through on its own: try its smallest example, redraw its picture, and give each symbol a job before worrying about its name. The six chapters run from the most self-contained material to the most interconnected, and inside each chapter the pages climb in three bands. The **foothills** introduce the objects and the language. The **footholds** give methods you can carry out and check. The **handholds** reach across several problems at once.

A coloured tab on the outer edge of each page marks its strand: **geometric** (shape, coordinates and transformation), **probabilistic** (what can happen and what an observation supports), and **networked–lattice** (how local connections organise a whole). The three dots on the tab show which band the page sits in. Each chapter opens with a short account of the questions it follows and a list of the pages it gathers, so you can read straight through, follow a single thread, or begin with one question that interests you.

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Chapter 1

Sequences, algebra and counting

What this chapter gathers.

Foothills

- Operations and their laws
- Distributivity: multiplying by parts
- Identity, inverse and undoing
- Letters as numbers: expanding, factorising and proof
- Actions that do not commute: composition
- Turning and flipping a triangle
- Matrices: machines that do not commute
- Rotations in the plane and in space
- The map: which laws hold where
- Sequences, rules and first differences
- Second differences and quadratic rules
- Cubics and higher differences
- Newton's formula: the table as a rule
- Power sequences and their sums
- Falling factorials: the powers of finite calculus
- From differences to the derivative
- From sums to the integral
- The dictionary and the exceptions

Foothold

- Binomial distribution from Galton board and Pascal triangle
- Partitioning a hand: Stirling, Bell and entropy
- Dealing cards into piles: partitions and finite differences
- The general sequencing law for triangles
- Bell numbers and Worpitzky's triangle
- From Stirling to Worpitzky: counting becomes differences
- Finding a quadratic from its sequence
- Finding a quartic with Worpitzky's triangle
- From differences to polynomial coefficients
- From binomials to bits and trits
- Pascal's triangle mod 2 and mod 3
- Polynomial interpolation: Vandermonde and Newton
- Delannoy numbers: three ways through a grid
- Rook, king and crow: chess moves and distance
- The characteristic equation beneath five presentations
- The determinant in its many guises
- Difference of two squares: determinants, cubes and i-time
- From cubes to Fibonacci: tridiagonal determinants
- Why i makes Fibonacci inevitable: norms and recurrence

Handhold

- Bernoulli numbers, step by step
- Regular and irregular primes: a Bernoulli numerator test
- A quartic interpolator reveals Stirling numbers

Why can we replace $3(4+2)$ by $3 \cdot 4 + 3 \cdot 2$? Three copies of a combined amount can be separated into three copies of each part. A rectangle divided into two smaller rectangles makes the same argument visible. Distributivity records that construction, while letters let us express it for any suitable numbers. We begin by examining the laws behind familiar calculations: what can be rearranged, what leaves a quantity unchanged, and what allows an operation to be undone.

These laws become more interesting when we change what we are operating on. Turn a triangle and then reflect it. Reverse the order, and its final position can change. The actions have become mathematical objects, with composition as their operation. Matrices give us a way to record and combine such actions. Rotations extend the investigation from the plane into space. Each extension invites the same question: which familiar laws still hold, and what does their survival or failure tell us?

Now write down $1, 4, 9, 16$ and ask for the next number. Twenty-five is a convincing guess because we recognise the squares. Draw them as growing squares of dots, and each new square adds an L-shaped border containing the next odd number. The identity $1+3+5+\cdots+(2n-1) = n^2$ records the construction. The first differences are consecutive odd numbers; their differences are all 2. A picture has supplied both a rule and a reason for its continuation.

Difference tables also let us work backwards from values. Under a stated polynomial assumption, successive differences help recover a rule and its coefficients. The assumption matters: finitely many values can agree with many different rules. Newton's formula organises the reconstruction, and falling factorials provide building blocks especially suited to taking differences. Adding terms reverses that process. Following change and accumulation in this discrete setting prepares us for the derivative and the integral.

Counting supplies another route to these structures. Paths through a Galton board lead to Pascal's triangle and binomial probabilities. Dividing a hand of cards into groups leads to Stirling and Bell numbers. Walks through a grid produce further recurrences. In each case, we organise a large collection of possibilities by describing how smaller cases combine. Even the remainders of these counts carry structure: colouring Pascal's triangle modulo 2 or 3 reveals patterns that the digits of the row and position can explain.

The useful surprise is that these constructions acquire further jobs. Stirling numbers count groupings and translate between ordinary powers and falling factorials. Worpitzky's triangle connects counting with the recovery of polynomial coefficients. Determinants enter interpolation and then reappear in recurrences leading to Fibonacci numbers. A characteristic equation captures how a repeated rule propagates its values. Recognising these connections gives us several ways to approach the same problem, with each representation making a different feature easier to see.

Even the fractions arising in sums of powers take us further. The Bernoulli numbers organise those sums; their numerators also provide a test separating odd primes into regular and irregular classes. Returning to a quartic interpolator brings the Stirling numbers back into view through another calculation. We are learning to reconstruct a rule and recognise a familiar construction when it appears in a new setting.

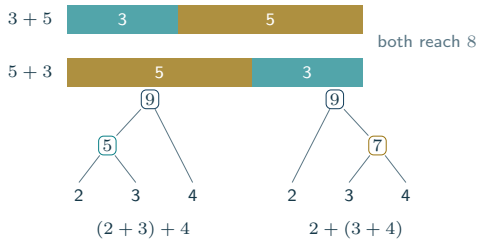
ALGEBRA LADDER 1 OF 9

Operations and their laws

An operation turns two numbers into one, and two laws ask whether the order and the grouping matter.

1 / THE RULE AND THE PICTURE

Commutative means the order does not matter, and associative means the grouping does not matter



Put two lengths end to end and the total is the same whichever comes first, which is the commutative law, and add three numbers in two steps and the answer is the same whichever pair is added first, which is the associative law:

$$a \circ b = b \circ a, \qquad (a \circ b) \circ c = a \circ (b \circ c)$$

The symbol $\circ$ stands for any operation, so each law is a question to ask of an operation, and a few operations pass and many fail.

2 / WHICH OPERATIONS PASS

Addition and multiplication pass both tests, and subtraction, division and powers fail both

operation	laws that hold	checking with numbers
$a + b$	commutative and associative	$3+5=5+3$ and $(2+3)+4=2+(3+4)=9$
$a \times b$	commutative and associative	$6\times7=7\times6$ and $(2\times3)\times4=2\times(3\times4)=24$
$a - b$	neither law	$8-3=5$ but $3-8=-5$; $(10-4)-3=3$ but $10-(4-3)=9$
$a \div b$	neither law	$12\div4=3$ but $4\div12=\frac{1}{3}$; $(24\div6)\div2=2$ but $24\div(6\div2)=8$
a^b	neither law	$2^3=8$ but $3^2=9$; $(2^3)^2=64$ but $2^{(3^2)}=512$
average of a, b	commutative only	$\frac{2+4}{2}=\frac{4+2}{2}$; but $\text{avg}(\text{avg}(2,4),6)=4.5$ and $\text{avg}(2,\text{avg}(4,6))=3.5$

3 / WHY THE LAWS ARE USEFUL

Reordering and regrouping make hard sums easy

To find $25 \times 17 \times 4$ the commutative law lets us swap to $25 \times 4 \times 17$, and the associative law groups $(25 \times 4) \times 17 = 100 \times 17 = 1700$. To find $98 + 47 + 2$ we can add $98 + 2 = 100$ first and then 47 to get 147. In both sums the two numbers that combine to something friendly are brought together, which is allowed only because addition and multiplication obey the laws, and the same moves would give wrong answers for $98 - 47 - 2$ or $24 \div 6 \div 2$.

4 / A CONVENTION IS DIFFERENT FROM A LAW

Brackets and the order of operations tell us how to read a line of symbols

Reading $2 + 3 \times 4$ as 14 and not 20 is an agreement that multiplication is done before addition, made so that a line of symbols has one meaning, and brackets override it: $(2 + 3) \times 4 = 20$.
The agreement and the laws work together, since the next sheet shows that the rule "multiply before adding" is exactly what lets us split 7×13 into $7 \times 10 + 7 \times 3$ without brackets.

5 / A LAW THAT FAILS IN DAILY LIFE

Doing one thing and then another can matter

Putting on socks and then shoes is a different action from putting on shoes and then socks, so the operation "do this then that" is not commutative. It is still associative, because doing three tasks in order gives the same result however we group them in our heads, and this pair of properties (order matters and grouping does not) returns for turns, flips and matrices later in the ladder.

6 / TRY THESE

Decide which laws hold and prove a failure with one example

(a) Is $a \circ b = a + 2b$ commutative? Take $a = 1, b = 2$: $1 + 4 = 5$ but $2 + 2 = 4$, so it fails.
(b) Is "the larger of a and b" associative? $\max(\max(2, 5), 3) = 5 = \max(2, \max(5, 3))$, and it holds for every triple.
(c) Mixing red and blue paint into purple has the same result in either order, so we say mixing is commutative. One counterexample breaks a law, and many matching examples can never prove one.

ASIDE / A LAW IS A PROMISE

A law has to hold for every number, so a single counterexample ends it while any number of agreeing examples leaves it unproved.

LINKS TO OTHER SHEETS

The next sheet uses these laws to multiply by splitting numbers into parts; the modular clock in the number theory ladder is a world where both laws still hold.

Convention. $\circ$ is a placeholder for any operation; $\max$ is the larger of two numbers; avg is the average.

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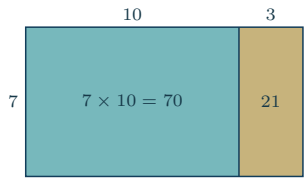
ALGEBRA LADDER 2 OF 9

Distributivity: multiplying by parts

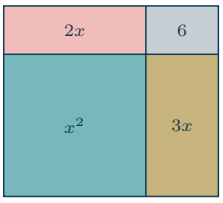
Multiplication spreads over addition, which turns a hard product into easy pieces and an expression into a factorised form.

1 / THE RULE AND THE PICTURE

A rectangle split into two pieces has the area of both pieces together



$7 \times 13 = 70 + 21 = 91$



$(x+3)(x+2) = x^2 + 5x + 6$

Splitting one side of a rectangle into b and c cuts the area into ab and ac, so

$$a(b+c) = ab + ac$$

and the same picture with two split sides gives the four pieces of $(x+3)(x+2)$.

2 / MENTAL ARITHMETIC

Choose the split that makes the pieces friendly

product	split one factor	answer
17×99	$17 \times 100 - 17$	1683
6×47	$6 \times 40 + 6 \times 7$	282
8×102	$8 \times 100 + 8 \times 2$	816
13×15	$13 \times 10 + 13 \times 5$	195
15×24	$15 \times 20 + 15 \times 4$	360

The same rule works for subtraction, as in $7 \times 9 = 7 \times 10 - 7 = 63$, which is the move behind 17×99, and the rule is the reason the column method for long multiplication works at all.

3 / EXPANDING AND FACTORISING

Reading the rule from left to right expands, and from right to left it factorises

Expanding $(x+3)(x+2)$ by the area picture gives $x^2+3x+2x+6 = x^2+5x+6$, and a check with $x = 4$ gives $7 \times 6 = 42$ and $16+20+6 = 42$. Squares work the same way: $13^2 = (10+3)^2 = 100 + 60 + 9 = 169$, with the middle 60 made of two rectangles 10×3, and the tempting $100 + 9 = 109$ misses exactly those two.

Reading right to left, $6x+9 = 3(2x+3)$ takes out the common factor 3, and this is the move used to factorise every expression in school algebra.

4 / WHAT DOES NOT SPREAD

Multiplication spreads over addition and the reverse is false

the false rule	left side	right side
$2 + (3 \times 4)$ against $(2+3) \times (2+4)$	14	30
$12 \div (2+4)$ against $12 \div 2 + 12 \div 4$	2	9
$(3+4)^2$ against $3^2 + 4^2$	49	25
$\sqrt{9+16}$ against $\sqrt{9} + \sqrt{16}$	5	7

Powers and roots do not spread over a sum either, though they do spread over products: $(2 \times 3)^2 = 2^2 \times 3^2 = 36$. Division spreads over a sum when the sum is on top, so $(12+6) \div 3 = 12 \div 3 + 6 \div 3 = 6$, and it fails when the sum is the divisor.

5 / TRY THESE

Split, expand and check by a number

(a) Find 35×102 as $35 \times 100 + 35 \times 2 = 3570$.
(b) Expand $(x+5)(x+1)$ as $x^2 + 6x + 5$ and check with $x = 2$: $7 \times 3 = 21$ and $4 + 12 + 5 = 21$.
(c) Factorise $10y + 15$ as $5(2y+3)$. **(d)** Show that $(a+b)^2 = a^2 + b^2$ is false by taking $a = b = 1$: the left side is 4 and the right side is 2.

ASIDE / ONE LAW, TWO DIRECTIONS

Because multiplication is commutative the law also holds with the sum on the left, so $(b+c)a = ba + ca$.

LINKS TO OTHER SHEETS

The previous sheet gives the laws used here; the matrix sheet later asks which of them survive when multiplication stops commuting.

Convention. $a(b+c) = ab + ac$ is the distributive law of multiplication over addition; squares and roots are written a^2 and $\sqrt{a}$.

Networked-lattice

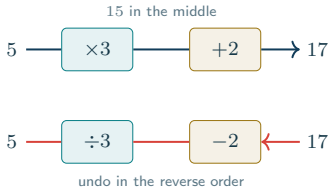
ALGEBRA LADDER 3 OF 9

Identity, inverse and undoing

Every operation has a do-nothing element and a way to reverse it, and solving an equation is the art of undoing.

1 / THE RULE AND THE PICTURE

An identity changes nothing, and an inverse brings you back to the identity



Adding 0 and multiplying by 1 leave a number alone, and every number can be brought back to those do-nothing values by an inverse:

$$a + 0 = a, \quad a + (-a) = 0, \quad a \times 1 = a, \quad a \times \tfrac{1}{a} = 1$$

The machine above takes 5 to 17 by tripling and adding 2, and the undo machine reverses the steps in the opposite order.

2 / IDENTITIES AND INVERSES

Each operation has its own pair

operation	identity	inverse of a	what the inverse does
addition	0, since $a + 0 = a$	$-a$	subtracting a
multiplication	1, since $a \times 1 = a$	$1/a$, for $a \neq 0$	dividing by a
doing one step then another	doing nothing	the step that reverses it	the undo machine

3 / SOLVING AN EQUATION IS UNDOING

Undo the last step first and keep the order reversed

To solve $3x + 2 = 17$ the machine triples and then adds 2, so we subtract 2 to get 15 and divide by 3 to get $x = 5$, which checks since $3 \times 5 + 2 = 17$.
For $\frac{x}{4} - 1 = 6$ the machine divides by 4 and subtracts 1, so we add 1 to get 7 and multiply by 4 to get $x = 28$, and indeed $\frac{28}{4} - 1 = 6$.
Doing the same thing to both sides of an equation is the balance method, and it is the undo machine written line by line.

4 / SUBTRACTION AND DIVISION IN DISGUISE

Taking away is adding the inverse, and dividing is multiplying by the reciprocal

Writing $8 - 3 = 8 + (-3)$ and $12 \div 4 = 12 \times \frac{1}{4}$ explains why the two laws failed on the first sheet: the inverse hides in front of the second number, so $3 - 8 = 3 + (-8) = -5$ and the order matters. In $10 - (4 - 3)$ the bracket flips the sign of the 3, giving $10 + (-4) + 3 = 9$, while $(10 - 4) - 3 = 10 + (-4) + (-3) = 3$.
With the inverse made visible only the commutative and associative laws of addition are in use.

5 / WHY THE NUMBERS GROW

Inverses that are missing force new kinds of number

Among the counting numbers $3 - 5$ has no answer, so the negative numbers are invented to give every number an additive inverse, and among the integers $3 \div 4$ has no answer, so fractions are invented to give every number except zero a reciprocal.
Zero has no reciprocal because $0 \times x = 0$ for every x, so no number multiplies with 0 to give 1, which is why $\frac{1}{0}$ is left undefined.

6 / TRY THESE

Find the inverse and undo the machine

(a) The inverse of $\div 5$ is $\times 5$, and the inverse of "add 7" is "subtract 7".
(b) Solve $4x - 3 = 21$: add 3 to get 24, divide by 4 to get $x = 6$.
(c) A machine squares and then adds 1: why can it not be undone uniquely? Both 3 and -3 give 10, so the undo machine would need to choose, which is why square roots carry a $\pm$.

ASIDE / THE SHAPE OF THE LADDER

A set with one operation, an identity and inverses is called a group, and the numbers under addition are the first example.

LINKS TO OTHER SHEETS

The previous sheet uses subtraction and division as examples of failing laws; the composition sheet returns to inverses that reverse the order of steps.

Convention. The inverse of a under addition is $-a$ and under multiplication is $1/a$; an equation is solved by applying inverses in reverse order.

Networked–lattice

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ALGEBRA LADDER 4 OF 9

Letters as numbers: expanding, factorising and proof

A letter stands for any number, so a line of algebra makes a claim about every number at once.

1 / THE RULE AND THE PICTURE

To show a claim holds for every number, write the number with a letter and use the laws



$5^2 = 25$ teal dots
$+ 11$ gold dots $= 6^2 = 36$

Going from a square of n dots a side to a square of $n+1$ adds an L-shaped strip of n dots down one side, n along the other and one in the corner, and the picture holds for every n:

$$(n+1)^2 = n^2 + 2n + 1$$

Expanding $(n+1)(n+1)$ with the distributive law gives the same three terms, so the dots and the algebra agree, and the algebra covers numbers too large to draw.

2 / EVEN AND ODD NUMBERS

Every even number is $2m$ and every odd number is $2m+1$

Two odd numbers add to $(2m+1)+(2n+1) = 2m+2n+2 = 2(m+n+1)$, which has a factor 2 and so is even. Two odd numbers multiply to $(2m+1)(2n+1) = 4mn+2m+2n+1 = 2(2mn+m+n)+1$, which is odd.
Each step uses only the commutative, associative and distributive laws, and each holds for every pair of odd numbers, which no list of examples could show.

3 / WHY A NUMBER TRICK WORKS

Follow the letter through the steps

step	in letters	with $n = 7$
think of a number	n	7
add 4	$n+4$	11
double it	$2n+8$	22
subtract 8	$2n$	14
halve it	n	7

Whatever number n is chosen, the steps $n \to n+4 \to 2n+8 \to 2n \to n$ return it, so the trick "think of a number, add 4, double, subtract 8, halve" always gives back the starting number. The doubling used distributivity, $2(n+4) = 2n+8$, and the halving used the inverse of doubling, so the trick is the undo machine of the previous sheet hidden behind a story.

4 / A PRODUCT FROM A DIFFERENCE

$a^2 - b^2 = (a-b)(a+b)$ turns squares into quick sums

Expanding the right side gives $a^2 + ab - ba - b^2 = a^2 - b^2$, since $ab = ba$.
This makes $51^2 - 49^2 = (51-49)(51+49) = 2 \times 100 = 200$ and $99^2 - 1 = 98 \times 100 = 9800$ immediate.
Taking $a = n+1$ and $b = n$ shows that $(n+1)^2 - n^2 = 2n+1$, so each new square adds an odd number of dots, as the gold strip showed: $8^2 - 7^2 = 15$.

5 / CHECKING IS DIFFERENT FROM PROVING

A pattern that holds many times can still fail

The expression $n^2 + n + 41$ gives a prime for every n from 0 to 39, yet at $n = 40$ it gives $40^2 + 40 + 41 = 1681 = 41^2$. Substituting numbers is a good way to catch a mistake, such as testing $(x+3)(x+2) = x^2 + 5x + 6$ at $x = 4$, and a proof with letters is what settles a claim for all numbers.

6 / TRY THESE

Use a letter and a law

(a) Show that three consecutive numbers add to a multiple of 3: $n + (n+1) + (n+2) = 3n + 3 = 3(n+1)$, as in $7 + 8 + 9 = 24 = 3 \times 8$.
(b) Show that the sum of an even and an odd number is odd: $2m + (2n+1) = 2(m+n) + 1$.
(c) Use $a^2 - b^2 = (a-b)(a+b)$ to find $1001^2 - 999^2 = 2 \times 2000 = 4000$.

ASIDE / THE POLYNOMIAL THAT FAILS

The expression $n^2 + n + 41$ is Euler's prime-generating polynomial, and the hexagonal spiral sheet explains why its run stops at 40.

LINKS TO OTHER SHEETS

The next sheet turns to operations that do not commute; the number theory ladder uses the same letters to prove statements about divisors and remainders.

Convention. n, m, a, b stand for integers; even is $2m$ and odd is $2m+1$; a proof uses letters, a check uses numbers.

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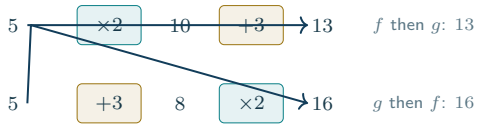
ALGEBRA LADDER 5 OF 9

Actions that do not commute: composition

Doing one thing and then another is an operation, and it keeps associativity while giving up commutativity.

1 / THE RULE AND THE PICTURE

Two machines in the opposite orders can give different answers



Let f double and g add 3. Writing $g \circ f$ for "f first, then g", the same starting number 5 ends at 13 one way and at 16 the other:

$$\boxed{g \circ f \neq f \circ g}$$

because $g(f(x)) = 2x + 3$ and $f(g(x)) = 2(x + 3) = 2x + 6$.

2 / WHICH LAWS SURVIVE

Order matters, grouping does not, and every reversible action can be undone

law for composition	verdict and example
associative	yes: $(h \circ g) \circ f = h \circ (g \circ f)$, both equal $(2x + 3)^2$ in the example
commutative	no: $g \circ f = 2x + 3$ but $f \circ g = 2x + 6$
identity	yes: the function that returns its input, $\mathrm{id}(x) = x$
inverse	yes for a function that can be reversed, and then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
distributes over $+$	only for functions like $f(x) = 2x$: $f(1 + 2) = f(1) + f(2) = 6$ but $g(1 + 2) = 6$ and $g(1) + g(2) = 9$

3 / WALKING AND TURNING

The same pair of moves in the other order ends somewhere else

Starting at the origin facing north, "turn right and then walk 10 steps" ends at the point 10 east and 0 north, while "walk 10 steps and then turn right" ends 10 north and 0 east, so the two orders land at different places.
The sock and shoe example is the same idea, since the second action only makes sense after the first.

4 / UNDOING A PAIR OF ACTIONS

The undo reverses the order

To undo "socks then shoes" we take the shoes off first and then the socks, and for the machines, $g \circ f$ turns x into $2x + 3$, so its inverse takes y to $\frac{y-3}{2}$, which is $f^{-1}(g^{-1}(y))$ with the subtraction first and the halving second. Reversing the order gives $g^{-1}(f^{-1}(y)) = \frac{y}{2} - 3$, a different answer, and the check $y = 13 \to 5$ works only for the first.

5 / TRY THESE

Compose in both orders and compare

(a) For $f(x) = x + 1$ and $g(x) = x^2$, find $g \circ f = (x + 1)^2$ and $f \circ g = x^2 + 1$, which differ at $x = 2$ as 9 and 5.
(b) Is "add 3 then add 5" the same as "add 5 then add 3"? Yes, because adding is commutative, so some pairs of actions do commute.
(c) Find a pair of machines that commute: $f(x) = 2x$ and $g(x) = 5x$ give $10x$ in either order.

ASIDE / A NEW KIND OF PRODUCT

Composition is a product of actions with an identity, inverses and associativity, so it behaves like numbers except for commutativity.

LINKS TO OTHER SHEETS

The next sheet lists every way of turning and flipping a triangle and writes the products in a table; the matrix sheet shows composition as multiplication.

Convention. $g \circ f$ means apply f first and then g; id is the identity function; f^{-1} undoes f.

Networked–lattice

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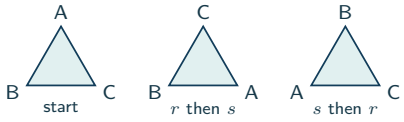
ALGEBRA LADDER 6 OF 9

Turning and flipping a triangle

Six moves of a triangle form a small world where the product of two moves is another move and the order matters.

1 / THE RULE AND THE PICTURE

Turn by a third of a circle and flip across the vertical, and the order changes the result



Let r turn the triangle by $120°$ anticlockwise and s flip it left to right. The symbols sr and rs mean the move made by doing the right one first, as with functions, and

$$sr \neq rs$$

because the labelled corners end in different places, though both products are again one of the six moves.

2 / THE TABLE OF ALL PRODUCTS

Every row and column contains each of the six moves exactly once

	e	r	r^2	s	sr	sr^2
e	e	r	r^2	s	sr	sr^2
r	r	r^2	e	sr^2	s	sr
r^2	r^2	e	r	sr	sr^2	s
s	s	sr	sr^2	e	r	r^2
sr	sr	sr^2	s	r^2	e	r
sr^2	sr^2	s	sr	r	r^2	e

row move done after column move

The six moves are doing nothing (e), the turns r and r^2 ($240°$), and the three flips s, sr and sr^2. Reading the table, $sr = r^2 s$ and $r^3 = e$ and $s^2 = e$, so a flip undoes itself and the turn r is undone by r^2.
The table is a Latin square, which is the footprint of inverses: to get any move from any other there is exactly one move to apply.

3 / THE LAWS THAT HOLD

Closed, associative, with identity and inverses

Every product of two moves is a move, so the set is closed, and doing nothing is the identity. Every move has an inverse, since $r^{-1} = r^2$ and each flip is its own inverse. Associativity holds for all 216 triples of moves, and it holds in general because moves are composed like functions.
A set with these three properties is a group, and this one is called D_3.

4 / WHO COMMUTES WITH WHOM

Exactly half of the ordered pairs commute

Of the 36 ordered pairs of moves exactly 18 commute: the three turns e, r, r^2 commute with each other, as flat rotations always do, and each move commutes with itself and with the identity, but a flip and a turn never commute unless the turn is the identity, and two different flips do not commute either.
The rule $sr = r^{-1}s$ says that passing a flip across a turn reverses the turn.

5 / TRY THIS

Cut out a paper triangle and label the corners

Colour the corners A, B, C and apply r then s, then start again with s then r, and compare the corner order with the picture.
A second task is to find which single move undoes sr: since $(sr)(sr) = e$ the move sr is its own inverse, because it is a flip, so doing it twice returns the triangle.

ASIDE / THE SMALLEST NON-COMMUTATIVE GROUP

With six elements this is the smallest group where order matters, and the same group is the six ways of arranging three objects.

LINKS TO OTHER SHEETS

The turns alone behave like clock arithmetic in the number theory ladder, and the next sheet writes turns and flips as matrices.

Convention. e is doing nothing, r a turn of $120°$ anticlockwise, s a left to right flip; sr means r first and then s; D_3 is the dihedral group of the triangle.

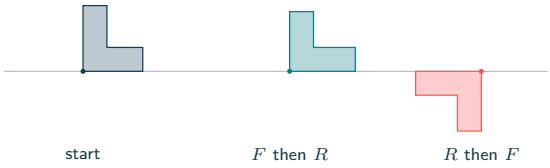
ALGEBRA LADDER 7 OF 9

Matrices: machines that do not commute

A matrix is a machine on the plane, the product of two matrices is one machine after another, and the order changes the machine.

1 / THE RULE AND THE PICTURE

Choose R for rotation, F for flip (reflection), and S for shear



Dots mark separate origins; the L-shapes are spaced for comparison.

Here R turns $90°$ anticlockwise about the origin; F reflects in the x-axis. Read products from right to left: RF means F first, then R.

$$\boxed{RF \neq FR}$$

RF reflects in $y = x$; FR reflects in $y = -x$. The two final shapes differ by a $180°$ turn.

2 / THREE MACHINES

The columns of a matrix say where the two unit steps go

name	machine	matrix	columns: where the unit steps go
R	turn $90°$ anticlockwise	$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	$(1,0) \to (0,1)$, $(0,1) \to (-1,0)$
F	reflect in the x-axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$(1,0) \to (1,0)$, $(0,1) \to (0,-1)$
S	shear to the right	$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	$(1,0) \to (1,0)$, $(0,1) \to (1,1)$

To multiply, go along a row of the left matrix and down a column of the right one: RF has top row $(0 \cdot 1 + (-1) \cdot 0,\ 0 \cdot 0 + (-1)(-1)) = (0,1)$ and bottom row $(1 \cdot 1 + 0 \cdot 0,\ 1 \cdot 0 + 0 \cdot (-1)) = (1,0)$.

3 / WHICH SCHOOL LAWS STILL HOLD

Associativity and distributivity survive and four other habits fail

school law	holds for matrices?	example
associative	yes	$(RF)S = R(FS) = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$
distributive	yes	$R(F+S) = RF + RS = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}$
commutative	no	$RF = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ but $FR = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$
$(A+B)^2 = A^2 + 2AB + B^2$	no	$(R+F)^2 = 0$ but $R^2 + 2RF + F^2 = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$
product zero forces a zero factor	no	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = 0$
cancel A from $AB = AC$	no	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ gives $AB = AC$ for $B \neq C$

4 / WHAT THE FAILURES COST

Expanding, cancelling and solving take care

Because $AB \neq BA$ the expansion of $(A+B)^2$ has two middle terms $AB + BA$, and in the example they cancel exactly, leaving $(R+F)^2 = 0$ for two machines that are far from zero. Because a product can vanish with both factors non-zero, $AB = 0$ does not give $A = 0$ or $B = 0$, and an equation $AX = B$ can be solved by multiplying by the inverse only when one exists, which is when the determinant is not zero: $\det R = 1$, $\det F = -1$, $\det S = 1$. When an inverse does exist the undo rule of the composition sheet returns, $(AB)^{-1} = B^{-1}A^{-1}$.

5 / TRY THESE

Multiply in both orders and compare

(a) Find $FS = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ and $SF = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$, which differ. **(b)** Find two matrices that commute: S and $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ commute because both are shears. **(c)** Check that $R^4 = I$, since four quarter turns return every point.

ASIDE / FROM TRIANGLES TO MATRICES

The six moves of the triangle are six matrices, and the same table of products appears with r a turn by $120°$ and s a reflection.

LINKS TO OTHER SHEETS

The linear algebra ladder develops matrices as transformations and inverses; the next sheet asks how rotations in space behave.

Convention. R: $90°$ anticlockwise rotation about the origin; F: flip in the x-axis; S: shear. AB means B first; I is the identity.

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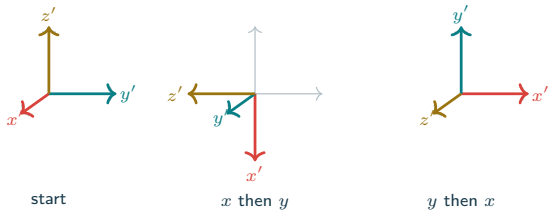
ALGEBRA LADDER 8 OF 9

Rotations in the plane and in space

Turns in the plane commute, turns in space do not, and quaternions keep track of the difference.

1 / THE RULE AND THE PICTURE

Turn a quarter about the x-axis and then a quarter about the y-axis, or the other way round



start x then y y then x

In the plane a turn by θ followed by a turn by φ is a turn by $\theta+\varphi$ in either order, but in space

$$R_y(90^\circ)R_x(90^\circ) \neq R_x(90^\circ)R_y(90^\circ)$$

and the pictures show where the three axes end up in each order.

2 / THE MATRICES

Quarter turns in space, and their two products

$$R_x=\begin{pmatrix}1&0&0\\0&0&-1\\0&1&0\end{pmatrix},\quad R_y=\begin{pmatrix}0&0&1\\0&1&0\\-1&0&0\end{pmatrix},\quad R_yR_x=\begin{pmatrix}0&1&0\\0&0&-1\\-1&0&0\end{pmatrix},\quad R_xR_y=\begin{pmatrix}0&0&1\\1&0&0\\0&1&0\end{pmatrix}.$$

In the plane, $R(30^\circ)R(60^\circ)=R(60^\circ)R(30^\circ)=R(90^\circ)$ for every order, because the angles simply add, and this is the reason complex numbers can be multiplied in either order: multiplying by $\cos\theta+i\sin\theta$ is a turn in the plane.

3 / ROTATION PLUS TRANSLATION

Turning about the origin and sliding do not commute either

Rotate the point $(1,0)$ by 90° about the origin and then slide it by $(2,0)$ and it lands at $(2,1)$, while sliding first to $(3,0)$ and then rotating sends it to $(0,3)$.

Moves that combine turns with slides, the motions of a rigid body, therefore form another non-commuting system, and a computer handles them as a single larger matrix.

4 / QUATERNIONS

Three square roots of -1 that do not commute

Hamilton's quaternions have $i^2=j^2=k^2=ijk=-1$, so that $ij=k$ and $ji=-k$. A turn of 90° about a unit axis is the quaternion $\frac{1}{\sqrt{2}}(1+\text{axis})$, and a vector v is turned to qvq^{-1}, which for $q=\frac{1}{\sqrt{2}}(1+k)$ sends i to j.

The two quarter turns above become $q_yq_x=\frac{1}{2}(1+i+j-k)$ and $q_xq_y=\frac{1}{2}(1+i+j+k)$, which differ only in the sign of k and are the two rotations of 120° about the axes $(1,1,-1)$ and $(1,1,1)$.

5 / TRY THESE

Compare the orders

(a) Check $ij=k$ and $ji=-k$ with the rule $(a+bi)(c+di)$ extended by $ij=k$; swapping the order flips the sign of k.
(b) Hold a book flat and give it a quarter turn about the vertical and then about the long edge, then reverse the order, and compare where the cover points.
(c) Check that two turns about the same axis commute: $R_x(90^\circ)R_x(180^\circ)=R_x(180^\circ)R_x(90^\circ)=R_x(270^\circ)$.

ASIDE / WHY THREE DIMENSIONS CHANGE THE RULES

The only ways to multiply arrays of real numbers with division are the reals, the complex numbers and the quaternions, and commutativity is lost at the quaternions (Frobenius).

LINKS TO OTHER SHEETS

The previous sheet shows two dimensional matrices that fail to commute, and the last sheet collects every structure met in the ladder and the laws each one keeps.

Convention. R_x, R_y turn about the axes; R_yR_x means R_x acts first; q unit quaternion; x', y', z' are the images of the axes after the turns.

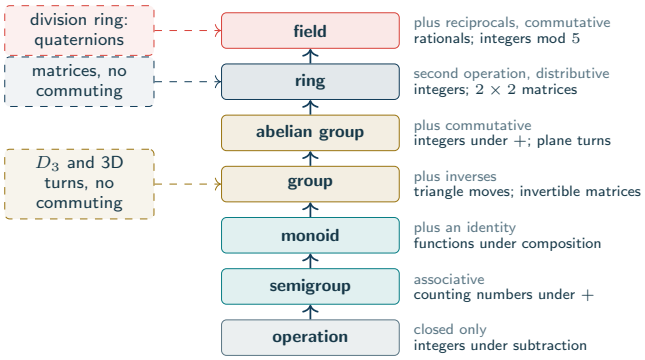
ALGEBRA LADDER 9 OF 9

The map: which laws hold where

Every structure in the ladder is a list of which laws it keeps, and each law buys one thing you can do.

1 / THE RULE AND THE PICTURE

Add one law at a time and a new kind of structure appears



Test a new operation by asking its questions in this order: is it associative, does it have an identity, does every element have an inverse, and does it commute, and with a second operation ask whether it distributes and whether non-zero elements have reciprocals.

laws kept $\Longrightarrow$ name

2 / THE STRUCTURES MET ON THE LADDER

Associativity is almost always kept, and commutativity is the first law to go

structure and operation	associative	commutative	inverses	name
counting numbers, $+$	yes	yes	no	semigroup, no identity
integers, $-$	no	no	no	not even associative
integers, $+$	yes	yes	yes	abelian group
non-zero rationals, $\times$	yes	yes	yes	abelian group
functions, composition	yes	no	yes	monoid; reversible ones form a group
triangle moves D_3	yes	no	yes	group, not abelian
plane turns	yes	yes	yes	abelian group
space turns	yes	no	yes	group, not abelian
2×2 matrices, $\times$	yes	no	only if $\det \neq 0$	monoid; invertible ones form a group
non-zero quaternions, $\times$	yes	no	yes	group, not abelian

3 / WHAT EACH LAW BUYS

A law is a licence to do something

Associativity lets us drop brackets in abc, an identity gives a place to start, and inverses let us solve $a \circ x = b$ by $x = a^{-1} \circ b$. Commutativity lets us reorder freely, and when it fails the side matters: for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ the equation $AX = B$ has solution $X = A^{-1}B = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$ while $XA = B$ gives $X = BA^{-1} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$. Distributivity ties two operations together, and reciprocals allow division.

4 / RINGS AND FIELDS IN SMALL WORLDS

Clock arithmetic shows how a field and a ring differ

On a clock with 5 hours every non-zero number has a reciprocal, since $2 \times 3 = 6 \equiv 1$, so it is a field and equations $ax = b$ are always solvable. On a clock with 6 hours $2 \times 3 = 6 \equiv 0$, so two non-zero numbers multiply to zero and 2 has no reciprocal, which makes it a ring only. The 2×2 matrices behave like the 6-hour clock in this respect, with zero products of non-zero matrices, and the quaternions behave like the 5-hour clock except that they do not commute.

5 / CLASSIFY A NEW OPERATION

Define $a \circ b = a + b + ab$ on the integers

It commutes and associates because $a \circ b = (1 + a)(1 + b) - 1$, which is ordinary multiplication seen through the shift $a \mapsto 1 + a$, and it has identity 0 since $a \circ 0 = a$. The inverse of a would be $-\frac{a}{1+a}$, which is an integer only for $a = 0$ and $a = -2$, so the operation forms a commutative monoid, and the shift explains each law at once.

6 / TRY THESE

Name the structure

(a) The six moves of a triangle under composition form a group that is not abelian. **(b)** The integers under $+$ and $\times$ are a commutative ring and not a field, since $\frac{1}{2}$ is missing. **(c)** Check the claim "associativity holds for matrices" with $(RF)S = R(FS)$ from the matrix sheet, and check that commutativity fails with $RF \neq FR$.

ASIDE / THE WHOLE LADDER

The nine sheets climb from addition and multiplication of whole numbers, through composition of actions and matrices, to rotations in space.

LINKS TO OTHER SHEETS

The ladder runs: 1 operations and their laws, 2 distributivity, 3 identity and inverse, 4 letters and proof, 5 composition, 6 triangle moves, 7 matrices, 8 rotations and quaternions, and this map.

Convention. Group: associative with identity and inverses; abelian: also commutative; ring: two operations with distributivity; field: a commutative ring where non-zero elements have reciprocals; D_3 is the symmetry group of the triangle.

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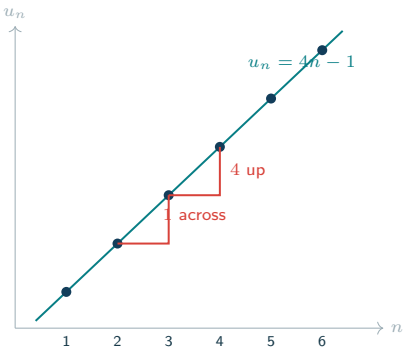
SEQUENCES LADDER 1 OF 9

Sequences, rules and first differences

The step between neighbouring terms is the first difference, and a constant step gives a straight-line rule.

1 / THE RULE AND THE PICTURE

A constant first difference d gives the position-to-term rule $u_n = dn + (u_1 - d)$



A sequence is a list $u_1, u_2, u_3, \ldots$ and its first differences are $\Delta u_n = u_{n+1} - u_n$. When every first difference equals d the terms climb by the same amount each step, so the points lie on a straight line of gradient d, and

$$\boxed{\Delta u_n = d \implies u_n = dn + (u_1 - d)}$$

For $3, 7, 11, 15, \ldots$ the difference is 4 and $u_1 - d = 3 - 4 = -1$, so $u_n = 4n - 1$.

2 / TWO KINDS OF RULE

Add the same step each time, or say the term from its position

sequence	term to term	position to term	first difference
$3, 7, 11, 15, \ldots$	add 4	$4n - 1$	$+4$
$10, 7, 4, 1, \ldots$	subtract 3	$13 - 3n$	-3
$2, 5, 8, 11, \ldots$	add 3	$3n - 1$	$+3$
$\frac{1}{2}, 1, 1\frac{1}{2}, 2, \ldots$	add $\frac{1}{2}$	$\frac{n}{2}$	$+\frac{1}{2}$

3 / FINDING THE RULE IN THREE STEPS

Differences first, then compare with dn

Write down the differences, and if they are all the same number d the rule begins with dn.
Then compare dn with the actual terms and note what must be added or taken away: for $5, 8, 11, 14, \ldots$ the differences are 3, the multiples of 3 are $3, 6, 9, 12$, and each term is 2 more, so $u_n = 3n + 2$.
Check with $n = 1$ and $n = 2$, which give 5 and 8.

4 / THE DIFFERENCE IS A GRADIENT

The first picture of calculus

On the graph of u_n against n the neighbouring points are 1 apart across, so the difference is the rise divided by the run, the gradient of the line between them.
For $4n - 1$ the gradient is 4 everywhere, but for $1, 4, 9, 16$ the differences $3, 5, 7$ keep growing, so the gradient changes and the graph bends, which is what the next sheet measures with second differences.

5 / A TRAP

Two or three terms can fit several rules

The terms $1, 2, 4$ continue as 8 if each is doubled, and as 7 if the differences go $1, 2, 3$, which gives $1, 2, 4, 7, 11$.
A rule found from differences must be tested on more terms than it uses, and a safe habit is to check that the differences really are constant over at least three steps.

6 / TRY THESE

Find the rule

(a) For $7, 10, 13, 16, \ldots$ the difference is 3 and the rule is $3n + 4$. **(b)** For $20, 17, 14, 11, \ldots$ the difference is -3 and the rule is $23 - 3n$. **(c)** Is $2, 5, 10, 17$ a straight-line sequence? The differences $3, 5, 7$ are not constant, so no.

ASIDE / TWO POINTS FIX A LINE

Two terms are enough to find a straight-line rule, and in general $k+1$ terms fix a polynomial of degree k, as the interpolation sheet shows.

LINKS TO OTHER SHEETS

The next sheet takes the differences of the differences; the polynomial interpolation sheet gives the same rules by Vandermonde and Newton.

Convention. u_n is the nth term; $\Delta u_n = u_{n+1} - u_n$ is the first difference; d is a constant difference.

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SEQUENCES LADDER 2 OF 9

Second differences and quadratic rules

A constant second difference D signals a quadratic rule $u_n = \frac{D}{2}n^2 + bn + c$.

1 / THE RULE AND THE PICTURE

If the second differences all equal D then $a = D/2$

u_n	2	6	12	20	30
Δ	4	6	8	10	
Δ^2	2	2	2		

The first differences $4, 6, 8, 10$ are not constant, but their own differences, the second differences, are all 2, and

$$\Delta^2 u_n = D \implies u_n = \tfrac{D}{2}n^2 + bn + c$$

Here $D = 2$, so the rule begins with $1 \cdot n^2$, and the remaining part is found by subtracting n^2 from every term.

2 / THE RECIPE

Subtract the square part and read off the straight line left over

For $2, 6, 12, 20, 30$ the second difference is 2, so $a = 1$, and subtracting $n^2 = 1, 4, 9, 16, 25$ leaves $1, 2, 3, 4, 5$, which is n, so $u_n = n^2 + n$.
For $5, 12, 23, 38, 57$ the first differences are $7, 11, 15, 19$ and the second differences are 4, so $a = 2$, and subtracting $2n^2 = 2, 8, 18, 32, 50$ leaves $3, 4, 5, 6, 7 = n + 2$, so $u_n = 2n^2 + n + 2$.

3 / WHY $D = 2a$

Differences of a quadratic are a straight line

	formula	meaning
$u_n = an^2 + bn + c$	$an^2 + bn + c$	
first difference	$a(2n+1) + b = 2an + (a+b)$	grows by $2a$ each step
second difference	$2a$	constant, so $D = 2a$

The step from an^2 to $a(n+1)^2$ is $a(2n+1)$, which itself grows by $2a$ each step, so the second difference is $2a$ whatever b and c are.

4 / FAMOUS QUADRATIC SEQUENCES

Triangular, square and pentagonal numbers

The triangular numbers $1, 3, 6, 10, 15$ have $D = 1$, so $a = \frac{1}{2}$, and subtracting $\frac{1}{2}n^2$ leaves $\frac{1}{2}n$, giving $\frac{n(n+1)}{2}$.
The squares $1, 4, 9, 16, 25$ have $D = 2$ and rule n^2. The pentagonal numbers $1, 5, 12, 22, 35$ have first differences $4, 7, 10, 13$ and $D = 3$, so $a = \frac{3}{2}$ and subtracting $\frac{3}{2}n^2$ leaves $-\frac{1}{2}n$, giving $\frac{3n^2-n}{2}$.

5 / WHAT THE SECOND DIFFERENCE MEASURES

How fast the gradient is changing

The first differences are the gradients between neighbouring points, and the second difference says how much the gradient grows each step, which for a quadratic is the same everywhere, $2a$.
This is the discrete form of the curvature of a parabola, and in calculus it becomes the second derivative.

6 / TRY THESE

Use the recipe

(a) For $4, 9, 16, 25, 36$ the second difference is 2, so $n^2 + 2n + 1 = (n+1)^2$. **(b)** For $0, 3, 8, 15, 24$ it is 2, and subtracting n^2 leaves $-1, -1, -1$, so $n^2 - 1$. **(c)**
For $3, 5, 9, 15, 23$ the differences $2, 4, 6, 8$ give $D = 2$, and subtracting n^2 leaves $2, 1, 0, -1, -2 = 3 - n$, so $n^2 - n + 3$.

ASIDE / TWO LEVELS DEEP

A quadratic needs three terms to fix its three constants and a fourth to confirm that the second differences really are constant.

LINKS TO OTHER SHEETS

The previous sheet gives first differences; the next sheet goes to third differences and cubics.

Convention. $\Delta^2 u_n = u_{n+2} - 2u_{n+1} + u_n$ is the second difference; D is its constant value; $a = D/2$ is the coefficient of n^2.

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SEQUENCES LADDER 3 OF 9

Cubics and higher differences

Differences of a degree k rule settle at the constant $k!$ times the leading coefficient.

1 / THE RULE AND THE PICTURE

A rule of degree k has constant k-th differences equal to $k!\,a$

u_n	4	15	48	115	228
Δ	11	33	67	113	
Δ^2	22	34	46		
Δ^3	12	12			

For the terms $4, 15, 48, 115, 228$ the third differences are the constant 12, and

$$\Delta^k u_n = k!\,a_k \quad \text{for degree } k$$

so $a_3 = 12/3! = 2$ and the rule starts $2n^3$. The same peeling as for quadratics finishes the job.

2 / PEELING OFF ONE DEGREE AT A TIME

Subtract the leading term and look at what is left

Subtracting $2n^3 = 2, 16, 54, 128, 250$ from $4, 15, 48, 115, 228$ leaves $2, -1, -6, -13, -22$, whose second differences are the constant -2, so $a = -1$ and the next term is $-n^2$.
Adding $n^2 = 1, 4, 9, 16, 25$ gives $3, 3, 3, 3, 3$, so the rule is $u_n = 2n^3 - n^2 + 3$. Each pass removes one power, and the process stops when a constant is left.

3 / WHY $k!$ APPEARS

Each difference lowers the degree and multiplies by the old degree

degree	rule	constant difference
degree 1	$u_n = an + b$	$\Delta u = a = 1!\,a$
degree 2	$u_n = an^2 + \dots$	$\Delta^2 u = 2a = 2!\,a$
degree 3	$u_n = an^3 + \dots$	$\Delta^3 u = 6a = 3!\,a$
degree k	$u_n = an^k + \dots$	$\Delta^k u = k!\,a$

The step from n^k to $(n+1)^k$ is $kn^{k-1} + \dots$, a polynomial of degree $k-1$ with leading coefficient k, and repeating k times multiplies the coefficients $k(k-1)\cdots 1 = k!$.

4 / A SUM THAT TURNS OUT CUBIC

Adding up squares

The running totals of the squares $1, 5, 14, 30, 55, 91$ have first differences $4, 9, 16, 25, 36$, second differences $5, 7, 9, 11$ and third differences $2, 2, 2$, so the totals follow a cubic with leading coefficient $2/3! = \frac{1}{3}$, which matches $\frac{1}{3}n^3$ in the formula $\frac{n(n+1)(2n+1)}{6}$ for the sum of the first n squares.

5 / HOW MANY TERMS ARE ENOUGH

One more than the number of unknowns

A cubic has four coefficients, and checking that the third differences are constant needs two of them, so 5 terms are the least that confirm a cubic.
If the differences never settle, as for $1, 2, 4, 8, 16$ whose differences are $1, 2, 4, 8$ again, the rule is not a polynomial, and the last sheet shows it is exponential.

6 / TRY THESE

Find the degree and the leading coefficient

(a) For $1, 8, 27, 64, 125, 216$ the third differences are 6, so $a = 6/6 = 1$ and the rule is n^3. **(b)** For $1, 16, 81, 256, 625$ the fourth powers have fourth differences 24, so $a = 24/4! = 1$. **(c)**
For $3, 10, 29, 66, 127$ the third difference is 6 and subtracting n^3 leaves $2, 2, 2, 2, 2$, so $n^3 + 2$.

ASIDE / THE FACTORIAL

The factor $k!$ is the same one that appears when a power is differentiated k times, $\frac{d^k}{dx^k}x^k = k!$.

LINKS TO OTHER SHEETS

The next sheet turns the difference table into a formula, and later sheets compare differences with derivatives.

Convention. Δ^k is the k-th difference; a_k is the coefficient of n^k; $k! = k(k-1)\cdots 1$.

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SEQUENCES LADDER 4 OF 9

Newton's formula: the table as a rule

The first entry of each difference row is a coefficient, and binomial coefficients turn the table into a formula.

1 / THE RULE AND THE PICTURE

Read down the left-hand diagonal and multiply by binomial coefficients

u_n	5	12	23	38	57
Δ	7	11	15	19	
Δ^2	4	4	4		

Take the first entry of each row, the red diagonal $5, 7, 4$, and use it with the binomial coefficients $\binom{n-1}{k}$:

$$u_n = u_1 + \Delta u_1 \binom{n-1}{1} + \Delta^2 u_1 \binom{n-1}{2} + \cdots$$

Here $u_n = 5 + 7(n-1) + 4\binom{n-1}{2} = 2n^2 + n + 2$.

2 / THE BINOMIAL BUILDING BLOCKS

Each $\binom{n-1}{k}$ is a column of Pascal's triangle

$n =$	1	2	3	4	5	6
$\binom{n-1}{0}$	1	1	1	1	1	1
$\binom{n-1}{1}$	0	1	2	3	4	5
$\binom{n-1}{2}$	0	0	1	3	6	10
$\binom{n-1}{3}$	0	0	0	1	4	10

The block $\binom{n-1}{0}$ is always 1, $\binom{n-1}{1} = n-1$, $\binom{n-1}{2} = \frac{(n-1)(n-2)}{2}$ is $0, 0, 1, 3, 6, 10$, and each block is zero until its own turn, which is why the formula fits the first terms one at a time.

3 / TWO WORKED EXAMPLES

Quadratic and cubic

For $5, 12, 23, 38, 57$ the top diagonal is $5, 7, 4$, so $u_n = 5 + 7(n-1) + 2(n-1)(n-2) = 2n^2 + n + 2$, which agrees with the peeling method.
For $4, 15, 48, 115, 228$ the diagonal is $4, 11, 22, 12$, so $u_n = 4 + 11(n-1) + 22\binom{n-1}{2} + 12\binom{n-1}{3} = 2n^3 - n^2 + 3$, found without ever subtracting a power.

4 / WHY IT WORKS

Differencing a binomial lowers its index

Pascal's rule $\binom{m+1}{k} - \binom{m}{k} = \binom{m}{k-1}$ says that taking the difference of $\binom{m}{k}$ gives $\binom{m}{k-1}$, so $\Delta\binom{n-1}{k} = \binom{n-1}{k-1}$, exactly as differentiating $x^k/k!$ gives $x^{k-1}/(k-1)!$.
Applying Δ to the whole formula reproduces the row of differences, so it must be the right rule.

5 / RUNNING THE TABLE BACKWARDS

The formula also gives u_0

Putting $n = 0$ in the first example uses $\binom{-1}{k} = (-1)^k$ and gives $5 - 7 + 4 = 2$, the term that comes before 5, since running the first differences $7, 11, \ldots$ backwards gives $7 - 4 = 3$ and $5 - 3 = 2$.
The same rule with a formula in n lets us ask for $u_{1.5}$ or u_{-3}, which a list of terms cannot.

6 / TRY THESE

Use the diagonal

(a) For $1, 4, 9, 16, 25$ the diagonal is $1, 3, 2$, so $u_n = 1 + 3(n-1) + 2\binom{n-1}{2} = n^2$. **(b)** For $3, 5, 9, 15, 23$ the diagonal is $3, 2, 2$, so $3 + 2(n-1) + 2\binom{n-1}{2} = n^2 - n + 3$.
(c) What is the top diagonal of $2, 2, 2, 2$? $2, 0$, so the rule is constant.

ASIDE / THE CALCULUS TWIN

Newton's formula in binomials is the discrete twin of the Taylor series in powers $x^k/k!$, with differences in place of derivatives.

LINKS TO OTHER SHEETS

The polynomial interpolation sheet derives the same Newton form from divided differences, and the Worpitsky page shows the jet at $n = 0$.

Convention. $\binom{n-1}{k} = \frac{(n-1)(n-2)\cdots(n-k)}{k!}$; the top diagonal is $u_1, \Delta u_1, \Delta^2 u_1, \ldots$; Newton's forward difference formula.

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SEQUENCES LADDER 5 OF 9

Power sequences and their sums

Adding the p-th powers gives a polynomial of degree $p+1$, found from a few terms by differences.

1 / THE RULE AND THE PICTURE

The running total of k^p is a polynomial of degree $p+1$ in n



$1^2 = 1$ $2^2 = 4$ $3^2 = 9$ $4^2 = 16$

layers of a square pyramid: $1 + 4 + 9 + 16 = 30$

Adding a new term k^p to the total changes it by k^p, so the first difference of the total is the polynomial n^p of degree p, and the total must have degree $p+1$:

$$\Delta \sum_{k \le n} k^p = (n+1)^p$$

The totals of the squares, $1, 5, 14, 30, \ldots$, therefore have constant third differences.

2 / THE STANDARD RESULTS

Four sums with their degrees

sum	formula	$n = 10$	degree
$\sum k$	$\frac{n(n+1)}{2}$	55	quadratic
$\sum k^2$	$\frac{n(n+1)(2n+1)}{6}$	385	cubic
$\sum k^3$	$\left(\frac{n(n+1)}{2}\right)^2$	3025	quartic
$\sum k^4$	$\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$	25333	degree 5

3 / FINDING $\sum k^2$ BY DIFFERENCES

Take the table of the totals and apply Newton's formula

The totals $1, 5, 14, 30, 55$ have first differences $4, 9, 16, 25$, second differences $5, 7, 9$ and third differences $2, 2$, so the top diagonal is $1, 4, 5, 2$ and $S_n = 1 + 4(n-1) + 5\binom{n-1}{2} + 2\binom{n-1}{3}$, which expands to $\frac{n(n+1)(2n+1)}{6}$.
A sum of p-th powers has degree $p+1$ and needs $p+3$ totals to confirm it.

4 / FINDING IT BY TELESCOPING

The cube trick

Since $(k+1)^3 - k^3 = 3k^2 + 3k + 1$, adding for $k = 1$ to n makes the left side collapse to $(n+1)^3 - 1$, so $(n+1)^3 - 1 = 3\sum k^2 + 3 \cdot \frac{n(n+1)}{2} + n$, and solving gives $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$.
The same step with fourth powers gives $\sum k^3$, and each sum comes from the sums below it.

5 / A FIRST GLIMPSE OF THE INTEGRAL

The leading term is $\frac{n^{p+1}}{p+1}$

The leading coefficients are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$, so $\sum k^p \approx \frac{n^{p+1}}{p+1}$ for large n, which is the same $\frac{x^{p+1}}{p+1}$ that integration gives for the area under x^p. The next sheets show why.

6 / TRY THESE

Use the formulae

(a) Find $1 + 2 + \cdots + 100 = \frac{100 \cdot 101}{2} = 5050$. **(b)** Find $1^2 + \cdots + 10^2 = \frac{10 \cdot 11 \cdot 21}{6} = 385$. **(c)** Check that $1^3 + \cdots + 5^3 = 225 = 15^2$, the square of $1 + \cdots + 5$.

ASIDE / A SURPRISE

The sum of the first n cubes is the square of the sum of the first n integers, which the difference table of the totals shows without any trick.

LINKS TO OTHER SHEETS

The previous sheet supplies Newton's formula; the next sheet finds a basis in which the sums are even simpler.

Convention. $S_n = \sum_{k=1}^{n} k^p$; the formulae are exact for every n; $\binom{n-1}{k}$ is a binomial coefficient.

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SEQUENCES LADDER 6 OF 9

Falling factorials: the powers of finite calculus

The products $n(n-1)(n-2)\cdots$ difference as neatly as x^k differentiates.

1 / THE RULE AND THE PICTURE

The difference of $n^{\underline{k}}$ is $k\,n^{\underline{k-1}}$

$n^{\underline{3}}$	0	0	0	6	24	60	120
Δ		0	0	6	18	36	60

$\Delta n^{\underline{3}} = 3n^{\underline{2}}$: $0, 0, 6, 18, 36, 60$

Write $n^{\underline{k}} = n(n-1)\cdots(n-k+1)$, a falling factorial with k factors. Then $(n+1)^{\underline{k}} - n^{\underline{k}} = n^{\underline{k-1}}\big[(n+1) - (n-k+1)\big]$, so

$$\boxed{\Delta\, n^{\underline{k}} = k\, n^{\underline{k-1}}}$$

the exact copy of $\frac{d}{dx}x^k = kx^{k-1}$, with no extra terms.

2 / THE TABLE OF VALUES

Each row is the difference of the row below, times the index

$n =$	0	1	2	3	4	5	6
$n^{\underline{1}}$	0	1	2	3	4	5	6
$n^{\underline{2}}$	0	0	2	6	12	20	30
$n^{\underline{3}}$	0	0	0	6	24	60	120
$n^{\underline{4}}$	0	0	0	0	24	120	360

For example the row of $n^{\underline{3}}$ has differences $0, 0, 6, 18, 36, 60$, which is 3 times the row of $n^{\underline{2}}$, namely $3\cdot 0, 3\cdot 0, 3\cdot 2, 3\cdot 6, 3\cdot 12, 3\cdot 20$.

3 / CHANGING BETWEEN POWERS AND FALLING FACTORIALS

The coefficients are the Stirling numbers of the second kind

Every power is a sum of falling factorials with whole-number coefficients: $n^2 = n^{\underline{2}} + n^{\underline{1}}$, $n^3 = n^{\underline{3}} + 3n^{\underline{2}} + n^{\underline{1}}$ and $n^4 = n^{\underline{4}} + 6n^{\underline{3}} + 7n^{\underline{2}} + n^{\underline{1}}$. A check at $n = 4$ gives $256 = 24 + 144 + 84 + 4$, and the numbers $1, 6, 7, 1$ are a row of the Stirling numbers of the second kind.

4 / SUMMING IS AS EASY AS INTEGRATING

$$\sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}$$

Because $\Delta \frac{n^{\underline{k+1}}}{k+1} = n^{\underline{k}}$, adding the terms up to $n-1$ collapses to $\frac{n^{\underline{k+1}}}{k+1}$, just as $\int x^k\,dx = \frac{x^{k+1}}{k+1}$.

To add squares, write $i^2 = i^{\underline{2}} + i^{\underline{1}}$ and sum: $\sum_{i=0}^{n-1} i^2 = \frac{n^{\underline{3}}}{3} + \frac{n^{\underline{2}}}{2}$, which at $n = 5$ is $20 + 10 = 30 = 0 + 1 + 4 + 9 + 16$.

5 / THE DICTIONARY BEGINS

Differences and derivatives share a rule

A falling factorial plays the part of a power, the difference plays the part of the derivative, and the sum $\sum_{i=0}^{n-1}$ plays the part of the integral from 0 to n.

The formulae look identical, which is why finite differences make a gentle first course in calculus.

6 / TRY THESE

Use the rules

(a) Find $\Delta\big(n(n-1)(n-2)\big) = 3n(n-1)$, which at $n = 4$ is 36, and indeed $60 - 24 = 36$ for $n = 4 \to 5$. **(b)** Write n^3 as falling factorials and sum: $\sum_{i=0}^{n-1} i^3 = \frac{n^{\underline{4}}}{4} + n^{\underline{3}} + \frac{n^{\underline{2}}}{2}$, which at $n = 4$ is $6 + 24 + 6 = 36 = 0 + 1 + 8 + 27$. **(c)** Check $\Delta(n^2) = 2n + 1$ with $n^2 = n^{\underline{2}} + n$: $2n + 1$.

ASIDE / STIRLING NUMBERS

The coefficients $1, 3, 1$ and $1, 6, 7, 1$ count the ways of splitting a set into blocks, and multiplying by $k!$ gives the Worpitzky numbers $1, 14, 36, 24$ of the interpolation sheet.

LINKS TO OTHER SHEETS

The previous sheet gives sums of powers by differences; the next sheets let the step between terms shrink to zero.

Convention. $n^{\underline{k}} = n(n-1)\cdots(n-k+1)$ is a falling factorial; $\Delta f(n) = f(n+1) - f(n)$; the sum runs from $i = 0$ to $n-1$.

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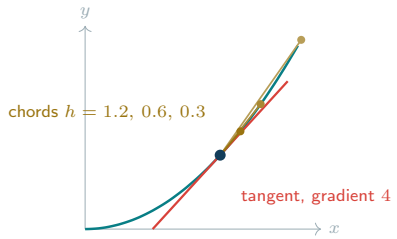
SEQUENCES LADDER 7 OF 9

From differences to the derivative

Shrinking the step in a difference quotient to zero turns the gradient of a chord into the gradient of the curve.

1 / THE RULE AND THE PICTURE

The difference quotient with a smaller and smaller step h gives the derivative



For a rule $f(x)$ the first difference with step h divided by h is the gradient of the chord from x to $x + h$, and as the step shrinks the chords settle on the tangent:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Up to now h was 1, so the first difference of a sequence is the gradient of a chord of step 1.

2 / WATCHING THE GRADIENTS SETTLE

Three examples with smaller and smaller steps

step h	x^2 at $x = 3$	x^3 at $x = 2$	$5t^2$ at $t = 2$
1	7	19	25
0.1	6.1	12.61	20.5
0.01	6.01	12.0601	20.05
0.001	6.001	12.006001	20.005

For x^2 at $x = 3$ the quotient is $6 + h$, for x^3 at $x = 2$ it is $12 + 6h + h^2$, and for the distance $s = 5t^2$ metres the average speed over $[2, 2 + h]$ is $20 + 5h$ metres per second, so the speed at $t = 2$ is 20 m/s, matching $v = 10t$.

3 / THE POWER RULE FROM THE BINOMIAL THEOREM

The quotient is a polynomial in h whose first term survives

Expanding $(x + h)^n = x^n + nx^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \cdots$ gives $\frac{(x+h)^n - x^n}{h} = nx^{n-1} + \binom{n}{2}x^{n-2}h + \cdots$, and every term except the first has a factor h and vanishes as $h \to 0$, so $\frac{d}{dx}x^n = nx^{n-1}$. With $h = 1$ the same expansion gives the whole difference, $\Delta n^3 = 3n^2 + 3n + 1$, whose first term $3n^2$ is the derivative.

4 / SECOND DIFFERENCES BECOME THE SECOND DERIVATIVE

The change of the gradient

The second difference with step h over h^2, $\frac{f(x+2h) - 2f(x+h) + f(x)}{h^2}$, tends to $f''(x)$, and for $f = x^3$ at $x = 1$ with $h = 0.1$ it is $\frac{1.728 - 2.662 + 1}{0.01} = 6.6$, heading to 6.

For a quadratic $an^2 + bn + c$ it equals $2a$ for every h, which is why a constant second difference means a constant second derivative.

5 / WHAT CALCULUS ADDS

Every step size, not only whole numbers

Differences only compare terms a whole step apart, while the derivative asks about the instant, by letting the step run through $0.1, 0.01, 0.001$ and beyond.

The rules are the ones met in the earlier sheets, since the power rule is the leading term of the difference rule, and the second difference of a quadratic is its second derivative.

6 / TRY THESE

Find the gradient numerically

(a) For x^4 at $x = 1$ with $h = 0.01$ the quotient is 4.060401, heading to 4. **(b)** For a car with distance $s = 5t^2$ the average speed from $t = 1$ to $t = 2$ is 15 m/s, and over $[2, 2.1]$ it is 20.5 m/s, so the speed at $t = 2$ is 20 m/s. **(c)** Show that for $f = x^2$ the quotient is exactly $2x + h$, so the limit is $2x$.

ASIDE / THE FIRST LESSON

The derivative is a limit of gradients of chords, and the sequences met so far are the chords of step 1.

LINKS TO OTHER SHEETS

The next sheet does the same for sums and areas; the physics hinge sheets use the derivative as a rate.

Convention. h is the step; $f'(x)$ the derivative; $\Delta f = f(x + h) - f(x)$; the chord gradient is $\Delta f / h$.

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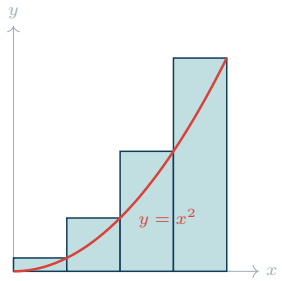
SEQUENCES LADDER 8 OF 9

From sums to the integral

Adding more and thinner rectangles turns the sums of powers into the area under a curve.

1 / THE RULE AND THE PICTURE

Thin rectangles under $y = x^2$ add up to $\frac{1}{3}$



four rectangles: $\frac{1}{4}\left(\frac{1}{1}6 + \frac{4}{1}6 + \frac{9}{1}6 + 1\right) = 0.469$

Split $0 \le x \le 1$ into n strips of width $\frac{1}{n}$ and take the height at the right end of each, so the area is $\frac{1}{n}\sum_{k=1}^{n}\left(\frac{k}{n}\right)^2 = \frac{n(n+1)(2n+1)}{6n^3}$, and as n grows

$$\boxed{\frac{(n+1)(2n+1)}{6n^2} \to \frac{1}{3} = \int_0^1 x^2\,dx}$$

The formula for the sum of squares from the earlier sheet does all the work.

2 / NUMBERS

The totals settle on $\frac{1}{3}$

rectangles n	total area	excess over $\frac{1}{3}$
4	0.46875	0.1354
10	0.385	0.0517
100	0.33835	0.0050
1000	0.3338335	0.0005
$\to \infty$	$\frac{1}{3}$	0

The excess over $\frac{1}{3}$ is $\frac{1}{2n} + \frac{1}{6n^2}$, so it shrinks like $\frac{1}{2n}$, and each tenfold increase in the number of rectangles gives one more correct digit.

3 / OTHER POWERS

The leading coefficient of the sum gives the area

For $y = x$ the sum is $\frac{n+1}{2n}$, which is 0.625 for $n = 4$ and tends to $\frac{1}{2}$, and for $y = x^3$ it is $\frac{(n+1)^2}{4n^2}$, which is 0.3906 for $n = 4$ and tends to $\frac{1}{4}$. In general $\frac{1}{n^{p+1}}\sum k^p \to \frac{1}{p+1}$ because the sum has leading term $\frac{n^{p+1}}{p+1}$, so $\int_0^1 x^p\,dx = \frac{1}{p+1}$.

4 / TELESCOPING BECOMES THE FUNDAMENTAL THEOREM

Adding differences gives the end value minus the start

Adding the first differences of any sequence collapses: $\sum_{k=a}^{b-1}\Delta f(k) = f(b) - f(a)$, as $\sum_{k=0}^{n-1}(2k+1) = n^2$ shows with $f = n^2$. The integral version is the same statement with differences replaced by derivatives, $\int_a^b f'(x)\,dx = f(b) - f(a)$, so finding an area means finding a function whose derivative is the height.

5 / THE AREA FUNCTION

Its gradient is the height

Let $A(x) = \int_0^x t^2\,dt = \frac{x^3}{3}$. Increasing x by a small h adds a thin strip of area about $x^2 h$, so $\frac{A(x+h)-A(x)}{h} \to x^2$, which is $A'(x) = x^2$, and in the discrete case the totals of squares differ by exactly n^2 from one total to the next.

6 / TRY THESE

Work out the area

(a) Find the area under $y = x$ from 0 to 1 with 10 rectangles: $\frac{11}{20} = 0.55$, heading to $\frac{1}{2}$. **(b)** Check with the triangle: its area is $\frac{1}{2}\cdot 1\cdot 1$. **(c)** Using $\sum k^3 = \left(\frac{n(n+1)}{2}\right)^2$, show that the rectangles under x^3 give $\frac{(n+1)^2}{4n^2} \to \frac{1}{4}$.

ASIDE / ONE SIDED SUMS

Taking the height at the right end overestimates a rising curve and the left end underestimates it, and the true area sits between them.

LINKS TO OTHER SHEETS

The previous sheet gives the derivative from differences; the last sheet collects the whole dictionary.

Convention. n is the number of strips; the area is the limit of the rectangle sums; $\int_a^b$ is the integral from a to b.

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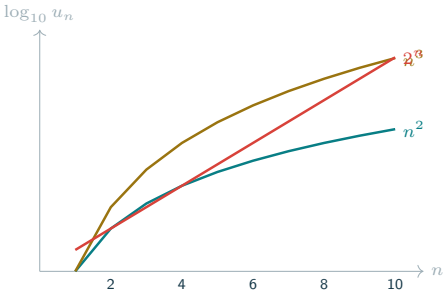
SEQUENCES LADDER 9 OF 9

The dictionary and the exceptions

Finite differences mirror calculus term by term, and sequences whose differences never settle are the exponentials.

1 / THE RULE AND THE PICTURE

Polynomials bend on a log plot and exponentials run straight



A polynomial of degree k has differences that fall to a constant, while an exponential keeps its differences proportional to itself, and on a plot of the logarithm the polynomial flattens and the exponential is a straight line:

$$\boxed{\Delta\, r^n = (r-1)\, r^n}$$

The sequence 2^n has $\Delta 2^n = 2^n$ exactly, so its differences are the sequence again.

2 / THE DICTIONARY

Each finite idea has a continuous twin

finite world	continuous world	link
difference Δ	derivative $\frac{d}{dx}$	chord gradient against tangent gradient
sum $\sum$	integral $\int$	rectangles against area
falling factorial $n^{\underline{k}}$	power x^k	$\Delta n^{\underline{k}} = k n^{\underline{k-1}}$ against $k x^{k-1}$
binomial $\binom{n}{k}$	$x^k/k!$	the basis of Newton and Taylor series
2^n, with $\Delta 2^n = 2^n$	e^x, with $\frac{d}{dx} e^x = e^x$	the function equal to its own rate of change
telescoping $\sum \Delta f = f(b) - f(a)$	$\int f' = f(b) - f(a)$	the fundamental theorem

3 / HOW TO CLASSIFY A SEQUENCE

Look at the differences and the ratios

what the differences do	what the rule is	example
constant k-th differences	polynomial of degree k	$1, 4, 9, 16$
constant ratio r	geometric, $u_n = u_1 r^{n-1}$	$3, 6, 12, 24$
Δu_n is a multiple of u_n	exponential, $u_n = c r^n$	$\Delta 3^n = 2 \cdot 3^n$
differences reproduce the sequence shifted	linear recurrence	Fibonacci: $\Delta F_n = F_{n-2}$

4 / WHERE e COMES FROM

The sequence whose difference quotient reproduces it

Take $f(x) = (1+h)^{x/h}$. Then $f(x+h) - f(x) = f(x) \cdot h$, so the difference quotient is exactly $f(x)$ for every step h, and as $h \to 0$ the function tends to e^x, the function that is its own derivative.
At $x = 1$ the values $(1 + \frac{1}{n})^n$ are $2,\ 2.5937,\ 2.7048,\ 2.7169$ for $n = 1, 10, 100, 1000$, heading to $e = 2.71828$.

5 / THE FIBONACCI NUMBERS

Differences that return the sequence

For $1, 1, 2, 3, 5, 8, 13$ the differences $0, 1, 1, 2, 3, 5$ are the same sequence moved two places along, since $F_{n+1} - F_n = F_{n-1}$, so no row of differences ever becomes constant.
Such sequences are exponential in disguise, $F_n \approx \varphi^n/\sqrt{5}$, and the characteristic equation $r^2 = r + 1$ finds the rate $\varphi = 1.618$.

6 / TRY THESE

Classify and predict

(a) For $3, 6, 12, 24$ the ratio is 2, so geometric with $u_n = 3 \cdot 2^{n-1}$. **(b)** For $1, 8, 27, 64$ the third differences are 6, so polynomial, n^3. **(c)** For $2, 5, 10, 17, 26$ the second differences are 2, so $n^2 + 1$, and for $5, 15, 45, 135$ the ratio is 3.

ASIDE / THE WHOLE LADDER

Starting from the step between neighbouring terms, the ladder reaches polynomials, sums of powers, the derivative, the integral and the exponential.

LINKS TO OTHER SHEETS

The ladder runs: 1 first differences, 2 second differences, 3 cubics, 4 Newton's formula, 5 sums of powers, 6 falling factorials, 7 the derivative, 8 the integral, and this dictionary; the functions ladder continues with the exponential.

Convention. Δ difference; r the common ratio; φ the golden ratio; $e = 2.71828\ldots$; a log plot shows $\log_{10} u_n$ against n.

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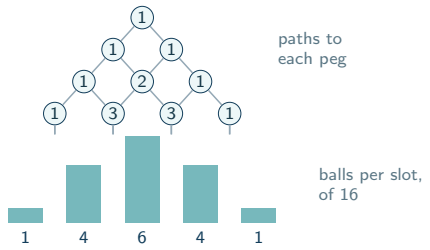
BINOMIAL ON ONE PAGE

From the Galton board to the binomial

Trees count paths, Pascal's triangle counts them for free, and the board runs the experiment.

1 / THE RULE AND THE PICTURE

Each peg sends a ball left or right, and the slot counts the rights



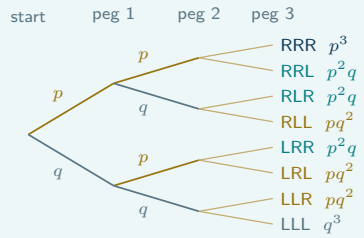
At each peg a ball goes right with probability p and left with $q = 1 - p$, independently, so after n rows the slot is k, the number of rights. Routes to a slot are the ways to choose which k pegs deflect right, and Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ adds the two parents:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

For a fair board every route has probability 2^{-n}, so the slot probabilities are the Pascal row over 2^n.

2 / TREE DIAGRAM / THREE ROWS OF PEGS

Multiply along each path, then add the paths that reach the same slot



rights k	routes	$P(X = k)$	$p = 0.3$
3	1	p^3	0.027
2	3	$3p^2q$	0.189
1	3	$3pq^2$	0.441
0	1	q^3	0.343

Probabilities multiply along a path and routes to one slot add, which gives the counts $1, 3, 3, 1$. A fair board gives $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, and tilting to $p = 0.3$ keeps the counts and shifts the weights.

3 / PASCAL'S TRIANGLE / PROBABILITY AND ODDS

The row gives the routes, 2^n gives the total, and the odds follow

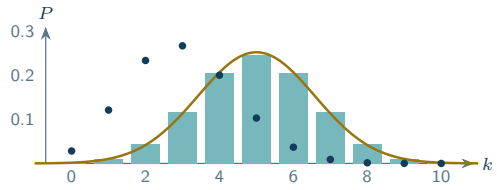
								sum
			1					2^0
		1		1				2^1
	1		2		1			2^2
1		3		3		1		2^3
	1		4		6		4		1	2^4
1		5		10		10		5		1	2^5
1		6		15		20		15		6		1	2^6

k	0	1	2	3	4	5	6
routes $\binom{6}{k}$	1	6	15	20	15	6	1
$P = \binom{6}{k}/64$	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$
odds for	1:63	3:29	15:49	5:11	15:49	3:29	1:63

Probability is favourable over total, $\binom{n}{k}/2^n$; the odds for are favourable to unfavourable, $\binom{n}{k} : (2^n - \binom{n}{k})$. The centre slot of a 6-row board has $P = \frac{5}{16}$ and odds $5 : 11$, and the two outer slots together have odds $1 : 31$.

4 / THE DISTRIBUTION / SHAPE, CENTRE AND SPREAD

With n pegs the slots follow $\mathrm{Bin}(n, p)$ with mean np and variance npq



Bars show the fair board with $n = 10$: mean $np = 5$ and $\sqrt{npq} \approx 1.58$, with the bell curve of the same mean and spread over them. The dots show $p = 0.3$, with mean 3.0 and $\sqrt{npq} \approx 1.45$, skewed. As n grows the spread rises as $\sqrt{n}$ and the bars settle onto the normal curve.

$$P(k) = \binom{n}{k} p^k q^{n-k}, \qquad \mu = np, \qquad \sigma^2 = npq$$

5 / SYNTHESIS

One count, three pictures: tree, triangle and board

picture	what it does	what it gives
tree diagram	lists every route, multiplying branch probabilities	the weight $p^k q^{n-k}$ of one route
Pascal triangle	counts the routes by adding parents	the coefficient $\binom{n}{k}$ without drawing the tree
Galton board	runs the experiment with many balls	the histogram that approaches $\mathrm{Bin}(n, p)$

ASIDE / WHY A BELL

Each peg adds an independent step, so variances add and the width grows as $\sqrt{n}$, the dice sheet's rule.

ASIDE / CHECK IN PYTHON

`[comb(6,k) for k in range(7)]` gives $1, 6, 15, 20, 15, 6, 1$, and `binom.pmf(1,3,0.3)` gives 0.441, the $3pq^2$ entry.

Convention. X counts rights, pegs are independent and $p + q = 1$; odds are favourable to unfavourable. Companions: the sqrt-time dice sheet.

Probabilistic

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COMBINATORICS ON ONE PAGE

Partitioning a hand: Stirling, Bell and entropy

A five-card hand splits into groups of equal rank, and that pattern is a macrostate whose microstates are the actual hands.

1 / THE RULE AND THE PICTURE

A full house is a hand whose five cards split into a group of three and a group of two



pair of 3s — three 8s

The hand $3\heartsuit$, $3\clubsuit$, $8\clubsuit$, $8\diamondsuit$, $8\spadesuit$ sorts into the partition $\{3\heartsuit, 3\clubsuit \mid 8\clubsuit, 8\diamondsuit, 8\spadesuit\}$ of its five cards. The macrostate keeps only the pattern of shared ranks, here $(3, 2)$, and the microstates are all hands with that pattern:

$$S = k \ln W, \qquad W_{\text{full house}} = 3744$$

2 / THE STIRLING TRIANGLE / EACH ROW BUILT FROM THE ONE ABOVE

Row n counts the ways to group n cards by rank, and its sum is the Bell number B_n

$n = 1$			1				$B_1 = 1$
$n = 2$			1	1			$B_2 = 2$
$n = 3$		1	3	1			$B_3 = 5$
$n = 4$	1	7	6	1			$B_4 = 15$
$n = 5$	1	15	25	10	1		$B_5 = 52$
$n = 6$	1	31	90	65	15	1	$B_6 = 203$

$$S(n, k) = k\, S(n-1, k) + S(n-1, k-1)$$

Deal out one more card: it can join any of the k groups already formed, in k ways, or start a new group after $k - 1$. So $S(5, 3) = 3 \cdot 6 + 7 = 25$. The row $1, 15, 25, 10, 1$ is the five-card hand, with k the number of different ranks it shows, and the row sums $1, 2, 5, 15, 52, 203$ are the Bell numbers.

3 / PARTITIONING FIVE CARDS / THE STIRLING AND BELL COUNTS

The cards of a hand can be grouped by rank in $B_5 = 52$ ways, in seven patterns

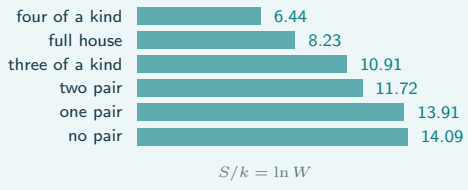
groups	pattern	name	partitions	how counted
x x x x x	(5)	five of a kind	1	all five alike, impossible in a real pack
x x x x y	$(4, 1)$	four of a kind	5	choose the odd card out
x x x y y	$(3, 2)$	full house	10	choose which two cards form the pair, $\binom{5}{2}$
x x x y z	$(3, 1, 1)$	three of a kind	10	choose the three alike, $\binom{5}{3}$
x x y y z	$(2, 2, 1)$	two pair	15	choose the single card, then split the other four into pairs in 3 ways
x x y z w	$(2, 1, 1, 1)$	one pair	10	choose the pair, $\binom{5}{2}$
x y z w v	$(1, 1, 1, 1, 1)$	no pair	1	all five different

By number of groups, $S(5, 2) = 5 + 10 = 15$ and $S(5, 3) = 10 + 15 = 25$. The letters show only which cards share a rank.

4 / MICROSTATES AND ENTROPY / THE ACTUAL HANDS OF A REAL PACK

Choose the ranks, then the suits, and the count W of each pattern gives its entropy

pattern	name	rank choices	suit choices	W	P	S/k
$(4, 1)$	four of a kind	$13 \cdot 12$	$1 \cdot 4$	624	0.024%	6.44
$(3, 2)$	full house	$13 \cdot 12$	$4 \cdot 6$	3 744	0.144%	8.23
$(3, 1, 1)$	three of a kind	$13\binom{12}{2}$	$4 \cdot 4^2$	54 912	2.113%	10.91
$(2, 2, 1)$	two pair	$\binom{13}{2} \cdot 11$	$6^2 \cdot 4$	123 552	4.754%	11.72
$(2, 1, 1, 1)$	one pair	$13\binom{12}{3}$	$6 \cdot 4^3$	1 098 240	42.257%	13.91
(1^5)	no pair	$\binom{13}{5}$	4^5	1 317 888	50.708%	14.09
total				$2\,598\,960 = \binom{52}{5}$	100%	



The full house takes the rank of the three in 13 ways and its suits in $\binom{4}{3} = 4$, then the pair's rank in 12 and suits in $\binom{4}{2} = 6$, so $13 \cdot 4 \cdot 12 \cdot 6 = 3744$. Five different ranks has the most microstates, and looking at suits and straights as well splits it into high card 1 302 540, straight 10 200, flush 5 108, straight flush 36 and royal flush 4, with $S/k = \ln 4 = 1.39$.

5 / WHERE THE STIRLING COUNT ENTERS / ORDERED DEALS

Partitions of the five positions times rank choices times suit orders gives every ordered deal

For a full house, 10 partitions of the positions times $13 \cdot 12 = 156$ rank choices times $4 \cdot 3 \cdot 2 \cdot 4 \cdot 3 = 288$ suit orders gives $449\,280 = 3744 \times 5!$, and summing over all seven patterns gives $311\,875\,200 = 52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$, every ordered deal.

ASIDE / BELL AND THE PACK

The partitions of a five-card hand number $B_5 = 52$, the size of the pack, which looks like a coincidence.

LINKS TO OTHER SHEETS

Entropy, microstates and a growing pack of cards; *Cards, piles and finite differences*.

Convention. Hands are unordered sets of five cards from a pack of 52; ln is the natural logarithm and $k_B = 1$; (1^5) means five different ranks.

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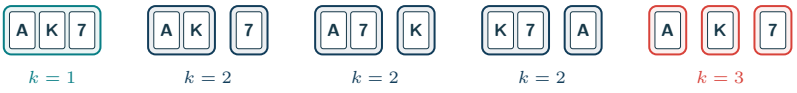
COMBINATORICS ON ONE PAGE

Cards, piles and finite differences

The x^n deals of n cards into x piles are Newton's difference expansion with Worpitzky coefficients.

1 / THE RULE AND THE PICTURE

Partition the cards into heaps, hand the heaps to piles, and sum over the number of piles used



the five partitions of three cards, $B_3 = 5$, with $S(3,k) = 1, 3, 1$ for $k = 1, 2, 3$

Deal n distinct cards into x labelled piles, any pile allowed to stay empty, in x^n ways. If exactly k piles are used, choose them in $\binom{x}{k}$ ways, partition the cards into k heaps in $S(n,k)$ ways and hand the heaps to the chosen piles in $k!$ ways:

$$x^n = \sum_k \binom{x}{k} k!\, S(n,k) = \sum_k \binom{x}{k} W(n,k)$$

2 / A COUNT TO CHECK / THREE CARDS, THREE PILES

The twenty-seven deals split as $3 + 18 + 6$ by the number of piles used

k	$S(3,k)$	$\times\, k! = W(3,k)$	$\binom{3}{k}$ piles	deals	example
1	1	1	3	3	all three cards on one pile
2	3	6	3	18	two cards together, one apart
3	1	6	1	6	one card on each pile
total	$B_3 = 5$			$27 = 3^3$	

3 / FINITE DIFFERENCES / THE SAME NUMBERS ON THE LEFT EDGE

The kth difference of x^n at 0 is $W(n,k)$, and a difference is inclusion and exclusion

x^4	0	1	16	81	256	625
Δ	1	15	65	175	369	
Δ^2	14	50	110	194		
Δ^3	36	60	84			
Δ^4	24	24				
Δ^5	0					

The values of x^4 at $x = 0, \ldots, 5$ and their successive differences have the left edge $0, 1, 14, 36, 24$, which is row 4 of the Worpitzky numbers, and the fourth difference is the constant $24 = 4!$. The formula

$$\Delta^k x^n\big|_0 = \sum_j (-1)^{k-j} \binom{k}{j} j^n = W(n,k)$$

counts the deals onto j piles, removes those that leave a pile empty, and adds back the overlaps, so each difference removes the empty piles. Newton's expansion $p(x) = \sum_k \Delta^k p(0) \binom{x}{k}$ then reproduces $x^n = \sum_k W(n,k) \binom{x}{k}$.

4 / THE TWO EXTREMES / COARSEST AND FINEST PARTITION

The first and last non-zero differences are all cards in one pile and all cards alone

For $k = 1$ every card lies on the same pile and $W(n,1) = 1$, the coarsest partition, while for $k = n$ every card is alone and the n piles are ordered, giving $W(n,n) = n!$, the finest. The next difference is zero because n cards cannot occupy $n+1$ piles, which is why a polynomial of degree n has a constant nth difference $n!$ times its leading coefficient and nothing beyond.

5 / A POKER VERSION / RANK SEQUENCES OF FIVE CARDS

Ignoring suits, $13^5 = \sum_k \binom{13}{k} W(5,k)$ with k the number of distinct ranks

k	$S(5,k)$	$W(5,k)$	$\binom{13}{k}$	ordered rank sequences	hand patterns
1	1	1	13	13	(5), impossible with four suits
2	15	30	78	2 340	$(4,1)$ four of a kind; $(3,2)$ full house
3	25	150	286	42 900	$(3,1,1)$ three of a kind; $(2,2,1)$ two pair
4	10	240	715	171 600	$(2,1,1,1)$ one pair
5	1	120	1287	154 440	$(1,1,1,1,1)$ no pair, straight, flush

The terms sum to $371\,293 = 13^5$, and the hand patterns are the integer partitions of 5 that refine each number of distinct ranks. With four suits the pattern (5) cannot occur, so real hand counts need that cap.

ASIDE / BELL AND THE PACK

The partitions of the five positions of a hand into equal-rank groups number $B_5 = 52$, the size of the pack, which looks like a coincidence.

LINKS TO OTHER SHEETS

The triangles and their gauge: *The general sequencing law for triangles*; hand counts with four suits: the poker hands sheet.

Convention. $W(n,k) = k!\, S(n,k)$, S the Stirling numbers of the second kind, $\Delta f(x) = f(x+1) - f(x)$. The cards are distinct and the piles are labelled.

Networked–lattice

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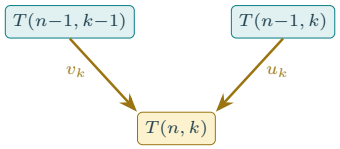
COMBINATORICS ON ONE PAGE

The general sequencing law for triangles

One bidiagonal rule with two sets of weights produces the Pascal, Stirling, Worpitzky and Eulerian triangles.

1 / THE RULE AND THE PICTURE

Each entry is a weighted sum of the two entries above it



$$\boxed{T(n,k) = u_k\,T(n-1,k) + v_k\,T(n-1,k-1)}$$

Stacking a row as a column vector, one step of the law is multiplication by a lower bidiagonal matrix:

$$\begin{pmatrix} T(n,0) \\ T(n,1) \\ T(n,2) \end{pmatrix} = \begin{pmatrix} u_0 & & \\ v_1 & u_1 & \\ & v_2 & u_2 \end{pmatrix} \begin{pmatrix} T(n-1,0) \\ T(n-1,1) \\ T(n-1,2) \end{pmatrix}.$$

The weights are the only choices, and a triangular matrix has its diagonal u_k as its eigenvalues.

2 / FOUR TRIANGLES / THE SAME LAW, DIFFERENT WEIGHTS

Pascal, Stirling and Worpitzky with fixed weights, and Eulerian with weights that depend on the row

	Pascal $\binom{n}{k}$ weights $(1,1)$	**Stirling** $S(n,k)$ weights $(k,1)$	**Worpitzky** $W(n,k)$ weights (k,k)	**Eulerian** $A(n,k)$ weights $(k+1,\,n-k)$
$n=0$	1	1	1	1
$n=1$	1 1	1 1	1 2	1 1
$n=2$	1 2 1	1 3 1	1 6 6	1 4 1
$n=3$	1 3 3 1	1 7 6 1	1 14 36 24	1 11 11 1
$n=4$	1 4 6 4 1	1 15 25 10 1	1 30 150 240 120	1 26 66 26 1
$n=5$	1 5 10 10 5 1	1 31 90 65 15 1	1 62 540 1560 1800 720	1 57 302 302 57 1
$n=6$	1 6 15 20 15 6 1	$25 = 3\cdot 6 + 7$	$36 = 3(6+6)$	$66 = 3\cdot 11 + 3\cdot 11$
	$15 = 10 + 5$			

3 / WHAT EACH TRIANGLE COUNTS / AND ITS ROW SUMS

Subsets, set partitions, surjections and permutations by descents

triangle	counts	row sums	closed form
Pascal $\binom{n}{k}$	k-subsets of an n-set	$1, 2, 4, 8, 16, 32, 64$, that is 2^n	$\dfrac{n!}{k!\,(n-k)!}$
Stirling $S(n,k)$	partitions of n things into k blocks	Bell $1, 2, 5, 15, 52, 203$	$\dfrac{1}{k!}\sum_j (-1)^{k-j}\binom{k}{j}\,j^n$
Worpitzky $W(n,k)$	surjections from n things onto k labelled boxes	Fubini $1, 3, 13, 75, 541, 4683$	$\sum_j (-1)^{k-j}\binom{k}{j}\,j^n$
Eulerian $A(n,k)$	permutations of n with k descents	$n!$: $1, 2, 6, 24, 120, 720$	$\sum_j (-1)^j\binom{n+1}{j}(k+1-j)^n$

Labelling the blocks of a partition in $k!$ ways turns Stirling into Worpitzky, so $W(n,k) = k!\,S(n,k)$, and for example $W(4,2) = 2^4 - 2 = 14$ counts the maps from four things onto two boxes.

4 / HOW THE TRIANGLES LINK / GAUGE, BASIS AND THE BELL RECURRENCE

Pascal feeds Stirling through the weights, Worpitzky through a gauge, and Bell through the binomials

Replacing the Pascal weights $(1,1)$ by $(k,1)$ gives Stirling. Scaling row k by $k!$, the diagonal gauge $D = \mathrm{diag}(k!)$, sends v_k to $k\,v_k$ and carries Stirling to Worpitzky, while the diagonal $u_k = k$ is untouched, so both have the eigenvalues $1, \dots, n$ and the characteristic polynomial $(\lambda-1)(\lambda-2)\cdots(\lambda-n)$. In the language of bases,

$$x^n = \sum_k S(n,k)\,x^{\underline{k}} = \sum_k W(n,k)\binom{x}{k}, \qquad B_{n+1} = \sum_k \binom{n}{k} B_k,$$

so the Worpitzky numbers are the coefficients of a power in Pascal's own basis, the falling factorial $x^{\underline{k}} = k!\binom{x}{k}$ is the change of basis between them, and the Bell numbers $B_n = \sum_k S(n,k)$ are built from binomials: $B_4 = 1+3\cdot 1+3\cdot 2+5 = 15$. The Eulerian numbers expand the same power in shifted binomials, $x^n = \sum_k A(n,k)\binom{x+k}{n}$, which is Worpitzky's identity proper.

ASIDE / WEIGHTS THAT DEPEND ON n

The Eulerian weights $(k+1,\,n-k)$ depend on the row, as do those of the unsigned Stirling numbers of the first kind, $(n-1,1)$, whose row sums are also $n!$.

LINKS TO OTHER SHEETS

The characteristic equation and the gauge: *The characteristic equation beneath five presentations*; the determinant as a product of eigenvalues: *The determinant in its many guises*.

Convention. $T(n,k)$ is the entry in row n, column k; $x^{\underline{k}} = x(x-1)\cdots(x-k+1)$; S and W are indexed from $k=1$.

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GROUPING, ADDITION AND DIFFERENCES ON ONE PAGE

Bell numbers and Worpitzky's triangle

Count every way to form groups, then find that total inside a difference table.

1 / WHAT DOES BELL COUNT?

A partition is a way to split a set into groups

The **Bell number** B_n counts the partitions of n distinct objects into non-empty, unlabelled groups. Each object belongs to exactly one group; any number of groups is allowed. For A, B, C, the five ways are

$$\{A,B,C\}, \quad \{A,B\} \mid \{C\}, \quad \{A,C\} \mid \{B\}, \quad \{A\} \mid \{B,C\}, \quad \{A\} \mid \{B\} \mid \{C\}.$$

Bars separate groups. Their order does not matter, so $\{A,B\} \mid \{C\}$ and $\{C\} \mid \{A,B\}$ count once. Thus $B_3 = 1+3+1 = 5$. Also $B_0 = 1$: there is one empty partition.

2 / CONSTRUCT BELL'S TRIANGLE

Copy the last entry, then add your way across

n	entries in row n					
0	**1**					
1	**1**	2				
2	**2**	3	5			
3	**5**	7	10	15		
4	**15**	20	27	37	52	
5	**52**	67	87	114	151	203

The gold left edge gives $B_0, B_1, B_2, \ldots$.
The right edge gives $B_1, B_2, B_3, \ldots$.

Start with the single entry 1. For each new row:

First: copy the previous row's last entry.
Then: add the new entry on your left to the old entry above-left.

For example, after the row $2, 3, 5$:

$$5, \qquad 5+2=7,$$
$$7+3=10, \qquad 10+5=15.$$

The new row is $5, 7, 10, 15$.

3 / THE BRIDGE THROUGH STIRLING NUMBERS

Divide a Worpitzky row by factorials, then add

The Stirling number $S(n,r)$ counts partitions into **exactly** r **groups**, so $B_n = \sum_{r=0}^{n} S(n,r)$. Using the difference-sheet convention, $\Delta f(x) = f(x+1) - f(x)$ and

$$W(m,k) = \Delta^k(x^m)\Big|_{x=1} = k!\,S(m+1,k+1), \qquad 0 \le k \le m.$$

For the quartic row, $m = 4$:

k	0	1	2	3	4
$W(4,k)$	1	15	50	60	24
$k!$	1	1	2	6	24
$W(4,k)/k! = S(5,k+1)$	1	15	25	10	1

The last row counts partitions of **five** objects into $1, 2, 3, 4, 5$ groups. Adding gives $B_5 = 1+15+25+10+1 = 52$.

$$B_{m+1} = \sum_{k=0}^{m} \frac{W(m,k)}{k!} \qquad (m \ge 0)$$

4 / WHY A BINOMIAL SUM APPEARS

Choose who stays outside the new object's group

Add a distinguished object $\star$ to n existing objects. Choose the j objects outside its group in $\binom{n}{j}$ ways, and partition those j objects in B_j ways. Everyone else joins $\star$. Each partition is counted exactly once, giving

$$\boxed{B_{n+1} = \sum_{j=0}^{n} \binom{n}{j} B_j}.$$

For example, Pascal's row $1, 4, 6, 4, 1$ gives

$$B_5 = 1\cdot B_0 + 4\cdot B_1 + 6\cdot B_2 + 4\cdot B_3 + 1\cdot B_4 = 1+4+12+20+15 = 52.$$

Keep the indices. A degree-m Worpitzky row gives B_{m+1}. For $m = 2$: $1+3+2/2! = 5 = B_3$; take $B_0 = 1$ separately.
References. NIST DLMF 26.7; OEIS A011971 (Bell triangle); OEIS A028246 (Worpitzky triangle).

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STIRLING NUMBERS OF THE SECOND KIND ON ONE PAGE

From Stirling to Worpitzky

Counting groups builds a triangle; factorials turn it into the weights of a difference table.

1 / WHAT DOES A STIRLING NUMBER COUNT?

Split distinct objects into non-empty groups

$S(n,r)$ counts the ways to partition n distinct objects into exactly r non-empty groups. The groups have no order. For three objects and two groups, the possibilities are

$$\{A\} \mid \{B,C\}, \qquad \{B\} \mid \{A,C\}, \qquad \{C\} \mid \{A,B\}, \qquad \boxed{S(3,2)=3.}$$

Here the vertical bar separates groups; swapping the groups gives the same partition.

2 / BUILD THE TRIANGLE BY ADDING ONE OBJECT

Join a group or start a new one

Consider the last object in a partition of n objects.

Join: it enters one of r existing groups. This gives $rS(n-1,r)$ possibilities.

Start: it forms a singleton group. The other objects form $r-1$ groups, giving $S(n-1,r-1)$ possibilities.

$$S(n,r) \;=\; rS(n-1,r) + S(n-1,r-1)$$

Start with $S(0,0)=1$: the empty set has one partition. For $n>0$, $S(n,0)=0$; also use zero when $r<0$ or $r>n$.

$n\backslash r$	0	1	2	3	4	5
0	1					
1	0	1				
2	0	1	1			
3	0	1	3	1		
4	0	1	**7**	**6**	1	
5	0	1	15	25	10	1

$$S(5,3) = 3\,S(4,3) + S(4,2)$$
$$= 3\cdot 6 + 7 = \boxed{25}.$$

3 / BUILD WORPITZKY FROM STIRLING, OR GO BACK

Shift the indices and multiply by factorials

For the power x^m, take Stirling row $n=m+1$. Match group count $r=k+1$, then multiply by $k!$:

$$W(m,k) \;=\; k!\,S(m+1,k+1), \qquad S(n,r) \;=\; \frac{W(n-1,r-1)}{(r-1)!}.$$

difference order k	0	1	2	3	4
group count $r=k+1$	1	2	3	4	5
Stirling $S(5,r)$	1	15	25	10	1
Multiply by $k!$	1	1	2	6	24
Worpitzky $W(4,k)$	1	15	50	60	24

Going back uses division by $0!, 1!, 2!, 3!, 4!$ and restores the Stirling row $1, 15, 25, 10, 1$.

4 / WHY DO THESE BECOME DIFFERENCE WEIGHTS?

One marked object explains the shift; labelling groups explains the factorial

Add a marked object $\star$ to m ordinary objects. Partition all $m+1$ objects into $k+1$ groups in $S(m+1,k+1)$ ways. Call the group containing $\star$ box 0, and label the other k groups $1,\dots,k$ in $k!$ ways.

Now remove $\star$. The m objects fill $k+1$ labelled boxes: boxes $1,\dots,k$ must be non-empty, while box 0 may be empty. So their number is $k!S(m+1,k+1)$.

For $m=4, k=2$, there are 3^4 assignments to three boxes. Subtract 2^4 for each required box being empty; add back 1^4 for both being empty. This is exactly a second difference:

$$W(4,2) = 2!\,S(5,3) = 3^4 - 2\cdot 2^4 + 1^4 = 81 - 32 + 1 = \boxed{50}.$$

In general, the same inclusion-exclusion count is the alternating sum in a kth forward difference, with $\Delta f(x) = f(x+1) - f(x)$:

$$\Delta^k(x^m)\Big|_{x=1} = \sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} (j+1)^m = k!\,S(m+1,k+1).$$

The quartic weights $1, 15, 50, 60, 24$ therefore count labelled groupings as well as entries in a difference table.

Conventions. S means Stirling numbers of the second kind. $W(m,k)$ uses unit differences starting at $x=1$. The inverse formula applies for $n\geq 1$, $1\leq r\leq n$; retain the seed and zero column above.

Bell link. Summing Stirling row 5 gives $B_5 = 1+15+25+10+1 = 52$. **References:** NIST DLMF §26.8; OEIS A028246.

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QUADRATIC SEQUENCES AND BACKTRACKING ON ONE PAGE

Finding a quadratic from its sequence

Second differences give a. Work back to the first difference, then the first term.

1 / BEGIN WITH A SEQUENCE

Make the difference table

Let s_x mean the term at position x, starting at $x = 1$. The symbol Δ means "take the next value minus this one":

$$\Delta s_x = s_{x+1} - s_x, \qquad \Delta^2 s_x = \Delta s_{x+1} - \Delta s_x.$$

x	1	2	3	4	5
s_x	**10**	19	32	49	70
Δs_x	**9**	13	17	21	
$\Delta^2 s_x$	**4**	4	4		

For example,

$$\Delta s_1 = 19 - 10 = 9,$$
$$\Delta^2 s_1 = 13 - 9 = 4.$$

The 2 means taking differences twice. The second differences are constant.

2 / EXPLAIN THE PATTERN WITH LETTERS

Start with $s_x = ax^2 + bx + c$

Substitute $x = 1, 2, 3$, then subtract neighbours just as before:

x	s_x	Δs_x	$\Delta^2 s_x$
1	$a + b + c$	$3a + b$	$2a$
2	$4a + 2b + c$	$5a + b$	
3	$9a + 3b + c$		

The first two differences are

$$\Delta s_1 = (4a + 2b + c) - (a + b + c) = 3a + b,$$
$$\Delta s_2 = (9a + 3b + c) - (4a + 2b + c) = 5a + b.$$

Subtract again: $(5a + b) - (3a + b) = 2a$.

$$a = \frac{\Delta^2 s_1}{2}, \quad b = \Delta s_1 - 3a, \quad c = s_1 - a - b.$$

3 / BACKTRACK THROUGH OUR EXAMPLE

Find a, then b, then c

First: halve the second difference.

$$2a = 4 \implies a = 2.$$

Next: use the first difference. The first jump is 9, so

$$3a + b = 9 \implies b = 9 - 3(2) = 3.$$

Finally: use the first term. The first term is 10, so

$$a + b + c = 10 \implies c = 10 - 2 - 3 = 5.$$

$$s_x = 2x^2 + 3x + 5$$

The constant $c = 5$ is the value at $x = 0$; the first listed term, at $x = 1$, is 10. Check the fifth term: $2(5^2)+3(5)+5 = 70$.

4 / THE SAME BACKTRACKING AS A SUBTRACTION TABLE

Remove the quadratic part, then the linear part

x	1	2	3	4	5
Original s_x	10	19	32	49	70
Subtract $2x^2$	2	8	18	32	50
Remainder $s_x - 2x^2$	8	11	14	17	20
Subtract $3x$	3	6	9	12	15
Remainder $s_x - 2x^2 - 3x$	5	5	5	5	5

After removing $2x^2$, constant steps of 3 give $b = 3$; removing $3x$ leaves $c = 5$.

5 / A FIRST LOOK AT WORPITZKY'S TRIANGLE

Construct the triangle from the simplest building blocks

Take the sequences 1, x and x^2. Keep the first entry at each difference level, shown in gold:

rule	terms at $1, 2, 3$	first differences	second
1	**1**, 1, 1	**0**, 0	**0**
x	**1**, 2, 3	**1**, 1	**0**
x^2	**1**, 4, 9	**3**, 5	**2**

Omit the trailing zeros:

1	1		
x	1	1	
x^2	1	3	2

Weight these rows by c, b, a and add columns:

$$(s_1, \Delta s_1, \Delta^2 s_1) = (a + b + c,\ 3a + b,\ 2a).$$

Extending this construction to $x^3, x^4, \ldots$ gives the larger Worpitzky triangle in the next sheet.

Conventions. Positions start at $x = 1$ and increase by 1. We assume a quadratic rule; a non-zero constant second difference equals $2a$. **Triangle reference:** OEIS A028246, with rows starting at the constant power.

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QUARTIC SEQUENCES AND WORPITZKY'S TRIANGLE ON ONE PAGE

Finding a quartic, one coefficient at a time

The quadratic method grows naturally: build the power rows, then backtrack from $24a$.

1 / BEGIN WITH A SEQUENCE

Take differences until they are constant

Use unit steps from $x = 1$. Each Δ means "next value minus this one"; gold marks the first entry at each level.

x	1	2	3	4	5	6	7
s_x	**15**	65	233	645	1475	2945	5325
Δs_x	**50**	168	412	830	1470	2380	
$\Delta^2 s_x$	**118**	244	418	640	910		
$\Delta^3 s_x$	**126**	174	222	270			
$\Delta^4 s_x$	**48**	48	48				

For a quartic $s_x = ax^4 + bx^3 + cx^2 + dx + e$,

$$\Delta^4 s_x = 4!a = 24a.$$

$$a = \frac{48}{24} = 2$$

2 / CONSTRUCT WORPITZKY'S TRIANGLE

Each power has its own difference pattern

Build a row by taking the first entries from the difference table of a power. For x^4, start with $1, 16, 81, 256, 625$:

x	1	2	3	4	5
x^4	**1**	16	81	256	625
Δx^4	**15**	65	175	369	
$\Delta^2 x^4$	**50**	110	194		
$\Delta^3 x^4$	**60**	84			
$\Delta^4 x^4$	**24**				

Repeat for the lower powers:

power	0	1	2	3	4
1	1				
x	1	1			
x^2	1	3	2		
x^3	1	7	12	6	
x^4	1	15	50	60	24

Column 0 gives the original value; column k gives Δ^k at $x = 1$. The diagonal is $0!, 1!, 2!, 3!, 4!$.

3 / LINK THE TRIANGLE TO THE COEFFICIENTS

Read a column, then subtract known parts

For $s_x = ax^4 + bx^3 + cx^2 + dx + e$, weight the rows $x^4, x^3, x^2, x, 1$ by a, b, c, d, e and add columns. For example, $\Delta^3 s_1 = 60a + 6b$.

$$a = \frac{\Delta^4 s_1}{24},$$

$$b = \frac{\Delta^3 s_1 - 60a}{6},$$

$$c = \frac{\Delta^2 s_1 - 50a - 12b}{2},$$

$$d = \Delta s_1 - 15a - 7b - 3c,$$

$$e = s_1 - a - b - c - d.$$

Start at the highest difference. Each line then has only one unknown coefficient left. The divisors are $4!, 3!, 2!, 1!, 0!$ in that order.

4 / BACKTRACK THROUGH THE EXAMPLE

Use the gold edge: $15, 50, 118, 126, 48$

Begin with the fourth difference and move upwards:

$$a = \frac{48}{24} = 2,$$

$$b = \frac{126 - 60(2)}{6} = 1,$$

$$c = \frac{118 - 50(2) - 12(1)}{2} = 3,$$

$$d = 50 - 15(2) - 7(1) - 3(3) = 4,$$

$$e = 15 - 2 - 1 - 3 - 4 = 5.$$

$$s_x = 2x^4 + x^3 + 3x^2 + 4x + 5$$

The first term is $s_1 = 15$; the constant term is $e = 5$, the value at $x = 0$. Check: $s_2 = 32 + 8 + 12 + 8 + 5 = 65$.

5 / THE SAME METHOD AS SUCCESSIVE SUBTRACTION

Remove one recovered term at a time

Each Δ below applies to the current remainder. The table shows the first five positions:

sequence left	$x = 1$	2	3	4	5	read off
Original s_x	15	65	233	645	1475	$\Delta^4 = 48 \Rightarrow a = 2$
Remove $2x^4$	13	33	71	133	225	$\Delta^3 = 6 \Rightarrow b = 1$
Then remove x^3	12	25	44	69	100	$\Delta^2 = 6 \Rightarrow c = 3$
Then remove $3x^2$	9	13	17	21	25	$\Delta = 4 \Rightarrow d = 4$
Then remove $4x$	5	5	5	5	5	constant $= 5 \Rightarrow e = 5$

Conventions. Exact polynomial rule, unit spacing, first position $x = 1$. A quartic has constant fourth differences $24a$ and zero fifth differences. **Triangle reference:** OEIS A028246, with rows starting at the constant power.

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WORPITZKY'S TRIANGLE AND FINITE DIFFERENCES ON ONE PAGE

From differences to polynomial coefficients

A sequence hides its polynomial. Repeated subtraction reveals its coefficients.

1 / FROM VALUES TO DIFFERENCES

Subtract neighbours, then repeat

For samples $s_x = P(x)$ at $x = 1, 2, 3, \ldots$, use forward differences. The superscript counts how many times we take differences:

$$\Delta s_x = s_{x+1} - s_x, \qquad \Delta^2 s_x = \Delta s_{x+1} - \Delta s_x = s_{x+2} - 2s_{x+1} + s_x.$$

x	1	2	3	4	5	6
s_x	**8**	17	44	101	200	353
Δs_x	**9**	27	57	99	153	
$\Delta^2 s_x$	**18**	30	42	54		
$\Delta^3 s_x$	**12**	12	12			

Read the gold edge: $D_k = \Delta^k s_1$, with $D_0 = s_1$.

The third differences are constant. For a cubic rule $P(x) = ax^3 + bx^2 + cx + d$,

$$\Delta^3 s_x = 3!a = 12.$$

$$a = \frac{12}{6} = 2$$

The fourth differences are zero.

2 / WORPITZKY'S TRIANGLE

Precompute the powers once

Row j records the first entries in the difference table of $1^j, 2^j, 3^j, \ldots$. Our indexing starts at zero:

$$W_{j,k} = \Delta^k (x^j)\Big|_{x=1}.$$

power	$k=0$	1	2	3	4
1	1				
x	1	1			
x^2	1	3	2		
x^3	1	7	12	6	
x^4	1	15	50	60	24

For example, the cubic row begins with $1^3 = 1$ and $2^3 - 1^3 = 7$. Its next differences begin with 12 and 6.

Build further rows using

$$W_{j,k} = (k+1)W_{j-1,k} + kW_{j-1,k-1}.$$

$W_{0,0} = 1$, $W_{j,j} = j!$; missing entries are zero.

3 / RECOVER THE CUBIC

Solve from the highest difference down

For $P(x) = ax^3 + bx^2 + cx + d$, combine the power rows using their unknown coefficients. Read each column:

$$D_3 = 6a = 12 \qquad \Rightarrow a = 2,$$

$$D_2 = 12a + 2b = 18 \qquad \Rightarrow b = -3,$$

$$D_1 = 7a + 3b + c = 9 \qquad \Rightarrow c = 4,$$

$$D_0 = a + b + c + d = 8 \qquad \Rightarrow d = 5.$$

The cubic contributes $12a = 24$ to D_2. Removing that contribution exposes $2b$:

$$b = \frac{D_2 - 12a}{2} = \frac{18 - 24}{2} = -3.$$

$$P(x) = 2x^3 - 3x^2 + 4x + 5$$

Check: $P(1) = 8$, $P(2) = 17$ and $P(6) = 353$.

4 / THE SAME METHOD FOR ANY DEGREE

A triangular system has a natural solving order

Write $P(x) = \sum_{j=0}^{n} a_j x^j$ and let $D_k = \Delta^k P(1)$. Linearity gives $D_k = \sum_{j=k}^{n} W_{j,k} a_j$. Since $W_{k,k} = k!$, recover the coefficients in descending order:

$$a_n = \frac{D_n}{n!}, \qquad a_k = \frac{D_k - \sum_{j=k+1}^{n} W_{j,k} a_j}{k!} \quad (k = n-1, n-2, \ldots, 0).$$

5 / A BASIS BUILT FOR DIFFERENCES

Newton's expansion reads the same edge directly

The difference edge is already a set of coefficients, in the **binomial basis** rather than the ordinary powers:

$$P(x) = \sum_{k=0}^{n} D_k \binom{x-1}{k}, \qquad \binom{x-1}{k} = \frac{(x-1)(x-2)\cdots(x-k)}{k!}, \quad \binom{x-1}{0} = 1.$$

For our sequence this gives

$$P(x) = 8 + 9(x-1) + \frac{18}{2!}(x-1)(x-2) + \frac{12}{3!}(x-1)(x-2)(x-3).$$

Pascal's identity gives $\Delta\binom{x-1}{k} = \binom{x-1}{k-1}$ for $k \geq 1$: each difference lowers the binomial index by one. Expanding the products recovers $2x^3 - 3x^2 + 4x + 5$. Worpitzky's triangle converts between these two ways of writing the same polynomial.

Conventions. These tables start at $x = 1$ with unit spacing. For a degree-n polynomial and step h, $\Delta_h^n P = n! a_n h^n$. Given a degree bound n, $n+1$ exact samples uniquely determine the polynomial.

Sources. OEIS A028246 (rows reindexed from $j = 0$); J. K. Sikora (2018), *On Calculating the Coefficients of a Polynomial Generated Sequence Using the Worpitzky Number Triangles*.

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BINOMIAL COEFFICIENTS, PLACE VALUE AND REPEATING PATTERNS

Lucas's theorem: read the digits

A large entry in Pascal's triangle can leave a remainder you can find by hand.

1 / BEGIN WITH PASCAL'S TRIANGLE

Add neighbours, then keep only their remainders

The entry $\binom{n}{k}$ counts ways to choose k objects from n. It is the coefficient of x^k in $(1+x)^n$. Rows and positions start at 0.

row 6	1	6	15	20	15	6	1
remainder on division by 2	1	0	1	0	1	0	1

Modulo p means keeping the remainder on division by p. Lucas connects these remainders to place value.

2 / THE DIGIT RULE

Write in base p, pair the digits, multiply

Let p be **prime** and $0 \le k \le n$. Write both numbers in base p, adding leading zeros to line up the columns:

$$n = (n_m \cdots n_1 n_0)_p = n_0 + n_1 p + \cdots + n_m p^m, \qquad k = (k_m \cdots k_1 k_0)_p.$$

Each digit lies between 0 and $p-1$. Lucas's theorem says

$$\binom{n}{k} \equiv \binom{n_m}{k_m} \cdots \binom{n_1}{k_1}\binom{n_0}{k_0} \pmod{p}.$$

Here $\equiv$ means "same remainder"; use $\binom{a}{b} = 0$ if $b > a$, and $\binom{a}{0} = 1$. **A larger lower digit makes the result divisible by p.**

3 / BITS: THE PRIME 2

Which entries are odd?

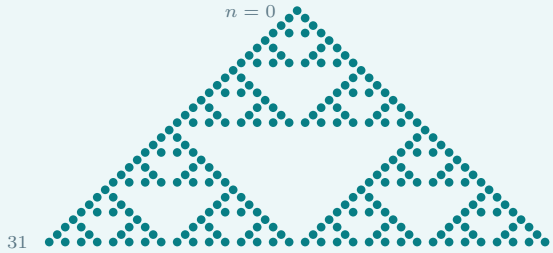
Write $12 = (1100)_2 = 8+4$ and $4 = (0100)_2$:

place value	8	4	2	1
$n = 12$	1	1	0	0
$k = 4$	0	1	0	0

Pairing the columns gives

$$\binom{12}{4} \equiv \binom{1}{0}\binom{1}{1}\binom{0}{0}\binom{0}{0} = 1 \pmod 2.$$

So $\binom{12}{4} = 495$ is odd. For $5 = (0101)_2$, the units column gives $\binom{0}{1} = 0$, so $\binom{12}{5} = 792$ is even.



Rows 0 to 31. Teal: remainder 1 (odd). Blank: 0.

Odd entries have every 1 in k under a 1 in n. Repeating this at each place gives the **Sierpiński triangle** pattern.

4 / TRITS: THE PRIME 3

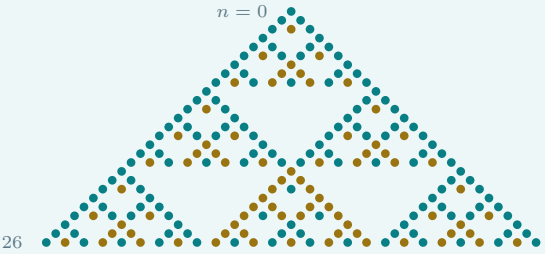
Three remainders, the same rule

Write $14 = (112)_3 = 9+3+2$ and $5 = (012)_3$:

place value	9	3	1
$n = 14$	1	1	2
$k = 5$	0	1	2

$$\binom{14}{5} \equiv \binom{1}{0}\binom{1}{1}\binom{2}{2} = 1 \pmod 3.$$

Indeed, $2002 = 3 \cdot 667 + 1$. For $6 = (020)_3$, the middle column gives $\binom{1}{2} = 0$, so $\binom{14}{6} = 3003$ is divisible by 3.



Rows 0 to 26. Teal: 1. Gold: 2. Blank: 0.

The digit rule repeats at scales $3, 9, 27, \ldots$, building a **ternary cousin** of the binary picture.

5 / WHY THE DIGITS SEPARATE

A short binomial explanation

For prime p, all the middle coefficients of $(1+x)^p$ are divisible by p. Keeping remainders,

$$(1+x)^p \equiv 1 + x^p \pmod p.$$

Repeat this and use the base-p digits of n:

$$(1+x)^n \equiv \prod_{r=0}^{m} (1+x^{p^r})^{n_r} \pmod p.$$

To make x^k, choose k_r copies of x^{p^r} in factor r. Since $n_r < p$, these are digits with no carries. The numbers of choices multiply, giving Lucas's formula.

6 / COUNT A ROW WITHOUT BUILDING IT

How many entries survive?

An entry is non-zero modulo p exactly when every k_r lies in $0, 1, \ldots, n_r$. There are $n_r + 1$ choices in that column. Multiply to count the entries not divisible by p:

$$(n_m + 1) \cdots (n_1 + 1)(n_0 + 1).$$

Row $12 = (1100)_2$ has $2 \cdot 2 \cdot 1 \cdot 1 = 4$ odd entries, at $k = 0, 4, 8, 12$. Row $14 = (112)_3$ has $2 \cdot 2 \cdot 3 = 12$ entries not divisible by 3.

Try it. How many odd entries are in row 100? Answer: $100 = (1100100)_2$ has three 1s, so $2^3 = 8$.

Connections. Binomial expansion ⟶ prime remainders ⟶ base-p digits ⟶ patterns at repeated scales. **Further reading:** F. E. Su et al., *Lucas' Theorem*, Math Fun Facts; P. J. Cameron, *Notes on Combinatorics*, Appendix 1.6.

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PASCAL'S TRIANGLE MOD 2 AND MOD 3

From binomials to bits and trits

Colour the remainders: the number base becomes visible as geometry.

1 Expand, add, reduce

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k, \qquad (1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4.$$

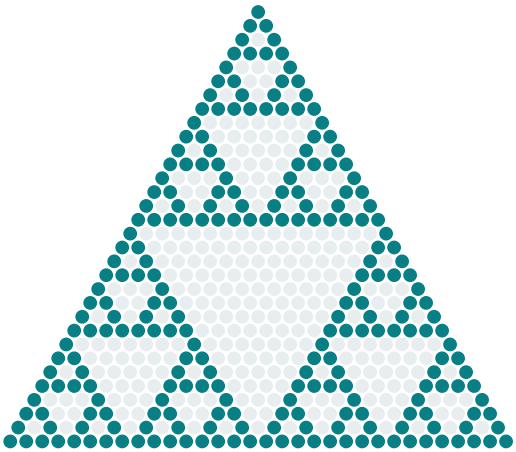
Row n contains the coefficients $\binom{n}{k}$. Replace each coefficient by its remainder $R_p(n,k)$:

$$R_p(n,k) = \binom{n}{k} \bmod p, \qquad R_p(n,k) = \big[R_p(n-1,k-1) + R_p(n-1,k)\big] \bmod p.$$

Start with $R_p(0,0) = 1$ and zeros outside the triangle. Each entry adds its two parents, then keeps only the remainder.

2a Mod 2: Sierpiński's triangle

$2^5 = 32$ rows: $0 \le n < 32$.

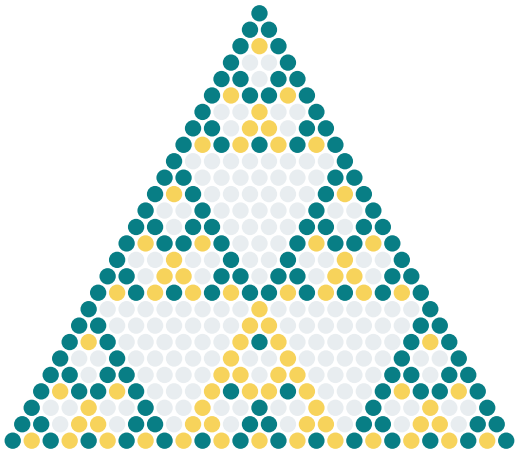


0: even 1: odd

The seed is $1;\ 1,1$. Three copies repeat at scale $1/2$. Parent addition mod 2 is the bit operation XOR.

2b Mod 3: a ternary cousin

$3^3 = 27$ rows: $0 \le n < 27$.



0 1 2

The seed is $1;\ 1,1;\ 1,2,1$. Six copies repeat at scale $1/3$. The central copy multiplies by 2, swapping colours 1 and 2.

3 Lucas's theorem: read the digits

For prime p, write $n = \sum_i n_i p^i$ and $k = \sum_i k_i p^i$, with $0 \le n_i, k_i < p$. Pad with leading zeros.

$$\boxed{\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod p}$$

Use $\binom{a}{b} = 0$ when $b > a$. An entry is nonzero precisely when **every digit of k fits beneath the corresponding digit of n.**

4 One coefficient, two readings: $\binom{14}{4} = 1001$

Binary: bits

$$14 = (1110)_2, \qquad 4 = (0100)_2.$$

$$\binom{14}{4} \equiv \binom{1}{0}\binom{1}{1}\binom{1}{0}\binom{0}{0} = 1 \pmod 2.$$

Every 1-bit of k fits: remainder 1.

Ternary: trits

$$14 = (112)_3, \qquad 4 = (011)_3.$$

$$\binom{14}{4} \equiv \binom{1}{0}\binom{1}{1}\binom{2}{1} = 2 \pmod 3.$$

The factor $\binom{2}{1} = 2$ gives remainder 2.

5 Why the base controls the pattern

For prime p, the middle coefficients of $(1+x)^p$ are divisible by p, so $(1+x)^p \equiv 1 + x^p \pmod p$. Consequently,

$$(1+x)^n = \prod_i \big[(1+x)^{p^i}\big]^{n_i} \equiv \prod_i (1+x^{p^i})^{n_i} \pmod p.$$

Choosing $x^{k_i p^i}$ from each factor gives Lucas's product. Each extra digit repeats the seed at a new scale: **the same bits and trits that encode the indices also determine the colours and the gaps.**

Lucas's theorem: R. Meštrović (2014), arXiv:1409.3820. Self-similarity: R. P. Kubelka (2004), *Fibonacci Quarterly* 42(1), 70–75.

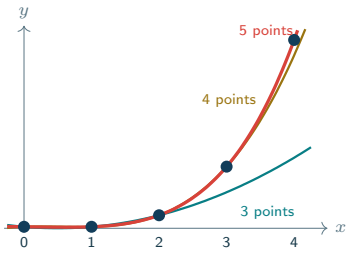
MATHS ON ONE PAGE

Polynomial interpolation: Vandermonde and Newton

Five points fix one quartic; the Vandermonde system finds it all at once, and Newton's differences build it one point at a time.

1 / THE RULE AND THE PICTURE

Through $n+1$ points with distinct x there is exactly one polynomial of degree at most n



Newton's path: each added point adds one term

Two such polynomials would differ by a polynomial of degree at most n with $n+1$ zeros, which must vanish, so every method that works returns the same curve. The data of your Worpitsky page,

$$q(0),\dots,q(4) = 7,\ 7,\ 61,\ 289,\ 883 \implies q(x) = 3x^4 + 2x^3 - 5x + 7$$

are used throughout, and the methods differ only in the basis they use: monomials x^k for Vandermonde, the products $x(x-1)\cdots$ for Newton, binomials $\binom{x}{k}$ for the jet at 0, and cardinal polynomials for Lagrange.

2 / VANDERMONDE: ALL POINTS AT ONCE

Write $q = a_0 + a_1x + \cdots + a_4x^4$ and solve the linear system for the coefficients

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 3 & 9 & 27 & 81 \\ 1 & 4 & 16 & 64 & 256 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \\ 61 \\ 289 \\ 883 \end{pmatrix} \implies a = (7, -5, 0, 2, 3).$$

The determinant is $\prod_{i<j}(x_j - x_i) = 1!\,2!\,3!\,4! = 288$, non-zero exactly when the nodes are distinct. The price is conditioning: the condition number is 2.6×10^3 for these five nodes but 9×10^{10} for ten and 3×10^{18} for fifteen nodes $0, 1, 2, \dots$, so the monomial coefficients lose all accuracy in double precision. Changing $q(4)$ by δ changes four coefficients, $a_1, \dots, a_4$ by $\delta\left(-\frac{1}{4}, \frac{11}{24}, -\frac{1}{4}, \frac{1}{24}\right)$.

3 / NEWTON: ONE POINT AT A TIME

Divided differences give the coefficients of the nested products $1,\ x,\ x(x-1),\ \dots$

x_i	y_i	first	second	third	fourth
0	7				
		0			
1	7		27		
		54		20	
2	61		87		3
		228		32	
3	289		183		
		594			
4	883				

Each entry is the difference of its two neighbours divided by the span of x values, and the top edge $7, 0, 27, 20, 3$ gives

$$q = 7 + 0\,x + 27\,x(x-1) + 20\,x(x-1)(x-2) + 3\,x(x-1)(x-2)(x-3).$$

Adding a point appends one term and leaves the others alone: changing $q(4)$ by δ adds $\delta\,x(x-1)(x-2)(x-3)/24$ only, and a sixth point $(5, 2107)$ on the same quartic gives a new coefficient 0.

4 / WORPITSKY AND LAGRANGE AS NEIGHBOURS

Equal spacing makes divided differences into forward differences over $k!$

The coefficients $\Delta^k q(0) = 7, 0, 54, 120, 72$ give $q(n) = 7 + 0\binom{n}{1} + 54\binom{n}{2} + 120\binom{n}{3} + 72\binom{n}{4}$, the discrete jet at $n = 0$ of your Worpitsky page. The Worpitzky numbers convert from the monomial coefficients: $n^4 = \binom{n}{1} + 14\binom{n}{2} + 36\binom{n}{3} + 24\binom{n}{4}$ and $n^3 = \binom{n}{1} + 6\binom{n}{2} + 6\binom{n}{3}$, so $54 = 3 \cdot 14 + 2 \cdot 6$, $120 = 3 \cdot 36 + 2 \cdot 6$ and $72 = 3 \cdot 24$. Lagrange writes $q = \sum y_i L_i(x)$ with $L_i = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$, for example $L_0 = \frac{(x-1)(x-2)(x-3)(x-4)}{24}$, and the barycentric weights for these nodes are $\frac{1}{24}, -\frac{1}{6}, \frac{1}{4}, -\frac{1}{6}, \frac{1}{24}$.

5 / CHOOSING A METHOD, AND A WARNING

Newton suits data that arrive in sequence, and the nodes matter more than the method

method	build	evaluate	add a point
Vandermonde	solve a 5×5 system, $O(n^3)$	$O(n)$ by Horner	all coefficients change
Newton	difference table, $O(n^2)$	$O(n)$ nested	one new term
Worpitsky jet	differences at 0, equal spacing	$\sum \Delta^k q(0) \binom{n}{k}$	one new term
Lagrange	weights, $O(n^2)$	$O(n)$ barycentric	weights recomputed

Every method reproduces the data exactly and can still misbehave between the nodes: interpolating $1/(1+25x^2)$ on $[-1, 1]$ at 11 equally spaced points has a largest error of 1.92, while the 11 Chebyshev nodes give 0.109.

ASIDE / THE SAME POLYNOMIAL

The four methods are changes of basis for one polynomial, and the four views on your page agree as $q(4)$ moves.

LINKS TO OTHER SHEETS

The interactive version is *Four Interpolation Methods: Deep Dive*; the Worpitzky numbers are in *Cards, piles and differences*.

Convention. $\Delta q(n) = q(n+1) - q(n)$; condition numbers are for Vandermonde matrices on the nodes $0, 1, 2, \dots$.

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PATHS, PASCAL AND DIAGONAL STEPS ON ONE PAGE

Delannoy: three ways through a grid

Allow a diagonal move, and familiar grid walks produce a new counting triangle.

1 / BEGIN WITH PASCAL'S GRID WALKS

Right, up, or diagonally up-right

With only right and up steps, a walk from $(0,0)$ to (m,n) has $\binom{m+n}{m}$ possible step orders.

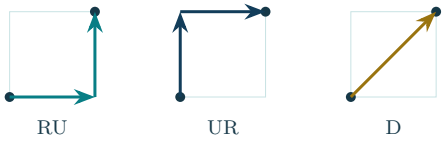
Now allow three unit moves:

$$\mathrm{R} = (1,0), \quad \mathrm{U} = (0,1), \quad \mathrm{D} = (1,1).$$

The **Delannoy number** $D(m,n)$ counts all such paths. Even one square has three routes:

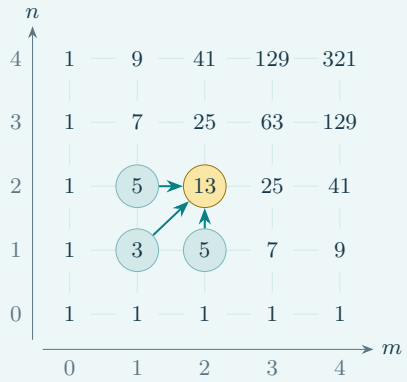
$$\mathrm{RU}, \quad \mathrm{UR}, \quad \mathrm{D}, \qquad D(1,1) = 3.$$

Each diagram goes from $(0,0)$ to $(1,1)$.



2 / COUNT BY THE FINAL STEP

Every entry adds three neighbours



A path reaches (m,n) from the left, from below, or along the diagonal. These last-step possibilities do not overlap:

$$D(m,n) = D(m-1,n) + D(m,n-1) + D(m-1,n-1), \qquad m,n \geq 1.$$

There is one route along either axis, so start with

$$D(m,0) = D(0,n) = 1.$$

At the origin, $D(0,0) = 1$ counts the empty walk. The three shaded neighbours give

$$\boxed{D(2,2) = 5 + 5 + 3 = 13.}$$

Swapping right and up reflects every path: $D(m,n) = D(n,m)$.

3 / CHECK THE SAME ANSWER ANOTHER WAY

Sort the paths by their number of diagonal steps

With exactly k diagonals, reaching (m,n) requires $m-k$ right steps and $n-k$ up steps: $m+n-k$ steps altogether. Count their different orders. For a 2×2 grid:

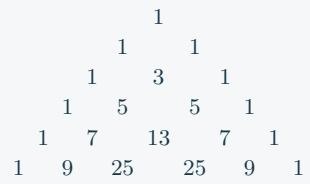
k	steps to arrange	number of steps	orders
0	2R, 2U	4	$4!/(2!\,2!) = 6$
1	R, U, D	3	$3! = 6$
2	2D	2	$2!/2! = 1$

Thus $D(2,2) = 6 + 6 + 1 = 13$. In general, the same reasoning gives

$$D(m,n) \quad = \quad \sum_{k=0}^{\min(m,n)} \frac{(m+n-k)!}{(m-k)!\,(n-k)!\,k!}$$

4 / TURN THE ARRAY INTO A TRIANGLE

Read along coordinate sums $m + n = s$



Rows $s = 0, 1, \ldots, 5$ list $D(s-r, r)$ for $r = 0, \ldots, s$. The square endpoints give the **central Delannoy numbers**:

$$D(n,n) = 1, 3, 13, 63, 321, \ldots \quad (n = 0, 1, \ldots).$$

These paths also encode alignments of two ordered lists: R advances one list, U the other, and D pairs their next entries.

Conventions. m, n are non-negative integers; only the three stated unit moves are allowed. A diagonal counts as one step.
References. OEIS A008288; A001850 (central numbers); Banderier & Schwer, Why Delannoy numbers?

CHESS MOVES AND DISTANCE ON ONE PAGE

Rook, king and crow

Three ways to measure a journey across the board, and the shortest paths that each one allows.

THE RULE

One displacement, three distances

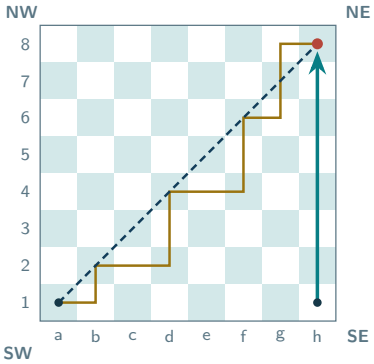
Give each square coordinates, files a to h as $x = 1$ to 8 and ranks as $y = 1$ to 8, and write a journey as the displacement $(\Delta x, \Delta y)$. The three measures are the cases $p = 1, 2, \infty$ of $d_p = \big(|\Delta x|^p + |\Delta y|^p\big)^{1/p}$:

rook, one square at a time	$d_1 = \lvert\Delta x\rvert + \lvert\Delta y\rvert$
crow, in a straight line	$d_2 = \sqrt{\Delta x^2 + \Delta y^2}$
king, counting moves	$d_\infty = \max\big(\lvert\Delta x\rvert, \lvert\Delta y\rvert\big)$

The rook's d_1 is the Manhattan or taxicab distance and the king's d_∞ is the Chebyshev distance, and for every journey

$$d_\infty \;\le\; d_2 \;\le\; d_1 \;\le\; 2\,d_\infty,$$

with all three equal exactly when the journey runs along a single rank or file.



SE to NE along the h-file, and SW to NE with a rook staircase and the crow's line.

TWO JOURNEYS TO THE NE CORNER

Along an edge the measures agree, across the diagonal they split

From h1 in the SE corner to h8 in the NE corner the displacement is $(0, 7)$, so all three distances are 7; from a1 in the SW corner the displacement is $(7, 7)$, and the three measures separate as far as the inequality allows. The geodesic counts, the number of different shortest paths, behave in the opposite way to the distances.

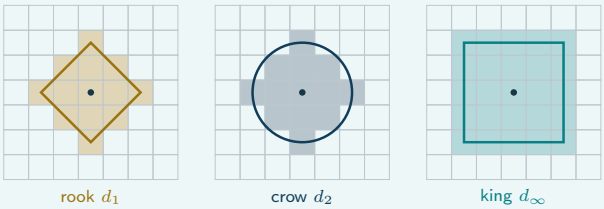
journey	d_1	d_2	d_∞	shortest rook-step paths	shortest king paths
h1 to h8, SE to NE	7	7	7	1	127
a1 to h8, SW to NE	14	$7\sqrt{2} \approx 9.90$	7	$\binom{14}{7} = 3432$	1

A real rook covers either journey in at most two moves, since it slides any number of squares; its path *length* is d_1, while its move *count* is 0, 1 or 2.

CIRCLES

Every square within distance 2

A circle is the set of points at a fixed distance from a centre, so each measure draws its own circle. The rook's is a diamond, the king's is a square and only the crow's is round; the king's square of radius 2 holds 25 squares, the rook's diamond 13, and these are exactly the squares a king reaches in two moves and a rook reaches in two single steps. At radius 2 the crow's squares happen to coincide with the rook's diamond, and the two part company from radius 3, where d6 to f8 lies within $2\sqrt{2}$ by air but 4 by rook.



Counting the shortest paths

A geodesic is a path whose length equals the distance. The crow's geodesic is the unique straight segment, while a rook-step geodesic from a1 to h8 is any order of 7 steps right and 7 steps up, giving $\binom{14}{7}$. A king's geodesic up the h-file makes 7 moves upward, each drifting one file left, right or not at all, without leaving the board and returning to the h-file, which is counted by the Motzkin number $M_7 = 127$.

Geodesics on a curved surface

On a flat board the crow's geodesic is a straight line, but on the Earth the shortest route between two places is an arc of a great circle, with length $R\,\Delta\sigma$ for the central angle $\Delta\sigma$; flying between two cities on the same latitude therefore bends towards the pole, which a flat map draws as a curve.

Conventions. Distances are measured between square centres in units of one square, with White's side at the south, so a1 is SW, h1 is SE and h8 is NE. A rook-step means a rook moving one square at a time along a rank or file, and d_∞ is the limit of d_p as $p \to \infty$.

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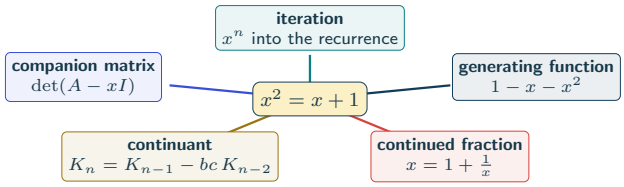
THE CHARACTERISTIC EQUATION ON ONE PAGE

The characteristic equation beneath five presentations

Five presentations of a recurrence are forced to one quadratic, and the diagonal gauge leaves it alone.

1 / THE RULE AND THE PICTURE

Five descriptions of the Fibonacci sequence are forced to the same quadratic



The recurrence $F_n = F_{n-1} + F_{n-2}$ has one characteristic equation, whose roots are $\varphi = 1.618$ and $\psi = -0.618$:

$$x^2 - x - 1 = 0$$

Every presentation carries these two roots: eigenvalues, Binet bases, convergent limit and pole reciprocals.

2 / THE FIVE PRESENTATIONS / HOW THE EQUATION APPEARS

Each presentation arrives at $x^2 - x - 1$ by its own route

presentation	route	what the roots become
iteration	try $F_n = x^n$, so $x^n = x^{n-1} + x^{n-2}$ and the equation appears	bases of the Binet formula
companion matrix	$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has $\det(A - xI) = x^2 - x - 1$, the origin of the name	eigenvalues of A
continuant	$K_n = K_{n-1} - bc\,K_{n-2}$ with $bc = -1$ inherits the recurrence	growth rate of K_n
continued fraction	$[1; 1, 1, \ldots]$ is the fixed point of $x \mapsto 1 + \frac{1}{x}$, so $x^2 = x + 1$	limit of the convergents
generating function	$\sum F_n x^n = \dfrac{x}{1 - x - x^2}$, the reversed polynomial	poles at $\frac{1}{\varphi}$ and $\frac{1}{\psi}$

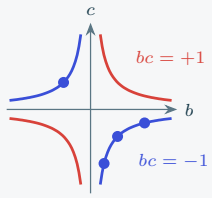
3 / HARD TO VARY / CHANGE THE EQUATION AND EVERYTHING MOVES

Altering a coefficient of $x^2 = px + q$ moves roots, bases, limits and poles together

equation	recurrence	first terms	dominant root
$x^2 = x + 1$	$a_n = a_{n-1} + a_{n-2}$	$1, 1, 2, 3, 5, 8$	$\varphi = 1.618$
$x^2 = 2x + 1$	$a_n = 2a_{n-1} + a_{n-2}$	$1, 2, 5, 12, 29, 70$ (Pell)	$1 + \sqrt{2} = 2.414$
$x^2 = x + 2$	$a_n = a_{n-1} + 2a_{n-2}$	$0, 1, 1, 3, 5, 11, 21$ (Jacobsthal)	2

4 / THE GAUGE FREEDOM / ONLY THE PRODUCT bc MATTERS

Diagonal conjugation slides (b, c) along a hyperbola $bc = \text{const}$



For the continuant with 1 on the diagonal, b above and c below, $\det M_n = \det M_{n-1} - bc \det M_{n-2}$, so the pairs $(1, -1)$, $(2, -\frac{1}{2})$ and (i, i) all give Fibonacci because $bc = -1$. Conjugating by $D = \mathrm{diag}(d_i)$ sends $b \mapsto \frac{d_i}{d_{i+1}} b$ and $c \mapsto \frac{d_{i+1}}{d_i} c$, which slides the pair along its hyperbola. The only symmetric member is the complex pair (i, i), since a real symmetric or Hermitian continuant has $bc = b^2$ or $|b|^2 \geq 0$. Changing the product, $(i, -i)$ is Hermitian with $bc = +1$, giving $x^2 - x + 1 = 0$ and the period-six sequence $1, 1, 0, -1, -1, 0, \ldots$

5 / THE GENERAL SEQUENCING LAW / PASCAL, STIRLING, WORPITZKY AND BELL

We meet a list, then methods, then one equation, and the triangles climb the same steps

Each row of a triangle is the previous row acted on by a bidiagonal operator, and the weights u_k and v_k are the only choices:

$$T(n, k) = u_k\, T(n-1, k) + v_k\, T(n-1, k-1)$$

triangle	(u_k, v_k)	row $n = 4$	sum	row sums
Pascal $\binom{n}{k}$	$(1,\ 1)$	$1,\ 4,\ 6,\ 4,\ 1$	16	powers of 2
Stirling $S(n, k)$	$(k,\ 1)$	$1,\ 7,\ 6,\ 1$	15	Bell numbers $1, 2, 5, 15, 52$
Worpitzky $W(n, k) = k!\, S(n, k)$	$(k,\ k)$	$1,\ 14,\ 36,\ 24$	75	Fubini numbers $1, 3, 13, 75, 541$

Scaling row k by $k!$, the diagonal gauge $D = \mathrm{diag}(k!)$, turns v_k into $k\,v_k$ and carries Stirling to Worpitzky while the diagonal $u_k = k$, which holds the eigenvalues $1, \ldots, n$, stays put, the same invariance as the continuant. Worpitzky's numbers are the coefficients of x^n in Pascal's own basis, $x^n = \sum_k W(n, k)\binom{x}{k}$, and Pascal feeds Bell through $B_{n+1} = \sum_k \binom{n}{k} B_k$, so $B_4 = 1 + 3 + 6 + 5 = 15$.

ASIDE / THE NAME

The name comes from linear algebra, where $\det(A - \lambda I) = 0$ gives the eigenvalues of the companion matrix.

LINKS TO OTHER SHEETS

The gauge (i, i) and Pell, Jacobsthal: *Why i makes Fibonacci inevitable*; *Metallic means as differences of squares*. The continuant and eigenvalues: *The determinant in its many guises*; *Solving a quadratic as a line meeting a hyperbola*.

Convention. bc is the product of the off-diagonal entries, $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$. Compresses the essay *The Characteristic Equation*.

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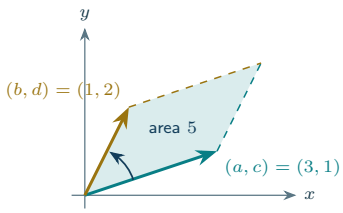
LINEAR ALGEBRA ON ONE PAGE

The determinant in its many guises

One number that is an oriented area, a difference of two squares, a product of eigenvalues and a count of tilings.

1 / THE RULE AND THE PICTURE

The determinant is the oriented area of the parallelogram spanned by the columns



For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the columns (a, c) and (b, d) span a parallelogram whose signed area is

$$\det A = ad - bc$$

Here $3 \cdot 2 - 1 \cdot 1 = 5$. The sign records orientation, positive when the turn from the first column to the second is anticlockwise, so swapping the columns gives -5, and a determinant of 0 means the columns lie on one line and the parallelogram has collapsed.

2 / FAMILIES OF MATRICES / ONE FORMULA, MANY NAMES

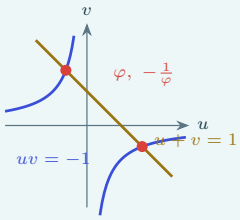
Restricting the shape of A turns $ad - bc$ into a norm

matrix	determinant	name	example
$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$	$a^2 - b^2 = (a-b)(a+b)$	difference of two squares	$\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix} \to 16 = 2 \cdot 8$
$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$	$a^2 + b^2$	Gaussian norm	$\begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix} \to 13$
$\begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$	$a^2 - 2b^2$	Pell norm	$\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \to 1$
3×3 circulant (a, b, c)	$a^3 + b^3 + c^3 - 3abc$	Eisenstein norm times $a + b + c$	$(1, 2, 3) \to 18 = 6 \cdot 3$

Each 2×2 row is multiplication by $a + bj$ with $j^2 = +1,\ -1,\ +2$, so its determinant is the norm of $a + bj$.

3 / THE CHARACTERISTIC POLYNOMIAL / EIGENVALUES ON A HYPERBOLA

The eigenvalues are where $u + v = \mathrm{tr}A$ meets $uv = \det A$



Since $\det(\lambda I - A) = \lambda^2 - \mathrm{tr}(A)\,\lambda + \det A$, the determinant is the product of the two eigenvalues and the trace is their sum. For $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ they are $a \pm b$, which is why $a^2 - b^2$ factors, and for $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ they are $a \pm ib$, with product $a^2 + b^2$. The matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has trace 1 and determinant -1, so its eigenvalues are the points in the picture, φ and $-\frac{1}{\varphi}$.

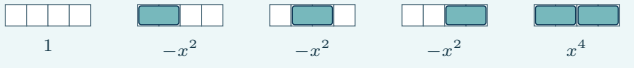
4 / TWO PROPERTIES THAT MAKE IT WORK / PRODUCTS AND SHEARS

$\det(AB) = \det A \det B$, and adding a multiple of one row to another leaves $\det$ unchanged

Multiplying rotation-scaling matrices (a, b) and (c, d) turns a matrix product into a number product, $(a^2+b^2)(c^2+d^2) = (ac-bd)^2+(ad+bc)^2$, so $5 \cdot 13 = 65 = (-4)^2+7^2$. Because row shears keep $\det$, elimination reads it off as the product of the pivots: for $3x+2y = 16$ and $2x+5y = 19$ the pivots are 3 and $\frac{11}{3}$, and the determinant is 11.

5 / A DETERMINANT THAT COUNTS / THE COMPLEX FIBONACCI GENERATOR

With 1 on the diagonal and i beside it, the determinant becomes the Fibonacci number F_{n+1}



n	1	2	3	4	5	6	7
$K_n(i)$	1	2	3	5	8	13	21

Expanding the determinant with 1 on the diagonal and x beside it gives $K_n = K_{n-1} - x^2 K_{n-2}$, and $x = i$ turns the minus into the plus of Fibonacci. Each term is a tiling of a row of squares by squares and dominoes, each domino weighing $-x^2 = +1$, so for $n = 4$ the five tilings give $1 + 3 + 1 = 5$.

6 / LINKS TO OTHER SHEETS / WHERE EACH GUISE IS DEVELOPED

in this sheet	developed in
oriented area, shears and pivots	*Linear simultaneous equations and the LCM*; *The LCM method as Gaussian elimination*
eigenvalues on a hyperbola	*Solving a quadratic as a line meeting a hyperbola*; *Cardano's cubic as a hyperbola*
Gaussian, Pell and Eisenstein norms	*Cyclotomic fields and the arcs of inscribed polygons*
tridiagonal determinant and tilings	*Why i makes Fibonacci inevitable*; *Metallic means as differences of squares*

ASIDE / FOR THE CURIOUS

The eigenvalues of the Fibonacci matrix are $1 + 2i\cos\frac{k\pi}{n+1}$, and their product is F_{n+1}.

ASIDE / CHECK IN PYTHON

`np.linalg.det` of the 4×4 matrix with 1 and i gives 5.

Convention. tr is the trace, φ the golden ratio, $F_1 = F_2 = 1$.

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FROM DIAGONAL PRODUCTS TO A CHANGE OF SIGN

Difference of two squares

Squares, cubes and a change of sign through two small determinants.

1 / START WITH A TWO-BY-TWO DETERMINANT

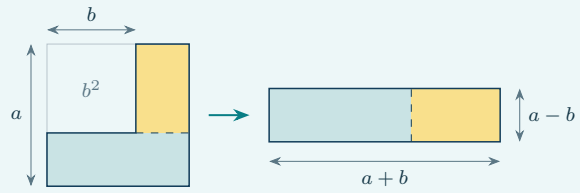
The difference of squares is a determinant

Subtract the products of the two diagonals:

$$D_2 = \begin{vmatrix} a & b \\ b & a \end{vmatrix} = a \cdot a - b \cdot b.$$

$$\boxed{D_2 = a^2 - b^2 = (a+b)(a-b)}.$$

The same area can be seen in two ways. For $a > b > 0$, remove the smaller square and rearrange the pieces.



$$8^2 - 5^2 = (8+5)(8-5) = 13 \times 3 = 39.$$

2 / MOVE EACH ROW ROUND ONE PLACE

And for a 3×3 matrix?

$$D_3 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}.$$

The six determinant terms give

$$D_3 = a^3 + b^3 + c^3 - abc - abc - abc.$$

$$\boxed{D_3 = a^3 + b^3 + c^3 - 3abc}.$$

Set $c = 0$: this becomes $a^3 + b^3$. Now replace b by $-b$ to obtain the difference:

$$\boxed{a^3 - b^3 = (a-b)(a^2 + ab + b^2)}.$$

The middle ab is still needed:

$$(a-b)(a^2+b^2) \neq a^3 - b^3 \quad \text{in general.}$$

3 / LET THE SECOND TERM BECOME IMAGINARY

A sum of squares can factorise too

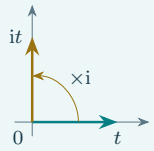
The imaginary unit is defined by $\mathrm{i}^2 = -1$. Hence

$$(\mathrm{i}b)^2 = \mathrm{i}^2 b^2 = -b^2.$$

Replace b by $\mathrm{i}b$ in the familiar identity:

$$a^2 + b^2 = a^2 - (\mathrm{i}b)^2$$
$$= \boxed{(a + \mathrm{i}b)(a - \mathrm{i}b)}.$$

$$\begin{vmatrix} a & \mathrm{i}b \\ \mathrm{i}b & a \end{vmatrix} = a^2 + b^2.$$



Multiplying by i makes a quarter-turn in the complex plane.

Squaring gives the sign change:

$$(\mathrm{i}t)^2 = -t^2.$$

4 / A SMALL GLIMPSE OF PHYSICS

From t to $\mathrm{i}t$: Wick rotation

Try $t \mapsto \mathrm{i}t$ in a difference of squares:

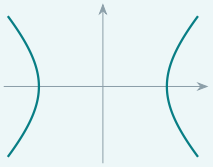
$$x^2 - t^2 \longrightarrow x^2 - (\mathrm{i}t)^2 = \boxed{x^2 + t^2}.$$

In physics, space and time can appear with opposite signs. Writing $t = \mathrm{i}\tau$, with τ real, gives

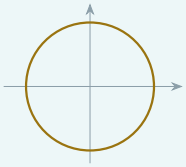
$$x^2 - c^2 t^2 \longrightarrow x^2 + c^2 \tau^2.$$

Here c is the speed of light. This is the sign change behind **Wick rotation**, which relates spacetime formulas to Euclidean geometry.

For real coordinates, compare these two curves:



$x^2 - y^2 = 1$
hyperbola



$x^2 + y^2 = 1$
circle

The same matrix pattern, with one change of sign

$$\begin{vmatrix} a & b \\ b & a \end{vmatrix} = a^2 - b^2 \qquad \begin{vmatrix} a & \mathrm{i}b \\ \mathrm{i}b & a \end{vmatrix} = a^2 + b^2.$$

Physics link. An imaginary-time substitution is a mathematical change of coordinate used in calculations. Further reading: J. Penedones and S. Reymond, *Quantum Physics IV*, EPFL, Lecture 5.

Networked–lattice

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TRIDIAGONAL DETERMINANTS AND THE FIBONACCI MATRIX

From cubes to Fibonacci

Change the entries, grow the matrix, and discover an addition rule.

1 / PICK UP THE CUBIC EXAMPLE

Set $c = 0$, then change b

The cyclic matrix from the previous page becomes

$$C_3(a,b) = \begin{pmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{pmatrix}, \qquad \det C_3 = a^3 + b^3.$$

The power of b decides what a substitution does:

Replace b by	New determinant
b	$a^3 + b^3$
$-b$	$a^3 - b^3$
$\mathrm{i}b$	$a^3 - \mathrm{i}b^3$

$$\mathrm{i}^2 = -1, \qquad \mathrm{i}^3 = -\mathrm{i}.$$

Gold marks the closing link. Next, use both neighbouring diagonals and zero far corners.

2 / KEEP ONLY THE THREE CENTRAL DIAGONALS

A tridiagonal pattern

Only the main diagonal and its two neighbours can have non-zero entries:

$$T_3(a,b) = \begin{pmatrix} a & b & 0 \\ b & a & b \\ 0 & b & a \end{pmatrix}.$$

Write $D_n = \det T_n(a,b)$ as this pattern grows. Use $D_0 = 1$, $D_1 = a$. The first cases are

$$D_2 = a^2 - b^2,$$
$$D_3 = a(a^2 - b^2) - b(ab).$$

Expanding along the first row gives the same rule at every size:

$$\boxed{D_n = aD_{n-1} - b^2 D_{n-2}} \quad (n \geq 2).$$

Here $b \mapsto -b$ leaves D_n unchanged.

3 / LET AN IMAGINARY ENTRY CHANGE THE RULE

A minus becomes a plus

Choose $a = 1$ and $b = \mathrm{i}$. For example,

$$T_4(1,\mathrm{i}) = \begin{pmatrix} 1 & \mathrm{i} & 0 & 0 \\ \mathrm{i} & 1 & \mathrm{i} & 0 \\ 0 & \mathrm{i} & 1 & \mathrm{i} \\ 0 & 0 & \mathrm{i} & 1 \end{pmatrix}.$$

Since $b^2 = \mathrm{i}^2 = -1$, the determinant rule becomes

$$D_n = D_{n-1} - (-1)D_{n-2}.$$

$$\boxed{D_n = D_{n-1} + D_{n-2}}.$$

Start with $D_0 = 1$, $D_1 = 1$, and add the last two:

n	0	1	2	3	4	5
D_n	1	1	2	3	5	8

With $F_0 = 0$, $F_1 = 1$, these are Fibonacci numbers shifted one place: $D_n = F_{n+1}$.

4 / STORE TWO SUCCESSIVE VALUES

The same addition in a 2×2 matrix

Let

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Multiplication does two things: add the entries, then keep the previous first entry:

$$Q \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u+v \\ u \end{pmatrix}.$$

Starting with $\binom{1}{1}$, repeated multiplication gives

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 3 \\ 2 \end{pmatrix} \xrightarrow{Q} \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

The same values appear inside the powers of Q:

$$Q^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad Q^3 = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} D_n \\ D_{n-1} \end{pmatrix} = Q \begin{pmatrix} D_{n-1} \\ D_{n-2} \end{pmatrix} \quad (n \geq 2).$$

The link: one recurrence, two matrix constructions

$$\det T_n(1,\mathrm{i}) = F_{n+1} \qquad Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix} \quad (n \geq 1).$$

Companion to the difference-of-squares page. $D_0 = 1$ is the empty-determinant convention. The cyclic matrix C_3 and the tridiagonal family T_n use different entry patterns.

Further reading. Cahill and Narayan, *Fibonacci and Lucas Numbers as Tridiagonal Matrix Determinants*; H. W. Gould, the Fibonacci Q-matrix.

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DETERMINANTS ON ONE PAGE

Why i makes Fibonacci inevitable

A determinant subtracts, and the imaginary unit turns its subtraction into the addition that defines Fibonacci.

1 / THE RULE AND THE PICTURE

The tridiagonal continuant subtracts x^2, and $x = i$ makes it add

$$\begin{bmatrix} 1 & x & 0 & 0 \\ x & 1 & x & 0 \\ 0 & x & 1 & x \\ 0 & 0 & x & 1 \end{bmatrix}$$

$K_4(x)$

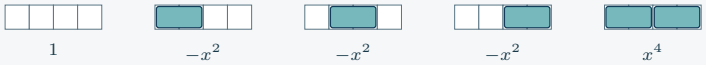
Expanding along the first row gives the continuant recurrence, whose minus sign comes from the alternating character of the determinant, and setting $x = i$ gives $-x^2 = -i^2 = +1$:

$$K_n = K_{n-1} - x^2 K_{n-2} \xrightarrow{x=i} K_n = K_{n-1} + K_{n-2}$$

n	1	2	3	4	5	6	7
$K_n(i)$	1	2	3	5	8	13	21

2 / WHY THE SUM IS A COUNT / SQUARES AND DOMINOES

Each term of the determinant is a tiling of a strip, and every domino weighs $-x^2$



$K_4 = 1 - 3x^2 + x^4$, and at $x = i$ the five tilings each weigh $+1$, so $K_4(i) = 1 + 3 + 1 = 5$.

In general $K_n(x) = \sum_k \binom{n-k}{k}(-x^2)^k$, one term per tiling with k dominoes, so $K_n(i) = \sum_k \binom{n-k}{k} = F_{n+1}$, the shallow diagonals of Pascal's triangle. The planar version of this trick is Kasteleyn's: weighting the vertical edges by i makes a determinant count domino tilings of a rectangle, the model behind Catalan's constant.

3 / THE METALLIC FAMILY / TWO WAYS TO CHANGE THE MATRIX

The amplitude $\sqrt{m}\,i$ weighs dominoes, and the diagonal d weighs squares

matrix	recurrence	first terms	growth
diagonal 1, off-diagonal i	$D_n = D_{n-1} + D_{n-2}$	$1, 2, 3, 5, 8, 13$	$\varphi = 1.618$
diagonal 1, off-diagonal $\sqrt{2}\,i$	$D_n = D_{n-1} + 2D_{n-2}$	$1, 3, 5, 11, 21, 43$ (Jacobsthal)	2
diagonal 1, off-diagonal $\sqrt{3}\,i$	$D_n = D_{n-1} + 3D_{n-2}$	$1, 4, 7, 19, 40, 97$	2.303
diagonal 2, off-diagonal i	$D_n = 2D_{n-1} + D_{n-2}$	$2, 5, 12, 29, 70$ (Pell)	$1 + \sqrt{2} = 2.414$
diagonal 3, off-diagonal i	$D_n = 3D_{n-1} + D_{n-2}$	$3, 10, 33, 109$	$\frac{3+\sqrt{13}}{2} = 3.303$

The recurrence $D_n = dD_{n-1} + D_{n-2}$ has growth $\lambda^2 = d\lambda + 1$, the d-th metallic mean (golden, silver, bronze), so Pell and the silver mean come from the diagonal 2 and not from the amplitude $\sqrt{2}\,i$. Tridiagonal matrices only link neighbours and so give second order; Tribonacci needs a 3×3 companion matrix.

4 / CASSINI / THE DYNAMIC DETERMINANT

Consecutive Fibonacci numbers span a parallelogram of area one

With $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ we have $Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$ and $\det Q = -1$, so

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n, \qquad \text{for example } 8 \cdot 3 - 5^2 = -1, \quad 13 \cdot 5 - 8^2 = +1$$

The columns stretch as the numbers grow, yet the signed area stays at ± 1 and its sign records a flip of handedness at every step.

5 / THE LADDER OF NORMS / MATRIX SYMMETRY SELECTS THE NUMBER SYSTEM

Change the structure of the matrix, and the determinant lands in a different ring

matrix	determinant	ring	integers produced
$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$	$a^2 - b^2$	split	odd and multiples of 4
$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$	$a^2 + b^2$	$\mathbb{Z}[i]$	sums of two squares
3×3 circulant	$a^3 + b^3 + c^3 - 3abc = (a+b+c)\,N$	$\mathbb{Z}[\omega]$	not $\equiv 3, 6 \pmod 9$

Here $N = a^2 + b^2 + c^2 - ab - bc - ca = p^2 - pq + q^2$ with $p = a - c$, $q = b - c$ is the Eisenstein norm; the triple $(n+1, n, n-1)$ gives $9n$, and if $3 \mid \det$ then $3 \mid s = a + b + c$ and $q \equiv s^2 \equiv 0$, so $9 \mid \det$.

ASIDE / WHAT i DOES

It does not create the recurrence; it selects the additive branch of a structure that was already there.

ASIDE / CHECK IN PYTHON

`det(np.eye(4)+1j*(np.eye(4,k=1)+np.eye(4,k=-1)))` gives 5.

Convention. $K_0 = K_1 = 1$ and $F_1 = F_2 = 1$, so $K_n(i) = F_{n+1}$. After *Why i Makes Fibonacci Inevitable* (McMurmerings).

Networked-lattice

SUMS OF POWERS, EVEN AND ODD INDICES, AND TWO CLASSES OF PRIME

Bernoulli numbers, step by step

Start by adding whole numbers. Follow the fractions that appear.

1 / SOMETHING FAMILIAR

Pair a sum with its reverse

Throughout, use n terms, starting at 0 and ending at $n-1$. For example, when $n=4$, the sum is $0+1+2+3=6$.

$$S_1(n) = 0 + 1 + 2 + \cdots + (n-1),$$
$$S_1(n) = (n-1) + (n-2) + \cdots + 0.$$

Each column adds to $n-1$. There are n columns, so

$$2S_1(n) = n(n-1), \qquad S_1(n) = \frac{n^2}{2} - \frac{1}{2}n.$$

The subscript 1 means that we are adding first powers.



Two equal sums fill a rectangle.

2 / BUILD THE NEXT SUM

To add squares, subtract cubes

Write $S_2(n) = 0^2+1^2+\cdots+(n-1)^2$. The binomial expansion gives

$$(k+1)^3 - k^3 = 3k^2 + 3k + 1.$$

Add this for $k = 0, 1, \ldots, n-1$. On the left, all the middle cubes cancel:

$$(1^3 - 0^3) + (2^3 - 1^3) + \cdots = n^3.$$

On the right, we have three times the squares, three times the first powers, and n ones. Thus

$$n^3 = 3S_2(n) + 3S_1(n) + n.$$

Substitute our formula for $S_1(n)$ and rearrange:

$$S_2(n) = \frac{n^3}{3} - \frac{n^2}{2} + \frac{1}{6}n.$$

Check $n=4$: $0^2+1^2+2^2+3^2 = 14$.

3 / NAME THE RECURRING NUMBERS

Read the coefficient of n

The same method adds cubes, fourth powers and beyond. Each answer is a polynomial in n. The number multiplying the **plain n term** is the Bernoulli number B_m for the mth powers.

$$S_1(n) = \tfrac{1}{2}n^2 + \left(-\tfrac{1}{2}\right)n,$$
$$S_2(n) = \tfrac{1}{3}n^3 - \tfrac{1}{2}n^2 + \tfrac{1}{6}n,$$
$$S_3(n) = \tfrac{1}{4}n^4 - \tfrac{1}{2}n^3 + \tfrac{1}{4}n^2 + 0\,n.$$

So $B_1 = -1/2$, $B_2 = 1/6$ and $B_3 = 0$. A missing n term means its coefficient is zero.

Continuing the calculation gives

$$S_4(n) = \tfrac{1}{5}n^5 - \tfrac{1}{2}n^4 + \tfrac{1}{3}n^3 - \tfrac{1}{30}n,$$

so $B_4 = -1/30$. Also set $B_0 = 1$: adding n ones gives n.

4 / TWO DIFFERENT PATTERNS IN ONE SEQUENCE

Even and odd refer to the index

In B_m, the index m tells us the power being added. Read the even and odd positions separately:

even index m	0	2	4	6	8	10	12
B_m	1	$\frac{1}{6}$	$-\frac{1}{30}$	$\frac{1}{42}$	$-\frac{1}{30}$	$\frac{5}{66}$	$-\frac{691}{2730}$
odd index m	1	3	5	7	9	11	$\cdots$
B_m	$-\frac{1}{2}$	0	0	0	0	0	$\cdots$

The general rule: every odd-indexed B_m after B_1 is zero. At positive even indices the values are non-zero and their signs alternate $+, -, +, -, \ldots$. For example, the sum of cubes has no plain n term, while the sum of fourth powers has coefficient $-1/30$.

5 / A FIRST GLIMPSE OF A NEW PRIME CLASSIFICATION

Let the even-indexed numerators sort the odd primes

Choose an **odd prime** p. Reduce the fractions $B_2, B_4, \ldots, B_{p-3}$ to lowest terms. Check whether p divides any **numerator**, the integer above the fraction bar. The sign does not affect divisibility.

Regular: no numerator is divisible by p.
For $p=7$, check only $B_2 = 1/6$ and $B_4 = -1/30$. Neither 1 nor -1 is divisible by 7, so 7 is regular.

Irregular: at least one numerator is.
For $p = 691$, $B_{12} = -691/2730$ is on the checklist because $12 \le 691-3$. Its numerator is divisible by 691, so 691 is irregular.

The stopping point matters. Although 5 appears in the numerator of B_{10}, the test for $p=5$ stops at B_2; 5 is regular.

Next: two classes, two staircases. The only irregular primes below 100 are 37, 59 and 67. Every other odd prime below 100 is regular. We can count each class separately, then add them to count all the odd primes. These are two labels within the odd primes; 2 is outside this test.

Convention. Sums run from 0 to $n-1$, giving $B_1 = -1/2$. **Sources.** NIST DLMF, §24.2 (values and parity); §24.4 (power sums); The Bernoulli Number Page (prime classification).

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BERNOULLI NUMBERS AND PRIME CLASSIFICATION ON ONE PAGE

Regular and irregular primes

How sums of powers lead to a new way of classifying primes.

1 / START WITH FAMILIAR SUMS

Fractions appear when we add powers

The **Bernoulli numbers**, written $B_0, B_1, B_2, \ldots$, organise the coefficients in formulas for sums of powers. For example,

$$1^2 + 2^2 + \cdots + n^2 = \frac{n^3}{3} + \frac{n^2}{2} + \frac{1}{6}\,n, \qquad 1^4 + 2^4 + \cdots + n^4 = \frac{n^5}{5} + \frac{n^4}{2} + \frac{n^3}{3} - \frac{1}{30}\,n.$$

Here the coefficients of n are $B_2 = 1/6$ and $B_4 = -1/30$. For every positive even power m, that coefficient is B_m. The beginning of the even-indexed list, with fractions in lowest terms, is

$$B_2 = \frac{1}{6}, \quad B_4 = -\frac{1}{30}, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}, \quad B_{10} = \frac{5}{66}, \quad B_{12} = -\frac{691}{2730}, \quad B_{14} = \frac{7}{6}, \quad \ldots$$

The index tells us *which Bernoulli number*; the numerator is the signed whole number above the fraction bar.

2 / KUMMER'S TEST

Give each odd prime its own finite checklist

Choose an odd prime p. Write down the Bernoulli numbers at the **even indices from 2 to $p-3$**, in lowest terms:

$$B_2,\ B_4,\ B_6,\ \ldots,\ B_{p-3}.$$

Check whether p divides any of their **numerators** exactly. The denominators play no part in this test.

Regular: no numerator in the checklist is divisible by p.
Irregular: at least one numerator in the checklist is divisible by p.

The stopping point changes with the prime. For $p = 13$ it is B_{10}; for $p = 37$ it is B_{34}.

3A / CHECK EVERY ENTRY

Why 13 is regular

Here $p-3 = 10$, so the whole checklist is $B_2, B_4, B_6, B_8, B_{10}$.

index	2	4	6	8	10
numerator	1	−1	1	−1	5

None is a multiple of 13. We have checked every entry required, so 13 is **regular**.

Regular needs the entire checklist.

3B / FIND ONE COUNTEREXAMPLE

Why 37 is irregular

The first irregular prime is 37. Its checklist runs to B_{34}, and includes

$$B_{32} = -\frac{7709321041217}{510}.$$

Its numerator has a factor 37:

$$7709321041217 = 37 \times 208360028141.$$

This one entry proves that 37 is **irregular**.

Irregular needs only one counterexample.

4 / A LARGE PRIME WITH A SHORT EXPLANATION

The numerator 691 is already a clue

The prime 691 appears in $B_{12} = -691/2730$. Its checklist runs all the way to B_{688}, so B_{12} is included. Since 691 divides that numerator exactly, 691 **is irregular**. We can stop as soon as one counterexample appears.

5 / SEE THE FIRST EXCEPTIONS

Only 37, 59 and 67 are irregular below 100

3	5	7	11	13	17	19	23	29	31	**37**	41
43	47	53	**59**	61	**67**	71	73	79	83	89	97

Gold marks the irregular primes; the others shown are regular. We classify odd primes here. For $p = 3$ the checklist is empty, so it is regular.

A FAMILIAR CONNECTION / HEEGNER AND 163

A small bridge back to unique factorisation

The Heegner number 163 gives a quadratic integer system with **class number** 1, where factorisation is unique. Kummer studies a different family of integer systems, built using pth roots of unity: numbers whose pth power is 1. He proved that p is regular exactly when p does *not* divide that system's class number. That class number can still exceed 1. The Bernoulli checklist gives us an arithmetic way to detect this property.

Sources. NIST DLMF, §24.2 (Bernoulli values); The Bernoulli Number Page (Kummer's test and first exceptions); J. S. Milne, *Algebraic Number Theory*, §§3,6 (class numbers and regularity).

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FROM SEQUENCES TO GROUPS ON ONE PAGE

A quartic reveals Stirling numbers

Recover coefficients, uncover a counting triangle, then give its entries another job.

1 / BEGIN WITH AN UNKNOWN QUARTIC

Separate the contribution of each power

Let $P(x) = ax^4 + bx^3 + cx^2 + dx + e$, sampled at $x = 1, 2, 3, 4, 5$. Write $\Delta f(x) = f(x+1) - f(x)$ and $D_k = \Delta^k P(x)\Big|_{x=1}$.

Subtract adjacent values repeatedly. For x^4, the values $1, 16, 81, 256, 625$ give first entries $1, 15, 50, 60, 24$. Doing this for each power constructs the universal weights:

power	$\Delta^0\|_1$	$\Delta^1\|_1$	$\Delta^2\|_1$	$\Delta^3\|_1$	$\Delta^4\|_1$
1	1	0	0	0	0
x	1	1	0	0	0
x^2	1	3	2	0	0
x^3	1	7	12	6	0
x^4	1	15	50	60	24

These are Worpitzky weights $W(m,k) = \Delta^k(x^m)\Big|_{x=1}$. For P, combine the rows with weights e, d, c, b, a respectively.

2 / USE THE TABLE AS A SEQUENCE INTERPOLATOR

Work backwards from the highest difference

The last column gives $D_4 = 24a$; the next gives $D_3 = 60a + 6b$. Continue leftwards, subtracting the contributions already found:

$$
\begin{aligned}
a &= D_4/24,\\
b &= (D_3 - 60a)/6,\\
c &= (D_2 - 50a - 12b)/2,\\
d &= D_1 - 15a - 7b - 3c,\\
e &= D_0 - a - b - c - d.
\end{aligned}
$$

Example. The values

$$15,\ 65,\ 233,\ 645,\ 1475$$

give D_0, D_1, D_2, D_3, D_4 equal to

$$15,\ 50,\ 118,\ 126,\ 48.$$

Backtracking gives $(a, b, c, d, e) = (2, 1, 3, 4, 5)$:

$$\boxed{P(x) = 2x^4 + x^3 + 3x^2 + 4x + 5.}$$

3 / GIVE THE UNIVERSAL WEIGHTS A NEW INTERPRETATION

Divide the kth difference weights by $k!$

Apply the division to the **power rows** above. In the quartic row:

$$\left(\frac{1}{0!}, \frac{15}{1!}, \frac{50}{2!}, \frac{60}{3!}, \frac{24}{4!}\right) = (1, 15, 25, 10, 1).$$

These are **Stirling numbers of the second kind**: $S(n, r)$ counts partitions of n distinct objects into r non-empty, unlabelled groups. The rows inferred from powers $1, x, \ldots, x^4$ are

$n\backslash r$	1	2	3	4	5
1	1				
2	1	1			
3	1	3	1		
4	1	7	6	1	
5	1	15	25	10	1

Why 25? Assign four objects to three named boxes, requiring boxes $1, 2$ to be occupied; box 0 may be empty. Inclusion–exclusion gives

$$3^4 - 2 \cdot 2^4 + 1^4 = 50 = W(4, 2).$$

Add a marked fifth object to box 0. Its group is now distinguished; forgetting the labels on the other two groups divides by $2!$, giving $S(5, 3) = 50/2! = 25$.

$$W(m, k) \quad = \quad k!\, S(m \ + \ 1, k \ + \ 1)$$

4 / PUT THOSE STIRLING NUMBERS TO ANOTHER USE

Five pupils: groups, named activities and Bell numbers

Split five pupils into **three unlabelled discussion groups**: $S(5, 3) = 25$ ways. Assign them instead to **three named activities**, with every activity occupied: label the groups in $3!$ ways, giving $3!\, S(5, 3) = 150$ assignments.

Allow any number of unlabelled groups and add the entire row:

$$\boxed{B_5 = \sum_{r=1}^{5} S(5, r) = 1 + 15 + 25 + 10 + 1 = 52.}$$

Conventions. Unit spacing, starting at $x = 1$. Five samples determine a polynomial of degree at most four.
References. NIST DLMF 26.8 (Stirling); 26.7 (Bell); OEIS A028246 (Worpitzky).

Chapter 2

Number, ratio and algorithms

What this chapter gathers.

Foothills

- Primes, the sieve and divisibility tests to 13
- Bases and place value: binary, ternary, qubits and qutrits
- HCF, LCM, the cicada beetle and Euler's totient function
- Set notation and the triple intersection: unions, intersections and brackets
- Long division is a walk through remainders
- Linear simultaneous equations and the LCM
- Euclid's algorithm and Bézout's identity
- Prime factorisation, divisors and squares
- Arithmetic on a clock: congruences and inverses

Foothold

- The Chinese remainder theorem
- Fermat, Euler and repeating decimals
- Greedy Tarts: Egyptian fractions
- The mediant: why stop there?
- Mediants and double angles
- Fractions that touch: Ford circles
- Merging and boosting: mediants and rapidities
- Metallic means as differences of consecutive square roots
- Nearly twice a square: Pell's equation
- The LCM method as Gaussian elimination

Write the whole numbers in rows and begin crossing out multiples. The sieve gradually separates the primes from the composite numbers. A simple instruction has revealed the numbers from which every whole number greater than one can be built by multiplication. Divisibility tests make this investigation quicker, but they also invite explanation. Why should the last digit, or the sum of the digits, tell us anything about the factors of the whole number?

Place value supplies part of the answer. The symbols 100 represent four in binary, nine in ternary and one hundred in decimal. The quantity and its written representation need to be distinguished. Factors and multiples then help us compare quantities: a common factor provides a shared measuring unit, while a common multiple tells us when repeating cycles meet again. Set notation gives us a language for collecting numbers that satisfy a condition and describing how several conditions overlap.

Long division brings representation and divisibility together. One eighth ends after three decimal places; one seventh returns to the same six digits. Follow the remainders and the difference becomes understandable. Each remainder determines the next digit and the next remainder, and only finitely many remainders are possible. A zero ends the calculation. Returning to an earlier remainder makes the same future unfold again. The decimal records a process whose behaviour we can explain.

Euclid's algorithm offers another way to follow what a division preserves. Cut the largest square from a rectangle thirteen units long and eight units wide, then repeat with the smaller rectangle. A length that measures both original sides also measures their difference. The common measuring units therefore survive each cut. Bézout's identity strengthens the conclusion: the greatest common divisor can be expressed as a whole-number combination of the original numbers. Here, $5 \cdot 8 - 3 \cdot 13 = 1$. The calculation has left a record of how its answer was constructed.

Clock arithmetic makes remainders the objects of study. We keep the position on the clock while setting aside the number of completed turns. Some multiplications can be reversed, and repeated multiplication can return us to the starting position. Fermat's and Euler's theorems help explain these returns and the repeating decimals they produce. The Chinese remainder theorem asks a complementary question: when several clocks give us partial information, can their readings identify a single number within a larger cycle?

Fractions open further routes through arithmetic. Egyptian fractions assemble an amount from unit fractions, with a greedy choice reducing what remains at each step. The mediant combines numerators and denominators to place a new fraction between two positive neighbours. Ford circles make an exact calculation visible as a statement about distance. Connections with double angles and rapidities give us further reasons to examine how ratios combine.

Approximation also becomes more precise when it leaves an exact relationship behind. The equation $17^2 - 2 \cdot 12^2 = 1$ explains why $17/12$ lies close to $\sqrt{2}$, and Pell's equation develops this connection between whole numbers and irrational ratios. Elsewhere, eliminating a variable using common multiples leads towards Gaussian elimination. Across these examples, the lasting skill is to examine a procedure: identify what each step preserves, explain why progress is possible, and establish what makes the calculation finish or repeat.

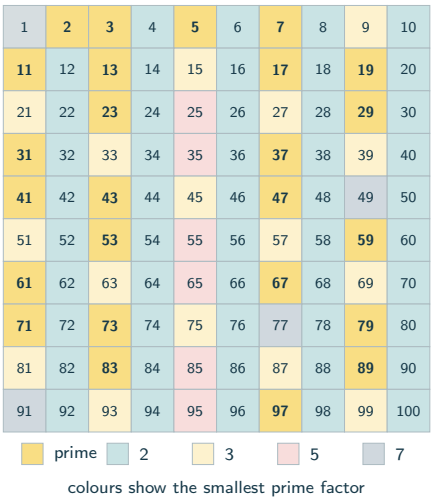
NUMBER THEORY ON ONE PAGE

Primes, the sieve and divisibility tests to 13

The sieve needs only primes up to the square root, and one ansatz explains the tests for 3, 7, 9, 11 and 13.

1 / THE RULE AND THE PICTURE

A composite number has a prime factor at most its square root, so the sieve stops there



colours show the smallest prime factor

If $N = pq$ with $p \le q$ then $p^2 \le N$, so a composite N has a prime factor $p \le \sqrt{N}$:

$$N \text{ is prime} \iff \text{no prime } p \le \sqrt{N} \text{ divides } N$$

Eratosthenes strikes out the multiples of each prime p, beginning at p^2 because the smaller multiples $2p, 3p, \dots$ were struck by smaller primes. The primes $2, 3, 5, 7$ clear everything up to 100 and leave its 25 primes, since the next prime 11 first acts at $11^2 = 121$. To settle every number below $17^2 = 289$ we need the primes up to 13, which leave 61 primes, and the tests below decide divisibility by each of them quickly.

prime p	2	3	5	7	11	13
first strike p^2	4	9	25	49	121	169
primes up to p settle $N <$	9	25	49	121	169	289

2 / THE ANSATZ / MULTIPLY BY THE INVERSE OF 10

Write $n = 10a + b$ and multiply by m with $10m \equiv 1 \pmod p$, so that $mn \equiv a + mb$

Because m is invertible modulo p, $p \mid n$ exactly when $p \mid a + mb$, which shortens n by one digit each time. The primes 2 and 5 divide 10 itself and use the last digit.

p	m with $10m \equiv 1$	test on $n = 10a + b$	why	example
2	none	last digit even	10 is even	2024
3, 9	1	digit sum, $a + b$	$10 \equiv 1$	$2025 \to 9$
4	none	last two digits	$4 \mid 100$	$2024 \to 24$
5	none	last digit 0 or 5	$5 \mid 10$	2015
7	5 or -2	$a - 2b$ (or $a + 5b$)	$10(a - 2b) + 21b = n$	2023
11	-1	$a - b$	$11a + (b - a) = n$	2024
13	4	$a + 4b$	$4n = 39a + (a + 4b)$	2015

3 / WHY THE TEST FOR 7 WORKS / $n = 2023$

Since $10a + b = 10(a - 2b) + 21b$ and $7 \mid 21b$, the number n is a multiple of 7 exactly when $a - 2b$ is

Subtracting twice the last digit from the rest shortens the number, and n differs from $10(a - 2b)$ by a multiple of 7 while 10 is invertible modulo 7. For 2023 we have $a = 202$ and $b = 3$, so $202 - 6 = 196$, then $19 - 12 = 7$, and 7 divides 7, so $7 \mid 2023$ and indeed $2023 = 7 \times 17^2$. The multiplier 5 gives the second test, since $5n = 49a + (a + 5b)$ and $7 \mid 49a$: $202 + 15 = 217 = 7 \times 31$.

4 / WHY THE TEST FOR 11 WORKS / $n = 2024$

Since $10 \equiv -1 \pmod{11}$, the number $n = 10a + b$ is $b - a$ modulo 11

Writing $10a + b = 11a + (b - a)$ shows that $11 \mid n$ exactly when $11 \mid a - b$. For 2024 we have $a = 202$ and $b = 4$, so $202 - 4 = 198 = 11 \times 18$, and $2024 = 8 \times 11 \times 23$. Applying the same step to every digit gives the alternating sum $4 - 2 + 0 - 2 = 0$, because $10^k \equiv (-1)^k \pmod{11}$, so the digits alternate in sign from the right.

5 / ONE TEST FOR 7, 11 AND 13 / THE NUMBER 1001

As $1001 = 7 \times 11 \times 13$ and $1000 \equiv -1$, alternating three-digit blocks decide all three

n	blocks	alternating sum	divisible by	factorisation
2023	$2 \mid 023$	$23 - 2 = 21 = 3 \cdot 7$	7	$7 \cdot 17^2$
2024	$2 \mid 024$	$24 - 2 = 22 = 2 \cdot 11$	11	$2^3 \cdot 11 \cdot 23$
2015	$2 \mid 015$	$15 - 2 = 13$	13	$5 \cdot 13 \cdot 31$

Powers of 10 modulo 7 run $1, 3, 2, 6, 4, 5$, modulo 13 they run $1, 10, 9, 12, 3, 4$, and modulo 11 they alternate $1, -1$; in both 7 and 13 the third power is -1, which is where the blocks of three come from.

ASIDE / THE NEXT PRIME

For 17 the multiplier is $m = 12$ and the test is $a + 12b \equiv a - 5b$, found the same way.

CHECK IN PYTHON

All three reductions agree with `n % p == 0` for every $n < 100\,000$.

Convention. $n = 10a + b$ with b the last digit; $\equiv$ is congruence; $\pi(100) = 25$ and 61 primes lie below $289 = 17^2$.

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BASES AND PLACE VALUE ON ONE PAGE

Each digit counts a power of the base

Binary, ternary and decimal share one rule, which reappears in standard form and in the states of qubits and qutrits.

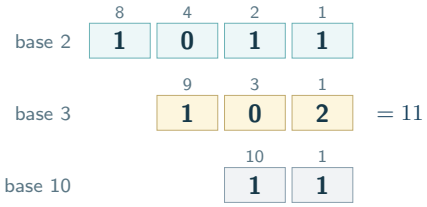
THE RULE

A numeral is a sum of digits times powers

In base b the digits run from 0 to $b-1$, and the digit in position k counts copies of b^k:

$$(d_n \ldots d_1 d_0 \,.\, d_{-1} d_{-2} \ldots)_b \quad = \quad \sum_k d_k \, b^k$$

so eleven is written $1011_2 = 8+2+1$ in binary, $102_3 = 9+0+2$ in ternary and $11_{10} = 10+1$ in decimal. Positions after the point carry negative powers, which is why $\frac{1}{3}$ is the tidy 0.1_3 in ternary while it repeats as $0.333\ldots_{10}$ and $0.0101\ldots_2$.



Eleven in three bases; the small numbers are the place values b^k.

CONVERTING

Divide by the base, keep the remainders

Each division by b peels off the last digit, so reading the remainders from the bottom up gives the numeral:

$$\begin{aligned} 11 \div 3 &= 3 \text{ r } \mathbf{2} \quad \uparrow \\ 3 \div 3 &= 1 \text{ r } \mathbf{0} \qquad \Longrightarrow \qquad 11 = 102_3. \\ 1 \div 3 &= 0 \text{ r } \mathbf{1} \end{aligned}$$

COUNTING

What n digits can hold

With n digits in base b there are b^n numerals, from 0 to $b^n - 1$, and N needs $\lfloor \log_b N \rfloor + 1$ digits. One ternary digit, a **trit**, carries $\log_2 3 \approx 1.585$ bits. Charging b for each digit, writing every number up to a million costs about 40 in binary, 60 in decimal and 37.7 in ternary, the cheapest whole base.

STANDARD FORM

Slide the point until one digit sits in front of it

Standard form records where the point belongs as a power of the base, writing $N = m \times b^e$ with $1 \le m < b$, where the exponent $e = \lfloor \log_b N \rfloor$ is one less than the number of digits:

$$3200 = 3.2 \times 10^3, \qquad 102_3 = 1.02_3 \times 3^2, \qquad 1011_2 = 1.011_2 \times 2^3.$$

Computers store floating-point numbers in exactly this binary standard form, keeping a sign, an exponent and the digits of m; since the leading binary digit of m is always 1, it is left implicit, and a 64-bit double keeps 52 further bits after it. Fractions with a factor other than 2 in the denominator never terminate in binary, so $0.1 = 0.000110011\ldots_2$ is stored approximately and Python reports `0.1 + 0.2 == 0.3` as `False`.

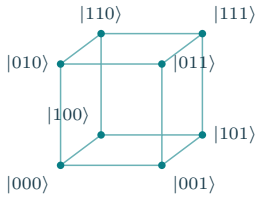
QUBITS AND QUTRITS

Place value labels the basis states

A classical bit holds 0 or 1; a qubit holds a superposition $\alpha|0\rangle + \beta|1\rangle$, and a qutrit uses the three ternary digits, $\alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle$, with complex amplitudes whose squared sizes add to 1. A register of n qudits of dimension d has one basis state for every n-digit numeral in base d:

n qubits: 2^n states $|0\ldots0\rangle, \ldots, |1\ldots1\rangle$
n qutrits: 3^n states $|0\ldots0\rangle, \ldots, |2\ldots2\rangle$

so three qubits label the eight numbers 0 to 7 in binary and two qutrits label the nine numbers 0 to 8 in ternary. A general state needs d^n amplitudes to describe it, yet measuring it returns a single numeral, so n qudits deliver at most $n \log_2 d$ bits, the same capacity as n classical digits in base d.



3 qubits: $2^3 = 8$



2 qutrits: $3^2 = 9$

Conventions. A subscript names the base, so 102_3 is read "one zero two in base three"; an unmarked numeral is decimal. The ket $|x\rangle$ names the basis state labelled by the numeral x, a qudit is a quantum digit with d levels, and $\log_2 d$ converts one base-d digit into bits.

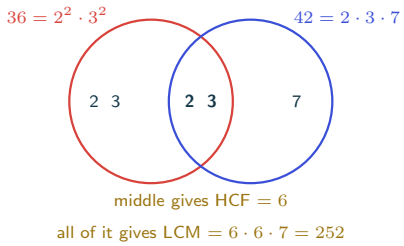
NUMBER THEORY ON ONE PAGE

HCF, LCM, cicadas and the totient

Shared factors decide when cycles realign, and Euler's $\varphi(n)$ counts the cycle lengths that share none.

1 / THE RULE AND THE PICTURE

Shared prime factors give the HCF, and all of them give the LCM



HCF is the largest d dividing both numbers, the product of shared primes at their smaller powers, and LCM is the first common multiple, the product of every prime at its larger power:

$$\boxed{\mathrm{HCF}(a,b) \times \mathrm{LCM}(a,b) = a \times b}$$

For 36 and 42, $6 \times 252 = 1512 = 36 \times 42$.

2 / METHODS / EUCLID AND PRIME FACTORS

Divide, keep the remainder, and repeat until it vanishes

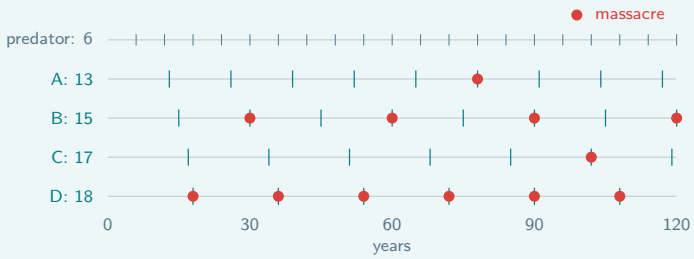
$133 \bmod 56$	21
$56 \bmod 21$	14
$21 \bmod 14$	7
$14 \bmod 7$	0
$\mathrm{HCF}(133, 56)$	7

numbers	HCF	LCM
36, 42	6	252
225, 81	9	2025
452, 226	226	452
360, 100	20	1800
30, 42, 63	3	630

HCF answers what divides evenly into both and LCM answers when cycles realign, so three walkers taking 2, 3 and 4 steps meet again at step $\mathrm{LCM}(2, 3, 4) = 12$.

3 / THE CICADA BEETLE / LCM AS THE TIME BETWEEN MASSACRES

A predator surfaces every 6 years, and colonies emerge every 13, 15, 17 and 18

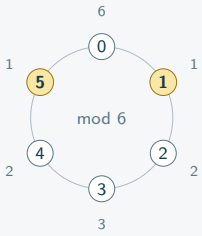


cycle	HCF	LCM	massacre every
13	1	78	6th emergence
15	3	30	2nd emergence
17	1	102	6th emergence
18	6	18	every emergence

The next encounter is $\mathrm{LCM}(c, 6) = 6c/\mathrm{HCF}(c, 6)$ years, so colony C at 17 years waits 102 years while D at 18 is hit every time because $6 \mid 18$. The argument is sound but selective: we only see the lineages that survived.

4 / EULER'S TOTIENT / COUNTING THE CYCLE LENGTHS THAT SHARE NO FACTOR

$\varphi(n)$ counts $k \le n$ with $\mathrm{HCF}(k, n) = 1$, and $\varphi(6) = 2$



Only cycle lengths with $\mathrm{HCF}(c, 6) = 1$ achieve the full LCM $6c$, and since the HCF depends only on $c \bmod 6$, the safe lengths are $c \equiv 1$ or $5 \pmod 6$: $13 \equiv 1$ and $17 \equiv 5$ are safe, $15 \equiv 3$ and $18 \equiv 0$ are not. The numbers on the dial are $\mathrm{HCF}(k, 6)$, and sorting the six residues by it gives $\varphi(1)+\varphi(2)+\varphi(3)+\varphi(6) = 1+1+2+2 = 6$. The totient is multiplicative for coprime arguments, $\varphi(6) = \varphi(2)\varphi(3) = 1 \times 2$, and Euler's theorem $a^{\varphi(n)} \equiv 1 \pmod n$ gives $5^2 = 25 \equiv 1 \pmod 6$.

predator period m	6	12	30	7
$\varphi(m)$	2	4	8	6
safe fraction $\varphi(m)/m$	33%	33%	27%	86%

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right), \qquad \varphi(p) = p - 1, \qquad \varphi(6) = 6 \cdot \tfrac{1}{2} \cdot \tfrac{2}{3} = 2$$

ASIDE / WHY PRIMES

A prime larger than every prime factor of the predator's period is coprime to it, so its LCM with the predator is the full product.

ASIDE / CHECK IN PYTHON

`sum(gcd(k,n)==1 for k in range(1,n+1))` gives 2 for `n=6`, and `lcm(c,6)` gives $78, 30, 102, 18$.

Convention. HCF is the GCD; φ is Euler's totient. After the HCF, LCM and cicada beetle learner.

Networked–lattice

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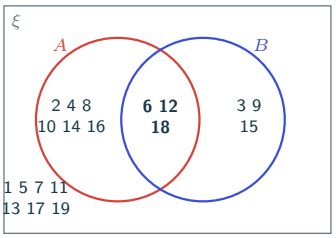
SETS ON ONE PAGE

Set notation: unions, intersections and brackets

From two sets of numbers to the eight regions of three, with the brackets that pick each region out.

1 / THE NOTATION AND A WORKED EXAMPLE

Take $\xi = \{1, \dots, 19\}$, $A =$ even numbers, $B =$ multiples of 3



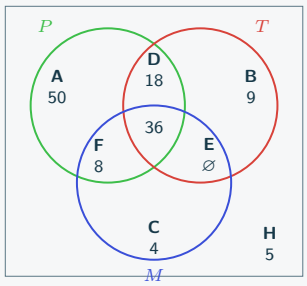
$A \cap B$ in the middle

symbol	read as	meaning
ξ	universal set	everything under discussion
$x \in A$, $x \notin A$	is, is not an element of	$6 \in A$, $5 \notin A$
$A \subseteq B$	subset	every element of A is in B
$\varnothing$	empty set	no elements
A'	complement	in ξ but not in A
$A \cup B$	union	in A or B or both
$A \cap B$	intersection	in both A and B
$n(A)$ or $\lvert A \rvert$	number of elements	$\lvert A \rvert = 9$
$\{x : \dots\}$	set builder	$A = \{x \in \xi : x \text{ even}\}$

$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$, $B = \{3, 6, 9, 12, 15, 18\}$, $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18\}$ and $A \cap B = \{6, 12, 18\}$, with $\lvert A \cup B \rvert = \lvert A \rvert + \lvert B \rvert - \lvert A \cap B \rvert = 9 + 6 - 3 = 12$ and $(A \cup B)' = \{1, 5, 7, 11, 13, 17, 19\}$.

2 / THREE SETS / THE EIGHT REGIONS

Let $\xi = \{1, \dots, 100\}$ and ask what belongs in each region



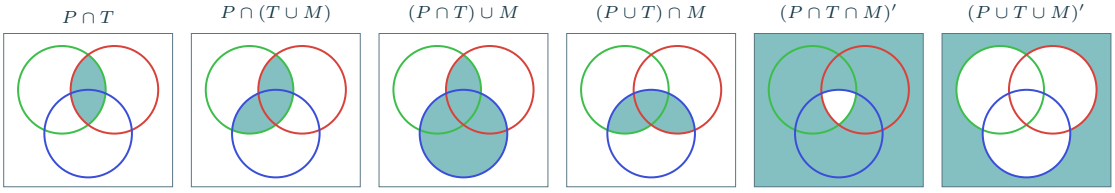
region	expression	example	members
A	$P \cap (T \cup M)'$	$50 = 2 \cdot 5^2$	3
B	$T \cap (P \cup M)'$	$9 = 3^2$	12
C	$M \cap (P \cup T)'$	$4 = 2^2$	1
D	$(P \cap T) \cap M'$	$18 = 2 \cdot 3^2$	13
E	$(T \cap M) \cap P'$	none	0
F	$(P \cap M) \cap T'$	$8 = 2^3$	16
centre	$P \cap T \cap M$	$36 = 4 \times 3^2$	8
H	$(P \cup T \cup M)'$	5	47

Here P is "a product of at least 3 primes" (repeats count, so $8 = 2 \cdot 2 \cdot 2$), T is "has 3 as a factor" and M is "a multiple of 4". **Region E is empty:** a number divisible by 3 and 4 is a multiple of $12 = 2 \cdot 2 \cdot 3$, which already has three prime factors, so $T \cap M \subseteq P$. **Region C is just $\{4\}$:** a multiple of 4 has two 2s, and staying out of P allows no third prime factor.

$T \cap M \subseteq P$, so $(T \cap M) \cap P' = \varnothing$; $\qquad P \cap T \cap M = \{12, 24, 36, \dots\}$, the multiples of 12

3 / BRACKETS / THE SAME LETTERS, DIFFERENT REGIONS

The shaded region changes when the brackets move



21 members | 37 members | 38 members | 24 members | 92 members | 47 members

Moving one bracket turns $P \cap (T \cup M)$ into $(P \cap T) \cup M$, which gains the single extra member 4 from region C. The distributive law $P \cap (T \cup M) = (P \cap T) \cup (P \cap M)$ holds, and so do De Morgan's laws, $(P \cup T)' = P' \cap T'$ and $(P \cap T)' = P' \cup T'$.

4 / COUNTING THE UNION

Inclusion and exclusion keeps each region counted once

$$\lvert P \cup T \cup M \rvert = \lvert P \rvert + \lvert T \rvert + \lvert M \rvert - \lvert P \cap T \rvert - \lvert P \cap M \rvert - \lvert T \cap M \rvert + \lvert P \cap T \cap M \rvert$$
$$= 40 + 33 + 25 - 21 - 24 - 8 + 8 = 53, \qquad \lvert \xi \rvert - 53 = 47 = \lvert (P \cup T \cup M)' \rvert$$

Adding the three sets counts the overlaps twice, subtracting the pairs removes the centre three times over, and the triple term puts it back once.

ASIDE / ALWAYS WRITE THE BRACKETS

Unlike $\times$ before $+$, no rule ranks $\cup$ against $\cap$, so $P \cup T \cap M$ is ambiguous and the two bracketings give different regions.

ASIDE / CHECK IN PYTHON

With `P, T, M` built as Python sets over `range(1,101)`, `len(P|T|M)` gives 53 and `T&M-P` gives `set()`.

Convention. Regions A to H follow the slide; the sets are named P, T, M so that the letters do not clash with the regions.

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DECIMALS, MODULAR CYCLES AND FERMAT ON ONE PAGE

Long division is a walk through remainders

Keep track of what is left over, and a repeating decimal becomes a finite walk.

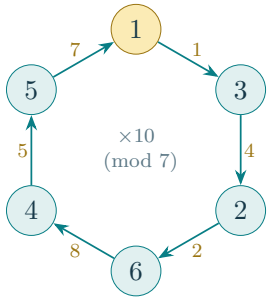
1 / CALCULATE ONE SEVENTH

Record the digit and keep the remainder

Start with $r_0 = 1$. At each step, multiply the remainder by 10, divide by 7, write the digit d_k, and keep the remainder r_k.

k	division step	digit d_k	remainder r_k
1	$10 = 7 \cdot 1 + 3$	1	3
2	$30 = 7 \cdot 4 + 2$	4	2
3	$20 = 7 \cdot 2 + 6$	2	6
4	$60 = 7 \cdot 8 + 4$	8	4
5	$40 = 7 \cdot 5 + 5$	5	5
6	$50 = 7 \cdot 7 + 1$	7	1

Read down the digit column: 142857.



Circles: remainders. Arrow labels: digits.

$$\frac{1}{7} = 0.\overline{142857}$$ Remainder 1 returns: the six steps repeat.

2 / ONLY FINITELY MANY POSSIBILITIES

A finite walk must stop or repeat

With denominator q, the possible remainders are $0, 1, \ldots, q-1$. Each remainder determines the next digit and the next remainder:

$$10r_{k-1} = qd_k + r_k, \qquad 0 \le r_k < q.$$

A zero remainder ends the decimal. Repeating a non-zero remainder forces the same following digits, so the decimal repeats.

A cycle can have a short lead-in:

$$1 \xrightarrow{1} 4 \xrightarrow{6} 4, \qquad \frac{1}{6} = 0.1\overline{6}.$$

For comparison, $1/8 = 0.125$ reaches remainder 0 and terminates.

3 / FOLLOW THE WALK ALGEBRAICALLY

Powers of ten retrace the walk

Writing $x \equiv y \pmod 7$ says that x, y leave the same remainder. Each step gives $r_k \equiv 10r_{k-1} \pmod 7$, so, from $r_0 = 1$,

$$r_k \equiv 10^k \pmod 7.$$

k	0	1	2	3	4	5	6
$10^k \bmod 7$	1	3	2	6	4	5	1

Returning to remainder 1 after L steps means $10^L \equiv 1 \pmod 7$. The first such positive L is the **multiplicative order** of 10 modulo 7:

$$\operatorname{ord}_7(10) = 6.$$

4 / A PRIME FORCES A RETURN

Fermat's little theorem

For prime p and $p \nmid b$, multiplying $1, 2, \ldots, p-1$ by b modulo p rearranges that list: no two distinct entries acquire the same remainder. Taking products gives

$$b^{p-1}(p-1)! \equiv (p-1)! \pmod p.$$

The product $(p-1)! = 1 \cdot 2 \cdots (p-1)$ has no factor p, so it can be cancelled:

$$b^{p-1} \equiv 1 \pmod p$$

With $b = 10$, the theorem applies to every prime except 2 and 5.

5 / COUNT COMPLETE LAPS

The period divides $p - 1$

Here p is prime and $p \neq 2, 5$. The return after $p-1$ steps completes whole laps, so the first return L satisfies

$$L = \operatorname{ord}_p(10), \qquad L \mid (p-1).$$

p	repeating block	L	$p-1$
7	142857	6	6
11	09	2	10
13	076923	6	12
17	0588235294117647	16	16

Leading zeros are part of the repeating block. Full length $p-1$ occurs when the walk visits every non-zero remainder.

Change the base. In base $b \ge 2$, use $br_{k-1} = qd_k + r_k$. For prime $p \nmid b$, the period of $1/p$ in base b divides $p-1$. Here $p \nmid b$ means that p does not divide b.

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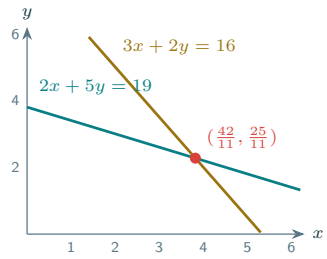
ALGEBRA ON ONE PAGE

Linear simultaneous equations and the LCM

Each equation is a line and the solution is where they cross; the LCM matches a coefficient and the GCD shrinks the numbers first.

1 / THE RULE AND THE PICTURE

To eliminate x from $ax + by = c$ and $dx + ey = f$, scale both equations to the coefficient $\mathrm{LCM}(a, d)$



Multiplying an equation by a constant leaves its line unchanged, so scaling to match coefficients and then subtracting only changes how the two lines are written. The multipliers are $\mathrm{LCM}(a, d)/a$ for the first equation and $\mathrm{LCM}(a, d)/d$ for the second:

$$\frac{\mathrm{LCM}(a,d)}{a}\,(1) - \frac{\mathrm{LCM}(a,d)}{d}\,(2) \quad \text{leaves an equation in } y \text{ alone}$$

The same recipe on y uses $\mathrm{LCM}(b, e)$, and a variable with opposite signs on its coefficients is eliminated by adding the scaled equations.

2 / A WORKED EXAMPLE / $3x + 2y = 16$ AND $2x + 5y = 19$

$\mathrm{LCM}(3, 2) = 6$ removes x, and $\mathrm{LCM}(2, 5) = 10$ removes y

	eliminate x	eliminate y
multipliers	$(1) \times 2,\ (2) \times 3$	$(1) \times 5,\ (2) \times 2$
matched equations	$6x + 4y = 32$ and $6x + 15y = 57$	$15x + 10y = 80$ and $4x + 10y = 38$
subtract	$11y = 25$	$11x = 42$
result	$y = \frac{25}{11}$	$x = \frac{42}{11}$

Each elimination gives one coordinate, and substituting back confirms the pair: $3 \cdot \frac{42}{11} + 2 \cdot \frac{25}{11} = \frac{176}{11} = 16$ and $2 \cdot \frac{42}{11} + 5 \cdot \frac{25}{11} = \frac{209}{11} = 19$. The LCM keeps the numbers small when the coefficients share a factor, as in $4x + 3y = 10$ and $6x + 5y = 16$, where $\mathrm{LCM}(4, 6) = 12$ (multipliers 3 and 2) replaces the product 24 and gives $y = 2$, $x = 1$.

3 / SIMPLIFY FIRST / THE GCD OF EACH EQUATION

Dividing an equation by the GCD of its coefficients shrinks the LCM you need

$6x + 9y = 15$	GCD $= 3$, so $2x + 3y = 5$	The x-coefficients of 6 and 4 would have needed LCM $= 12$, whereas after simplifying they are both 2 and the equations subtract at once.
$4x + 2y = 10$	GCD $= 2$, so $2x + y = 5$	
subtract	$2y = 0$, so $y = 0$ and $x = \frac{5}{2}$	

For $4x + 6y = 12$ and $3x + 2y = 8$ the first equation divides by 2 to give $2x + 3y = 6$, and $\mathrm{LCM}(2, 3) = 6$ gives $6x + 9y = 18$ and $6x + 4y = 16$, so $5y = 2$ and $y = \frac{2}{5}$. Substituting into $3x + 2y = 8$ gives $3x = 7.2$, hence

$$x = 2.4 = \tfrac{12}{5}, \qquad y = 0.4 = \tfrac{2}{5}$$

4 / WHEN ELIMINATION FAILS / PARALLEL AND COINCIDENT LINES

The number $ae - bd$ decides between one crossing and none or infinitely many

For $ax + by = c$ and $dx + ey = f$ the solution is $x = \dfrac{ce - bf}{ae - bd}$ and $y = \dfrac{af - cd}{ae - bd}$, so a unique crossing needs $ae - bd \neq 0$, and the example above has $ae - bd = 15 - 4 = 11$, which is the 11 in $11y = 25$. When $ae - bd = 0$ the lines share a slope, and they are parallel with no solution if the constants disagree, or the same line with infinitely many solutions if they agree; simplifying $6x + 4y = 20$ and $9x + 6y = 30$ by their GCDs of 2 and 3 gives $3x + 2y = 10$ twice, which is the second case.

5 / PRACTICE / ANSWERS TO CHECK AGAINST THE GRAPH

Four systems, three unique crossings and one repeated line

system	key step	solution	check
$2x + y = 7,\ x - y = 2$	add the equations: $3x = 9$	$(3, 1)$	$6 + 1 = 7$
$3x + 2y = 12,\ x + 4y = 10$	$(1) \times 2$ against (2): $5x = 14$	$(\frac{14}{5}, \frac{9}{5})$	$8.4 + 3.6 = 12$
$5x - 3y = 11,\ 2x + y = 5$	$(2) \times 3$ then add: $11x = 26$	$(\frac{26}{11}, \frac{3}{11})$	$\frac{130-9}{11} = 11$
$6x + 4y = 20,\ 9x + 6y = 30$	both reduce to $3x + 2y = 10$	a whole line	$ae - bd = 0$

ASIDE / THE HINT IN PROBLEM 4

The question asks what you notice after dividing by the GCDs, and the answer is that two equations have become one.

ASIDE / CHECK IN PYTHON

`Fraction(c*e-b*f,a*e-b*d)` gives `42/11` for `(3,2,16,2,5,19)`.

Convention. Equations are $ax + by = c$ and $dx + ey = f$. After the Year 9 Algebra: Path to Simultaneous Equations page.

Networked–lattice

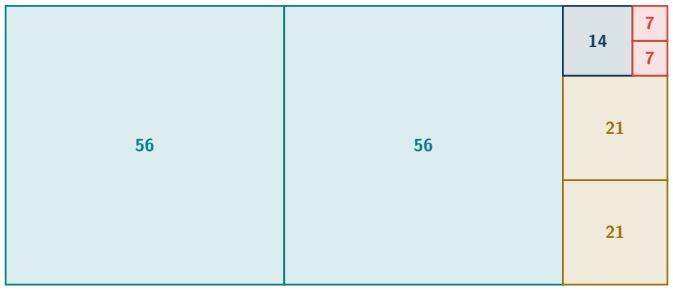
NUMBER THEORY LADDER 1 OF 9

Euclid's algorithm and Bézout's identity

Repeated remainders find the HCF, and running them backwards writes it as $ax + by$.

1 / THE RULE AND THE PICTURE

The HCF of two numbers is unchanged when the larger is replaced by its remainder



a 133×56 rectangle cut into the largest squares, and the smallest square is $7 = \mathrm{HCF}$

Any number dividing a and b also divides $a - qb$, and conversely, so the common divisors of a, b and of b, r coincide:

$$\mathrm{HCF}(a, b) = \mathrm{HCF}(b,\ a \bmod b)$$

Cutting off squares removes qb from a at each step.

2 / THE ALGORITHM / 133 AND 56

Divide, keep the remainder, and stop when it reaches zero

a	b	q	r	division
133	56	2	21	$133 = 2 \cdot 56 + 21$
56	21	2	14	$56 = 2 \cdot 21 + 14$
21	14	1	7	$21 = 1 \cdot 14 + 7$
14	7	2	0	$14 = 2 \cdot 7 + 0$

The last non-zero remainder, 7, is $\mathrm{HCF}(133, 56)$, and since $133 \cdot 56/7 = 1064$ the lowest common multiple follows from $\mathrm{HCF} \times \mathrm{LCM} = ab$.

3 / BÉZOUT'S IDENTITY / BACK-SUBSTITUTION

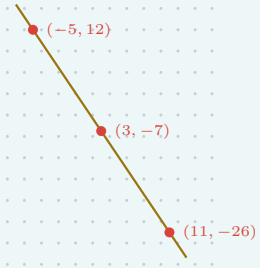
Working backwards writes the HCF as $ax + by$, and its solutions form a line of lattice points

Using each remainder in turn,

$$\begin{aligned} 7 &= 21 - 14 = 21 - (56 - 2 \cdot 21) = 3 \cdot 21 - 56 \\ &= 3(133 - 2 \cdot 56) - 56 = 3 \cdot 133 - 7 \cdot 56, \end{aligned}$$

so $x = 3$ and $y = -7$ solve $133x + 56y = 7$, and adding $(56/7,\ -133/7) = (8, -19)$ gives all solutions:

$$\boxed{(x, y) = (3 + 8t,\ -7 - 19t), \quad t \in \mathbb{Z}}$$



4 / WHAT IT GIVES / SOLVABILITY, INVERSES AND THE LCM

$ax + by = c$ has integer solutions exactly when $\mathrm{HCF}(a, b)$ divides c

So $133x + 56y = c$ is solvable when $7 \mid c$ and never otherwise.
If $\mathrm{HCF}(a, n) = 1$ then $ax + ny = 1$ and x is the inverse of a modulo n: from $10 \cdot 4 - 13 \cdot 3 = 1$ the inverse of 10 modulo 13 is 4, the multiplier behind the divisibility test for 13, and $10 \cdot 5 - 7 \cdot 7 = 1$ gives the multiplier 5 for 7.
The inverse of 56 modulo 133 does not exist, because the HCF is 7.

5 / A SECOND EXAMPLE / 1071 AND 462

The same steps give $\mathrm{HCF} = 21$ and the identity $21 = 7 \cdot 462 - 3 \cdot 1071$

$$\begin{aligned} 1071 &= 2 \cdot 462 + 147 \\ 462 &= 3 \cdot 147 + 21 \\ 147 &= 7 \cdot 21 + 0 \end{aligned}$$

Back-substituting, $21 = 462 - 3 \cdot 147 = 462 - 3(1071 - 2 \cdot 462) = 7 \cdot 462 - 3 \cdot 1071$, and a check gives $3234 - 3213 = 21$. Every linear combination $1071x + 462y$ is a multiple of 21, and every multiple of 21 occurs.

ASIDE / FAST IN PRACTICE

The remainders at least halve every two steps, so the algorithm takes at most about $\log_2$ of the number of steps in digits.

LINKS TO OTHER SHEETS

The ladder continues with *Prime factorisation, divisors and squares*; the divisibility multipliers appear in *Primes, the sieve and divisibility tests to 13*.

Convention. HCF is the highest common factor (GCD); $a \bmod b$ is the remainder; t ranges over the integers.

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NUMBER THEORY LADDER 2 OF 9

Prime factorisation, divisors and squares

Every number is a product of primes in one way only, and its divisors are the combinations of those primes.

1 / THE RULE AND THE PICTURE

Divisors are the products $2^i 3^j$, so $36 = 2^2 3^2$ has $3 \times 3 = 9$ of them

	3^0	3^1	3^2
2^0	1	3	9
2^1	2	6	18
2^2	4	12	36

d and $36/d$ sit opposite each other, and 6 pairs with itself

Writing $n = p_1^{e_1} p_2^{e_2} \cdots$, a divisor takes each exponent from 0 to e_i, so

$$d(n) = \prod (e_i + 1), \qquad \sigma(n) = \prod \frac{p_i^{e_i+1} - 1}{p_i - 1}$$

where d counts the divisors and σ adds them. For 36 we get $d = 3 \cdot 3 = 9$ and $\sigma = (1+2+4)(1+3+9) = 7 \cdot 13 = 91$.

2 / HCF AND LCM FROM EXPONENTS / 36 AND 90

The HCF takes the smaller exponent of each prime and the LCM the larger

number	2	3	5	value
$36 = 2^2 \cdot 3^2 \cdot 5^0$	2	2	0	36
$90 = 2^1 \cdot 3^2 \cdot 5^1$	1	2	1	90
HCF, the minimum	1	2	0	18
LCM, the maximum	2	2	1	180

The identity $\mathrm{HCF} \times \mathrm{LCM} = ab$ holds because $\min + \max =$ the sum of the two exponents, so $18 \times 180 = 3240 = 36 \times 90$.

3 / COUNTING DIVISORS / $d(n)$ AND $\sigma(n)$

The count is odd only when every exponent is even, that is when n is a square

n	factorisation	$d(n)$	$\sigma(n)$	note
12	$2^2 \cdot 3$	6	28	
28	$2^2 \cdot 7$	6	56	perfect
36	$2^2 \cdot 3^2$	9	91	square, odd d
60	$2^2 \cdot 3 \cdot 5$	12	168	
100	$2^2 \cdot 5^2$	9	217	square, odd d
360	$2^3 \cdot 3^2 \cdot 5$	24	1170	

A number with $\sigma(n) = 2n$ is perfect, and $28 = 2^2 \cdot 7$ has $\sigma = 7 \cdot 8 = 56$. Divisors pair as d and n/d, so they come in twos unless $d = n/d$, and that needs $n = d^2$.

4 / THE LOCKER PUZZLE / ONE HUNDRED DOORS

Locker k is toggled once per divisor of k, so it ends open exactly when k is a square

1	2	3	**4**	5	6	7	8	**9**	10
11	12	13	14	15	**16**	17	18	19	20
21	22	23	24	**25**	26	27	28	29	30
31	32	33	34	35	**36**	37	38	39	40
41	42	43	44	45	46	47	48	**49**	50
51	52	53	54	55	56	57	58	59	60
61	62	63	**64**	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	**100**

Students walk a corridor of 100 closed lockers and student s toggles every sth locker. Locker 12 is toggled by students $1, 2, 3, 4, 6, 12$, six times, and ends closed, while locker 36 is toggled nine times and ends open. The open lockers are the ten squares $1, 4, 9, \ldots, 100$.

5 / WHY FACTORISATION IS UNIQUE / EUCLID'S LEMMA

If a prime p divides ab then p divides a or b, and uniqueness follows

Suppose $p \mid ab$ and $p \nmid a$. Then $\mathrm{HCF}(p, a) = 1$, so Bézout gives $px + ay = 1$ for some integers, and multiplying by b gives $b = pbx + aby$, which p divides because $p \mid ab$. Cancelling primes one at a time from two factorisations of the same number then shows they agree. Among the even numbers alone the argument fails, since 6 and 18 cannot be split inside that set and $36 = 6 \cdot 6 = 2 \cdot 18$ factors in two ways.

ASIDE / HIGHLY COMPOSITE

360 has 24 divisors, more than any smaller number, which is why it rules circles and clocks.

LINKS TO OTHER SHEETS

The proof uses Bézout from *Euclid's algorithm and Bézout's identity*; the next sheet is *Arithmetic on a clock*.

Convention. $d(n)$ is the number of divisors, $\sigma(n)$ their sum; $\min$ and $\max$ are taken exponent by exponent.

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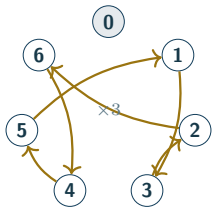
NUMBER THEORY LADDER 3 OF 9

Arithmetic on a clock: congruences and inverses

Two numbers are congruent when they differ by a multiple of n, and sums and products respect that.

1 / THE RULE AND THE PICTURE

Working modulo 7 keeps only the remainder, and multiplying by 3 walks round the whole clock



We write $a \equiv b \pmod{n}$ when $n \mid a-b$, for example $9 \equiv 2 \pmod{7}$, and congruences can be added and multiplied:

$$a \equiv a',\ b \equiv b' \implies a+b \equiv a'+b',\quad ab \equiv a'b'$$

Starting at 1 and multiplying by 3 repeatedly visits $1, 3, 2, 6, 4, 5$ and returns to 1, using every non-zero remainder, so 3 generates the clock.

2 / MULTIPLICATION TABLES / A PRIME CLOCK AND A COMPOSITE ONE

Modulo a prime every row is a shuffle of the clock, while $2 \cdot 3 \equiv 0 \pmod{6}$ breaks that

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

modulo 7, with the 1s in gold

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	0	2	4
3	3	0	3	0	3
4	4	2	0	4	2
5	5	4	3	2	1

modulo 6, with the zeros in red

In the table for 7 each non-zero number meets 1 exactly once, so it has an inverse: $2 \cdot 4 \equiv 1$, $3 \cdot 5 \equiv 1$, $6 \cdot 6 \equiv 1$. In the table for 6 the numbers $2, 3, 4$ make zeros and 5 alone has an inverse besides 1.

3 / WHEN AN INVERSE EXISTS / UNITS

a has an inverse modulo n exactly when $\mathrm{HCF}(a, n) = 1$

n	units	$\varphi(n)$	inverses
7	$1, 2, 3, 4, 5, 6$	6	all non-zero, a field
12	$1, 5, 7, 11$	4	each is its own inverse
10	$1, 3, 7, 9$	4	$3 \cdot 7 \equiv 1$, $9 \cdot 9 \equiv 1$
15	$1, 2, 4, 7, 8, 11, 13, 14$	8	$2 \cdot 8 \equiv 1$, $4 \cdot 4 \equiv 1$

If $\mathrm{HCF}(a, n) = 1$ then $ax + ny = 1$ by Bézout, so $ax \equiv 1$ and x is the inverse, and if the HCF is larger then every product ax shares that factor with n and can never be 1.
The number of units is Euler's $\varphi(n)$, and modulo 12 we have $5^2 = 25$, $7^2 = 49$, $11^2 = 121$, all $\equiv 1$.

4 / USING CLOCKS TO CHECK ARITHMETIC / CASTING OUT NINES AND ELEVENS

A product must survive reduction modulo 9 and modulo 11

Since $10 \equiv 1 \pmod{9}$, a number is congruent to its digit sum, and since $10 \equiv -1 \pmod{11}$ to its alternating digit sum.
For $1234 \times 5678 = 7\,006\,652$ the digit sums are $1 \times 8 = 8$ on the left and $7+0+0+6+6+5+2 = 26 \to 8$ on the right, and the alternating sums from the right are $2 \times 2 = 4$ and $2-5+6-6+0-0+7 = 4$, so both clocks agree. A wrong answer would have to be wrong by a multiple of 99 to survive both checks.

ASIDE / ARITHMETIC WITHOUT LARGE NUMBERS

Powers reduce at every step: $3^{100} \bmod 7$ needs only the cycle of length 6, and $100 \equiv 4$, so it is $3^4 = 81 \equiv 4$.

LINKS TO OTHER SHEETS

Units and inverses use *Euclid's algorithm and Bézout's identity*; the next sheet is *Fermat, Euler and repeating decimals*.

Convention. $\equiv$ is congruence, $\varphi(n)$ the number of units modulo n; tables list products reduced modulo n.

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NUMBER THEORY LADDER 5 OF 9

The Chinese remainder theorem

Residues modulo coprime numbers combine, so a number modulo 15 is a pair of numbers modulo 3 and modulo 5.

1 / THE RULE AND THE PICTURE

Each pair of remainders modulo 3 and 5 occurs for exactly one number below 15

$n \bmod 3$ \ $n \bmod 5$	0	1	2	3	4
0	**0**	**6**	**12**	**3**	**9**
1	**10**	**1**	**7**	**13**	**4**
2	**5**	**11**	**2**	**8**	**14**

the cell for $n \equiv 2 \pmod 3$ and $n \equiv 3 \pmod 5$ holds 8

If $\mathrm{HCF}(m, n) = 1$ the map $x \mapsto (x \bmod m,\ x \bmod n)$ is a bijection from the mn residues onto the mn pairs:

$$\boxed{\mathbb{Z}/mn \ \cong\ \mathbb{Z}/m \times \mathbb{Z}/n}$$

Every cell of the table is filled once, so the system $x \equiv 2 \pmod 3$, $x \equiv 3 \pmod 5$ has exactly one solution modulo 15, namely $x \equiv 8$.

2 / SUNZI'S PUZZLE / THREE MODULI

Find x with remainders $2, 3, 2$ on division by $3, 5, 7$

With $M = 105$ take $M_i = M/m_i = 35, 21, 15$ and the inverses $35^{-1} \equiv 2 \pmod 3$, $21^{-1} \equiv 1 \pmod 5$, $15^{-1} \equiv 1 \pmod 7$, found by Bézout. Each term $r_i M_i M_i^{-1}$ is $1 \cdot r_i$ for its own modulus and 0 for the others, so

$$x = 2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1 = 233 \equiv 23 \pmod{105},$$

and $23 = 7 \cdot 3 + 2 = 5 \cdot 4 + 3 = 3 \cdot 7 + 2$ checks all three conditions.

3 / UNITS SPLIT TOO / $\varphi(15) = \varphi(3)\varphi(5)$

Invertible numbers modulo 15 are the pairs of invertible numbers modulo 3 and 5

a	$(a \bmod 3,\ a \bmod 5)$	a	$(a \bmod 3,\ a \bmod 5)$
1	$(1, 1)$	8	$(2, 3)$
2	$(2, 2)$	11	$(2, 1)$
4	$(1, 4)$	13	$(1, 3)$
7	$(1, 2)$	14	$(2, 4)$

There are $2 \times 4 = 8$ pairs and 8 units. The equation $x^2 \equiv 1$ has 2 solutions modulo 3 and 2 modulo 5, so it has 4 solutions modulo 15, namely $1, 4, 11, 14$, and 8 modulo 105.

4 / READING 2023, 2024 AND 2015 / MODULO 1001

Since $1001 = 7 \cdot 11 \cdot 13$, a number is a triple of remainders, and blocks of three reveal it

n	$n \bmod 1001$	mod7	mod11	mod13
2023	21	0	10	8
2024	22	1	0	9
2015	13	6	2	0

Because $1000 \equiv -1 \pmod{1001}$, the alternating sum of the three-digit blocks of 2023, which is $23 - 2 = 21$, is congruent to 2023 modulo 1001 and so shares its residues: $21 \equiv (0, 10, 8)$, which is why $7 \mid 2023$.

5 / WHEN THE MODULI SHARE A FACTOR

A system is solvable only if its conditions agree modulo the HCF

For $x \equiv 2 \pmod 4$ and $x \equiv 0 \pmod 6$ the HCF is 2 and both conditions say x is even, so the solutions form one class modulo the LCM: $x \equiv 6 \pmod{12}$. For $x \equiv 1 \pmod 4$ and $x \equiv 0 \pmod 6$ the first makes x odd and the second even, so there is no solution.

ASIDE / ARITHMETIC IN PIECES

To compute $2^{100} \bmod 15$ work modulo 3 and 5: $2^{100} \equiv 1$ and $\equiv 1$, so it is 1.

LINKS TO OTHER SHEETS

The inverses come from *Euclid's algorithm and Bézout's identity*; the test for $7, 11, 13$ is in *Primes, the sieve and divisibility tests to 13*.

Convention. $\mathbb{Z}/n$ is the integers modulo n; $\cong$ means a bijection that respects addition and multiplication.

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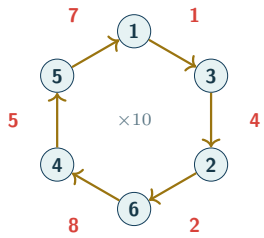
NUMBER THEORY LADDER 4 OF 9

Fermat, Euler and repeating decimals

Powers of a number modulo n cycle, and the cycle length explains both Fermat's theorem and the digits of 1/7.

1 / THE RULE AND THE PICTURE

Long division of $1/7$ is the clock of remainders, multiplying by 10 each step



Each step turns the remainder r into the digit $\lfloor 10r/7 \rfloor$ and the new remainder $10r \bmod 7$, so the remainders run $1, 3, 2, 6, 4, 5$ and the digits $1, 4, 2, 8, 5, 7$ in red:

$$\frac{1}{7} = 0.\overline{142857}$$

The remainders are the powers $10^k \bmod 7$, and since they return to 1 after 6 steps the decimal repeats with period $6 = 7 - 1$.

2 / FERMAT'S LITTLE THEOREM

If p is prime and $p \nmid a$ then $a^{p-1} \equiv 1 \pmod p$

Multiplying the numbers $1, 2, \ldots, p-1$ by a only shuffles them modulo p, because a has an inverse. For $p = 7$ and $a = 3$ the list $1, 2, 3, 4, 5, 6$ becomes $3, 6, 2, 5, 1, 4$.

Taking products, $a^{p-1}(p-1)! \equiv (p-1)!$, and cancelling the unit $(p-1)!$ gives

$$\boxed{a^{p-1} \equiv 1 \pmod p}$$

Here $3^6 = 729 = 7 \cdot 104 + 1$, and Wilson's theorem $(p-1)! \equiv -1$ comes from pairing each number with its inverse, as in $6! = 720 \equiv -1 \pmod 7$.

3 / REPTEND LENGTHS / THE ORDER OF 10

The decimal of $1/p$ repeats with the smallest k for which $10^k \equiv 1 \pmod p$

p	$1/p$	period k	$p-1$	
3	$0.\overline{3}$	1	2	
7	$0.\overline{142857}$	6	6	full reptend
11	$0.\overline{09}$	2	10	
13	$0.\overline{076923}$	6	12	
17	$0.\overline{0588235294117647}$	16	16	full reptend
19	$0.\overline{052631578947368421}$	18	18	full reptend
23	$0.0434782608\ldots$ (22 digits repeat)	22	22	full reptend
29	$0.0344827586\ldots$ (28 digits repeat)	28	28	full reptend
31	$0.\overline{032258064516129}$	15	30	
37	$0.\overline{027}$	3	36	

The period always divides $p-1$, and primes where it equals $p-1$ are the full reptend primes, $7, 17, 19, 23, 29, 47, 59, 61, 97$ below 100, whose cycles of digits are the cyclic numbers.
For 11 the period 2 gives $0.\overline{09}$, and the primes 2 and 5 divide 10 and give terminating decimals.

4 / EULER'S THEOREM / COMPOSITE MODULI

If $\mathrm{HCF}(a, n) = 1$ then $a^{\varphi(n)} \equiv 1 \pmod n$, and the cycle length divides $\varphi(n)$

For $n = 10$ we have $\varphi(10) = 4$, and the last digits of the powers cycle: $3, 9, 7, 1$ for 3^k, $7, 9, 3, 1$ for 7^k, and $9, 1$ for 9^k, with cycle lengths $4, 4, 2$ all dividing 4. So $3^4 = 81$, $7^4 = 2401$ and $9^4 = 6561$ all end in 1, and the last digit of 7^{2023} is $7^{2023 \bmod 4} = 7^3 \equiv 3$. Fermat's theorem is the case $n = p$, where $\varphi(p) = p - 1$.

ASIDE / THE CYCLIC NUMBER

Multiplying 142857 by 1 to 6 only rotates its digits, and by 7 gives 999999.

LINKS TO OTHER SHEETS

The cycle of powers uses *Arithmetic on a clock*; the totient appears in *HCF, LCM, cicadas and the totient*.

Convention. — marks the repeating block; $\varphi(n)$ is Euler's totient; k is the multiplicative order of 10.

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EGYPTIAN FRACTIONS AND THE GREEDY ALGORITHM ON ONE PAGE

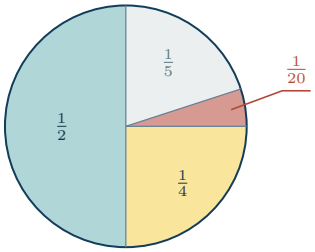
Greedy Tarts

Take the biggest unit-fraction slice that fits, then repeat with the share still owed.

1 / SHARE FOUR TARTS AMONG FIVE PEOPLE

Each plate needs four fifths of a whole tart

A **unit fraction** $1/n$ is one of n equal parts of a whole. An **Egyptian fraction** expresses an amount as a finite sum of **distinct** unit fractions. Here is one person's share:



Colour totals $4/5$; grey is $1/5$.
All fractions refer to a whole tart.

Share still owed	Serve the largest slice	Still owed
$\frac{4}{5}$	$\frac{1}{2}$	$\frac{3}{10}$
$\frac{3}{10}$	$\frac{1}{4}$	$\frac{1}{20}$
$\frac{1}{20}$	$\frac{1}{20}$	0

$$\frac{4}{5} = \frac{1}{2} + \frac{1}{4} + \frac{1}{20}$$

2 / TURN THE SERVING RULE INTO AN ALGORITHM

Largest slice means smallest denominator

For $0 < p < q$ in lowest terms, we want the largest $1/n$ satisfying $1/n \le p/q$. Thus $n \ge q/p$, and the first allowed integer is

$$n = \left\lceil \frac{q}{p} \right\rceil, \qquad R_{\text{new}} = \frac{pn - q}{qn}.$$

The **ceiling** $\lceil x \rceil$ rounds up to an integer. Subtract $1/n$, reduce the remaining fraction, and repeat until zero.

At $3/10$, the ratio $10/3 = 3.33\ldots$ rounds up to 4. Serve $1/4$: it fits, while $1/3$ is too large.

Greedy means taking the largest possible slice now.

3 / A SHRINKING INTEGER PROVES THAT IT STOPS

The numerator falls at every step

The ceiling condition gives

$$n - 1 < \frac{q}{p} \le n \quad \Longrightarrow \quad 0 \le pn - q < p.$$

The new numerator $pn - q$ is smaller than p. Cancelling common factors cannot increase it. A strictly falling sequence of positive integers must end.

So greedy uses **at most the original p terms**. In our example, the reduced numerators fall $4 \to 3 \to 1 \to 0$.

The denominators are distinct too. If the remainder is non-zero,

$$0 < R_{\text{new}} < \frac{1}{n(n-1)} \le \frac{1}{n}.$$

The next slice is smaller, so its denominator is larger.

4 / COMPARE A LOCAL CHOICE WITH THE WHOLE RESULT

Greedy can use more terms

For $9/20$, the greedy recipe gives three pieces:

$$\frac{9}{20} = \frac{1}{3} + \frac{1}{9} + \frac{1}{180}.$$

The remainders are $7/60$, then $1/180$, then 0. Choosing $1/4$ first gives a shorter expansion:

$$\frac{9}{20} = \frac{1}{4} + \frac{1}{5}.$$

Greedy guarantees a finite answer. **Finding the fewest terms** is a separate optimisation problem: the biggest slice now can leave an awkward remainder.

5 / KEEP SPLITTING THE SMALLEST PIECE

One amount, infinitely many recipes

For any integer $n \ge 2$, one unit fraction splits into two:

$$\frac{1}{n} = \frac{1}{n+1} + \frac{1}{n(n+1)}.$$

For example,

$$\frac{1}{3} = \frac{1}{4} + \frac{1}{12} = \frac{1}{4} + \frac{1}{13} + \frac{1}{156}.$$

Split the term with the **largest denominator**. Both new denominators are larger, so no duplicates appear. Repeat to get infinitely many different *finite* expansions of the same positive proper fraction.

YOUR TURN / ONE SHARE, TWO RECIPES

Find the greedy expansion of $7/12$, then a different two-term expansion.

Check: $\frac{1}{2} + \frac{1}{12}$ is greedy; $\frac{1}{3} + \frac{1}{4}$ is another recipe.

History. Unit-fraction arithmetic was used in ancient Egypt; Fibonacci described this greedy algorithm in 1202.

Sources: David Eppstein, *Algorithms for Egyptian Fractions*; Jukka Pihko (2001), introduction to the greedy algorithm, §1.

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COUNTS, FAREY FRACTIONS AND RATIONAL GEOMETRY ON ONE PAGE

The mediant: why stop there?

Adding tops and bottoms opens a path from pooled data to trees, touching circles and triples.

1 / CHANGE THE OPERATION AND CHANGE THE QUESTION

Two scores become one pooled fraction

One success in 3 trials and one in 4 trials give 2 successes in 7 trials. Ordinary addition and pooling answer different questions:

$$\frac{1}{3} + \frac{1}{4} = \frac{7}{12} \text{ (sum)}, \qquad \frac{1}{3} \oplus \frac{1}{4} = \frac{2}{7} \text{ (mediant)}.$$

For positive denominators, the mediant is a **weighted average**:

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d} = \frac{b}{b+d}\,\frac{a}{b} + \frac{d}{b+d}\,\frac{c}{d}.$$

The weights add to 1, so distinct inputs put the mediant strictly between them: $\frac{1}{4} < \frac{2}{7} < \frac{1}{3}$.

2 / A DETERMINANT OF ONE GIVES SPECIAL NEIGHBOURS

The smallest denominator in the gap

For reduced Farey neighbours $a/b < c/d$, we have $bc - ad = 1$. Their mediant is already in lowest terms.

Any fraction x/y strictly between them gives positive integers

$$U = bx - ay \geq 1, \qquad V = cy - dx \geq 1.$$

Using $bc - ad = 1$, we obtain

$$y = dU + bV \geq b + d.$$

The mediant achieves this bound: it has the **smallest possible denominator** between the neighbours.

$$\frac{1}{4} < \frac{2}{7} < \frac{1}{3}, \qquad 7 = 4 + 3.$$

No fraction with denominator below 7 lies in that gap. This makes the mediant a useful tool for rational approximation.

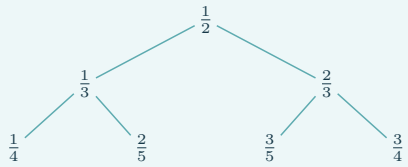
3 / CHOOSE HOW TO ORGANISE THE INSERTIONS

Farey sequences and a rational tree

F_N lists the reduced fractions in $[0,1]$ with denominator at most N, in numerical order. For example,

$$F_4 = \left\{\frac{0}{1}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{1}\right\}.$$

Start with $0/1, 1/1$. Insert adjacent mediants whenever $b+d \leq N$; repeat until none qualify. This builds F_N.



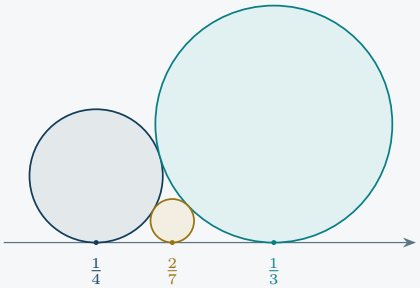
Unrestricted insertion gives the Stern–Brocot refinement on $[0,1]$; F_N selects fractions by denominator.

4 / MAKE THE NEIGHBOUR TEST VISIBLE

Ford circles: the mediant fits between

At each reduced p/q, put a circle above the number line, tangent to it, with centre $(p/q, 1/(2q^2))$ and radius

$$r_{p/q} = \frac{1}{2q^2}.$$



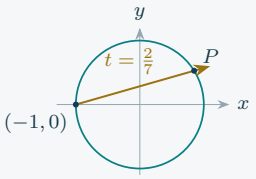
Two distinct Ford circles touch exactly when $|ad - bc| = 1$. The mediant circle touches both parent circles: $\frac{1}{4} \oplus \frac{1}{3} = \frac{2}{7}$.

5 / LET A RATIONAL FRACTION BECOME A SLOPE

A route to Pythagorean triples

Draw $y = t(x+1)$ through $(-1,0)$ with rational slope $t = p/q$, $0 < p < q$. Its second intersection with $x^2 + y^2 = 1$ is

$$(x, y) = \left(\frac{q^2 - p^2}{q^2 + p^2}, \frac{2pq}{q^2 + p^2}\right).$$



With $p/q = 2/7$, $P = (45/53, 28/53)$, giving

$$45^2 + 28^2 = 53^2.$$

Coprime p, q of opposite parity give a **primitive** triple: its three side lengths have no common factor.

6 / KEEP THE DENOMINATOR'S MEANING

Pooling $1/3$ and $1/4$ gives $2/7$; pooling $2/6$ and $1/4$ gives $3/10$. Doubling both counts doubles their weight. Use actual counts for pooling, and lowest terms for rational trees.

Explore: *Why Stop There?* **Sources:** Hatcher, *Topology of Numbers*, Ch. 1; Ainsworth et al., *The Farey Sequence*; Johnson, *A Formula for Pythagorean Triples*.

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MEDIANTS AND DOUBLE ANGLES ON ONE PAGE

Adding fractions the Farey way

Adding numerators and denominators adds vectors, while the angle-addition formula multiplies them.

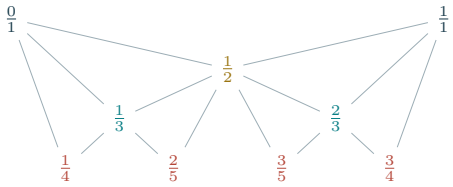
THE RULE

The mediant lands between, and only between

The mediant of two fractions adds tops and bottoms separately,

$$\frac{a}{b} \oplus \frac{c}{d} = \frac{a+c}{b+d} = \frac{b\cdot\frac{a}{b}+d\cdot\frac{c}{d}}{b+d}$$

so it is the average of the two values weighted by their denominators, and it always lies strictly between them. Two fractions are Farey neighbours when $|ad-bc|=1$; then their mediant is the simplest fraction between them, and starting from $\frac{0}{1}$ and $\frac{1}{1}$ and taking mediants of neighbours produces every fraction in $[0,1]$ exactly once, in lowest terms. The mediant of a fraction with itself, $\frac{2a}{2b}$, is the same number, so mediants can split an interval but can never double anything.



Each new fraction is the mediant of the two neighbours it is joined to.

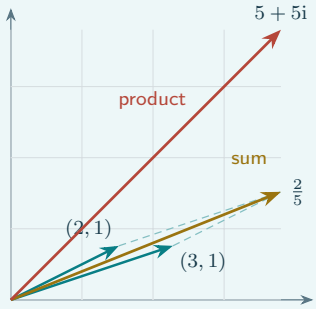
TWO WAYS TO COMBINE SLOPES

Mediants add vectors, angle addition multiplies them

Read the slope $\frac{p}{q}$ as the vector (q,p), or the Gaussian integer $q+\mathrm{i}p$, at angle $\arctan\frac{p}{q}$. Adding two vectors gives the mediant, whose angle lies between theirs, pulled towards the longer vector; multiplying them as complex numbers adds the angles exactly:

$$\frac{p}{q}\oplus\frac{c}{d}=\frac{p+c}{q+d},\qquad \tan(\alpha+\beta)=\frac{pd+qc}{qd-pc}.$$

With $\frac{1}{2}$ and $\frac{1}{3}$, the sum $(2,1)+(3,1)=(5,2)$ gives the mediant $\frac{2}{5}$ at 0.3805 rad, between 0.3218 and 0.4636, while $(2+\mathrm{i})(3+\mathrm{i})=5+5\mathrm{i}$ gives slope 1, the classic identity $\arctan\frac{1}{2}+\arctan\frac{1}{3}=\frac{\pi}{4}$.



The sum lies between the two; the product swings out to $45°$.

DOUBLING

Squaring doubles the angle; a mediant with itself stays put

Squaring $q+\mathrm{i}p$ gives $(q^2-p^2)+2pq\,\mathrm{i}$, the traditional double-angle formula in fraction form, and its three numbers are a Pythagorean triple. The two operations divide the work: mediants of Farey neighbours reach every slope $\frac{p}{q}$, and doubling the angle of each slope turns it into a triple.

	mediant	angle addition
on vectors (q,p)	add: $(q+d,\,p+c)$	multiply as $(q+\mathrm{i}p)(d+\mathrm{i}c)$
on angles	lands strictly between	adds exactly
combined with itself	$\frac{p}{q}\oplus\frac{p}{q}=\frac{p}{q}$	$\tan 2\alpha=\frac{2pq}{q^2-p^2}=\frac{2t}{1-t^2}$
depends on the form of the fraction	yes, $\frac{1}{2}\oplus\frac{1}{3}\neq\frac{2}{4}\oplus\frac{1}{3}$	no, only on the value
what it builds	every fraction, once, by Farey	every Pythagorean triple, by doubling

slope $\frac{p}{q}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{2}{5}$	$\frac{3}{5}$
doubled: $q^2-p^2,\ 2pq,\ p^2+q^2$	$3,4,5$	$8,6,10$	$5,12,13$	$15,8,17$	$7,24,25$	$21,20,29$	$16,30,34$

FRACTIONAL RADIANS

Rational angles or rational slopes, not both

Taking mediants of angles measured in radians gives a weighted mean angle: $\frac{1}{2}$ rad $\oplus$ $\frac{1}{3}$ rad $=$ $\frac{2}{5}$ rad, the average of 0.5 and 0.333 weighted $2:3$. But Lambert proved in 1761 that $\tan r$ is irrational for every rational $r\neq 0$, so a rational angle never has a rational slope; Farey arithmetic can live on the angles or on the slopes, and only the slopes lead to triples.

WHERE MEDIANTS EARN THEIR KEEP

Best approximations and gear trains

The convergents of a continued fraction are reached by repeated mediants, which is why $\frac{22}{7}$ and $\frac{355}{113}$ appear on the Stern–Brocot path to π; a clockmaker choosing gear teeth to approximate a ratio walks the same tree. A season's success rate is the mediant of its parts, weighted by attempts, so one player can lead in every half-season yet trail overall, which is Simpson's paradox.

Conventions. $\oplus$ is mediant & slopes $\frac{p}{q}$ are in lowest terms; when p,q are both odd the doubled triple halves to a primitive one, as $8,6,10$ becomes $4,3,5$.

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PYTHAGORAS, MEDIANTS AND FAREY NEIGHBOURS ON ONE PAGE

Fractions that touch: Ford circles

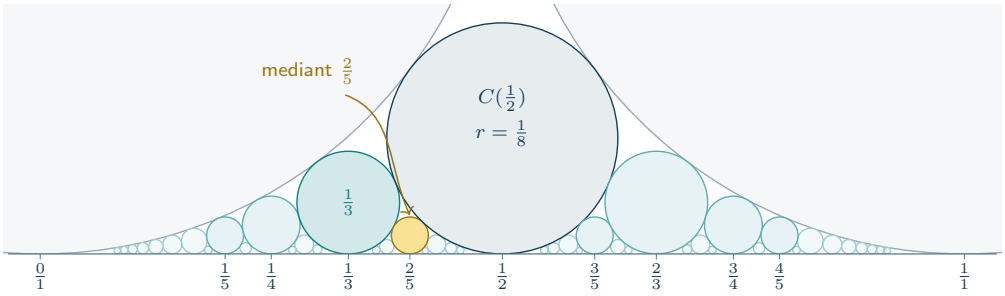
Give each fraction a circle, and the mediant becomes a picture of tangency.

1 / TURN FRACTIONS INTO A PICTURE

A denominator sets the size

For a reduced fraction p/q, with $q > 0$, draw a circle above the number line:

$$\text{radius } r_q = \frac{1}{2q^2}, \qquad \text{centre } \left(\frac{p}{q}, \frac{1}{2q^2}\right).$$



Labels show the Farey sequence of order 5; finer circles have denominators up to 12. The large endpoint circles continue above the window.

2 / PYTHAGORAS SUPPLIES THE TEST

When do two circles touch?

Let $R = 1/(2b^2)$ and $S = 1/(2d^2)$ be the radii at distinct reduced fractions a/b and c/d. Let D be the distance between their centres. Pythagoras, then subtraction, give

$$D^2 = \left(\frac{a}{b} - \frac{c}{d}\right)^2 + (R - S)^2,$$

$$D^2 - (R + S)^2 = \frac{(ad - bc)^2 - 1}{b^2 d^2}.$$

Here $4RS = 1/(b^2 d^2)$.

The non-zero integer $ad - bc$ has absolute value at least 1, so $D \geq R + S$: distinct Ford circles never overlap.

circles touch $\iff$ $|bc - ad| = 1$

3 / ADD ACROSS TO FILL THE GAP

The mediant touches both neighbours

Now suppose $a/b < c/d$ and $bc - ad = 1$. Their mediant lies between them:

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

The two new pairs pass exactly the same touching test:

$$b(a+c) - a(b+d) = bc - ad = 1,$$
$$c(b+d) - d(a+c) = bc - ad = 1.$$

Any common divisor of $a + c$ and $b + d$ must divide 1, so the mediant is already reduced. Its circle fits in the gap, tangent to both original circles and the number line.

$\frac{1}{3}$ and $\frac{1}{2}$ $\longrightarrow$ $\frac{1+1}{3+2} = \frac{2}{5}$

4 / WHICH FRACTION ARRIVES NEXT?

Farey neighbours and the next circle

The **Farey sequence** F_N orders all reduced fractions in $[0, 1]$ whose denominators are at most N. For example,

$$F_4: \quad \frac{0}{1}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{1}{1}.$$

Neighbours obey $bc - ad = 1$. Any fraction p/q strictly between them has positive integers $bp - aq$ and $cq - dp$. Thus

$$q = b(cq - dp) + d(bp - aq) \geq b + d.$$

The mediant attains the minimum denominator $b + d$: it is the first fraction to enter that interval and has the largest Ford circle there.

So $1/3$ and $1/2$ stay neighbours in F_4; their mediant $2/5$ arrives in F_5.

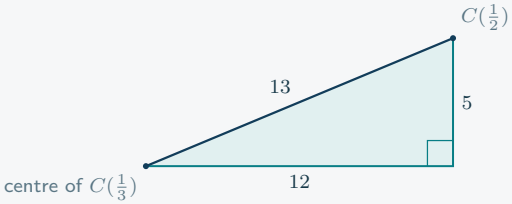
5 / A FAMILIAR TRIANGLE IS HIDING HERE

The circles conceal 5–12–13

The centres of $C(1/3)$ and $C(1/2)$ have offsets

$$\Delta x = \frac{1}{2} - \frac{1}{3} = \frac{12}{72}, \qquad \Delta y = \frac{1}{8} - \frac{1}{18} = \frac{5}{72}.$$

Because the circles touch, $D = R + S = 1/18 + 1/8 = 13/72$.



Scale every length by 72 and the touching proof becomes the familiar Pythagorean triple:

$$\boxed{5^2 + 12^2 = 13^2}.$$

Companion to the mediant sheet. Adding numerators and denominators inserts a fraction between its parents; Ford circles show the insertion as tangency. Always reduce fractions before drawing.

Source. L. R. Ford, *Fractions*, *American Mathematical Monthly* **45** (1938), 586–601.

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MEDIANTS AND BOOSTS ON ONE PAGE

Merging and boosting

Write each speed as momentum over energy: the mediant merges two bodies and halves rapidities, Einstein's rule changes frame and adds them.

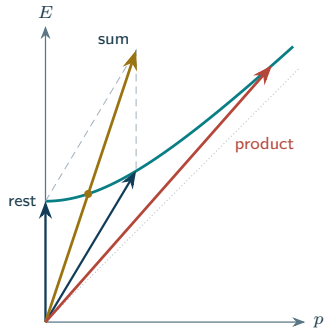
THE RULE

A speed is a fraction $\beta = p/E$

With $c = 1$, a body of mass m and rapidity φ has energy $E = m\cosh\varphi$ and momentum $p = m\sinh\varphi$, so its speed is the fraction $\beta = \frac{p}{E} = \tanh\varphi$ and its energy–momentum is a vector on the hyperbola $E^2 - p^2 = m^2$. Two such vectors combine in two ways:

$$\frac{p_1}{E_1} \oplus_{\mathrm{m}} \frac{p_2}{E_2} = \frac{p_1 + p_2}{E_1 + E_2}, \qquad \beta_1 \oplus_{\mathrm{E}} \beta_2 = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$$

The mediant adds the vectors and gives the velocity of the merged pair, its centre of momentum, which always lies between the two; Einstein's rule composes boosts, changing frame, and adds the rapidities.



Navy: at rest and at $\varphi = \ln 2$. Their sum points along $\frac{1}{2}\ln 2$; the product of the moving one with itself reaches $\ln 4$.

HALF AND DOUBLE

Equal masses: the mediant averages rapidities exactly

For equal masses the sum-to-product identities give

$$\frac{\sinh\varphi_1 + \sinh\varphi_2}{\cosh\varphi_1 + \cosh\varphi_2} = \tanh\frac{\varphi_1 + \varphi_2}{2},$$

so the merged pair moves at the average rapidity, while a boost moves to the sum. Merging with an identical body at rest therefore gives the half-rapidity formula and boosting twice gives the double-rapidity formula:

$$\frac{0}{1} \oplus_{\mathrm{m}} \frac{\gamma\beta}{\gamma} = \frac{\gamma\beta}{1+\gamma} = \tanh\frac{\varphi}{2}, \qquad \beta \oplus_{\mathrm{E}} \beta = \frac{2\beta}{1+\beta^2} = \tanh 2\varphi$$

starting from $\beta = \frac{3}{5}$, $\varphi = \ln 2$	merge with rest	merge with a twin	boost by itself	boost by its reverse
operation	$\frac{0}{1} \oplus_{\mathrm{m}} \frac{3/4}{5/4}$	$\frac{3/4}{5/4} \oplus_{\mathrm{m}} \frac{3/4}{5/4}$	$\frac{3}{5} \oplus_{\mathrm{E}} \frac{3}{5}$	$\frac{3}{5} \oplus_{\mathrm{E}} \left(-\frac{3}{5}\right)$
rapidity	$\frac{1}{2}\ln 2$	$\ln 2$	$\ln 4$	0
speed	$\frac{1}{3}$	$\frac{3}{5}$	$\frac{15}{17}$	0
mass afterwards	$\sqrt{4.5}\,m = 2.12m$	$2m$	m	m

UNEQUAL MASSES AND THE ENERGY RELEASED

Denominators are weights, and merging costs nothing but makes mass

For unequal masses the mediant is weighted by energy, so writing $\frac{3}{5}$ as $\frac{0.75}{1.25}$ or $\frac{7.5}{12.5}$ matters: the denominator says how heavy the body is. The merged mass comes from the Minkowski length of the summed vector,

$$M^2 = (E_1 + E_2)^2 - (p_1 + p_2)^2 = m_1^2 + m_2^2 + 2m_1m_2\cosh(\varphi_1 - \varphi_2) \;\geq\; (m_1 + m_2)^2,$$

the reverse triangle inequality: the faster the two approach, the more kinetic energy becomes rest mass. Two protons meeting head-on at $\pm\frac{3}{5}$ have relative rapidity $\ln 4$ and give $M = 2.5\,m_p$; a beam hitting a proton at rest needs the same relative rapidity, the Einstein double $\frac{3}{5} \oplus_{\mathrm{E}} \frac{3}{5} = \frac{15}{17}$, to reach the same M, which is why colliders beat fixed targets.

THE CIRCULAR TWIN

Adding unit vectors bisects the angle

On the circle the same move gives the half-angle formula, $\frac{0}{1} \oplus_{\mathrm{m}} \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{1+\cos\theta} = \tan\frac{\theta}{2}$, because two unit vectors span a rhombus whose diagonal bisects their angle; multiplying them as complex numbers doubles the angle instead. In both geometries the mediant of unit vectors halves and the product doubles.

COMPOSING THE TWO MOVES

Boosts commute with merging

Boosting every body by ψ adds ψ to every rapidity and to the merged one, so merge-then-boost equals boost-then-merge: $(\beta_1 \oplus_{\mathrm{m}} \beta_2) \oplus_{\mathrm{E}} \beta_\psi = (\beta_1 \oplus_{\mathrm{E}} \beta_\psi) \oplus_{\mathrm{m}} (\beta_2 \oplus_{\mathrm{E}} \beta_\psi)$ with each fraction carried as p/E. Galileo's limit keeps both moves, with mass-weighted mean velocity for merging and plain addition for boosts.

Conventions. Units with $c = 1$; $\oplus_{\mathrm{m}}$ is the mediant of fractions written as momentum over energy and $\oplus_{\mathrm{E}}$ is Einstein's rule; $\gamma = \cosh\varphi$ and $\gamma\beta = \sinh\varphi$.

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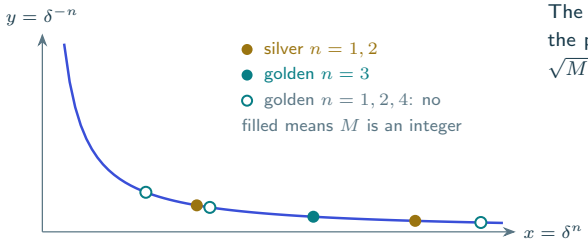
NUMBER THEORY ON ONE PAGE

Metallic means as differences of squares

When is $\delta_k^{-n} = \sqrt{M+1} - \sqrt{M}$ for an integer M? Every n if k is even, only multiples of 3 if k is odd.

1 / THE RULE AND THE PICTURE

Powers of $\delta_k = \frac{k+\sqrt{k^2+4}}{2}$ lie on the hyperbola $xy = 1$



The metallic mean δ_k is the positive root of $x^2 - kx - 1 = 0$, so $\delta_k^{-1} < 1$ and the powers δ_k^{-n} shrink towards 0. The question is when $y = \delta_k^{-n}$ can be written $\sqrt{M+1} - \sqrt{M}$ with $M \in \mathbb{Z}_{\geq 0}$.

$$M_n(k) = \frac{\delta_k^{2n} + \delta_k^{-2n} - 2}{4} = \frac{S_n - 2}{4}$$

2 / THE MECHANISM / RATIONALISE, THEN SQUARE

The formula for M comes from y and $1/y$, and S_n is a Lucas sequence

If $y = \sqrt{M+1} - \sqrt{M}$ then $1/y = \sqrt{M+1} + \sqrt{M}$, so $1/y - y = 2\sqrt{M}$ and $M = \left(\frac{1/y - y}{2}\right)^2 = \frac{y^{-2} + y^2 - 2}{4}$. With $y = \delta_k^{-n}$ this gives $M_n(k)$, and since $\delta_k^2 + \delta_k^{-2} = k^2 + 2$,

$$S_{n+1} = (k^2+2)S_n - S_{n-1}, \qquad S_0 = 2,\ S_1 = k^2 + 2, \qquad M_n \in \mathbb{Z} \iff S_n \equiv 2 \pmod 4.$$

S_n	$n=0$	1	2	3	4	5	6
$k=1$	2	3	7	18	47	123	322
$k=2$	2	6	34	198	1154	6726	39 202
$k=3$	2	11	119	1298	14 159	154 451	1 684 802

3 / THE PARITY THEOREM / WORKING MODULO 4

Even k always works, and odd k works exactly when $3 \mid n$

Put $A = k^2 + 2$. For even k, $A \equiv 2$ and $S_{n+1} \equiv 2S_n - S_{n-1} \pmod 4$ from $S_0 \equiv S_1 \equiv 2$ keeps $S_n \equiv 2$ for ever. For odd k, $A \equiv 3$ and $S_{n+1} \equiv -S_n - S_{n-1}$, which cycles with period three:

n	0	1	2	3	4	5	6	7	8
$S_n \bmod 4$, k even	2	2	2	2	2	2	2	2	2
$S_n \bmod 4$, k odd	2	3	3	**2**	3	3	**2**	3	3

k even: $M_n(k) \in \mathbb{Z}\ \forall n$; $\quad$ k odd: $M_n(k) \in \mathbb{Z} \iff 3 \mid n$

4 / WHY PERIOD THREE / THE CUBE TRICK AND SQUARES

Cubing turns an odd metallic mean into an even one

Cubing gives $\delta_k^3 = \delta_{k^3+3k}$, and $k^3 + 3k = k(k^2+3)$ is even exactly when k is odd: $\varphi^3 = 2 + \sqrt{5} = \delta_4$ and $\delta_3^3 = 18 + \sqrt{325} = \delta_{36}$. An even-index mean has $\delta_j^{-1} = \sqrt{(j/2)^2 + 1} - \frac{j}{2}$, so $\varphi^{-3} = \sqrt{5} - \sqrt{4}$ and $\delta_3^{-3} = \sqrt{325} - \sqrt{324}$. Equivalently $M_n = \left(\frac{V_n}{2}\right)^2$ for odd n and $\frac{D\,U_n^2}{4}$ for even n, with $D = k^2 + 4$ and U_n, V_n the Lucas sequences, and these are integers when k is even or when U_n, V_n are even, which for odd k happens exactly when $3 \mid n$. For $k = 2$ this gives $M = 1,\ 8,\ 49,\ 288$: alternately a square and twice a square.

5 / EXAMPLES / THE REPRESENTATIONS

Silver always, golden and bronze at multiples of three

mean	power	difference of square roots	M
golden, $k=1$	φ^{-3}, φ^{-6}	$\sqrt{5} - \sqrt{4}$, $\sqrt{81} - \sqrt{80}$	4, 80
silver, $k=2$	$(\sqrt{2}-1)^{1,2,3,4}$	$\sqrt{2} - \sqrt{1}$, $\sqrt{9} - \sqrt{8}$, $\sqrt{50} - \sqrt{49}$, $\sqrt{289} - \sqrt{288}$	1, 8, 49, 288
bronze, $k=3$	δ_3^{-3}, δ_3^{-6}	$\sqrt{325} - \sqrt{324}$, $\sqrt{421\,201} - \sqrt{421\,200}$	324, 421 200
$k=4$	δ_4^{-1}	$\sqrt{5} - \sqrt{4}$ (the golden cube)	4

Numerically $\varphi^{-3} = 0.23607 = \sqrt{5} - 2$ and $(\sqrt{2}-1)^4 = 0.029437 = \sqrt{289} - \sqrt{288}$.

ASIDE / THE GALOIS PICTURE

δ_k is a unit of norm -1 with conjugate $-\delta_k^{-1}$, and modulo 4 the odd-k recurrence has characteristic polynomial $x^2 + x + 1$, whose roots are cube roots of unity.

ASIDE / CHECK IN PYTHON

Iterating `a,b = b,(k*k+2)*b-a` from `(2,k*k+2)` and testing `(S-2)%4==0` reproduces the table.

Convention. $S_n = \delta_k^{2n} + \delta_k^{-2n}$; for $k = 1$ these are the Lucas numbers L_{2n}. After the Metallic Means: Consecutive Square Roots page.

Geometric

PELL'S EQUATION ON ONE PAGE

Nearly twice a square

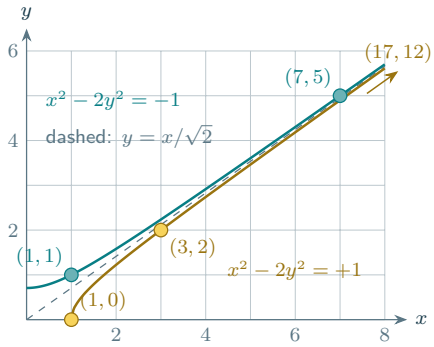
From $3^2 - 2(2^2) = 1$ to the best fractions for $\sqrt{2}$, integer points on a hyperbola, and the norm form $a^2 - 2b^2$.

1 / THE RULE AND THE PICTURE

Squares that miss twice a square by one

Two small coincidences start everything, $3^2 - 2(2^2) = 9 - 8 = 1$ and $17^2 - 2(12^2) = 289 - 288 = 1$, and Pell's equation asks for every such pair, together with its twin that misses by -1:

$$x^2 - 2y^2 = \pm 1$$



x	y	$x^2 - 2y^2$	x/y	error $x/y - \sqrt{2}$
1	1	-1	1	-0.414
3	2	$+1$	1.5	$+0.0858$
7	5	-1	1.4	-0.0142
17	12	$+1$	$1.41\dot{6}$	$+0.00245$
41	29	-1	1.413793…	-0.000420
99	70	$+1$	1.4142857…	$+0.0000722$
239	169	-1	1.4142012…	-0.0000124
577	408	$+1$	1.4142157…	$+0.00000212$

The fractions land on alternate sides of $\sqrt{2} = 1.41421356\ldots$, each error about a sixth of the one before, since y grows by $1 + \sqrt{2}$ and so y^2 by $(1 + \sqrt{2})^2 \approx 5.83$.

$$\sqrt{2} \approx \frac{17}{12} \approx \frac{99}{70} \approx \frac{577}{408} \qquad +1 \text{ rows overshoot, } -1 \text{ rows undershoot}$$

2 / WHY THE FRACTIONS ARE SO GOOD

Divide by y^2 and the miss shrinks

Dividing $x^2 = 2y^2 + 1$ by y^2 shows that the square of x/y misses 2 by only $1/y^2$, as in $\left(\frac{17}{12}\right)^2 = \frac{289}{144} = 2 + \frac{1}{144}$. Factorising the difference of two squares turns this into a bound on x/y itself,

$$\frac{x}{y} - \sqrt{2} = \frac{x^2 - 2y^2}{y\,(x + y\sqrt{2})} = \frac{\pm 1}{y\,(x + y\sqrt{2})},$$

so $|x/y - \sqrt{2}| < 1/(2\sqrt{2}\,y^2)$, far smaller than the $1/(2y)$ that the nearest fraction with an arbitrary denominator y can promise.

$$\frac{17}{12} - \sqrt{2} = \frac{1}{12(17 + 12\sqrt{2})} \approx \frac{1}{408}$$

3 / THE NORM FORM

Multiply, and the norm comes along

In $\mathbb{Z}[\sqrt{2}]$ the number $a + b\sqrt{2}$ times its conjugate $a - b\sqrt{2}$ gives the norm

$$\mathrm{N}(a + b\sqrt{2}) = (a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2.$$

The norm is multiplicative, $\mathrm{N}(uv) = \mathrm{N}(u)\,\mathrm{N}(v)$, so powers of the smallest unit generate every solution,

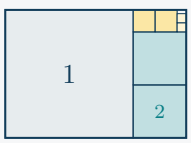
$$(1 + \sqrt{2})^n = x_n + y_n\sqrt{2}, \qquad x_n^2 - 2y_n^2 = (-1)^n,$$

each row of the table following from the last by $(x, y) \mapsto (x + 2y,\ x + y)$. The Gaussian form $a^2 + b^2$ and the Eisenstein form $a^2 - ab + b^2$ belong to the same family; the minus sign here makes the level sets hyperbolas in place of closed curves, so the units never run out.

4 / CUT SQUARES FROM THIS PAGE

The silver rectangle

Euclid's square cutting applied to a $\sqrt{2} \times 1$ rectangle removes one square and from then on always two, because the remainder reproduces itself, $\sqrt{2} - 1 = 1/(2 + (\sqrt{2} - 1))$, so $\sqrt{2} = [1; 2, 2, 2, \ldots]$, whose convergents are exactly the table's fractions.



A4 is cut so that halving keeps the shape, hence ratio $\sqrt{2}$, and this page measures 297×210 mm:

$$\frac{297}{210} = \frac{99}{70}$$
$$99^2 - 2(70^2) = 1$$

The golden rectangle sheds one square at every step; the silver one sheds two.

5 / STEPS ALONG THE HYPERBOLA

Each solution is a boost of the last

The step $(x, y) \mapsto (3x + 4y,\ 2x + 3y)$ is multiplication by $3 + 2\sqrt{2}$, a matrix of determinant $9 - 8 = 1$ that slides the hyperbola $x^2 - 2y^2 = 1$ along itself, as a Lorentz boost slides the mass shell $E^2 - p^2 = m^2$. Writing $x = \cosh(n\ell)$ and $y\sqrt{2} = \sinh(n\ell)$, the integer points sit at equal steps of rapidity

$$\ell = \ln(3 + 2\sqrt{2}) = 2\ln(1 + \sqrt{2}) \approx 1.7627,$$

so the rapidity adds while the coordinates multiply, which is why the points race off along the asymptote.

$$x_n + y_n\sqrt{2} = e^{n\ell}, \qquad x_n - y_n\sqrt{2} = e^{-n\ell}$$

The second factor shrinking towards zero is the error of panel 2 seen from the other side.

Convention. Pell's equation $x^2 - dy^2 = 1$ has infinitely many solutions for every positive non-square d (Lagrange, 1768), the smallest for $d = 61$ being $x = 1\,766\,319\,049$; Brahmagupta and Bhaskara II solved such equations centuries before Euler's misattribution to Pell. **Companions:** cutting squares; hyperbola triples; mass shell.

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LINEAR ALGEBRA ON ONE PAGE

The LCM method as Gaussian elimination

Matching coefficients with the LCM is one row operation on the augmented matrix, repeated down the columns.

1 / THE RULE AND THE PICTURE

Write the system as a matrix, and each LCM step becomes one row operation

$$\left(\begin{array}{cc|c} 3 & 2 & 16 \\ 2 & 5 & 19 \end{array}\right) \xrightarrow{R_2 \leftarrow 3R_2 - 2R_1} \left(\begin{array}{cc|c} 3 & 2 & 16 \\ 0 & 11 & 25 \end{array}\right) \xrightarrow{\text{back-substitute}} \left(\begin{array}{cc|c} 1 & 0 & \frac{42}{11} \\ 0 & 1 & \frac{25}{11} \end{array}\right)$$

$3x + 2y = 16,\ 2x + 5y = 19$ $\qquad$ $11y = 25$ $\qquad$ $x = \frac{42}{11},\ y = \frac{25}{11}$

The coefficients sit left of the bar and the constants right of it:

$$R_2 \leftarrow \frac{L}{d} R_2 - \frac{L}{a} R_1$$

with $L = \mathrm{LCM}(a, d)$ the first column of row 2 becomes 0.

2 / THE THREE ALLOWED MOVES / NONE OF THEM CHANGES THE SOLUTION

Scale a row, add a multiple of one row to another, or swap two rows

row operation	what it does to the lines	in the LCM method
$R_i \leftarrow kR_i$	rewrites a line, as $6x + 4y = 32$ is $3x + 2y = 16$	multiply by L/a or L/d
$R_i \leftarrow R_i + kR_j$	replaces a line by another through the crossing point	subtract the matched rows
$R_i \leftrightarrow R_j$	relabels the lines	put the easier row on top

Every move keeps the common point of the two lines, which is why the final matrix has the same solution as the first. The two multipliers 3 and 2 in the picture are $6/2$ and $6/3$ with $L = \mathrm{LCM}(3, 2) = 6$, and the surviving pivot $11 = ae - bd$ is the determinant.

3 / ECHELON FORM AND BACK-SUBSTITUTION / FINISHING THE JOB

Forward elimination makes a staircase, and back-substitution climbs it

From $11y = 25$ we get $y = \frac{25}{11}$, and the first row $3x + 2y = 16$ gives $3x = 16 - \frac{50}{11} = \frac{126}{11}$, so $x = \frac{42}{11}$.
Continuing to divide the rows by their pivots and clear above them (Gauss–Jordan) leaves the identity on the left, with the solution in the right-hand column.

$$x = \frac{42}{11} \approx 3.82, \qquad y = \frac{25}{11} \approx 2.27$$

When the coefficients share a factor the LCM is smaller than the product: for $4x + 3y = 10$ and $6x + 5y = 16$ take $L = \mathrm{LCM}(4, 6) = 12$, so $R_2 \leftarrow 2R_2 - 3R_1$ turns $(6, 5 \mid 16)$ into $(0, 1 \mid 2)$, giving $y = 2$ and then $x = 1$, with multipliers 2 and 3 where the product 24 would have needed 4 and 6.

4 / WHEN THE STAIRCASE BREAKS / A ZERO ROW

A row of zeros means infinitely many solutions, and $0 = 1$ means none

system	operation	result	meaning
$6x + 4y = 20,\ 9x + 6y = 30$	$L = 18$: $R_2 \leftarrow 2R_2 - 3R_1$	$\left(\begin{array}{cc\|c} 6 & 4 & 20 \\ 0 & 0 & 0 \end{array}\right)$	the same line
$2x + y = 3,\ 4x + 2y = 7$	$R_2 \leftarrow R_2 - 2R_1$	$\left(\begin{array}{cc\|c} 2 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right)$	parallel, $0 = 1$

5 / THREE UNKNOWNS / THE SAME MOVE, APPLIED TWICE

Clear the first column, then the second, and read z first

$x + y + z = 6$, $2x - y + z = 3$, $x + 2y - z = 2$. The moves $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$ give rows $(0, -3, -1 \mid -9)$ and $(0, 1, -2 \mid -4)$, and then $R_3 \leftarrow 3R_3 + R_2$ gives $(0, 0, -7 \mid -21)$, so $z = 3$; back-substitution gives $-3y - 3 = -9$, so $y = 2$, and then $x = 1$. Each equation is a plane, and the solution $(1, 2, 3)$ is the point where three planes meet.

ASIDE / WHY LCM

The LCM is the smallest multiplier that makes the coefficients equal, so the numbers in the matrix stay as small as possible.

ASIDE / CHECK IN PYTHON

`np.linalg.solve([[3,2],[2,5]],[16,19])` gives $3.818,\ 2.273$.

Convention. The augmented matrix lists coefficients then constants; $L = \mathrm{LCM}(a, d)$ is taken on the column being cleared.

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Chapter 3

Geometry of integers

What this chapter gathers.

Foothills

- Cutting squares from a rectangle: the Euclidean algorithm
- The line ax+by=c
- Solving a quadratic as a line meeting a hyperbola
- Solving intersections: roots as where lines meet curves
- When simultaneous equations give a quadratic
- From the quadratic formula to Ramanujan's constant
- The point at infinity
- Homogeneous coordinates
- Homogenising curves
- Vectors, combinations and the two pictures
- Dot products, length and projection
- Matrices as transformations
- Inverses and elementary matrices
- Determinants: area, volume and invertibility
- Span, basis and the four subspaces
- Eigenvalues and eigenvectors
- Diagonalisation and powers
- Least squares and orthogonal projection

Foothold

- Primes mod 4 and the Gaussian integers
- Sums of two squares and Pythagorean triples
- Fibonacci modulo p and the golden ratio
- Bézout's theorem and multiplicity
- Projective maps, matrices and cross-ratio
- Pappus and Desargues by cross products
- Pythagorean triples from the projective line
- Cubics: chords, tangents and Bézout
- Finite fields, the Fano plane and the whole picture
- A number can have a direction
- Bits and trits in the Gaussian and Eisenstein integers
- A hexagonal Ulam spiral centred on 41
- Complex powers, real answers: from i to the Eisenstein root
- Cyclotomic fields and the arcs of inscribed polygons
- From polygons to the circle
- Cardano's cubic as a hyperbola
- Pythagorean triples on the circle and on the hyperbola
- Spirals in parametric form
- The isoperimetric problem in a prime box
- Two ratios, one best rectangle
- Cuboids: information, size and rank
- Normalisation separates size from shape
- Kirchhoff's laws and squared rectangles
- The period of a stiff spring: Gaussian elimination

Handhold

- Primes mod 3 and the Eisenstein integers
- $E^2 - p^2 = m^2$ as Pythagoras, Gauss and Eisenstein
- Euler's $n^2 + n + 41$ and the Heegner numbers
- Pick's theorem: Gaussian to Eisenstein triangles
- Swapping Eisenstein powers: magnitude, phase and 23
- The cosine rule at 60 and 120 degrees
- The Legendre symbol and quadratic reciprocity
- Triples, circles and ellipses

Draw a rectangle with whole-number side lengths and begin cutting away squares. The cuts reveal a common measuring unit and give Euclid's algorithm a visible form. Now place the lengths on a coordinate grid. The equation $ax + by = c$ describes a line, while requiring x and y to be integers selects particular points. Arithmetic has become a question about where points lie.

Intersections give equations another meaning. Solving a quadratic can locate where a line meets a hyperbola; simultaneous equations can describe a line meeting a circle. A known point on a curve provides a starting place for drawing further lines. Their slopes can organise whole families of solutions, including the rational points that produce Pythagorean triples.

Some constructions invite us to extend the picture. Parallel lines share a direction, which projective geometry records as a point at infinity. Homogeneous coordinates express this extension through ratios that survive a common rescaling. We gain a language for following intersections through changes of viewpoint and for recognising when an apparently exceptional case belongs to a larger pattern.

Vectors and matrices make these changes calculable. Vectors describe movements and combinations; matrices describe transformations. A determinant measures how area or volume changes and tells us whether a linear transformation can be reversed. Eigenvectors can simplify repeated actions. Projection supplies a different tool: when several conditions cannot all be satisfied exactly, least squares finds the closest fit according to a stated measure of error.

Return to the square grid and write a point as $a + bi$. Its squared distance from the origin is $a^2 + b^2$, so complex multiplication connects arithmetic with length and rotation. The factorisation $5 = (2+i)(2-i)$ shows why being prime depends on which numbers we permit as factors. Sums of squares and whole-number triangles now belong to a common investigation.

The meaning of measurement also needs attention. A longer cuboid diagonal may reflect greater size, a different shape, or both. Normalisation helps separate these effects. Lattice area requires us to specify a unit cell, while squared rectangles connect lengths to a network obeying Kirchhoff's laws. Each representation becomes useful when we identify what its quantities measure and preserve.

A triangular grid extends the arithmetic again. Angles of 60° and 120° introduce the cross terms behind Eisenstein norms and new families of triples. Later pages investigate how primes behave in these number systems and how lattice geometry meets quadratic residues and prime-producing expressions. The purpose of learning these languages is to make numerical patterns constructible, comparable and explainable.

EUCLID'S ALGORITHM ON ONE PAGE

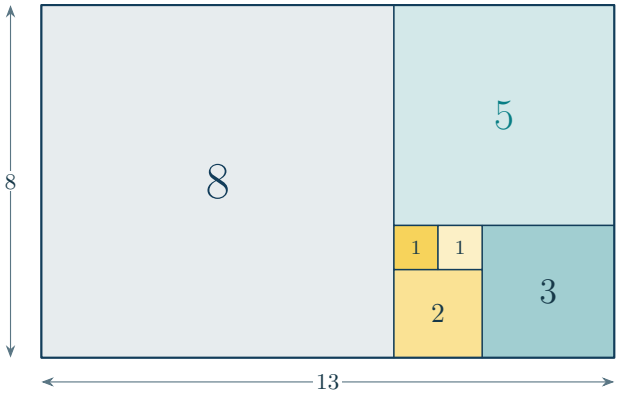
Cutting squares from a rectangle

From 13×8 to common factors, continued fractions and the golden ratio.

1 / THE PICTURE IS A DIVISION

Remove the biggest square, then repeat on the remainder

Begin with a 13×8 rectangle. Cut parallel to its edges and repeat on the rectangle left over.



Square side lengths: $8, 5, 3, 2, 1, 1$.

Division is repeated square cutting.

$$\begin{aligned} 13 &= \mathbf{1} \times 8 + 5, \\ 8 &= \mathbf{1} \times 5 + 3, \\ 5 &= \mathbf{1} \times 3 + 2, \\ 3 &= \mathbf{1} \times 2 + 1, \\ 2 &= \mathbf{2} \times 1 + 0. \end{aligned}$$

The quotients $1, 1, 1, 1, 2$ count the squares at each size: one of each, then two unit squares.

$$\mathrm{HCF}(13, 8) = 1$$

The last non-zero remainder.

2 / WHY THE ALGORITHM WORKS

The common factor survives every cut

Write $\gcd$ for the highest common factor. If $a = qb + r$, any common divisor of a, b divides $r = a - qb$. Conversely, any common divisor of b, r divides $a = qb + r$. Thus

$$\boxed{\gcd(a, b) = \gcd(b, r)}.$$

For our rectangle,

$$\begin{aligned} \gcd(13, 8) &= \gcd(8, 5) = \cdots \\ &= \gcd(1, 0) = 1. \end{aligned}$$

Geometrically, 1 is the largest square side for a regular grid tiling of the whole rectangle. Doubling every length gives a 26×16 rectangle and $\gcd(26, 16) = 2$.

3 / READ THE QUOTIENTS

A continued fraction

Turn the remainder fraction into a reciprocal:

$$\frac{13}{8} = 1 + \frac{5}{8} = 1 + \frac{1}{8/5}.$$

Repeat inside the denominator:

$$\frac{13}{8} = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{2}}}} = [1; 1, 1, 1, 2].$$

The bracket lists the five quotients. Since $2 = 1 + 1/1$, the same number can also be written $[1; 1, 1, 1, 1, 1]$: a finite fraction containing only ones.

4 / RUN THE CONSTRUCTION BACKWARDS

Fibonacci numbers build the rectangle

Read the square sides in reverse, $1, 1, 2, 3, 5, 8, 13, \ldots$, and each new term is the sum of the preceding two:

$$F_{n+1} = F_n + F_{n-1}, \qquad F_1 = F_2 = 1.$$

Starting from a unit square, attach a square along the longer side at each step. A rectangle with sides a, b becomes one with sides $a + b, a$, so its aspect ratio changes by

$$x = \frac{a}{b} \quad \longmapsto \quad 1 + \frac{1}{x}.$$

The successive ratios converge:

$$1,\ 2,\ \frac{3}{2},\ \frac{5}{3},\ \frac{8}{5},\ \frac{13}{8},\ \frac{21}{13}, \ldots \longrightarrow \phi.$$

5 / A SHAPE THAT REPEATS

The golden rectangle

Remove a unit square from an $x \times 1$ rectangle, $1 < x < 2$. The remainder is $1 \times (x - 1)$. For the same shape after a quarter-turn,

$$x = \frac{1}{x - 1} \quad \Longleftrightarrow \quad x = 1 + \frac{1}{x}.$$

$$\phi^2 - \phi - 1 = 0, \qquad \phi = \frac{1 + \sqrt{5}}{2}$$

$$\phi = [1; 1, 1, 1, \ldots] \approx 1.618034$$

Our $13/8 = 1.625$ is a rational approximation. A true golden rectangle repeats the cutting process indefinitely: each remainder is a smaller copy of the original.

Companion: squared rectangles. **Explore:** NRICH: Euclid and continued fractions; NRICH: Fibonacci and the golden ratio.

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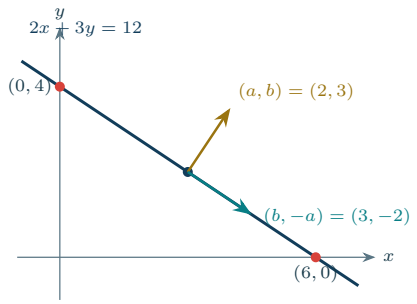
ALGEBRAIC GEOMETRY ON ONE PAGE

The line $ax + by = c$

The set of all points whose coordinates satisfy one linear equation is a straight line, and its coefficients carry the geometry.

1 / THE RULE AND THE PICTURE

An equation $ax + by = c$ picks out exactly the points (x, y) that satisfy it, and they form a line



Take $2x + 3y = 12$. A point such as $(3, 2)$ lies on the line because $2 \cdot 3 + 3 \cdot 2 = 12$, while $(3, 3)$ does not because it gives 15. The numbers a, b and c then read off the geometry:

$$ax + by = c: \quad x\text{-intercept } \tfrac{c}{a}, \quad y\text{-intercept } \tfrac{c}{b}, \quad \text{gradient } -\tfrac{a}{b}$$

The vector (a, b) is perpendicular to the line and $(b, -a)$ points along it, so $2x+3y = 12$ has intercepts 6 and 4, gradient $-\frac{2}{3}$ and normal $(2, 3)$.

2 / SOLUTIONS / ONE EQUATION, INFINITELY MANY POINTS

With one equation in two unknowns, one variable is free and the other follows

x	-3	0	3	6
$y = \dfrac{12 - 2x}{3}$	6	4	2	0
check $2x + 3y$	12	12	12	12

Writing $x = t$ gives the whole line as $(t, \frac{12-2t}{3})$, and rearranging to $y = -\frac{2}{3}x + 4$ is the familiar gradient form with $m = -a/b$ and $k = c/b$. In the language of algebraic geometry, the line is the zero set of the degree one polynomial $ax + by - c$.

3 / SCALING AND SPECIAL CASES / THE SAME LINE IN MANY DRESSES

Multiplying all three numbers by a constant changes the equation and leaves the line unchanged

$4x + 6y = 24$ is $2x + 3y = 12$ doubled, and $-2x - 3y = -12$ is the same line multiplied by -1, so the simplest form divides by the GCD of the coefficients. This is why elimination is allowed to scale an equation before combining it with another.

$c = 0$	line through the origin
$a = 0$	$by = c$, a horizontal line $y = \frac{c}{b}$
$b = 0$	$ax = c$, a vertical line $x = \frac{c}{a}$
$a = b = 0$	no line: all or no points

4 / TWO LINES / CROSSING, PARALLEL OR IDENTICAL

Two equations $ax + by = c$ and $dx + ey = f$ give a crossing, parallel lines or one repeated line



one point | no point | every point

The normals (a, b) and (d, e) are parallel exactly when $ae - bd = 0$. If $ae - bd \neq 0$ the lines cross once, at $x = \dfrac{ce - bf}{ae - bd}$, $y = \dfrac{af - cd}{ae - bd}$. If $ae - bd = 0$ the lines are parallel with no common point, or the same line when the constants agree too, as in $2x + 3y = 12$ and $4x + 6y = 24$.

5 / BUILDING THE EQUATION / THREE ROUTES TO THE SAME LINE

A point with a normal, two intercepts or two points each produce $2x + 3y = 12$

given	working	result
point $(3, 2)$ and normal $(2, 3)$	$2(x - 3) + 3(y - 2) = 0$	$2x + 3y = 12$
intercepts 6 and 4	$\frac{x}{6} + \frac{y}{4} = 1$, times 12	$2x + 3y = 12$
points $(6, 0)$ and $(0, 4)$	$6a = c$ and $4b = c$, so $a : b : c = 2 : 3 : 12$	$2x + 3y = 12$

The first route uses $c = ax_0 + by_0$ for any point (x_0, y_0) on the line, which is also the quickest check that a point lies on it.

6 / THE NORMAL VECTOR / DISTANCE FROM THE ORIGIN

The coefficients also give the distance from the origin to the line

The foot of the perpendicular from the origin lies along (a, b), and the distance is $\dfrac{|c|}{\sqrt{a^2 + b^2}}$; for $2x + 3y = 12$ it is $\dfrac{12}{\sqrt{13}} = 3.33$. A line $y = mx + k$ is at distance $|k|/\sqrt{1 + m^2}$, and a circle of radius r about the origin is touched by the line exactly when this distance equals r.

ASIDE / WHY GEOMETRY

Algebraic geometry begins here, with a polynomial equation in two unknowns defining a curve, and degree one gives the line.

ASIDE / CHECK IN PYTHON

`12/sqrt(13)` gives 3.328, and `2*3+3*2` gives 12.

Convention. Lines are written $ax + by = c$ with $(a, b) \neq (0, 0)$; the gradient is $-a/b$ when $b \neq 0$.

Geometric

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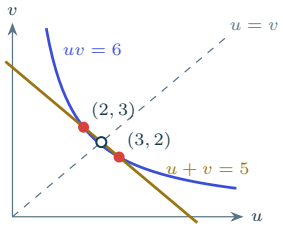
ALGEBRA ON ONE PAGE

Solving a quadratic as a line meeting a hyperbola

Factorising asks for two numbers with a given sum and product, which is where the line $u+v=s$ meets the hyperbola $uv=P$.

1 / THE RULE AND THE PICTURE

Factorising ax^2+bx+c as $a(x-p)(x-q)$ needs $p+q=-\frac{b}{a}$ and $pq=\frac{c}{a}$



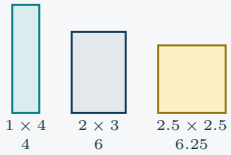
Expanding $(x-p)(x-q)=x^2-(p+q)x+pq$ and matching with $x^2+\frac{b}{a}x+\frac{c}{a}$ turns the quadratic into a pair of simultaneous equations, one linear and one reciprocal:

$$u+v=s=-\tfrac{b}{a}, \qquad uv=P=\tfrac{c}{a}$$

The roots are the coordinates of the points where the line meets the hyperbola, and since swapping u and v changes neither equation the points come in mirror pairs (p,q) and (q,p) across the diagonal $u=v$. For x^2-5x+6 they are $(2,3)$ and $(3,2)$, with the midpoint $(2.5,2.5)$ on the diagonal.

2 / THE AREA PICTURE / SUM AND PRODUCT OF A RECTANGLE

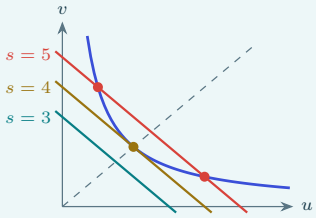
A rectangle with sides p and q has semi-perimeter s and area P, and the square has the most area



With $s=5$ the rectangles 1×4, 2×3 and 2.5×2.5 have areas 4, 6 and 6.25, so no rectangle with semi-perimeter 5 can have area above $6.25=(s/2)^2$, which is the arithmetic and geometric mean inequality. Real roots therefore exist exactly when $P\le s^2/4$, that is $s^2-4P\ge0$, the discriminant b^2-4ac once $a=1$; x^2-5x+7 would need an area of 7 and has none.

3 / SLIDING THE LINE / THE DISCRIMINANT AS A PICTURE

On $uv=4$ the lines $u+v=5,\ 4,\ 3$ cross twice, touch once and miss

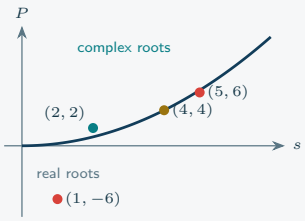


s	quadratic	Δ	picture
5	$x^2-5x+4=(x-1)(x-4)$	9	crossings $(1,4),(4,1)$
4	$x^2-4x+4=(x-2)^2$	0	tangent at $(2,2)$
3	x^2-3x+4	-7	line passes below

The tangent line touches at the square $(2,2)$ on the diagonal. For $s=3$ the midpoint $(1.5,1.5)$ is nearer the origin than the hyperbola's vertex, so the roots $\frac{3\pm i\sqrt{7}}{2}$ are complex.

4 / THE (s,P) PLANE / EVERY QUADRATIC IS A POINT

The parabola $P=s^2/4$ separates real roots below it from complex roots above it



The point $(5,6)$ is x^2-5x+6 with real roots, $(4,4)$ is the double root on the boundary, and $(2,2)$ is x^2-2x+2 with roots $1\pm i$. A negative product, such as $(1,-6)$ for $x^2-x-6=(x-3)(x+2)$, puts the hyperbola in the second and fourth quadrants where the line always crosses it, so the roots are real with opposite signs, at $(3,-2)$ and $(-2,3)$.

5 / THE FORMULA AS A PICTURE / MIDPOINT PLUS OR MINUS THE HALF GAP

The roots sit symmetrically about $s/2=-\frac{b}{2a}$, the vertex of the parabola

$$p,\ q=\frac{s}{2}\pm\sqrt{\frac{s^2}{4}-P}=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

For $2x^2-10x+12=0$ divide by 2 to get $s=5$ and $P=6$, so the roots are 2 and 3. For $x^2-4x+1=0$ we have $s=4$, $P=1$, midpoint 2 and half gap $\sqrt{4-1}=\sqrt{3}$, so the roots are $2\pm\sqrt{3}\approx3.73,\ 0.27$, the points $(2+\sqrt{3},\ 2-\sqrt{3})$ and its mirror image.

ASIDE / WHY SYMMETRIC

The pair (p,q) is unordered, so the equations can only ask for $p+q$ and pq, the two quantities that do not change when p and q swap.

ASIDE / CHECK IN PYTHON

`np.roots([1,-4,1])` gives $3.732,\ 0.268$, and `np.roots([1,-3,4])` gives $1.5\pm1.32j$.

Convention. $s=-b/a$ and $P=c/a$; real roots need $s^2\ge4P$. After the earlier sheets on lines meeting curves.

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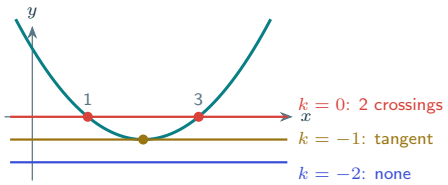
ALGEBRA ON ONE PAGE

Solving intersections: lines meet curves

Every quadratic is a line crossing a curve, and the discriminant decides how and what kind of number results.

1 / THE RULE AND THE PICTURE

Set the curve equal to the line and the crossings are the roots



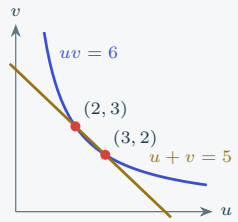
The parabola $y = x^2 - 4x + 3$ meets the line $y = k$ where the two y-values agree. For $y = x^2 + bx + c$ this reads

$$x^2 + bx + (c - k) = 0, \qquad \Delta = b^2 - 4(c - k)$$

so raising the line by 1 raises Δ by 4. Here $\Delta = 4 + 4k$: two crossings at $k = 0$ (roots $1, 3$), one repeated at the vertex $k = -1$ (root 2) and none at $k = -2$.

2 / VIETA AND THE DISCRIMINANT / SUM, PRODUCT AND SEPARATION

Roots as a point where a line meets a hyperbola, and Δ as a distance test

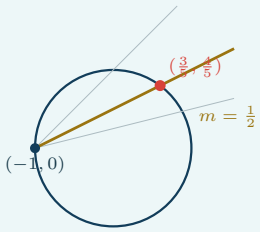


If the roots of $x^2 + Bx + C = 0$ are x_1, x_2 then $x_1 + x_2 = -B$ and $x_1 x_2 = C$, so with $s = -B$ and $p = C$ the roots are the coordinates where the line $u + v = s$ meets the hyperbola $uv = p$, at (x_1, x_2) and (x_2, x_1); $s = 5$, $p = 6$ gives 2 and 3, and a line that misses the hyperbola means complex roots. For $ax^2 + bx + c$ the roots sit at $-b/2a \pm \sqrt{\Delta}/2a$, so Δ measures their separation:

$\Delta > 0$	two real crossings, a perfect square giving rational roots
$\Delta = 0$	one repeated root, the line is tangent
$\Delta < 0$	no real crossing, complex conjugate roots

3 / A KNOWN ROOT / THE CIRCLE COLLAPSES THE QUADRATIC

Lines through $(-1, 0)$ meet the unit circle again at a rational point



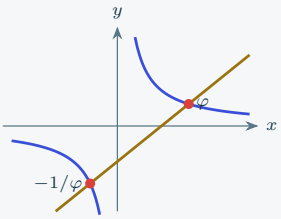
Substituting $y = m(x + 1)$ into $x^2 + y^2 = 1$ gives a quadratic that must contain the factor $(x + 1)$, because $(-1, 0)$ is already on the circle. Dividing it out leaves a linear equation with rational coefficients, so

$$x = \frac{1 - m^2}{1 + m^2}, \qquad y = \frac{2m}{1 + m^2}$$

and $m = \frac{1}{2}$ gives $(\frac{3}{5}, \frac{4}{5})$, the Pythagorean triple $3, 4, 5$ once the denominators are cleared.

4 / NO KNOWN ROOT / THE HYPERBOLA KEEPS THE QUADRATIC

The line $y = x - k$ meets $xy = 1$ where $x^2 - kx - 1 = 0$



No root is known in advance, so the full quadratic survives, $x = \frac{k}{2} \pm \frac{1}{2}\sqrt{k^2 + 4}$. Since $k^2 < k^2 + 4 < (k + 1)^2$ for $k \geq 2$ and 5 is not a square, $\Delta = k^2 + 4$ is never a perfect square and the roots are always irrational:

k	1	2	3	4
$\Delta = k^2 + 4$	5	8	13	20
positive root	golden φ	silver $1 + \sqrt{2}$	bronze $\frac{3+\sqrt{13}}{2}$	$2 + \sqrt{5}$

5 / SUMMARY / WHAT THE GRAPH IS SAYING

A known root gives rational points; no known root gives metallic means

construction	equation	Δ	roots	meaning
parabola at $y = 0$	$x^2 - 4x + 3 = 0$	4	$1, 3$	two rational crossings
same parabola at $y = -1$	$x^2 - 4x + 4 = 0$	0	2 repeated	line tangent at the vertex
circle, $m = \frac{1}{2}$	root -1, then $\frac{3}{5}$	linear	rational	the point $(\frac{3}{5}, \frac{4}{5})$
hyperbola, $k = 1$	$x^2 - x - 1 = 0$	5	$\varphi,\ -1/\varphi$	golden mean and conjugate
hyperbola, $k = 2$	$x^2 - 2x - 1 = 0$	8	$1 \pm \sqrt{2}$	silver mean and conjugate
Vieta, $s = 5$, $p = 6$	$x^2 - 5x + 6 = 0$	1	$2, 3$	$u + v = 5$ meets $uv = 6$

ASIDE / MONIC FORM

Any $ax^2 + bx + c$ divides by a to become monic, so taking $a = 1$ loses nothing and the leading coefficient leaves every formula.

ASIDE / CHECK IN PYTHON

`np.roots([1,-1,-1])` gives $1.618, -0.618$, and `(1-m**2)/(1+m**2)` at `m=0.5` gives 0.6.

Convention. Roots are where the curve and line share a point; $\Delta = b^2 - 4ac$. After the Lines Meet Curves preamble.

Geometric

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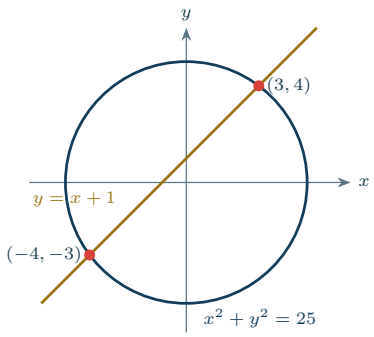
ALGEBRA ON ONE PAGE

When simultaneous equations give a quadratic

Substitute a line into a curve of degree two and the crossings become the roots of a quadratic, counted by the discriminant.

1 / THE RULE AND THE PICTURE

Substitute the linear equation into the curved one, and solve the quadratic that results



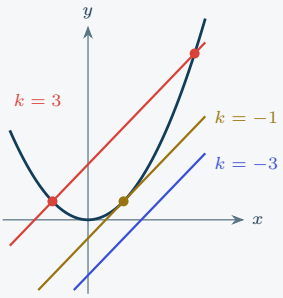
To solve $y = x + 1$ together with $x^2 + y^2 = 25$, replace y in the circle:

$$x^2 + (x+1)^2 = 25 \implies 2x^2 + 2x - 24 = 0 \implies x^2 + x - 12 = (x+4)(x-3) = 0.$$

The roots $x = 3$ and $x = -4$ give $y = 4$ and $y = -3$ from the line, and both points satisfy the circle ($9+16 = 25$ and $16+9 = 25$). The number of crossings is the number of real roots, which the discriminant $\Delta = b^2 - 4ac$ decides: here $\Delta = 1 + 48 = 49$.

2 / LINE AND PARABOLA / A FAMILY OF PARALLEL LINES

The lines $y = 2x + k$ meet $y = x^2$ twice, once or never as k falls



Setting $x^2 = 2x + k$ gives $x^2 - 2x - k = 0$ with $\Delta = 4 + 4k$. At $k = 3$ the roots are $x = -1$ and $x = 3$, so the points are $(-1, 1)$ and $(3, 9)$. At $k = -1$ the equation is $(x-1)^2 = 0$, a double root, so the line touches the parabola at $(1, 1)$ and is the tangent. At $k = -3$ we get $\Delta = -8 < 0$ and no real crossing.

3 / SUM AND PRODUCT / A SYMMETRIC PAIR

The system $x + y = s$, $xy = p$ gives $t^2 - st + p = 0$, whose roots are x and y

Substituting $y = s - x$ into $xy = p$ gives $x(s-x) = p$, that is $x^2 - sx + p = 0$. For $x + y = 5$ and $xy = 6$ this is $x^2 - 5x + 6 = (x-2)(x-3) = 0$, so $(x, y) = (2, 3)$ or $(3, 2)$, and the symmetry of the pair means that both orders appear. When $\Delta = s^2 - 4p < 0$ the line misses the hyperbola: $x + y = 2$ and $xy = 2$ give $x^2 - 2x + 2 = 0$ with roots $1 \pm i$.

4 / WHY A QUADRATIC / COUNTING DEGREES

A linear expression inside a degree two equation leaves degree two in one unknown

curve	after substituting	Δ	crossings
circle $x^2 + y^2 = 25$, line $y = x + 1$	$x^2 + x - 12 = 0$	49	2
parabola $y = x^2$, line $y = 2x - 1$	$x^2 - 2x + 1 = 0$	0	1 (tangent)
parabola $y = x^2$, line $y = 2x - 3$	$x^2 - 2x + 3 = 0$	-8	0
hyperbola $xy = 6$, line $x + y = 5$	$x^2 - 5x + 6 = 0$	1	2

Two lines give a linear equation after substitution, hence one crossing, a line and a conic give a quadratic and so at most two, and in general a line meets a curve of degree n in at most n points. Tangency is the double root.

5 / TANGENT TO A CIRCLE / WHEN DOES THE QUADRATIC HAVE A DOUBLE ROOT

The line $y = mx + k$ touches $x^2 + y^2 = r^2$ exactly when $k^2 = r^2(1 + m^2)$

Substituting gives $(1 + m^2)x^2 + 2mkx + k^2 - r^2 = 0$ with $\Delta = 4\big(r^2(1 + m^2) - k^2\big)$, which vanishes at $k^2 = r^2(1 + m^2)$, the condition that the distance $|k|/\sqrt{1 + m^2}$ from the centre equals r. For $r = 5$ and $m = 1$ the tangents are $y = x \pm 5\sqrt{2} \approx x \pm 7.07$, so the line $k = 1$ cuts the circle, with $\Delta = 4(50 - 1) = 196$ and roots $x = \dfrac{-2 \pm 14}{4} = 3,\ -4$.

ASIDE / COMPLEX ROOTS

A negative discriminant means no real crossing; the complex roots are crossings that the real plane cannot show.

ASIDE / CHECK IN PYTHON

`np.roots([1,1,-12])` gives 3 and -4, and `np.roots([1,-2,2])` gives $1 \pm 1j$.

Convention. Substitute the line into the curve, collect to $ax^2 + bx + c = 0$, and read the crossings from $\Delta = b^2 - 4ac$.

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FROM THE QUADRATIC FORMULA TO RAMANUJAN'S CONSTANT ON ONE PAGE

-163 and the almost-integer

One discriminant explains forty primes in a row and a power of e that misses an integer by less than 10^{-12}.

THE RULE

The discriminant decides the roots

The roots of $ax^2+bx+c=0$ are

$$x \;=\; \frac{-b\pm\sqrt{\Delta}}{2a}, \qquad \Delta \;=\; b^2 \;-\; 4ac$$

and they are complex when $\Delta<0$. Euler's prime-producing polynomial n^2+n+41 has $\Delta = 1-164=-163$, so its roots $n=\frac{1}{2}\left(-1\pm\mathrm{i}\sqrt{163}\right)$ sit far from the real line. A prime q can divide a value only if -163 is a square modulo q, which fails for every prime below 41, and so the values are prime for every n from 0 to 39, failing first at $n=40$, where $40^2+40+41=41^2$.

41	43	47	53	61	71
83	97	113	131	151	173
197	223	251	281	313	347
383	421	461	503	547	593
641	691	743	797	853	911
971	1033	1097	1163	1231	1301
1373	1447	1523	1601	1681	$=41^2$

n^2+n+41 for $n=0$ to 40: forty primes, then the first composite.

ALL NINE HEEGNER NUMBERS

Unique factorisation in $\mathbb{Q}(\sqrt{-d})$

Rabinowitsch proved in 1913 that n^2+n+p is prime for every n from 0 to $p-2$ exactly when the integers of $\mathbb{Q}(\sqrt{1-4p})$ have unique factorisation, or class number one. Gauss conjectured the list of such fields, Heegner proved it complete in 1952 and Baker and Stark confirmed it in 1966–67: $d=1,2,3,7,11,19,43,67,163$, with $d=4p-1$ for the odd ones.

d	3	7	11	19	43	67	163
Euler-type polynomial	n^2+n+1	n^2+n+2	n^2+n+3	n^2+n+5	n^2+n+11	n^2+n+17	n^2+n+41
$-j\left(\frac{1+\sqrt{-d}}{2}\right)$	0	15^3	32^3	96^3	960^3	5280^3	640320^3
$\mathrm{e}^{\pi\sqrt{d}}$ − nearest integer	-0.24	-0.068	0.14	-0.22	-2.2×10^{-4}	-1.3×10^{-6}	-7.5×10^{-13}

The remaining two, $d=1$ and $d=2$, give the Gaussian integers $\mathbb{Z}[\mathrm{i}]$ and $\mathbb{Z}[\sqrt{-2}]$, with $j(\mathrm{i})=12^3$ and $j(\sqrt{-2})=20^3$.

RAMANUJAN'S CONSTANT

Why $\mathrm{e}^{\pi\sqrt{163}}$ is nearly a whole number

$$\mathcal{R} \;=\; \mathrm{e}^{\pi\sqrt{163}} \;=\; 262\,537\,412\,640\,768\,743.999\,999\,999\,999\,250\,072\,597\,198\,185\,688\ldots$$

The j-function satisfies $j(\tau)=q^{-1}+744+196884\,q+\ldots$ with $q=\mathrm{e}^{2\pi\mathrm{i}\tau}$, and at $\tau=\frac{1}{2}\left(1+\mathrm{i}\sqrt{163}\right)$, a root of n^2-n+41, $q=-1/\mathcal{R}$ and class number one forces $j(\tau)$ to be the integer -640320^3. Rearranging,

$$\mathcal{R} = 640320^3+744-\frac{196884}{\mathcal{R}}+\cdots \approx 640320^3+744-7.4993\times10^{-13},$$

which predicts the shortfall to twelve decimal places; smaller Heegner numbers miss by more because their q is less tiny.

POWERS OF $\mathcal{R}$ AND A NUMEROLOGICAL COINCIDENCE

$\mathcal{R}^7$ and the vacuum energy problem

Since $\log_{10}\mathcal{R}=17.4192$, each power of $\mathcal{R}$ adds 17.42 decades, and $\mathcal{R}^7\approx 8.60\times10^{121}$, equal to 640320^{21} within two parts in 10^{14}. The observed cosmological constant in Planck units is $\Lambda\ell_{\mathrm{P}}^2\approx 2.9\times10^{-122}$, whose reciprocal 3.5×10^{121} lies within a factor of 2.5 of $\mathcal{R}^7$, and the mismatch between the quantum vacuum estimate and observation is quoted as 10^{120}, 10^{122} or 10^{123} depending on the cutoff and on factors of 8π.

The coincidence is striking but carries little weight as evidence. A fixed target is landed within 0.39 decades by some power of $\mathcal{R}$ with probability about $2\times0.39/17.42\approx4.5\%$, and the freedom to choose the power, the constant and which of several quoted targets to aim at makes such near misses easy to find; no mechanism links the j-function to the vacuum.

n	$n\log_{10}\mathcal{R}$	$\mathcal{R}^n$
1	17.42	2.63×10^{17}
2	34.84	6.89×10^{34}
3	52.26	1.81×10^{52}
4	69.68	4.75×10^{69}
5	87.10	1.25×10^{87}
6	104.52	3.27×10^{104}
7	121.93	8.60×10^{121}
8	139.35	2.26×10^{139}

Conventions and history. The constant is named for Srinivasa Ramanujan (1887–1920), whose 1914 paper on modular equations and approximations to π studies near-integers of the form $\mathrm{e}^{\pi\sqrt{n}}$; Hermite had observed the 163 case in 1859, and the name was popularised by Martin Gardner's April 1975 hoax claiming $\mathcal{R}$ was an exact integer. The polynomial n^2+n+41 is due to Euler (1772), and j is Klein's modular invariant normalised so that $j(\mathrm{i})=1728$.

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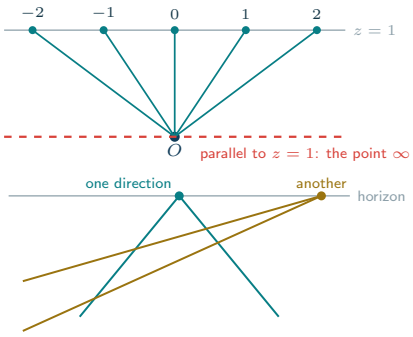
PROJECTIVE GEOMETRY LADDER 1 OF 9

The point at infinity

Parallel lines never meet in the plane, and adding one point at infinity for each direction removes the exception.

1 / THE RULE AND THE PICTURE

Add one point for every direction, and every two lines meet in exactly one point



A point of the projective line is a line through the origin, and each such line meets the line $z = 1$ at one number x, except the one line parallel to it, which supplies the single new point ∞:

$$\mathbb{P}^1(F) = F \cup \{\infty\}$$

In the plane the same idea adds a whole line of new points, one per direction, as railway lines show when they appear to meet on the horizon.

2 / THREE WAYS TO MEET A POINT AT INFINITY

As a limit, as a direction and as a line through the origin

Walk along the line $y = 2x + 1$ and the point $(t, 2t + 1)$ runs off to infinity, but in the coordinates $(t : 2t + 1 : 1) = (1 : 2 + \frac{1}{t} : \frac{1}{t})$ it settles down to $(1 : 2 : 0)$, the point at infinity of that direction.
Every line with slope 2 shares that limit, which is why parallel lines meet there.
The same point is the direction $(1, 2)$ and the line through the origin of three-dimensional space with direction $(1, 2, 0)$.

3 / THE LINE AT INFINITY

One point for each slope, so nothing is left over

The new points $(a : b : 0)$ form a line, because they all satisfy $Z = 0$, and two lines of the plane meet at infinity exactly when they have the same slope.
Two distinct lines now always share exactly one point: the lines $y = 2x + 1$ and $y = 2x + 5$ meet at $(1 : 2 : 0)$, and the lines $y = 2x + 1$ and $y = -x$ meet at the ordinary point $(-\frac{1}{3}, \frac{1}{3})$.

4 / THE PROJECTIVE LINE ON ITS OWN

A circle over the reals, a sphere over the complex numbers

Over the real numbers the line $\mathbb{P}^1(\mathbb{R})$ is a circle, since turning a direction through $180°$ gives the same line and the two ends of the number line join at ∞.
Over the complex numbers it is the Riemann sphere, and over a field with p elements it has $p + 1$ points, the p numbers and ∞.

5 / TRY THESE

Convert between the two descriptions

(a) The triple $(4 : 6 : 2)$ is the point $(2, 3)$, found by dividing by the last entry. **(b)** The triple $(3 : 6 : 0)$ has last entry 0, so it is the direction of slope 2, the same point as $(1 : 2 : 0)$. **(c)**
The triple $(0 : 0 : 0)$ is not a point, since it describes no line through the origin.

ASIDE / WHY BOTHER

Every statement that used to need the exception "unless the lines are parallel" now holds without it, which is what makes the counting theorems on the later sheets possible.

LINKS TO OTHER SHEETS

The next sheet turns the idea into coordinates; the algebra ladder shows the same laws of arithmetic that this geometry rests on.

Convention. F is a field; $\mathbb{P}^1(F)$ is the projective line over F; the point $(a : b : c)$ is the set of non-zero multiples of (a, b, c).

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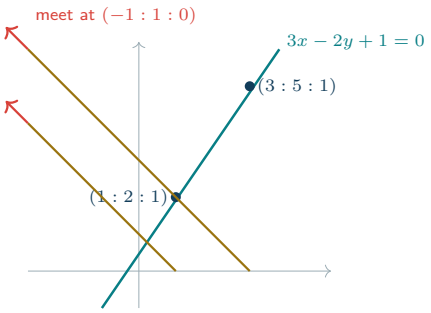
PROJECTIVE GEOMETRY LADDER 2 OF 9

Homogeneous coordinates

A triple up to scale names a point, and one cross product both joins points and meets lines.

1 / THE RULE AND THE PICTURE

Points and lines are both triples up to scale, and incidence is a zero dot product



A point is $(X:Y:Z)$, meaning any non-zero multiple of the triple, and a line is $[a:b:c]$, meaning the equation $aX+bY+cZ=0$, so

$$\text{join } P,Q:\ P\times Q, \qquad \text{meet } \ell,m:\ \ell\times m$$

The same formula serves both jobs, because points and lines enter symmetrically, and it returns a point with $Z=0$ when the lines are parallel.

2 / THE DICTIONARY

Ordinary points, points at infinity and lines

object	coordinates	example
ordinary point (x,y)	$(x:y:1)$, and $(X:Y:Z)$ with $Z\neq 0$ is $(X/Z,\ Y/Z)$	$(4:6:2)=(2:3:1)$ is $(2,3)$
point at infinity	$(a:b:0)$, the direction (a,b)	$(3:6:0)=(1:2:0)$, slope 2
line $ax+by+c=0$	the coefficients $[a:b:c]$, with $aX+bY+cZ=0$	$x+y=1$ is $[1:1:-1]$
line through two points	the cross product $P\times Q$	$(1:2:1)\times(3:5:1)=[-3:2:-1]$
meeting point of two lines	the cross product $\ell\times m$	$[1:1:-1]\times[1:1:-3]=(-2:2:0)=(-1:1:0)$

3 / WHY THE CROSS PRODUCT WORKS

It is the unique triple orthogonal to both inputs

For points P and Q the vector $P\times Q$ is perpendicular to both, so the line $[a:b:c]=P\times Q$ satisfies $aX+bY+cZ=0$ at P and at Q.
The join of $(1,2)$ and $(3,5)$ is $[-3:2:-1]$, which is $3x-2y+1=0$, and indeed $3-4+1=0$ and $9-10+1=0$.
For two intersecting lines the cross product returns an ordinary point, as $[1:1:-1]\times[1:-1:-1]=(-2:0:-2)=(1:0:1)$, the point $(1,0)$ where $x+y=1$ meets $x-y=1$.

4 / COLLINEAR AND CONCURRENT

A determinant detects both

Three points are collinear exactly when the determinant of their coordinate triples is zero, and three lines are concurrent exactly when the determinant of their coefficient triples is zero. This turns incidence statements of geometry into polynomial identities, which is how the theorems of Pappus and Desargues are checked by computer on later sheets.

5 / TRY THESE

Compute with the triples

(a) Find the line through $(0,0)$ and $(2,1)$: $(0:0:1)\times(2:1:1)=(-1,2,0)$, so $[-1:2:0]$, which is $y=x/2$. **(b)** Where do $x=1$ and $x=3$ meet? $[1:0:-1]\times[1:0:-3]=(0,2,0)$, so $(0:1:0)$, the vertical direction. **(c)** Why is $(2:4:6)$ the same point as $(1:2:3)$? They are multiples of one another.

ASIDE / DUALITY

Swap the words point and line in a true statement and it stays true, because the same cross product underlies both.

LINKS TO OTHER SHEETS

The previous sheet explains the point at infinity; the next sheet writes curves in these coordinates.

Convention. $(X:Y:Z)$ is a point; $[a:b:c]$ is a line; $P\times Q$ is the cross product of the coordinate triples; the determinant test for collinearity is $\det(P,Q,R)=0$.

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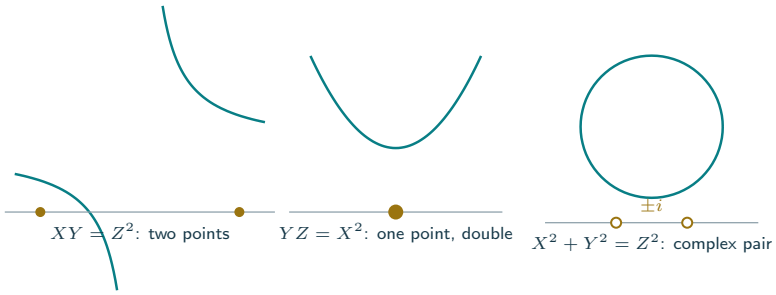
PROJECTIVE GEOMETRY LADDER 3 OF 9

Homogenising curves

Padding terms with powers of Z makes a curve projective and exposes its points at infinity.

1 / THE RULE AND THE PICTURE

A curve of degree d becomes $F = Z^d f(X/Z, Y/Z)$, and $Z = 0$ finds its points at infinity



Replace x by X/Z and y by Y/Z and clear denominators, so that every term has degree d:

$$F(X, Y, Z) = Z^d f\left(\tfrac{X}{Z}, \tfrac{Y}{Z}\right)$$

Setting $Z = 0$ in F finds where the curve meets the line at infinity.

2 / THE STANDARD CURVES

Each curve has its own pattern at infinity

curve	homogeneous equation	points with $Z = 0$
line $y = 2x + 1$	$Y = 2X + Z$	$(1 : 2 : 0)$
parabola $y = x^2$	$YZ = X^2$	$(0 : 1 : 0)$, a double point
hyperbola $xy = 1$	$XY = Z^2$	$(1 : 0 : 0)$ and $(0 : 1 : 0)$
circle $x^2 + y^2 = 1$	$X^2 + Y^2 = Z^2$	$(1 : i : 0)$ and $(1 : -i : 0)$
cubic $y^2 = x^3 - 2$	$Y^2Z = X^3 - 2Z^3$	$(0 : 1 : 0)$, a triple point

3 / ONE CURVE IN THREE DISGUISES

The conics differ only in how they meet the line at infinity

The hyperbola crosses the line at infinity in two points, which are the directions of its asymptotes, the parabola touches it at one point counted twice, and the circle meets it in two complex points $(1 : \pm i : 0)$, the circular points that every circle passes through.
Over the complex numbers a parabola, a hyperbola and a circle are the same curve seen from different positions of the line at infinity, and a projective map can carry one to another.

4 / MULTIPLICITY APPEARS

A root of F at infinity can repeat

For the parabola $Z = 0$ forces $X^2 = 0$, a double root, which records that the line at infinity is tangent to the curve there, and for the cubic $Z = 0$ forces $X^3 = 0$, a triple root, so the point at infinity is a flex.
Repeated roots are how the algebra remembers tangency, and counting them is what makes the intersection counts of the next sheets exact.

5 / TRY THESE

Homogenise and look at infinity

(a) Homogenise $y = x^3$: degree 3 gives $YZ^2 = X^3$, and $Z = 0$ gives $X^3 = 0$, so the point $(0 : 1 : 0)$ with multiplicity 3.
(b) Homogenise $x^2 - y^2 = 1$: $X^2 - Y^2 = Z^2$, with $Z = 0$ giving $(1 : 1 : 0)$ and $(1 : -1 : 0)$, so it is a hyperbola with asymptote slopes ± 1.
(c) Homogenise $x + y = 1$: $X + Y = Z$, a line with the point $(1 : -1 : 0)$ at infinity.

ASIDE / ONE EQUATION, ALL POINTS

The homogeneous equation does not distinguish the plane from the line at infinity, so one polynomial and one count handle all of the curve at once.

LINKS TO OTHER SHEETS

The next sheet counts how many points two such curves share, using exactly these equations.

Convention. d is the degree of f; F is homogeneous of degree d; the circular points are $(1 : \pm i : 0)$.

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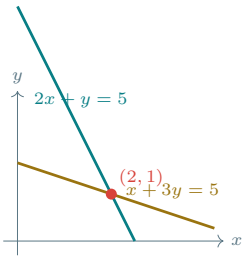
LINEAR ALGEBRA LADDER 1 OF 9

Vectors, combinations and the two pictures

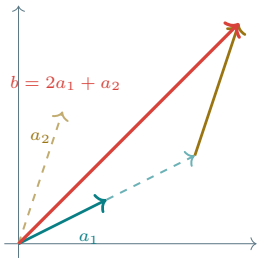
A system of equations is both a pair of lines and a combination of columns.

1 / THE RULE AND THE PICTURE

The system $2x + y = 5$, $x + 3y = 5$ is two lines meeting, and also the vector b built from two columns



row picture



column picture

Reading the equations by rows gives two lines through $(2, 1)$. Reading them by columns asks how much of each column is needed to reach b:

$$x\begin{pmatrix}2\\1\end{pmatrix} + y\begin{pmatrix}1\\3\end{pmatrix} = \begin{pmatrix}5\\5\end{pmatrix}$$

and $x = 2$, $y = 1$ does it: two steps of a_1, one of a_2.

2 / VECTORS AND WHAT WE CAN DO TO THEM

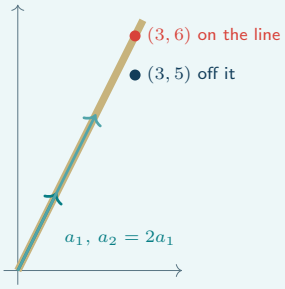
Add vectors, scale them, and combine the two

operation	rule	example with $u = (2, 1)$, $v = (1, 3)$
sum	$u + v$	$(2, 1) + (1, 3) = (3, 4)$
scalar multiple	$3u$	$3(2, 1) = (6, 3)$
linear combination	$2u - v$	$2(2, 1) - (1, 3) = (3, -1)$
length	$\lvert u\rvert = \sqrt{u_1^2 + u_2^2}$	$\lvert(2, 1)\rvert = \sqrt{5}$
unit vector	$u/\lvert u\rvert$	$(2, 1)/\sqrt{5}$

Adding vectors places them head to tail, and a linear combination $c_1u + c_2v$ is every point reachable by stretching each and adding.

3 / SPAN / WHICH RIGHT-HAND SIDES CAN BE REACHED

Independent columns reach every point of the plane, and dependent columns only a line



For $A = \begin{pmatrix}1 & 2\\2 & 4\end{pmatrix}$ the second column is twice the first, so every combination $xa_1 + ya_2 = (x + 2y)(1, 2)$ lies on one line. Then $Ax = (3, 6)$ has infinitely many solutions, such as $x + 2y = 3$, while $Ax = (3, 5)$ has none. The columns of $\begin{pmatrix}2 & 1\\1 & 3\end{pmatrix}$ are independent, with determinant $5 \neq 0$, so every b has exactly one solution.

4 / THE SAME IDEA IN THREE DIMENSIONS

Three equations in three unknowns ask for b as a combination of three columns in space

The span of one non-zero vector is a line, of two independent vectors a plane, of three independent vectors all of space.
Three equations therefore give one solution when the columns span space, none or infinitely many when they collapse to a plane or a line, and the vectors $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$, $\mathbf{k} = (0, 0, 1)$ reach every point as $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

ASIDE / ROWS OR COLUMNS

The row picture shows where the solution is, and the column picture shows what it is made of, which is the one that generalises.

LINKS TO OTHER SHEETS

Elimination on these systems is in *Linear simultaneous equations and the LCM*; the next sheet is *Dot products, length and projection*.

Convention. Vectors are columns; a_1, a_2 are the columns of A; the span is the set of all linear combinations.

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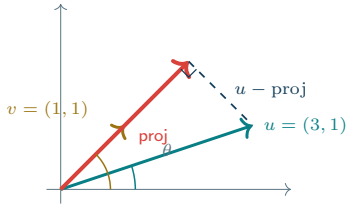
LINEAR ALGEBRA LADDER 2 OF 9

Dot products, length and projection

The dot product measures how far two vectors point the same way, and projection keeps only that part.

1 / THE RULE AND THE PICTURE

The dot product $u \cdot v$ multiplies matching entries and equals $|u||v| \cos\theta$



For $u = (3,1)$ and $v = (1,1)$ we have $u \cdot v = 3 \cdot 1 + 1 \cdot 1 = 4$, while $|u| = \sqrt{10}$ and $|v| = \sqrt{2}$, so

$$\cos\theta = \frac{u \cdot v}{|u||v|} = \frac{4}{\sqrt{10} \cdot \sqrt{2}} = 0.894$$

which gives $\theta = 26.6°$, matching the directions $18.4°$ and $45°$ of the two vectors.

2 / PROJECTION / THE PART OF u ALONG v

Dropping a perpendicular from u onto the line of v gives the projection

$$\text{proj}_v u = \frac{u \cdot v}{v \cdot v} v = \frac{4}{2}(1,1) = (2,2)$$

The remainder $u - \text{proj}_v u = (1,-1)$ is perpendicular to v, since $(1,-1) \cdot (1,1) = 0$, so u splits into a part along v and a part across it.

The length of the projection is $|u| \cos\theta = \sqrt{10} \cdot 0.894 = 2.83 = 2\sqrt{2}$, which is the length of $(2,2)$, and a second case $u = (4,0)$ onto $v = (1,1)$ gives $\frac{4}{2}(1,1) = (2,2)$ as well, with the right-angled triangle of sides $2\sqrt{2}$, $2\sqrt{2}$, 4.

3 / ORTHOGONALITY AND THE RULES

Perpendicular vectors have dot product zero, and lengths obey Pythagoras

rule	statement	example
perpendicular	$u \cdot v = 0$	$(1,2) \cdot (2,-1) = 0$
Pythagoras	$\lvert u+v\rvert^2 = \lvert u\rvert^2 + \lvert v\rvert^2 + 2u \cdot v$	$\lvert(4,2)\rvert^2 = 20 = 10 + 2 + 8$
Cauchy and Schwarz	$\lvert u \cdot v\rvert \le \lvert u\rvert\lvert v\rvert$	$4 \le \sqrt{10} \cdot \sqrt{2} = 4.47$
triangle inequality	$\lvert u+v\rvert \le \lvert u\rvert + \lvert v\rvert$	$4.47 \le 3.16 + 1.41 = 4.57$

For perpendicular vectors the cross term $2u \cdot v$ vanishes, which is Pythagoras' theorem, and $\cos\theta \le 1$ is the Cauchy and Schwarz inequality.

4 / THE PROJECTION MATRIX / AND THE NORMAL OF A LINE

Projection onto v is a matrix, and the line $ax + by = c$ has normal (a,b)

Writing the projection as $P = \dfrac{vv^T}{v^T v} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ for $v = (1,1)$, we have $Pu = (2,2)$ for $u = (3,1)$ and $P^2 = P$, since projecting twice changes nothing.

The equation $ax + by = c$ says $(a,b) \cdot (x,y) = c$, so every point on the line has the same component $c/|(a,b)|$ along the normal (a,b), which is why that vector is perpendicular to the line and why the distance from the origin is $c/\sqrt{a^2+b^2}$.

ASIDE / WORK AND ANGLES

Work is force dot displacement, $W = F \cdot d$, so a force perpendicular to the motion does none.

LINKS TO OTHER SHEETS

The normal vector is developed in *The line $ax + by = c$*; the previous sheet is *Vectors, combinations and the two pictures*, and the next is *Matrices as transformations*.

Convention. $u \cdot v = \sum u_i v_i$; $|u| = \sqrt{u \cdot u}$; θ is the angle between u and v.

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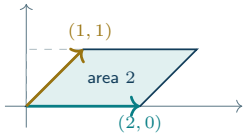
LINEAR ALGEBRA LADDER 3 OF 9

Matrices as transformations

A matrix moves the plane, its columns are where the basis vectors land, and its determinant is the area scale.

1 / THE RULE AND THE PICTURE

The columns of A are the images of $(1,0)$ and $(0,1)$, and everything else follows by combination



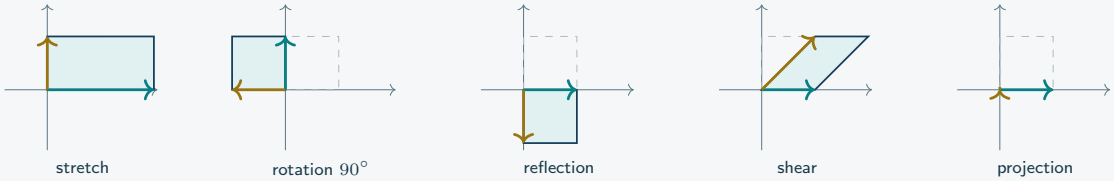
For $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ the unit square is carried to a parallelogram, because $A(1,0) = (2,0)$ and $A(0,1) = (1,1)$ are its two columns. A general point follows from linearity:

$$A\begin{pmatrix} x \\ y \end{pmatrix} = x\begin{pmatrix} 2 \\ 0 \end{pmatrix} + y\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

The area of the parallelogram is 2, which is $\det A = 2 \cdot 1 - 1 \cdot 0$.

2 / A GALLERY OF 2×2 MAPS

Stretching, turning, flipping, shearing and flattening are five different matrices



stretch $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ | rotation $90°$ $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ | reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ | shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ | projection $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

In each picture the dashed unit square becomes the shaded shape, with $(1,0)$ in teal and $(0,1)$ in gold. The determinants are $2,\ 1,\ -1,\ 1,\ 0$: the reflection reverses orientation and the projection flattens the square to a segment of area zero.

3 / COMPOSITION / PRODUCTS OF MATRICES

Doing one map after another multiplies the matrices, and the order matters



SR: rotate, then shear



RS: shear, then rotate

With the shear $S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and the rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$,

$$SR = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad RS = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix},$$

so rotating first sends $(1,0)$ to $(1,1)$, while shearing first sends it to $(0,1)$. The determinants agree, $\det SR = \det RS = 1$, since $\det(AB) = \det A \det B$.

4 / SUMMARY / WHAT THE DETERMINANT SAYS

The determinant is the signed area scale, and zero means the map flattens the plane

map	det	effect on area	inverse
stretch $\mathrm{diag}(2,1)$	2	doubles areas	invertible
rotation, shear	1	keeps areas	invertible
reflection	-1	keeps areas, flips orientation	invertible
projection	0	flattens to a line	not invertible

An invertible matrix can be undone, and the inverse sheet shows how, while a matrix with $\det = 0$ loses a direction for ever.

ASIDE / ROWS AND COLUMNS

Multiplying A by a vector takes combinations of the columns, while each entry of Ax is a row dotted with x.

LINKS TO OTHER SHEETS

The area scale is developed in *The determinant in its many guises*; the previous sheet is *Dot products, length and projection* and the next is *Inverses and elementary matrices*.

Convention. A acts on column vectors; $(1,0)$ and $(0,1)$ are the standard basis; $\det$ is the signed area scale factor.

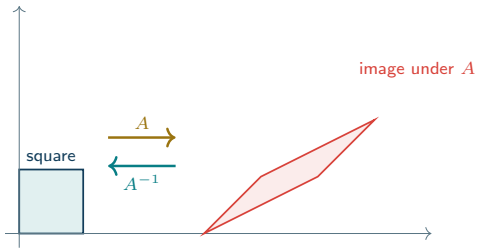
LINEAR ALGEBRA LADDER 4 OF 9

Inverses and elementary matrices

An inverse undoes a matrix, and the row operations of elimination are themselves matrices.

1 / THE RULE AND THE PICTURE

A^{-1} is the matrix that undoes A, so $A^{-1}A = AA^{-1} = I$



For $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ with $\det A = 2 - 1 = 1$ the inverse swaps the diagonal, negates the other entries and divides by the determinant:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

so $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, and an invertible map is reversed by its inverse.

2 / SOLVING WITH THE INVERSE

$Ax = b$ has the solution $x = A^{-1}b$ whenever A^{-1} exists

For $b = (5, 3)$ we get $x = A^{-1}b = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, and indeed $2 \cdot 2 + 1 = 5$ and $2 + 1 = 3$.

The same inverse solves every right-hand side at once, which is why it is worth computing when many b occur, although elimination is cheaper for a single system.

3 / ELEMENTARY MATRICES / ELIMINATION AS MULTIPLICATION

Each row operation is multiplication by an elementary matrix, and the product of them is A^{-1}

Reducing $[A \mid I]$ to $[I \mid A^{-1}]$ takes two steps:

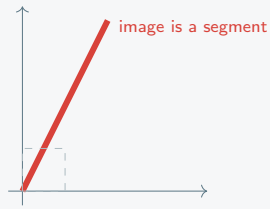
$$R_1 \leftarrow R_1 - R_2 : \; E = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \; EA = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$R_2 \leftarrow R_2 - R_1 : \; F = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \; FEA = I$$

Hence $A^{-1} = FE = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$, found by the same row operations applied to the identity beside A.

4 / WHEN THERE IS NO INVERSE / SINGULAR MATRICES

A matrix with $\det = 0$ squashes the plane and cannot be undone



For $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ the determinant is $4 - 4 = 0$ and the whole plane lands on the line through $(1, 2)$, so many points share each image and the map cannot be reversed.
Elimination shows the same thing as a zero row, and $Ax = b$ then has no solution or infinitely many.
For products the order reverses, $(AB)^{-1} = B^{-1}A^{-1}$, as in taking off shoes before socks.

ASIDE / INVERSE OF A PRODUCT

Undoing two steps means undoing the last one first, which is why $(AB)^{-1} = B^{-1}A^{-1}$.

LINKS TO OTHER SHEETS

Elimination is in *The LCM method as Gaussian elimination*; the previous sheet is *Matrices as transformations* and the next is *Determinants: area, volume and invertibility*.

Convention. I is the identity matrix; E, F are elementary matrices; $[A \mid I]$ is A augmented by the identity.

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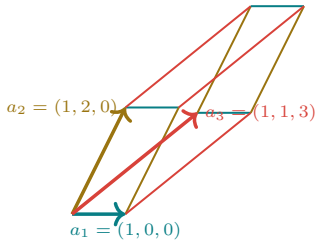
LINEAR ALGEBRA LADDER 5 OF 9

Determinants: area, volume and invertibility

The determinant measures how a matrix scales area or volume, and it vanishes exactly when the matrix flattens space.

1 / THE RULE AND THE PICTURE

The determinant of a 3×3 matrix is the volume of the box spanned by its columns



For the columns $a_1 = (1, 0, 0)$, $a_2 = (1, 2, 0)$, $a_3 = (1, 1, 3)$ the matrix is triangular with diagonal $1, 2, 3$, so the box has volume

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{pmatrix} = 1 \cdot 2 \cdot 3 = 6$$

In two dimensions the same idea gives area, $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$, and the sign records whether orientation is kept or reversed.

2 / COMPUTING A 3×3 DETERMINANT / TWO WAYS

Expand along a row with signs $+, -, +$, or use the rule of Sarrus

For $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}$, expanding along the first row,

$$\det A = 2 \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} - 0 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 2 \cdot 1 - 0 + 1 \cdot 1 = 3.$$

The rule of Sarrus adds the three down-right diagonals and subtracts the three down-left ones: $(2{\cdot}3{\cdot}1 + 0{\cdot}2{\cdot}0 + 1{\cdot}1{\cdot}1) - (1{\cdot}3{\cdot}0 + 2{\cdot}2{\cdot}1 + 0{\cdot}1{\cdot}1) = 7 - 4 = 3$.

3 / THE RULES THAT MAKE IT WORK

Row operations change the determinant in a predictable way

operation on $A = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}$	effect on $\det$	example
swap two rows	sign changes	-3
multiply a row by k	determinant times k	6 for $k = 2$
add a multiple of one row to another	unchanged	3
two rows proportional	0	the $1 \ldots 9$ matrix gives 0
product	$\det(AB) = \det A \det B$	$\det A^2 = 9 = 3^2$
triangular matrix	product of the diagonal	$1 \cdot 2 \cdot 3 = 6$

These rules explain why elimination computes determinants: reduce to a triangular matrix, track the sign of each swap and the factor of each scaling, and multiply the diagonal.

4 / CRAMER'S RULE / RATIOS OF AREAS

Each unknown is a ratio of two determinants

For $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$ we have $\det A = 5$, and replacing a column by b gives

$$x = \frac{\begin{vmatrix} 5 & 1 \\ 5 & 3 \end{vmatrix}}{5} = \frac{10}{5} = 2, \qquad y = \frac{\begin{vmatrix} 2 & 5 \\ 1 & 5 \end{vmatrix}}{5} = \frac{5}{5} = 1,$$

because the parallelogram spanned by b and one column is x or y times the parallelogram of the two columns, and the formula needs $\det A \neq 0$.

5 / ZERO DETERMINANT / FLATTENING

$\det A = 0$ exactly when the columns are dependent, and then A has no inverse

The matrix $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$ has $\det = 0$ because the third row is $2 \cdot (4, 5, 6) - (1, 2, 3)$, so the box collapses to a plane and every image lies in it. Then $Ax = b$ has no solution or infinitely many, and an invertible matrix is exactly one whose determinant is non-zero.

ASIDE / SHEARS KEEP VOLUME

Adding a multiple of one row to another slides the box sideways without changing its volume, which is the rule used in elimination.

LINKS TO OTHER SHEETS

The guises of the determinant are in *The determinant in its many guises*; the previous sheet is *Inverses and elementary matrices* and the next is *Span, basis and the four subspaces*.

Convention. $\det A$ is the signed volume scale; $|\ |$ around a matrix means its determinant.

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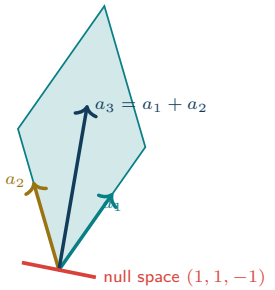
LINEAR ALGEBRA LADDER 6 OF 9

Span, basis and the four subspaces

The columns of a matrix span a space, the vectors sent to zero form another, and their dimensions add to the number of columns.

1 / THE RULE AND THE PICTURE

The columns of A fill a plane, and a line of vectors is sent to zero



column space: the plane $z = x + y$

For $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}$ the third column is the sum of the first two, so the columns span only the plane $z = x + y$, and $A(1, 1, -1) = a_1 + a_2 - a_3 = 0$ puts the line through $(1, 1, -1)$ in the null space:

$$\boxed{\text{rank} + \text{nullity} = n : \quad 2 + 1 = 3}$$

The rank is the dimension of the column space and the nullity that of the null space.

2 / FINDING THEM / BY ELIMINATION

Row reduction exposes the pivot columns and the free variable

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 - R_1} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The pivot columns 1 and 2 of the original matrix, a_1 and a_2, are a basis of the column space.
Column 3 has no pivot, so $x_3 = t$ is free and the equations $x_1 + x_3 = 0$, $x_2 + x_3 = 0$ give $x = t(-1, -1, 1)$, the null space.

3 / INDEPENDENCE, BASIS AND DIMENSION

Vectors are independent when the only combination giving zero is the trivial one

The columns a_1, a_2, a_3 are dependent because $a_1 + a_2 - a_3 = 0$, while a_1, a_2 alone are independent, since $xa_1 + ya_2 = (x, y, x + y) = 0$ forces $x = y = 0$. A basis is an independent set that spans, here $\{a_1, a_2\}$ for the plane, and every basis of a space has the same number of vectors, its dimension, which is 2 for the plane and 1 for the null line.

4 / THE FOUR SUBSPACES

Each matrix with m rows and n columns carries four spaces whose dimensions come from the rank r

space	definition	lives in	dimension	here
column space	combinations of the columns	$\mathbb{R}^m$	r	plane $z = x + y$
null space	solutions of $Ax = 0$	$\mathbb{R}^n$	$n - r$	line through $(1, 1, -1)$
row space	combinations of the rows	$\mathbb{R}^n$	r	plane $x + y - z = 0$
left null space	solutions of $A^T y = 0$	$\mathbb{R}^m$	$m - r$	line through $(1, 1, -1)$

The null space is perpendicular to the row space, since each row dotted with $(1, 1, -1)$ gives 0, and the left null space is perpendicular to the column space; this matrix is symmetric, so the row and column spaces coincide.

5 / SOLVABILITY / WHEN $Ax = b$ HAS AN ANSWER

$Ax = b$ is solvable exactly when b lies in the column space

For $b = (1, 2, 3)$ we have $1 + 2 = 3$, so b is on the plane and $x = (1, 2, 0)$ works, with the general solution $x = (1, 2, 0) + t(1, 1, -1)$.
For $b = (1, 2, 4)$ the plane equation fails and there is no solution. The pattern is a particular solution plus anything from the null space.

ASIDE / FREE AND PIVOT

A column without a pivot is a free variable, and the number of free variables is the nullity.

LINKS TO OTHER SHEETS

The vectors in space come from *Vectors, combinations and the two pictures*; the previous sheet is *Determinants: area, volume and invertibility* and the next is *Eigenvalues and eigenvectors*.

Convention. $\mathbb{R}^m$ is m-dimensional space; r is the rank; A^T is the transpose; a basis is an independent spanning set.

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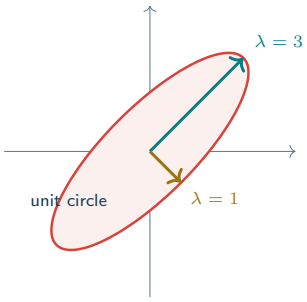
LINEAR ALGEBRA LADDER 7 OF 9

Eigenvalues and eigenvectors

An eigenvector keeps its direction under a matrix and is only stretched, by a factor called the eigenvalue.

1 / THE RULE AND THE PICTURE

A matrix turns the unit circle into an ellipse whose axes are its eigenvectors



For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ the circle becomes an ellipse with semi-axes 3 along $(1,1)$ and 1 along $(1,-1)$, because

$$Av = \lambda v: \quad A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad A\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Along these two directions the matrix acts as plain multiplication by 3 and by 1.

2 / FINDING THEM / THE CHARACTERISTIC EQUATION

The eigenvalues are the roots of $\det(A - \lambda I) = 0$, a quadratic in λ

Here $\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = (1-\lambda)(3-\lambda)$, a difference of squares, so $\lambda = 1$ and $\lambda = 3$. Expanding, $\lambda^2 - 4\lambda + 3 = 0$, whose coefficients are the trace $2 + 2 = 4$ and the determinant $4 - 1 = 3$: the eigenvalues add to the trace and multiply to the determinant.
Each eigenvector solves $(A - \lambda I)v = 0$, for example $\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} v = 0$ gives $v = (1,1)$ at $\lambda = 3$.

3 / A GALLERY / WHAT EIGENVALUES LOOK LIKE

Some matrices have two real eigen-directions, some one, and some none

matrix	trace	det	eigenvalues	eigenvectors
$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	4	3	$3,\ 1$	$(1,1),\ (1,-1)$
stretch $\mathrm{diag}(2,1)$	3	2	$2,\ 1$	axes
projection $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$	1	0	$1,\ 0$	x-axis, y-axis
reflection $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	0	-1	$1,\ -1$	axes
shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	2	1	$1,\ 1$	only $(1,0)$
rotation $90°$	0	1	$\pm i$	none real

The shear has the repeated eigenvalue 1 but only one direction, so it cannot be diagonalised, and the rotation turns every direction, so its eigenvalues $\pm i$ are complex.

4 / POWERS / WHY THEY MATTER

Repeated application multiplies each eigen-direction by λ^n

Since $A\,v = \lambda v$ we have $A^n v = \lambda^n v$, so $A^2(1,1) = 9(1,1)$. Any vector splits along the eigenvectors, $(3,1) = 2(1,1) + 1(1,-1)$, and then $A^n(3,1) = 2 \cdot 3^n(1,1) + 1 \cdot 1^n(1,-1)$; for $n = 1$ this is $(6+1,\ 6-1) = (7,5) = A(3,1)$.
After many steps the direction with the largest $|\lambda|$ dominates, which is why the iteration settles along $(1,1)$.

ASIDE / THE QUADRATIC AGAIN

The two eigenvalues are where the line $u + v = \mathrm{trace}$ meets the hyperbola $uv = \det$, the picture for any quadratic.

LINKS TO OTHER SHEETS

The characteristic equation is developed in *The characteristic equation beneath five presentations*; the previous sheet is *Span, basis and the four subspaces* and the next is *Diagonalisation and powers*.

Convention. λ is an eigenvalue and $v \neq 0$ an eigenvector; trace is the sum of the diagonal entries.

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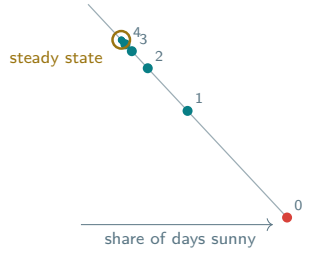
LINEAR ALGEBRA LADDER 8 OF 9

Diagonalisation and powers

Writing $A = PDP^{-1}$ turns repeated multiplication into raising the eigenvalues to a power.

1 / THE RULE AND THE PICTURE

In the eigenvector basis A is diagonal, so $A^n = PD^nP^{-1}$



a weather chain settling to its steady state

If A has n independent eigenvectors, put them in the columns of P and the eigenvalues in $D = \operatorname{diag}(\lambda_i)$. Then $AP = PD$, so

$$A = PDP^{-1}, \qquad A^n = P\,D^n\,P^{-1}$$

because the $P^{-1}P$ factors cancel in the middle of the product, and D^n just raises each diagonal entry to the nth power.

2 / A WORKED EXAMPLE / $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

With eigenvalues $3, 1$ and eigenvectors $(1,1), (1,-1)$ the powers have a closed form

Taking $P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $D = \operatorname{diag}(3,1)$ and $P^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$,

$$A^n = \frac{1}{2}\begin{pmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{pmatrix}.$$

For $n = 2$ this is $\frac{1}{2}\begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = A^2$, and for $n = 0$ it is the identity, so one formula covers every power.

3 / FIBONACCI / POWERS OF ONE MATRIX

The Fibonacci matrix Q has eigenvalues φ and $-1/\varphi$

For $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ the characteristic equation is $\lambda^2 - \lambda - 1 = 0$, with roots $\varphi = 1.618$ and $\psi = -0.618$, and diagonalising gives

$$Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}, \qquad F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}.$$

For $n = 10$ the entries are $89, 55, 55, 34$, and the ratio of neighbours tends to the larger eigenvalue φ, since $\psi^n \to 0$.

4 / A MARKOV CHAIN / STEADY STATES

A column of probabilities that sums to 1 always has the eigenvalue 1

Each day is sunny after a sunny day with probability 0.9 and after a rainy day with probability 0.5, so $M = \begin{pmatrix} 0.9 & 0.5 \\ 0.1 & 0.5 \end{pmatrix}$ has eigenvalues 1 and 0.4.
The eigenvector for 1 is $(5,1)/6$, so the chain settles to $\frac{5}{6}$ sunny, and starting from a sunny day the share after n days is $\frac{5}{6} + \frac{1}{6}(0.4)^n$, giving $0.9,\ 0.86,\ 0.844, \ldots$ in the picture, with the gap shrinking by the factor 0.4 each step.

5 / WHEN IT WORKS

Diagonalisation needs enough independent eigenvectors

A matrix with n different eigenvalues always has n independent eigenvectors. A repeated eigenvalue may fail, as for the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ with only one direction, and a rotation needs complex numbers. Symmetric matrices such as the first example always diagonalise with perpendicular eigenvectors, which is the spectral theorem.

ASIDE / FASTER THAN MULTIPLYING

Computing A^{1000} by multiplying takes a thousand products, while $PD^{1000}P^{-1}$ needs only two.

LINKS TO OTHER SHEETS

The Fibonacci matrix is also in *Why i makes Fibonacci inevitable*; the previous sheet is *Eigenvalues and eigenvectors* and the last is *Least squares and orthogonal projection*.

Convention. P holds eigenvectors as columns, D the eigenvalues on the diagonal; $\varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$.

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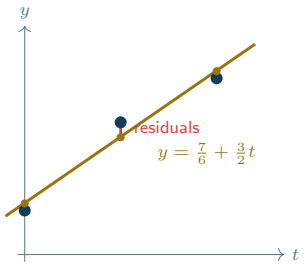
LINEAR ALGEBRA LADDER 9 OF 9

Least squares and orthogonal projection

When $Ax = b$ has no solution, the best answer projects b onto the column space and leaves a perpendicular error.

1 / THE RULE AND THE PICTURE

The best-fit line makes the squared vertical errors as small as possible



Three points $(0,1)$, $(1,3)$, $(2,4)$ do not lie on a line, so $\begin{pmatrix}1&0\\1&1\\1&2\end{pmatrix}\begin{pmatrix}c\\m\end{pmatrix} = \begin{pmatrix}1\\3\\4\end{pmatrix}$ has no solution.

The least squares answer $\hat{x}$ minimises $|b - Ax|^2$ and satisfies the normal equations

$$\boxed{A^TA\hat{x} = A^Tb}$$

which give the line $y = \frac{7}{6} + \frac{3}{2}t$.

2 / SOLVING IT

The two-by-two normal equations have a unique answer

Here $A^TA = \begin{pmatrix}3&3\\3&5\end{pmatrix}$ and $A^Tb = \begin{pmatrix}8\\11\end{pmatrix}$, with $\det A^TA = 15 - 9 = 6$, so

$$\hat{x} = \frac{1}{6}\begin{pmatrix}5&-3\\-3&3\end{pmatrix}\begin{pmatrix}8\\11\end{pmatrix} = \frac{1}{6}\begin{pmatrix}7\\9\end{pmatrix} = \begin{pmatrix}7/6\\3/2\end{pmatrix}.$$

The fitted values are $A\hat{x} = (\frac{7}{6},\ \frac{8}{3},\ \frac{25}{6})$ and the errors $e = b - A\hat{x} = (-\frac{1}{6},\ \frac{1}{3},\ -\frac{1}{6})$, with squared length $\frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6}$.

3 / THE GEOMETRY / PROJECTION ONTO THE COLUMN SPACE

The error is perpendicular to every column, which is exactly what the normal equations say

Writing $A^T(b - A\hat{x}) = 0$ says the error e is perpendicular to each column of A: dotting with the ones column gives $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$ and with $(0,1,2)$ gives $\frac{1}{3} - \frac{1}{3} = 0$.

The fitted vector $A\hat{x}$ is therefore the projection of b onto the column space.

$$P = A(A^TA)^{-1}A^T = \frac{1}{6}\begin{pmatrix}5&2&-1\\2&2&2\\-1&2&5\end{pmatrix}$$

has $P^2 = P$ and $P^T = P$, and $Pb = A\hat{x}$.

4 / WHY THE RESIDUALS SUM TO ZERO

A column of ones among the columns forces it

Because the first column of A is all ones, the equation $A^Te = 0$ contains $\sum e_i = 0$, so the best-fit line passes through the average point: the means are $\bar{t} = 1$ and $\bar{y} = \frac{8}{3}$, and indeed $\frac{7}{6} + \frac{3}{2}\cdot 1 = \frac{8}{3}$.

The matrix A^TA is symmetric with eigenvalues $4 \pm \sqrt{10}$, both positive, which is why it is invertible whenever the columns of A are independent.

ASIDE / ONE IDEA THROUGHOUT

Projection onto a line from the dot product sheet is this problem with one column.

LINKS TO OTHER SHEETS

The projection formula is in *Dot products, length and projection*; the symmetric eigenvalues are in *Eigenvalues and eigenvectors*; the previous sheet is *Diagonalisation and powers*.

Convention. $\hat{x}$ is the least squares solution, $e = b - A\hat{x}$ the error, P the projection onto the column space of A.

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NUMBER THEORY LADDER 6 OF 9

Primes mod 4 and the Gaussian integers

A prime is a sum of two squares exactly when it is 1 mod 4, which is exactly when it splits in the Gaussian integers.

1 / THE RULE AND THE PICTURE

Odd primes fall into two classes, $1 \bmod 4$ and $3 \bmod 4$, and only the first are sums of two squares

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60
61	62	63	64	65	66	**67**	68	69	70
71	72	**73**	74	75	76	77	78	**79**	80
81	82	**83**	84	85	86	87	88	**89**	90
91	92	93	94	95	96	**97**	98	99	100

1 mod 4 (11) · 3 mod 4 (13) · 2

In the Gaussian integers $\mathbb{Z}[i] = \{a + bi\}$ the norm is $N(a + bi) = a^2 + b^2 = (a + bi)(a - bi)$, so a prime that is a sum of two squares factors:

$$\boxed{p = a^2 + b^2 = (a + bi)(a - bi)}$$

The eleven primes up to 100 that are $1 \bmod 4$ are the ones that factor, the thirteen that are $3 \bmod 4$ stay prime, and $2 = -i(1 + i)^2$ is the single ramified prime.

2 / SPLIT, INERT AND RAMIFIED

Every prime of $\mathbb{Z}$ behaves in one of three ways in $\mathbb{Z}[i]$

type	condition	examples	sum of squares
split	$p \equiv 1 \pmod 4$	$5 = (2 + i)(2 - i)$, $13 = (3 + 2i)(3 - 2i)$	$p = a^2 + b^2$
inert	$p \equiv 3 \pmod 4$	$3, 7, 11, 19$ stay prime	no representation
ramified	$p = 2$	$2 = -i(1 + i)^2$	$2 = 1^2 + 1^2$

The primes $1 \bmod 4$ up to 100 are $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $17 = 1^2 + 4^2$, $29 = 2^2 + 5^2$, $37 = 1^2 + 6^2$, $41 = 4^2 + 5^2$, $53 = 2^2 + 7^2$, $61 = 5^2 + 6^2$, $73 = 3^2 + 8^2$, $89 = 5^2 + 8^2$, $97 = 4^2 + 9^2$.

3 / WHY $1 \bmod 4$ / THE EQUATION $x^2 \equiv -1$

A prime splits exactly when -1 is a square modulo p

p	$x^2 \equiv -1$
5	$2^2 = 4$
13	$5^2 = 25$
17	$4^2 = 16$
29	$12^2 = 144$
7	squares $1, 2, 4$, no
11	squares $1, 3, 4, 5, 9$, no

If $x^2 \equiv -1$ then x has order 4 in the units modulo p, so $4 \mid p - 1$. Conversely for $p \equiv 1 \pmod 4$ the criterion $(-1)^{(p-1)/2} = 1$ makes -1 a square. Then $p \mid x^2 + 1 = (x + i)(x - i)$ while p divides neither factor, so p is not prime in $\mathbb{Z}[i]$, where factorisation is unique, and it splits.

4 / WHICH NUMBERS ARE SUMS OF TWO SQUARES

Primes $3 \bmod 4$ must appear to an even power

A number is a sum of two squares exactly when every prime $\equiv 3 \pmod 4$ in it has an even exponent: $45 = 3^2 \cdot 5 = 6^2 + 3^2$ and $50 = 2 \cdot 5^2 = 7^2 + 1^2 = 5^2 + 5^2$ qualify, while $21 = 3 \cdot 7$ and $63 = 3^2 \cdot 7$ do not. The norm is multiplicative, which gives the identity

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2,$$

so $5 \cdot 13 = 65 = (1 \cdot 2 - 2 \cdot 3)^2 + (1 \cdot 3 + 2 \cdot 2)^2 = 4^2 + 7^2$, and the conjugate choice gives $65 = 8^2 + 1^2$.

5 / THE RACE BETWEEN THE CLASSES

Primes $3 \bmod 4$ lead most of the way, though not for ever

	$1 \bmod 4$	$3 \bmod 4$	
up to 100	11	13	$3 \bmod 4$ ahead
up to 1000	80	87	$3 \bmod 4$ ahead
up to $10\,000$	609	619	$3 \bmod 4$ ahead

Each class holds half of the primes in the long run, yet the lead belongs to $3 \bmod 4$ almost always, and the first prime at which the $1 \bmod 4$ count overtakes is $26\,861$.

ASIDE / THE UNITS

The four units $1, i, -1, -i$ are the vertices of the square, so each prime has four associates, $a + bi$ and its rotations.

LINKS TO OTHER SHEETS

The order argument uses *Fermat, Euler and repeating decimals*; the Legendre symbol is in *The Legendre symbol and quadratic reciprocity*; the next sheet is *Sums of two squares and Pythagorean triples*.

Convention. $N(a + bi) = a^2 + b^2$; counts exclude 2; $\mathbb{Z}[i]$ has unique factorisation up to the four units.

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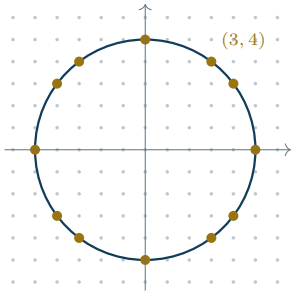
NUMBER THEORY LADDER 7 OF 9

Sums of two squares and Pythagorean triples

Counting lattice points on a circle, and squaring a Gaussian integer to build right triangles.

1 / THE RULE AND THE PICTURE

The number of integer points on the circle $x^2 + y^2 = n$ is $r_2(n) = 4(d_1 - d_3)$



$x^2 + y^2 = 25$ has 12 points

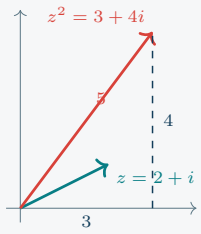
Here d_1 and d_3 count the divisors of n that are 1 and 3 modulo 4, and the factor 4 is the four units $\pm 1, \pm i$:

$$\boxed{r_2(n) = 4\left(d_1(n) - d_3(n)\right)}$$

For 25 the divisors $1, 5, 25$ are all $1 \bmod 4$, so $r_2 = 4 \cdot 3 = 12$, which are $(\pm 5, 0)$, $(0, \pm 5)$, $(\pm 3, \pm 4)$ and $(\pm 4, \pm 3)$. For 3 the divisors $1, 3$ cancel and $r_2 = 0$.

2 / EUCLID'S FORMULA / SQUARING $m + ni$

Squaring a Gaussian integer squares its length, so $(m + ni)^2$ is a right triangle



Since $(m + ni)^2 = (m^2 - n^2) + 2mn\,i$ and $|z^2| = |z|^2$,

$$\boxed{(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2}$$

With $m = 2$ and $n = 1$ the point $z^2 = 3 + 4i$ gives the $3, 4, 5$ triangle, and its angle is twice the angle of z. The triple is primitive when $\mathrm{HCF}(m, n) = 1$ and m, n have opposite parity.

3 / THE FIRST TRIPLES

Every primitive Pythagorean triple comes from exactly one pair (m, n)

(m, n)	triple (a, b, c)	check
$(2, 1)$	$(3,\ 4,\ 5)$	$3^2 + 4^2 = 5^2$
$(3, 2)$	$(5,\ 12,\ 13)$	$5^2 + 12^2 = 13^2$
$(4, 1)$	$(15,\ 8,\ 17)$	$15^2 + 8^2 = 17^2$
$(4, 3)$	$(7,\ 24,\ 25)$	$7^2 + 24^2 = 25^2$
$(5, 2)$	$(21,\ 20,\ 29)$	$21^2 + 20^2 = 29^2$
$(5, 4)$	$(9,\ 40,\ 41)$	$9^2 + 40^2 = 41^2$
$(6, 1)$	$(35,\ 12,\ 37)$	$35^2 + 12^2 = 37^2$
$(6, 5)$	$(11,\ 60,\ 61)$	$11^2 + 60^2 = 61^2$

The legs may swap, as $(4, 1)$ gives $15, 8, 17$, and for a primitive triple the hypotenuse $c = m^2 + n^2$ is odd with every prime factor $\equiv 1 \pmod 4$.

4 / COUNTING TRIPLES WITH A GIVEN HYPOTENUSE / $c = 65$

A hypotenuse needs a prime factor that is $1 \bmod 4$, and $65 = 5 \cdot 13$ has two

The triples with hypotenuse 65 are $(16, 63)$, $(33, 56)$, $(25, 60)$ and $(39, 52)$, the first two primitive and the others multiples, $5 \cdot (5, 12)$ and $13 \cdot (3, 4)$. They come from $65 = 5 \cdot 13 = (2 + i)(2 - i) \cdot (3 + 2i)(3 - 2i)$, since choosing $(2 + i)(3 + 2i) = 4 + 7i$ or $(2 + i)(3 - 2i) = 8 - i$ gives the two representations $65 = 4^2 + 7^2 = 8^2 + 1^2$, and $r_2(65) = 4 \cdot 4 = 16$ points. The primes $3 \bmod 4$ such as $3, 7, 11$ never occur in a primitive hypotenuse.

5 / WHY $(m + ni)^2$ GIVES EVERY PRIMITIVE TRIPLE

If $a^2 + b^2 = c^2$ with no common factor, then $a + bi$ is a square times a unit

Writing $c^2 = (a + bi)(a - bi)$ and noting that $a + bi$ and $a - bi$ share no prime factor in $\mathbb{Z}[i]$ when $\mathrm{HCF}(a, b) = 1$, unique factorisation forces each factor to be a unit times a square, so $a + bi = u\,(m + ni)^2$. The unit only swaps or changes the signs of the legs, which leaves the triangle unchanged.

ASIDE / RATIONAL POINTS

Dividing by c, the triple $(3, 4, 5)$ is the point $(\frac{3}{5}, \frac{4}{5})$ on the unit circle, with slope parameter $t = \frac{n}{m} = \frac{1}{2}$.

LINKS TO OTHER SHEETS

The factorisation in $\mathbb{Z}[i]$ is set up in *Primes mod 4 and the Gaussian integers*; the parametrisation by t is in *The line $ax + by = c$* and the circle sheet; the next sheet is *Primes mod 3 and the Eisenstein integers*.

Convention. $r_2(n)$ counts integer solutions of $x^2 + y^2 = n$ with signs and order; d_1, d_3 count divisors $\equiv 1, 3 \pmod 4$.

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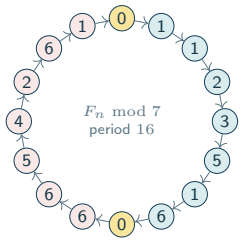
NUMBER THEORY LADDER 9 OF 9

Fibonacci modulo p and the golden ratio

Fibonacci repeats modulo every n, and whether a prime splits in the golden ratio field fixes the period.

1 / THE RULE AND THE PICTURE

Modulo 7 the sequence repeats every 16 terms, and its second half is the negative of its first



Reducing F_n modulo 7 gives $0, 1, 1, 2, 3, 5, 1, 6, 0, 6, 6, 5, 4, 2, 6, 1$ and then $0, 1, \ldots$ again, a cycle of period $\pi(7) = 16$, because a pair of consecutive terms determines the rest and there are finitely many pairs. The zero first reappears at $n = 8$ because $F_8 = 21 = 3 \cdot 7$, and then

$$F_{n+8} \equiv -F_n \pmod 7$$

so the sequence is the negative of itself half way round, and $16 = 2 \cdot 8$.

2 / THE CHARACTERISTIC EQUATION MODULO p

The roots of $x^2 - x - 1$ live in $\mathbb{F}_p$ or $\mathbb{F}_{p^2}$ according to the discriminant 5

Writing $F_n = (r_1^n - r_2^n)/(r_1 - r_2)$ with $r_{1,2} = \frac{1 \pm \sqrt{5}}{2}$ works modulo p whenever $\sqrt{5}$ exists.
If 5 is a square modulo p, that is $p \equiv \pm 1 \pmod 5$, the roots lie in $\mathbb{F}_p$ and $r^{p-1} = 1$, so the period divides $p - 1$. If 5 is not a square, $p \equiv \pm 2 \pmod 5$, the roots are conjugates with $r_1 r_2 = -1$, so $r_1^{p+1} = r_1 \cdot r_1^p = r_1 r_2 = -1$ and the period divides $2(p + 1)$. For $p = 5$ the double root 3 gives period 20.

3 / THE PERIODS / PRIMES UP TO 37

The period divides $p - 1$ for split primes and $2(p + 1)$ for inert ones

p	$p \bmod 5$	in $\mathbb{Z}[\varphi]$	period $\pi(p)$	first zero $\alpha(p)$	bound
2	2	inert	3	3	6
3	3	inert	8	4	8
5	0	ramified	20	5	20
7	2	inert	16	8	16
11	1	splits	10	10	10
13	3	inert	28	7	28
17	2	inert	36	9	36
19	4	splits	18	18	18
23	3	inert	48	24	48
29	4	splits	14	14	28
31	1	splits	30	30	30
37	2	inert	76	19	76

The first zero $\alpha(p)$, the rank of apparition, divides $p - \left(\frac{5}{p}\right)$ and the period is α, 2α or 4α: for example $p = 13$ has $\alpha = 7$ since $13 \mid F_7 = 13$, and $p = 11$ has $\alpha = 10$ since $F_{10} = 55 = 5 \cdot 11$.

4 / WHICH PRIMES DIVIDE WHICH FIBONACCI NUMBERS

Primes $\pm 1 \bmod 5$ split in the golden ratio field and the others stay prime

The ring $\mathbb{Z}[\varphi]$ has norm $a^2 + ab - b^2$, and a prime splits when $x^2 - x - 1$ has a root modulo p, which happens exactly when $\left(\frac{5}{p}\right) = 1$.
By reciprocity with $5 \equiv 1 \pmod 4$ this equals $\left(\frac{p}{5}\right)$, so the split primes below 100 are $11, 19, 29, 31, 41, 59, 61, 71, 79, 89$, and 5 is ramified.
Each prime divides some Fibonacci number: $17 \mid F_9 = 34$, $19 \mid F_{18} = 2584$, $23 \mid F_{24} = 46368$.

ASIDE / EVERY NUMBER DIVIDES ONE

Every integer divides infinitely many Fibonacci numbers, since 0 recurs in each period.

LINKS TO OTHER SHEETS

The Legendre symbol $\left(\frac{5}{p}\right)$ is in *The Legendre symbol and quadratic reciprocity*; the characteristic equation is in *The characteristic equation beneath five presentations*; the previous sheet is *Primes mod 3 and the Eisenstein integers*.

Convention. $\pi(p)$ is the Pisano period, $\alpha(p)$ the rank of apparition; $\varphi = \frac{1+\sqrt{5}}{2}$; $\mathbb{F}_p$ is the integers modulo p.

Geometric

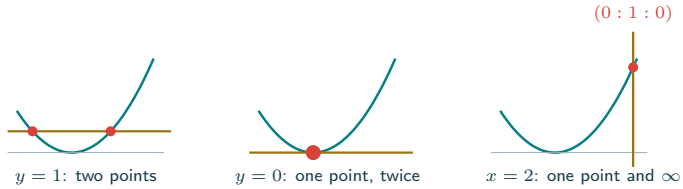
PROJECTIVE GEOMETRY LADDER 4 OF 9

Bézout's theorem and multiplicity

With infinity added and every root available, curves of degrees m and n meet in exactly mn points.

1 / THE RULE AND THE PICTURE

Two plane curves of degrees m and n with no common component meet in exactly mn points



$y = 1$: two points $\quad$ $y = 0$: one point, twice $\quad$ $x = 2$: one point and ∞

The count is made in the projective plane over an algebraically closed field, with each point counted by its multiplicity:

$$\#(C \cap D) = m\,n$$

A line meets the parabola twice in every picture, though the second point may be double, at infinity, or complex.

2 / EXAMPLES WITH THE COUNT

Every pair of curves reaches its full count

curves (degrees)	intersection points	count
$y = 1$ and $y = x^2$ $(1 \cdot 2)$	$(1, 1)$ and $(-1, 1)$	2
$y = 0$ and $y = x^2$	$(0, 0)$, a double root	2
$x = 2$ and $y = x^2$	$(2, 4)$ and $(0 : 1 : 0)$	$1 + 1$
$y = -1$ and $y = x^2$	$(i, -1)$ and $(-i, -1)$	2, complex
$x^2 + y^2 = 1$ and $y = x^2$ $(2 \cdot 2)$	$y = \frac{-1 \pm \sqrt{5}}{2}$, so $x = \pm 0.786$ and $x = \pm 1.272\,i$	$2 + 2$
$x^2 + y^2 = 1$ and $x^2 + y^2 = 4$	none, but $(1 : \pm i : 0)$ each twice	$2 + 2$
$y = 0$ and $y = x^3$ $(1 \cdot 3)$	$(0, 0)$, a triple root	3
$x + y = 1$ and $x + y = 3$	$(-1 : 1 : 0)$	1

3 / FOUR THINGS THE THEOREM NEEDS

Remove any one and the count fails

Without the line at infinity the lines $x + y = 1$ and $x + y = 3$ have no common point, and the line $x = 2$ meets the parabola only once. Without an algebraically closed field the line $y = -1$ misses the parabola over the reals, since $x^2 = -1$ has no real root.
Without multiplicity the tangent $y = 0$ meets the parabola once and not twice, and without the condition on components two copies of the same line meet in infinitely many points.

4 / HOW MULTIPLICITY IS COUNTED

The number of times a root repeats

Substitute one curve into the other and count repeated roots: putting $y = 0$ into $y = x^3$ gives $x^3 = 0$, a triple root, so the line and the cubic meet at the origin with multiplicity 3, which is the full count $1 \cdot 3$ for the cubic's flex.
For two circles with equal centres the homogeneous equations $X^2 + Y^2 = Z^2$ and $X^2 + Y^2 = 4Z^2$ subtract to $3Z^2 = 0$, so $Z = 0$ twice, and each of the two circular points $(1 : \pm i : 0)$ carries multiplicity 2, so the circles are tangent to each other at these two points.

5 / CONSEQUENCES

Counting lets us determine curves by points

A conic has six coefficients up to a common factor, so five points in general position fix a conic, and two different conics share at most 4 points. A line meets a cubic in 3 points, so a line through two rational points of a cubic meets it in a third point that is again rational, which the next sheets use.

6 / TRY THESE

Count the intersection

(a) How many points do two conics share? Four, counting complex points, points at infinity and multiplicity.
(b) A line and the hyperbola $xy = 1$: $y = 1$ gives $x = 1$ and the point $(1 : 0 : 0)$ at infinity, so $1 + 1 = 2$.
(c) Two parabolas $y = x^2$ and $y = 2x^2$ meet at the origin and at infinity: setting $YZ = X^2$ and $YZ = 2X^2$ gives $X^2 = 0$, so $X = 0$ twice, and $YZ = 0$ gives $(0 : 1 : 0)$ and the origin; each is counted twice for a total of 4.

ASIDE / WHY mn

The product counts the roots of one polynomial of degree mn in a single variable after eliminating the other, which is what the resultant does.

LINKS TO OTHER SHEETS

The previous sheet writes conics and cubics homogeneously; the sheet on chords and tangents uses the count 3 for a line and a cubic.

Convention. m, n are the degrees; multiplicity is the number of repeated roots at a point; the circular points are $(1 : \pm i : 0)$; algebraically closed means every polynomial has a root.

Geometric

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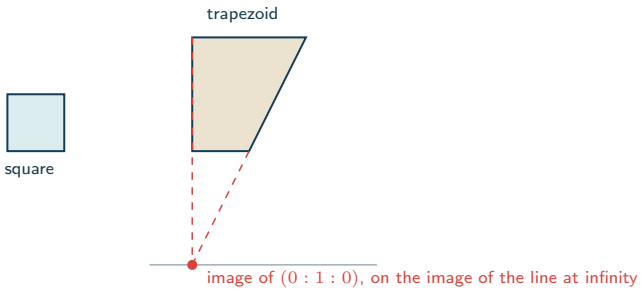
PROJECTIVE GEOMETRY LADDER 5 OF 9

Projective maps, matrices and cross-ratio

An invertible 3×3 matrix acts on the triples, moves the line at infinity, and keeps lines and cross-ratio.

1 / THE RULE AND THE PICTURE

Multiply the triple by a matrix, and parallel lines can converge



A projective map sends $(X:Y:Z)$ to the triple $M(X,Y,Z)^T$ for an invertible matrix M, and rescaling M changes nothing:

$$(X:Y:Z) \mapsto M\,(X:Y:Z)^T$$

With M below, the vertical sides of a square leave parallel and arrive pointing at $(0,-2)$.

2 / THE EXAMPLE

The matrix $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{pmatrix}$

A point (x,y) goes to $(x:y:1-\frac{y}{2})$, which is $\left(\frac{x}{1-y/2}, \frac{y}{1-y/2}\right)$, so the corners of the unit square $(0,0),(1,0),(1,1),(0,1)$ land on $(0,0),(1,0),(2,2),(0,2)$. The point at infinity $(0:1:0)$ of the vertical direction goes to $(0:1:-\frac{1}{2}) = (0,-2)$, where the vertical sides now meet, and the whole line at infinity goes to the horizontal line $y=-2$. Horizontal lines stay horizontal, since the point $(1:0:0)$ is fixed.

3 / WHAT IS KEPT AND WHAT IS LOST

Projective geometry studies what survives such maps

property	under a projective map	example
straight lines	kept	a line maps to a line
incidence	kept	points on a line stay on a line
cross-ratio	kept	$(0,1;2,3) = \frac{4}{3}$ becomes $(1,\frac{1}{2};\frac{1}{3},\frac{1}{4}) = \frac{4}{3}$
conics	kept, as a conic	a circle may become an ellipse or a hyperbola
parallelism	changed	parallel lines may meet in a finite point
distance and angle	changed	nothing measured is kept

4 / FOUR POINTS FIX THE MAP

Eight numbers are needed and four points supply them

A 3×3 matrix up to scale has $9-1=8$ free numbers, and each point requires 2 conditions, so four points, no three collinear, can be sent to any other four by exactly one projective map.
This is why a photograph of a rectangle, whose four corners are known, can be straightened by a single matrix.

5 / THE LINE, AND CROSS-RATIO

On the projective line the maps are $x \mapsto \frac{ax+b}{cx+d}$

The matrix $\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ sends x to $\frac{1}{x+1}$ and takes $0,1,2,3$ to $1,\frac{1}{2},\frac{1}{3},\frac{1}{4}$. The cross-ratio $(a,b;c,d) = \frac{(c-a)(d-b)}{(c-b)(d-a)}$ equals $\frac{2\cdot 2}{1\cdot 3} = \frac{4}{3}$ before and $\frac{(-\frac{2}{3})(-\frac{1}{4})}{(-\frac{1}{6})(-\frac{3}{4})} = \frac{4}{3}$ after, and this single number is what four collinear points keep. Composing two such maps multiplies their matrices, so the maps form a group that does not commute.

6 / TRY THESE

Apply the matrix

(a) Where does M send $(1,0)$? To $(1:0:1)$, so $(1,0)$ is fixed.
(b) Where does M send $(0,-2)$? To $(0:-2:2) = (0:-1:1)$, so $(0,-1)$.
(c) The map $x \mapsto x+1$ takes $0,1,2,3$ to $1,2,3,4$, and the cross-ratio is $\frac{(3-1)(4-2)}{(3-2)(4-1)} = \frac{4}{3}$ again.

ASIDE / PERSPECTIVE

A camera is a projective map from space to the plane, which is why parallel railway lines in a photograph meet at a vanishing point.

LINKS TO OTHER SHEETS

The algebra ladder shows why matrix products do not commute, which applies to these maps, and the linear algebra ladder develops matrices as transformations.

Convention. M is an invertible 3×3 matrix; cross-ratio of four collinear points is $(a,b;c,d)$; a projective map is determined by images of four points no three collinear.

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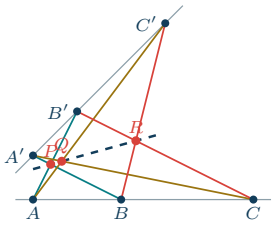
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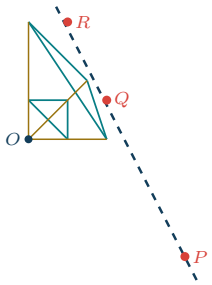
Pappus and Desargues by cross products

Two theorems about lines and meeting points become exact identities once points at infinity are allowed.

1 / THE RULE AND THE PICTURE

Six points on two lines give three meeting points on one line





Pappus: take A, B, C on one line and A', B', C' on another, then the points $AB' \cap A'B$, $AC' \cap A'C$ and $BC' \cap B'C$ are collinear.
Desargues: triangles ABC and $A'B'C'$ whose corresponding vertices are joined by concurrent lines have corresponding sides meeting on a line.

2 / PAPPUS IN COORDINATES

Every meeting point is a cross product and collinearity is a determinant

case	first line	second line	the three meeting points
finite case	$A = (0,0)$ $B = (2,0)$ $C = (5,0)$	$A' = (0,1)$ $B' = (1,2)$ $C' = (3,4)$	$P = (\frac{2}{5}, \frac{4}{5})$ $Q = (\frac{15}{23}, \frac{20}{23})$ $R = (\frac{1}{3}, \frac{4}{3})$ $\det(P,Q,R) = 0$
one point at infinity	$A = (0,0)$ $B = (1,0)$ $C = (3,0)$	$A' = (0,-1)$ $B' = (1,1)$ $C' = (2,3)$	$P = (1:1:0)$ $Q = (-\frac{6}{7}, -\frac{9}{7})$ $R = (\frac{9}{7}, \frac{6}{7})$ collinear along slope 1

3 / WHEN A MEETING POINT IS AT INFINITY

The theorem still holds

In the second case the lines AB' and $A'B$ both have slope 1, so in the plane they have no common point and the statement would have an exception.
Projectively they meet at $P = (1:1:0)$, and the line through Q and R has slope 1, so it passes through P and the three points are collinear. The computer check uses exact fractions and finds the determinant exactly zero.

4 / DESARGUES IN COORDINATES

Perspective from a point means perspective from a line

Take $A, B, C = (1,0), (0,1), (1,1)$ and $A', B', C' = (2,0), (0,3), (\frac{3}{2}, \frac{3}{2})$, which lie on lines through the origin O at ratios $2, 3, \frac{3}{2}$.
The corresponding sides meet at $(4,-3)$, $(2,1)$ and $(1,3)$, and the determinant of these three points is exactly zero, so they lie on the line $y = -2x + 5$.

5 / WHY THESE ARE PROJECTIVE THEOREMS

Their statements use only lines and points

Neither theorem mentions a length or an angle, only joining and meeting, so they hold in every projective plane that comes from a field and are unchanged by any projective map.
Pascal's theorem is the sibling for six points on a conic, whose three opposite-side meeting points are collinear, and Pappus is the case where the conic splits into two lines.

6 / TRY THESE

Compute a join and a meet

(a) Join $(1:0:1)$ and $(0:1:1)$: the cross product is $[-1:-1:1]$, the line $x + y = 1$.
(b) Meet $x = 0$ and $y = 0$: $[1:0:0] \times [0:1:0] = (0,0,1)$, the origin.
(c) Three points are collinear when $\det = 0$: for $(0,0), (1,1), (2,2)$ the triples are $(0,0,1), (1,1,1), (2,2,1)$ and the determinant is 0.

ASIDE / NO MEASUREMENT NEEDED

Because only joins and meets appear, the proofs can be done with cross products and determinants and need no distances, angles or coordinates adapted to the picture.

LINKS TO OTHER SHEETS

The homogeneous coordinates sheet gives the cross product rules, and the Bézout sheet explains why a degree count forces such coincidences.

Convention. P, Q, R are the three meeting points; collinear means $\det = 0$; a point $(a:b:0)$ lies on the line at infinity; computations use exact fractions.

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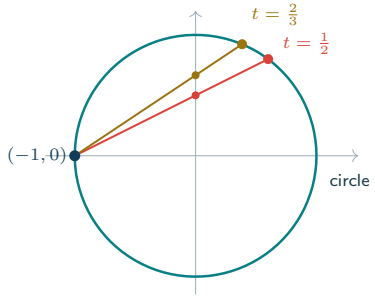
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Pythagorean triples from the projective line

A rational slope gives a rational point on the circle, and clearing denominators gives the triple.

1 / THE RULE AND THE PICTURE

Every line through $(-1,0)$ with rational slope t meets the circle again at a rational point



Putting $y = t(x+1)$ into $x^2 + y^2 = 1$ gives $(1+t^2)x^2 + 2t^2x + t^2 - 1 = 0$, a quadratic with one root $x = -1$, so the other root is rational too, by Bézout and the product of the roots:

$$(x, y) = \left(\tfrac{1-t^2}{1+t^2},\ \tfrac{2t}{1+t^2}\right)$$

With $t = \frac{n}{m}$ this is $\frac{m^2-n^2}{m^2+n^2}, \frac{2mn}{m^2+n^2}$, the legs over the hypotenuse of a right-angled triangle.

2 / HOMOGENEOUS COORDINATES

Clearing denominators turns the parameter $(m : n)$ into a point of the conic

(m,n)	m^2-n^2	$2mn$	m^2+n^2	point $(X : Y : Z)$
$(2,1)$	3	4	5	$(3:4:5)$
$(3,2)$	5	12	13	$(5:12:13)$
$(4,1)$	15	8	17	$(15:8:17)$
$(4,3)$	7	24	25	$(7:24:25)$
$(5,2)$	21	20	29	$(21:20:29)$
$(5,4)$	9	40	41	$(9:40:41)$

The map $(m : n) \mapsto (m^2 - n^2 : 2mn : m^2 + n^2)$ sends the projective line to the conic $X^2 + Y^2 = Z^2$, with $m = 0$ giving the base point $(-1 : 0 : 1)$, and the complex parameters $n = \pm im$ giving $Z = 0$ and the two circular points $(1 : \pm i : 0)$.
A Pythagorean triple is then a rational point of the conic written with coprime integer coordinates, and it is primitive when $\gcd(a,b,c) = 1$.

3 / EVERY PRIMITIVE TRIPLE APPEARS ONCE

Coprime $m > n$ of opposite parity

Every primitive triple has the form $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$ for a unique pair $m > n > 0$ with $\gcd(m,n) = 1$ and $m - n$ odd, and swapping the legs is the only freedom. Two pairs can give the same hypotenuse, as $(8,1)$ gives $(63,16,65)$ and $(7,4)$ gives $(33,56,65)$, and a pair of equal parity gives twice a smaller triple, as $(3,1)$ gives $(8,6,10) = 2(4,3,5)$.

4 / THE FAMILY TREE OF TRIPLES

Three matrices preserve the conic and generate all primitive triples from $(3,4,5)$

The matrices $\begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}$ and $\begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$ send $(3,4,5)$ to $(5,12,13)$, $(21,20,29)$ and $(15,8,17)$, and each satisfies $M^T Q M = Q$ for $Q = \mathrm{diag}(1,1,-1)$, so it is a projective map carrying the conic $X^2 + Y^2 = Z^2$ to itself. Berggren showed in 1934 that repeating them reaches every primitive triple exactly once.

5 / HOW MANY TRIPLES

About $N/2\pi$ primitive triples have hypotenuse up to N

Counting with the formula gives 16 primitive triples with $c \le 100$ against $100/2\pi = 15.9$, and 158 with $c \le 1000$ against 159.2, the first being $(3,4,5)$, $(5,12,13)$, $(8,15,17)$, $(7,24,25)$ and $(20,21,29)$.

6 / TRY THESE

Use (m,n)

(a) Find the triple for $(m,n) = (4,3)$: $(7,24,25)$, and check $49 + 576 = 625$.
(b) Which pair gives $(20,21,29)$? $(5,2)$, since $25 - 4 = 21$ and $2 \cdot 10 = 20$.
(c) Why does $(6,8,10)$ not come from a coprime pair of opposite parity? Because it is twice $(3,4,5)$.

ASIDE / A RATIONAL CIRCLE

The same line trick parametrises the rational points of every conic that has one, which is why conics have an easy theory of rational points and cubics do not.

LINKS TO OTHER SHEETS

The number theory ladder has the Pythagorean triples through Gaussian integers, and the next sheet shows what replaces this method for cubics.

Convention. (m,n) coprime with $m > n$ and $m - n$ odd gives a primitive triple; $Q = \mathrm{diag}(1,1,-1)$ is the quadratic form of the conic; Berggren, 1934.

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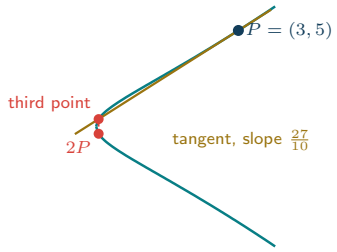
PROJECTIVE GEOMETRY LADDER 8 OF 9

Cubics: chords, tangents and Bézout

A line meets a cubic in three points, so two rational points force a third, and the points form a group.

1 / THE RULE AND THE PICTURE

A line meets a cubic in 3 points counted with multiplicity, so two rational points give a third



$y^2 = x^3 - 2$: the tangent at P meets the curve again, and reflecting gives $2P$

Substituting a line $y = \lambda x + \nu$ into the cubic gives a cubic equation in x whose three roots add to λ^2, so if two roots are rational so is the third. Bézout guarantees the three, counting a tangent as two:

$$x_1 + x_2 + x_3 = \lambda^2, \qquad P + Q = \text{reflection of the third point}$$

The reflection makes the operation commutative, with the point at infinity $O = (0 : 1 : 0)$ as the zero.

2 / DOUBLING AND TRIPLING A POINT

The coordinates stay rational and grow quickly

step	computation	check
P	$(3, 5)$, and $5^2 = 27 - 2$	$25 = 25$
tangent at P	slope $\frac{3x^2}{2y} = \frac{27}{10}$	$\lambda = \frac{27}{10}$
third point of the line	$x = \lambda^2 - 2 \cdot 3 = \frac{129}{100}$, $y = 5 + \lambda(\frac{129}{100} - 3) = \frac{383}{1000}$	on the curve
$2P$	reflect in the x-axis: $\left(\frac{129}{100}, -\frac{383}{1000}\right)$	$y^2 = x^3 - 2$ holds
$3P = P + 2P$	chord through P and $2P$: $\left(\frac{164323}{29241}, -\frac{66234835}{5000211}\right)$	on the curve

3 / THE POINT AT INFINITY

A flex, so the zero is its own negative

In homogeneous form the curve is $Y^2Z = X^3 - 2Z^3$, and $Z = 0$ forces $X^3 = 0$, so $(0 : 1 : 0)$ is a triple point of the intersection with the line at infinity and a flex of the curve.

The tangent there meets the curve three times at O itself, which says that $O + O = O$, and the line through P and its reflection $-P$ meets the curve a third time at O, which says $P + (-P) = O$.

4 / CONICS AGAINST CUBICS

A parametrisation against a group

On a conic one rational point and the line trick of the previous sheet give every rational point, while on a cubic the chord and tangent method only generates new points from old ones, and the point $(3, 5)$ generates infinitely many rational points $P, 2P, 3P, \ldots$ with ever larger numerators and denominators.

Mordell proved in 1922 that the rational points of such a curve are generated by finitely many of them, which is the beginning of the theory of elliptic curves.

5 / TRY THESE

Check by hand

(a) Verify that $2P = (\frac{129}{100}, -\frac{383}{1000})$ lies on the curve: $x^3 - 2 = 0.146689$ and $y^2 = 0.146689$.

(b) Find the tangent slope at $(3, 5)$: differentiating $y^2 = x^3 - 2$ gives $2y\,y' = 3x^2$, so $y' = \frac{27}{10}$.

(c) The roots of the cubic from the tangent are $3, 3, \frac{129}{100}$, which add to $\frac{729}{100} = \lambda^2$.

ASIDE / WHY THREE

The count 3 is Bézout for a line and a cubic, and it is the reason the cubic has a group law, while the count 2 for a conic gives only a parametrisation.

LINKS TO OTHER SHEETS

The Bézout sheet proves the count; the algebra ladder explains why this addition is associative with an identity and inverses, the group axioms.

Convention. λ is the slope of the chord or tangent; $O = (0 : 1 : 0)$ is the point at infinity; $P + Q$ is the reflection of the third intersection point; computations use exact fractions.

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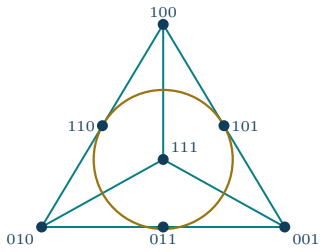
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Finite fields, the Fano plane and the whole picture

Over a field with p elements the plane has $p^2 + p + 1$ points, and the missing roots show why the field must grow.

1 / THE RULE AND THE PICTURE

Seven points and seven lines, every two lines meeting in exactly one point



the Fano plane, with points as binary vectors

Using $\mathbb{F}_2 = \{0, 1\}$ in homogeneous coordinates, the points are the seven non-zero vectors of $\mathbb{F}_2^3$ and a line is a set of three that add to zero, such as $110, 011, 101$, the circle in the picture. Every two points lie on one line and every two lines meet in one point, and in general

$$|\mathbb{P}^2(\mathbb{F}_p)| = p^2 + p + 1$$

with $p + 1$ points on each line and $p + 1$ lines through each point.

2 / THE SAME ALGEBRA MOD p

Cross products join and meet in any field

Over $\mathbb{F}_3$ the lines $x + y = 1$ and $x + y = 0$ are $[1 : 1 : -1]$ and $[1 : 1 : 0]$ and their cross product is $(1 : -1 : 0) = (1 : 2 : 0)$, a point at infinity, so parallel lines meet exactly as over the reals.
A line has $p + 1$ points, the p ordinary ones and one at infinity, and the whole plane has p^2 ordinary points together with $p + 1$ at infinity, which totals $p^2 + p + 1$.

3 / THE FIELD MATTERS

The circular points exist only when -1 is a square

Setting $Z = 0$ in the circle $X^2 + Y^2 = Z^2$ gives $X^2 + Y^2 = 0$, which has the roots $(1 : \pm i : 0)$ only if $i^2 = -1$ has a solution in the field. Modulo 5 it does, with $i = 2$, and modulo 13 with $i = 5$, but modulo $3, 7, 11$ it does not, so the two circular points exist only in the larger field with p^2 elements, which is how Bézout's count needs an algebraically closed field.

4 / COUNTING THE CIRCLE

Always $p + 1$ points, as many as the projective line

p	points of the plane	affine circle points	circular points	all circle points	$p \bmod 4$
3	13	4	0	4	3
5	31	4	2	6	1
7	57	8	0	8	3
11	133	12	0	12	3
13	183	12	2	14	1

The circle has $p - 1$ affine points when $p \equiv 1 \pmod 4$, with two at infinity, and $p + 1$ affine points when $p \equiv 3 \pmod 4$, with none at infinity, so the total is $p + 1$ either way, which is the parametrisation of the Pythagorean sheet working modulo p.

5 / WHAT EACH INGREDIENT REPAIRS

The ladder in one table

ingredient	what it repairs	example
points at infinity	parallel lines	$x + y = 1$ and $x + y = 3$ meet at $(-1 : 1 : 0)$
an algebraically closed field	missing roots	$y = -1$ meets $y = x^2$ only at $x = \pm i$
multiplicity	tangency	$y = 0$ touches $y = x^2$ twice at the origin
homogeneous coordinates	a separate rule for infinity	one cross product for every pair of lines
no common component	infinitely many points	two copies of one line

6 / TRY THESE

Count and check

(a) How many points and lines has $\mathbb{P}^2(\mathbb{F}_3)$? $9 + 3 + 1 = 13$ of each.
(b) Check that $(1 : 2 : 0)$ lies on $X^2 + Y^2 = 0$ over $\mathbb{F}_5$: $1 + 4 = 5 \equiv 0$.
(c) Why does the Fano plane have no parallel lines? Any two of its seven lines share a point, as the table of lines shows.

ASIDE / THE WHOLE LADDER

Points at infinity, homogeneous coordinates, an algebraically closed field and multiplicity together turn "usually mn" into "always mn".

LINKS TO OTHER SHEETS

The ladder runs: 1 point at infinity, 2 homogeneous coordinates, 3 homogenising curves, 4 Bézout, 5 projective maps, 6 Pappus and Desargues, 7 Pythagorean triples, 8 cubics, and this one; primes modulo 4 are in the number theory ladder.

Convention. $\mathbb{F}_p$ is the field with p elements; $\mathbb{P}^2(\mathbb{F}_p)$ its projective plane; the Fano plane is $\mathbb{P}^2(\mathbb{F}_2)$; points are non-zero vectors up to scale.

FOUNDATIONS: COMPLEX NUMBERS ON ONE PAGE

A number can have a direction

From the number line to the plane: a point, a length and a turn.

1 / A MULTIPLICATION CAN MAKE A TURN

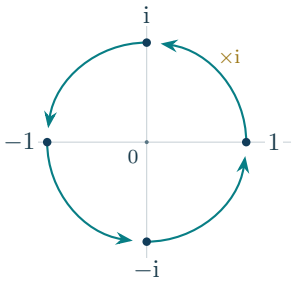
Two quarter-turns make a half-turn

Multiplying by -1 takes 1 to -1. In the plane, this is a half-turn about 0. A quarter-turn anticlockwise takes 1 to the upward unit i, called the **imaginary unit**.

Two quarter-turns give the same half-turn as multiplication by -1. We extend ordinary algebra using the rule

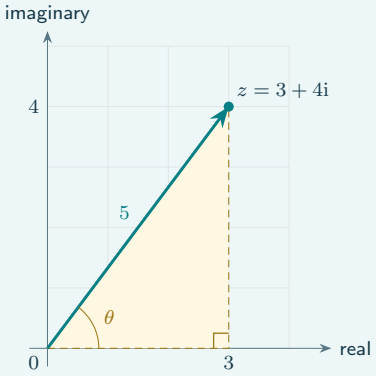
$$\mathrm{i}^2 \quad = \quad -1$$

Four quarter-turns bring us back to 1, so $\mathrm{i}^4 = 1$.



2 / READ A COMPLEX NUMBER AS TWO COORDINATES

$3 + 4\mathrm{i}$: three across and four up



A **complex number** has the form

$$z = a + b\mathrm{i} \quad \longleftrightarrow \quad (a, b),$$

where a, b are ordinary real numbers. Plot a horizontally and b vertically. For $3 + 4\mathrm{i}$, the real part is 3 and the imaginary part is 4.

The arrow from 0 records a length and a direction. Its length, written $|z|$, is the **modulus**. Pythagoras gives

$$|3 + 4\mathrm{i}| = \sqrt{3^2 + 4^2} = 5, \qquad |a + b\mathrm{i}| = \sqrt{a^2 + b^2}.$$

The angle θ from the positive real axis gives the direction, called the **argument**.

3 / A QUARTER-TURN FOR EVERY POINT

The algebra gives the quarter-turn

Expand the brackets and use $\mathrm{i}^2 = -1$:

$$\mathrm{i}(a + b\mathrm{i}) = a\mathrm{i} + b\mathrm{i}^2 = -b + a\mathrm{i}.$$

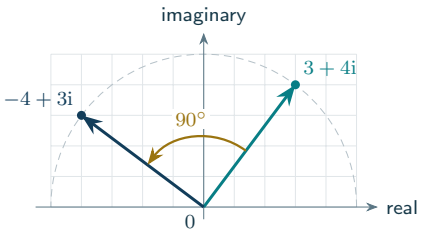
So the coordinate rule is

$$\boxed{(a, b) \longmapsto (-b, a).}$$

For our example,

$$\mathrm{i}(3 + 4\mathrm{i}) = 3\mathrm{i} - 4 = \boxed{-4 + 3\mathrm{i}}.$$

The point moves from $(3, 4)$ to $(-4, 3)$, a 90° anticlockwise turn. Its distance from 0 stays 5.



4 / GIVE ARITHMETIC A GEOMETRIC MEANING

Add displacements; multiply to scale and turn

Addition combines the two coordinates. For example, adding $2 - \mathrm{i}$ moves our point two right and one down:

$$\begin{aligned}(3 + 4\mathrm{i}) + (2 - \mathrm{i}) &= (3 + 2) + (4 - 1)\mathrm{i} \\ &= 5 + 3\mathrm{i}.\end{aligned}$$

Multiplication can change length and direction. With $z = 3 + 4\mathrm{i}$,

$$\begin{aligned}2z &= 6 + 8\mathrm{i} && \text{double the length,} \\ 2\mathrm{i}z &= -8 + 6\mathrm{i} && \text{double and turn } 90^\circ.\end{aligned}$$

Two quick checks. 1. Find $\mathrm{i}(2 + 3\mathrm{i})$. 2. Find $|-3 + 4\mathrm{i}|$.

Next. Whole-number coordinates a, b give the **Gaussian integers** $a + b\mathrm{i}$, the square lattice used for Pythagorean triples and Pick's theorem.

Remember. 0 has length 0 and no unique direction. **Check answers:** $-3 + 2\mathrm{i}$; 5.

References. OpenStax, Complex numbers; Emory University, The complex numbers.

Geometric

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BITS, TRITS AND CYCLOTOMIC INTEGERS ON ONE PAGE

Two colours on a square, three on a hexagon

A bit is a remainder in the Gaussian integers and a trit is a remainder in the Eisenstein integers.

THE RULE

Divide by the prime over 2 or over 3

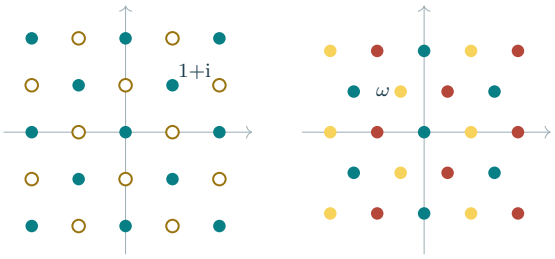
In $\mathbb{Z}[\mathrm{i}]$ the prime $1+\mathrm{i}$ divides 2, and in $\mathbb{Z}[\omega]$ the prime $1-\omega$ divides 3; each leaves exactly as many remainders as its norm, and those remainders are the digits:

$$a+b\,\mathrm{i} \equiv a+b \pmod 2, \qquad a+b\omega \equiv a+b \pmod 3$$
$$\mathbb{Z}[\mathrm{i}]/(1+\mathrm{i}) \;\cong\; \{0,1\} \qquad \mathbb{Z}[\omega]/(1-\omega) \;\cong\; \{0,1,2\}$$

The congruences come from $\mathrm{i} \equiv 1$ and $\omega \equiv 1$ modulo those primes. Both primes are special because they meet their rational prime twice,

$$2 = -\mathrm{i}\,(1+\mathrm{i})^2, \qquad 3 = -\omega^2(1-\omega)^2,$$

and 2 and 3 are the only primes that do this in their respective rings.



$\mathbb{Z}[\mathrm{i}]$: two classes $\qquad$ $\mathbb{Z}[\omega]$: three classes

Colouring by remainder gives a checkerboard on the square lattice and a three-colouring on the hexagonal one.

THE FOURTH AND THIRD CYCLOTOMIC FIELDS

Adjoin a root of unity and take its whole numbers

Each field adjoins a primitive root of unity to $\mathbb{Q}$, and its ring of integers is the lattice drawn above; both fields have degree $\varphi(4) = \varphi(3) = 2$, because each root satisfies a quadratic.

	fourth, the Gaussian side	third, the Eisenstein side
root and polynomial	$\mathrm{i} = \mathrm{e}^{2\pi\mathrm{i}/4}$, $\quad \Phi_4(x) = x^2+1$	$\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, $\quad \Phi_3(x) = x^2+x+1$
field and integers	$\mathbb{Q}(\mathrm{i})$, $\quad \mathbb{Z}[\mathrm{i}] = \{a+b\mathrm{i}\}$	$\mathbb{Q}(\omega)$, $\quad \mathbb{Z}[\omega] = \{a+b\omega\}$
norm	$N(a+b\mathrm{i}) = a^2+b^2$	$N(a+b\omega) = a^2-ab+b^2$
units	$\pm1, \pm\mathrm{i}$: the four 4th roots	$\pm1, \pm\omega, \pm\omega^2$: the six 6th roots
digit prime	$1+\mathrm{i}$, norm 2, gives a **bit**	$1-\omega$, norm 3, gives a **trit**

BINARY AND TERNARY INSIDE THE RINGS

A base whose norm is the number of digits

Repeatedly taking the remainder and dividing by the base, exactly as for ordinary numerals, writes every Gaussian integer in base $-1+\mathrm{i}$ with the digits $\{0,1\}$ and every Eisenstein integer in base $\omega-1$ with the digits $\{0,1,2\}$, each expansion finite and unique. Both bases are associates of the digit primes, and their squares are $(-1+\mathrm{i})^2 = -2\mathrm{i}$ and $(\omega-1)^2 = -3\omega$.

binary in base $-1+\mathrm{i}$

$$\mathrm{i} = 11, \qquad 2 = 1100, \qquad -1 = 11101$$

ternary in base $\omega-1$

$$\omega = 11, \qquad 3 = 1200, \qquad -1 = 12102$$

Negative and complex numbers need no sign or second coordinate, since the base itself points in a new direction at every power; checking $3 = (\omega-1)^3 + 2(\omega-1)^2 = (3+6\omega) - 6\omega$ is a good first exercise.

Digits as roots of unity

Sending a bit b to $(-1)^b$ and a trit t to ω^t turns adding digits into multiplying roots of unity, since $(-1)^{b+b'} = (-1)^b(-1)^{b'}$ and $\omega^{t+t'} = \omega^t\omega^{t'}$ with the exponents read modulo 2 and 3.

The same phases in qubits and qutrits

The phase gate on a qubit sends $|b\rangle$ to $(-1)^b|b\rangle$, and the clock gate on a qutrit sends $|t\rangle$ to $\omega^t|t\rangle$, so its phases live in $\mathbb{Z}[\omega]$; the qutrit Fourier transform is built from powers of ω just as the Hadamard is built from ±1.

Conventions. Throughout, $\omega = \mathrm{e}^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, so $\omega^2+\omega+1=0$, and Φ_n is the nth cyclotomic polynomial, whose roots are the primitive nth roots of unity; "degree" means the dimension of the field over $\mathbb{Q}$. Two integers are associates when they differ by a unit factor, as $1-\omega$ and $\omega-1$ do.

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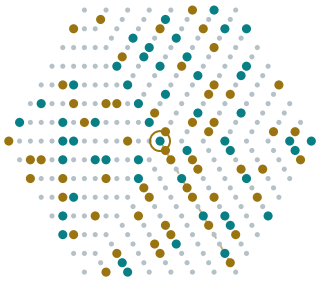
NUMBER THEORY ON ONE PAGE

A hexagonal Ulam spiral centred on 41

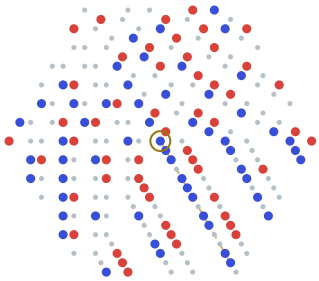
In each wedge a lattice line advances n by a fixed residue mod 6, giving solid, dashed and cycling diagonals.

1 / THE RULE AND THE PICTURE

Ring k holds $41 + 3k^2 - 3k + 1$ to $41 + 3k^2 + 3k$, and the six rays are the quadratics $3k^2 + ak + 41$



types mod 4: teal 1, gold 3



types mod 6: red 1 (split), blue 5 (inert)

The integers wind anticlockwise from 41 at the centre, ring k having $6k$ cells, and the corner cells along the six rays (E, NE, NW, W, SW, SE) are $n = 3k^2 + ak + 41$ with $a = -2, \ldots, 3$. Grey dots are unprimed survivors of the pre-screen, the dashed line is the SE ray $a = 3$, and the 631 cells hold 109 primes.

2 / THE SIX RAYS

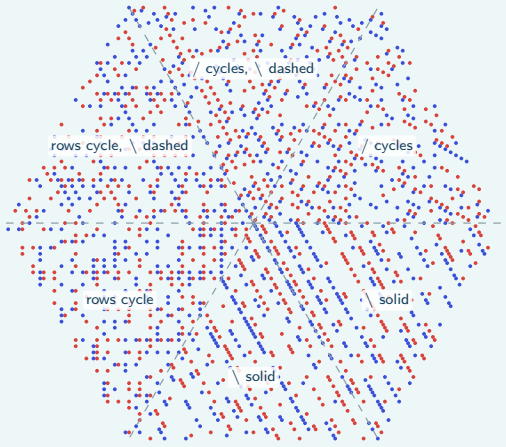
Only the ray $3k^2 + 3k + 41$ avoids the multiples of 2 and 3 throughout

a	ray	$n \bmod 6$ for $k = 0, 1, 2, \ldots$	primes, $k < 40$
-2	E	$5, 0, 1, 2, 3, 4, 5, 0$	10
-1	NE	$5, 1, 3, 5, 1, 3, 5, 1$	9
0	NW	$5, 2, 5, 2, 5, 2, 5, 2$	13
1	W	$5, 3, 1, 5, 3, 1, 5, 3$	8
2	SW	$5, 4, 3, 2, 1, 0, 5, 4$	5
3	SE	$5, 5, 5, 5, 5, 5, 5, 5$	24

The SE ray has $n \equiv 5 \pmod 6$ throughout and no root mod 5, giving 24 primes in 40 terms but only $41, 47, 59$ in a row. Euler's $n^2 + n + 41$ gives 40 because -163 is a Heegner discriminant, while this ray has -483.

3 / THE WEDGE MAP

In each wedge every lattice direction advances n by a fixed residue mod 6, which organises the diagonals



15 000 numbers from 41: red split (1 mod 3), blue inert (2 mod 3)

lines	wedge	step of n mod 6
\ (NW–SE)	240° to 360°	0: solid one-colour lines, about 1 in 3 alive, the rest empty
\ (NW–SE)	60° to 180°	3: dashed one-colour lines, every second cell
/ (NE–SW)	0° to 120°	2 or 4: red, blue, gap in turn, or empty lines
rows (E–W)	120° to 240°	2 or 4: red, blue, gap in turn, or empty lines

Out to ring 70 the step is constant for 97% to 100% of cells in a wedge. Step 0 keeps one residue, so a line is alive (all ± 1 mod 6) or empty; step 3 alternates $r, r + 3$, so half the cells are even; steps 2 and 4 cycle $r, r + 2, r + 4$, giving type 1, type 5 and a multiple of 3 for odd r. Elsewhere the step is ± 1 and lines are sparse. The first step of the walk fixes the angles.

4 / THE TWO SCREENS COMPARED

The hexagon suits the six-fold screen, and the four-fold screen shows up as pairs

	mod 6 (red 1, blue 5)	mod 4 (teal 1, gold 3)
fit to the lattice	$6k$ cells per ring: the 6-cycle repeats k times	$6k \equiv 2k \bmod 4$: the 4-cycle shifts each ring
solid lines	every NW–SE line in the lower wedge	none (types run $1, 3, 3, 1$)
neighbour types	same type 50.2% of adjacent prime pairs	same type 49.5%
look	stripes, dashes and cycles	pairs and moiré

ASIDE / THE CENTRE

The centre 41 is teal mod 4 ($41 \equiv 1$) and blue mod 6 (inert, $41 \equiv 5$), so the SE ray is a column of blue in the mod 6 picture.

LINKS TO OTHER SHEETS

The pre-screens are in *Prime types: the mod 4 and mod 6 races*; split and inert primes are in *Primes mod 3 and the Eisenstein integers*.

Convention. Primes above 3; the spiral goes anticlockwise from east; wedge angles are measured anticlockwise from east.

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COMPLEX POWERS, REAL ANSWERS

From i^{i} to ω^{ω}

One rule decides when a power of a unit-circle number lands back on the real line.

THE RULE

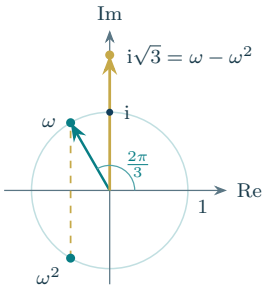
Imaginary part sets the size, real part sets the angle

Take a base on the unit circle, $u = \mathrm{e}^{\mathrm{i}\theta}$, with a chosen argument θ, so that $\operatorname{Log} u = \mathrm{i}\theta$ and $u^w := \mathrm{e}^{w \operatorname{Log} u}$. Writing the exponent as $w = a + \mathrm{i}b$,

$$u^{a+\mathrm{i}b} \;=\; \underbrace{\mathrm{e}^{-b\theta}}_{\text{size}}\;\underbrace{\mathrm{e}^{\mathrm{i}a\theta}}_{\text{angle}}$$

so the result is real exactly when $a\theta \in \pi\mathbb{Z}$, positive when $a\theta/\pi$ is even and negative when it is odd.

base	θ	real when
i	$\pi/2$	$a \in 2\mathbb{Z}$
ω	$2\pi/3$	$a \in \frac{3}{2}\mathbb{Z}$, sign $(-1)^{2a/3}$



CASE $a = 0$: PURELY IMAGINARY EXPONENTS GIVE POSITIVE REALS

i^{i}, ω^{i} and $\omega^{1+2\omega}$

With no real part in the exponent there is no rotation at all, so every one of these is a positive real; the third needs only the observation that doubling ω and adding 1 cancels its real part, $1 + 2\omega = \omega - \omega^2 = \mathrm{i}\sqrt{3}$.

$$\mathrm{i}^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}\cdot\mathrm{i}\pi/2} = \mathrm{e}^{-\pi/2} \approx 0.207880, \qquad \omega^{\mathrm{i}} = \mathrm{e}^{-2\pi/3} \approx 0.123145, \qquad \omega^{1+2\omega} = \mathrm{e}^{\mathrm{i}\sqrt{3}\cdot\mathrm{i}(2\pi/3)} = \mathrm{e}^{-2\pi/\sqrt{3}} \approx 0.0265799.$$

CASE $a \neq 0$

Powers of ω^{ω}

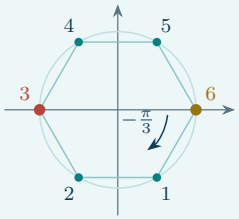
Since $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, the exponent ω has $a = -\frac{1}{2}$, so

$$\omega^{\omega} = \mathrm{e}^{\omega\cdot\mathrm{i}(2\pi/3)} = \mathrm{e}^{-\pi/\sqrt{3}}\,\mathrm{e}^{-\mathrm{i}\pi/3}.$$

For an integer n the power $(\omega^{\omega})^n = \omega^{n\omega}$ has $a = -n/2$, so its angle is $-n\pi/3$ and it is real exactly when $3 \mid n$, negative for odd multiples of 3 and positive for even ones.

$$\boxed{\omega^{3\omega} = -\mathrm{e}^{-\pi\sqrt{3}}} \approx -0.00433342$$

$$\boxed{\omega^{6\omega} = \mathrm{e}^{-2\pi\sqrt{3}}} \approx 1.87785 \times 10^{-5}$$



Direction of $\omega^{n\omega}$; each step also shrinks the size by $\mathrm{e}^{-\pi/\sqrt{3}} \approx 0.163$.

Where the three comes from

The step angle is $a\theta = \operatorname{Re}(\omega)\cdot\frac{2\pi}{3} = -\frac{\pi}{3}$, so the three is inherited from ω being a cube root of unity. The same recipe for i gives $\operatorname{Re}(\mathrm{i})\cdot\frac{\pi}{2} = 0$, which is why i^{i} needs no power at all, and for $\zeta = \mathrm{e}^{2\pi\mathrm{i}/m}$ the step is $\frac{2\pi}{m}\cos\frac{2\pi}{m}$.

An irrational component

The real number $\sqrt{3}$ is irrational. The number $\mathrm{i}\sqrt{3} \notin \mathbb{Q}$ is **purely imaginary and algebraic**, with irrational imaginary part: $(\mathrm{i}\sqrt{3})^2 + 3 = 0$. In standard terminology, "irrational" describes real numbers. Notice that $\mathrm{i}\sqrt{3} = 1 + 2\omega$ is nevertheless an **Eisenstein integer**.

ANY BASE, WITH THE NATURAL LOG

Read the answer through $\ln$

For a base $z = r\mathrm{e}^{\mathrm{i}\theta}$ the logarithm is $\ln z = \ln r + \mathrm{i}\theta$, and since $z^w = \mathrm{e}^{w\ln z}$ with $w = a + \mathrm{i}b$, the product $w\ln z = (a\ln r - b\theta) + \mathrm{i}\,(b\ln r + a\theta)$ has real part $\ln|z^w|$ and imaginary part equal to the angle, so z^w is real exactly when $\operatorname{Im}(w\ln z) \in \pi\mathbb{Z}$. On the unit circle $\ln r = 0$ and this is the rule above; off it, the imaginary part of the exponent also turns the result through $b\ln r$, so $2^{\mathrm{i}} = \mathrm{e}^{\mathrm{i}\ln 2}$ is not real while $2^{\mathrm{i}\pi/\ln 2} = \mathrm{e}^{\mathrm{i}\pi} = -1$.

Logarithm convention. Values use the principal logarithms $\operatorname{Log}\mathrm{i} = \mathrm{i}\pi/2$ and $\operatorname{Log}\omega = \mathrm{i}(2\pi/3)$. On another branch, $\theta \mapsto \theta + 2\pi k$, the numbers change: i^{i}, ω^{i} and $\omega^{1+2\omega}$ stay positive, $\omega^{3\omega}$ stays real with sign $(-1)^{k+1}$, and integer powers such as $(\omega^{\omega})^n = \omega^{n\omega}$ remain valid on every branch.

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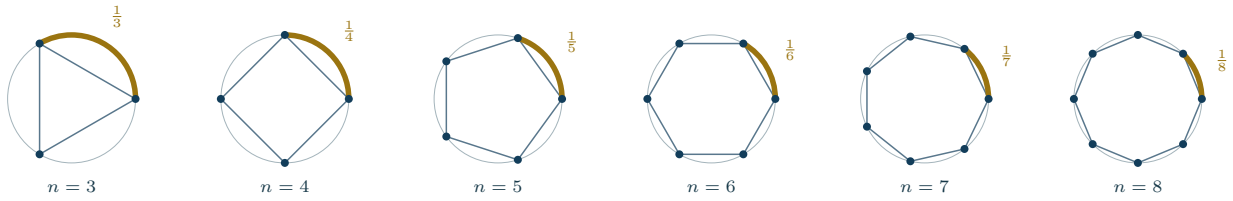
CYCLOTOMIC FIELDS ON ONE PAGE

Cyclotomic fields and the arcs of inscribed polygons

The vertex of a regular n-gon is $\zeta_n = e^{2\pi i/n}$, one arc of $\frac{1}{n}$ of the circle, and it generates the field $\mathbb{Q}(\zeta_n)$.

1 / THE RULE AND THE PICTURE

Each step round the inscribed n-gon is an arc of $\frac{1}{n}$ of the circle, multiplication by ζ_n



The roots of $x^n = 1$ are the vertices ζ_n^k, the smallest field containing ζ_n is the cyclotomic field $\mathbb{Q}(\zeta_n)$, and its degree over $\mathbb{Q}$ is Euler's totient $\varphi(n)$:

$$\zeta_n = e^{2\pi i/n}, \qquad [\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n), \qquad \text{ring of integers } \mathbb{Z}[\zeta_n]$$

2 / THE GEOMETRY / ARC AND CHORD

The arc shrinks as $2\pi/n$ and the chord $|1-\zeta_n| = 2\sin\frac{\pi}{n}$ follows it from below

n	arc fraction	angle	arc length $2\pi/n$	chord $2\sin\frac{\pi}{n}$	chord/arc	product of chords
3	1/3	120°	2.094	$\sqrt{3} = 1.732$	0.827	$\sqrt{3}\cdot\sqrt{3} = 3$
4	1/4	90°	1.571	$\sqrt{2} = 1.414$	0.900	$\sqrt{2}\cdot 2\cdot\sqrt{2} = 4$
5	1/5	72°	1.257	1.176	0.935	$1.176^2\cdot 1.902^2 = 5$
6	1/6	60°	1.047	1	0.955	$1\cdot\sqrt{3}\cdot 2\cdot\sqrt{3}\cdot 1 = 6$
7	1/7	51.43°	0.898	0.868	0.967	7
8	1/8	45°	0.785	0.765	0.974	8

The product of the $n-1$ chords from one vertex to the others is always n, because $\prod_{k=1}^{n-1}(1-\zeta_n^k)$ is $\frac{x^n-1}{x-1}$ evaluated at $x=1$.

3 / THE ALGEBRA / WHICH FIELD EACH POLYGON GENERATES

Degree $\varphi(n)$ is 2 for $n = 3, 4, 6$ and larger for $5, 7, 8$

n	$\Phi_n(x)$	$\varphi(n)$	field and ring	a familiar subfield	$\Phi_n(1)$	compass
3	x^2+x+1	2	$\mathbb{Q}(\omega)$, $\mathbb{Z}[\omega]$ Eisenstein	itself	3	yes
4	x^2+1	2	$\mathbb{Q}(i)$, $\mathbb{Z}[i]$ Gaussian	itself	2	yes
5	$x^4+x^3+x^2+x+1$	4	$\mathbb{Q}(\zeta_5)$	$\mathbb{Q}(\sqrt{5})$, golden ratio	5	yes
6	x^2-x+1	2	$\mathbb{Q}(\omega)$, the same as $n=3$	itself	1	yes
7	$x^6+x^5+\cdots+1$	6	$\mathbb{Q}(\zeta_7)$	$\mathbb{Q}(\sqrt{-7})$ and a real cubic	7	no
8	x^4+1	4	$\mathbb{Q}(\zeta_8) = \mathbb{Q}(i,\sqrt{2})$	$\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$	2	yes

The value $\Phi_n(1)$ is the norm of $1-\zeta_n$, so $1-\zeta_6$ is a unit (its chord is the radius) while $1-\zeta_3$ and $1-\zeta_4$ have norms 3 and 2. Gauss showed that the polygon can be built with ruler and compass exactly when $\varphi(n)$ is a power of 2, which excludes only the heptagon here.

4 / GAUSSIAN AND EISENSTEIN INTEGERS / THE TWO LATTICES

The square arc $\frac{1}{4}$ gives $\mathbb{Z}[i]$ and the sixth arc $\frac{1}{6}$ gives $\mathbb{Z}[\omega]$, and their units are the polygons



In $\mathbb{Z}[i]$ the norm is a^2+b^2 and in $\mathbb{Z}[\omega]$ it is a^2-ab+b^2. A prime p splits according to $p \bmod n$: in $\mathbb{Z}[i]$ the prime $5 \equiv 1 \pmod 4$ splits as $(2+i)(2-i)$, 3 stays prime and $2 = -i(1+i)^2$ ramifies; in $\mathbb{Z}[\omega]$ the prime $7 \equiv 1 \pmod 3$ splits as $(3+\omega)(3+\omega^2)$ with norm $9-3+1=7$, while 2 and 5 stay prime and 3 ramifies.

5 / BEYOND THE LATTICES / UNITS AND THE CRYSTALLOGRAPHIC RESTRICTION

When $\varphi(n) > 2$ the ring is dense in the plane, and the extra units are the metallic means

For $n = 5, 7, 8$ the ring $\mathbb{Z}[\zeta_n]$ has rank $4, 6, 4$ over $\mathbb{Z}$ and fills the plane densely instead of forming a lattice, so the arcs $\frac{1}{5}, \frac{1}{7}, \frac{1}{8}$ cannot be rotation symmetries of a crystal, while $\frac{1}{3}, \frac{1}{4}, \frac{1}{6}$ can ($\varphi(n) \le 2$); the five-fold and eight-fold cases appear in quasicrystals. Their unit groups are infinite: $n=5$ has the golden ratio $\varphi = -(\zeta_5^2+\zeta_5^3) = 1.618$, $n=8$ has the silver ratio $1+\zeta_8+\zeta_8^{-1} = 1+\sqrt{2} = 2.414$, and $n=7$ has $2\cos\frac{2\pi}{7} = 1.247$, a root of t^3+t^2-2t-1, while for $n = 3, 4, 6$ the only units are the roots of unity themselves.

ASIDE / THE SAME FIELD TWICE

Since $-\omega^2 = \zeta_6$ the fields for $n=3$ and $n=6$ coincide, and in general $\mathbb{Q}(\zeta_n) = \mathbb{Q}(\zeta_{2n})$ for odd n.

ASIDE / CHECK IN PYTHON

`sympy.cyclotomic_poly(8,x)` gives x^4+1, and `prod(2*sin(pi*k/7))` over $k = 1..6$ gives 7.

Convention. $\omega = e^{2\pi i/3}$, $\varphi(n)$ is Euler's totient, Φ_n the cyclotomic polynomial. After the cyclotomic fields project and the earlier sheets on the totient and the Legendre symbol.

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Geometric

GEOMETRY ON ONE PAGE

From polygons to the circle

Inscribed polygons squeeze a circle from inside, circumscribed polygons from outside, and doubling the sides closes the gap.

1 / THE RULE AND THE PICTURE

A circle of radius 1 lies between its inscribed and circumscribed regular polygons



$n = 4$
$5.657 < 2\pi < 8.000$

$n = 8$
$6.123 < 2\pi < 6.627$

$n = 16$
$6.243 < 2\pi < 6.365$

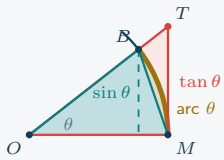
$n = 32$
$6.273 < 2\pi < 6.303$

The inscribed polygon (green) has its vertices on the circle and the circumscribed polygon (red) has its sides touching it, so the circle's circumference 2π is trapped between the two perimeters p_n and P_n, and the trap narrows as the number of sides n doubles:

$$p_n = 2n \sin \tfrac{\pi}{n} \;<\; 2\pi \;<\; 2n \tan \tfrac{\pi}{n} = P_n$$

2 / THE PROOF OF THE BOUNDS / ONE TRIANGLE AND ONE SECTOR

Comparing three areas shows $\sin\theta < \theta < \tan\theta$ for the half-angle $\theta = \pi/n$



Take one side of the polygon, seen from the centre O as half of the central angle $2\pi/n$. The green triangle OMB has area $\frac{1}{2}\sin\theta$, the gold sector has area $\frac{1}{2}\theta$, and the red triangle OMT has area $\frac{1}{2}\tan\theta$, and each region contains the one before it, so

$$\tfrac{1}{2}\sin\theta < \tfrac{1}{2}\theta < \tfrac{1}{2}\tan\theta.$$

Multiplying by $2n$ and using $n\theta = \pi$ turns the three areas into $p_n/2 < \pi < P_n/2$: the inscribed side is shorter than its arc, which is shorter than the tangent segment.

3 / THE PROGRESSION / DOUBLING WITH SQUARE ROOTS ONLY

Start with the hexagon, $p_6 = 6$ and $P_6 = 4\sqrt{3}$, and double the sides with two means

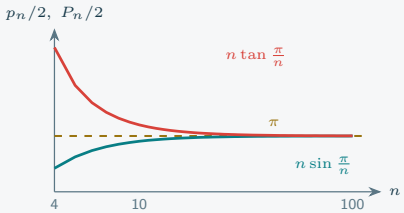
$$P_{2n} = \frac{2\,p_n P_n}{p_n + P_n} \text{ (harmonic mean)}, \qquad p_{2n} = \sqrt{p_n P_{2n}} \text{ (geometric mean)}$$

n	p_n	P_n	$p_n/2 < \pi < P_n/2$	gap $P_n - p_n$	gap ratio
6	6.00000	6.92820	$3.00000 < \pi < 3.46410$	0.9282	
12	6.21166	6.43078	$3.10583 < \pi < 3.21539$	0.2191	4.24
24	6.26526	6.31932	$3.13263 < \pi < 3.15966$	0.0541	4.05
48	6.27870	6.29217	$3.13935 < \pi < 3.14609$	0.0135	4.01
96	6.28206	6.28543	$3.14103 < \pi < 3.14271$	0.0034	4.00

The means reproduce $4n\tan\frac{\theta}{2}$ and $4n\sin\frac{\theta}{2}$ exactly, and at $n = 96$ the bounds sit inside Archimedes' $3\frac{10}{71} < \pi < 3\frac{1}{7}$, that is $3.1408 < \pi < 3.1429$. Each doubling cuts the gap by a factor of about 4.

4 / THE SQUEEZE / THE CIRCLE AS THE LIMITING CASE

Both perimeters tend to 2π, so the polygon with infinitely many sides is the circle



Writing $\theta = \pi/n$ we have $p_n = 2\pi \cdot \dfrac{\sin\theta}{\theta}$ and $P_n = 2\pi \cdot \dfrac{\tan\theta}{\theta}$, and as $n \to \infty$ the angle $\theta \to 0$ and both ratios tend to 1, so both perimeters tend to 2π. The ratio $P_n/p_n = 1/\cos\frac{\pi}{n}$ also tends to 1. The areas behave the same way, the inscribed area $\frac{n}{2}\sin\frac{2\pi}{n}$ and the circumscribed area $n\tan\frac{\pi}{n}$ both tending to π (giving $A = \pi r^2$), with 3.1394 and 3.1427 at $n = 96$.

5 / WHAT THE PROOF SHOWS / EXHAUSTION

A circle is never a polygon, and it is the one shape that every polygon sequence approaches

The argument needs only two facts, that a straight segment is the shortest path between two points and that the bounds close in on a single number, which is the definition of π. The perimeter $p_n = n\,|1 - \zeta_n|$ is n times the chord to the next root of unity, so the inscribed polygons here are those of the cyclotomic sheet, with arcs $\frac{1}{n}$ of the circle.

ASIDE / MEANS AGAIN

The doubling step uses a harmonic mean and then a geometric mean, the same two means H and G that measured the shapes of boxes.

ASIDE / CHECK IN PYTHON

`2*96*sin(pi/96)` gives 6.28206 and `2*96*tan(pi/96)` gives 6.28543.

Convention. Circle of radius 1, circumference 2π; p_n and P_n are perimeters of inscribed and circumscribed regular n-gons. After Archimedes' method of exhaustion.

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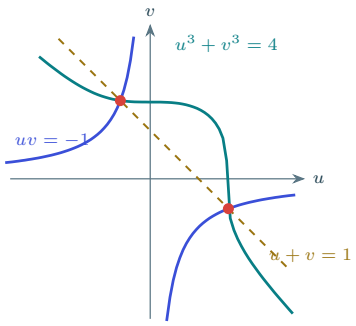
ALGEBRA ON ONE PAGE

Cardano's cubic as a hyperbola

Writing $t = u + v$ with $uv = -p/3$ turns the depressed cubic into the line and hyperbola of the quadratic, followed by a cube root.

1 / THE RULE AND THE PICTURE

The root of $t^3 + pt + q = 0$ is $t = u + v$ at a point on $uv = -\frac{p}{3}$ and on $u^3 + v^3 = -q$



Putting $t = u + v$ gives $t^3 = u^3 + v^3 + 3uv\,t$, that is $t^3 - 3uv\,t - (u^3 + v^3) = 0$, which is the reduced cubic when

$$\boxed{uv = -\frac{p}{3}, \qquad u^3 + v^3 = -q}$$

For $t^3 + 3t - 4 = 0$ the hyperbola is $uv = -1$ and the curve is $u^3 + v^3 = 4$, and they meet at $(\varphi, -\frac{1}{\varphi})$ and its mirror image, both on the line $u + v = 1$, so the root is $t = \varphi - \frac{1}{\varphi} = 1$. The two real points are mirror images in $u = v$, and the four other crossings of a conic with a cubic are complex.

2 / THE TWO LAYERS / A QUADRATIC, THEN A CUBE ROOT

Cubing sends the hyperbola $uv = m$ to $z_1 z_2 = m^3$, so u^3 and v^3 solve the line and hyperbola quadratic

step	rule	$t^3 + 3t - 4 = 0$	$t^3 - 3t - 1 = 0$
1. constants	$m = -\frac{p}{3},\ c = -q$	$m = -1,\ c = 4$	$m = 1,\ c = 1$
2. resolvent	$z^2 - cz + m^3 = 0$	$z^2 - 4z - 1 = 0$	$z^2 - z + 1 = 0$
3. roots	$z_{1,2} = \frac{c}{2} \pm \sqrt{\frac{c^2}{4} - m^3}$	$2 \pm \sqrt{5}$	$e^{\pm i\pi/3}$
4. cube roots	$u = \sqrt[3]{z_1},\ v = m/u$	$u = \varphi,\ v = -\frac{1}{\varphi}$	$u = e^{i\pi/9},\ v = e^{-i\pi/9}$
5. the root	$t = u + v$	1	$2\cos 20^\circ = 1.879$

Steps 2 and 3 are the quadratic sheet with $s = c$ and $P = m^3$, a line $z_1 + z_2 = c$ meeting the hyperbola $z_1 z_2 = m^3$, and step 4 is a cube root chosen so that $uv = m$, which uses the hyperbola to pair the two cube roots correctly.

3 / THE DISCRIMINANT / THREE CASES

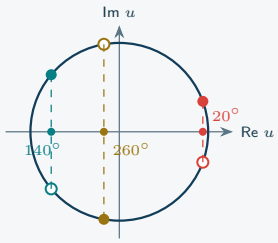
The line $z_1 + z_2 = c$ meets $z_1 z_2 = m^3$ when $c^2 \geq 4m^3$, which is $\frac{q^2}{4} + \frac{p^3}{27} \geq 0$

case	line against hyperbola	the cubic	example
> 0	two real crossings	one real root, two complex	$t^3 + 3t - 4$: $1,\ -\frac{1}{2} \pm \frac{\sqrt{15}}{2}i$
$= 0$	tangent, $u = v$	a repeated root	$t^3 - 3t + 2$: $u = v = -1$, roots $-2, 1, 1$
< 0	no real crossing	three real roots	$t^3 - 3t - 1$: $1.879,\ -0.347,\ -1.532$

The third row is the casus irreducibilis, where Cardano's formula needs the cube root of a complex number although every root is real, and the hyperbola picture shows why: the six crossings of $uv = m$ with $u^3 + v^3 = c$ all leave the real plane.

4 / THREE REAL ROOTS / THE HYPERBOLA BECOMES A CIRCLE

For $m = 1$ the points $(u, v) = (e^{i\theta}, e^{-i\theta})$ give $t = 2\cos\theta$ and $t^3 - 3t = 2\cos 3\theta$



When the line misses the hyperbola in the z-plane the numbers u^3 and v^3 are complex conjugates of modulus $m^{3/2}$, so $u = \sqrt{m}\,e^{i\theta}$ and $v = \sqrt{m}\,e^{-i\theta}$ lie on a circle and $t = u + v = 2\sqrt{m}\cos\theta$ is real. Then $t^3 - 3mt = 2m^{3/2}\cos 3\theta$ is the triple angle formula, so solving $t^3 - 3t = 1$ asks for $\cos 3\theta = \frac{1}{2}$, whose solutions $\theta = 20^\circ,\ 140^\circ,\ 260^\circ$ give the roots

$$t_k = 2\sqrt{m}\cos\frac{\theta_0 + 2\pi k}{3}, \text{ i.e. } 1.879,\ -1.532,\ -0.347$$

The open points are the conjugate partners (v, u), and each root is the common real part of a pair.

5 / THE OTHER ROOTS / ROTATING ALONG THE HYPERBOLA

Multiplying u by ω and v by ω^2 keeps uv fixed, which gives all three roots

There are nine sums $\omega^i u + \omega^j v$, and only the three with $i + j \equiv 0 \pmod 3$ keep $uv = m$, namely $u + v$, $\omega u + \omega^2 v$ and $\omega^2 u + \omega v$. For $t^3 + 3t - 4 = 0$ these are 1 and $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}(u - v)i = -\frac{1}{2} \pm \frac{\sqrt{15}}{2}i$, since $u - v = \sqrt{5}$, and their sum is 0 and their product $4 = -q$, as it must be.

ASIDE / VIETA'S SUBSTITUTION

Setting $t = u + \frac{m}{u}$ is Vieta's version of the trick, and $u^3 + \frac{m^3}{u^3} = t^3 - 3mt$ is a Dickson polynomial.

ASIDE / CHECK IN PYTHON

`np.roots([1,0,3,-4])` gives $1,\ -0.5 \pm 1.936j$, and $\varphi - 1/\varphi = 1$.

Convention. The reduced cubic is $t^3 + pt + q = 0$, $m = -p/3$, $c = -q$, $\omega = e^{2\pi i/3}$. After the earlier sheets on the quadratic as a line meeting a hyperbola.

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TWO CONICS, ONE PENCIL OF LINES ON ONE PAGE

Triples on the circle and the hyperbola

The same rational slope through $(-1,0)$ meets both curves, and both meetings read the same triple.

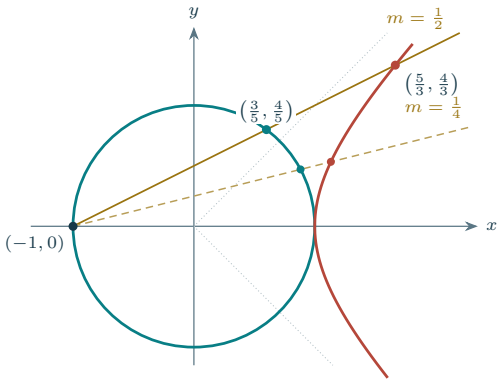
THE RULE

Divide $a^2+b^2=c^2$ by c^2, or by a^2

Lines $y=m(x+1)$ with $m=\frac{p}{q}$ all pass through $(-1,0)$, which lies on both the unit circle and the unit hyperbola. With $a=q^2-p^2$, $b=2pq$ and $c=p^2+q^2$, so that $a^2+b^2=c^2$, the second meetings are

circle $x^2+y^2=1$: $\left(\frac{a}{c},\frac{b}{c}\right)$ hyperbola $x^2-y^2=1$: $\left(\frac{c}{a},\frac{b}{a}\right)$

The circle point is the triple divided by its hypotenuse and the hyperbola point is the same triple divided by a leg, since $(\frac{a}{c})^2+(\frac{b}{c})^2=1$ and $(\frac{c}{a})^2-(\frac{b}{a})^2=1$ are two rearrangements of one identity. The slope $\frac{1}{2}$ gives $(\frac{3}{5},\frac{4}{5})$ and $(\frac{5}{3},\frac{4}{3})$, both carrying $3,4,5$.



One line, two meetings: $3,4,5$ from $m=\frac{1}{2}$ and $15,8,17$ from $m=\frac{1}{4}$; dotted asymptotes $y=\pm x$.

ON THE CIRCLE

The known root does the work

Substituting $y=m(x+1)$ into $x^2+y^2=1$ gives

$$(1+m^2)x^2+2m^2x+(m^2-1)=0.$$

One root is the base point $x=-1$, and the product of the roots is $\frac{m^2-1}{1+m^2}$, so the other root is rational:

$$x=\frac{1-m^2}{1+m^2},\qquad y=\frac{2m}{1+m^2}.$$

Every slope gives a point, and every rational point except $(-1,0)$ comes from exactly one rational slope.

ON THE HYPERBOLA

The same move with one sign changed

Substituting into $x^2-y^2=1$ gives

$$(1-m^2)x^2-2m^2x-(1+m^2)=0,$$

and dividing the product of roots by the known root -1 leaves

$$x=\frac{1+m^2}{1-m^2},\qquad y=\frac{2m}{1-m^2}.$$

Slopes with $|m|<1$ land on the right-hand branch; steeper ones land back on the left-hand branch that holds $(-1,0)$.

SIDE BY SIDE

Circular and hyperbolic trigonometry from the half-angle slope

The slope is a half-angle tangent in both cases, $m=\tan\frac{\theta}{2}$ on the circle and $m=\tanh\frac{\varphi}{2}$ on the hyperbola, which is why the coordinates are the half-angle formulae for $\cos,\sin$ and for $\cosh,\sinh$.

	unit circle	unit hyperbola
curve and invariant	$x^2+y^2=1$, distance	$x^2-y^2=1$, spacetime interval
point from slope m	$(\cos\theta,\sin\theta)=\left(\frac{1-m^2}{1+m^2},\frac{2m}{1+m^2}\right)$	$(\cosh\varphi,\sinh\varphi)=\left(\frac{1+m^2}{1-m^2},\frac{2m}{1-m^2}\right)$
slopes that work	every m	$\lvert m\rvert<1$
triple divided by	the hypotenuse c	the leg $a=q^2-p^2$
symmetry	rotations $\left(\begin{smallmatrix}\cos & -\sin\\ \sin & \cos\end{smallmatrix}\right)$	Lorentz boosts $\left(\begin{smallmatrix}\cosh & \sinh\\ \sinh & \cosh\end{smallmatrix}\right)$
combining two triples	angles add: $(3+4\mathrm{i})(5+12\mathrm{i})=-33+56\mathrm{i}$	rapidities add: $\frac{4}{5}\oplus\frac{3}{5}=\frac{35}{37}$

Every triple is a rational boost

On the hyperbola $\cosh\varphi=\gamma$ and $\tanh\varphi=\beta=v/c$, so each triple gives a speed with a rational Lorentz factor: $m=\frac{1}{2}$ gives $\beta=\frac{4}{5}$, $\gamma=\frac{5}{3}$, and $m=\frac{2}{3}$ gives $\beta=\frac{12}{13}$, $\gamma=\frac{13}{5}$. Relativistic velocity addition, $\beta_1\oplus\beta_2=\frac{\beta_1+\beta_2}{1+\beta_1\beta_2}$, sends $\frac{4}{5}$ and $\frac{3}{5}$ to $\frac{35}{37}$, the triple $12,35,37$.

When the base point leaves the curve

Both constructions work because $(-1,0)$ is on the curve, so the known rational root forces the other to be rational. Cut $xy=1$ with $y=x-k$ instead and no root is known in advance: $x^2-kx-1=0$ gives the metallic means $\frac{1}{2}\left(k+\sqrt{k^2+4}\right)$, the golden ratio at $k=1$ and the silver ratio $1+\sqrt{2}$ at $k=2$.

Conventions. Coprime $p<q$ of opposite parity give primitive triples; if both are odd, halve the sides, so $m=\frac{1}{3}$ gives $8,6,10$ and hence $4,3,5$.

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SPIRALS ON ONE PAGE

Spirals in parametric form

The same pair $(r\cos t,\ r\sin t)$ draws every spiral, and the gap between successive turns measures how tightly each one is wound.

1 / THE RULE AND THE PICTURE

A spiral is $(x, y) = (r(t)\cos t,\ r(t)\sin t)$, and only the function $r(t)$ changes from spiral to spiral



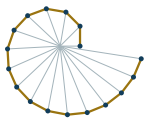
Archimedean
$r = at$



Fermat
$r = a\sqrt{t}$



logarithmic (golden)
$r = ae^{bt}$



Theodorus
$r_n = \sqrt{n}$

The Theodorus (square root) spiral stacks right triangles with a leg of 1, so its radii are the hypotenuses $\sqrt{2}, \sqrt{3}, \sqrt{4}, \ldots$; the first sixteen triangles are drawn, and the others show three turns of t.

2 / THE PARAMETRIC FORMS / ONE PARAMETER, TWO COORDINATES

Writing x and y separately in terms of t lets a curve wind round the origin as often as it likes

spiral	$r(t)$	$x(t)$	$y(t)$
Archimedean	at	$at\cos t$	$at\sin t$
Fermat	$a\sqrt{t}$	$a\sqrt{t}\cos t$	$a\sqrt{t}\sin t$
logarithmic	ae^{bt}	$ae^{bt}\cos t$	$ae^{bt}\sin t$
Theodorus	$\sqrt{n}$ at $t_n = \sum_{k<n}\arctan\frac{1}{\sqrt{k}}$	$\operatorname{Re} z_n$	$\operatorname{Im} z_n$, with $z_{n+1} = z_n(1+\frac{i}{\sqrt{n}})$

No equation $y = f(x)$ describes these curves, since a vertical line meets them many times, but the parameter t records how far round we are. The Theodorus rule is a complex multiplication: a step of length 1 at a right angle, $i\,z_n/|z_n|$, multiplies z_n by $1 + i/\sqrt{n}$ and takes the modulus from $\sqrt{n}$ to $\sqrt{n+1}$.

3 / PLOTTING FROM THE PARAMETRIC FORM / FIVE POINTS OF EACH SPIRAL

Choose $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ and evaluate $(r\cos t,\ r\sin t)$

	$t=0$	$t=\frac{\pi}{2}$	$t=\pi$	$t=\frac{3\pi}{2}$	$t=2\pi$
Archimedean, $r=\frac{t}{2}$	$(0,\ 0)$	$(0,\ 0.785)$	$(-1.571,\ 0)$	$(0,\ -2.356)$	$(3.142,\ 0)$
Fermat, $r=\sqrt{t}$	$(0,\ 0)$	$(0,\ 1.253)$	$(-1.772,\ 0)$	$(0,\ -2.171)$	$(2.507,\ 0)$
golden, $r=e^{bt}$	$(1,\ 0)$	$(0,\ \varphi)$	$(-\varphi^2,\ 0)$	$(0,\ -\varphi^3)$	$(\varphi^4,\ 0)$

The golden spiral lands on powers of $\varphi = 1.618$ at each quarter turn because $e^{b\pi/2} = \varphi$, while the Archimedean radii add a constant 0.785 and the Fermat radii add less each time.

4 / THREE MEASURES OF TIGHTNESS / GAP, RATIO AND PITCH

The gap $g = r(t+2\pi) - r(t)$, the ratio $\rho = r(t+2\pi)/r(t)$ and the pitch $\tan\psi = r'/r$

spiral	gap g	ratio ρ	pitch $\tan\psi$	verdict
Archimedean	$2\pi a$	$\to 1$	$\frac{1}{t} \to 0$	constant spacing
Fermat	$\approx \frac{\pi a}{\sqrt{t}} \to 0$	$\to 1$	$\frac{1}{2t} \to 0$	ever tighter packing
logarithmic	$r(e^{2\pi b} - 1)$ grows	$e^{2\pi b}$ constant	b constant	self-similar
Theodorus	$\to \pi$	$\to 1$	$\to 0$	Archimedean for large n

The golden spiral has $b = \frac{2}{\pi}\ln\varphi = 0.3063$ and $\rho = \varphi^4 = 6.854$ per turn. Since $\theta_n \approx 2\sqrt{n} - 2.158$ for the Theodorus spiral, $r \approx \frac{t}{2} + 1.08$, an Archimedean spiral with $a = \frac{1}{2}$ and gap π.

5 / THE NUMBERS / GAP BETWEEN TURN k AND TURN $k+1$

The Theodorus gaps settle on π, the Fermat gaps shrink and the golden gaps grow sevenfold per turn

gap between turns	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
Archimedean, $r=\theta/2$	3.142	3.142	3.142	3.142	3.142
Theodorus, $r_n=\sqrt{n}$	3.150	3.145	3.143	3.143	3.142
Fermat, $r=\sqrt{\theta}$	1.038	0.797	0.672	0.592	0.535
golden, $r=e^{b\theta}$	40.125	275	1 885	12 920	88 555

The Theodorus gaps are interpolated between vertices at $t = 2\pi k$, and its first turn nearly closes at $n = 17$, at 6.129 radians.

6 / WHICH TIGHTNESS IS THE RIGHT ONE / WHAT EACH SPIRAL MODELS

The measure to use depends on what the spiral is for

A record groove keeps the constant Archimedean gap $2\pi a$, a sunflower places seed n at radius $c\sqrt{n}$ and angle $n \times 137.5°$ (Fermat's spiral at the golden angle, every seed equally crowded), and a nautilus grows by a fixed ratio per turn, the logarithmic spiral, which looks the same at every scale.

ASIDE / WHY PARAMETRIC

The parameter t is the angle here, and arc length is another choice of parameter for the same curves.

ASIDE / CHECK IN PYTHON

Iterating `z *= 1+1j/sqrt(k)` for $k = 1..16$ gives $|z| = \sqrt{17} = 4.123$ and `angle(z)` $= 6.129$ radians.

Convention. t is the angle in radians, $b = \frac{2}{\pi}\ln\varphi$; gaps use $a = \frac{1}{2}$ (Archimedean) and $a = 1$ (Fermat, golden).

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THE ISOPERIMETRIC PROBLEM IN A PRIME BOX ON ONE PAGE

$2 \times 3 \times 5$ and the harmonic mean

A box's surface per unit volume is set by the harmonic mean of its sides, and its departure from a cube by how far that mean falls below the geometric one.

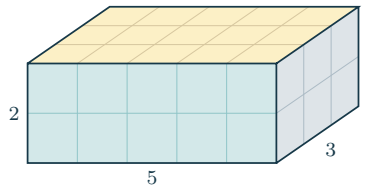
THE RULE

Surface to volume is six over the harmonic mean

A box with sides a, b, c has $V = abc$ and $S = 2(ab + bc + ca)$, so dividing turns the products into reciprocals. With the harmonic mean $H = 3/(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$ and the geometric mean $G = \sqrt[3]{abc}$,

$$\frac{S}{V} = 2\Big(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\Big) = \frac{6}{H}, \qquad \frac{S}{V^{2/3}} = 6\frac{G}{H} \geq 6$$

The first says every box has the surface-to-volume ratio of a cube of side H. The second, which does not change when the box is scaled, measures shape alone, and since $H \leq G$ with equality only when $a = b = c$, the cube uses the least surface for its volume among all boxes.



30 unit cubes behind 62 unit faces: $S/V = \frac{62}{30} = \frac{6}{H}$ with $H = \frac{90}{31}$.

THE SMALLEST PRIME BOX

$2 \times 3 \times 5$, the first box with three distinct prime sides

Its three means fall in the order the mean inequality demands,

$$H = \tfrac{90}{31} \approx 2.903 \;<\; G = \sqrt[3]{30} \approx 3.107 \;<\; A = \tfrac{10}{3} \approx 3.333,$$

and the shape measure is $G/H \approx 1.070$, so the box needs 7.0% more surface than a cube of the same volume, 62 against 57.93. Among the five boxes that 30 unit cubes can make, $S = 6V/H$ with V fixed means the largest harmonic mean wins, and the all-prime box wins outright.

In general $1 \times p \times qr$ loses to $p \times q \times r$ because their surfaces differ by $2p(q-1)(r-1) > 0$.

box	S	H
$1 \times 1 \times 30$	122	1.475
$1 \times 2 \times 15$	94	1.915
$1 \times 3 \times 10$	86	2.093
$1 \times 5 \times 6$	82	2.195
$2 \times 3 \times 5$	62	2.903

Every box of volume 30: surface falls as the harmonic mean rises.

OTHER PRIME BOXES

Close primes make nearly cubic boxes, but never a cube

Numbers pqr with three distinct prime factors are the **sphenic** numbers, and each builds exactly one box with prime sides. Because the sides always differ, G/H is always strictly above 1, and it approaches 1 only as the three primes crowd together relative to their size.

box	2×3×5	2×3×7	2×5×7	2×3×11	3×5×7	5×7×11	11×13×17	101×103×107
S/V	2.067	1.952	1.686	1.848	1.352	0.868	0.453	0.058
G/H	1.070	1.131	1.158	1.245	1.063	1.052	1.016	1.0003

Absolute size and shape separate cleanly: S/V falls steadily as boxes grow, while G/H ranks them by shape, so $3 \times 5 \times 7$ is rounder than $2 \times 3 \times 5$ even though both use three consecutive primes.

Why the harmonic mean

The harmonic mean averages quantities that add as reciprocals. Driving equal distances at 30 and 60 km/h gives an average speed of 40, the harmonic mean, because the times $1/30$ and $1/60$ per kilometre are what add. Each pair of faces contributes $2/a$, $2/b$ or $2/c$ of surface per unit volume, so the box's sides combine in the same way.

The isoperimetric problem

Among all solids of a given volume the sphere has least surface. The quotient $Q = 36\pi V^2/S^3$ equals 1 for a sphere and for a box becomes $Q = \frac{\pi}{6}(H/G)^3$, giving 0.524 for a cube, 0.427 for $2 \times 3 \times 5$ and 0.523 for $101 \times 103 \times 107$. In the plane, a $p \times q$ rectangle has $P/A = 4/H$ and the circle plays the sphere's part.

Conventions. For positive a, b, c the three means are $H = 3/(\frac{1}{a} + \frac{1}{b} + \frac{1}{c})$, $G = \sqrt[3]{abc}$ and $A = \frac{1}{3}(a + b + c)$, and they always satisfy $H \leq G \leq A$, with equality throughout only when $a = b = c$. Lengths are in units of one cube edge, so S counts unit faces and V counts unit cubes.

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ISOPERIMETRIC BALANCE ON ONE PAGE

Two ratios, one best rectangle

A product of shares connects fixed perimeter, three classical means, cos and cosh.

The combination: multiply the two shares

For $a, b > 0$, take their shares of the same total:

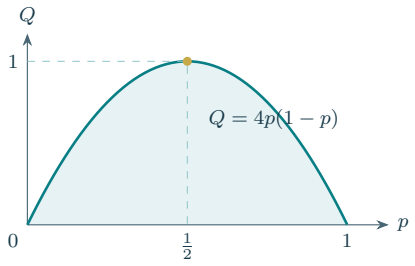
$$p = \frac{a}{a+b}, \qquad q = \frac{b}{a+b}, \qquad p + q = 1.$$

Multiply and normalise the product to have maximum 1:

$$\boxed{Q = 4pq = \frac{4ab}{(a+b)^2} = 1 - (p-q)^2 \le 1.}$$

Maximum at $a = b$; the loss $1 - Q$ is squared imbalance.

Balance wins



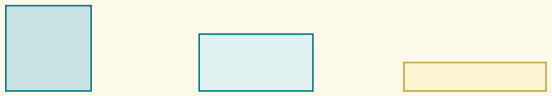
Geometry: fix the perimeter, maximise the area

A rectangle has perimeter $P = 2(a+b)$ and area $S = ab$. Thus

$$Q = \frac{16S}{P^2} \le 1, \qquad S \le \frac{P^2}{16}.$$

Equality gives the square: $a = b = P/4$.

Each rectangle has $P = 12$.



$(a, b) = (3, 3)$, $S = 9$, $Q = 1$ — $(4, 2)$, $S = 8$, $Q = \frac{8}{9}$ — $(5, 1)$, $S = 5$, $Q = \frac{5}{9}$

The same score, written through the three means

$$A = \frac{a+b}{2}, \qquad G = \sqrt{ab}, \qquad H = \frac{2ab}{a+b}.$$

$$\boxed{Q = \frac{H}{A} = \left(\frac{G}{A}\right)^2 = \frac{16S}{P^2}, \qquad AH = G^2.}$$

Hyperbolic: positive ratios

For $x = \frac{1}{2}\ln(a/b)$, write $a = \ell e^{x}$, $b = \ell e^{-x}$, with $\ell = \sqrt{ab}$.

$$A = \ell \cosh x, \quad G = \ell, \quad H = \ell \operatorname{sech} x.$$

The imbalance is $p - q = \tanh x$, so

$$\boxed{Q = \operatorname{sech}^2 x = 1 - \tanh^2 x.}$$

At fixed P, the scale adjusts to $\ell = P/(4\cosh x)$:

$$S = \frac{P^2}{16}\operatorname{sech}^2 x.$$

Circular: the same algebra

Use the complex pair $a = \ell e^{ix}$, $b = \ell e^{-ix}$, with real x and $\ell > 0$. Formally,

$$A = \ell \cos x, \quad G = \ell, \quad H = \ell \sec x.$$

Now $p - q = i \tan x$. The identity $Q = 1 - (p-q)^2$ gives

$$\boxed{Q = \sec^2 x = 1 + \tan^2 x.}$$

For these complex shares, $Q \le 1$ need not hold. This is an algebraic extension, defined when $\cos x \neq 0$.

If the two contributions carry different weights

For positive weights α, β, fix the weighted budget $C = \alpha a + \beta b$ and use

$$p_w = \frac{\alpha a}{C}, \qquad q_w = \frac{\beta b}{C}, \qquad Q_w = 4p_w q_w = \frac{4\alpha\beta S}{C^2} \le 1.$$

Thus $S_{\max} = \dfrac{C^2}{4\alpha\beta}$, attained when $\boxed{\alpha a = \beta b}$, or $a/b = \beta/\alpha$. The optimum balances the **weighted contributions**.

Normalisation. $Q = 16S/P^2$ is normalised to the best rectangle, the square. The usual planar quotient is $4\pi S/P^2 = (\pi/4)Q$, so it equals $\pi/4$ for a square. Except in the circular panel, all lengths and weights are positive real numbers.

Geometric

Information, size and rank

Identifying a cuboid, ordering its volume and comparing its shape ask different questions.

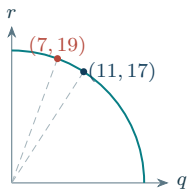
1 / WHAT DOES ONE DIAGONAL TELL US?

One measurement leaves many possible cuboids

Knowing D fixes $p^2+q^2+r^2$, while capacity depends on the product pqr. Even with three distinct prime edges,

(p,q,r)	D^2	V
$(2,7,19)$	414	266
$(2,11,17)$	414	374

The same diagonal can accompany different volumes. AM–GM gives a bound $V \le (D/\sqrt{3})^3$, attained by a cube, rather than a unique volume. An ordering by diagonal therefore need not be the ordering by capacity. In the diagram, $p=2$ and moving along the arc changes qr while keeping D fixed.



$p=2$ fixed
$q^2+r^2=410$

2 / COMPARE ORDERS FOR A DECLARED PURPOSE

Capacity and reach; surface cost and relative reach

The first eight prime-sided bricks by volume are listed below. Rank 1 means smallest; every rank is computed before rounding. Here $I = S/(6V^{2/3})$ and $J_V = D/(\sqrt{3}V^{1/3})$.

row	edges	V	D	r_D	I	r_I	J_V	r_J
1	$(2,3,5)$	30	6.164	1	1.0703	2	1.1454	2
2	$(2,3,7)$	42	7.874	2	1.1311	3	1.3078	4
3	$(2,3,11)$	66	11.576	5	1.2450	5	1.6538	6
4	$(2,5,7)$	70	8.832	3	1.1579	4	1.2372	3
5	$(2,3,13)$	78	13.491	7	1.2964	7	1.8230	7
6	$(2,3,17)$	102	17.378	8	1.3895	8	2.1474	8
7	$(3,5,7)$	105	9.110	4	1.0634	1	1.1149	1
8	$(2,5,11)$	110	12.247	6	1.2632	6	1.4758	5

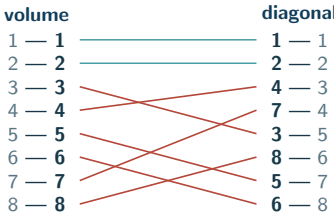
Rows $3,4$ show a volume increase $66 \to 70$ with a diagonal decrease $11.576 \to 8.832$. Likewise rows $6,7$ go from volume 102 to 105 while the diagonal almost halves. Size has several meanings.

For shape, I and J_V share a volume reference but measure different things. Expanding $E=(p-q)^2+(q-r)^2+(r-p)^2$ gives $2D^2 = S+E$, hence $J_V^2 = I + E/(6V^{2/3})$. This extra term measures edge inequality and explains the reversed relative-diagonal order for rows $2,4$ and rows $3,8$.

3 / READ THE RANKING DIAGRAMS

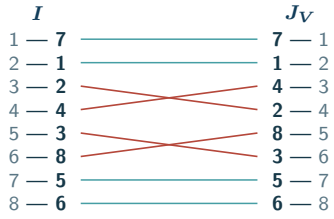
Each line follows one row; a crossing reverses a pair

Volume rank versus diagonal rank



6 reversed pairs; 22 agree.

Surface-cost rank versus J_V rank



2 reversed pairs; 26 agree.

Bold numbers identify rows; grey numbers give ranks. There are $\binom{8}{2}=28$ pairs. Kendall's τ uses these pair counts; Spearman's ρ uses rank displacements, whose squared sums here are 26 and 4, respectively. The two comparisons answer different questions.

4 / INFORMATION AND ORDER

Mutual information needs a specified population

For discrete variables, write H for uncertainty in bits and use MI to distinguish mutual information from surface cost I: $\mathrm{MI}(V;D) = H(V) - H(V \mid D)$.

If the only possibilities are the two equal-diagonal bricks above, chosen equally, D removes none of the 1 bit of uncertainty about volume: $\mathrm{MI}=0$. If instead one of the eight table rows is chosen equally, every diagonal identifies its row: $H(V \mid D)=0$ and $\mathrm{MI}=\log_2 8 = 3$ bits, even with six reversed pairs. This is catalogue lookup; it does not establish a volume rule for other cuboids. **Mutual information measures uncertainty removed; rank correlation measures agreement of order.**

Information definition: MIT 6.441, Chapter 2, Theorem 2.4. Cuboid examples and rank counts calculated from the stated edges.

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NORMALISATION AND CUBOIDS · 1 OF 2

Normalisation separates size from shape

Choose a common unit of comparison, then keep track of what the ratio preserves.

1 / DECIDE WHAT WE MEAN BY BIGGER

Capacity, surface and reach are different measurements

For edges $p, q, r > 0$, volume measures capacity, surface area measures the boundary, and the longest diagonal measures corner-to-corner reach:

$$V = pqr, \qquad S = 2(pq + qr + rp), \qquad D = \sqrt{p^2 + q^2 + r^2}.$$

Doubling every edge multiplies D, S, V by $2, 4, 8$. Consequently S/V halves: it measures **surface per unit capacity**, including the effect of size. It is useful for that purpose, and at fixed volume it gives the surface-area order. To compare *shape alone*, we need a score unchanged by enlarging an individual cuboid.

2 / THE OPERATION OF NORMALISING

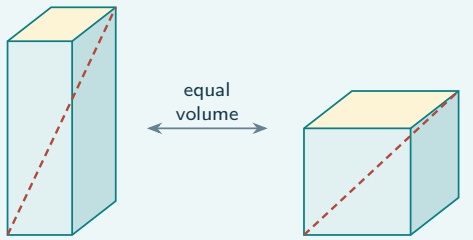
Divide every edge by the same reference length

Choose $G = V^{1/3}$, the side of the equal-volume cube. Measuring all lengths in units of G gives

$$(x, y, z) = \left(\frac{p}{G}, \frac{q}{G}, \frac{r}{G}\right), \qquad xyz = 1.$$

The edge ratios are preserved and every cuboid now has unit volume. A cube becomes $(1, 1, 1)$.

For the example $(2, 3, 6)\,\mathrm{cm}$, $G = \sqrt[3]{36}\,\mathrm{cm}$. The drawing compares its normalised shape with the reference cube.



normalised cuboid
$\approx (0.606, 0.909, 1.817)$
volume 1, diagonal ≈ 2.120

reference cube
$(1, 1, 1)$
volume 1, diagonal $= \sqrt{3}$

Dashed lines are body diagonals. Both drawings use the same normalised scale, so the comparison now describes shape.

3 / MAKE THE REFERENCE CUBE SCORE ONE

I normalises surface area; J_V normalises diagonal length

Areas divide by G^2 and lengths by G. The unit cube has surface area 6 and diagonal $\sqrt{3}$, which explain the final factors:

	original	after dividing edges by G	relative to the unit cube
surface	S	S/G^2	$I = \frac{S}{6G^2}$
diagonal	D	D/G	$J_V = \frac{D}{\sqrt{3}\,G}$

For the example, $S/G^2 \approx 6.604$ and $D/G \approx 2.120$. Thus $I \approx 1.101$ means 10.1% **extra surface**, while $J_V \approx 1.224$ means a 22.4% **longer diagonal**, each compared with the cube of the same volume. Among cuboids both scores are at least 1, with equality exactly for a cube.

4 / CHOOSING AND RETAINING THE SCALE

A different constraint gives a different useful ratio

For a fixed surface budget, choose the equal-surface-area cube instead. Its side is $\ell_S = \sqrt{S/6}$ and its diagonal is $\sqrt{S/2}$:

$$J_S \quad = \quad \frac{D}{\sqrt{S/2}} \quad = \quad \frac{J_V}{\sqrt{I}}.$$

For the same $2 \times 3 \times 6$ cuboid, $D = 7$ and $S = 72$, so $J_S = 7/6 \approx 1.167$: its diagonal is 16.7% longer than that of the cube with the same surface area.

Normalisation separates information about size from information about shape. Keeping G with (x, y, z) allows full reconstruction: $p = Gx$, $q = Gy$, $r = Gz$. Reporting only J_V compresses shape into one number and also omits the absolute scale. Restore that scale using

$$D = \sqrt{3}\,G J_V, \qquad V = G^3 \quad \Longrightarrow \quad V = \left(\frac{D}{\sqrt{3}\,J_V}\right)^3.$$

So J_V alone gives relative reach; D together with J_V determines volume.

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RESISTORS, KIRCHHOFF AND SQUARED RECTANGLES ON ONE PAGE

Every square is a one-ohm resistor

Parallel resistance is a harmonic mean, and Kirchhoff's laws for a network of equal resistors tile a rectangle with squares.

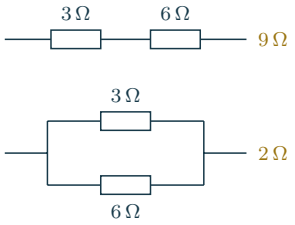
THE RULE

Series adds resistances, parallel adds their reciprocals

In series the same current passes through each resistor and the voltages add; in parallel each resistor feels the same voltage and the currents add, so it is the conductances $1/R$ that add. For n resistors the two totals are means in disguise:

$$R_{\text{series}} = \sum R_i = n \times \text{AM}, \qquad \frac{1}{R_{\text{parallel}}} = \sum \frac{1}{R_i}, \quad R_{\text{parallel}} = \frac{\text{HM}}{n}$$

So $3\,\Omega$ and $6\,\Omega$ give $9\,\Omega$ in series and $2\,\Omega$ in parallel, half their harmonic mean of $4\,\Omega$; equal resistors R give nR and R/n.



Series: voltages add.
Parallel: currents add.

KIRCHHOFF'S LAWS IN A TILING

Height is voltage, width is current

Kirchhoff's current law says the currents into any junction sum to the currents out, and his voltage law says the drops around any closed loop sum to zero. A rectangle of uniform conducting sheet, with current entering along its top and leaving along its bottom, has resistance $R = h/w$, so every square, of any size, is exactly $1\,\Omega$, with voltage equal to its height and current equal to its width. Squares side by side share their top and bottom edges and sit in parallel; squares stacked one on another sit in series. Each horizontal segment of a tiling becomes a junction at potential equal to its height, and each square becomes a $1\,\Omega$ resistor joining the segment on its top to the segment beneath it.

MOROŃ'S RECTANGLE, 1925

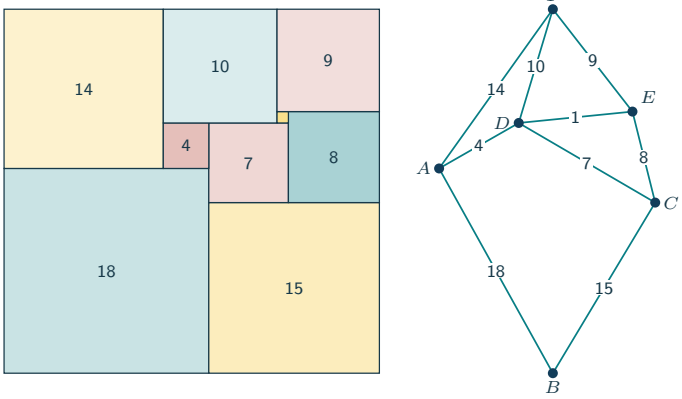
Nine squares, nine resistors

The 33×32 rectangle is tiled by squares of sides $18, 15, 14, 10, 9, 8, 7, 4, 1$, all different. Reading it as a network gives six junctions, the top T at height 32, the bottom B at 0, and four segments at heights $18, 15, 22, 23$, joined by nine $1\,\Omega$ resistors whose currents are the side lengths. Kirchhoff's current law holds at every junction because the widths above a segment fill the same length as the widths below it; at D, for instance,

$$10 + 1 = 4 + 7.$$

The whole network carries 33 units of current under a drop of 32, so its equivalent resistance is

$$R_{\text{eq}} = \frac{\text{height}}{\text{width}} = \frac{32}{33}\,\Omega$$



The tiling and its network, drawn with each junction at the height of its segment.

RUNNING IT BACKWARDS

Draw a network, solve it, get a tiling

Brooks, Smith, Stone and Tutte turned the dictionary round in 1940: choose a planar network of equal resistors, solve Kirchhoff's laws, and the currents are the sides of a squared rectangle. Feeding in a current equal to the number of spanning trees, 66 for Moroń's network, makes every current a whole number, here twice each side; distinct currents give a perfect rectangle, and the method went on to find perfect squared squares, down to Duijvestijn's 21 squares of 1978.

WHY THE CURRENTS ARE UNIQUE

Least energy picks the flow

Among all flows that respect the current law, the one Kirchhoff's voltage law selects is the one that dissipates the least power $\sum I^2 R$, Thomson's principle. With every $R = 1$ and the total current fixed at 33, the squares minimise the sum of their areas, $\sum s^2 = 33 \times 32 = 1056$, against every other split; a linear programme searches the corners of the same set of flows instead of its energy minimum.

Conventions. R is resistance in ohms, conductance is $1/R$, AM and HM are the arithmetic and harmonic means of the n resistances, and a squared rectangle is perfect when all its squares differ in size.

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GAUSSIAN ELIMINATION ON ONE PAGE

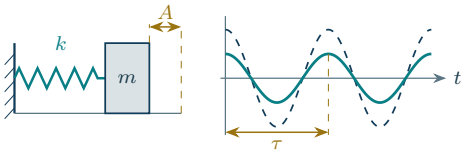
The period of a stiff spring

Dimensional analysis as a linear system, solved with every pivot and row swap in view, delivering $\tau \propto \sqrt{m/k}$.

1 / THE RULE AND THE PICTURE

Units become a matrix, and its solutions become formulas

A product $\Pi = q_1^{x_1} \cdots q_n^{x_n}$ has no units exactly when the exponents of mass, length and time all cancel, which is the linear system $\boxed{A\mathbf{x} = \mathbf{0}}$ whose column j holds the unit vector of q_j, and Buckingham's theorem counts its solutions: $n - \operatorname{rank} A$ independent dimensionless groups.



dashed: larger A
solid: smaller A
same τ

quantity	defining equation	units	$(\mathrm{M}, \mathrm{L}, \mathrm{T})$
m mass	base unit	M	$(1, 0, 0)$
k stiffness	$F = kx$	$(\mathrm{MLT}^{-2})/\mathrm{L}$	$(1, 0, -2)$
τ period	time to repeat	T	$(0, 0, 1)$
A amplitude	largest x	L	$(0, 1, 0)$

2 / THE CLASSICAL ROUTE

Equate exponents, then substitute

Try $\tau = C\, m^a k^b A^c$ with C a pure number. The right side has units $\mathrm{M}^{a+b}\mathrm{L}^{c}\mathrm{T}^{-2b}$, which must equal T^1, so

$$\mathrm{M}: \ a + b = 0, \qquad \mathrm{L}: \ c = 0, \qquad \mathrm{T}: \ -2b = 1.$$

Substituting upwards from the last equation gives $b = -\frac{1}{2}$, $a = \frac{1}{2}$, $c = 0$; the order of attack is a judgement, which the matrix route turns into rules.

3 / A DICTIONARY FOR THE MATRIX ROUTE

Every row operation is what you already do

swap two rows	write the equations in a different order
scale a row	multiply an equation by a non-zero constant
add a multiple	subtract equations to cancel an unknown
pivot	the unknown that a row solves for
free column	an unknown left as a parameter

A pivot must be non-zero, so a 0 on the diagonal is cured by swapping a non-zero entry up from below, which is the whole of pivoting.

4 / THE MATRIX ROUTE WITH THE PIVOTS IN VIEW

Four moves reduce the matrix, and the row labels travel with their rows

Rows are listed $\mathrm{L}, \mathrm{M}, \mathrm{T}$ so that the first column forces a swap; established pivots are gold and entries needing attention are red.

Step 1 / swap $R_1 \leftrightarrow R_2$

	m	k	τ	A
L	**0**	0	0	1
M	1	1	0	0
T	0	−2	1	0

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
L	0	0	0	1
T	0	−2	1	0

The corner entry is 0, so the pivot must come from lower down the column, and the M row supplies a 1.

Step 2 / swap $R_2 \leftrightarrow R_3$

	m	k	τ	A
M	**1**	1	0	0
L	0	**0**	0	1
T	0	−2	1	0

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
T	0	**−2**	1	0
L	0	0	0	1

The second pivot position also holds 0 while the -2 beneath it does not, so those rows trade places.

Step 3 / scale $R_2 \leftarrow -\frac{1}{2}R_2$

	m	k	τ	A
M	**1**	1	0	0
T	0	**−2**	1	0
L	0	0	0	1

$\Rightarrow$

	m	k	τ	A
M	**1**	1	0	0
T	0	**1**	−0.5	0
L	0	0	0	1

Dividing the pivot row by its pivot makes the pivot 1 and keeps the later arithmetic simple.

Step 4 / clear above $R_1 \leftarrow R_1 - R_2$

	m	k	τ	A
M	**1**	**1**	0	0
T	0	**1**	−0.5	0
L	0	0	0	1

$\Rightarrow$

	$\mathbf{m}$	$\mathbf{k}$	τ	$\mathbf{A}$
M	**1**	0	0.5	0
T	0	**1**	−0.5	0
L	0	0	0	**1**

Column τ has no non-zero entry left in row 3, so it never gets a pivot and is free, while column A already holds its 1.

5 / READ THE ANSWER AND COUNT WITH BUCKINGHAM

Each pivot row solves for its own unknown

The rows read $m + \frac{1}{2}\tau = 0$, $k - \frac{1}{2}\tau = 0$, $A = 0$, so setting the free exponent to the wanted quantity, $x_\tau = 1$, gives $\mathbf{x} = (-\frac{1}{2}, \frac{1}{2}, 1, 0)$. With $n = 4$ columns and 3 pivots, $n - r = 1$ free column gives one group, $\Pi = \tau\sqrt{k/m}$, which is Buckingham's count.

$$\tau = \Pi\sqrt{m/k} \quad \text{with } A \text{ absent and } \Pi \text{ a pure number}$$

6 / CHECK AGAINST THE PHYSICS

What the pure number turns out to be

Newton's law $m\ddot{x} = -kx$ gives $\tau = 2\pi\sqrt{m/k}$, so $\Pi = 2\pi \approx 6.28$; for 0.50 kg on 200 $\mathrm{N\,m^{-1}}$, $\sqrt{m/k} = 0.050$ s and $\tau = 0.314$ s. Quadrupling k halves the period, quadrupling m doubles it, and the zero exponent of A makes the period independent of amplitude, as the two waves showed.

ASIDE / HANG THE MASS AND ADD g

Adding $g = (0, 1, -2)$ keeps three pivots but frees both τ and g, so $n - r = 2$ groups appear: $\Pi_1 = \tau\sqrt{k/m}$ and $\Pi_2 = mg/(kA)$, the static sag set against the amplitude. Units cannot discard Π_2; physics does, since gravity only shifts the equilibrium.

ASIDE / PIVOTING IN PYTHON

`sympy.Matrix(A).rref()` returns the reduced matrix and the pivot columns `(0, 1, 3)`, namely m, k, A. `scipy.linalg.lu(A)` returns P, L, U with $A = PLU$, where P records the row swaps; software takes the largest entry as pivot to limit rounding error.

Convention. Rank is the number of pivots, and the groups are the null space of A, one basis vector for each free column.
Follow-up: the density of the universe from units alone, where five quantities of rank three leave two groups.

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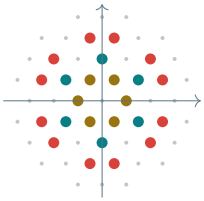
NUMBER THEORY LADDER 8 OF 9

Primes mod 3 and the Eisenstein integers

A prime is a value of $a^2 - ab + b^2$ exactly when it is 1 mod 3, which is exactly when it splits in the Eisenstein integers.

1 / THE RULE AND THE PICTURE

In $\mathbb{Z}[\omega] = \{a + b\omega\}$ the norm is $a^2 - ab + b^2$, and points of equal norm form hexagonal rings



norm 1: 6 units
norm 3: 6 points
norm 7: 12 points

With $\omega = e^{2\pi i/3}$ we have $N(a + b\omega) = (a + b\omega)(a + b\bar{\omega}) = a^2 - ab + b^2$, and the number of lattice points of norm n is

$$r(n) = 6\left(d_1(n) - d_2(n)\right)$$

where d_1, d_2 count divisors that are 1 and 2 modulo 3. For 7 the divisors are $1, 7$, both $1 \bmod 3$, giving 12 points, and for 3 we get 6.

2 / SPLIT, INERT AND RAMIFIED

Every prime of $\mathbb{Z}$ again behaves in one of three ways

type	condition	examples	norm form
split	$p \equiv 1 \pmod 3$	$7 = (3+\omega)(3+\bar{\omega})$, $13 = (4+\omega)(4+\bar{\omega})$	$p = a^2 - ab + b^2$
inert	$p \equiv 2 \pmod 3$	$2, 5, 11, 17$ stay prime	no representation
ramified	$p = 3$	$3 = -\omega^2(1-\omega)^2$	$3 = 2^2 - 2 + 1$

The primes $1 \bmod 3$ up to 100 are $7 = 3^2 - 3 + 1$, $13 = 4^2 - 4 + 1$, $19 = 5^2 - 10 + 4$, $31 = 6^2 - 6 + 1$, $37 = 7^2 - 21 + 9$, $43 = 7^2 - 7 + 1$, $61 = 9^2 - 36 + 16$, $67 = 9^2 - 18 + 4$, $73 = 9^2 - 9 + 1$, $79 = 10^2 - 30 + 9$, $97 = 11^2 - 33 + 9$.

3 / WHY $1 \bmod 3$ / THE EQUATION $x^2 + x + 1 \equiv 0$

A prime splits exactly when $x^2 + x + 1$ has a root modulo p

p	roots of $x^2 + x + 1$
7	2, 4
13	3, 9
5	none
11	none

A root is a cube root of unity modulo p, which exists exactly when $3 \mid p - 1$. Then $p \mid x^2 + x + 1 = (x - \omega)(x - \bar{\omega})$ while p divides neither factor, so p is not prime in $\mathbb{Z}[\omega]$ and splits. For $p = 7$ the root 2 gives $2^2 + 2 + 1 = 7$.

4 / WHICH NUMBERS ARE NORMS

Primes $2 \bmod 3$ must appear to an even power, and norms multiply

The values of $a^2 - ab + b^2$ are $1, 3, 4, 7, 9, 12, 13, 16, 19, 21, \ldots$ while $2, 5, 6, 8, 10, 11$ never occur. Norms multiply, so $7 \cdot 13 = 91$ with $(3+\omega)(4+\omega) = 11 + 6\omega$ gives $91 = 11^2 - 11 \cdot 6 + 6^2$. The same form appears in the cosine rule for a $60°$ angle, $c^2 = a^2 - ab + b^2$, so a triangle with sides $5, 8, 7$ has a $60°$ angle opposite 7.

5 / THE RACE BETWEEN THE CLASSES

Primes $2 \bmod 3$ lead for a very long way

	$1 \bmod 3$	$2 \bmod 3$	
up to 100	11	13	$2 \bmod 3$ ahead
up to 1000	80	87	$2 \bmod 3$ ahead
up to 10 000	611	617	$2 \bmod 3$ ahead

The prime 2 is counted with the inert class and 3 is excluded. The $1 \bmod 3$ count first overtakes at 608 981 813 029, far beyond the mod 4 case at 26 861.

ASIDE / THE HEXAGON

The six units $\pm 1, \pm\omega, \pm\omega^2$ are the vertices of the hexagon, so each prime has six associates.

LINKS TO OTHER SHEETS

The square lattice version is *Primes mod 4 and the Gaussian integers*; the cube root of unity appears in *Cyclotomic fields and the arcs of inscribed polygons*; the last sheet is *Fibonacci modulo p and the golden ratio*.

Convention. $N(a + b\omega) = a^2 - ab + b^2$; counts exclude 3; $\mathbb{Z}[\omega]$ has unique factorisation up to the six units.

Geometric

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THE MASS SHELL AS A NORM ON ONE PAGE

$E^2 - p^2 = m^2$, Gauss and Eisenstein

Without a cross term the mass shell is Pythagoras; with one it is Eisenstein, and a cross term only means something when there are two metrics.

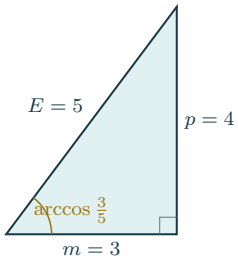
THE RULE

Energy is the hypotenuse

With $c = 1$, rearranging the mass shell as $m^2 + p^2 = E^2$ makes rest mass and momentum the legs of a right triangle with energy as hypotenuse, which is the norm of a Gaussian integer; read the other way it is the norm of a split-complex number:

$$E^2 = N(m + \mathrm{i}p) = m^2 + p^2, \qquad m^2 = N(E + jp) = E^2 - p^2$$

Integer states are Pythagorean triples: $(m, p, E) = (3, 4, 5)$ moves at $\beta = \frac{p}{E} = \frac{4}{5}$ with $\gamma = \frac{E}{m} = \frac{5}{3}$, and $(5, 12, 13)$ at $\beta = \frac{12}{13}$. The angle between the mass leg and the hypotenuse is $\arccos \frac{1}{\gamma}$, and the Gaussian product $(m_1 + \mathrm{i}p_1)(m_2 + \mathrm{i}p_2)$ makes new integer states from old.



Rest mass and momentum at right angles; energy across the diagonal.

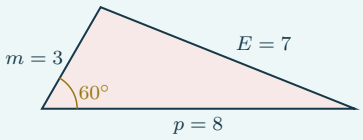
THE EISENSTEIN VERSION

A cross term sets the legs at $60°$

The Eisenstein norm keeps a cross term, $N(a + b\omega) = a^2 - ab + b^2$, which is the law of cosines for a triangle whose sides a and b meet at $60°$. Copying the mass shell into it gives a hypothetical relation

$$E^2 = m^2 - mp + p^2,$$

whose integer solutions are Eisenstein triples such as $(m, p, E) = (3, 8, 7)$ and $(5, 8, 7)$. The units grow from the four $\pm 1, \pm\mathrm{i}$ to the six $\pm 1, \pm\omega, \pm\omega^2$, and the integer states sit on a hexagonal lattice in place of a square one.



$9 - 24 + 64 = 49$: the legs meet at $60°$ and $E = 7$ closes it.

THE DIFFICULTY WITH ONE METRIC

A cross term in a single form can always be completed away

Completing the square removes any cross term by a change of basis, $a^2 - ab + b^2 = \left(a - \frac{b}{2}\right)^2 + 3\left(\frac{b}{2}\right)^2$, so a single quadratic form is fixed, up to relabelling, by the sign of its discriminant $b^2 - 4ac$ alone. The cross term changes which integer points are available but not the geometry, and in physics it fares worse: in $E^2 = m^2 + \kappa mp + p^2$ the invariant m multiplies one component of a vector, so the term flips sign under $p \to -p$ and changes under every boost, defining a preferred frame.

form	discriminant	geometry	integer points
$x^2 + y^2$	-4	Euclidean, elliptic	Gaussian, square lattice
$x^2 - xy + y^2$	-3	Euclidean, elliptic	Eisenstein, hexagonal lattice
$x^2 - y^2$	$+4$	Minkowski, hyperbolic	light cone at slopes ± 1
$x^2 + xy - y^2$	$+5$	Minkowski, hyperbolic	light cone at golden slopes $\frac{-1 \pm \sqrt{5}}{2}$

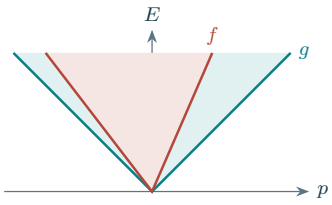
TWO METRICS

A relative cross term cannot be removed

With two metrics, g giving $E^2 - p^2$ and f giving $E^2 + 2bEp - kp^2$, one change of basis can diagonalise either, but the invariant content is the eigenvalues of $g^{-1}f$, the relative stretch between the two light cones. A common real frame exists exactly when

$$(1 - k)^2 \geq 4b^2$$

With equal light speeds, $k = 1$, any cross term makes the eigenvalues complex, $1 \pm \mathrm{i}b$, so the Eisenstein-type choice $b = -\frac{1}{2}$ gives $1 \pm \frac{1}{2}$; with $k = 3$ the same cross term survives as real eigenvalues $2 \pm \frac{\sqrt{3}}{2}$. Ghost-free bimetric gravity builds its interaction from $\sqrt{g^{-1}f}$ and needs that square root to be real, so the cross term carries physics only in the bimetric setting and only within this bound.



Light cones of g and of f with $b = -\frac{1}{2}$, $k = 3$: narrower and tilted, with a real common frame.

Conventions. $c = 1$; $\omega = \frac{1}{2}(-1 + \mathrm{i}\sqrt{3})$, $j^2 = +1$ is the split-complex unit, and N is the norm of each number system. The Eisenstein relation is a mathematical analogue, not an observed dispersion relation.

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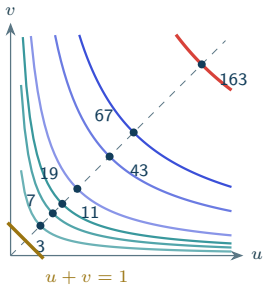
HEEGNER NUMBERS ON ONE PAGE

Euler's n^2+n+41 and the Heegner numbers

Each Heegner number is a line $u+v=1$ that misses the hyperbola $uv=k$, and it misses it modulo every prime below k.

1 / THE RULE AND THE PICTURE

$f(n)=n^2+n+k$ with $k=\frac{d+1}{4}$ is the value $k-uv$ at the lattice points $(n+1,-n)$ of the line $u+v=1$



Writing $g(x)=x^2-x+k$ we have $f(n)=g(n+1)=k-uv$ for $(u,v)=(n+1,-n)$, so the roots of f are where the line $u+v=1$ meets the hyperbola $uv=k$, and $\Delta=1-4k=-d$:

$$\boxed{d=4k-1,\qquad \text{line } u+v=1 \text{ misses } uv=k}$$

In the first quadrant the line reaches only $uv\le\frac14$, at $(\frac12,\frac12)$, so every hyperbola $uv=k$ with $k=1,2,3,5,11,17,41$ passes above it, with its vertex $(\sqrt{k},\sqrt{k})$ on the diagonal labelled by the Heegner number d.

n	0	1	2	3	4	5
(u,v)	$(1,0)$	$(2,-1)$	$(3,-2)$	$(4,-3)$	$(5,-4)$	$(6,-5)$
$41-uv$	41	43	47	53	61	71

2 / WHY THE VALUES ARE PRIME / THE LINE MISSES THE HYPERBOLA MODULO p

A prime p divides $f(n)$ exactly when the lattice point $(n+1,-n)$ lies on $uv\equiv k \pmod p$



For $d=163$ and $p=5$ we have $k=41\equiv1$, so the hyperbola $uv\equiv1$ has the squares $(1,1),(2,3),(3,2),(4,4)$ and the line $u+v\equiv1$ has $(0,1),(1,0),(2,4),(3,3),(4,2)$, and the two sets are disjoint, so $5\nmid f(n)$ for every n. The number of crossings in $\mathbb{F}_p^2$ is $1+\left(\frac{-d}{p}\right)$, and the Heegner property is that $-d$ is a nonresidue for every odd prime $p<k$, so there are no crossings below k. Then $f(n)$ has no prime factor below k, and since $f(n)<k^2$ for $0\le n\le k-2$ each value is prime.

$\mathbb{F}_5^2$ for $d=163$

0	0	0	0	0	0	0	0	0	0	0	2	2	2	2
3	5	7	11	13	17	19	23	29	31	37	41	43	47	53

crossings of the line and hyperbola in $\mathbb{F}_p^2$, for each prime p

At $p=41=k$ the hyperbola degenerates to the axes $uv\equiv0$, which the line meets at $(1,0)$ and $(0,1)$, that is $n=40$ and $n=0$, giving $f(40)=41^2$ and $f(0)=41$.

3 / ALL NINE HEEGNER NUMBERS / THE SAME PICTURE WITH $k=\frac{d+1}{4}$

Seven odd Heegner numbers give Euler-type polynomials, and 1 and 2 use the line $u+v=0$

d	(s,P)	Δ	polynomial	consecutive primes	next value
1	$(0,1)$	-4	n^2+1	ring $\mathbb{Z}[i]$	
2	$(0,2)$	-8	n^2+2	ring $\mathbb{Z}[\sqrt{-2}]$	
3	$(1,1)$	-3	n^2+n+1	$f(1)=3,\ f(2)=7,\ f(3)=13$	$f(4)=21$
7	$(1,2)$	-7	n^2+n+2	$n=0$: 2	$f(1)=4=2^2$
11	$(1,3)$	-11	n^2+n+3	$n=0,1$: 3, 5	$f(2)=9=3^2$
19	$(1,5)$	-19	n^2+n+5	$n=0..3$: 5, 7, 11, 17	$f(4)=25=5^2$
43	$(1,11)$	-43	n^2+n+11	$n=0..9$: 11, ..., 101	$f(10)=121=11^2$
67	$(1,17)$	-67	n^2+n+17	$n=0..15$: 17, ..., 257	$f(16)=289=17^2$
163	$(1,41)$	-163	n^2+n+41	$n=0..39$: 41, ..., 1601	$f(40)=1681=41^2$

4 / WHY THE LIST STOPS / COUNTER-EXAMPLES AND THE THEOREM

A line that meets the hyperbola modulo a small prime breaks the run

Take $k=101$, where $\Delta=-403$ is not a Heegner discriminant: $f(4)=121=11^2$, because the lattice point $(5,-4)$ has $5\cdot(-4)=-20\equiv2\equiv101 \pmod{11}$ and so lies on the hyperbola modulo 11. Likewise $k=47$ fails at $f(1)=49=7^2$. The theorem of Baker, Heegner and Stark says exactly nine d have class number one, and Rabinowitsch's criterion says n^2+n+k is prime for $0\le n\le k-2$ exactly when $\mathbb{Q}(\sqrt{1-4k})$ has class number one, which leaves $k=2,3,5,11,17,41$. In the (s,P) plane each Heegner point sits above the parabola $P=s^2/4$ by $d/4$.

ASIDE / THE EVEN CASES

For $d=1$ and 2 the discriminant is $-4d$ and the line is $u+v=0$, through the origin, with hyperbolas $uv=1$ and $uv=2$.

ASIDE / CHECK IN PYTHON

`all(isprime(n*n+n+41) for n in range(40))` gives `True`, and `1681 == 41**2`.

Convention. d is a Heegner number, $k=(d+1)/4$ for odd d, and $\Delta=1-4k=-d$. After the earlier sheets on the Legendre symbol and on lines meeting hyperbolas.

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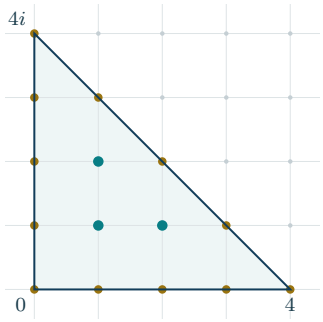
COUNTING AREA ON TWO LATTICES ON ONE PAGE

Pick: from squares to Eisenstein triangles

Keep the point counts; change the area of a lattice cell.

1 / START ON THE GAUSSIAN INTEGER LATTICE

Area from interior and boundary points



Teal: interior. Gold: boundary.

The **Gaussian integers** $a+bi$, with $a,b\in\mathbb{Z}$, form a square lattice with unit cells of area 1. For a simple polygon with lattice vertices and no holes, let I count points strictly inside and B count all boundary points, including each corner once. **Pick's theorem** says

$$A_{\square} \quad = \quad I \; + \; \frac{B}{2} \; - \; 1$$

The triangle with vertices $0, 4, 4i$ has $I=3$ and $B=12$:

$$A_{\square} = 3 + \frac{12}{2} - 1 = 8.$$

The ordinary check is $\frac{1}{2}\times 4\times 4 = 8$.

2 / MOVE TO THE EISENSTEIN INTEGER LATTICE

A unit square becomes a unit rhombus

Set

$$\omega = \mathrm{e}^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \qquad \eta = 1+\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2}.$$

The **Eisenstein integers** $\mathbb{Z}[\omega]$ can equally be written $a+b\eta$, with $a,b\in\mathbb{Z}$. They form a triangular lattice with six nearest neighbours, also called a hexagonal lattice.

The real linear map

$$\Phi(a+bi) = a+b\eta$$

preserves interior and boundary point counts. It scales every area by the new cell area:

$$\delta = 1\cdot 1\cdot \sin 60^\circ = \frac{\sqrt{3}}{2}.$$



area 1 area $\sqrt{3}/2$

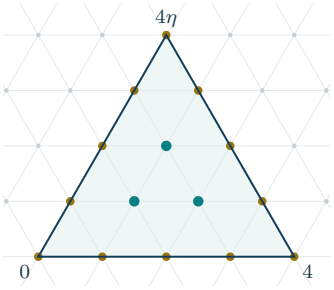
One rhombus contains two unit equilateral triangles, each of area $\sqrt{3}/4$.

$$A_{\mathrm{E}} \quad = \quad \frac{\sqrt{3}}{2}\left(I + \frac{B}{2} - 1\right) \quad = \quad \frac{\sqrt{3}}{4}(2I \; + \; B \; - \; 2)$$

3 / COUNT ON THE TRIANGULAR LATTICE

An equilateral triangle and a regular hexagon

The image of $0, 4, 4i$ is $0, 4, 4\eta$.

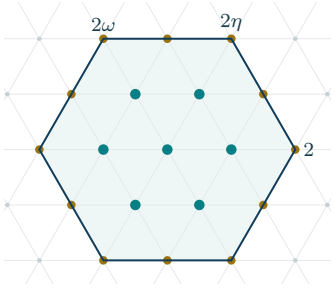


Side 4; the counts remain $I=3$, $B=12$.

$$A_{\mathrm{E}} = \frac{\sqrt{3}}{2}(3+6-1) = 4\sqrt{3}.$$

Check: $\frac{\sqrt{3}}{4}\,4^2 = 4\sqrt{3}$.

Six equilateral triangles of side 2.



Side 2; count $I=7$, $B=12$.

$$A_{\mathrm{E}} = \frac{\sqrt{3}}{2}(7+6-1) = 6\sqrt{3}.$$

Check: $6\times\frac{\sqrt{3}}{4}\,2^2 = 6\sqrt{3}$.

Area units. Nearest-neighbour spacing is 1. $I + B/2 - 1$ measures area in lattice cells; $2I + B - 2$ measures Eisenstein area in units of $\sqrt{3}/4$.

References. Harrison, A formal proof of Pick's theorem; Wolfram, Eisenstein integers.

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EISENSTEIN POWERS ON ONE PAGE

ω^η or η^ω: size, angle and 23

Swap two Eisenstein integers: one power is positive real; the other has a phase.

Choose the pair $\omega = e^{i(2\pi/3)} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad \eta = 1 + 2\omega = i\sqrt{3} = \sqrt{3}\,e^{i\pi/2}.$

Both lie in $\mathbb{Z}[\omega] = \{a + b\omega : a, b \in \mathbb{Z}\}$. The principal logarithms are

$$\operatorname{Log}\omega = i\frac{2\pi}{3}, \qquad \operatorname{Log}\eta = \frac{1}{2}\ln 3 + i\frac{\pi}{2}.$$

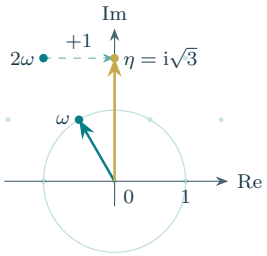
The rule: separate magnitude and phase

Write $z = re^{i\theta} \neq 0$, with principal argument θ, and $w = a + ib$. Then

$$z^w := e^{w \operatorname{Log} z} = \underbrace{e^{a\ln r - b\theta}}_{\text{magnitude}}\underbrace{e^{i(b\ln r + a\theta)}}_{\text{phase}}.$$

The result is real exactly when $b\ln r + a\theta \in \pi\mathbb{Z}$. Complex numbers have no usual $<$ ordering, so compare their **magnitudes** and record their angles.

Two points on the lattice



Forward: ω^η is positive real

The unit-circle base has $r = 1$; its logarithm is purely imaginary. So is the exponent:

$$\omega^\eta = e^{(i\sqrt{3})\cdot i(2\pi/3)} = e^{-2\pi/\sqrt{3}}.$$

The phase is zero: there is no residual rotation.

$$\boxed{\omega^\eta = e^{-2\pi/\sqrt{3}}}$$
$$\omega^\eta \approx 0.0265799$$

Reverse: η^ω has a phase

Now the logarithm has both a real and an imaginary part:

$$\eta^\omega = e^{\omega(\frac{1}{2}\ln 3 + i\pi/2)} = \rho\,e^{i\phi},$$

where

$$\rho = 3^{-1/4}e^{-\pi\sqrt{3}/4} \approx 0.1949519,$$
$$\phi = \frac{\sqrt{3}\ln 3 - \pi}{4} \approx -0.309685 \text{ rad}.$$
$$\eta^\omega \approx 0.185678 - 0.059413i$$

Which has the larger magnitude?

$$\boxed{|\omega^\eta| \approx 0.02658 \;<\; 0.19495 \approx |\eta^\omega|}$$

Both are proved transcendental

Both inputs are algebraic:

$$\omega^2 + \omega + 1 = 0, \qquad \eta^2 + 3 = 0.$$

By Gelfond–Schneider [1], an algebraic base outside $\{0, 1\}$, raised to an algebraic exponent outside $\mathbb{Q}$, is transcendental. Thus **both powers are transcendental on every branch**.

The imaginary part $\sqrt{3}$ of η is irrational.

The arithmetic coda: 23 stays prime

The Eisenstein ring $\mathbb{Z}[\omega]$ is Euclidean, so it has unique factorisation. An ordinary prime $p \equiv 2 \pmod 3$ remains prime here; in particular, **23 remains prime** [2].

Modulo 23, $x^2 + x + 1$ has no root: a root would be a nontrivial cube root of unity in the group $\mathbb{F}_{23}^\times$ of order 22, impossible since $3 \nmid 22$. Hence

$$\boxed{\mathbb{Z}[\omega]/(23) \cong \mathbb{F}_{23^2} = \mathbb{F}_{529}.}$$

Residues of $a + b\omega$ give $23 \times 23 = 529$ field elements.

Convention. The numerical values and magnitude comparison use principal logarithms. Other branches change the values; ω^η remains positive real on every branch.

Sources. [1] University of South Carolina, Math 785 notes, §8, Theorem 19. [2] Northeastern University, *Quadratic Integer Rings*, pp. 14, 31.

PYTHAGORAS • SIXFOLD SYMMETRY • EISENSTEIN INTEGERS

The cosine rule at $60°$ and $120°$

From school trigonometry to integer triangles on a hexagonal lattice.

1 One rule, three angles

Let sides a and b meet at angle θ, with side c opposite that angle:

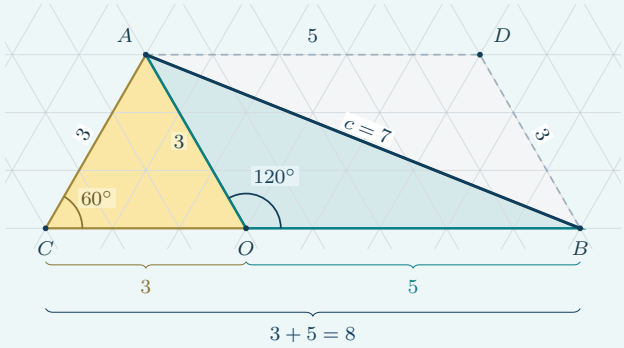
$$\boxed{c^2 = a^2 + b^2 - 2ab\cos\theta}.$$

Included angle	$\cos\theta$	Cosine rule becomes	Example (a,b,c)
$60°$	$\frac{1}{2}$	$c^2 = a^2 - ab + b^2$	$(3,8,7)$
$90°$	0	$c^2 = a^2 + b^2$	$(3,4,5)$
$120°$	$-\frac{1}{2}$	$c^2 = a^2 + ab + b^2$	$(3,5,7)$

The $90°$ case gives Pythagorean triples. The $60°$ and $120°$ cases give **Eisenstein triples**. Here c is opposite the named angle; in $(3,8,7)$, the longest side is $b = 8$.

2 Two triples, one shared side

All marked vertices lie on a unit triangular lattice. The darker triangle OAB has sides $3, 5, 7$.



Gold COA is equilateral. CAB is the $60°$ triangle. The faint copy ADB completes a parallelogram.

$120°$ at O:

$$3^2 + 5^2 + 3\cdot 5 = 49 = 7^2.$$

$60°$ at C:

$$3^2 + 8^2 - 3\cdot 8 = 49 = 7^2.$$

Attaching the equilateral triangle extends the base from 5 to $3+5 = 8$. The same construction works generally:

$$a^2 + (a+b)^2 - a(a+b) = a^2 + ab + b^2.$$

Hence

$$\underbrace{(a,b,c)}_{120°} \longmapsto \underbrace{(a,a+b,c)}_{60°}.$$

Why it tiles. Six $60°$ cells meet at each lattice point. Pair any triangle with its half-turn to make a parallelogram, then repeat by translations. Here the half-turn about the midpoint of AB pairs OAB with ADB, keeping all vertices on the same sixfold lattice.

3 A recipe for more triples

Like Euclid's recipe for Pythagorean triples, this uses two integers. For $m > n > 0$, set

$$a = m^2 - n^2,$$
$$b = 2mn + n^2,$$
$$c = m^2 + mn + n^2.$$

Expanding verifies

$$\boxed{a^2 + ab + b^2 = c^2}.$$

These are $120°$ triples; the construction above supplies their $60°$ companions.

(m,n)	$120°$ triple	$60°$ companion
$(2,1)$	$(3,5,7)$	$(3,8,7)$
$(3,1)$	$(8,7,13)$	$(8,15,13)$

Scaling all 3 sides by same positive integer preserves angle.

4 The Eisenstein connection

The unit step at $120°$ is

$$\omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad \omega^3 = 1.$$

Lattice points are **Eisenstein integers**

$$z = m + n\omega, \qquad m, n \in \mathbb{Z}.$$

Their norm is ordinary squared distance:

$$N(z) = |z|^2 = \left(m - \frac{n}{2}\right)^2 + \frac{3n^2}{4} = m^2 - mn + n^2.$$

For a $120°$ triangle, the side c joins the endpoints a and $b\omega$:

$$c^2 = N(a - b\omega) = a^2 + ab + b^2.$$

In the diagram, $A = 3\omega$ and $B = 5$, so

$$AB = |5 - 3\omega| = \sqrt{25 + 15 + 9} = 7.$$

The cross term records the oblique lattice axes.

Reading: OpenStax: law of cosines; S. Chang (2019): integer triples; G. McShane (2024), Section 1.1: Eisenstein integers and their norm.

Geometric

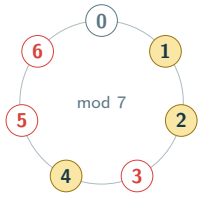
NUMBER THEORY ON ONE PAGE

The Legendre symbol and quadratic reciprocity

A square-root test modulo a prime, the law that swaps its top and bottom, and the run of primes behind -163.

1 / THE RULE AND THE PICTURE

For an odd prime p, the symbol (D/p) records whether D is a square modulo p



The symbol asks whether $x^2 \equiv D \pmod p$ has a solution x, where the unknown is squared and D is the required remainder:

$$\left(\frac{D}{p}\right) = \begin{cases} +1, & x^2 \equiv D \text{ has a nonzero solution (residue)}, \\ -1, & x^2 \equiv D \text{ has no solution (nonresidue)}, \\ 0, & p \mid D. \end{cases}$$

x	0	1	2	3	4	5	6
$x^2 \bmod 7$	0	1	4	2	2	4	1

The clock shows $1, 2, 4$ as residues (gold) and $3, 5, 6$ as nonresidues (red), so $(2/7) = (4/7) = 1$, $(3/7) = -1$ and $(7/7) = 0$.

2 / COUNTING / EXACTLY HALF ARE SQUARES

Each nonzero square has two roots, x and $-x$, so $(p-1)/2$ residues exist

p	nonzero residues	count
5	1, 4	2
7	1, 2, 4	3
11	1, 3, 4, 5, 9	5
13	1, 3, 4, 9, 10, 12	6

Modulo 7 the inputs 3 and $4 = -3$ both give 2, the inputs 2 and 5 both give 4, and 1 and 6 both give 1, so the 6 nonzero inputs land on 3 distinct squares and the other 3 remainders are nonresidues.

3 / TWO RULES / MULTIPLYING THE TOP, AND TESTING -1

The top entry multiplies like signs, and -1 depends only on $p \bmod 4$

a	b	ab	labels
R	R	R	$(+1)(+1) = +1$
R	N	N	$(+1)(-1) = -1$
N	R	N	$(-1)(+1) = -1$
N	N	R	$(-1)(-1) = +1$

So $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$, and for example $(15/7) = (3/7)(5/7) = (-1)(-1) = +1$.

Since $-1 \equiv p - 1$, the sign is $(-1)^{(p-1)/2}$, which is $+1$ for $p = 4k+1$ and -1 for $p = 4k+3$:

$$\boxed{\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}}$$

Modulo 5 we have $-1 \equiv 4 = 2^2$, so $(-1/5) = +1$, whereas modulo 7 the value 6 is absent from the squares and $(-1/7) = -1$. Euler's criterion $(a/p) \equiv a^{(p-1)/2} \pmod p$ gives both rules.

4 / RECIPROCITY / SWAPPING TOP AND BOTTOM

For distinct odd primes the two answers agree unless both primes are $3 \bmod 4$

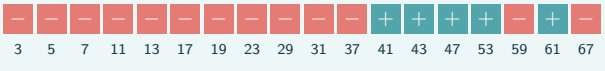
$$\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}} \left(\frac{p}{q}\right)$$

$p \bmod 4,\ q \bmod 4$	sign	relation	example
$1,1$ or $1,3$ or $3,1$	$+$	answers agree	$(5/13) = (13/5) = -1$
$3,3$	$-$	answers are opposite	$(3/7) = -1,\ (7/3) = +1$

The law turns a hard question into an easy one by reducing the bigger prime. For $D = -163$ the factor $163 \equiv 3 \pmod 4$ gives $(163/p) = (-1)^{(p-1)/2}(p/163)$, so $\left(\frac{-163}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{163}{p}\right) = (-1)^{p-1}\left(\frac{p}{163}\right) = \left(\frac{p}{163}\right)$: whether -163 is a square modulo p depends only on $p \bmod 163$, a square among 81 of the 162 classes, and primes spread evenly over those classes.

5 / AN UNEVEN START / WHY -163 BEGINS WITH ELEVEN NONRESIDUES

Every odd prime below 41 gives -1, and Euler's polynomial $n^2 + n + 41$ explains it



the sign of $(-163/p)$ for each prime p

Completing the square gives $4f(n) = (2n+1)^2 + 163$, so a prime dividing $f(n)$ makes -163 a square modulo that prime, which is impossible below 41. The forty values $f(0), \ldots, f(39)$ are all below 41^2 and so are prime, and $f(40) = 41^2$.

At $p = 41$ we have $-163 \equiv 1$, so the symbol turns $+1$. The deeper reason is that 163 is the largest Heegner number, so $\mathbb{Z}[\omega]$ with $\omega = \frac{1+\sqrt{-163}}{2}$ has unique factorisation and no element of prime norm below 41. With Mertens' $\sum_{p \le x} (\log p)/p = \log x + O(1)$ and the weighted imbalance $o(\log x)$, the nonresidue primes carry half the weight eventually.

Convention. p, q are distinct odd primes; R and N mean residue and nonresidue. After the student preamble *Squares, primes and 163*.

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GAUSSIAN AND EISENSTEIN INTEGERS ON ONE PAGE

Triples, circles and ellipses

Draw a rational line, find a rational point, then clear denominators.

1 / GEOMETRY AND ALGEBRA TELL THE SAME STORY

A known point turns a quadratic into a linear problem

Divide a triple by c: $(x, y) = (a/c, b/c)$ lies on a unit conic. Draw $y = t(x + 1)$ through $(-1, 0)$. Substitution gives a quadratic with the known root $x = -1$; factoring it out leaves a rational second intersection when t is rational.

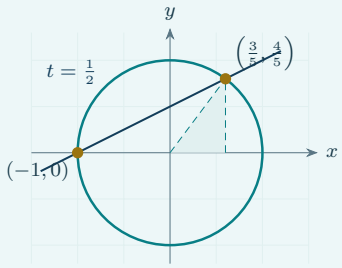
The same formulas arise by squaring complex integers: $N(z^2) = N(z)^2$, where the **norm** is $N(z) = z\bar{z} = |z|^2$.

2 / GAUSSIAN: A SQUARE LATTICE

The familiar unit circle

$\mathbb{Z}[\mathrm{i}] = \{m + n\mathrm{i} : m, n \in \mathbb{Z}\}$, with $\mathrm{i}^2 = -1$.

$$N(x + y\mathrm{i}) = x^2 + y^2, \qquad x^2 + y^2 = 1.$$



For a rational slope $t = n/m$, the other point is

$$(x, y) = \left(\frac{1 - t^2}{1 + t^2}, \frac{2t}{1 + t^2}\right).$$

Clearing denominators gives

$$\boxed{(a, b, c) = (m^2 - n^2,\ 2mn,\ m^2 + n^2).}$$

Algebraically, $(m + n\mathrm{i})^2 = a + b\mathrm{i}$ and $N(m + n\mathrm{i}) = c$.

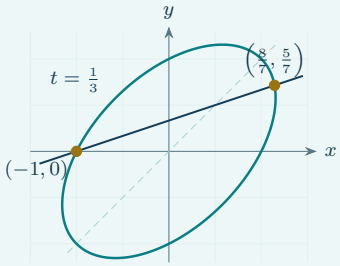
$$(m, n) = (2, 1) : \quad 3^2 + 4^2 = 5^2.$$

3 / EISENSTEIN: A TRIANGULAR LATTICE

A unit ellipse in coefficient coordinates

$\mathbb{Z}[\omega] = \{m + n\omega : m, n \in \mathbb{Z}\}$, $\omega = \mathrm{e}^{2\pi\mathrm{i}/3} = \frac{-1+\mathrm{i}\sqrt{3}}{2}$.

$$N(x + y\omega) = x^2 - xy + y^2, \quad \omega^2 = -1 - \omega.$$



The same line now meets $x^2 - xy + y^2 = 1$ at

$$(x, y) = \left(\frac{1 - t^2}{1 - t + t^2}, \frac{t(2 - t)}{1 - t + t^2}\right).$$

With $t = n/m$, clear denominators:

$$\boxed{(a, b, c) = (m^2 - n^2,\ 2mn - n^2,\ m^2 - mn + n^2).}$$

Again, $(m + n\omega)^2 = a + b\omega$ and $N(m + n\omega) = c$.

$$(m, n) = (3, 1) : \quad 8^2 - 8 \cdot 5 + 5^2 = 7^2.$$

4 / WHY THE ELLIPSE IS ALSO A CIRCLE

The coefficients are not the real and imaginary parts

In $z = x + y\omega$, the basis vectors $1, \omega$ meet at $120°$. Its actual complex-plane coordinates are

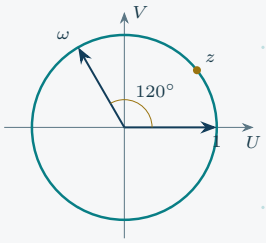
$$U = \Re z = x - \frac{y}{2}, \qquad V = \Im z = \frac{\sqrt{3}\,y}{2}.$$

Thus $x^2 - xy + y^2 = U^2 + V^2$: the **coefficient ellipse** maps to the **unit circle in the complex plane**.

On perpendicular coefficient axes, the ellipse is tilted $45°$, with semi-axes $\sqrt{2}, \sqrt{2/3}$ and eccentricity $e = \sqrt{2/3}$. Completing the square also gives an integer identity:

$$a^2 - ab + b^2 = c^2 \quad \Longrightarrow \quad (2a - b)^2 + 3b^2 = (2c)^2.$$

So $X^2 + 3Y^2 = Z^2$ with $(X, Y, Z) = (2a - b, b, 2c)$: our $(8, 5, 7)$ becomes $(11, 5, 14)$, since $121 + 75 = 196$.



$z = \frac{8}{7} + \frac{5}{7}\omega, \quad |z| = 1.$

5 / MOVE THE PENCIL POINT

Moving off the circle introduces a square-root test

For the circle, shift the starting point to $(-h, 0)$, with rational h, and use $y = t(x + h)$. The quadratic now has discriminant

$$\Delta = 4\left[1 + (1 - h^2)t^2\right].$$

Rational intersections require the bracket to be a rational square. At $h = 1$ it is always 1; at $h = 0, t = 1$ it is 2, giving $(x, y) = (\pm 1/\sqrt{2}, \pm 1/\sqrt{2})$ with matching signs.

Triples. Take integers $m > n > 0$; divide out any common factor. Gaussian triples describe $90°$ triangles; Eisenstein triples obey $c^2 = a^2 + b^2 - 2ab\cos 60°$.
Explore. Pencils on the circle · From circles to ellipses. **Further reading.** Conrad, *Pythagorean Triples*; Cuoco–McCallum, *The Double Continuity of Algebra*, §4.2.

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Chapter 4

Growth, logarithms and scale

What this chapter gathers.

Foothills

- Powers forwards, logarithms backwards
- Logarithms on one page
- Logarithms
- Price changes, simple returns, log returns and annualised growth
- Arithmetic and geometric: sequences, series, means and graphs
- Same mean, different spreads
- Fixed-point iteration to the edge of chaos
- The rightful fear of loss
- Geometric series and saving a thousand pounds a month

Foothold

- Five constants from 0 to 5: gamma, phi, e, pi, delta
- Irrational powers, π^e against e^π, and the integer 23
- How the primes thin out
- Regular, irregular and combined: cumulative prime staircases
- From prime staircases to prime waves
- Kepler's third law on log-log axes
- How long is a lifetime? A logarithmic time ruler
- The density of the universe from units alone
- Bode's rule: from planetary data to a straight line
- Points on a graph: pure numbers and measured quantities
- Combining falloffs: inverse square, dust and the primes
- Counting clicks, square-root precision and radiocarbon
- How big can a human be? Stress, strain and the quarter power
- Surface to volume, and why small animals run hot

Handhold

- The making of Goldilocks: a story of clumping

The equation $2^3 = 8$ describes repeated multiplication. Reading it backwards gives $\log_2 8 = 3$: the logarithm asks which exponent produces the answer. This reversal supplies a language for growth and scale. Products become sums of logarithms, allowing successive multiplicative changes to be recorded as additive steps.

A gain of ten per cent followed by an equal percentage loss leaves $1.1 \times 0.9 = 0.99$ of the starting amount. Each percentage acts on a different quantity. Growth factors keep track of this changing base, while geometric means summarise their combined effect. The spread of the factors matters as well as their arithmetic average. Savings calculations develop the same principle through repeated deposits and compounding.

Repeated rules can also change their character. An iteration may approach a fixed value, settle into a cycle, or produce behaviour that is highly sensitive to its starting point. Following these possibilities takes a familiar numerical method towards chaos. The constants encountered along the way acquire meaning through the processes that produce them: a limiting ratio, a growth rate or a threshold between different behaviours.

The primes offer another test of how we describe growth. They do not arrive at fixed intervals, yet their cumulative count follows an increasingly recognisable trend. We can count selected families separately and compare their contributions. Giving jumps logarithmic weights changes the quantity being accumulated; moving their horizontal positions onto a logarithmic axis changes the scale. Keeping these choices distinct prepares us to examine the fluctuations that lead towards prime waves.

Logarithmic axes give equal distances to equal ratios. They let a human lifetime share a legible scale with durations far beyond ordinary experience. They can also reveal relationships: a power law becomes a straight line on log–log axes, whose slope records the exponent. Kepler's law and Bode's rule then invite a further question: what supports the pattern? A physical derivation and a fit to planetary data provide different reasons for confidence.

Measurement requires attention to units, size and uncertainty. Dimensional analysis constrains possible laws, while changes in length affect areas and volumes by different factors. These distinctions matter when comparing animals, interpreting falloffs or estimating a cosmic density. Counting introduces another scale: under a Poisson model, four times as many clicks halves the fractional counting uncertainty. The appropriate comparison depends on what we measure.

The final story of clumping brings these questions together by asking which conditions allow structure to form. Growth, competing effects and characteristic scales become parts of one explanation. Throughout the chapter, the aim is to choose a representation whose steps we can interpret, then explain why it suits the phenomenon.

FOUNDATIONS: POWERS, LOGARITHMS AND ONE STEP BEYOND

Powers forwards, logarithms backwards

Read a power backwards, then follow repeated operations up to power towers.

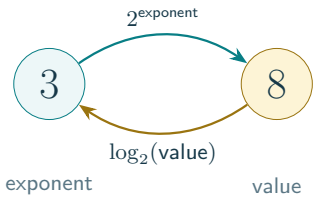
1 / READ THE SAME FACT IN BOTH DIRECTIONS

A logarithm names an exponent

The equation $2^3 = 8$ says that raising 2 to the power 3 gives 8. Reading it backwards, $\log_2 8 = 3$ answers: *what exponent on* 2 *gives* 8*?*

$$\log_b N \quad = \quad p \qquad \Longleftrightarrow \qquad b^p \quad = \quad N$$

Real logarithms use $b > 0$, $b \neq 1$ and $N > 0$.



2 / COMPACT A REPEATED OPERATION

Multiplication collects sums; powers collect products

For a positive whole number n, count n equal terms or n equal factors:

$$a\,n = \underbrace{a + a + \cdots + a}_{n \text{ terms}}, \quad 3 \times 4 = 3 + 3 + 3 + 3 = 12,$$

$$a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}}, \qquad 3^4 = 3 \times 3 \times 3 \times 3 = 81.$$

Extending the exponent rules gives zero, negative and fractional exponents:

$$2^0 = 1 \iff \log_2 1 = 0, \qquad 2^{-2} = \tfrac{1}{4} \iff \log_2 \tfrac{1}{4} = -2, \qquad 2^{1/2} = \sqrt{2} \iff \log_2 \sqrt{2} = \tfrac{1}{2}.$$

3 / COUNT FACTORS TO EXPLAIN THE RULES

Multiplying powers adds exponents

With a common base, we can join the factors:

$$2^3 \cdot 2^2 = (2 \cdot 2 \cdot 2)(2 \cdot 2) = 2^5,$$

$$b^m b^n = \boxed{b^{m+n}}.$$

Put $x = b^m$, $y = b^n$ and read backwards:

$$\log_b(xy) \quad = \quad \log_b x \; + \; \log_b y$$

For example, $8 \times 4 = 32$ becomes

$$\log_2 32 = \log_2 8 + \log_2 4 = 3 + 2 = 5.$$

Raising a power multiplies exponents

Two copies of a product with three factors give six:

$$(2^3)^2 = (2^3)(2^3) = 2^{3\times 2},$$

$$(b^m)^n = \boxed{b^{mn}}.$$

With $x = b^m$, this becomes a logarithm rule:

$$\log_b(x^n) \quad = \quad n \log_b x$$

Brackets decide which operation happens first:

$$(2^3)^2 = 2^6 = 64, \qquad 2^{(3^2)} = 2^9 = 512.$$

Thus logarithms turn products into sums and powers into multiples. These rules hold for positive x, y and real exponents; $b > 0$, $b \neq 1$.

4 / REPEAT EXPONENTIATION ITSELF

Tetration repeats exponentiation

Fix an integer base $a \geq 2$. Start with $T_0 = 1$ and repeatedly raise a to the previous result:

$$T_{n+1} = a^{T_n}, \qquad T_n = a \uparrow\uparrow n.$$

For $n \geq 1$, this is a tower of n copies of a. It is called **tetration**. Knuth's arrows distinguish $a \uparrow n = a^n$ from $a \uparrow\uparrow n$.

A tower is evaluated from the top down:

$$3 \uparrow\uparrow 3 = 3^{(3^3)} = 3^{27}, \qquad (3^3)^3 = 3^9.$$

height n	tower	value
1	2	2
2	2^2	4
3	$2^{(2^2)}$	16
4	$2^{\left(2^{(2^2)}\right)}$	$65\,536$

Each new level makes the previous tower the exponent.

One logarithm removes one level, since $\log_a T_{n+1} = T_n$:

$$65\,536 \xrightarrow{\log_2} 16 \xrightarrow{\log_2} 4 \xrightarrow{\log_2} 2 \xrightarrow{\log_2} 1.$$

Repeating tetration gives **pentation**, written $a \uparrow\uparrow\uparrow n$. For example, $3 \uparrow\uparrow\uparrow 2 = 3 \uparrow\uparrow 3 = 3^{27}$. Further repetitions continue the **hyperoperation** hierarchy.

Two quick checks. Find $\log_3 81$ and $2 \uparrow\uparrow 3$. **Answers:** 4, since $3^4 = 81$; 16, since $2^{(2^2)} = 16$.

Next connections. Logarithmic scales, compound growth and the size of algorithms.

Sources. OpenStax, *Algebra and Trigonometry 2e*, §§1.2, 6.3, 6.5; D. E. Knuth, *Coping with Finiteness* (1976).

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LOGARITHMS ON ONE PAGE

$\log_b x$ undoes b^x

A logarithm answers one question: which power of the base gives this number?

THE DEFINITION

A logarithm is an exponent

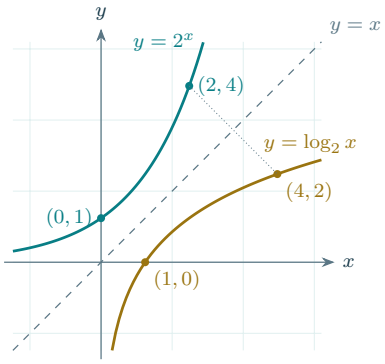
For a base $b > 0$ with $b \neq 1$ and a number $x > 0$,

$$y \;=\; \log_b x \qquad \Longleftrightarrow \qquad b^{\,y} \;=\; x$$

so $\log_2 8 = 3$ because $2^3 = 8$, $\log_{10} 0.01 = -2$ because $10^{-2} = 0.01$, and $\log_9 3 = \frac{1}{2}$ because $\sqrt{9} = 3$. Applying one function after the other gets you back where you started:

$$b^{\log_b x} = x, \qquad \log_b\left(b^x\right) = x, \qquad \log_b 1 = 0, \qquad \log_b b = 1.$$

The two special bases have their own names: $\ln x = \log_{\mathrm{e}} x$ is the natural logarithm and $\lg x = \log_{10} x$ is the common one.



Swapping input and output reflects each graph onto the other in $y = x$.

THREE LAWS, ONE SOURCE

Every log law is an index law read backwards

Writing $x = b^m$ and $y = b^n$, so that $m = \log_b x$ and $n = \log_b y$, each index law turns into a log law by taking $\log_b$ of both sides.

$$b^m\, b^n \;=\; b^{m+n}$$
$$\log_b(xy) \;=\; \log_b x + \log_b y$$

$$b^m / b^n \;=\; b^{m-n}$$
$$\log_b\left(\frac{x}{y}\right) \;=\; \log_b x - \log_b y$$

$$(b^m)^k \;=\; b^{mk}$$
$$\log_b\left(x^k\right) \;=\; k \log_b x$$

Watch for: there is no law for $\log_b(x+y)$; the expression $(\log_b x)^2$ is a different thing from $\log_b x^2 = 2\log_b x$; and $\dfrac{\ln x}{\ln y}$ is a different thing from $\ln \dfrac{x}{y}$.

CHANGE OF BASE

Any log from $\ln$

Taking $\ln$ of $b^{\,y} = x$ gives $y \ln b = \ln x$, so

$$\log_b x \;=\; \frac{\ln x}{\ln b} \;=\; \frac{\log_a x}{\log_a b}$$

for any convenient base a. For example $\log_2 10 = \dfrac{\ln 10}{\ln 2} \approx 3.32$, which says that ten lies a little past 2^3.

SOLVING

Bring the unknown down from the exponent

Isolate the power, then take logs:

$$3 \cdot 2^x = 50 \;\Rightarrow\; 2^x = \tfrac{50}{3} \;\Rightarrow\; x = \frac{\ln(50/3)}{\ln 2} \approx 4.06.$$

Combining logs can create false roots, so check the domain at the end:

$$\log_2 x + \log_2(x-2) = 3 \;\Rightarrow\; x(x-2) = 8 \;\Rightarrow\; x = 4 \text{ or } x = -2,$$

and only $x = 4$ survives, since $\log_2(-2)$ is undefined.

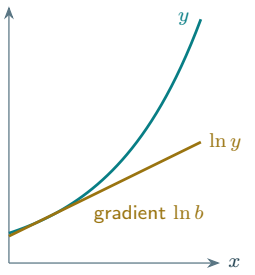
LOGS STRAIGHTEN CURVES

Turning exponentials and powers into straight lines

Taking $\ln$ of a model turns multiplication into addition, which turns a curve into a line whose gradient and intercept you can read off:

$$y = A\,b^x \;\Rightarrow\; \ln y = \ln A + x \ln b, \qquad y = A\,x^n \;\Rightarrow\; \ln y = \ln A + n \ln x.$$

Plot $\ln y$ against x for growth and decay, where the gradient is $\ln b$, and $\ln y$ against $\ln x$ for a power law, where the gradient is the power n; in both cases the intercept is $\ln A$.



Conventions. On most calculators the `log` key means $\log_{10}$ and the `ln` key means $\log_{\mathrm{e}}$. Only positive numbers have real logarithms, and the base must be positive and different from 1, because no power of 1 can give anything except 1.

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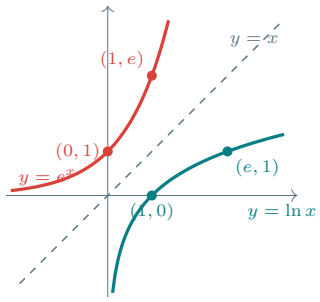
FUNCTIONS LADDER 5 OF 10

Logarithms

The logarithm undoes the exponential, turning products into sums and powers into multiples.

1 / THE RULE AND THE PICTURE

$\ln x$ is the inverse of e^x, so its graph is the mirror image in the line $y = x$



Swapping the coordinates of every point on $y = e^x$ gives the curve $y = \ln x$, defined for $x > 0$ only because e^y is always positive:

$$\ln(e^x) = x, \qquad e^{\ln x} = x, \qquad \ln 1 = 0, \quad \ln e = 1$$

The base-10 logarithm has $\log_{10} 10^x = x$, and $\log_a x$ is the power to which a must be raised to give x.

2 / THE LAWS / MULTIPLICATION BECOMES ADDITION

Each law is an exponential law read backwards

law	statement	example
product	$\log(ab) = \log a + \log b$	$\log_{10} 6 = 0.30103 + 0.47712 = 0.77815$
quotient	$\log(a/b) = \log a - \log b$	$\log_{10} 1.5 = 0.47712 - 0.30103$
power	$\log a^n = n \log a$	$\log_{10} 2^{10} = 3.0103$, and $2^{10} = 1024$
change of base	$\log_a x = \dfrac{\ln x}{\ln a}$	$\log_3 2 = \frac{0.6931}{1.0986} = 0.6309$

Since $e^{a+b} = e^a e^b$, taking logarithms gives the product law, and before calculators this turned long multiplications into additions read from tables, as on a slide rule.
Common values are $\log_{10} 2 = 0.30103$, $\log_{10} 3 = 0.47712$, $\log_{10} 5 = 0.69897$ and $\log_{10} 7 = 0.84510$.

3 / CALCULUS AND SMALL VALUES

The derivative of $\ln x$ is $1/x$, and $\ln(1 + x) \approx x$ for small x

Differentiating $e^{\ln x} = x$ gives $e^{\ln x} (\ln x)' = 1$, so $(\ln x)' = 1/x$, and hence $\int \frac{dx}{x} = \ln|x| + C$. For small x the series $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$ gives $\ln 1.05 \approx 0.0488$ and $\ln 1.07 = 0.0677$ against $0.07 - 0.0025 = 0.0676$, which is why a growth of 7% is about 0.07 in logarithmic terms.

4 / LOGARITHMIC SCALES

Equal steps on the scale are equal ratios

quantity	formula	meaning
decibels	$10 \log_{10}(P/P_0)$ for power	doubling the power adds 3.01 dB, and a factor 1000 adds 30 dB
amplitude in dB	$20 \log_{10}(A/A_0)$	doubling the amplitude adds 6.02 dB
Richter magnitude	$\log_{10}$ of amplitude	one unit is $10\times$ the amplitude and $10^{1.5} = 31.6\times$ the energy
pH	$-\log_{10}[\mathrm{H}^+]$	$[\mathrm{H}^+] = 10^{-7}$ gives 7; 3.2×10^{-4} gives 3.49

Such scales compress huge ranges of values into manageable numbers, and they match the way hearing and brightness respond to ratios.

5 / LOG-LOG PLOTS AND POWERS

A power law $y = ax^b$ becomes a straight line of slope b when both axes are logarithmic

For $y = 2x^{3/2}$ the values at $x = 1, 10, 100, 1000$ are $2,\ 63.2,\ 2000,\ 63\,246$, and $\log_{10} y = 0.301,\ 1.801,\ 3.301,\ 4.801$ rise by exactly 1.5 for each factor of ten in x.
The same trick solves $1.07^t = 2$ as $t = \ln 2 / \ln 1.07 = 10.24$, and counts digits: 2^{100} has $\lfloor 100 \log_{10} 2 \rfloor + 1 = 31$ digits.

ASIDE / SEMI-LOG

Plotting $\ln y$ against t turns an exponential $y = ae^{kt}$ into a straight line of slope k, the quickest test for exponential growth.

LINKS TO OTHER SHEETS

The inverse relation starts in *Exponential functions and the number e*; the logarithm of a complex number is in *Euler's formula and the complex exponential*.

Convention. $\ln$ is $\log_e$; $\log_{10}$ the common logarithm; $\lfloor\ \rfloor$ rounds down; P is power and A amplitude.

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RETURNS AND GROWTH RATES ON ONE PAGE

Differences, ratios and logarithms

Several honest ways to say how fast a price grew, and why the arithmetic mean of returns flatters a fluctuating one.

THE RULE

Change, ratio, and the logarithm of the ratio

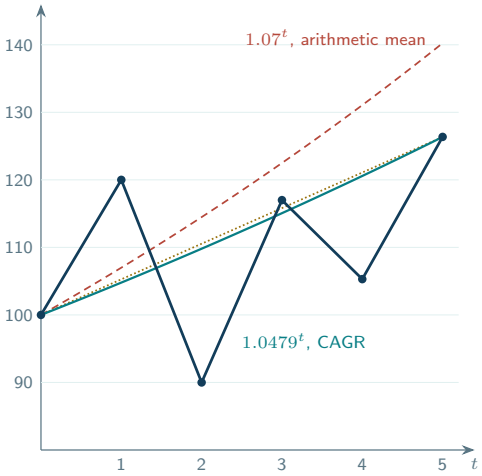
For prices $P_0, P_1, \dots, P_T$ there are three ways to measure each step:

$$\Delta P_t = P_t - P_{t-1}, \qquad R_t = \frac{P_t - P_{t-1}}{P_{t-1}}, \qquad r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t)$$

Each has its own natural way of combining over time:

$$\sum_t \Delta P_t = P_T - P_0, \qquad \prod_t (1 + R_t) = \frac{P_T}{P_0}, \qquad \sum_t r_t = \ln \frac{P_T}{P_0}.$$

First differences add but depend on the price level, simple returns multiply, and log returns add, which is why logs turn compounding back into ordinary sums; for small moves $r_t \approx R_t$.



The price path, the compound curve that fits its ends, and the straight line of first differences.

A FLUCTUATING PRICE

Five years of $+20\%$, -25%, $+30\%$, -10%, $+20\%$

year t	0	1	2	3	4	5	combined
price P_t	100	120	90	117	105.30	126.36	
first difference ΔP_t		$+20$	-30	$+27$	-11.70	$+21.06$	sum $+26.36$
simple return R_t		$+20\%$	-25%	$+30\%$	-10%	$+20\%$	product of $1 + R_t$: 1.2636
log return r_t		$+0.1823$	-0.2877	$+0.2624$	-0.1054	$+0.1823$	sum $0.2340 = \ln 1.2636$

The two $+20\%$ years have the same log return wherever they happen, while their first differences, $+20$ and $+21.06$, depend on the price they start from.

ANNUALISING

Five average rates for one price path

rate	formula	value	what it answers
arithmetic mean	$\frac{1}{T}\sum R_t$	7.00%	the expected one-year return, if each year is a fresh draw
simple average	$\frac{1}{T}\left(\frac{P_T}{P_0} - 1\right)$	5.27%	straight-line growth, the first differences spread evenly
geometric mean, CAGR	$\left(\frac{P_T}{P_0}\right)^{1/T} - 1$	4.79%	the steady yearly rate that reaches the same end price
mean log return	$\frac{1}{T}\sum r_t = \frac{1}{T}\ln\frac{P_T}{P_0}$	4.68%	the same growth as a continuously compounded rate, $\mathrm{e}^{0.0468} = 1.0479$
harmonic mean of $1 + R_t$	$\frac{T}{\sum 1/(1+R_t)} - 1$	2.45%	the lowest of the three classical means of the growth factors

Only the geometric mean and the mean log return reproduce the actual end price; $100 \times 1.07^5 = 140.26$ overshoots 126.36 badly. The order $7.00\% \geq 4.79\% \geq 2.45\%$ is the inequality AM $\geq$ GM $\geq$ HM applied to the growth factors $1 + R_t$.

VOLATILITY DRAG

Fluctuation costs growth

A gain of 50% followed by a loss of 50% has arithmetic mean 0 but leaves $1.5 \times 0.5 = 0.75$ of the start, a geometric mean of -13.4% a year. In general

$$\text{GM} \approx \text{AM} - \tfrac{1}{2}\sigma^2,$$

and here $\sigma = 20.9\%$ gives $7.00\% - 2.18\% = 4.82\%$, close to the true 4.79%. The wider the swings, the further the arithmetic mean runs ahead of what the price actually did.

WHERE THE HARMONIC MEAN EARNS ITS PLACE

Pound-cost averaging

Investing the same sum at each price from 100 to 105.30 buys more shares when the price is low, so the average cost per share is the harmonic mean of the prices, 105.29, below their arithmetic mean of 106.46. The harmonic mean belongs to prices bought with fixed money, just as it belonged to the sides of a box on the isoperimetric sheet.

Conventions. T is the number of years, σ the population standard deviation of the simple returns and CAGR the compound annual growth rate; the figures illustrate the arithmetic of returns and are not investment advice.

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SEQUENCES, SERIES AND MEANS ON ONE PAGE

Add a step, or multiply by one

Taking logarithms turns every geometric pattern on this page into an arithmetic one.

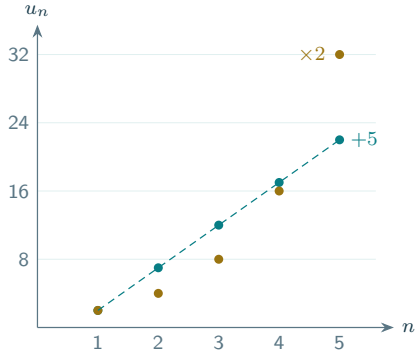
THE TWO FAMILIES

A common difference d, or a common ratio r

	arithmetic	geometric
nth term	$u_n = u_1 + (n-1)d$	$u_n = u_1 r^{\,n-1}$
sum of n	$S_n = \frac{n}{2}(u_1 + u_n)$	$S_n = \dfrac{u_1(1-r^{\,n})}{1-r}$
to infinity	diverges, unless every term is 0	$S_\infty = \dfrac{u_1}{1-r}$ when $\lvert r\rvert < 1$

$$u_n = u_1 r^{\,n-1} \implies \ln u_n = \ln u_1 + (n-1)\ln r$$

the logs of a positive geometric sequence form an arithmetic sequence with $d = \ln r$



$2, 7, 12, 17, 22$ and $2, 4, 8, 16, 32$: equal steps up, or equal ratios.

ARITHMETIC SUM

Pair the ends

Write the sum forwards and backwards; each column adds to the same total $u_1 + u_n$, and there are n columns, so

$$2S_n = n(u_1 + u_n) \implies S_n = \tfrac{n}{2}(u_1 + u_n),$$

which for $1 + 2 + \cdots + 100$ gives $50 \times 101 = 5050$.

GEOMETRIC SUM

Shift and subtract

Multiplying by r slides every term one place along, so only two terms survive the subtraction:

$$S_n - rS_n = u_1 - u_1 r^{\,n} \implies S_n = \frac{u_1(1-r^{\,n})}{1-r},$$

and when $\lvert r\rvert < 1$ the term r^n dies away, leaving $S_\infty = \dfrac{u_1}{1-r}$.

THE MEANS

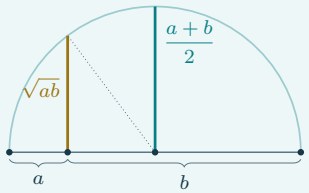
The middle term of each family

Solving $m - a = b - m$ and $m/a = b/m$ for the middle term gives the two means of positive a and b:

$$\text{AM} = \frac{a+b}{2}, \qquad \text{GM} = \sqrt{ab}.$$

Between 2 and 18 these give $2, 10, 18$ and $2, 6, 18$. The geometric mean is the arithmetic mean of the logs, since $\ln\sqrt{ab} = \frac{1}{2}(\ln a + \ln b)$, and expanding $\left(\sqrt{a} - \sqrt{b}\right)^2 \geq 0$ gives

$$\frac{a+b}{2} \geq \sqrt{ab}, \quad \text{with equality only when } a = b.$$



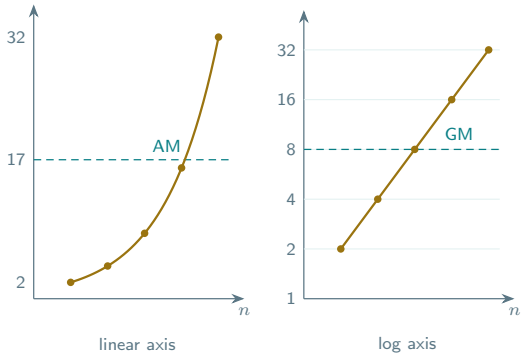
The radius is the AM and the half-chord is the GM, so the half-chord is never longer.

LINEAR AND SEMI-LOG GRAPHS

Each family has an axis that straightens it

On linear axes an arithmetic sequence is a straight line of gradient d and a geometric one curves upward; a semi-log graph spaces the vertical axis by $\log u$, so equal ratios sit at equal heights and the geometric sequence straightens to gradient $\log r$, which is the bridge above drawn as a picture.

Halfway up the linear axis between 2 and 32 is the AM, 17, and halfway up the log axis is the GM, 8, which is also the term at the middle step $n = 3$.



The same data, $u_n = 2^n$, on both axes.

Conventions. Here u_1 is the first term (some texts write a) and S_n is the sum of the first n terms. A semi-log graph has one logarithmic axis and a log-log graph has two; plotting $\lg u$ or $\ln u$ on an ordinary axis gives the same straight line, with gradient $\lg r$ or $\ln r$ respectively.

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FOUR MEANS, VARIANCE AND GAUSSIAN REGRESSION ON ONE PAGE

Same mean, different spreads

What does an average preserve, and why do squared errors appear in a best-fit line?

1 / CHOOSE WHAT THE REPLACEMENT VALUE MUST PRESERVE

Four means answer four different questions

Take n positive values $x_1, \dots, x_n$, weighted equally. Here $\sum_i$ adds over all n values and $\prod_i$ multiplies them.

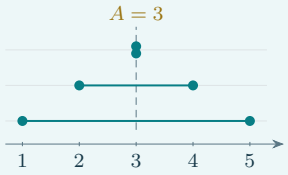
mean	definition	preserved quantity and a use
Arithmetic	$A = \frac{1}{n}\sum_i x_i$	**Sum.** Totals and equally weighted measurements.
Geometric	$G = \left(\prod_i x_i\right)^{1/n}$	**Product.** Growth factors over equal time intervals.
Harmonic	$H = \frac{n}{\sum_i 1/x_i}$	**Sum of reciprocals.** Speeds over equal distances.
RMS	$Q = \sqrt{\frac{1}{n}\sum_i x_i^2}$	**Sum of squares.** Heating by current through a fixed resistance.

$$H \le G \le A \le Q$$ Equal throughout exactly when every value is the same.

2 / HOLD THE CENTRE AT THREE AND MOVE THE VALUES APART

The same arithmetic mean can conceal a different spread

values	H	G	A	Q
$(3,3)$	3	3	3	3
$(2,4)$	$8/3$	$\sqrt{8}$	3	$\sqrt{10}$
$(1,5)$	$5/3$	$\sqrt{5}$	3	$\sqrt{13}$



For $(1,5)$: $H \approx 1.67$, $G \approx 2.24$, $Q \approx 3.61$.

For the positive pair $a-d,\ a+d$, where $0 \le d < a$, substitution gives

$$H = a - \frac{d^2}{a}, \qquad G = \sqrt{a^2 - d^2}, \qquad A = a, \qquad Q = \sqrt{a^2 + d^2}.$$

In this family, increasing d lowers H and G, raises Q, and leaves A fixed.

3 / MEASURE DEVIATIONS FROM THE MEAN

RMS includes the centre; standard deviation isolates the spread

Let $v = \frac{1}{n}\sum_i (x_i - A)^2$ be the descriptive variance and $\mathrm{SD} = \sqrt{v}$. For any proposed centre c,

$$\frac{1}{n}\sum_i (x_i - c)^2 = v + (A - c)^2.$$

Expanding uses $\sum_i (x_i - A) = 0$. Hence $\boxed{Q^2 = A^2 + v}$, and RMS error is smallest at $c = A$.

The pairs $(3,3)$, $(2,4)$, $(1,5)$ have SDs 0, 1, 2 respectively, although all three have $A = 3$.

4 / REPLACE THE CONSTANT BY A LINE

Gaussian errors turn maximum likelihood into least squares

For a trial line $mt + c$, define the signed vertical residuals $r_i = y_i - (mt_i + c)$ and $\mathrm{SSE} = \sum_i r_i^2$.

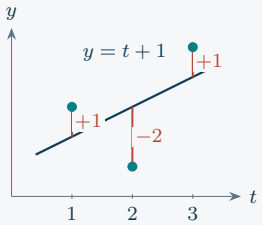
Assume independent Gaussian errors, each with mean zero and the same SD σ. Multiplying their densities gives the **likelihood**, which scores the line's parameters:

$$L(m,c) \;\propto\; \prod_i \exp\left(-\frac{r_i^2}{2\sigma^2}\right) = \exp\left(-\frac{\mathrm{SSE}}{2\sigma^2}\right).$$

For fixed σ, taking logarithms gives

$$-\ln L = C + \frac{\mathrm{SSE}}{2\sigma^2},$$

where C does not depend on m, c.



$r = (1, -2, 1)$ has mean 0, but $\mathrm{SSE} = 6$ and $\mathrm{RMSE} = \sqrt{2}$.

maximise L $\iff$ minimise SSE $\iff$ minimise $\mathrm{RMSE} = \sqrt{\mathrm{SSE}/n}$

Squaring prevents cancellation; the Gaussian density supplies the squared-error penalty. Doubling a residual quadruples its contribution. The root restores the units of y.

Conventions. v divides by n. For a fitted line with $n > 2$ and varying inputs, the usual error-variance estimate is $\mathrm{SSE}/(n-2)$. Least squares can also be used without Gaussian errors.

Sources. Université de Montréal, *Inequalities*, §§1–2; NIST, *Normal Distribution*; NIST, *Parameter Estimation*.

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NORMALISATION AND CUBOIDS · 1 OF 2

Normalisation separates size from shape

Choose a common unit of comparison, then keep track of what the ratio preserves.

1 / DECIDE WHAT WE MEAN BY BIGGER

Capacity, surface and reach are different measurements

For edges $p, q, r > 0$, volume measures capacity, surface area measures the boundary, and the longest diagonal measures corner-to-corner reach:

$$V = pqr, \qquad S = 2(pq + qr + rp), \qquad D = \sqrt{p^2 + q^2 + r^2}.$$

Doubling every edge multiplies D, S, V by $2, 4, 8$. Consequently S/V halves: it measures **surface per unit capacity**, including the effect of size. It is useful for that purpose, and at fixed volume it gives the surface-area order. To compare *shape alone*, we need a score unchanged by enlarging an individual cuboid.

2 / THE OPERATION OF NORMALISING

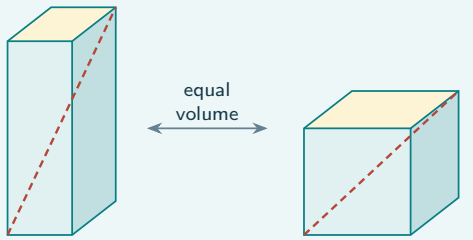
Divide every edge by the same reference length

Choose $G = V^{1/3}$, the side of the equal-volume cube. Measuring all lengths in units of G gives

$$(x, y, z) = \left(\frac{p}{G}, \frac{q}{G}, \frac{r}{G}\right), \qquad xyz = 1.$$

The edge ratios are preserved and every cuboid now has unit volume. A cube becomes $(1, 1, 1)$.

For the example $(2, 3, 6)\,\mathrm{cm}$, $G = \sqrt[3]{36}\,\mathrm{cm}$. The drawing compares its normalised shape with the reference cube.



normalised cuboid
$\approx (0.606, 0.909, 1.817)$
volume 1, diagonal ≈ 2.120

reference cube
$(1, 1, 1)$
volume 1, diagonal $= \sqrt{3}$

Dashed lines are body diagonals. Both drawings use the same normalised scale, so the comparison now describes shape.

3 / MAKE THE REFERENCE CUBE SCORE ONE

I normalises surface area; J_V normalises diagonal length

Areas divide by G^2 and lengths by G. The unit cube has surface area 6 and diagonal $\sqrt{3}$, which explain the final factors:

	original	after dividing edges by G	relative to the unit cube
surface	S	S/G^2	$I = \frac{S}{6G^2}$
diagonal	D	D/G	$J_V = \frac{D}{\sqrt{3}\,G}$

For the example, $S/G^2 \approx 6.604$ and $D/G \approx 2.120$. Thus $I \approx 1.101$ means 10.1% **extra surface**, while $J_V \approx 1.224$ means a 22.4% **longer diagonal**, each compared with the cube of the same volume. Among cuboids both scores are at least 1, with equality exactly for a cube.

4 / CHOOSING AND RETAINING THE SCALE

A different constraint gives a different useful ratio

For a fixed surface budget, choose the equal-surface-area cube instead. Its side is $\ell_S = \sqrt{S/6}$ and its diagonal is $\sqrt{S/2}$:

$$J_S \quad = \quad \frac{D}{\sqrt{S/2}} \quad = \quad \frac{J_V}{\sqrt{I}}.$$

For the same $2 \times 3 \times 6$ cuboid, $D = 7$ and $S = 72$, so $J_S = 7/6 \approx 1.167$: its diagonal is 16.7% longer than that of the cube with the same surface area.

Normalisation separates information about size from information about shape. Keeping G with (x, y, z) allows full reconstruction: $p = Gx$, $q = Gy$, $r = Gz$. Reporting only J_V compresses shape into one number and also omits the absolute scale. Restore that scale using

$$D = \sqrt{3}\,G J_V, \qquad V = G^3 \quad \Longrightarrow \quad V = \left(\frac{D}{\sqrt{3}\,J_V}\right)^3.$$

So J_V alone gives relative reach; D together with J_V determines volume.

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PERCENTAGES, GEOMETRIC AVERAGES AND LOGARITHMS ON ONE PAGE

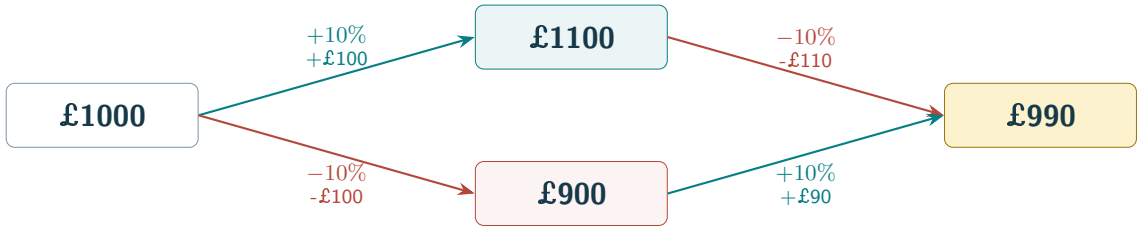
The rightful fear of loss

Why an equal percentage gain cannot undo a percentage loss.

1 / FOLLOW THE SAME MONEY

The percentage base changes at each step

Start with £1000. Each change is a percentage of the pot *at that moment*.



$$1.1 \times 0.9 = 0.99, \qquad (1+p)(1-p) = 1-p^2 \quad (0<p<1)$$

Either order loses 1% of the starting pot. In general, an equal up-and-down pair loses the fraction p^2; here $p = 0.10$.

2 / WHAT WOULD ACTUALLY UNDO THE LOSS?

The climb back is steeper

A loss of fraction p leaves $(1-p)P$. To recover the original P, the next gain q must satisfy

$$(1-p)(1+q) = 1.$$

$$q = \frac{1}{1-p} - 1 = \frac{p}{1-p}$$

loss	gain to recover
10%	11.11%
20%	25%
25%	33.33%
50%	100%
75%	300%
90%	900%

At 100% loss, every finite multiplier still gives zero.

3 / WHICH AVERAGE FOLLOWS THE POT?

Average the growth factors

The arithmetic average of $+10\%$ and -10% is 0%. The constant rate g giving the same two-period outcome instead solves $(1+g)^2 = 0.99$. Its return per period is

$$g = \sqrt{0.99} - 1 \approx -0.5013\%.$$

For n equal-length periods with fractional returns $r_1, \ldots, r_n$, take the geometric mean of their *multipliers*:

$$1 + g = \left(\prod_{k=1}^{n}(1+r_k)\right)^{1/n}$$

Then $P_n = P_0(1+g)^n$. This geometric rate never exceeds the arithmetic mean of the same returns; equality holds only when all the period returns are identical.

4 / PRODUCTS BECOME SUMS

Logarithmic returns add exactly

A fractional return r_k multiplies the pot by $1 + r_k$. Define its **logarithmic return** by $L_k = \ln(1+r_k)$. Then

$$\ln\left(\frac{P_n}{P_0}\right) = \sum_{k=1}^{n} L_k.$$

For our pair, $\ln(1.1) \approx 0.09531$ and $\ln(0.9) \approx -0.10536$. These add to

$$\ln(1.1) + \ln(0.9) = \ln(0.99) \approx -0.01005.$$

A true reversal uses *reciprocal* multipliers:

$$0.9 \times \frac{10}{9} = 1, \qquad \ln(0.9) + \ln\left(\frac{10}{9}\right) = 0$$

Average the log returns to get $\overline{L}$. Exponentiating gives the geometric rate: $g = \exp(\overline{L}) - 1$.

5 / THE GEOMETRY OF RECOVERY

Bigger losses demand much bigger gains

The solid curve gives the recovery gain; the dashed line shows an equal percentage gain.



As $p \to 1$, the recovery gain $p/(1-p)$ grows without bound.

Small shortfalls compound too. Fifty $+10\%, -10\%$ pairs leave $0.99^{50} \approx 0.605$ of the starting pot: about £605 from £1000.

Conventions. One pot, no payments in or out; $P_0 > 0$ and $r_k > -1$ for logarithms. Equal cash gains and losses cancel; equal percentage changes act on different bases.

Further reading. Will Morrison, *The Myth of Volatility Drag (Part 1)*, CFA Institute, 23 March 2015.

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GEOMETRIC SERIES AND SAVING ON ONE PAGE

£1000 a month to a billion

A savings account is a finite geometric series, and the interest rate decides whether the wait is a lifetime or a geological age.

THE RULE

Multiply by the ratio and subtract

For $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$, the shifted copy rS_n shares every term except the first and last, so $S_n - rS_n = a - ar^n$, and when $|r| < 1$ the last term dies away:

$$S_n = a\,\frac{r^n - 1}{r - 1}, \qquad S_\infty = \frac{a}{1 - r} \quad (|r| < 1)$$

Depositing D at the end of every month into an account paying a nominal annual rate r, credited monthly at $i = r/12$, makes the oldest deposit grow by $(1+i)^{n-1}$ and the newest not at all, so after n months the balance is a finite geometric series with ratio $1+i$:

$$B_n = D\big(1 + (1+i) + \cdots + (1+i)^{n-1}\big) = D\,\frac{(1+i)^n - 1}{i}.$$

ANNUALISING

Nominal, effective and geometric

A quoted 7% credited monthly is a nominal rate. Twelve months of $\frac{7}{12}\%$ compound to the effective annual rate, the AER on a UK savings advert,

$$(1 + \tfrac{r}{12})^{12} - 1 = 7.229\%,$$

and as the crediting becomes continuous the limit is $\mathrm{e}^r - 1 = 7.251\%$. When yearly returns vary, the rate that reproduces the final balance is their geometric mean, which falls below their arithmetic mean by roughly half the variance.

THE INFINITE SERIES

What pot pays £1000 a month forever?

At a fixed nominal annual rate of 5%, credited monthly, $i = 0.05/12$. A pot of £240 000 earns £1000 each month. Withdraw the interest and the original pot remains.

We can also build this pot payment by payment. To provide £1000 in k months, set aside $1000/(1+i)^k$ today. Adding these amounts gives

$$P = \frac{1000}{1+i} + \frac{1000}{(1+i)^2} + \cdots = \frac{1000/(1+i)}{1 - 1/(1+i)} = \frac{1000}{i} = £240\,000.$$

This is the **present value of a perpetuity**: the capital needed now to fund those payments indefinitely, assuming a constant rate and ignoring tax and inflation.

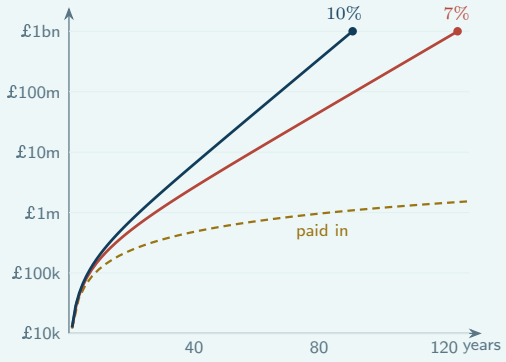
HOW LONG TO A BILLION

Solve the series for n with a logarithm

Setting $B_n = £10^9$ with $D = £1000$ and taking logarithms,

$$n = \frac{\ln\big(1 + 10^9\, i/D\big)}{\ln(1+i)} \text{ months}$$

nominal r	AER	months	years	paid in
0%	0%	1 000 000	83 333	£1000m
3%	3.04%	3134	261	£3.13m
5%	5.12%	2005	167	£2.01m
7%	7.23%	1491	124	£1.49m
10%	10.47%	1088	91	£1.09m
15%	16.08%	759	63	£0.76m



Balance on a logarithmic scale: the sum paid in flattens out while compound interest keeps climbing.

A WORKING LIFETIME

Forty years at 7%

After 480 months the balance is $1000 \times \frac{1.005833^{480} - 1}{0.005833} \approx £2.62\text{m}$ from £480 000 paid in. Money at 7% doubles in about $\ln 2 / \ln 1.07 \approx 10.2$ years, the rule of 72, so each extra decade roughly doubles the pot, which is why the last stretch to a billion is so much faster than the first million.

WHAT THE ARITHMETIC LEAVES OUT

Inflation, volatility and tax

A billion pounds in 124 years' time buys far less than today; at 2% inflation the real rate is about 5%, so a billion in today's money takes about 167 years. Variable investment returns reach a lower final balance than a steady account with the same average, and tax takes a share of the interest along the way. The sums illustrate geometric series and are not financial advice.

Conventions. r is nominal annual rate, $i = r/12$ monthly rate, deposits are made at end of each month, and AER is annual equivalent rate $(1+i)^{12} - 1$.

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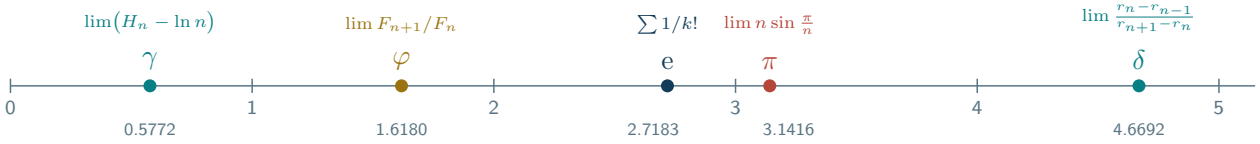
FIVE GREAT CONSTANTS FROM 0 TO 5 ON ONE PAGE

γ, φ, e, π, δ, one in each unit interval

Each is the limit of a simple repeated process, from a root of a quadratic to numbers not yet proved irrational.

THE RULE

Every one of them is a limit



BETWEEN 0 AND 1

Euler–Mascheroni $\gamma = 0.577\,215\,66\ldots$

The harmonic sum outruns $\ln n$ by a gap that settles, and splitting it term by term gives a convergent series and an integral:

$$\gamma = \sum_{k\geq 1}\left(\frac{1}{k} - \ln\left(1+\frac{1}{k}\right)\right) = -\int_0^\infty \mathrm{e}^{-x}\ln x\,\mathrm{d}x.$$

It has no known trigonometric form, and it also appears as $-\Gamma'(1)$.

BETWEEN 1 AND 2

Golden ratio $\varphi = 1.618\,033\,98\ldots$

The positive root of $x^2 = x + 1$, so $x = 1 + \frac{1}{x}$ repeated gives the continued fraction $[1;1,1,1,\ldots]$ and the Fibonacci ratios $\frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \ldots$. In trigonometric form it is the diagonal of a regular pentagon of side 1,

$$\varphi = \frac{1+\sqrt{5}}{2} = 2\cos\frac{\pi}{5} = 2\cos 36^\circ.$$

BETWEEN 2 AND 3

Euler's number $\mathrm{e} = 2.718\,281\,83\ldots$

Compound growth and factorials give the same limit,

$$\mathrm{e} = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = \sum_{k\geq 0}\frac{1}{k!} = 1 + 1 + \tfrac{1}{2} + \tfrac{1}{6} + \ldots,$$

where eight terms already give 2.71825. Its trigonometric face is $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$, which ties it to π through $\mathrm{e}^{\mathrm{i}\pi} = -1$.

BETWEEN 3 AND 4

Archimedes' constant $\pi = 3.141\,592\,65\ldots$

Inscribed polygons give $n\sin\frac{\pi}{n}$, or $n\sin\frac{180^\circ}{n}$, which is 3 for a hexagon, 3.1058 for a 12-gon and 3.1410 for Archimedes' 96-gon. Arctangent series give it exactly,

$$\frac{\pi}{4} = \arctan 1 = 1 - \tfrac{1}{3} + \tfrac{1}{5} - \cdots = 4\arctan\tfrac{1}{5} - \arctan\tfrac{1}{239},$$

the last being Machin's formula of 1706.

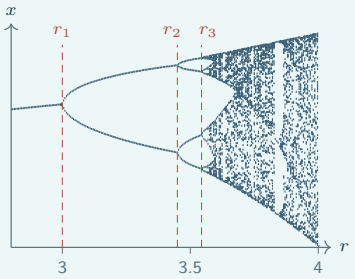
BETWEEN 4 AND 5

Feigenbaum's $\delta = 4.669\,201\,61\ldots$

Iterate the logistic map $x \mapsto rx(1-x)$. Its long-run behaviour settles on one value until $r_1 = 3$, on a cycle of two until $r_2 = 1+\sqrt{6} \approx 3.4495$, of four until $r_3 \approx 3.5441$, of eight until $r_4 \approx 3.5644$, and the gaps shrink by a ratio that converges:

$$\frac{r_2 - r_1}{r_3 - r_2} = 4.751, \quad \frac{r_3 - r_2}{r_4 - r_3} = 4.656, \quad \ldots \quad \to \quad \delta = 4.6692\ldots$$

Feigenbaum found in 1975 that every smooth one-humped map gives the same δ, a universal constant of the route to chaos with no known series or trigonometric form.



Long-run values of the logistic map: each fork arrives about 4.67 times sooner.

From algebraic to unknown

φ	irrational but algebraic, a root of $x^2 - x - 1$, so not transcendental
e and π	transcendental, proved by Hermite in 1873 and Lindemann in 1882
γ and δ	believed transcendental, though neither has yet been proved even irrational

Other constants share these intervals, such as $\sqrt{2} \approx 1.414$ and Apéry's $\zeta(3) \approx 1.202$, proved irrational in 1978 but not yet transcendental, and even $\mathrm{e} + \pi$ is not known to be irrational.

Conventions. A number is algebraic if it solves a polynomial equation with integer coefficients and transcendental otherwise; H_n is the nth harmonic number, F_n the nth Fibonacci number, and r_n the parameter where the logistic cycle first reaches length 2^n.

IRRATIONAL POWERS ON ONE PAGE

π^{e} or e^{π}, and the 23 between them

An irrational power is defined through $\ln$, and the same logarithm settles which of two famous powers is larger.

THE RULE

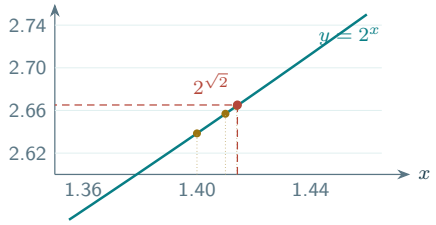
An irrational power is a limit, or a logarithm

A rational power is a root of a whole power, $2^{7/5} = \sqrt[5]{2^7}$. An irrational exponent such as $\sqrt{2}$ is approached by its decimal truncations, and because 2^x increases steadily the powers settle on one value,

$$2^{1.4} = 2.639, \quad 2^{1.41} = 2.657, \quad 2^{1.414} = 2.6647, \quad \ldots \to 2^{\sqrt{2}} \approx 2.66514.$$

The same number comes from a single formula for every real y and base $x > 0$, which turns any comparison of powers into one of logarithms:

$$x^y \quad = \quad \mathrm{e}^{\,y \ln x}$$



Truncations of $\sqrt{2}$ climb the graph of 2^x towards $2^{\sqrt{2}}$.

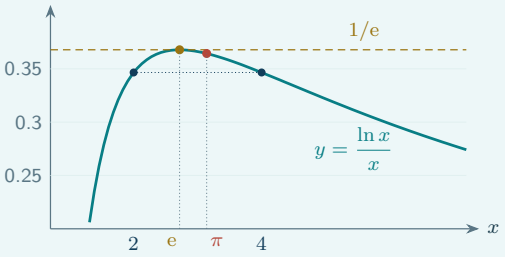
WHICH IS BIGGER?

Compare $\dfrac{\ln x}{x}$, which peaks at $x = \mathrm{e}$

Taking $\ln$ of both numbers gives $\ln(\pi^{\mathrm{e}}) = \mathrm{e}\ln\pi$ and $\ln(\mathrm{e}^{\pi}) = \pi$, and dividing both by $\mathrm{e}\pi$ reduces the question to comparing $\frac{\ln\pi}{\pi}$ with $\frac{\ln \mathrm{e}}{\mathrm{e}}$. The function $f(x) = \frac{\ln x}{x}$ has $f'(x) = \frac{1-\ln x}{x^2}$, which vanishes only at $x = \mathrm{e}$, so its single maximum is $f(\mathrm{e}) = \frac{1}{\mathrm{e}}$ and every other $x > 0$ has $f(x) < \frac{1}{\mathrm{e}}$, that is $x^{\mathrm{e}} < \mathrm{e}^x$. Putting $x = \pi$,

$$\pi^{\mathrm{e}} \quad \approx \quad 22.459 \quad < \quad \mathrm{e}^{\pi} \quad \approx \quad 23.141$$

The margin is thin because π sits just past the peak, where $f(\pi) = 0.3644$ against $f(\mathrm{e}) = 0.3679$. The same graph explains the only whole-number coincidence of this kind, $2^4 = 4^2$, since $f(2) = f(4)$.



The peak at e beats every other value, π only narrowly; 2 and 4 share a height.

GOOD GUESSES

Both by hand, from $\mathrm{e}^3 \approx 20.09$

Splitting each exponent at 3 leaves a small t for $\mathrm{e}^t \approx 1 + t + \frac{t^2}{2}$:

$$\mathrm{e}^{\pi} = \mathrm{e}^3\,\mathrm{e}^{0.1416} \approx 20.09 \times 1.152 \approx 23.14,$$

and with $\ln\pi \approx \ln 3 + 0.046 \approx 1.145$, so that $\mathrm{e}\ln\pi \approx 3.112$,

$$\pi^{\mathrm{e}} = \mathrm{e}^3\,\mathrm{e}^{0.112} \approx 20.09 \times 1.118 \approx 22.46.$$

WHAT IS KNOWN

One is proved transcendental, one is open

Since $\mathrm{e}^{\pi} = (-1)^{-\mathrm{i}}$, an algebraic number raised to a non-rational algebraic power, the Gelfond–Schneider theorem makes Gelfond's constant e^{π} transcendental. Nobody has yet proved that π^{e} is even irrational. A near miss worth knowing is $\mathrm{e}^{\pi} - \pi = 19.99910\ldots$, which has no known explanation.

THE ONE INTEGER IN THE GAP

$22.459 < 23 < 23.141$, and 23 is where cyclotomic factorisation first fails

The interval between the two powers is only 0.68 wide and holds exactly one whole number, 23. In 1847 Lamé announced a proof of Fermat's Last Theorem that factorised $x^p + y^p = \prod_k (x + \zeta^k y)$ in the cyclotomic integers $\mathbb{Z}[\zeta_p]$, with $\zeta_p = \mathrm{e}^{2\pi\mathrm{i}/p}$, and silently assumed that factorisation there is unique. Kummer showed the assumption holds for every prime $p \le 19$ and fails first at $p = 23$, where the class number of $\mathbb{Q}(\zeta_{23})$ is 3; he rescued unique factorisation by inventing ideal numbers, the ancestors of modern ideals.

$$h\big(\mathbb{Q}(\zeta_p)\big) \quad = \quad 1 \text{ for } p \quad = \quad 2, 3, 5, 7, 11, 13, 17, 19, \qquad h\big(\mathbb{Q}(\zeta_{23})\big) \quad = \quad 3$$

The class number is the one from the Heegner sheet, where $h = 1$ for $\mathbb{Q}(\sqrt{-163})$ gave forty primes in a row; that 23 also falls between π^{e} and e^{π} is an accident of size.

Conventions. For a positive base, $x^y = \mathrm{e}^{y\ln x}$ is the unique real value and agrees with roots and integer powers whenever y is rational; for a negative or complex base the logarithm has many branches and the value depends on the branch chosen. The class number h measures how far a ring of integers is from unique factorisation, with $h = 1$ meaning it has it.

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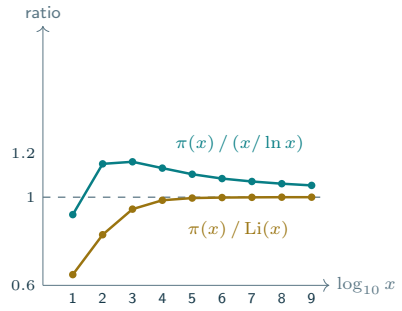
NUMBER THEORY ON ONE PAGE

How the primes thin out

The density of primes near x is 1/ln x, so the count is x/ln x and the typical gap is ln x, a falloff with no characteristic scale.

1 / THE RULE AND THE PICTURE

The prime number theorem says $\pi(x) \sim x/\ln x$, and the logarithmic integral fits far better



ratios of the true count $\pi(x)$ to two approximations

Near a large x a randomly chosen number is prime with probability $1/\ln x$, so counting up to x,

$$\pi(x) \sim \tfrac{x}{\ln x}, \qquad \pi(x) \approx \mathrm{Li}(x) = \int_2^x \tfrac{dt}{\ln t}$$

The ratio to $x/\ln x$ tends to 1 very slowly, still 1.054 at 10^9, while $\mathrm{Li}(x)$ agrees to four figures there, since it integrates the local density $1/\ln t$ instead of freezing it at x.

2 / THE COUNTS

The density of primes falls from 17% at a thousand to 5% at a billion

x	$\pi(x)$	$x/\ln x$	ratio	$\mathrm{Li}(x)$	π/Li
10^3	168	145	1.161	178	0.9459
10^4	1 229	1 086	1.132	1 246	0.9862
10^5	9 592	8 686	1.104	9 630	0.9961
10^6	78 498	72 382	1.085	78 628	0.9984
10^7	664 579	620 421	1.071	664 918	0.9995
10^8	5 761 455	5 428 681	1.061	5 762 209	0.9999
10^9	50 847 534	48 254 942	1.054	50 849 235	1.0000

The density $\pi(x)/x$ is 0.168 at 10^3, 0.0785 at 10^6 and 0.0508 at 10^9.

3 / THE GAPS

The mean gap is $\ln x$, and the largest gap below x stays of the order $(\ln x)^2$

x	$\ln x$	largest gap below x	follows the prime	$(\ln x)^2$	gap/$\ln x$
10^3	6.91	20	887	48	2.9
10^4	9.21	36	9551	85	3.9
10^5	11.51	72	31397	133	6.3
10^6	13.82	114	492113	191	8.3
10^7	16.12	154	4652353	260	9.6
10^8	18.42	220	47326693	339	11.9

The record gaps grow only logarithmically: 220 below 10^8 against a mean of 18.4, below Cramér's heuristic $(\ln x)^2 = 339$.

4 / LOCAL EXPONENTS / HOW FAST EACH QUANTITY FALLS

The log-log slope of every prime quantity tends to a whole number, but only as $1/\ln x$

quantity	exponent $n = \frac{d\ln f}{d\ln x}$	$x = 10^2$	10^6	10^9	limit
density $1/\ln x$	$-1/\ln x$	-0.217	-0.072	-0.048	0
count $x/\ln x$	$1 - 1/\ln x$	0.783	0.928	0.952	1
mean gap $\ln x$	$+1/\ln x$	0.217	0.072	0.048	0
relative gap $\ln x/x$	$-1 + 1/\ln x$	-0.783	-0.928	-0.952	-1

A power law has a constant exponent and an exponential has an exponent that grows without bound, so the primes sit between them: slopes that drift towards 0 or ± 1 at the pace of $1/\ln x$, which is why the density has no scale at which it suddenly drops.

5 / A FIRST MODEL / THE SIEVE AS A STACK OF FILTERS

Each prime p removes a fraction $1/p$ of the integers, and the survivors are a product

The fraction of integers with no prime factor up to y is $\prod_{p \le y}(1 - \frac{1}{p})$, and Mertens showed it is asymptotic to $e^{-\gamma}/\ln y$ with $\gamma = 0.5772$: for $y = 1000$ the product is 0.0810 against the formula 0.0813.
Taking $y = \sqrt{n}$ so that survivors are primes gives $2e^{-\gamma}/\ln n = 1.123/\ln n$, which is 12% above the true density $1/\ln n$ because divisibility by different primes is not quite independent.

ASIDE / ONE LOG OR TWO

Thinning as $1/\ln x$ is slower than any power $x^{-\epsilon}$, which is why prime tables stay usefully dense far out.

LINKS TO OTHER SHEETS

The attenuation analogy continues in *Three attenuators*; the local exponent idea is introduced in *Combining falloffs*.

Convention. $\pi(x)$ counts primes up to x; $\mathrm{Li}(x)$ is the logarithmic integral; γ is the Euler constant; gaps are differences of consecutive primes.

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PRIME CLASSES, CUMULATIVE TOTALS AND TWO USES OF LOGARITHMS

Regular, irregular and combined

Follow each class separately, then add them to recover the whole.

1 / THREE RISING CURVES

Every prime contributes to its class and to the combined total

Use all 24 odd primes below 100: 21 are regular and 3 are irregular, 37, 59, 67, by the Bernoulli numerator test. Each prime belongs to exactly one class, so at every position

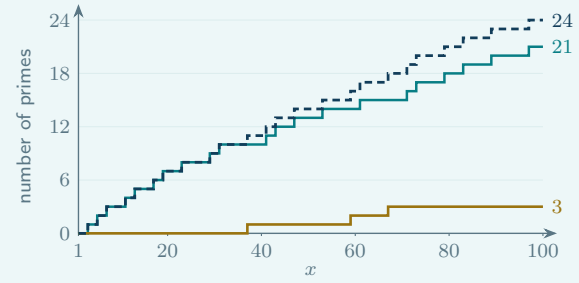
$$\text{combined height} = \text{regular height} + \text{irregular height}.$$

Two independent choices: unit jumps add 1 at each prime; log-weighted jumps add $\ln p$ at prime p. Using $t = \ln x$ on the horizontal axis changes the positions of steps while preserving their heights.

regular — irregular — combined

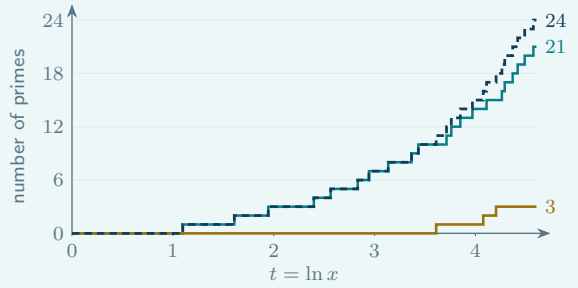
A / UNIT JUMPS

Linear position x



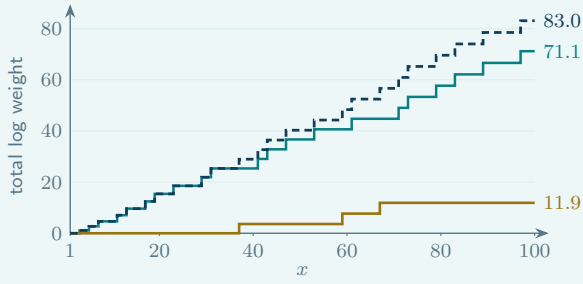
B / UNIT JUMPS

Logarithmic position $t = \ln x$



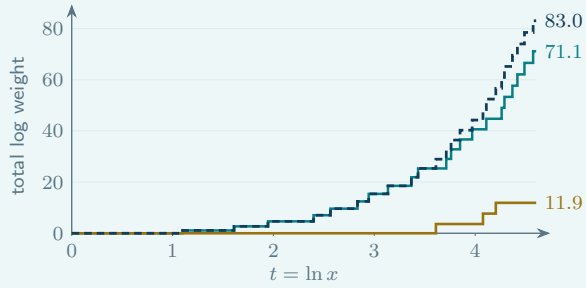
C / JUMPS OF HEIGHT $\ln p$

Linear position x



D / JUMPS OF HEIGHT $\ln p$

Logarithmic position $t = \ln x$



The two plots in each row use the same vertical scale. On the right, $x = 1, 10, 100$ correspond to $t = 0, 2.303, 4.605$.

2 / ONE EVENT, FOUR VIEWS

What happens at 37?

The irregular and combined curves rise together; the regular curve stays flat.

panel	position	jump
A	37	1
B	$\ln 37 \approx 3.611$	1
C	37	$\ln 37 \approx 3.611$
D	$\ln 37 \approx 3.611$	$\ln 37 \approx 3.611$

3 / READ THE ACCUMULATED TOTALS

Where the curves finish at $x = 100$

class	unit total	log total
regular	21	71.142
irregular	3	11.893
combined	24	83.035

Unit totals count primes. Log totals give the logarithm of the product of the primes reached. All logarithms here are natural logarithms.

4 / COMPARE EACH PART WITH THE WHOLE

The vertical gap from the regular curve to the combined curve is exactly the irregular curve's height, in both jump modes. The regular and combined curves coincide until the first irregular prime, 37. The combined curves contain all odd primes; 2 is outside this classification. Jumps occur at primes only, and the example is restricted to the stated window.

Sources. The Bernoulli Number Page (regularity and the first irregular primes); T. Tao, *246B, Notes 4* (2021) (logarithmic prime weights). Counts, weights and plots calculated for this finite example.

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COUNTING, LOGARITHMS AND THE FIRST HINT OF FOURIER ON ONE PAGE

From prime staircases to prime waves

Count the primes, change the jump sizes, then look at the remaining fluctuations.

1 / THE CLASSIC STAIRCASE

Add one whenever a prime is reached

Write $\pi(x)$ for the number of primes at or below x. It jumps by 1 at $2, 3, 5, 7, \ldots$ and stays flat between primes. Thus $\pi(10) = 4$. The horizontal gaps between steps are the prime gaps.

2 / THE FIRST RULE: EACH PRIME CONTRIBUTES ONCE

θ (theta): welcome each new prime

Keep a running total, adding $\ln 2$ when 2 is reached, $\ln 3$ when 3 is reached, and so on. Once a prime has contributed, it is finished: $4, 8, 16$ **produce no further contribution from** 2.

$$\theta(10) = \ln 2 + \ln 3 + \ln 5 + \ln 7 = \ln(2 \cdot 3 \cdot 5 \cdot 7) = \ln 210 \approx 5.35.$$

Here $\ln$ is the natural logarithm. The rule $\ln(ab) = \ln a + \ln b$ turns the product of **different primes** into a sum.

3 / THE SECOND RULE: EACH PRIME POWER CONTRIBUTES

ψ (psi): welcome each new power of a prime

Let 2 contribute at $2, 4, 8, 16, \ldots$, adding $\ln 2$ each time. Let 3 contribute at $3, 9, 27, \ldots$, adding $\ln 3$ each time. The table lists **new jumps at selected numbers**; 0 means the staircase stays flat:

number reached	2	3	4	5	6	7	8	9	16
jump in θ	$\ln 2$	$\ln 3$	0	$\ln 5$	0	$\ln 7$	0	0	0
jump in ψ	$\ln 2$	$\ln 3$	$\ln 2$	$\ln 5$	0	$\ln 7$	$\ln 2$	$\ln 3$	$\ln 2$

At $8 = 2^3$, the added weight is still $\ln 2$. At $6 = 2 \cdot 3$, neither staircase jumps: 6 is neither a prime nor a power of one prime.

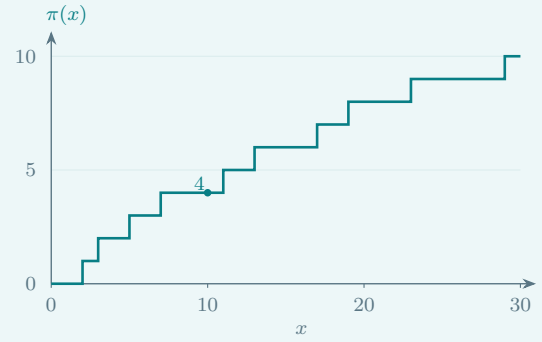
$$\psi(10) = 3\ln 2 + 2\ln 3 + \ln 5 + \ln 7 = \ln(2^3 \cdot 3^2 \cdot 5 \cdot 7) = \ln 2520 \approx 7.83.$$

The product $2^3 \cdot 3^2 \cdot 5 \cdot 7 = 2520$ is the **least common multiple of** $1, \ldots, 10$: every number in that list divides it.

4A / UNIT JUMPS

The ordinary count $\pi(x)$

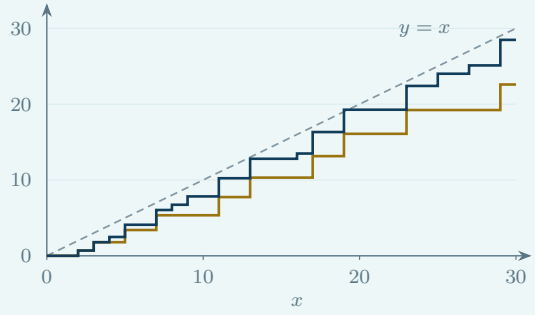
Ten primes have appeared by $x = 30$.



4B / LOGARITHMIC JUMPS

Two weighted counts

θ: each prime once ψ: each prime power



The extra jumps of ψ up to 30 occur at $4, 8, 9, 16, 25, 27$. The plots use the same horizontal scale and different vertical scales.

5 / THE PRIME NUMBER THEOREM

Logarithmic weights give a straight main trend

$$\pi(x) \sim \frac{x}{\ln x}, \qquad \theta(x) \sim x, \qquad \psi(x) \sim x.$$

The symbol $\sim$ means ratio $\to 1$ as x grows. Subtract x from $\psi(x)$ to isolate the fluctuations.

6 / A BRIDGE TO THE NEXT SHEET

Look for waves in logarithmic position

Set $t = \ln x$: multiplication of x becomes addition of t. A zeta zero $\beta + \mathrm{i}\gamma$ supplies terms involving

$$x^{\beta+\mathrm{i}\gamma} = x^{\beta} \mathrm{e}^{\mathrm{i}\gamma \ln x} = x^{\beta}\big(\cos(\gamma t) + \mathrm{i}\sin(\gamma t)\big).$$

These logarithmic waves have changing amplitudes; the full explicit formula also includes correction terms.

Sources. NIST DLMF, §25.16 (weighted counting and the prime number theorem); T. Tao, *246B, Notes 4* (2021); A. Granville, *The Explicit Formula*, Ch. 10. Examples and plots calculated for this sheet.

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KEPLER'S THIRD LAW ON LOG–LOG AXES

The planets draw a power law

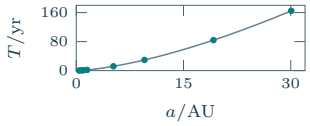
Observations reveal the slope. Dimensions explain the exponent. Dynamics supplies the coefficient.

1 Planetary data

Planet	a/AU	T/yr
Mercury	0.387	0.241
Venus	0.723	0.615
Earth	1.00	1.00
Mars	1.52	1.88
Jupiter	5.20	11.86
Saturn	9.54	29.46
Uranus	19.19	84.01
Neptune	30.06	164.82

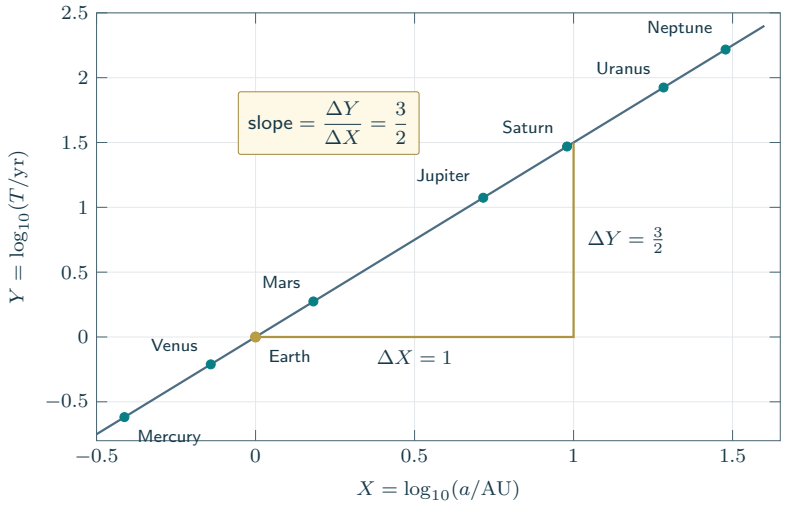
Rounded orbital values.

Linear axes



2 The same data, straightened

One decade across corresponds to one and a half decades up.



Data fit: slope $\simeq 1.500$. Reference line: $Y = \frac{3}{2}X$.

3 A power law becomes a straight line

$$\frac{T}{\mathrm{yr}} = K\left(\frac{a}{\mathrm{AU}}\right)^{p} \quad \Longleftrightarrow \quad \boxed{Y = pX + \log_{10} K}.$$

For planets around Sun, $p \simeq 3/2$ and $K \simeq 1$. Thus $T \propto a^{3/2}$, or equivalently $\boxed{T^2 \propto a^3}$. Here a is the ellipse's **semimajor axis**.

4 Dimensions predict the exponent

Hold the orbit's shape fixed, neglect the planet's mass, and suppose only a and GM set the period. Here M is the central mass.

$$T = C\,a^{\alpha}(GM)^{\beta}, \qquad [GM] = \mathsf{L}^3\mathsf{T}^{-2}, \qquad [T] = \mathsf{L}^{\alpha+3\beta}\mathsf{T}^{-2\beta}.$$

Matching length and time powers gives

$$\alpha + 3\beta = 0, \qquad -2\beta = 1 \quad \Longrightarrow \quad \boxed{T = C\sqrt{\frac{a^3}{GM}}}.$$

The $3/2$ exponent follows from these model inputs. For ellipses, dimensions could still allow C to depend on the dimensionless eccentricity e; the coefficient needs dynamics.

5 Newton supplies the coefficient

For a circular test-particle orbit, gravitational force provides the centripetal acceleration:

$$\frac{GMm}{a^2} = m\frac{4\pi^2 a}{T^2} \quad \Longrightarrow \quad \boxed{T = 2\pi\sqrt{\frac{a^3}{GM}}}.$$

The full elliptical solution gives the same result with a as the semimajor axis: $C(e) = 2\pi$, independent of eccentricity.

6 The architecture of scale

$$\boxed{a \mapsto sa \quad \Longrightarrow \quad T \mapsto s^{3/2}T} \qquad \text{at fixed central mass.}$$

$\times 2 \;\to\; \times 2.83 \qquad \times 4 \;\to\; \times 8 \qquad \times 10 \;\to\; \times 31.6$

Units and central mass shift the intercept. The exponent records the response to a change of scale.

Rounded data: NASA planetary fact sheets and OpenStax, *Astronomy 2e*, Table 3.2. Orbital law: NASA, *Orbits and Kepler's Laws*. Graphs and dimensional argument constructed from the stated data and assumptions.

SECONDS, LIFETIMES AND ORDERS OF MAGNITUDE ON ONE PAGE

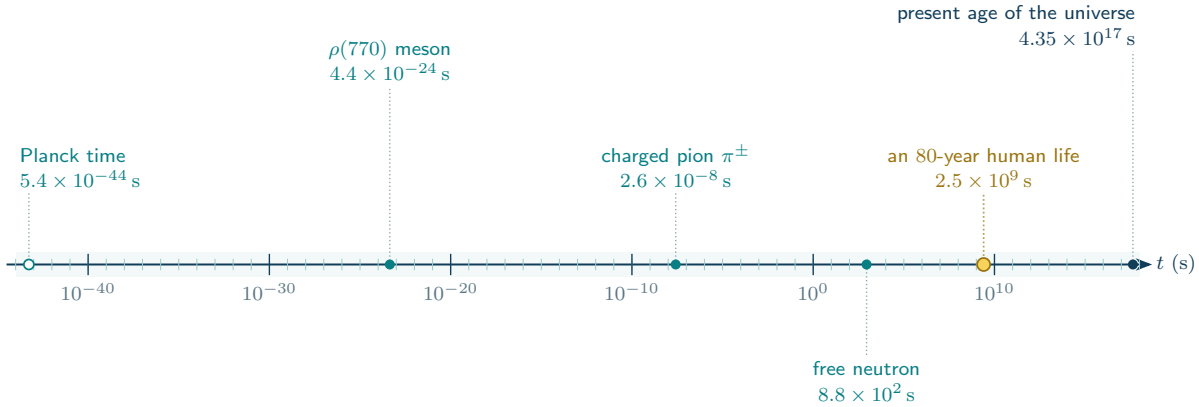
How long is a lifetime?

A logarithmic ruler from the rho meson to a human life and the age of the universe.

1 / PUT EVERY DURATION ON THE SAME AXIS

One ruler across about 61 orders of magnitude

$$x = \log_{10}\left(\frac{t}{1\,\mathrm{s}}\right), \qquad x_2 - x_1 = \log_{10}\left(\frac{t_2}{t_1}\right).$$



Each small tick is one **decade**: moving one tick right multiplies the duration by 10. The positions compare lengths of time.

2 / START WITH A FAMILIAR LIFESPAN

A human life is a few billion seconds

Using a 365.25-day year,

$$1\,\mathrm{yr} = 365.25 \times 24 \times 3600\,\mathrm{s} \approx 3.156 \times 10^{7}\,\mathrm{s}.$$

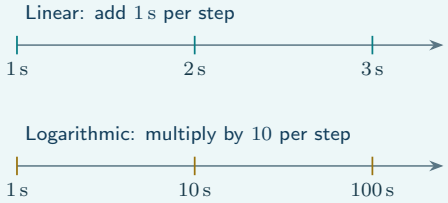
The mnemonic $1\,\mathrm{yr} \approx \pi \times 10^{7}\,\mathrm{s}$ is within about 0.5%. An illustrative human lifespan is therefore

$$T_H \;=\; 80\,\mathrm{yr} \;\approx\; 2.5 \times 10^{9}\,\mathrm{s}$$

Its log-position is $x_H \approx 9.40$, between 10^9 and 10^{10} seconds. Negative x means a positive duration shorter than one second.

3 / READ EQUAL SPACING CORRECTLY

A logarithmic midpoint is geometric



Halfway along a log interval means averaging the logarithms:

$$t_{\mathrm{mid}} = \sqrt{t_1 t_2}.$$

Between 1 s and 100 s, the midpoint is 10 s.

4 / GAUGING A LIFETIME IN RHO UNITS

How many rho lifetimes fit in ours?

Use $\tau_\rho \approx 4.4 \times 10^{-24}\,\mathrm{s}$ as a reference interval. The number of these intervals in an 80-year life is

$$N_H = \frac{T_H}{\tau_\rho} \approx \frac{2.5 \times 10^{9}}{4.4 \times 10^{-24}} \approx 5.7 \times 10^{32}.$$

The gap on the log ruler is therefore

$$\Delta x \;=\; \log_{10} N_H \;\approx\; 32.8 \quad \text{(about 33 decades)}$$

Even two tiny particle lifetimes can be enormously different:

$$\frac{\tau_{\pi^\pm}}{\tau_\rho} \approx 5.9 \times 10^{15},$$

about 16 orders of magnitude between the charged pion and the rho. Here a count means intervals of length τ_ρ.

5 / FROM PLANCK TIME TO COSMIC AGE

About 61 decades fit on one page

Compare the theoretical Planck time $t_P \approx 5.4 \times 10^{-44}\,\mathrm{s}$ with the universe's present age, about 13.8 billion years:

$$t_U \approx 4.35 \times 10^{17}\,\mathrm{s}.$$

$$\frac{t_U}{t_P} \;\approx\; 8.1 \times 10^{60}, \qquad \Delta x \;\approx\; 60.9$$

The Sun's main-sequence duration, roughly 10^{10} years or 3.2×10^{17} s, lies close to the cosmic-age mark on this ruler.

Changing to rho units shifts the zero:

$$x_\rho = \log_{10}\left(\frac{t}{\tau_\rho}\right) \approx x + 23.36.$$

The time ratios and their log-separations stay the same.

Conventions. All $t > 0$. Particle entries are approximate rest-frame mean lifetimes; Planck time is a theoretical scale, and the cosmic entry is an age. The human lifespan is a chosen example.

Origin. Lee McCulloch-James (2022), *Gauging a lifetime on skid rho*. **Values.** PDG mesons, PDG neutron, NIST/CODATA, NASA cosmos, NASA stars.

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DIMENSIONAL ANALYSIS ON ONE PAGE

The density of the universe from units alone

Gaussian elimination on a matrix of exponents delivers $\rho \sim H^2/G$, within a decade of the Friedmann critical density.

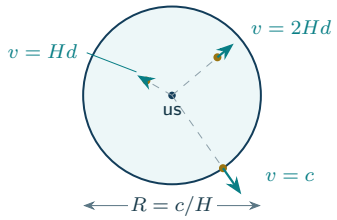
1 / THE RULE AND THE PICTURE

Every quantity is a vector of exponents of $\mathrm{M}, \mathrm{L}, \mathrm{T}$

A product $\rho^a H^b G^g c^k h^e$ is free of units exactly when the exponents of mass, length and time all cancel, which is the linear system

$$A\mathbf{x} = \mathbf{0}, \qquad \mathbf{x} = (a, b, g, k, e)^{\mathsf{T}}$$

Each column of A lists the $(\mathrm{M}, \mathrm{L}, \mathrm{T})$ exponents of one quantity, read from its own defining equation, and n quantities with $\operatorname{rank} A = r$ give $n - r$ dimensionless groups.



quantity	defining equation	rearranged	$(\mathrm{M}, \mathrm{L}, \mathrm{T})$
ρ density	$\rho = m/V$	$\mathrm{M/L^3}$	$(1, -3, 0)$
H Hubble	$v = Hd$	$(\mathrm{LT^{-1}})/\mathrm{L} = \mathrm{T^{-1}}$	$(0, 0, -1)$
G Newton	$F = Gm_1m_2/r^2$	$(\mathrm{MLT^{-2}})\mathrm{L^2/M^2}$	$(-1, 3, -2)$
c light	$c = d/t$	$\mathrm{L/T}$	$(0, 1, -1)$
h Planck	$E = h\nu$	$(\mathrm{ML^2T^{-2}})/\mathrm{T^{-1}}$	$(1, 2, -1)$

Force is $\mathrm{MLT^{-2}}$ and energy $\mathrm{ML^2T^{-2}}$ from $F = ma$ and $E = Fd$; $\hbar = h/2\pi$ changes no exponent. The picture shows H as a speed per distance, reaching c at $R = c/H$.

2 / ELIMINATE WITH h INCLUDED

Five quantities, rank 3, two groups

Columns ρ, H, G, c, h and rows $\mathrm{M}, \mathrm{L}, \mathrm{T}$. Add $3\times$ the M row to the L row, then swap the L and T rows and negate the new second row:

	ρ	H	G	c	h
M	1	0	−1	0	1
L	−3	0	3	1	2
T	0	−1	−2	−1	−1

$\Longrightarrow$

	ρ	H	G	c	h
	1	0	−1	0	1
	0	**1**	2	1	1
	0	0	0	**1**	5

The pivots (teal) sit in ρ, H, c, leaving G and h free (gold). Back-substitution from the bottom gives $k = -5e$, then $b = -2g + 4e$, then $a = g - e$, so that

$$\mathbf{x} = \mathbf{g}\,(1, -2, 1, 0, 0)^{\mathsf{T}} + \mathbf{e}\,(-1, 4, 0, -5, 1)^{\mathsf{T}}.$$

$$\Pi_A = \tfrac{G\rho}{H^2}, \quad \Pi_B = \tfrac{hH^4}{\rho c^5}, \quad \Pi_A\Pi_B = \tfrac{hGH^2}{c^5}$$

$$\rho = \frac{H^2}{G}\, F\!\left(\tfrac{hGH^2}{c^5}\right)$$

3 / CLOSE THE SYSTEM: LEAVE h OUT

Four quantities, rank 3, one group

Without Planck's constant the matrix loses its last column, and the same first step, $\mathrm{L} \leftarrow \mathrm{L} + 3\mathrm{M}$, leaves the row $(0, 0, 0, 1)$, which reads $k = 0$:

	ρ	H	G	c
M	1	0	−1	0
L	−3	0	3	1
T	0	−1	−2	−1

$\Longrightarrow$

	ρ	H	G	c
	1	0	−1	0
	0	**1**	2	1
	0	0	0	**1**

The speed of light is eliminated by its own row, because length can only be balanced between ρ and G. Only G is free, so $\mathbf{x} = \mathbf{g}\,(1, -2, 1, 0)^{\mathsf{T}}$ and the single group is $\Pi = G\rho/H^2$.

$$\rho = \Pi\,\frac{H^2}{G}$$

with Π a pure number that units cannot fix

Check: $[H^2/G] = \mathrm{T^{-2}} \cdot \mathrm{ML^{-3}T^2} = [\rho]$.

4 / PUT IN A NUMBER

What units alone predict

With $H_0 \approx 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}} = 2.27 \times 10^{-18}\ \mathrm{s^{-1}}$ and $G = 6.67 \times 10^{-11}$ in SI units,

$$\frac{H^2}{G} \approx 7.7 \times 10^{-26}\ \mathrm{kg\,m^{-3}} \approx 46 \text{ protons per cubic metre.}$$

Only H and G were used, yet the answer lands in the right decade.

5 / THE FRIEDMANN VALUE OF Π

The pure number from full general relativity

For a spatially flat universe $H^2 = \frac{8\pi G}{3}\rho$, so $\rho_c = \dfrac{3}{8\pi}\dfrac{H^2}{G}$ and $\Pi = \dfrac{3}{8\pi} \approx 0.119$, giving $\rho_c \approx 9.2 \times 10^{-27}\ \mathrm{kg\,m^{-3}}$, about 5.5 protons per cubic metre. The dimensional estimate therefore overshoots by $8\pi/3 \approx 8.4$, less than one decade.

ASIDE / THE HUBBLE SPHERE

Ask that the mass inside $R = c/H$ just reaches its Schwarzschild limit, $2GM/(Rc^2) = 1$. Then $M = c^3/(2GH)$ and $\rho = \dfrac{M}{\frac{4}{3}\pi R^3} = \dfrac{3H^2}{8\pi G}$ exactly, with the powers of c cancelling just as the elimination predicted.

ASIDE / WHY h STAYS OUT

The group $hGH^2/c^5 = (H\ell_P/c)^2$ is of order 10^{-122}, the ratio of H^2/G to the Planck density $c^5/(\hbar G^2)$. Friedmann's equation contains no h, so F is a constant; a quantum estimate with $F \sim \Pi^{-1}$ would instead return the Planck density, the familiar vacuum-energy discrepancy of about 122 orders of magnitude.

Convention. Buckingham's Π theorem: n quantities in r independent dimensions yield $n - r$ groups, the null space of A. In Python, `sympy.Matrix(A).nullspace()` returns the two vectors of panel 2. **Companions:** sqrt-time rule with dice; the Kepler log–log sheet; *How Many Scales Can You Build?*

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PLANETARY DATA, LOGARITHMS AND MODEL TESTING ON ONE PAGE

Bode's rule: turn the curve into a line

An exponential pattern in orbital spacing becomes visible after a shift and a logarithm.

1 / BEGIN WITH THE OBSERVATIONS

Measure orbit size, then choose the index

The semimajor axis a measures orbit size. Write $d = a/(1\,\mathrm{AU})$, so Earth has $d = 1$. The rounded distances below come from the supplied data.

body	m	d	$d - 0.4$
Mercury	–	0.39	−0.01
Venus	0	0.72	0.32
Earth	1	1.00	0.60
Mars	2	1.52	1.12
Ceres†	3	2.77	2.37
Jupiter	4	5.20	4.80
Saturn	5	9.55	9.15
Uranus	6	19.22	18.82
Neptune	7	30.11	29.71
Pluto†	8	39.54	39.14

† Ceres and Pluto are dwarf planets.

Number Venus $m = 0$, Earth $m = 1$, and continue outwards. Include Ceres at $m = 3$ in the asteroid-belt slot.

The gaps grow roughly geometrically. Try an offset exponential:

$$d = A + BQ^m.$$

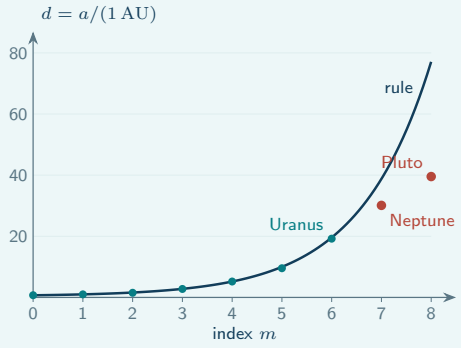
Test the *historical* offset $A = 0.4$ by plotting $d - 0.4$ on a logarithmic scale. The offset is an assumption; the straight-line fit estimates B and Q.

Mercury is a separate $d \approx 0.4$ case. Its observed $d - 0.4 < 0$ cannot enter the logarithmic plot.

2 / ORDINARY LINEAR AXES

The orbital gaps widen

Plot d against m. Points are observations; the curve is the historical Bode rule.

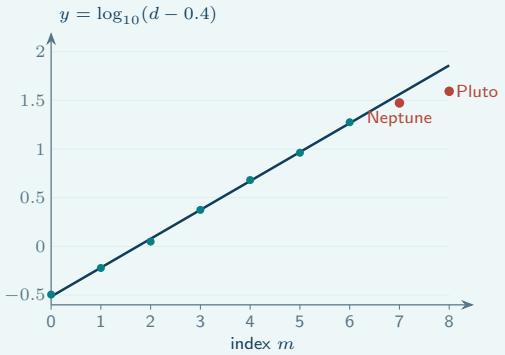


The inner bodies crowd near zero on this scale. Equal index steps correspond to increasingly large distance steps.

3 / LOG–LINEAR COORDINATES

Subtract the offset before taking logs

Plot $y = \log_{10}(d - 0.4)$ against m. Equal steps in y represent equal ratios in the shifted distance.



The line fits Venus through Uranus ($m = 0$ to 6) and is extrapolated outwards. Neptune and Pluto fall below it.

4 / READ THE SLOPE AND INTERCEPT

Recover the exponential from the line

With the offset fixed at 0.4, least squares gives

$$y = sm + c, \qquad s \approx 0.29668, \quad c \approx -0.51510.$$

Since $y = \log_{10}(d - 0.4)$, undo the logarithm:

$$d = 0.4 + 10^c(10^s)^m$$
$$\approx 0.4 + 0.3054(1.9801)^m.$$

The coefficients are close to the simple historical rule:

$$d_m \approx 0.4 + 0.3 \times 2^m$$

For that rule, the exact log-line has slope $\log_{10} 2 \approx 0.30103$ and intercept $\log_{10} 0.3 \approx -0.52288$.

5 / KEEP THE AWKWARD OBSERVATIONS

A useful pattern has limits

Keep the same index sequence when testing the historical rule:

body	m	observed d	rule d
Uranus	6	19.22	19.6
Neptune	7	30.11	38.8
Pluto	8	39.54	77.2

Skipping Neptune would move Pluto to $m = 7$, making its apparent fit much better. That changes the test.

Bode's rule is an **empirical spacing pattern**. Its offset and indexing are choices in the model. A nearly straight graph does not establish a physical cause or guarantee the next orbit.

Data. `Bodes&MilankowitzCycles Law.xlsx`, `student_Bode`, rows 4–13; observed distances and finite indices retained. The fitted line uses only $m = 0, \ldots, 6$, with equal weights in log space.

Further reading. W. Hayes and S. Tremaine (1997), *Fitting random stable solar systems to Titius–Bode laws*, arXiv:astro-ph/9710116.

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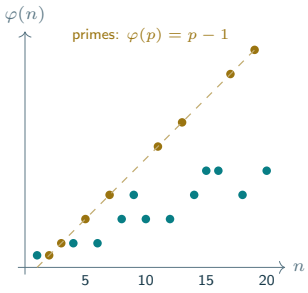
METROLOGY ON ONE PAGE

Points on a graph: numbers and quantities

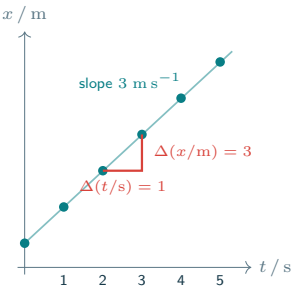
A physical point is a pair of ratios to units, so a graph plots numbers and its slope carries the ratio of the units.

1 / THE RULE AND THE PICTURE

On a physical graph the axis label $q/[q]$ says that the number plotted is the quantity divided by its unit



pure numbers: $\varphi(n)$ against n



measured: $x = 2\,\mathrm{m} + 3\,\mathrm{m\,s^{-1}}\,t$

Every plotted coordinate is already a ratio, so the point on the right is $(t/\mathrm{s},\ x/\mathrm{m}) = (2, 8)$, not "2 s and 8 m". Dividing the equation through by the units turns it into an ordinary number line:

$$\boxed{\frac{x}{m} = 2 + 3\,\frac{t}{\mathrm{s}}}$$

where the 3 is $v/(\mathrm{m\,s^{-1}})$.

2 / WHAT THE DIFFERENCE DOES TO A POINT

The same drawing means different things when the numbers carry units

feature	number theory graph	kinematic graph
axis label	n is the number itself	t/s is the time divided by one second
a point	a pair of integers $(n, \varphi(n))$	a pair of ratios $(t/\mathrm{s},\ x/\mathrm{m})$ standing for a time and a position
changing the scale	there is no unit to change	seconds to milliseconds multiplies the t coordinate by 1000 and leaves the physical point alone
slope	a pure number	a ratio of units, here $\mathrm{m\,s^{-1}}$
intercept	a pure number	carries the unit of the vertical axis, here 2 m
area under the curve	counts something	a product of units, $\mathrm{m\,s}$ for a velocity graph

The number theory points live in a world with one choice of scale, while the kinematic points depend on a pair of unit choices, so the coordinates are a chart on the physical situation and the units are the chart's basis.

3 / DIVIDING THROUGH / CHANGING UNITS

Rewriting $x = x_0 + vt$ in other units changes the coordinates and the slope, but not the motion

axes	dimensionless line	slope	speed
seconds and metres	$x/\mathrm{m} = 2 + 3\,(t/\mathrm{s})$	3	$3\ \mathrm{m\,s^{-1}}$
milliseconds and metres	$x/\mathrm{m} = 2 + 0.003\,(t/\mathrm{ms})$	0.003	$0.003\ \mathrm{m\,ms^{-1}}$
seconds and kilometres	$x/\mathrm{km} = 0.002 + 0.003\,(t/\mathrm{s})$	0.003	$0.003\ \mathrm{km\,s^{-1}}$
hours and kilometres	$x/\mathrm{km} = 0.002 + 10.8\,(t/\mathrm{h})$	10.8	$10.8\ \mathrm{km\,h^{-1}}$

The slope is a number only after a unit pair is chosen, and the speed $3\,\mathrm{m\,s^{-1}} = 10.8\,\mathrm{km\,h^{-1}}$ is the invariant; the line is also consistent because vt has the unit of a metre, which is why a time cannot be added to a length.

4 / GETTING AWAY FROM y AGAINST x / DIMENSIONLESS GROUPS

Dividing by the natural scales of the problem returns a pure number graph that every case shares

The range of a projectile launched at speed v and angle θ is $R = \dfrac{v^2}{g}\sin 2\theta$. At $v = 20\ \mathrm{m\,s^{-1}}$ and $\theta = 40°$ the range is 40.2 m on Earth ($g = 9.81$) and 243.2 m on the Moon ($g = 1.62$), and the two graphs of R/m against θ look different, but dividing each R by its own scale $v^2/g = 40.8$ m or 246.9 m gives 0.985 in both cases. The plot of $R/(v^2/g)$ against θ is the single curve $\sin 2\theta$ for every planet, a graph of pure numbers like the number theory one.

5 / LOGS AND POWER LAWS / WHAT SURVIVES

The exponent is a pure number and the intercept carries the units

planet	a/AU	T/yr
Mercury	0.387	0.2408
Earth	1	1
Mars	1.524	1.881
Jupiter	5.203	11.87
Saturn	9.537	29.45

On log axes these points lie on a line of slope $\frac{3}{2}$, since $T/\mathrm{yr} = (a/\mathrm{AU})^{3/2}$; in SI units $T/\mathrm{s} = 5.454 \times 10^{-10}\,(a/\mathrm{m})^{3/2}$, so the slope is still $\frac{3}{2}$ and the intercept $\log_{10}(5.454 \times 10^{-10}) = -9.263$ has changed because it hides the unit $\mathrm{s\,m^{-3/2}}$. One cannot take ln of a metre, only of x/m, and changing units shifts a log graph sideways without changing its slope.

ASIDE / ZEROS AND OFFSETS

Metres and seconds rescale about a true zero, but Celsius and kelvin differ by an offset of 273.15, so only temperature differences behave like the ratios above.

LINKS TO OTHER SHEETS

The exponents in power laws are the subject of *The Architecture of Scale* and *Kleiber, Jeans and the Goldilocks clumping*; the logarithm of a ratio is in *Logarithms*.

Convention. $q/[q]$ is a quantity divided by its unit; φ is Euler's totient; AU is the astronomical unit and yr the year; slopes are read as ratios of the axis units.

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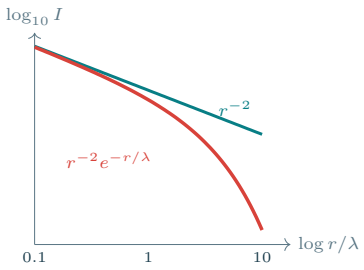
FUNCTIONS ON ONE PAGE

Combining falloffs

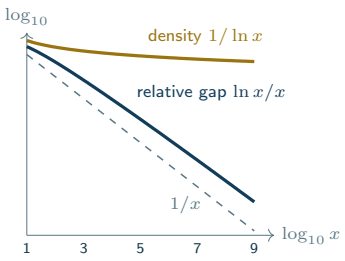
Multiplying falloffs adds their logarithms and their log-log slopes, and the local slope shows which falloff is winning.

1 / THE RULE AND THE PICTURE

For a product the logarithms add, so the local log-log slopes $n = d\ln f / d\ln x$ add too



light from a point source through dust



two ways the primes thin out

Light spreading from a point source dilutes as r^{-2}, and dust removes a fixed fraction per unit path, so $dI = -I\,dr/\lambda$ gives $e^{-r/\lambda}$:

$$I(r) = \frac{P}{4\pi r^2}\, e^{-r/\lambda}$$

The local exponent is $-2 - r/\lambda$: the power law sets the slope near the source and the exponential takes over beyond about λ.

2 / THE ATTENUATED INVERSE-SQUARE LAW

Measuring r in units of the attenuation length λ turns the table into pure numbers

$x = r/\lambda$	x^{-2}	e^{-x}	product	local exponent $-2-x$
0.1	100	0.9048	90.48	−2.1
0.5	4	0.6065	2.426	−2.5
1	1	0.3679	0.3679	−3
2	0.25	0.1353	0.03383	−4
5	0.04	0.006738	2.70×10^{-4}	−7
10	0.01	4.54×10^{-5}	4.54×10^{-7}	−12

At $x = 1$ the dust has already removed 63% of the beam and the slope has steepened from -2 to -3, by $x = 2$ the slope is -4, and at $x = 10$ the exponential has cut the intensity a further factor of 2.2×10^4 below the pure inverse-square value. For a street lamp in fog with $\lambda = 100$ m, doubling the distance from 50 m to 100 m costs a factor 4 from geometry and $e^{-0.5} = 0.61$ from the fog.

3 / ADDING AS WELL AS MULTIPLYING

Multiplying falloffs adds slopes, but adding intensities keeps the larger one

With a diffuse scattered background of fixed strength 10^{-3} in the same units, the total $x^{-2}e^{-x} + 10^{-3}$ follows the direct light until the two are equal at $x = 4.09$, where $x^{-2}e^{-x} = 10^{-3}$, and beyond that the background dominates and the curve on log axes flattens to a horizontal line. A sum of falloffs therefore looks like the upper envelope of its parts with a smooth knee at the crossover, while a product looks like the sum of its log curves.

4 / THE PRIMES THIN OUT MUCH MORE SLOWLY / DENSITY AND GAP

The density of primes near x is $1/\ln x$, and the average gap is $\ln x$, so the gap relative to x is $\ln x/x$

x	$\pi(x)$	$x/\ln x$	$\pi(x)/x$	$1/\ln x$
10^1	4	4	0.4000	0.4343
10^2	25	22	0.2500	0.2171
10^3	168	145	0.1680	0.1448
10^4	1 229	1 086	0.1229	0.1086
10^6	78 498	72 382	0.0785	0.0724
10^9	50 847 534	48 254 942	0.0508	0.0483

The prime number theorem says $\pi(x) \sim x/\ln x$, so the density $\pi(x)/x$ only falls from 0.25 at 100 to 0.051 at 10^9, slower than any power of x. It is the relative gap $\ln x/x$ that falls roughly like $1/x$ with a logarithmic correction.

5 / COMPARING THE FALLOFFS BY LOCAL EXPONENT

Exponentials have a scale, powers and logarithms do not

function	local exponent	sample values	large x
r^{-2}	-2	-2	-2
$r^{-2}e^{-r/\lambda}$	$-2 - r/\lambda$	-12 at $r = 10\lambda$	no limit
$\ln x/x$	$-1 + 1/\ln x$	-0.57 at 10, -0.78 at 100	-0.95 at 10^9
$1/\ln x$	$-1/\ln x$	-0.43 at 10, -0.22 at 100	-0.048 at 10^9

The dust law has a built-in length λ, so its exponent grows without bound once $r \gg \lambda$, while r^{-2} has the same slope everywhere and the prime quantities have slopes that creep towards 0 for the density and -1 for the relative gap as $1/\ln x$. None of the prime falloffs has a characteristic scale.

ASIDE / THE FUNCTION ln x / x

As a pure function it peaks at $x = e$ with value $1/e$ and then decays; for the primes it is the typical gap measured in units of position.

LINKS TO OTHER SHEETS

The ratio r/λ is the dimensionless variable of *Points on a graph: numbers and quantities*; see also *Exponential functions and the number e* and *Logarithms*.

Convention. I is intensity, P the source power, λ the attenuation length, $x = r/\lambda$ for the dust law; $\pi(x)$ counts primes up to x; n is the local log-log exponent.

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COUNTING CLICKS, SQUARE-ROOT PRECISION AND RADIOCARBON

How many clicks make a date?

Decay supplies the clock. Logarithms read it. Counting statistics set its precision.

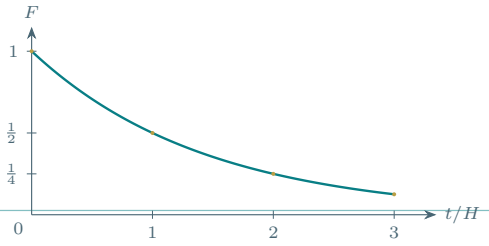
1 Decay and half-life

Activity is $A = \lambda n$, where n is the number of undecayed nuclei. The mean decay law gives

$$n(t) = n_0 e^{-\lambda t}, \qquad A(t) = A_0 e^{-\lambda t}.$$

Writing the half-life as H,

$$H = \frac{\ln 2}{\lambda}, \qquad F = \frac{A}{A_0} = 2^{-t/H}.$$

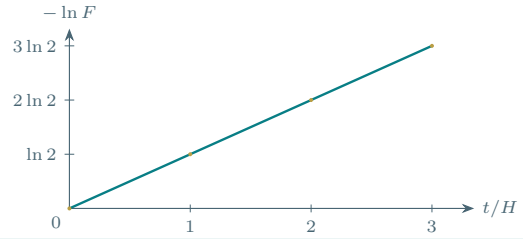


2 Logarithms read the clock

t: elapsed age. T: laboratory counting time.

$$\ln F = -\lambda t, \qquad \boxed{t = -\frac{\ln F}{\lambda}}.$$

For radiocarbon, compare activities for equal amounts of carbon, assuming a closed sample and a known initial activity.



Each half-life adds $\ln 2$ to $-\ln F$.

3 The dice rule returns as counting statistics

As with dice, variances of independent trials add. For M Bernoulli trials, $\mathrm{Var}(N) = Mp(1-p)$. Rare, independent detections give the Poisson limit:

$$N \sim \mathrm{Poisson}(\mu), \qquad \mu = \eta A T, \qquad \sigma_N = \sqrt{\mu} \approx \sqrt{N}.$$

Assume negligible background, known detection efficiency η, and steady activity during $T \ll H$. Using matching time units,

$$\widehat{A} = \frac{N}{\eta T}, \qquad \sigma_A \approx \frac{\sqrt{N}}{\eta T}, \qquad \boxed{\frac{\sigma_A}{\widehat{A}} \approx \frac{1}{\sqrt{N}}}.$$

Absolute count uncertainty grows as $\sqrt{N}$; relative activity uncertainty falls as $1/\sqrt{N}$.

4 Turn an activity error into an age error

$$\sigma_t \approx \left|\frac{\mathrm{d}t}{\mathrm{d}A}\right| \sigma_A = \frac{1}{\lambda}\frac{\sigma_A}{A} \approx \boxed{\frac{1}{\lambda\sqrt{N}}} \propto T^{-1/2}.$$

Here A_0 and λ are treated as known, and enough counts are collected for this local uncertainty approximation.

If an independent reference also has counting uncertainty, with N_0 clicks, then $\sigma_t \approx \frac{1}{\lambda}\sqrt{\frac{1}{N} + \frac{1}{N_0}}$.

5 An idealised radiocarbon age

Take $H = 5730$ years and $F = 1/4$ for a closed sample, with known A_0 and $N = 10\,000$ detected decays.

$$t = \frac{H}{\ln 2}\ln 4 = 11\,460 \text{ years}.$$

$$\sigma_t \approx \frac{5730}{\ln 2\sqrt{10\,000}} \approx 83 \text{ years}.$$

$$\boxed{t \approx 11\,460 \pm 83 \text{ years}}$$

1σ, counting uncertainty only; ideal decay-model age.

Same sample, longer counting

time	clicks	$\sigma_A/\widehat{A}$	σ_t
T_0	10 000	1%	83 yr
$4T_0$	40 000	0.5%	41 yr
$16T_0$	160 000	0.25%	21 yr

Typical counts at the same steady activity. The estimated central age stays the same.

Half the counting error needs four times as long.

6 What laboratory dates also contain

The model above describes decay counting. Modern accelerator mass spectrometry (AMS) measures rare ^{14}C atoms directly. Background, standards and sample preparation also contribute uncertainty.

Conventional reporting uses a corrected fraction modern F and $t_{RC} = -8033 \ln F$ years BP (before AD 1950), based on the 5568-year Libby half-life. Calibration converts this to calendar-age ranges, which can be asymmetric or split.

The $T^{-1/2}$ law describes counting noise; longer counting cannot remove contamination/uncertainty in sample's history.

Counting: NPL (2014), *Practical Radiation Monitoring*, §6.1.2. Radiocarbon: Oxford ORAU, dating and calibration; Historic England, radiocarbon methods.

ALLOMETRY ON ONE PAGE

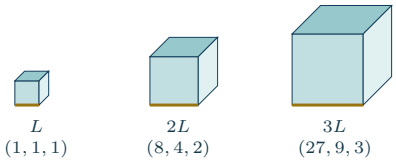
How big can a human be?

Stress, strain and a falling body: from the square–cube law to the quarter power of the mass–radius map.

1 / THE RULE AND THE PICTURE

Scale a body by λ and its forces outgrow its supports

Scaling a body by λ in every direction multiplies its weight by λ^3 and each supporting cross-section by λ^2, so the stress $\sigma = F/A$ (a pressure, in pascals) grows as λ, and with it the strain $\varepsilon = \Delta L/L = \sigma/E$ of a material of Young's modulus E: $\boxed{W \propto L^3,\ A \propto L^2 \ \Rightarrow\ \sigma = W/A \propto L}$



quantity	expression	scales as
weight W	$\rho g L^3$	L^3
cross-section A	shape $\times L^2$	L^2
stress $\sigma = W/A$	$\rho g L$	L
hydrostatic pressure P	$\rho g h$	L
fall energy E_{fall}	$WL = \rho g L^4$	L^4
fracture energy E_{break}	ΓL^2	L^2

Γ is the fracture energy per area ($\mathrm{J\,m^{-2}}$); cubes show (W, A, σ).

2 / THE STATIC LIMIT IS GENEROUS

Weight alone would allow a column of kilometres

The stress at the foot of a column is $\rho g L$, which reaches the crushing stress σ_c at $L_s = \sigma_c/(\rho g)$. For bone, with $\sigma_c \approx 1.7 \times 10^8$ Pa and $\rho \approx 1900\ \mathrm{kg\,m^{-3}}$, this is about 9 km. A 70 kg person standing on two femora of about 3 $\mathrm{cm^2}$ each carries roughly 1 MPa, under one per cent of σ_c, so standing is not what limits a human, and the binding constraint must be dynamic.

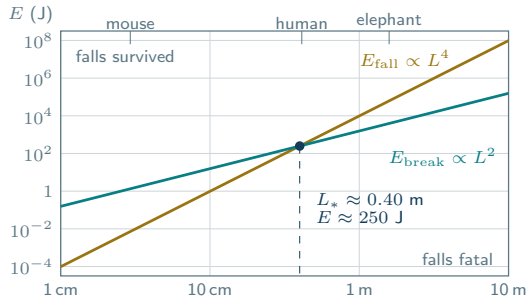
3 / A FALL AGAINST A FRACTURE

Energy in grows as L^4, energy absorbed as L^2

A fall through a height $\sim L$ releases $E_{\text{fall}} \sim WL = \rho g L^4$. A fracture crosses a plane of $(L/a_0)^2$ bonds, each worth $E_{\text{em}} = e^2/(4\pi\varepsilon_0 a_0) \approx 27$ eV, so $E_{\text{break}} = E_{\text{em}}(L/a_0)^2 = \Gamma L^2$ with $\Gamma = E_{\text{em}}/a_0^2 \approx 1.6\ \mathrm{kJ\,m^{-2}}$. In stress and strain terms the plane carries a force $\sigma_c L^2$, with $\sigma_c = E_{\text{em}}/a_0^3 \approx 3 \times 10^{13}$ Pa, through a rupture distance a_0, and work is force times distance.

4 / THE CROSSING AND THE QUARTER POWER

Equate the two energies and a limiting size appears



Setting $\rho g L^4 = \Gamma L^2$ gives $L_* = \sqrt{\Gamma/(\rho g)}$, which for water density and Earth gravity is 0.40 m and $M = \rho L_*^3 \approx 64$ kg; a mouse, human and elephant have mass-equivalent sides of 3 cm, 41 cm and 1.6 m.

In the units of the power graph, $\rho = m_p/a_0^3$, $\Gamma = E_{\text{em}}/a_0^2$ and the planetary ceiling supplies $g_P = GM_P/R_P^2 = Gm_pQ^{1/2}/a_0^2$ from $R_P = a_0 Q^{1/2}$ and $M_P = m_p Q^{3/2}$, so

$$\left(\frac{L}{a_0}\right)^2 = \frac{E_{\text{em}}}{m_p g_P a_0} = Q^{1/2}.$$

$$\frac{L}{a_0} \sim Q^{1/4}, \qquad \frac{M}{m_p} \sim Q^{3/4}, \qquad Q = \frac{e^2/4\pi\varepsilon_0}{Gm_p^2} \approx 1.2 \times 10^{36}$$

5 / WHY THE EXPONENT IS A QUARTER

The organism is the geometric mean of atom and planet

atom		organism		planet
a_0	$\xrightarrow{\times Q^{1/4}}$	$a_0 Q^{1/4}$	$\xrightarrow{\times Q^{1/4}}$	$a_0 Q^{1/2}$
0.53 Å		$\times 10^9$		6×10^7 m
m_p	$\xrightarrow{\times Q^{3/4}}$	$m_p Q^{3/4}$	$\xrightarrow{\times Q^{3/4}}$	$m_p Q^{3/2}$
1.7×10^{-27} kg		$\times 10^{27}$		2×10^{27} kg
		5.6 cm		
		2 kg		

Each step multiplies the length by $Q^{1/4}$, so $L^2 = a_0 R_P$. The static crushing length $\sigma_c/(\rho g_P) = a_0 Q^{1/2}$ is the planetary radius itself, since $\rho g_P R_P = e^2/(4\pi\varepsilon_0 a_0^4)$ is the atomic pressure, and the dynamic limit is the square root of atom times static length, $L = \sqrt{a_0 L_s}$.

6 / GRAVITY, MATERIAL AND SHAPE

What moves the limit, and what it leaves alone

Since $L_* = \sqrt{\Gamma/(\rho g)}$, the limiting size goes as $g^{-1/2}$ and the mass as $g^{-3/2}$, while the exponents stay put:

	g ($\mathrm{m\,s^{-2}}$)	L_*	M_*
Moon	1.6	$\times 2.5$	$\times 15$
Earth	9.8	$\times 1$	$\times 1$
Jupiter cloud tops	24.8	$\times 0.63$	$\times 0.25$

The bare powers give 5.6 cm and 2 kg because they use atomic density and $g_P \approx 44\ \mathrm{m\,s^{-2}}$; real chemistry and biomechanics, which Press's fuller estimate captures with a factor $\epsilon^{1/4}$, shift the prefactor and leave the quarter power.

ASIDE / PRESSURE IN A TALL BODY

Blood obeys $P = \rho g h \propto L$: head to foot of a 1.7 m human it is 17.7 kPa (133 mmHg), and a giraffe's brain, about 2 m above its heart, costs 156 mmHg, roughly why its arterial pressure is double ours. Long-bone diameters rise nearer $M^{3/8}$ than $M^{1/3}$, so large animals have stouter legs.

ASIDE / CHECK THE NUMBERS IN PYTHON

With `scipy.constants`, `Q = e**2/(4*pi*epsilon_0*G*m_p**2)` gives 1.24×10^{36}, and `a0*Q**0.25` and `m_p*Q**0.75` give 0.056 m and 1.96 kg. Replacing g_P by `9.80665` and ρ by `1000` in $\sqrt{\Gamma/(\rho g)}$ gives the 0.40 m above.

Convention. Section 5 of *The Architecture of Scale*, after Press (1980); $O(1)$ factors are dropped, so the powers locate a scale only.

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SCALING ON ONE PAGE

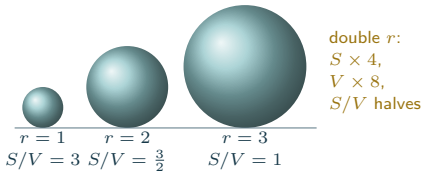
Surface to volume, and why small animals run hot

From the isoperimetric inequality to Kleiber's slope: a sphere gives $2/3$, and the observed $3/4$ asks for a fourth dimension.

1 / THE RULE AND THE PICTURE

The sphere has the least surface for its volume

The isoperimetric inequality $S^3 \geq 36\pi V^2$ becomes scale-free as a quotient, $\boxed{Q = \dfrac{36\pi V^2}{S^3} \leq 1}$ with $Q = 1$ for the sphere alone. Scaling every length by λ multiplies S by λ^2 and V by λ^3 and leaves Q unchanged, so Q records shape only, whereas for a sphere of radius r the ratio itself is $S/V = 3/r$.



double r: $S \times 4$, $V \times 8$, S/V halves

shape	$Q = \dfrac{36\pi V^2}{S^3}$	$S/V^{2/3}$
sphere	1	4.84
cylinder, $h = 2r$	$2/3$	5.54
cube	$\pi/6 \approx 0.52$	6.00
regular tetrahedron	$\pi/(6\sqrt{3}) \approx 0.30$	7.21
human, $70\,\mathrm{kg}$, $1.8\,\mathrm{m}^2$	≈ 0.095	≈ 10.6

2 / ANIMALS AS HOMOGENEOUS SPHERES

Heat leaves through the surface, so the slope is $2/3$

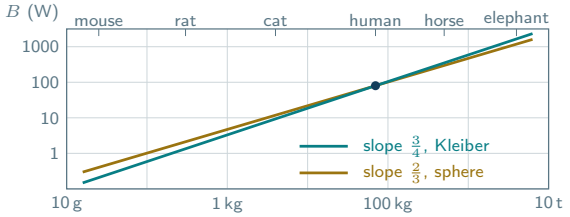
With uniform density ρ an animal of mass M has $r \propto M^{1/3}$. Heat escapes through the surface at a rate $\propto S \propto r^2 \propto M^{2/3}$, by Newtonian cooling $hS\,\Delta T$ or by radiation $\sigma\varepsilon S(T^4 - T_0^4)$, and a warm-blooded body holding its temperature burns at that same rate (Rubner's surface rule), so each kilogram burns at $B/M \propto S/V = 3/r$, giving $B \propto M^{2/3}$ and $B/M \propto 3/r \propto M^{-1/3}$, a straight line of slope $\frac{2}{3}$ in $\log B$ against $\log M$.

	M (kg)	r	S/V
mouse	0.025	$1.8\,\mathrm{cm}$	$165\ \mathrm{m}^{-1}$
human	70	$26\,\mathrm{cm}$	$11.7\ \mathrm{m}^{-1}$
elephant	4000	$98\,\mathrm{cm}$	$3.0\ \mathrm{m}^{-1}$

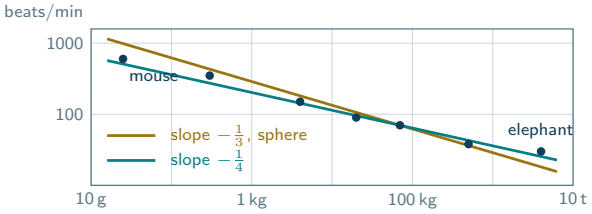
at $\rho = 1000\ \mathrm{kg\,m^{-3}}$; per kilogram the mouse loses heat $54 = (1.6 \times 10^5)^{1/3}$ times faster.

3 / KLEIBER'S OBSERVATION

Two straight lines on log–log axes, and the animals fall between them



Lines pass through a $70\,\mathrm{kg}$ human at $80\ \mathrm{W}$ and drift apart by 2.7 from mouse to elephant.



Approximate resting heart rates; the fitted slope -0.26 is nearer $-\frac{1}{4}$.

4 / COUNT DIMENSIONS

Which ball gives $3/4$?

A d-ball has $V = V_d r^d$ and $S = dV_d r^{d-1}$ ($V_3 = 4\pi/3$, $V_4 = \pi^2/2$), so $S/V = d/r$ in every dimension and, with $M \propto V$,

$$B \propto S \propto M^{(d-1)/d}, \qquad B/M \propto S/V \propto M^{-1/d}.$$

d	S/V	$B \propto$	$B/M \propto$
2	$2/r$	$M^{1/2}$	$M^{-1/2}$
3	$3/r$	$M^{2/3}$	$M^{-1/3}$
4	$4/r$	$M^{3/4}$	$M^{-1/4}$
5	$5/r$	$M^{4/5}$	$M^{-1/5}$

$$\frac{d-1}{d} = \frac{3}{4} \implies d = 4$$

5 / WHERE A FOURTH DIMENSION COULD COME FROM

The $1/r$ survives; only the mass law changes

Since $S/V = d/r$ keeps its $1/r$ form in every dimension, the difference between $2/3$ and $3/4$ lies in how mass relates to the length that sets the surface. West, Brown and Enquist (1997) found $B \propto M^{D/(D+1)}$ for a D-dimensional body fed by a space-filling, self-similar branching network, which gives $3/4$ at $D = 3$ and matches the ball formula once $d = D + 1$.

Measured exponents stay contested, with $3/4$ from Kleiber (1932) and values near $2/3$ from temperature-corrected mammal data (White and Seymour, 2003), so geometry shows which dimension $3/4$ needs and leaves open whether biology supplies it.

ASIDE / REAL ANIMALS ARE NOT SPHERES

Since $Q \leq 1$ the sphere is the best case for keeping warm. A human with $Q \approx 0.095$ exposes 2.2 times the sphere's area, which under geometric similarity shifts the prefactor of $B \propto S$ and leaves the exponent at $2/3$.

ASIDE / FIT THE SLOPE IN PYTHON

`np.polyfit(np.log10(M), np.log10(HR), 1)[0]` gives the slope of log rate against log mass, -0.265 for the seven points shown, since a power law $y = cx^p$ becomes the line $Y = pX + \log c$.

Convention. Heart rates and masses are approximate typical resting values, and both lines are anchored at a $70\,\mathrm{kg}$ human; Kleiber's fit is $B \approx 3.4\,M^{3/4}$ W for M in kg.
Companions: prime-box isoperimetric; logarithms lesson; scales essay.

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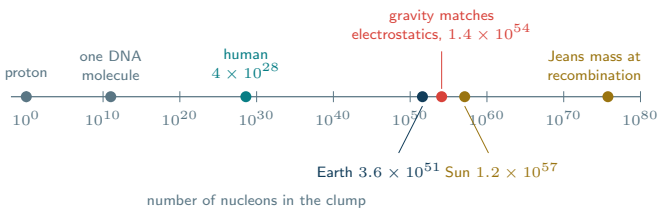
CLUMPING ON ONE PAGE

The making of Goldilocks: a story of clumping

Matter clumps when binding energy beats what pushes back, with Kleiber setting the terms for living bodies and Jeans for stars.

1 / THE RULE AND THE PICTURE

A merged whole weighs less than its parts, and the difference leaves as binding energy



Mergers of companies shed staff, economies of scale being economies of energy, and merging matter sheds binding energy in the same way. Three power laws then describe where a clump stops being passive:

$$W \propto M^{2/3}, \quad U \propto M^{3/4}, \quad M_J = \frac{p^{3/2}}{G^{3/2}\rho^2}$$

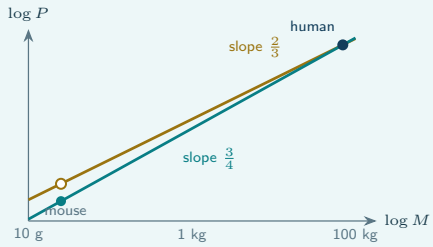
2 / INERT CLUMPS / WHEN SELF-GRAVITY MAKES A SPHERE

The work to remove one atom grows as $M^{2/3}$, so enough atoms overcome gravity's weakness

At constant density $M \propto r^3$, so $r \propto M^{1/3}$, and the work to lift one atom out of the body's own field is $W \propto M/r \propto M^{2/3}$. The electrostatic force between a proton pair exceeds gravity by $e^2/(4\pi\varepsilon_0 G m_p^2) = 1.24 \times 10^{36}$, so the gravitational gain $N^{2/3}$ catches up at $N \approx (10^{36})^{3/2} = 1.4 \times 10^{54}$ nucleons, which is 2.3×10^{27} kg, of Jupiter's order (the arithmetic $54 \times \frac{2}{3} = 36$). Sand grains and asteroids sit far below it and keep irregular shapes, while the Earth (3.6×10^{51} nucleons) and the Sun (1.2×10^{57}) bracket it.

3 / LIVING CLUMPS / THE INTERNAL CHEMICAL BUDGET OF KLEIBER

Heat made inside must leave through the surface, and the measured exponent $3/4$ fits a 3-sphere



A body that holds its temperature must metabolise at the rate it loses heat, so a 2-sphere with $A \propto r^2 \propto M^{2/3}$ predicts a $\frac{2}{3}$ power, whereas animals follow Kleiber's $U \propto M^{3/4}$. In d dimensions the boundary of a ball scales as $M^{(d-1)/d}$, so $\frac{3}{4}$ corresponds to $d = 4$, a 3-sphere, which suits the fractal-like internal structure of living bodies. Anchored at a 70 kg human using 82 W, the 25 g mouse draws 0.21 W by Kleiber against 0.41 W by the surface law, a human-to-mouse ratio of 385 where the area law gives 199.

4 / STELLAR CLUMPS / THE INTERNAL THERMAL PRESSURE OF JEANS

Gravity wins once GM^2/r exceeds the thermal energy pr^3, which fixes a critical mass

With $E_p \approx -GM^2/r$, internal energy $U \propto pr^3$ and $r = (M/\rho)^{1/3}$, clumping is favoured when $GM^2/r > pr^3$, giving $M_J = p^{3/2}/(G^{3/2}\rho^2) \propto \sqrt{p^3/\rho^4}$. At recombination, when the universe was 3000 K and nucleons and electrons first bound with the release of photons, $\rho = 9.9 \times 10^{-19}$ kg m^{-3} and the plasma pressure $p \approx \rho c^2/3 = 0.030$ Pa give

$$M_J = 9.6 \times 10^{48} \text{ kg} \approx 5 \times 10^{18} M_\odot$$

a ratio to the Sun of 4.8×10^{18}, comparable to the 5×10^{17} ratio of a 70 kg human to the mass of one of its chromosomes.

5 / TWO WAYS TO HOLD A CLUMP TOGETHER / THE PHOTON EATER AND THE KETOSIS

The human lives on photons from outside and the star lives on fuel stored inside

	human (Kleiber)	star (Jeans)
size	4×10^{28} nucleons	1.2×10^{57} nucleons
energy source	low-entropy photons through the food chain, a 500 nm photon in and thirty 15 μm photons out	hydrogen stored in the core and burnt from within, as in ketosis
what balances	heat produced against heat lost through the surface	gravitational pull against thermal and radiation pressure
scaling	$U \propto M^{3/4}$	$M_J \propto \sqrt{p^3/\rho^4}$

ASIDE / THE HUMAN IN THE MIDDLE

$\sqrt{m_p M_\odot} = 58$ kg, so a human is the geometric mean of a proton and the Sun, thriving between the quantum and the Newtonian.

ASIDE / CHECK IN PYTHON

`p**1.5/(G**1.5*rho**2)` with `p=rho*c**2/3` gives 9.56×10^{48} kg.

Convention. Counts are numbers of nucleons (M/m_p); scalings drop numerical factors. After *The making of Goldilocks: a story of Clumping* (McMurmerings, 2020).

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Chapter 5

Chance and measurement

What this chapter gathers.

Foothills

- Sample spaces and equally likely outcomes
- Counting: arrangements and choices
- Conditional probability and independence
- Bayes' theorem and updating
- Random variables, expectation and variance
- The binomial distribution
- Waiting times: geometric and Poisson
- The normal distribution and the central limit theorem
- Information and entropy

Foothold

- Prime types: the mod 4 and mod 6 races
- The square-root-of-time rule, told with dice
- The Poisson distribution: counting arrivals
- Are any composite types Poisson?
- Deficient, perfect, abundant and Poisson
- Entropy, microstates and the card-pack analogy
- Perudo: probability under a cup
- Poker hands from a pack: probability and combinatorics with workings
- Yahtzee: probability across three rolls
- Two-way tables meet Punnett squares
- Euler's gamma and Catalan's G

Handhold

- Apery's constant zeta(3), and Ramanujan's constant rebuilt from it
- Do the judges agree? From marks to rankings
- Means and the Kolmogorov distance
- Primes as a filtered signal
- Primes in a two-way table: splitting, inertness and ramification
- Ramanujan's constant, the Basel sum and the Riemann hypothesis
- Spearman rho versus Kendall tau: two ways to compare rankings
- The cubic, its discriminant, and a Ramanujan-like number from Apery
- The Hurst exponent of the composites
- Why the Kalman filter: guessing well from noisy data

Roll two fair dice. Their eleven possible sums are not equally likely: seven can be made in six ways, while twelve has only one. The equally likely outcomes are the thirty-six ordered pairs of faces. Probability begins by identifying what we are counting and why the outcomes should have the weights we assign.

New information changes the possibilities. Seeing one card affects the chances for the next; knowing our own dice changes how we assess a Perudo bid. Conditional probability records these restrictions, while independence describes situations where one event leaves another's probability unchanged. Bayes' theorem reverses the direction of a calculation, connecting the likelihood of evidence under an explanation with the support that evidence gives the explanation.

Random variables turn outcomes into quantities. Expectation describes a weighted average, and variance measures spread around it. Binomial models count successes, geometric models describe the wait for a first success, and Poisson models count arrivals in an interval. The normal distribution supplies an important approximation for many accumulated effects, with the central limit theorem specifying conditions under which it emerges.

These models also explain limits on precision. A Poisson count with mean N has standard deviation $\sqrt{N}$, giving relative variation $1/\sqrt{N}$. Four times the expected count halves this relative uncertainty. More observing time therefore brings a gradual improvement in precision. Background counts, calibration and the assumptions of the model introduce further questions about what the measurement supports.

Different summaries preserve different information. Entropy describes uncertainty across a probability distribution. Two-way tables organise overlapping categories, while Punnett squares organise possible inheritance under a biological model. Judges' marks introduce another distinction: agreement on numerical scores and agreement on an ordering are separate questions. Spearman's rho and Kendall's tau provide different ways to examine that ordering.

The number-theory examples test how carefully we use these tools. Primes and composites form deterministic sequences, which we can compare with specified random models. Counts, cumulative distributions and measures of dependence describe different features. Their interpretation depends on the range, scale and comparison chosen. A departure from a model gives us a pattern to investigate, with its assumptions available for examination.

Filtering brings the chapter back to updating. A Kalman filter repeatedly predicts a state and corrects that prediction using a noisy measurement. Under its model, the uncertainties determine how much weight each source receives. The progression from dice to filtering develops one habit: state the possibilities, account for the information available, and judge how strongly it supports the conclusion.

PROBABILITY LADDER 1 OF 9

Sample spaces and equally likely outcomes

Count the outcomes in an event and divide by all the outcomes, and combine events by adding and complementing.

1 / THE RULE AND THE PICTURE

With equally likely outcomes, $P(A)$ is the count of outcomes in A over the total count

first die \ second die	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

each cell is the sum, and the six gold cells are the sum 7

Two dice have a sample space of $6 \times 6 = 36$ equally likely outcomes, one for each cell. The event "the sum is 7" contains 6 of them, so

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$$

Every probability lies between 0 (an impossible event) and 1 (a certain one), and the probabilities of all outcomes add to 1.

2 / COUNTING EVENTS / A TABLE

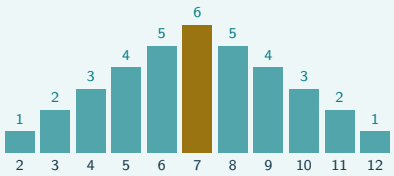
Each event is a set of cells, and its probability is the count over 36

event	outcomes	probability	decimal
sum is 7	$6/36$	$1/6$	0.167
doubles	$6/36$	$1/6$	0.167
sum at least 10	$6/36$	$1/6$	0.167
at least one six	$11/36$	$11/36$	0.306
both even	$9/36$	$1/4$	0.250
sum is prime	$15/36$	$5/12$	0.417

The probabilities of the sums $2, \dots, 12$ rise and fall as $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$ over 36, because the sum 7 can be made six ways and the sums 2 and 12 only one.

3 / THE SUMS / A BAR CHART

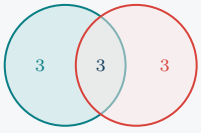
The sum 7 is the most likely, since it has the most cells



The height of each bar counts the cells with that sum, and dividing by 36 gives the probability, for example $P(\text{sum} = 4) = \frac{3}{36} = \frac{1}{12}$. The tempting guess that all sums are equally likely fails, since the outcomes and not the sums are the equally likely things.

4 / COMBINING EVENTS / ADDITION AND COMPLEMENT

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$, and $P(\text{not } A) = 1 - P(A)$



A: sum ≥ 10 B: first is 6

Let A be "sum at least 10" and B "first die is 6". Each has 6 outcomes and they share 3, namely $(6,4)$, $(6,5)$, $(6,6)$, so $P(A \text{ or } B) = \frac{6}{36} + \frac{6}{36} - \frac{3}{36} = \frac{9}{36} = \frac{1}{4}$, which counts 9 distinct cells.
The complement is quicker for "at least one six": $1 - \frac{25}{36} = \frac{11}{36}$, since 25 outcomes show no six.

5 / PROBABILITY AND FREQUENCY

Long runs of trials settle near the probability

In a seeded simulation of 6000 rolls of two dice the sum was 7 on 1001 occasions, a relative frequency of 0.167 against the exact 0.167, and the gap shrinks as the number of trials grows.
Probability is the limit this frequency approaches, and the next sheets show how fast it does.

ASIDE / EXCLUSIVE EVENTS

Events that cannot both happen, such as sum 7 and sum 8, simply add: $\frac{6}{36} + \frac{5}{36} = \frac{11}{36}$.

LINKS TO OTHER SHEETS

Counting without listing is in the next sheet, *Counting: arrangements and choices*; sums of dice return in *Random variables, expectation and variance*.

Convention. S is the sample space; an event is a subset of S; P is probability; the dice are fair and independent.

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PROBABILITY LADDER 2 OF 9

Counting: arrangements and choices

Probabilities of equally likely outcomes need counts, and the multiplication principle and binomial coefficients supply them.

1 / THE RULE AND THE PICTURE

Choosing 2 of 4 in order gives 12 pairs, and ignoring order halves each into 6

	second pick A	B	C	D
A		AB	AC	AD
B	BA		BC	BD
C	CA	CB		CD
D	DA	DB	DC	

first pick down the side, with the gold pairs counted once

There are 4 choices for the first pick and 3 for the second, so $4 \cdot 3 = 12$ ordered pairs (AB and BA differ). If order does not matter each pair is counted $2!$ times, so the number of unordered pairs is $12/2 = 6$:

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!} = \frac{n(n-1)\cdots(n-r+1)}{r!}$$

2 / ARRANGEMENTS / THE MULTIPLICATION PRINCIPLE

Independent choices multiply, and a repeated letter divides the count

situation	count	value
5 books on a shelf	$5!$	120
podium of 3 from 10 runners	$10 \cdot 9 \cdot 8$	720
plate: 3 letters, 2 digits	$26^3 \cdot 10^2$	$1\,757\,600$
arrangements of MISSISSIPPI	$\frac{11!}{1!\,4!\,4!\,2!}$	$34\,650$

If a task has a ways for the first stage and b for the second, there are ab ways in all. Swapping identical letters changes nothing, so MISSISSIPPI divides $11!$ by $4!$ for the I's, $4!$ for the S's and $2!$ for the P's.

3 / CHOICES / ORDER DOES NOT MATTER

Binomial coefficients count the ways to pick a group

situation	count	value
handshakes among 10 people	$\binom{10}{2}$	45
committee of 3 from 8	$\binom{8}{3}$	56
five-card poker hands	$\binom{52}{5}$	$2\,598\,960$
lottery, 6 from 49	$\binom{49}{6}$	$13\,983\,816$

The count is symmetric, $\binom{n}{r} = \binom{n}{n-r}$, since picking r people to include is the same as picking $n-r$ to leave out, and $\binom{49}{6} = 13\,983\,816$ is why the lottery is hard to win.

4 / PASCAL'S TRIANGLE / THE TABLE OF COEFFICIENTS

Each coefficient is the sum of the two above it, and each row sums to 2^n

$n=0$				1				$2^0 = 1$
$n=1$			1		1			$2^1 = 2$
$n=2$		1		2		1		$2^2 = 4$
$n=3$		1		3		3	1	$2^3 = 8$
$n=4$	1		4		6		4	1	$2^4 = 16$
$n=5$	1	5		10		10	5	1	$2^5 = 32$
$n=6$	1	6	15		20		15	6	1	$2^6 = 64$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}, \qquad \binom{6}{2} = \binom{5}{1} + \binom{5}{2} = 5 + 10 = 15.$$

Picking a person to be in or out gives two choices each, so the 2^n subsets of n people split by size as the row $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$, and the sum of the row is 2^n. The same coefficients expand $(a+b)^n$.

5 / FROM COUNTS TO PROBABILITIES

Probability is favourable choices over all choices

For a poker hand the full house has $13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2} = 3744$ choices out of $\binom{52}{5} = 2\,598\,960$, a probability of 0.144%.

Two cards drawn from a pack are both aces with probability $\binom{4}{2}/\binom{52}{2} = \frac{6}{1326} = \frac{1}{221}$, and a single lottery ticket wins with probability $1/13\,983\,816$.

ASIDE / ORDER MATTERS WHEN

Counting arrangements uses $n!$ and counting groups uses $inomnr$, and the only difference is dividing out the $r!$ orders.

LINKS TO OTHER SHEETS

The card-pack entropy sheet uses the same poker counts; conditional draws are in the next sheet, *Conditional probability and independence*.

Convention. $n! = n(n-1)\cdots 1$; $\binom{n}{r}$ counts unordered choices; the cards and people are distinct unless stated.

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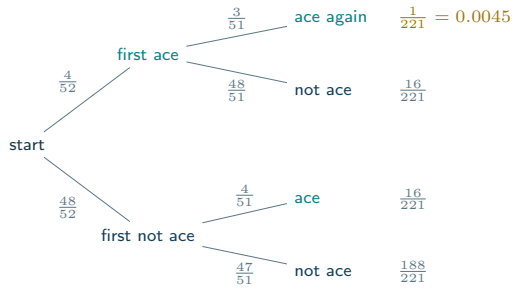
PROBABILITY LADDER 3 OF 9

Conditional probability and independence

Learning that one event happened shrinks the sample space, and independence says it leaves another event unchanged.

1 / THE RULE AND THE PICTURE

Along a branch of the tree the probabilities multiply, because each is conditional on the one before



two cards from a pack without replacement

The conditional probability of A given B restricts attention to the outcomes in B:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

so $P(A \text{ and } B) = P(B)\,P(A \mid B)$, and for two aces $\frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$.

2 / SHRINKING THE SAMPLE SPACE / TWO DICE

Given that the sum is 8, only five of the 36 outcomes remain

The outcomes with sum 8 are $(2,6)$, $(3,5)$, $(4,4)$, $(5,3)$, $(6,2)$, and exactly one of them has a six on the first die, so $P(\text{first is } 6 \mid \text{sum} = 8) = \frac{1}{5}$, against the unconditional $\frac{1}{6}$. By the formula the same number is $\dfrac{1/36}{5/36} = \frac{1}{5}$, and conditioning on a sum of 7 would give $\frac{1}{6}$ again.

3 / INDEPENDENCE

Events are independent when $P(A \text{ and } B) = P(A)\,P(B)$, so that conditioning changes nothing

events A, B	$P(A \text{ and } B)$	$P(A)P(B)$	verdict
first die even, sum 7	$\frac{3}{36}$	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	independent
first die even, sum ≥ 10	$\frac{4}{36}$	$\frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$	dependent
sum 7, sum 8	0	$\frac{1}{6} \cdot \frac{5}{36}$	exclusive, dependent

Exclusive events with positive probabilities are never independent, since knowing one happened rules the other out.

4 / THE GAMBLER'S FALLACY / AND THE COMPLEMENT

Past tosses do not change a fair coin, and "at least one" is easiest through the complement

After five heads the next toss is still $\frac{1}{2}$ heads, although six heads in a row had probability $\frac{1}{64}$ from the start.
For at least one six in four rolls, $1 - (\frac{5}{6})^4 = 0.518$, and for at least one double six in 24 rolls of two dice, $1 - (\frac{35}{36})^{24} = 0.491$, the two numbers that separated the bets of the Chevalier de Méré.

5 / TOTAL PROBABILITY

Splitting by the first card gives the chance of the second

The second card is an ace with probability $\frac{4}{52} \cdot \frac{3}{51} + \frac{48}{52} \cdot \frac{4}{51} = \frac{12+192}{2652} = \frac{1}{13}$, the same as for the first card by symmetry. In general $P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A)$, which the next sheet turns around into Bayes' theorem.

ASIDE / THE SAME ANSWER TWICE

Counting hands with $\binom{4}{2}/\binom{52}{2}$ and multiplying along the branch both give $\frac{1}{221}$.

LINKS TO OTHER SHEETS

The counting method is in *Counting: arrangements and choices*; the next sheet is *Bayes' theorem and updating*.

Convention. $P(A \mid B)$ is the probability of A given B; cards are dealt without replacement; the dice are fair.

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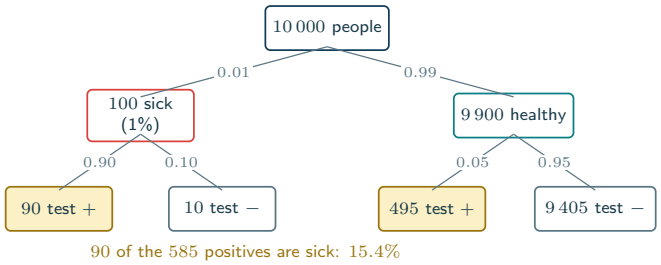
PROBABILITY LADDER 4 OF 9

Bayes' theorem and updating

Evidence turns a prior probability into a posterior one, and counting people in a table makes the arithmetic easy.

1 / THE RULE AND THE PICTURE

Most positives come from the large healthy group, so a positive test is far from conclusive



A disease affects 1% of people, the test catches 90% of the sick, and it wrongly flags 5% of the healthy.
Out of $10\,000$ people there are $90 + 495 = 585$ positives, so

$$P(\text{sick} \mid +) = \frac{90}{585} = 0.154$$

2 / BAYES' THEOREM / FROM THE MULTIPLICATION RULE

Since $P(A \text{ and } B) = P(A)P(B \mid A) = P(B)P(A \mid B)$, the two conditionals are related

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}, \qquad P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A).$$

Here $P(A) = 0.01$ is the prior, $P(B \mid A) = 0.90$ the sensitivity, and $P(B) = 0.9 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585$, so the posterior is $\dfrac{0.9 \cdot 0.01}{0.0585} = 0.154$, exactly the table's $\frac{90}{585}$.

3 / THE TABLE OF COUNTS

Rows and columns give every conditional probability at once

	test positive	test negative	total
sick	90	10	100
healthy	495	9 405	9 900
total	585	9 415	10 000

Reading along the row, $P(+ \mid \text{sick}) = 90/100$, and reading down the column, $P(\text{sick} \mid +) = 90/585$; mixing up these two is the base-rate fallacy, and it happens because $P(+ \mid \text{healthy}) = 5\%$ is small while the healthy group is 99 times larger.

4 / ODDS FORM / REPEATED EVIDENCE

Posterior odds equal prior odds times the likelihood ratio

The prior odds are $1 : 99$ and the likelihood ratio is $\frac{0.90}{0.05} = 18$, so the posterior odds are $18 : 99 = 2 : 11$, a probability of $\frac{2}{13} = 0.154$.
A second independent positive test multiplies by 18 again, giving $36 : 11$ and a probability of $\frac{36}{47} = 0.766$, which is why repeated testing is far more decisive than a single result.

5 / THE MONTY HALL PROBLEM

Switching wins with probability $\frac{2}{3}$

You pick door 1 and the host, who knows where the car is, opens a goat door. Suppose he opens door 3: if the car is behind door 1 he chooses door 3 with probability $\frac{1}{2}$, giving weight $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$, while if it is behind door 2 he is forced to open door 3, giving weight $\frac{1}{3} \cdot 1 = \frac{1}{3}$, and behind door 3 he cannot.
The weights $\frac{1}{6} : \frac{1}{3}$ are $1 : 2$, so the car is behind your door with probability $\frac{1}{3}$ and behind door 2 with probability $\frac{2}{3}$. Counting the three equally likely car positions agrees: staying wins in one and switching in two.

ASIDE / PRIORS MATTER

A test with the same accuracy applied to a group where the disease affects 50% of people would make a positive result about 95% reliable.

LINKS TO OTHER SHEETS

The conditional rules are in *Conditional probability and independence*; the next sheet is *Random variables, expectation and variance*.

Convention. Sensitivity is $P(+ \mid \text{sick})$; the false positive rate is $P(+ \mid \text{healthy})$; the figures are illustrative.

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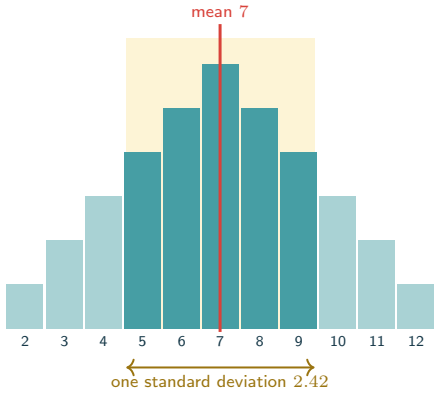
PROBABILITY LADDER 5 OF 9

Random variables, expectation and variance

A random variable attaches a number to each outcome, and its mean and spread summarise the whole distribution.

1 / THE RULE AND THE PICTURE

The mean is the weighted average, and the variance is the average squared distance from it



the sum of two dice, with bar heights proportional to $P(X = s)$

For a random variable X with probabilities $p(x)$,

$$E[X] = \sum x\,p(x), \quad \operatorname{Var} X = E[X^2] - \big(E[X]\big)^2$$

For the sum of two dice $E = 7$, $\operatorname{Var} = \frac{35}{6} = 5.83$ and the standard deviation is $\sqrt{35/6} = 2.42$.

2 / ONE DIE

The mean of a fair die is 3.5 though no roll shows it

With $p = \frac{1}{6}$ each, $E[X] = \dfrac{1+2+\cdots+6}{6} = \frac{7}{2}$ and $E[X^2] = \dfrac{1+4+9+16+25+36}{6} = \frac{91}{6}$, so

$$\operatorname{Var} X = \tfrac{91}{6} - \big(\tfrac{7}{2}\big)^2 = \tfrac{35}{12} = 2.92, \qquad \sigma = 1.71.$$

The mean is the balance point of the distribution, and the standard deviation measures how far a typical roll sits from it.

3 / COMBINING VARIABLES / LINEARITY

Means always add, and variances add for independent variables

rule	statement	example
mean of a sum	$E[X+Y] = E[X] + E[Y]$	two dice: $\frac{7}{2} + \frac{7}{2} = 7$
variance of a sum, independent	$\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$	$\frac{35}{12} + \frac{35}{12} = \frac{35}{6}$
scaling and shifting	$E[aX+b] = aE[X] + b$	double and add 1: $2 \cdot 3.5 + 1 = 8$
variance of a scaling	$\operatorname{Var}(aX+b) = a^2 \operatorname{Var}X$	doubling a die: $4 \cdot \frac{35}{12}$

The mean rule needs no independence at all, which is why it is so useful, while the standard deviation of the sum is $\sqrt{2}$ times that of one die and not twice as large.

4 / EXPECTED VALUE IN GAMES

A fair price for a game equals its expected winnings

game	expected winnings	meaning
roll a die, win its value in pounds	$\frac{7}{2}$	fair price £3.50
bet £1 on red at American roulette	$\frac{18}{38} - \frac{20}{38} = -\frac{1}{19}$	loses 5.3p on average
lottery jackpot £1 000 000 from $\binom{49}{6}$ tickets	$\frac{10^6}{13\,983\,816}$	7.2p from the jackpot alone

A negative expectation does not predict any single play, but over many plays the average result converges to it, which is how casinos make their money.

5 / INDICATOR VARIABLES / COUNTING WITHOUT LISTING

Write a count as a sum of 0-1 variables and add their means

The number of aces in a five-card hand is the sum of five indicators, each with mean $\frac{4}{52}$, so the expected number is $5 \cdot \frac{4}{52} = \frac{5}{13} = 0.385$, even though the cards are dependent.

The number of fixed points of a random permutation of n objects has mean $n \cdot \frac{1}{n} = 1$ for every n, and a simulation over all 120 permutations of five objects confirms the average of exactly 1. The expected number of heads in ten tosses is $10 \cdot \frac{1}{2} = 5$.

ASIDE / MEAN IS NOT TYPICAL

A mean can be a value that never occurs, such as 3.5 for a die, so it describes the long-run average and not a single outcome.

LINKS TO OTHER SHEETS

The dice sums come from *Sample spaces and equally likely outcomes*; the next sheet is *The binomial distribution*, which has mean np.

Convention. $E[X]$ is the expected value; $\operatorname{Var} X$ the variance; $\sigma = \sqrt{\operatorname{Var} X}$ the standard deviation; dice are fair and independent.

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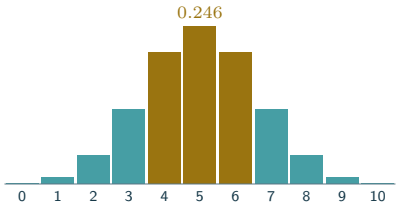
PROBABILITY LADDER 6 OF 9

The binomial distribution

Count the successes in a fixed number of independent trials, and the probabilities come from binomial coefficients.

1 / THE RULE AND THE PICTURE

In n independent trials with success probability p, the chance of k successes is $\binom{n}{k}p^k(1-p)^{n-k}$



heads in 10 fair tosses, with 4, 5 or 6 in gold

Each particular sequence with k successes has probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such sequences, so

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\quad E[X]=np,\quad \operatorname{Var}X=np(1-p)$$

For ten fair tosses $\binom{10}{5}/1024=\frac{252}{1024}=0.246$ is the most likely count.

2 / WORKED EXAMPLES

Exactly three sixes in ten rolls of a die

Here $n=10$, $p=\frac{1}{6}$ and $k=3$, so the probability is $\binom{10}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^7=120\cdot\frac{1}{216}\cdot 0.279=0.155$.
For at least one success the complement is quickest: at least one defect among ten items that are each defective with probability 0.1 has probability $1-0.9^{10}=0.651$.

3 / THE DISTRIBUTION FOR TEN FAIR TOSSES

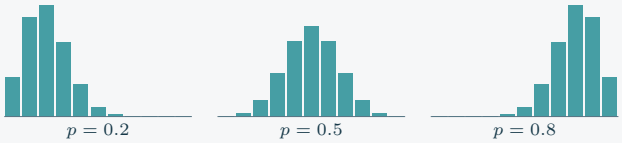
The probabilities are the row of Pascal's triangle divided by $2^{10}=1024$

k	$\binom{10}{k}$	probability	decimal
0	1	1/1024	0.0010
1	10	10/1024	0.0098
2	45	45/1024	0.0439
3	120	120/1024	0.1172
4	210	210/1024	0.2051
5	252	252/1024	0.2461

The distribution is symmetric about 5.
The chance of 4, 5 or 6 heads is $\frac{210+252+210}{1024}=\frac{672}{1024}=0.656$, and the chance of two or fewer is $\frac{1+10+45}{1024}=\frac{56}{1024}=0.055$.

4 / THE EFFECT OF p

Small p piles the probability at the left, large p at the right, and $p=\frac{1}{2}$ is symmetric



The means are $np=2,\ 5,\ 8$ and the variance $np(1-p)=1.6,\ 2.5,\ 1.6$, so the spread is greatest for $p=\frac{1}{2}$. For $n=20$ and $p=0.3$ the mean is 6, the standard deviation $\sqrt{4.2}=2.05$, and the most likely count 6 has probability 0.192.

5 / WHY np AND $np(1-p)$ / THE GALTON BOARD

A count of successes is a sum of n independent 0-1 trials

Each trial contributes mean p and variance $p(1-p)$, and means and independent variances add, giving np and $np(1-p)$.
In a Galton board with 8 rows each peg sends the ball left or right with probability $\frac{1}{2}$, so the final bins have probabilities $1, 8, 28, 56, 70, 56, 28, 8, 1$ out of 256, the binomial row with $n=8$.

ASIDE / INDEPENDENCE IS ESSENTIAL

Drawing cards without replacement is not binomial, because the success probability changes after each draw.

LINKS TO OTHER SHEETS

The coefficients come from *Counting: arrangements and choices*, and the mean np from *Random variables, expectation and variance*; the next sheet is *Waiting times: the geometric and Poisson distributions*.

Convention. X counts successes; n is the number of trials; p the success probability; trials are independent with equal p.

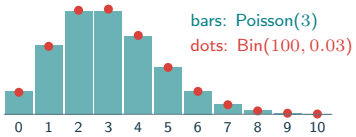
PROBABILITY LADDER 7 OF 9

Waiting times: geometric and Poisson

The geometric law times the first success, and the Poisson law counts rare events, each the limit of simpler binomial pictures.

1 / THE RULE AND THE PICTURE

When trials are many and successes rare, the binomial count settles to $e^{-\lambda}\lambda^k/k!$



If a count has mean λ and events occur independently at random, then

$$P(X=k)=\frac{e^{-\lambda}\lambda^k}{k!}, \qquad E[X]=\operatorname{Var}X=\lambda$$

It is the limit of $\mathrm{Bin}(n,\lambda/n)$ as $n\to\infty$, and already $\mathrm{Bin}(100,0.03)$ is close to $\mathrm{Poisson}(3)$.

2 / A WORKED POISSON EXAMPLE

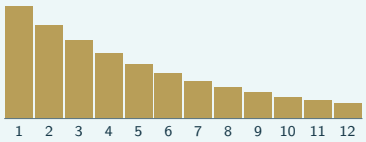
A switchboard receiving 3 calls an hour on average

calls k	Poisson (3)	$\mathrm{Bin}(100,0.03)$
0	0.0498	0.0476
1	0.1494	0.1471
2	0.2240	0.2252
3	0.2240	0.2275
4	0.1680	0.1706
5	0.1008	0.1013
6	0.0504	0.0496

The probability of no call in an hour is $e^{-3}=0.0498$, and of five or more calls $1-(0.0498+0.1494+0.2240+0.2240+0.1680)=0.185$.
The probabilities of 2 and 3 calls are equal because $\lambda=3$.

3 / THE GEOMETRIC DISTRIBUTION / WAITING FOR A SIX

The first success on trial k has probability $(1-p)^{k-1}p$



For a die, $p=\frac{1}{6}$ and the first six comes at roll k with probability $\frac{1}{6}\left(\frac{5}{6}\right)^{k-1}$: $0.167,\ 0.139,\ 0.116,\ldots$.
The mean wait is $1/p=6$ rolls, and the chance of a six within six rolls is $1-(\frac{5}{6})^6=0.665$, with a median wait of 4 rolls since $1-(\frac{5}{6})^4=0.518$.

4 / NO MEMORY / THE EXPONENTIAL WAITING TIME

Having waited without success does not shorten the remaining wait

For the geometric law $P(X>m+n\mid X>m)=P(X>n)$, because each trial starts afresh.
In continuous time the gap T between Poisson events at rate λ satisfies $P(T>t)=e^{-\lambda t}$, with mean $1/\lambda$; for $\lambda=3$ per hour the mean gap is 20 minutes and the median $\ln 2/3=0.231$ hours, about 14 minutes.
Radioactive decay is of this kind, and its half-life is $\ln 2$ divided by the decay rate.

5 / WHICH LAW TO USE

Match the question to the story

law	counts	mean	variance
binomial	successes in n trials	np	$np(1-p)$
geometric	trials until the first success	$1/p$	$(1-p)/p^2$
Poisson	events in a fixed interval	λ	λ

The Poisson law fits phone calls, typing errors, goals in a football match or radioactive counts, wherever events are rare in each small moment and independent of one another.

ASIDE / THE LIMIT

Writing $(1-\lambda/n)^n\to e^{-\lambda}$ and $\binom{n}{k}/n^k\to 1/k!$ turns the binomial formula into the Poisson formula.

LINKS TO OTHER SHEETS

The binomial starting point is *The binomial distribution*; the next sheet is *The normal distribution and the central limit theorem*.

Convention. λ is the mean number of events in the interval; p the success probability; T the waiting time between events.

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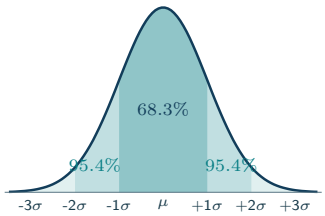
PROBABILITY LADDER 8 OF 9

The normal distribution and the central limit theorem

Sums and averages of many independent trials follow a bell curve, whatever the shape of a single trial.

1 / THE RULE AND THE PICTURE

About 68%, 95% and 99.7% lie within 1, 2 and 3 standard deviations of the mean



A normal variable with mean μ and standard deviation σ has density $\frac{1}{\sigma\sqrt{2\pi}}e^{-(x-\mu)^2/2\sigma^2}$, and its distances from the mean are measured in the z-score $z = (x-\mu)/\sigma$:

$$P(|X-\mu| < \sigma) = 0.683, \quad P(< 2\sigma) = 0.954, \quad P(< 3\sigma) = 0.997$$

2 / USING IT / A WORKED EXAMPLE

Scores with mean 100 and standard deviation 15

Between 85 and 115 lies one standard deviation either side, so 68.3% of scores. A score above 130 is two standard deviations up, with probability $\frac{1}{2}(1-0.954) = 0.023$.
The cumulative values to remember are $\Phi(1) = 0.841$, $\Phi(2) = 0.977$, and $\Phi(1.96) = 0.975$, which gives the familiar 95% range $\mu \pm 1.96\sigma$.

3 / THE CENTRAL LIMIT THEOREM / SUMS OF DICE

As the number of dice grows, the histogram of the sum approaches the normal curve in red



A single die is flat, two dice give a triangle, and by five dice the shape is already close to the curve with the same mean $3.5n$ and variance $\frac{35}{12}n$.
The theorem says the sum of n independent copies of any variable with finite variance is approximately normal for large n.

4 / THE NORMAL APPROXIMATION TO THE BINOMIAL

$\mathrm{Bin}(100, \frac{1}{2})$ has mean 50 and standard deviation 5

To find $P(X \geq 60)$ treat X as normal and use the boundary 59.5, the continuity correction, giving $z = \frac{59.5-50}{5} = 1.9$ and a probability of 0.0287, against the exact binomial value 0.0284.
Without the half-unit correction the estimate $z = 2$ gives 0.0228, noticeably worse.

5 / AVERAGES AND THE STANDARD ERROR

The average of n readings has spread $\sigma/\sqrt{n}$

The mean of 25 rolls of a die has mean 3.5 and standard deviation $\frac{1.708}{\sqrt{25}} = 0.342$, so an average above 4 is $\frac{0.5}{0.342} = 1.46$ standard deviations high, with probability 0.072.
Quadrupling the sample halves the spread, which is why polls and measurements gain accuracy slowly.

ASIDE / WHY THE BELL

Adding many small independent effects blurs every distribution into the same shape, which is why heights, measurement errors and exam scores look normal.

LINKS TO OTHER SHEETS

The sums of dice start in *Random variables, expectation and variance*, and the binomial in *The binomial distribution*; the last sheet is *Information and entropy*.

Convention. Φ is the cumulative standard normal distribution; μ, σ the mean and standard deviation; probabilities from the exact normal and binomial formulas.

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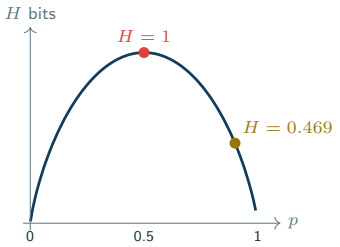
PROBABILITY LADDER 9 OF 9

Information and entropy

The surprise of an event is the logarithm of its improbability, and entropy is the average surprise.

1 / THE RULE AND THE PICTURE

An event of probability p carries $-\log_2 p$ bits of surprise, and the entropy is their average



entropy of a coin that lands heads with probability p

$$H = -\sum_i p_i \log_2 p_i = E[\text{surprise}]$$

A fair coin gives 1 bit, a coin that lands heads 90% of the time gives 0.469 bits because the outcome is usually predictable, and the entropy is largest when all outcomes are equally likely.

2 / SOME ENTROPIES

The more evenly spread the distribution, the more bits it carries

source	entropy	note
fair coin	1 bit	$\log_2 2$
coin with $P(\text{heads}) = 0.9$	0.469 bits	$-0.9\log_2 0.9 - 0.1\log_2 0.1$
fair die	2.585 bits	$\log_2 6$
sum of two dice	3.274 bits	less than $\log_2 11 = 3.459$, since the sums are not equally likely

For W equally likely outcomes the entropy is $\log_2 W$, which is the largest value possible for W outcomes.

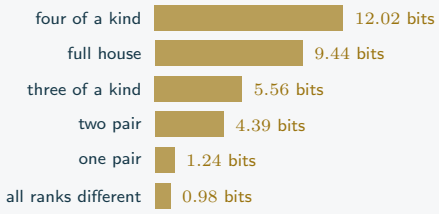
3 / ENTROPY AS THE LENGTH OF A CODE

No code can use fewer bits per symbol on average than the entropy

For four symbols with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ the codewords $0,\ 10,\ 110,\ 111$ have lengths $1, 2, 3, 3$, so the average length is $\frac{1}{2}\cdot 1 + \frac{1}{4}\cdot 2 + \frac{1}{8}\cdot 3 + \frac{1}{8}\cdot 3 = 1.75$ bits, which equals the entropy because each codeword length is exactly $\log_2(1/p)$.
Likely symbols get short codewords and the code needs no separators, which is the idea behind data compression.

4 / A POKER HAND / SURPRISE AND THE NUMBER OF MICROSTATES

A hand pattern with W hands has probability $W/2\,598\,960$ and surprise $\log_2(2\,598\,960/W)$



The six rank patterns have entropy $H = 1.365$ bits, while specifying the exact hand takes $\log_2 \binom{52}{5} = 21.31$ bits, and the difference 19.94 bits is the average $\log_2 W$ left within a pattern. The rare patterns are the surprising ones: a full house has probability 0.00144, a surprise of 9.44 bits.

5 / CONNECTION TO PHYSICS

Boltzmann's entropy is the entropy of a uniform distribution over microstates

pattern	hands W	probability	surprise (bits)	$\log_2 W$
four of a kind	624	0.00024	12.02	9.29
full house	3 744	0.00144	9.44	11.87
three of a kind	54 912	0.02113	5.56	15.74
two pair	123 552	0.04754	4.39	16.91
one pair	1 098 240	0.42257	1.24	20.07
all ranks different	1 317 888	0.50708	0.98	20.33

For a uniform choice among W microstates the entropy is $\ln W = 14.77$ nats for the whole pack of hands, and Boltzmann's $S = k \ln W$ is this quantity in physical units; the surprise of a pattern equals $\log_2(2\,598\,960) - \log_2 W$, so patterns with more hands are less surprising.

ASIDE / BITS AND NATS

Using $\ln$ in place of $\log_2$ changes the unit from bits to nats, with 1 nat $= 1.443$ bits.

LINKS TO OTHER SHEETS

The poker counts and the entropy of the card pack are developed in the card-pack entropy sheet; the previous sheet is *The normal distribution and the central limit theorem*.

Convention. H is the Shannon entropy; $\log_2$ gives bits; W is the number of equally likely hands in a pattern; k is Boltzmann's constant.

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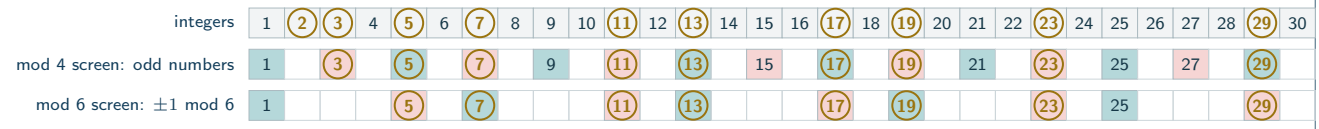
NUMBER THEORY ON ONE PAGE

Prime types: the mod 4 and mod 6 races

Two pre-screens cut the integers before the sieve finishes, and each sorts its survivors into two types.

1 / THE RULE AND THE PICTURE

The mod 4 screen removes the multiples of 2, and the mod 6 screen removes the multiples of 2 and 3

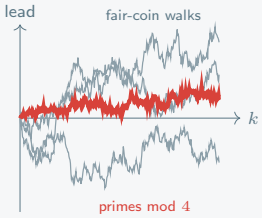


pre-screen	sieves out	survivors	primes among survivors	types
mod 4	multiples of 2	1 in 2	$2/\ln x$	1 or 3 mod 4
mod 6	multiples of 2 and 3	1 in 3	$3/\ln x$	1 or 5 mod 6

Primes are ringed in gold, and survivors are coloured by type, teal for 1 and red for the other. The mod 4 screen is the cruder, since a third of its odd survivors are multiples of 3 and never prime (apart from 3), while the mod 6 screen removes those too. Both colour the mod 12 classes $1, 5, 7, 11$ differently: mod 4 calls $1, 5$ type 1 and $7, 11$ type 3, mod 6 calls $1, 7$ type 1 and $5, 11$ type 5. Two consecutive primes share a type exactly when their gap is a multiple of the modulus, so $P(\text{same type}) = P(\text{gap} \equiv 0 \bmod 4)$ or $P(\text{gap} \equiv 0 \bmod 6)$, which is $\frac{1}{2}$ for a fair coin.

2 / THE TWO RACES

The two types are equally likely in the long run, but the counts keep a small lead



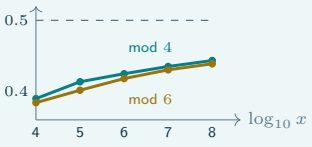
lead of 3 over 1 mod 4, to 10^7

to x	1 mod 4	3 mod 4	lead	1 mod 6	5 mod 6	lead
10^2	11	12	1	11	12	1
10^3	80	86	6	80	86	6
10^4	609	618	9	611	616	5
10^5	4 783	4 807	24	4 784	4 806	22
10^6	39 175	39 321	146	39 231	39 265	34
10^7	332 180	332 397	217	332 194	332 383	189
10^8	2 880 504	2 880 949	445	2 880 517	2 880 936	419

Counts exclude 2 and 3; the lead is the second type minus the first. The red walk, up for a 3 and down for a 1, never exceeds 375, while fair-coin walks reach ± 700 to ± 1100, and the 1 mod 4 primes lead for stretches, first at 26 861 in the usual count that includes 3.

3 / TRANSITIONS / WHAT FOLLOWS WHAT

After a prime of one type the next is more often of the other



$P(\text{same type})$ against $\log_{10} x$

mod 4	next 1	next 3	mod 6	next 1	next 5
after 1	0.443	0.557	after 1	0.439	0.561
after 3	0.556	0.444	after 5	0.561	0.439

Up to 10^8 a type repeats with probability 0.4435 mod 4 and 0.4389 mod 6, rising towards $\frac{1}{2}$ from 0.390 and 0.384 at 10^4, with the shortfall times $\ln x$ near 1.0 and 1.1, so the bias fades roughly like $1/\ln x$. The mean run of one type is 1.80 against 2 for a fair coin, and runs of length 1 are 54% against 50%. These are biases within each modulus, one table for mod 4 and a separate one for mod 6.

4 / BOTH MODULI AT ONCE / THE MOD 12 TABLE

The four classes 1, 5, 7, 11 mod 12 carry both types, and the two repeats are linked

from \ to	1	5	7	11	stay mod 4	stay mod 6
1	0.174	0.306	0.264	0.255	0.481	0.439
5	0.232	0.174	0.329	0.264	0.406	0.439
7	0.265	0.255	0.174	0.306	0.481	0.439
11	0.329	0.265	0.232	0.174	0.406	0.439

The diagonal, a repeat of the whole class, has probability 0.174 against $\frac{1}{4}$ for a fair choice and against $0.195 = 0.4435 \times 0.4389$ if the mod 4 and mod 6 repeats were independent, a ratio 0.77, 0.84, 0.88, 0.90 at 10^5 to 10^8. The link runs one way: the mod 4 repeat is 0.481 after a 1 mod 6 prime and 0.406 after a 5 mod 6 prime, while the mod 6 repeat is 0.439 whichever mod 4 type came first.

After a 1 mod 6 prime the gap 4, which is common, keeps the mod 4 type, whereas after a 5 mod 6 prime the matching gap is 8, which is rarer (5.8% against 7.6%).

5 / WHY, AND WHAT IT MEANS

Small gaps like 2, 6, 10 outnumber 4, 8, 12

Staying in a type mod 4 needs a gap of $4, 8, 12, \ldots$, which together are 44.3% of gaps up to 10^8, while the gaps $2, 6, 10, \ldots$ that switch type are led by 6 (13.3%) over 2 and 4 (7.6% each), a tilt that follows from Hardy and Littlewood's prime pair estimates. So the pre-screens do not turn irregular order into regular order: they remove the forced template, and what remains is mildly anti-persistent, visible only in consecutive pairs, in line with the primes being quieter than Poisson at long scales.

ASIDE / TWO NAMES

The within-modulus pair bias was found by Lemke Oliver and Soundararajan in 2016, and the lead of 3 mod 4 over 1 mod 4 is the Chebyshev bias of the 1850s.

LINKS TO OTHER SHEETS

The two types are the split and inert primes of *Primes mod 4 and the Gaussian integers* and *Primes mod 3 and the Eisenstein integers*; the low-frequency quiet is in *Primes as a filtered signal*.

Convention. Primes above 3; the lead is the count of the second type minus the first; all data are exact counts to 10^8.

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THE SQUARE-ROOT-OF-TIME RULE ON ONE PAGE

Spread grows like $\sqrt{n}$, told with dice

Add up many independent rolls and the total drifts in proportion to n while its scatter grows only as $\sqrt{n}$.

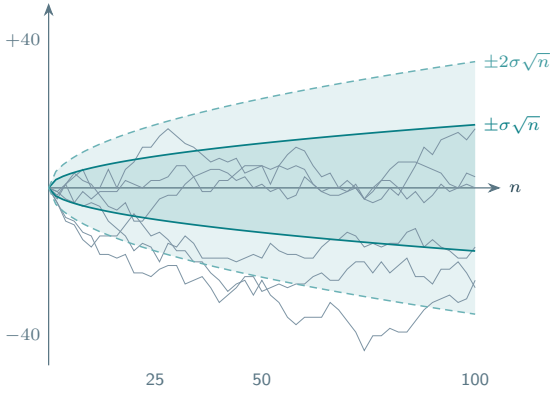
THE RULE

Variances add, so standard deviations go as $\sqrt{n}$

One fair die has mean 3.5 and variance $\frac{35}{12}$, so its standard deviation is $\sigma = \sqrt{35/12} \approx 1.71$. For independent rolls the variances add, and the total of n rolls has

$$\text{mean} \;=\; 3.5\,n, \qquad \text{SD} \;=\; \sigma\sqrt{n} \;\approx\; 1.71\sqrt{n}$$

Quadrupling the number of rolls doubles the spread. Subtracting 3.5 from each roll leaves a fair random walk whose typical distance from zero after n steps is $1.71\sqrt{n}$; the fan beside it shows six such walks of 100 rolls inside bands at one and two standard deviations.



Running total of (roll − 3.5) for six walks; the bands widen as $\sqrt{n}$, never as n.

LONG-RUN DICE

The total spreads, the average settles

rolls n	mean total	SD of total	95% of totals in	SD of average	spread ÷ mean
1	3.5	1.71	1 to 6, all	1.71	49%
4	14	3.42	7.3 to 20.7	0.85	24%
16	56	6.83	42.6 to 69.4	0.43	12%
100	350	17.1	316.5 to 383.5	0.17	4.9%
400	1400	34.2	1333 to 1467	0.085	2.4%

Every factor of 4 in rolls doubles the spread of the total and halves the spread of the average, which is the law of large numbers seen at its proper rate. By the central limit theorem the totals pile into a bell curve whatever the shape of one die, which is what lets ± 1.96 SD capture 95%.

FROM DICE TO PRICES

Daily log returns play the part of rolls

If each day's log return is an independent draw with standard deviation σ_{day}, the log return over T days is a sum of T draws, so

$$\sigma_T = \sigma_{\text{day}}\sqrt{T}, \qquad \sigma_{\text{year}} \approx \sigma_{\text{day}}\sqrt{252},$$

using 252 trading days. A typical daily move of 1% becomes about 2.2% a week, 4.6% a month and 15.9% a year. Any steady drift μ grows in proportion to T, so it overtakes one standard deviation of scatter only after $T = (\sigma/\mu)^2$, about five years for $\mu = 7\%$ and $\sigma = 15.9\%$ a year.

horizon	days	σ
day	1	1.0%
week	5	2.2%
month	21	4.6%
year	252	15.9%

Scaling a 1% daily standard deviation by $\sqrt{T}$.

WHY VARIANCES ADD

Independent steps are at right angles

Squaring a sum of deviations gives the squares plus cross terms $2\,d_i d_j$, and for independent rolls each cross term averages to zero, leaving $\mathrm{Var}\left(\sum d_i\right) = \sum \mathrm{Var}(d_i)$. Independent steps combine like perpendicular sides in Pythagoras, so n steps of size σ reach $\sigma\sqrt{n}$, the diagonal of an n-dimensional cube; the standard deviation is the RMS of the deviation, the same mean that measures alternating current.

WHEN THE RULE BREAKS

Real prices are not quite dice

The rule needs independent, identically distributed steps. Momentum makes long-horizon spread grow faster than $\sqrt{T}$ and mean reversion slower; volatility comes in clusters, so a calm month predicts a calm month; and daily returns have fatter tails than a bell curve, with crashes far more frequent than $\pm 1.96\,\sigma$ suggests. The $\sqrt{T}$ rule is the benchmark against which these departures are measured, and it is not investment advice.

Conventions. SD is standard deviation, variance is its square, T counts trading days, and log returns are used so that returns over successive days add.

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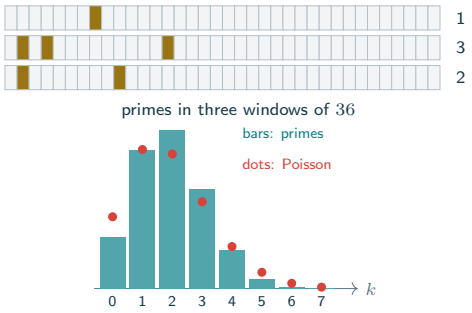
PROBABILITY ON ONE PAGE

The Poisson distribution: counting arrivals

A Poisson count tallies rare independent arrivals with variance equal to the mean; primes nearly fit it and composites cannot.

1 / THE RULE AND THE PICTURE

If events arrive independently at a steady rate, the number in a window of mean size λ is Poisson



counts in 1.1 million windows against the Poisson law

With arrival rate ρ per unit and a window of length h, so $\lambda = \rho h$, the count N satisfies

$$P(N = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad E[N] = \operatorname{Var} N = \lambda$$

Near 10^8 the density of primes is $\rho = 1/\ln x = 0.0543$ per integer, so a window of 36 integers expects $\lambda = 1.94$ primes, and the Poisson law gives no prime with probability $e^{-1.94} = 0.144$ and exactly one with probability 0.279. The law is the limit of splitting the window into n cells, each with arrival probability p, as $n \to \infty$ with $np = \lambda$ fixed.

2 / THREE CONDITIONS

Rare, independent and steady

A count is Poisson when arrivals are rare in each small cell, independent of one another, and come at a constant rate over the window. Then the waiting time between arrivals is exponential with mean $1/\rho$, so for the primes the gap has mean $\ln x$, and the variance equal to the mean is the signature to test: the ratio $\mathrm{Var}/\mathrm{mean}$ is the index of dispersion, equal to 1 for a Poisson count, below 1 for arrivals more regular than chance and above 1 for clumped ones.

3 / PRIMES IN WINDOWS OF 36

The mean matches, but the counts are more bunched round it than Poisson

primes k	observed	Poisson 1.94
0	0.104	0.144
1	0.279	0.279
2	0.318	0.270
3	0.200	0.174
4	0.077	0.084
5	0.018	0.033
6	0.003	0.011

Over 1.1 million windows from 10^8 the mean count is 1.936 against the predicted $36/\ln x = 1.935$, but the variance is 1.42 and not 1.94, an index of 0.73.
There are fewer empty windows (0.104 against 0.144) and fewer with five or more primes, and more with two or three.
Primes arrive more regularly than a random scatter would, because the small primes have already removed most of the integers in a fixed pattern.

4 / THE COMPOSITES IN THE SAME WINDOWS

The common event is not Poisson, because it is the complement of a rare one

A window of 36 integers holds $36 - N$ composites, so their mean is 34.06 and their variance is exactly that of the primes, 1.42, an index of 0.042: far below Poisson, since a count that nearly fills the window cannot fluctuate much.
The wait for the next composite is also nearly fixed, as each integer is composite with probability 0.946, so the mean wait is 1.06 integers, while the wait for the next prime averages 18.4. Poisson describes the rare arrival, and its complement is a nearly deterministic background.

5 / HOW THE REGULARITY GROWS WITH THE WINDOW

The index of dispersion of the prime count falls steadily as windows lengthen

window h	mean λ	variance	observed index	$1 - \frac{\ln h}{\ln x}$
36	1.94	1.42	0.73	0.81
108	5.81	3.91	0.67	0.75
360	19.4	11.8	0.61	0.68
1080	58.1	31.9	0.55	0.63
3600	193.6	96.6	0.50	0.56

Windows near 10^8; the last column is an estimate from pair correlation (Goldston and Montgomery), which follows the shape of the decline though it sits about 0.07 above the measured values.
The mod 6 screen alone, with only ± 1 mod 6 cells able to hold a prime, would give an index of $1 - 3/\ln x = 0.84$ for windows of 36; sieving by 5, 7 and the larger primes pushes it lower, and long windows lower still.

ASIDE / THE COMPARISON

Counts that fit a Poisson law look like random scatter at the scale of the mean gap, and the shortfall in their variance is a measure of hidden order.

LINKS TO OTHER SHEETS

The same shortfall appears as the quieter filtered signal in *Primes as a filtered signal* and as the Kolmogorov distance in *Means and the Kolmogorov distance*; the waiting times are in *Waiting times: geometric and Poisson*.

Convention. N number of arrivals in a window; $\lambda = \rho h$ its mean; ρ arrival rate; the index of dispersion is $\mathrm{Var}/\mathrm{mean}$; windows are non-overlapping stretches of integers from 10^8 to 1.4×10^8.

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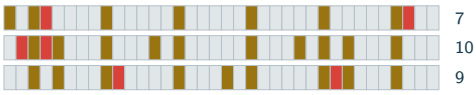
PROBABILITY ON ONE PAGE

Are any composite types Poisson?

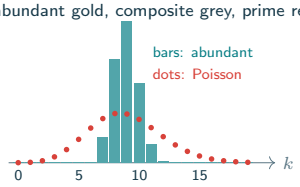
Only the perfect numbers, counted per doubling of their exponent, arrive like a Poisson process.

1 / THE TEST AND THE PICTURE

A Poisson count has variance equal to its mean, and the abundant counts have a tenth of that



abundant gold, composite grey, prime red



abundant counts in windows of 36 against the Poisson law

Every integer $n \geq 2$ is a prime, deficient ($s(n) < n$), perfect ($s(n) = n$) or abundant ($s(n) > n$), where s is the sum of proper divisors, so the composites split into three types. A type is Poisson-like when its arrivals are rare, independent and steady, which shows as an index of dispersion $\mathrm{Var}/\mathrm{mean}$ near 1:

$$N \text{ in a window of } h: \quad \mathrm{Var}\, N \approx E[N] \iff \text{Poisson-like}$$

For $h = 36$ from 10^6 the abundant numbers have mean count 8.91 but variance 1.00, an index of 0.11: windows hold almost the same number every time, as the three rows show $(7, 10, 9)$.

2 / ABUNDANT NUMBERS: DENSE AND NEARLY PERIODIC

They arrive at gaps of 2, 4 or 6 and almost never otherwise

gap to next abundant	observed	geometric, mean 4.04
1	0.002	0.248
2	0.325	0.186
3	0.015	0.140
4	0.305	0.106
5	0.000	0.079
6	0.354	0.060
7 or more	0.000	0.181

A quarter of the integers (0.2476) are abundant, and 99.2% of those are even and 67% are multiples of 6, because every multiple of 6 except 6 itself is abundant, as is every multiple of 20, 28, 30 and so on. The gaps are almost entirely 2, 4 and 6 (33%, 30%, 35%) against a geometric law for random arrivals, and the index of dispersion falls from 0.18 at $h = 12$ to 0.025 at $h = 3600$, far below both Poisson (1) and a random thinning ($1 - 0.25 = 0.75$).

3 / DEFICIENT COMPOSITES: THE BACKGROUND

The common type is the complement of the others, so it cannot be Poisson

Of the integers from 10^6 to 5×10^6, 68.5% are deficient composites, with a mean gap of 1.46 and two thirds of the gaps equal to 1.
Their window counts have index 0.15 at $h = 12$, 0.11 at 36 and 0.06 at 3600.
A type that fills most of the line has almost no room to fluctuate: a window of h integers holds h minus the abundant, prime and perfect counts, so its variance is that of the abundant and prime counts together, which is small.

4 / PERFECT NUMBERS: RARE, BUT ON THE WRONG SCALE

Per integer there is nothing to count, and per doubling of the exponent the count is Poisson-like

Below 5×10^6 there are four: 6, 28, 496, 8128.
They are $2^{p-1}(2^p - 1)$ for the Mersenne exponents $p = 2, 3, 5, 7, 13, 17, 19, 31, \ldots$, so the natural arrival scale is p, taken in doublings. For the 26 doublings $[2^j, 2^{j+1})$ up to $p \approx 1.3 \times 10^8$ the 51 known Mersenne primes give counts $0, 1, 2, 3, 4, 5$ on $3, 6, 9, 6, 1, 1$ occasions against $3.7,\ 7.2,\ 7.0,\ 4.6,\ 2.3,\ 0.9$ expected for Poisson with mean 1.96; the variance is 1.48, an index of 0.75, within the scatter expected from 26 windows.
The Lenstra, Pomerance and Wagstaff heuristic gives a rate of $e^\gamma = 1.78$ per doubling.

5 / VERDICT

Only the perfect numbers, counted per doubling, arrive like a Poisson process

type	density or rate	index $h = 36$	arrivals	verdict
abundant	0.248	0.11	2, 4, 6 only	not Poisson: dense and nearly periodic
deficient composite	0.685	0.11	66% are 1	not Poisson: the dense background
prime (for reference)	0.054	0.73	mean 18.4	nearly Poisson, a little regular
perfect, per integer	≈ 0	none	4 below 10^4	too rare to count in windows
perfect, per doubling of p	1.96	0.75	per doubling of p	Poisson-like on a log scale

ASIDE / CAUTION

The Mersenne list may be incomplete in the highest doublings, where not every exponent has been tested, and with only 26 windows the index of 0.75 is uncertain by about 0.3.

LINKS TO OTHER SHEETS

The Poisson law and the primes are in *The Poisson distribution: counting arrivals*; the Euclid and Euler form of the perfect numbers uses the characteristic ideas of *Prime factorisation, divisors and squares*.

Convention. $s(n) = \sigma(n) - n$ sum of proper divisors; the index of dispersion is $\mathrm{Var}/\mathrm{mean}$ of counts in non-overlapping windows of h integers over 10^6 to 5×10^6; the prime row is from the windows near 10^8 of the previous sheet.

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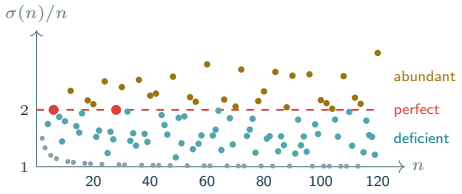
NUMBER THEORY ON ONE PAGE

Deficient, perfect, abundant and Poisson

Sort the integers by their divisor sums, then ask which kinds arrive like a Poisson process.

1 / THE RULE AND THE PICTURE

Add up the divisors of n other than n itself, and compare the total with n



$\sigma(n)/n$ for $n \le 120$, with the perfect numbers 6 and 28 on the line 2

Write $s(n)$ for the sum of the proper divisors of n, so $s(8) = 1+2+4 = 7$ and $s(12) = 1+2+3+4+6 = 16$. An integer is deficient if it is short of its divisors ($s < n$), perfect if it equals them, and abundant if they overshoot, and since $\sigma(n) = s(n) + n$ this is the same as comparing $\sigma(n)/n$ with 2:

$\frac{\sigma(n)}{n} < 2$ deficient,	$= 2$ perfect,	> 2 abundant

Primes sit at the bottom of the picture because $s(p) = 1$, and the abundant numbers rise in bands, the lowest being the multiples of 6.

2 / THE THREE TYPES AT A GLANCE

Most integers are deficient, a quarter are abundant and perfect numbers are very rare

type	test	examples	share up to 10^6
deficient	$s(n) < n$	$1, 2, 3, 4, 5, 7, 8, 9, 10, 11, \ldots$ (all primes and prime powers)	75.2%
perfect	$s(n) = n$	6, 28, 496, 8128	0 (four below 10^4)
abundant	$s(n) > n$	12, 18, 20, 24, 30, …; first odd one 945	24.8%

Multiples of an abundant or perfect number are abundant, which is why the abundant numbers are common: every multiple of 6 except 6 itself, every multiple of 20, and so on. The even perfect numbers are $2^{p-1}(2^p - 1)$ whenever $2^p - 1$ is prime, and no odd perfect number is known.

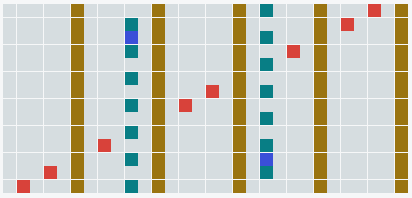
3 / POISSON REFRESHER

A Poisson count is the tally of rare, independent, steady arrivals in a window

If arrivals come at rate ρ per unit and a window has length h, the count N has $P(N = k) = e^{-\lambda}\lambda^k/k!$ with $\lambda = \rho h$, and its variance equals its mean, so the index of dispersion $\mathrm{Var}/\mathrm{mean}$ is 1. An index below 1 means the arrivals are more regular than chance and above 1 means they clump. The primes are the model case: near 10^8 a window of 36 integers expects $\lambda = 1.94$ primes and the counts have index 0.73, so primes are nearly Poisson, a little regular. A type that fills a large share of the line is the opposite case, since the count of a common type in a window barely varies.

4 / WHY DEFICIENT AND ABUNDANT ARE NOT POISSON, AND A BETTER VIEW

A better view: the deficient numbers are the survivors of a sieve of generators



the 420 residues, in rows of 30: multiples of 6 gold, 20 teal, 28 red, 70 blue, others grey

In windows of 36 near 10^6 the abundant numbers have index 0.11, the deficient composites 0.11 and the deficient numbers with primes 0.04, all far below 1, because the types are dense and nearly periodic. Every non-deficient number is a multiple of a generator, a non-deficient number with no smaller non-deficient divisor above 1: 6, 20, 28, 70, 88, 104, 272, 304, …. The first four have least common multiple 420, so they produce an exactly periodic skeleton in which 94 of every 420 residues are non-deficient.

generators used	non-deficient numbers explained	pattern mod 420
6	67.3%	70 of 420
6, 20	80.8%	84
6, 20, 28	88.5%	92
6, 20, 28, 70	90.4%	94
first 6 (…, 104)	94.1%	
first 12 (…, 550)	96.3%	
first 137 (to 20 000)	99.2%	

5 / WHAT IS LEFT TO COUNT

The sparse remainder and the primes are where Poisson tools apply

Removing the multiples of 6, 20, 28, 70 leaves non-deficient numbers of density 0.0238 and mean gap 42.0, whose window counts have index 0.73 at 36 (the same as the primes), 0.58 at 120 and 0.25 at 3600, so this remainder is the nearest thing to a Poisson stream among the composites. The primes are only about 9% of the deficient numbers near 10^6, so including them barely changes the dense background and they are best treated as their own rare component. The perfect numbers are Poisson-like only when counted per doubling of the Mersenne exponent, with mean 1.96 and index 0.75.

ASIDE / THE IDEA TO TAKE AWAY

To study a dense type, strip the periodic skeleton first and then test what remains, just as the small primes are stripped to study the primes.

LINKS TO OTHER SHEETS

The Poisson law and the primes are in *The Poisson distribution: counting arrivals*; the full test of the three types is in *Are any composite types Poisson?*; divisors are in *Prime factorisation, divisors and squares*.

Convention. $s(n)$ sum of proper divisors; $\sigma(n) = s(n) + n$; shares and indices are exact counts over 10^6 to 5×10^6 except the prime index, taken near 10^8.

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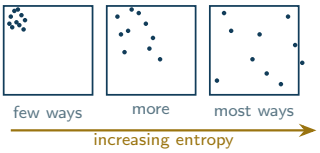
ENTROPY ON ONE PAGE

Entropy, microstates and a growing pack of cards

Counting hands as microstates, then adding cards to see how the counts grow, and whether they grow in proportion.

1 / THE RULE AND THE PICTURE

Entropy is the logarithm of the number of microstates behind a macrostate

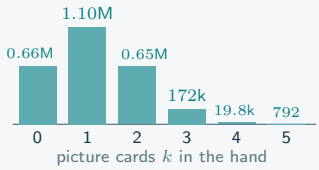


	gas	pack of cards
microstate	where each molecule is	the five exact cards
macrostate	molecules in the left half	the hand type
W	arrangements of that count	hands of that type

$\boxed{S = k \ln W}$ with $k = 1$ below; systems drift towards the largest W.

2 / A MACROSTATE YOU CAN SEE / PICTURE CARDS IN A FIVE-CARD HAND

The count peaks in the middle, as for molecules in a box; five picture cards is the gas in one corner

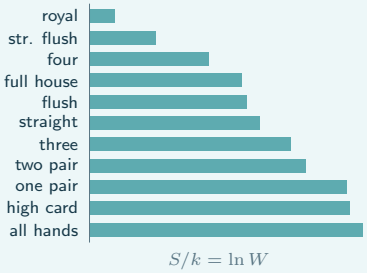


picture cards k	$W = \binom{12}{k}\binom{40}{5-k}$	$S/k = \ln W$	P
0	658 008	13.40	25.3%
1	1 096 680	13.91	42.2%
2	652 080	13.39	25.1%
3	171 600	12.05	6.6%
4	19 800	9.89	0.76%
5	792	6.67	0.03%

3 / EVERY HAND TYPE / ENTROPY OF EACH MACROSTATE

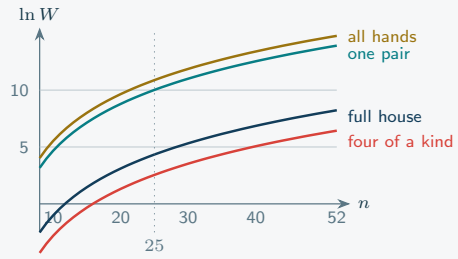
A royal flush is a low-entropy state and one pair is close to equilibrium

hand	W	S/k	P
royal flush	4	1.39	0.00015%
straight flush	36	3.58	0.0014%
four of a kind	624	6.44	0.024%
full house	3744	8.23	0.144%
flush	5108	8.54	0.197%
straight	10 200	9.23	0.392%
three of a kind	54 912	10.91	2.11%
two pair	123 552	11.72	4.75%
one pair	1 098 240	13.91	42.26%
high card	1 302 540	14.08	50.12%
all hands	2 598 960	14.77	100%



4 / GROWING THE PACK / RANDOM CARDS ADDED, COUNTS SCALE TOGETHER

From 25 random cards to 52, every macrostate gains the same 3.89 in $\ln W$



Deal five-card hands from n cards picked at random from the 52. Any hand survives into the pack with probability $\binom{n}{5}/\binom{52}{5}$, so every macrostate's expected count is its 52-card count times that one factor: full houses $3744 \times 53\,130/2\,598\,960 = 76.5$ at $n = 25$. The curves stay parallel, each gaining 3.89, and probabilities stay put on average (0.144%). Single 25-card packs scatter ($0.144 \pm 0.075\%$ over 4000 packs), and the scatter falls to $\pm 0.020\%$ at 40 cards.

$$\langle W_{\text{macro}}(n) \rangle = W_{\text{macro}}(52)\, \binom{n}{5}/\binom{52}{5}, \qquad \Delta(S/k) = \ln 48.9 = 3.89$$

5 / ADDING WHOLE RANKS / THE COUNTS NO LONGER SCALE TOGETHER

New ranks keep raising the entropy but concentrate probability on the common hands

ranks R	cards $4R$	full houses $24R(R-1)$	all hands $\binom{4R}{5}$	P(full house)	$\ln(W_{\text{all}}/W_{\text{FH}})$
6	24	720	42 504	1.69%	4.08
13	52	3744	2 598 960	0.144%	6.54
20	80	9120	24 040 016	0.038%	7.88
26	104	15 600	91 962 520	0.017%	8.68

Choosing the two ranks of a full house gives $W_{\text{FH}} \propto R^2$ while all hands grow as R^5, so $P \propto R^{-3}$ and the gap widens by $3 \ln R$: proportional growth needs new cards drawn from the same mix.

ASIDE / GAS VERSUS GRAVITY

For a gas the uniform macrostate has most microstates, but gravitating bodies gain microstates by clumping, which releases energy into motion; the interaction decides what is counted.

ASIDE / CHECK IN PYTHON

`log(comb(52,5)/comb(25,5))` gives 3.890; counting full houses per `random.sample` pack reproduces the scatter.

Convention. $S = k \ln W$ with $k = 1$; microstates are unordered five-card hands, macrostates are hand types.

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DICE, EXPECTATION, CONDITIONAL PROBABILITY AND BLUFF ON ONE PAGE

Perudo: probability under a cup

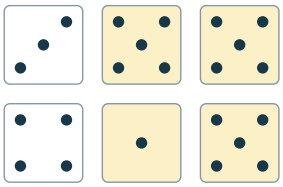
Your own dice are evidence. The hidden dice are uncertainty. A bid turns both into a claim.

1 / SEE YOUR DICE, REASON ABOUT EVERYBODY'S

A bid means "at least", and ones are wild

Roll and conceal the dice; inspect only your own. Take turns raising a bid or calling **Dudo** ("I doubt"). Reveal: if the claim fails, the bidder loses a die; if it holds, the challenger loses one. Re-roll; the last player with dice wins. For ordinary bids, raise the quantity, or keep it and raise the face.

Classroom example: six players with six dice each, so 36 dice in all. You see the hand below. A bid of "fourteen fives" needs at least 14 dice showing either 5 or a wild 1, across *all* cups.



Four known matches: three fives and one ace.

There are $36-6=30$ hidden dice. Each supports a five-bid with probability $p=\frac{1}{6}+\frac{1}{6}=\frac{1}{3}$. Let X count those hidden matches.

$$T=4+X, \qquad \mathbb{E}(X)=30\times\tfrac{1}{3}=10.$$

$$\mathbb{E}(T \mid \text{your hand}) = 4+10 = 14$$

Without wild ones, the same hand gives $3+30/6=8$ expected fives.

Before seeing any dice, the wild-one expectation was $36/3=12$. Your unusually helpful hand raises it to 14; **the mean is not a guarantee**.

2 / TURN THE CLAIM INTO A PROBABILITY

Count successes: the binomial model

For n independent hidden dice, each with match probability p,

$$X\sim \mathrm{Bin}(n,p), \qquad \mathbb{E}(X)=np.$$

Exactly j matches can occupy $\binom{n}{j}$ sets of positions. Each arrangement has probability $p^j(1-p)^{n-j}$, so

$$\mathbb{P}(X=j)=\binom{n}{j}p^j(1-p)^{n-j}.$$

In this hand, "fourteen fives" needs $14-4=10$ or more hidden matches:

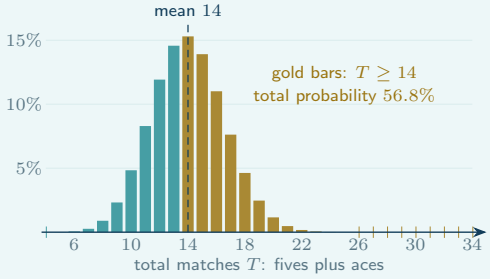
$$\mathbb{P}(T\geq 14 \mid \text{hand})=\sum_{j=10}^{30}\binom{30}{j}\left(\tfrac{1}{3}\right)^j\left(\tfrac{2}{3}\right)^{30-j}$$

$$\mathbb{P}(T \geq 14 \mid \text{hand}) \approx 56.8\%$$

Exactly 14 has probability only 15.3%. Perudo asks for a **tail probability**: add every outcome that makes the bid true.

3 / READ THE CHANCES FROM THE DISTRIBUTION

Higher bids leave less probability



Bid	Hidden needed	Chance true
12 fives	≥ 8	83.3%
14 fives	≥ 10	56.8%
16 fives	≥ 12	27.6%
18 fives	≥ 14	9.0%

4 / SEPARATE DICE EVIDENCE FROM PLAYERS' BEHAVIOUR

Independence, information and bluff

Seeing three fives in your hand does **not** make a hidden five less likely. Independent dice do not use up a fixed supply of each face. Only your known contribution has changed.

But another player's bid may reveal information about their hand. Updating on bids needs a model of how that player bids or bluffs; the binomial calculation above uses no such information.

If the chance a bid is true is r, calling Dudo loses you a die with probability r. Choosing between challenging and raising also depends on what happens next; 50% is no universal strategy rule.

5 / MAKE A PREDICTION, THEN TEST IT

Keep the hand fixed; re-roll the unknowns

Keep the six displayed dice unchanged. Re-roll the other 30 dice 50 times. Each time, record $T=4+X$, then compare:

$$\text{mean of } T \text{ with } 14; \qquad \frac{\#\{T\geq 14\}}{50} \text{ with } 0.568.$$

Expect roughly 28 successful claims in 50 trials, with random variation. Re-rolling all 36 dice would instead test the unconditional model, whose mean is 12.

Generalise. If s visible dice support the bid and n are hidden, then $T=s+X$, $X\sim\mathrm{Bin}(n,p)$:

$$\mathbb{E}(T \mid \text{hand})=s+np, \qquad \mathbb{P}(T\geq q)=\mathbb{P}(X\geq q-s).$$

Conventions. Standard Perudo starts with five dice per player; six each is the classroom variant here. Fair, independent dice; ordinary non-ace bids with wild ones. An ace-only bid has $p=1/6$. Special bidding and Palifico rules are outside this sheet's scope.

Rules: The official home of Perudo; Asmodee rulebook. Probabilities are calculated for the stated hand and assumptions.

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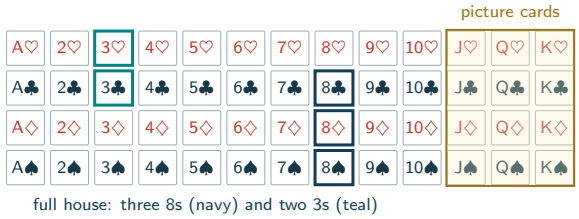
PROBABILITY ON ONE PAGE

Poker hands from a pack

Five cards dealt from a well-shuffled pack, counted two ways, with the workings shown.

1 / THE RULE AND THE PICTURE

Count the favourable hands and divide by all the equally likely hands



full house: three 8s (navy) and two 3s (teal)

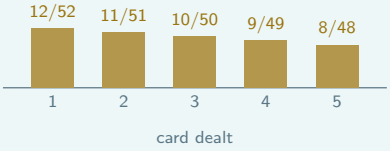
A hand is 5 cards in any order, so there are

$$\binom{52}{5} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5!} = 2\,598\,960$$

equally likely hands, and $P = \dfrac{\text{favourable hands}}{\binom{52}{5}}$

2 / FIVE PICTURE CARDS / TWO ROUTES, ONE ANSWER

$$\frac{12}{52} \times \frac{11}{51} \times \frac{10}{50} \times \frac{9}{49} \times \frac{8}{48} = \binom{12}{5} \Big/ \binom{52}{5}$$



Sequential. Each draw needs a picture card from those left: $\frac{12}{52}$, then $\frac{11}{51}$ given the first, then $\frac{10}{50}$, $\frac{9}{49}$, $\frac{8}{48}$, so $P = \dfrac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48} = \dfrac{95\,040}{311\,875\,200} \approx 3.05 \times 10^{-4}$.

Counting. Choose 5 of the 12 picture cards, $\binom{12}{5} = 792$ ways, from 2 598 960 hands. The routes agree since both ordered counts carry the factor 5!, which cancels.

$$P = \frac{792}{2\,598\,960} \approx 3.05 \times 10^{-4} \approx \frac{1}{3281}$$

3 / FULL HOUSE / THREE OF ONE RANK AND TWO OF ANOTHER

$$13 \times \binom{4}{3} \times 12 \times \binom{4}{2} = 3744 \text{ hands}$$

rank of the triple: 13
× its suits: $\binom{4}{3} = 4$
× rank of the pair: 12
× its suits: $\binom{4}{2} = 6$
= $13 \cdot 4 \cdot 12 \cdot 6 = 3744$

Counting. Choose the rank of the triple (13) and 3 of its 4 suits (4), then a *different* rank for the pair (12) and 2 of its 4 suits (6): $13 \cdot 4 \cdot 12 \cdot 6 = 3744$ hands, so $P = 3744/2\,598\,960 \approx 0.144\%$, about 1 in 694.

Sequential check. For the order XXXYY the first card is anything, then $\frac{3}{51} \cdot \frac{2}{50} \cdot \frac{48}{49} \cdot \frac{3}{48} = \frac{18}{124\,950}$, and the triple can sit in any of $\binom{5}{3} = 10$ positions: $10 \times \frac{18}{124\,950} = 0.001441$, the same value.

$$P = \frac{13\binom{4}{3} \cdot 12\binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2\,598\,960} \approx 0.144\%$$

4 / EVERY HAND TYPE / THE COUNTS PARTITION ALL 2 598 960 HANDS

The same method, one line per hand

hand	workings	hands	probability	1 in
royal flush	4 suits ×1	4	0.00015%	649 740
straight flush	$10 \times 4 - 4$ (royal excluded)	36	0.0014%	72 193
four of a kind	$13 \times \binom{4}{4} \times 48$	624	0.024%	4165
full house	$13 \times \binom{4}{3} \times 12 \times \binom{4}{2}$	3744	0.144%	694
flush	$4\binom{13}{5} - 40$	5108	0.197%	509
straight	$10 \times 4^5 - 40$	10 200	0.392%	255
three of a kind	$13\binom{4}{3} \times \binom{12}{2} \times 4^2$	54 912	2.11%	47
two pair	$\binom{13}{2}\binom{4}{2}^2 \times 44$	123 552	4.75%	21
one pair	$13\binom{4}{2} \times \binom{12}{3} \times 4^3$	1 098 240	42.26%	2.4
high card	$\left(\binom{13}{5} - 10\right)(4^5 - 4)$	1 302 540	50.12%	2.0
total		2 598 960	100%	
five picture cards	$\binom{12}{5}$ (slide 2)	792	0.0305%	3281

ASIDE / WHY THE 5! CANCELS

Ordered deals number $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48 = 311\,875\,200$ and every hand appears $5! = 120$ times, so counting ordered favourable deals over ordered deals gives the same probability as counting hands.

ASIDE / CHECK IN PYTHON

`comb(52,5)` gives 2598960, `comb(12,5)/comb(52,5)` gives 3.047×10^{-4}, and `13*comb(4,3)*12*comb(4,2)` gives 3744.

Convention. A well-shuffled 52-card pack, five cards dealt, order ignored; straights include A-2-3-4-5 and 10-J-Q-K-A, and each hand is counted once, in its highest category.

Probabilistic

YAHTZEE ON ONE PAGE

Yahtzee: the probability of each hand

Roll, keep, reroll, keep, reroll, then score: counting turns a game of luck into a choice of odds.

1 / THE RULE AND THE PICTURE

Five dice, three rolls, and a choice of category at the end

Roll five → Keep multiples or runs → Reroll the rest → Keep the promising → Reroll leftovers → Score a category



reroll

keep the three 5s

card: Fives 15, 3 of a kind 22, Chance 22

One roll has $6^5 = 7776$ equally likely outcomes, so

$$P = \frac{\text{favourable outcomes}}{7776}$$

and rerolls act only on the free dice, so probabilities multiply along a tree and add across branches.

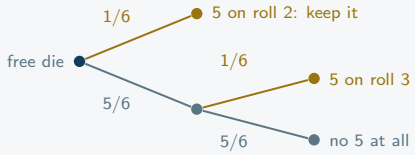
2 / ONE ROLL / COUNTING EACH HAND AMONG THE 7776

Choose the face, choose the places, then fill the other dice

hand	workings	outcomes	probability
Yahtzee	6 faces	6	0.077%
four of a kind (exactly)	$6 \times 5 \times 5$: the face, the odd die's place, its value	150	1.93%
full house	$6 \times 5 \times \binom{5}{3}$: triple face, pair face, places of the triple	300	3.86%
three of a kind (exactly)	$6 \times \binom{5}{3} \times 5 \times 4$	1200	15.43%
two pair	$\binom{6}{2} \times \frac{5!}{2!\,2!} \times 4$	1800	23.15%
one pair	$6 \times \binom{5}{2} \times 5 \times 4 \times 3$	3600	46.30%
five different	$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$	720	9.26%
small straight (a run of 4)	$3 \times 480 - 240$ for the runs counted twice	1200	15.43%
large straight	$2 \times 5!$	240	3.09%

3 / KEEPING AND REROLLING / THE DICE ON THE SCREEN

With $3, 4, 5, 5, 5$ keep the three 5s: each free die gets two chances



A free die ends as a 5 with probability $\frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} = \frac{11}{36}$, and the two free dice are independent, so keeping every 5 and rerolling the rest gives:

five 5s, a Yahtzee	$(11/36)^2 = 121/1296$	9.3%
exactly four 5s	$2 \cdot (11/36)(25/36) = 550/1296$	42.4%
exactly three 5s	$(25/36)^2 = 625/1296$	48.2%
four of a kind or better	$671/1296$	51.8%

4 / CHOOSING A CATEGORY / PROBABILITY TIMES POINTS

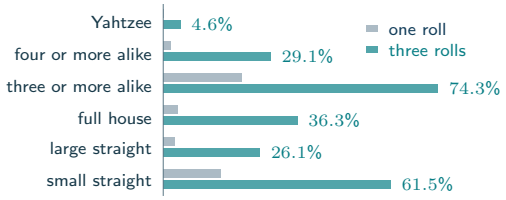
The same five dice support several plans, and expected score ranks them

aim	keep	P(made)	expected score
Yahtzee, 50 points	the three 5s	9.3%	4.7
four of a kind	the three 5s	51.8%	12.3
full house, 25	the three 5s first, then best play	30.6%	7.6
large straight, 40	$3, 4, 5$	27.8%	11.1
small straight, 30	$3, 4, 5$	80.2%	24.1
Fives	the three 5s	always	18.1
3 of a kind or Chance	the three 5s, reroll low dice	always	23.5

The card's 22 is close to the best figure here, so the choice turns on which boxes are still empty.

5 / A WHOLE TURN / BEST PLAY FOR ONE TARGET

Keeping dice lifts every hand, most of all the rare ones



Choosing the best keep at every step for one target, by searching all subsets of the five dice, gives 4.6% for a Yahtzee in a turn against 0.077% in one roll, and 26% for a large straight against 3.1%. Rare hands gain most in ratio, since a kept die is a success that need not be earned twice.

ASIDE / WHY KEEPING WORKS

A chased die succeeds within r rolls with probability $1 - (\frac{5}{6})^r$: $\frac{1}{6}$, $\frac{11}{36}$, $\frac{91}{216}$.

ASIDE / CHECK IN PYTHON

A memoised search over the 252 multisets of five dice, trying every kept subset, returns 0.046029 for Yahtzee and `(11/36)**2` gives 0.0934.

Convention. Dice and card follow the screenshot: three 5s held, two dice free. Fair dice, straights counted as runs, and best-play figures aim for one hand alone, ignoring the other categories.

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EYE COLOUR, INHERITANCE AND PROBABILITY ON ONE PAGE

Two-way tables meet Punnett squares

The same mathematics connects family observations to a model of inheritance.

1 / START WITH A TWO-WAY TABLE OF OBSERVATIONS

A cell combines two categories; a margin collects a row or column

An invented sample of 100 parent–child pairs, using only brown and blue categories. Choose one pair uniformly at random. Let $A =$ parent blue and $C =$ child blue.

	Child's eyes		
Parent's eyes	Brown	Blue	Total
Brown	36	24	60
Blue	4	36	40
Total	40	60	100

Divide every count by 100 to obtain a joint probability table.

Joint: both the parent and child have blue eyes.

$$\mathbb{P}(A \cap C) = \frac{36}{100} = 0.36.$$

Marginal: ignore the other classification.

$$\mathbb{P}(A) = 0.40, \qquad \mathbb{P}(C) = 0.60.$$

Conditional: restrict to the 40 blue-eyed parents.

$$\mathbb{P}(C \mid A) = \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(A)} = \frac{36}{40} = 0.90.$$

Here $0.36 \neq 0.40 \times 0.60$: these events are not independent for this random draw.

2 / REPLACE OBSERVED COLOURS WITH INHERITED ALLELES

A Punnett square is a joint table

An **allele** is a version of a gene; a child inherits one allele from each parent. In a one-gene **teaching model**, assign brown to BB, Bb and blue to bb: B is dominant over b. Genotype means the allele combination; phenotype means the observed trait.

For $Bb \times Bb$, an egg or sperm carries B or b, each with probability $\frac{1}{2}$. Assume independent gamete draws.

Egg \ sperm	B $(\frac{1}{2})$	b $(\frac{1}{2})$
B $(\frac{1}{2})$	BB $\frac{1}{4}$	Bb $\frac{1}{4}$
b $(\frac{1}{2})$	bB $\frac{1}{4}$	bb $\frac{1}{4}$

Bb and bB are the same genotype. Add their cells:

$$\mathbb{P}(BB) = \tfrac{1}{4}, \quad \mathbb{P}(Bb) = \tfrac{1}{2}, \quad \mathbb{P}(bb) = \tfrac{1}{4}.$$

The model gives $\frac{3}{4}$ brown, $\frac{1}{4}$ blue per child; four births need not produce a $3{:}1$ split.

3 / MAKE THE WEIGHTS EXPLICIT

Multiplication needs independence

Fix the parents' genotypes. Let p and q be the respective chances that the egg and sperm carry B. With independent draws:

	B (q)	b $(1-q)$
B (p)	pq	$p(1-q)$
b $(1-p)$	$(1-p)q$	$(1-p)(1-q)$

Cells are products. Row sums are $p, 1-p$; column sums are $q, 1-q$. All four cells sum to 1.

$$\mathbb{P}(\text{blue}) \;=\; (1-p)(1-q)$$

For $BB \times bb$ instead, $p = 1, q = 0$: every child is Bb, so the blue probability is 0.

Brown eyes alone do not distinguish BB from Bb. Genotypes set the weights.

4 / WHERE A FAMILIAR CHART GETS ITS FRACTIONS

Why 75%, 18.75% and 6.25%?

A **two-gene toy model** uses two imaginary gene positions (loci). Both parents are $BbGg$; assume the loci assort independently and assign colours by these rules:

Rule	Probability	Colour
At least one B	$\frac{3}{4} = 75\%$	brown
bb, at least one G	$\frac{1}{4} \times \frac{3}{4} = \frac{3}{16}$	green
$bbgg$	$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$	blue

The last two fractions are 18.75% and 6.25%. The 16 equally likely cells of a 4×4 Punnett square have been grouped into 12, 3 and 1.

These percentages assume this cross and these colour rules. Other brown-eyed parents can have different genotypes and different odds.

5 / KEEP ENOUGH BIOLOGY TO KNOW THE MODEL'S LIMITS

Real eye colour is polygenic

Eye colour depends largely on **melanin** in the iris. Brown eyes have more in its front layers; blue eyes have less. Colours vary continuously, often grouped as brown, green/hazel and blue.

OCA2 affects melanin; a region within **HERC2** regulates OCA2 activity. Other genes contribute too. OCA2 and HERC2 are neighbours on chromosome 15; nearby variants can be inherited together (**linkage**). Independent colour switches are an inadequate model.

Visible colour does not identify all these variants. Predictions need more genetic information and a model tested against data. Even two blue-eyed parents can occasionally have a brown-eyed child.

The mathematician's question: what are the outcomes, what are their weights, and which independence assumptions justify multiplying them?

One framework. Counts divided by a total give empirical probabilities; Punnett cells give model probabilities. Sum cells for combined outcomes; divide by the relevant margin to condition.

Biology sources. MedlinePlus Genetics: Is eye color determined by genetics?; NHGRI: Mendelian inheritance; NHGRI: Linkage.

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TWO CONSTANTS FROM RECIPROCALS ON ONE PAGE

Euler's γ and Catalan's G

One is what a sum of reciprocals leaves over its integral, the other is what alternating signs make of odd squares.

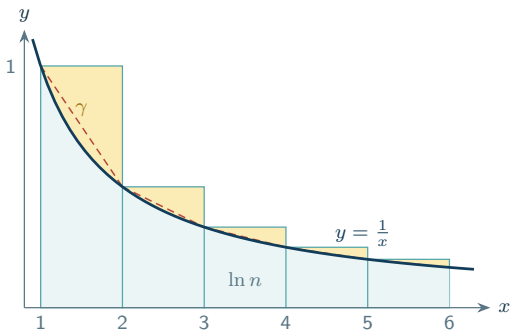
THE RULE

A sum minus its integral

The harmonic sum $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ grows without limit, and so does $\ln n = \int_1^n \frac{\mathrm{d}x}{x}$, but their difference settles down:

$$\gamma \;=\; \lim_{n\to\infty}\big(H_n - \ln n\big) \;=\; 0.577\,215\,664\,9\ldots$$

Each $\frac{1}{k}$ is a rectangle of width 1 on $[k, k+1]$ with the curve $y = \frac{1}{x}$ beneath its top, so γ is the total of the gold gaps; the gaps shrink fast enough for the total to converge, and the dashed chords of the trapezium rule split each one into a triangle and a thin sliver.



Rectangles total H_n, the area under the curve is $\ln n$, and the gold gaps add up to γ.

THE TRAPEZIUM RULE

Triangles give one half, slivers give the rest

The trapezium rule, the simplest Newton–Cotes rule, estimates $\int_a^b f\,\mathrm{d}x \approx h\big(\frac{1}{2}y_0 + y_1 + \cdots + y_{n-1} + \frac{1}{2}y_n\big)$; on a convex curve such as $\frac{1}{x}$ every chord lies above the curve, so each strip is overestimated by about $\frac{h^3}{12}f''$. With $h = 1$ on $\int_1^n \frac{\mathrm{d}x}{x}$ it reads

$$H_n - \tfrac{1}{2} - \tfrac{1}{2n} \;=\; \ln n + (\text{sliver area}), \qquad \text{so} \qquad \gamma = \underbrace{\tfrac{1}{2}}_{\text{triangles}} + \underbrace{0.0772\ldots}_{\text{slivers}},$$

and the error formula with $f'' = 2/x^3$ predicts slivers of about $\frac{1}{6}(7\zeta(3) - 8) \approx 0.069$. One more Euler–Maclaurin term gives $H_n \approx \ln n + \gamma + \frac{1}{2n} - \frac{1}{12n^2}$, and from $n = 10$ alone

$$H_{10} - \ln 10 - \tfrac{1}{20} + \tfrac{1}{1200} \;=\; 2.928968 - 2.302585 - 0.05 + 0.000833 \;=\; 0.577216,\ \gamma \text{ to six places}$$

ALTERNATING SIGNS

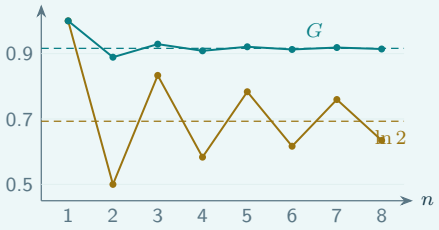
$\ln 2$ and Catalan's G from a geometric series

Alternating signs make reciprocals converge with no logarithm to subtract. Integrating $\frac{1}{1+x} = 1 - x + x^2 - \ldots$ from 0 to 1 gives $\ln 2$, and integrating $\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \ldots$ gives G:

$$\ln 2 \;=\; \int_0^1 \frac{\mathrm{d}x}{1+x} \;=\; 1 - \tfrac{1}{2} + \tfrac{1}{3} - \cdots \;=\; 0.693\,147\ldots$$

$$G \;=\; \int_0^1 \frac{\arctan x}{x}\,\mathrm{d}x \;=\; 1 - \tfrac{1}{3^2} + \tfrac{1}{5^2} - \cdots \;=\; 0.915\,965\ldots$$

Partial sums straddle the limit, so the error is less than the next term, and averaging two neighbours, the trapezium rule's move, does much better.



Partial sums zigzag about their limits; squares in the denominator damp G faster.

One family of reciprocal series

$1 + \frac{1}{2} + \frac{1}{3} + \ldots$	diverges, as $\ln n + \gamma$
$1 - \frac{1}{2} + \frac{1}{3} - \ldots$	$\ln 2$
$1 - \frac{1}{3} + \frac{1}{5} - \ldots$	$\pi/4$, Leibniz
$1 - \frac{1}{3^2} + \frac{1}{5^2} - \ldots$	G, Catalan
$1 + \frac{1}{3^2} + \frac{1}{5^2} + \ldots$	$\pi^2/8$

What is known

Both $\ln 2$ and π are transcendental, so the second, third and fifth limits are settled. Whether γ is even irrational is still open, and G had the same status until a 2026 preprint claimed a proof of its irrationality, which awaits the community's checking.

Conventions. γ is the Euler–Mascheroni constant, G is Catalan's constant (Eugène Catalan also names the Catalan numbers $1, 1, 2, 5, 14, \ldots$), H_n is the nth harmonic number and $\zeta(3) \approx 1.202$.

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APÉRY'S CONSTANT ON ONE PAGE

$\zeta(3)$ in place of $\zeta(2)$

The sum of reciprocal cubes, and what happens to Ramanujan's constant when π is rewritten through it.

THE RULE

The sum of reciprocal cubes, and Apéry's faster series

$$\zeta(3) \;=\; \sum_{n\ge1}\frac{1}{n^3} \;=\; \frac{5}{2}\sum_{n\ge1}\frac{(-1)^{n-1}}{n^3\binom{2n}{n}} \;=\; 1.202\,056\,903\,159\,594\ldots$$

Five terms of the plain sum leave an error of 0.016, while the central binomial coefficient $\binom{2n}{n}$, roughly quadrupling at each step, makes Apéry's series gain about 0.6 of a digit per term. Roger Apéry used series of this kind in 1978 to prove that $\zeta(3)$ is irrational, and Beukers later recast the proof through the integral

$$\zeta(3) = -\frac{1}{2}\int_0^1\!\!\int_0^1 \frac{\ln(xy)}{1-xy}\,\mathrm{d}x\,\mathrm{d}y.$$

n	$\sum 1/k^3$	Apéry's sum
1	1	1.25
2	1.125	1.197 9
3	1.162 0	1.202 55
4	1.177 7	1.201 99
5	1.185 7	1.202 068
∞		1.202 056 9 …

Five terms of each: the plain sum is still in the second decimal place.

EVEN VALUES ARE EASY

Odd values are hard

Euler showed that every even zeta value is a rational multiple of a power of π,

$$\zeta(2)=\frac{\pi^2}{6},\quad \zeta(4)=\frac{\pi^4}{90},\quad \zeta(6)=\frac{\pi^6}{945},$$

but no such form is known for any odd value. Nobody knows whether $\zeta(3)/\pi^3 \approx 0.038\,77$ is rational, whether $\zeta(5)$ is irrational, or whether $\zeta(3)$ is transcendental; it is known that infinitely many odd values are irrational and that at least one of $\zeta(5),\zeta(7),\zeta(9),\zeta(11)$ is.

WHERE IT APPEARS

Coprime triples and black-body light

Three whole numbers chosen at random share no common prime factor with probability $1/\zeta(3) \approx 0.832$, the cube analogue of $6/\pi^2$ for pairs. In physics, a black body at temperature T holds

$$n_\gamma = \frac{2\,\zeta(3)}{\pi^2}\left(\frac{kT}{\hbar c}\right)^3$$

photons per unit volume, which is about 411 per cubic centimetre in the cosmic microwave background; $\zeta(3)$ also appears in the electron's magnetic moment.

THE EXPERIMENT

Rewrite π through $\zeta(3)$ and leave 163 alone

In $\mathcal{R} = \mathrm{e}^{\sqrt{978\,\zeta(2)}}$ the factor 978 is 6×163, and the 6 belongs to $\zeta(2)$ through $\pi^2 = 6\,\zeta(2)$. The faithful rewrite for $\zeta(3)$ uses $\pi^3 = \kappa_3\,\zeta(3)$ with $\kappa_3 = \pi^3/\zeta(3)$, which changes the formula without changing the number:

$$\mathcal{R} \;=\; \mathrm{e}^{\sqrt{163}\,(\kappa_3\,\zeta(3))^{1/3}}, \qquad \kappa_3 \;=\; 25.794\,350\,166\,618\,684\ldots$$

So 640320^3 survives untouched, and the short form now runs backwards to give $\kappa_3 \approx \left[\ln(640320^3+744)\right]^3/\left(163^{3/2}\zeta(3)\right)$, correct to 29 decimal places. The table shows the reverse dependence: each further digit of κ_3 pins down more of 640320.

κ_3 used	$\sqrt[3]{\mathcal{R}-744}$
25.79	639 838.89
25.794	640 281.26
25.79435	640 319.98
exact	640 320

Truncating κ_3 rebuilds a slightly wrong 640320.

WHY $\zeta(2)$ GAVE A WHOLE NUMBER

Even κ_s are rational, odd ones are unexplained

For every even s Euler's formula makes $\kappa_s = \pi^s/\zeta(s)$ rational, so Ramanujan's constant has a family of exact rewrites,

$$\mathcal{R} = \mathrm{e}^{\sqrt{163}\,(6\,\zeta(2))^{1/2}} = \mathrm{e}^{\sqrt{163}\,(90\,\zeta(4))^{1/4}} = \mathrm{e}^{\sqrt{163}\,(945\,\zeta(6))^{1/6}},$$

each as clean as the original. For odd s nobody knows whether κ_s is rational; if κ_3 were, $\zeta(3)$ would be a rational multiple of π^3, and nothing points that way. Ramanujan's constant can therefore carry $\zeta(3)$ only numerically, through the unexplained $25.794\ldots$, while the integer 640320^3 is fixed by π and 163 alone.

s	$\kappa_s = \pi^s/\zeta(s)$	
2	6	rational
3	25.794 350 …	unknown
4	90	rational
5	295.121 509 …	unknown
6	945	rational
7	2995.284 764 …	unknown

The factor that converts $\zeta(s)$ into a power of π.

Conventions. $\kappa_s = \pi^s/\zeta(s)$, so that $\pi = (\kappa_s\zeta(s))^{1/s}$ for every $s \ge 2$; all figures were computed to 50 digits.

MARKS, AVERAGES AND RANKINGS ON ONE PAGE

Do the judges agree?

A skating-inspired route from combining scores to comparing orders.

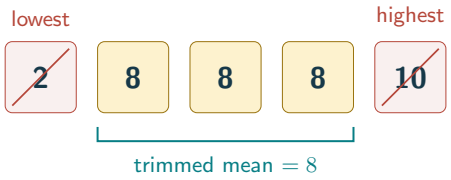
1 / TURN SEVERAL OPINIONS INTO ONE RESULT

The mean, the middle and the trimmed mean

Classroom case: five judges give one overall mark out of 10. All scores on this page are invented for teaching.

The **mean** uses every mark: $(2+8+8+8+10)/5 = 7.2$. The **median** is the middle mark in the sorted list: 8.

A **trimmed mean** removes one lowest and one highest mark, then averages the remaining three: $(8+8+8)/3 = 8$.



Skating provides a real example: the ISU discards the highest and lowest judges' marks for each element or programme component before averaging. This limits the influence of an isolated extreme mark.

2 / COMPARE THE JUDGES, RATHER THAN COMBINE THEM

Different marks can express the same preference

Now three judges assess four routines. Read down a column to see one judge's view; higher marks mean a better performance.

routine	Judge 1	Judge 2	Judge 3
A	6	4	6
B	7	5	8
C	8	6	7
D	9	7	9
mean mark	7.5	5.5	7.5

Judge 2 gives exactly two marks fewer than Judge 1 every time, yet both prefer D to C to B to A. Judge 3 has the same average mark as Judge 1, but prefers B to C. **Overall strictness and agreement about order are different questions.**

3 / ONE SIMPLE NORMALISATION: CENTRING

Put each judge's average at zero

Subtract each judge's mean from *each* of their marks, choosing that mean as their new zero. The means above are 7.5, 5.5 and 7.5:

$$\begin{aligned} \text{Judges 1 and 2:} &\quad (-1.5, -0.5,\ \ 0.5,\ \ 1.5) \\ \text{Judge 3:} &\quad (-1.5,\ \ 0.5, -0.5,\ \ 1.5) \end{aligned}$$

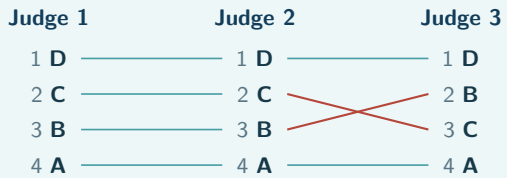
Routine A gives $6 - 7.5 = 4 - 5.5 = -1.5$: Judges 1 and 2 put it 1.5 marks below their own average.

Every new list has mean 0: the judges now share a reference level. Use this when a uniform difference in strictness should be set aside. Marks keep their units and each judge's ordering is preserved; Judge 3's B–C disagreement remains.

4 / KEEP ONLY THE ORDER

Ranks reveal two ways of describing agreement

Give the best routine rank 1, the next rank 2, and so on. Use the same convention for every judge. Ranks retain order while discarding the size of the gaps between marks.



Judges 1 and 2 agree on every position and every pair. Compare Judges 1 and 3:

Positions: B and C each move one place.

Pairs: of $3+2+1 = 6$ pairs, only B–C reverses; the other five agree. The crossing marks that reversal.

Next: Spearman's ρ summarises differences in rank positions; Kendall's τ summarises agreement and reversal between pairs.

5 / AGREEMENT AND FAIRNESS

Trimming can limit extreme scores; centring can remove a uniform offset; ranks can compare preferences. None establishes the correct judgement. A panel can agree and still share a bias or overlook the same fault. Fairness also needs clear criteria and independent evidence.

Real-world anchor: ISU figure-skating scoring explainer (2024). Further reading: Carroll, Rykken and Sorensen, *The Canadians Should Have Won!?*, Math Horizons (2003).

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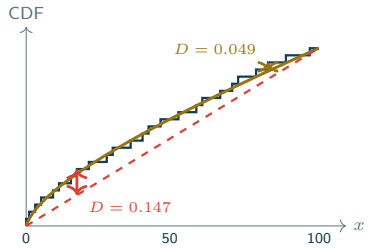
STATISTICS ON ONE PAGE

Means and the Kolmogorov distance

A mean compares one number, and the largest gap between two cumulative curves compares whole distributions.

1 / THE RULE AND THE PICTURE

The distance to a model is the largest vertical gap between the two cumulative curves



staircase: primes; gold: Li; red dashed: uniform

Let $F_n(x)$ be the fraction of the n primes up to X that are at most x, and $G(x)$ the model's cumulative distribution. The Kolmogorov and Smirnov distance is

$$D = \sup_x |F_n(x) - G(x)|$$

Here the model $G(x) = \mathrm{Li}(x)/\mathrm{Li}(X)$, from the thinning density $1/\ln t$, has $D = 0.049$, while a model with no thinning, $G = x/X$, has $D = 0.147$.

2 / WHAT MEANS CAN SEE

Means compare level, and for the primes they agree closely

up to x	mean gap observed	$\ln x - 1$	$\ln x$	mean of $g/\ln p$
10^3	5.958	5.908	6.908	1.0651
10^5	10.425	10.513	11.513	1.0041
10^7	15.047	15.118	16.118	1.0005

The mean gap below x is $x/\pi(x)$, which matches $\ln x - 1$, the average of the local gap $\ln t$ over $t \le x$, and the mean of the scaled gaps $g/\ln p$ is 1.0005 at 10^7. A mean also reveals a level error: the sieve estimate $2e^{-\gamma}\mathrm{Li}(x)$ overshoots the count by 12%, since $\pi/(2e^{-\gamma}\mathrm{Li}) \to 0.890$.

3 / WHAT THE DISTANCE ADDS

It is non-parametric, and for the primes it measures a gap and does not test a hypothesis

The distance D needs no assumption about shape, is unchanged by any monotone rescaling of x, and lies between 0 and 1; against the uniform law it is the star discrepancy of number theory. A classical significance test would ask how often n random draws from the model give so large a D, with $\sqrt{n}\,D$ following the Kolmogorov distribution (mean 0.869, 5% point 1.358), but the primes are one fixed set and not a sample, successive gaps are dependent, the gaps are even integers, and with a huge n every real deviation would be significant.

4 / POSITIONS: THE THINNING MODEL AGAINST THE PRIMES

The Li model fits the shape within 2×10^{-4} by 10^7

up to X	$n = \pi(X)$	D uniform	D Li model	$\sqrt{n}\,D_{\mathrm{Li}}$
10^3	168	0.0815	0.02292	0.30
10^5	9 592	0.0396	0.00260	0.25
10^7	664 579	0.0265	0.00023	0.19

The uniform model improves only slowly, as the thinning weakens like $1/\ln X$, while the Li model's D falls roughly like $1/\sqrt{X}$.

5 / GAPS: THE EXPONENTIAL MODEL AGAINST THE PRIMES

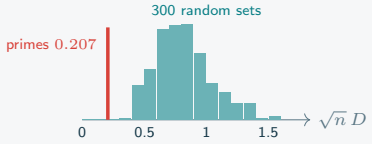
The means agree, but the distributions do not

primes between	gaps	mean of $g/\ln p$	variance	D against $\mathrm{Exp}(1)$
10^3 to 10^4	1 060	1.0043	0.515	0.195
10^4 to 10^5	8 362	1.0024	0.582	0.172
10^5 to 10^6	68 905	1.0013	0.646	0.151
10^6 to 10^7	586 080	1.0004	0.694	0.138

The exponential model has mean and variance 1, and the scaled gaps have a mean of 1.0004 but a variance of only 0.69, narrower than the model, and $D = 0.138$. At least 0.117 of that comes from the floor: no gap is shorter than 2, so no scaled gap is below $2/\ln 10^7 = 0.124$, where the model has already placed 11.7% of its mass.

6 / A MEANINGFUL TEST / AGAINST THE RANDOM MODEL

Against Cramér's random model the comparison has a genuine null distribution



In 300 sets up to 10^6, each including every integer n with probability $1/\ln n$, the distance to the Li curve has median 0.0029 and $\sqrt{n}\,D$ has median 0.80, but the actual primes give $D = 0.00074$ and $\sqrt{n}\,D = 0.207$, below all 300 simulations. Against the random model the verdict is that the primes are too regular to be typical draws, and this is a legitimate statement because the simulation supplies the sampling variability that the primes themselves lack.

ASIDE / RELATIVES

Cramér and von Mises integrate the squared gap, Anderson and Darling weight the tails, and Kuiper adds largest gap on each side.

LINKS TO OTHER SHEETS

The models compared here are in *How the primes thin out* and *Three attenuators*; the Cramér random model is described in first.

Convention. F_n empirical CDF; G model CDF; D Kolmogorov and Smirnov distance; $\mathrm{Exp}(1)$ the exponential law with mean 1; g is the gap after prime p.

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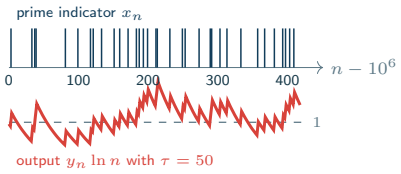
SIGNAL ANALYSIS ON ONE PAGE

Primes as a filtered signal

A low-pass filter turns the prime indicator into a density, and the noise left over measures how regular the primes are.

1 / THE RULE AND THE PICTURE

The RC low-pass $\tau y' + y = x$ averages the past exponentially, turning pulses into a density



the 34 primes from 10^6 to $10^6 + 420$, and the filtered signal

The circuit has corner $f_c = 1/(2\pi\tau)$, gain $1/\sqrt{1+(f/f_c)^2}$ and phase lag $\arctan(f/f_c)$, and its impulse response $\frac{1}{\tau}e^{-t/\tau}$ has area 1. The sampled version, applied to the prime indicator x_n (1 if n is prime, else 0), is

$$y_n = (1-a)\,y_{n-1} + a\,x_n, \qquad a = \tfrac{1}{\tau}$$

so y_n is primes per integer, about $1/\ln n$ once τ exceeds the mean gap $\ln n$. The picture scales y_n by $\ln n$ so that its mean is 1.

2 / THE PARTS OF THE PRIME SIGNAL

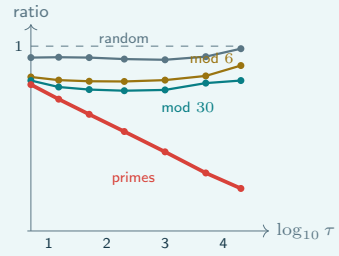
Smooth density, periodic template and irregular remainder sit in different frequency bands

part	where in frequency	what the filter does
smooth density $1/\ln n$	zero frequency	passes unchanged, gain 1
sieve template	lines at $f = \frac{1}{6}, \frac{1}{3}, \frac{1}{2}$	each line has strength equal to the mean; at $\tau = 50$ the gains are 0.019, 0.0095, 0.0064
irregular remainder	broad band	passes in proportion to $\lvert H\rvert^2$, the noise power

Every prime above 3 is ± 1 modulo 6, so the indicator contains the period-6 pattern $0, 1, 0, 0, 0, 1$, a pulse train whose Fourier lines are its harmonics, exactly as the harmonics of a square wave, and a filter with $\tau \gg 6$ removes them.

3 / THE NOISE LEVEL / REGULAR AGAINST IRREGULAR

For random events the filtered variance has a known baseline, and the primes fall below it as τ grows



variance of $y_n \ln n$ divided by $\ln n/(2\tau - 1)$

Independent events of density $p = 1/\ln n$ through this filter give $\mathrm{Var}(y_n \ln n) \approx \ln n/(2\tau)$, the baseline. Over $n = 2\times10^6$ to 10^7 a random set with that density follows it (ratio 0.94), sets that respect the mod 6 and mod 30 templates sit lower at 0.81 and 0.76 and stay flat, while the primes give 0.79 at $\tau = 5$, 0.63 at 50, 0.54 at 200, 0.43 at 10^3, 0.31 at 5×10^3 and 0.23 at 2×10^4.

4 / HOW TO READ IT

The short-scale deficit is arithmetic, and the long-scale deficit is rigidity

At $\tau \approx 5$ the primes (0.79) are close to the template-aware controls (0.81 to 0.83), so the extra regularity at short scales is the divisibility pattern. Beyond that the controls stay flat while the primes keep falling, so the filtered variance, which equals $\int S(f)|H(f)|^2 df$, shows that the power spectrum S of the primes is suppressed at low frequency: at $\tau = 2\times10^4$ the corner is $f_c = 8\times10^{-6}$, a wavelength of about 8000 mean gaps, and the spectrum at and below that frequency averages only about 0.23 of the white level.
This is the rigidity that the Kolmogorov distance saw from the cumulative side, now resolved by scale.

5 / CHOICES AND CAUTIONS

Measure τ in mean gaps, keep the complement, and respect the statistics

Choosing τ as a multiple of $\ln n$ keeps the comparison the same across ranges, and $x_n - y_n$ is the high-pass complement that holds the irregular part.
A cascade of two filters rolls off more sharply, and the window must be much longer than τ: at $\tau = 2\times10^4$ the window of 8×10^6 holds only about 400 independent stretches, so that ratio is uncertain by roughly 10%, and each control is a single random realisation.

ASIDE / FREQUENCY VERSUS CUMULATIVE

The Kolmogorov distance compares cumulative curves and the filter compares power at each scale, so the two views agree and the filter locates where the regularity lives.

LINKS TO OTHER SHEETS

The filter and the harmonics are in *Waves, superposition and the square wave*; the cumulative comparison is in *Means and the Kolmogorov distance*; the density is in *How the primes thin out*.

Convention. x_n prime indicator; y_n filtered signal; τ time constant in integers; f_c corner frequency; H filter response; S power spectrum; controls use probabilities $1/\ln n$ on all integers, $3/\ln n$ on residues ±1 mod 6, and $3.75/\ln n$ on residues coprime to 30.

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TWO CLASSIFICATIONS OF THE SAME PRIMES ON ONE PAGE

Primes in a two-way table

Regularity meets splitting, inertness and ramification.

1 / ENLARGE THE INTEGERS

New numbers make new factors available

$$\underbrace{a+bi}_{\text{Gaussian integers}}, \quad \mathrm{i}^2=-1; \qquad \underbrace{a+b\omega}_{\text{Eisenstein integers}}, \quad \omega=\frac{-1+\mathrm{i}\sqrt{3}}{2}.$$

Here a, b are ordinary integers, and $\omega^2+\omega+1=0$. These are written $\mathbb{Z}[\mathrm{i}]$ and $\mathbb{Z}[\omega]$. Both retain unique factorisation, but the available primes change.

2A / SPLITS

Two different primes

An ordinary prime acquires two different prime factors:

$$5=(2+\mathrm{i})(2-\mathrm{i}).$$

So 5 **splits** in $\mathbb{Z}[\mathrm{i}]$.

2B / IS INERT

Stays prime

An ordinary prime remains prime in the enlarged integer system:

$$3 \text{ stays prime in } \mathbb{Z}[\mathrm{i}].$$

So 3 is **inert** there.

2C / RAMIFIES

A prime repeats

The same prime factor occurs twice, allowing a rotation:

$$2=-\mathrm{i}(1+\mathrm{i})^2.$$

So 2 **ramifies** in $\mathbb{Z}[\mathrm{i}]$.

An invertible rotation in these systems is called a **unit**. Prime factors related by a unit count as the same prime. For example, $1-\mathrm{i}=-\mathrm{i}(1+\mathrm{i})$, explaining the repeated factor in 2.

3 / PREDICT THE BEHAVIOUR FROM A REMAINDER

The rules depend on which integers we allow

For a rational prime p, division by 4 or 3 gives these rules; k is an integer.

Gaussian integers $\mathbb{Z}[\mathrm{i}]$

$p=4k+1$	splits
$p=4k+3$	is inert
$p=2$	ramifies

Eisenstein integers $\mathbb{Z}[\omega]$

$p=3k+1$	splits
$p=3k+2$	is inert
$p=3$	ramifies

Eisenstein examples: $7=(3+\omega)(2-\omega)$ and $3=-\omega^2(1-\omega)^2$. Thus 3 stays prime in the Gaussian system and ramifies in the Eisenstein one.

4 / PAIR THE TWO CLASSIFICATIONS

Rows record regularity; columns record factorisation

Recall: an odd prime p is **regular** when none of the numerators of $B_2, B_4, \ldots, B_{p-3}$, in lowest terms, is divisible by p. Otherwise it is **irregular**.

Use **all 24 odd primes below 100**. Only $37, 59, 67$ are irregular. Every prime enters exactly one cell in each table; each cell counts primes with *both* its row and column labels.

Gaussian integers

	split	inert	ramified	total
regular	10	11	0	21
irregular	1	2	0	3
total	11	13	0	24

Eisenstein integers

	split	inert	ramified	total
regular	9	11	1	21
irregular	2	1	0	3
total	11	12	1	24

Place 67: it is irregular. Since $67=4\cdot16+3$, it enters Gaussian *irregular/inert*; since $67=3\cdot22+1$, it enters Eisenstein *irregular/split*. The Gaussian ramification column is zero because its only ramified prime, 2, is outside our set. The Eisenstein ramified entry is 3, which is regular.

5 / READ A CELL, THEN RESTRICT A ROW

The denominator tells us which group we are considering

Choose one of these 24 primes uniformly. The **Eisenstein table** gives 11 regular and inert primes; 2 of its 3 irregular primes split:

$$P(\text{regular and inert})=\frac{11}{24}, \qquad P(\text{split} \mid \text{irregular})=\frac{2}{3}.$$

The bar means **given**: divide by the irregular row total, 3. Irregularity alone does not determine factorisation: 37 splits while 59 is inert.

Sources. J. S. Milne, *Algebraic Number Theory*, §3 (prime factorisation in quadratic fields); O. Knill, *Eisenstein Primes* (Eisenstein classification); The Bernoulli Number Page (regularity and the first exceptions). Counts and worked examples calculated for this sheet.

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RAMANUJAN'S CONSTANT AND THE ZETA FUNCTION ON ONE PAGE

The Basel sum hidden in $e^{\pi\sqrt{163}}$

Writing π through $\zeta(2)$ turns Ramanujan's near-integer into thirty digits of the Basel sum.

THE RULE

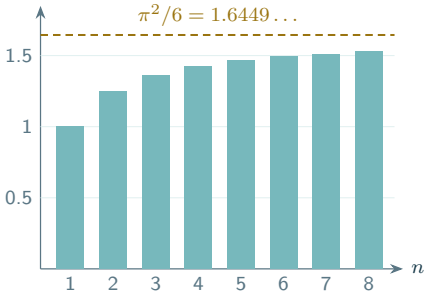
Replace π by the value of a zeta function at 2

Euler's solution of the Basel problem, and Riemann's completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ evaluated at the same point, give

$$\zeta(2) = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots = \frac{\pi^2}{6}, \qquad \xi(2) = \frac{\pi}{6},$$

so $\pi = \sqrt{6\,\zeta(2)} = 6\,\xi(2)$. Substituting into Ramanujan's constant moves $\zeta(2)$ inside the square root, or brings $\xi(2)$ out of it entirely:

$$\mathcal{R} \;=\; e^{\pi\sqrt{163}} \;=\; e^{\sqrt{978\,\zeta(2)}} \;=\; e^{6\sqrt{163}\,\xi(2)}$$



Partial sums of $\sum 1/n^2$ creep up towards Euler's $\pi^2/6$.

THE SHORT FORM OF RAMANUJAN'S CONSTANT

Thirty digits of π and of $\zeta(2)$ from one logarithm

Since $\mathcal{R}$ falls short of $640320^3 + 744$ by only 7.5×10^{-13}, a relative error of 3×10^{-30}, taking logarithms of the short form $\mathcal{R} \approx 640320^3 + 744$ loses almost nothing:

$$\zeta(2) \;\approx\; \frac{\left[\ln(640320^3+744)\right]^2}{978}, \qquad \xi(2) \;\approx\; \frac{\ln(640320^3+744)}{6\sqrt{163}}, \qquad \pi \;\approx\; \frac{\ln(640320^3+744)}{\sqrt{163}}$$

short form	$3.141\,592\,653\,589\,793\,238\,462\,643\,383\,279\,726\ldots$
true π	$3.141\,592\,653\,589\,793\,238\,462\,643\,383\,279\,502\ldots$

The two agree to thirty decimal places, and the expression for $\zeta(2)$ matches $1.644\,934\,066\,848\ldots$ just as far. The integer 640320^3 is $-j\left(\frac{1+i\sqrt{163}}{2}\right)$, so the digits of the Basel sum are being read off a value of the j-function.

FROM SQUARES TO PRIMES

Euler's product puts the primes inside ζ

For $s > 1$, unique factorisation turns the sum into a product over primes,

$$\zeta(s) = \sum_{n\geq 1} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}, \qquad \text{so} \qquad \frac{\pi^2}{6} = \frac{4}{3}\cdot\frac{9}{8}\cdot\frac{25}{24}\cdot\frac{49}{48}\cdots,$$

where primes up to 29 already give 1.633, and $6/\pi^2 \approx 0.608$ is the chance that two random whole numbers share no prime factor.

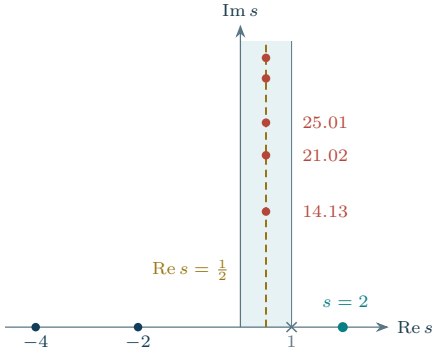
THE RIEMANN HYPOTHESIS

The zeros of ξ all lie on one line

Riemann extended $\zeta(s)$ to complex s in 1859; it vanishes at the trivial zeros $-2, -4, \ldots$ and otherwise only in the strip $0 < \operatorname{Re} s < 1$. The completed ξ, the function that gave $\xi(2) = \pi/6$, cancels the pole and the trivial zeros and satisfies $\xi(s) = \xi(1-s)$, so its zeros are exactly the non-trivial zeros of ζ.

Riemann hypothesis: every zero of $\xi(s)$ has real part $\frac{1}{2}$.

The first zeros sit at heights 14.1347, 21.0220 and 25.0109, and more than 10^{13} have been checked. If it holds, the count of primes up to x differs from $\int_2^x \frac{dt}{\ln t}$ by at most about $\sqrt{x}\,\ln x$.



Trivial zeros, the pole at 1, Basel's $s = 2$ and the first zeros on the critical line.

Back to 163. The generalised hypothesis concerns the zeta functions of fields such as $\mathbb{Q}(\sqrt{-163})$; Hecke, Deuring and Heilbronn showed that the class numbers of $\mathbb{Q}(\sqrt{-d})$ grow without limit whether or not it is true, which is why the list of Heegner numbers, and of near-integers like $\mathcal{R}$, had to be finite. Here Γ is the gamma function, with $\Gamma(1) = 1$.

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SPEARMAN'S RHO AND KENDALL'S TAU ON ONE PAGE

How closely do two orders agree?

Start with ranks, define two kinds of agreement, then calculate both for four bricks.

1 / REPLACE MEASUREMENTS BY POSITIONS

For $n \geq 2$ objects, measure two quantities x_i, y_i and rank each list: r_i is the rank of x_i, s_i the rank of y_i. **Rank 1 means smallest. Begin with no ties.** Both coefficients lie in $[-1, 1]$: $+1$ means the same order and -1 the reverse order. Near 0 means little monotonic agreement; other dependence can remain. An increasing change of measurement preserves both scores.

2A / SPEARMAN'S ρ

Correlate the rank positions

Let $\bar{r}, \bar{s}$ be the mean ranks. Spearman's coefficient is the Pearson correlation of the two rank lists:

$$\rho = \frac{\sum_i (r_i - \bar{r})(s_i - \bar{s})}{\sqrt{\sum_i (r_i - \bar{r})^2 \, \sum_i (s_i - \bar{s})^2}}.$$

Without ties, each rank list has mean $(n+1)/2$ and sum of squared deviations $A = n(n^2-1)/12$. Expanding with $d_i = r_i - s_i$ gives

$$\sum_i d_i^2 = 2A(1-\rho),$$

so the definition becomes the familiar shortcut:

$$\rho \quad = \quad 1 \; - \; \frac{6 \sum_i d_i^2}{n(n^2-1)}$$

Large rank displacements contribute through their **squares**.

2B / KENDALL'S τ

Average agreement between pairs

Choose two objects i, j and compare their order in the two measurements. The pair is **concordant** if $(x_i - x_j)(y_i - y_j) > 0$, and **discordant** if the product is negative.

Give an agreeing pair $+1$ and a reversed pair -1. Let P, Q count these pairs. There are $M = \binom{n}{2} = n(n-1)/2$ pairs altogether, so their average sign is

$$\tau \quad = \quad \frac{P - Q}{M} \quad = \quad 1 \; - \; \frac{2Q}{M}$$

Every reversed pair counts equally. If a pair is chosen uniformly from the observations, its chance of agreement is

$$\frac{P}{M} = \frac{1+\tau}{2}.$$

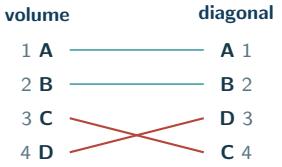
Order observations by x. Then Q is the **inversion count** of the y ranks, equal to the fewest adjacent swaps needed to sort them.

3 / A NUMBER-THEORY EXAMPLE: FOUR PRIME-SIDED CUBOIDS

Does greater volume usually mean a longer diagonal?

For edges p, q, t, volume is $V = pqt$ and the body diagonal satisfies $D^2 = p^2 + q^2 + t^2$. Ranking D^2 gives the same order as ranking D.

	edges	V	D^2	r_i	s_i	d_i	d_i^2
A	$(2,3,5)$	30	38	1	1	0	0
B	$(2,3,7)$	42	62	2	2	0	0
C	$(2,3,11)$	66	134	3	4	-1	1
D	$(2,5,7)$	70	78	4	3	1	1



Only C–D reverses its order, so $P = 5$, $Q = 1$ out of 6 pairs. Also $\sum d_i^2 = 2$. Substitution gives

$$\rho \quad = \quad 1 \; - \; \frac{6 \cdot 2}{4(4^2-1)} \quad = \quad \frac{4}{5} \quad = \quad 0.8, \qquad \tau \quad = \quad \frac{5-1}{6} \quad = \quad \frac{2}{3}.$$

Here $(1+\tau)/2 = 5/6$ is the fraction of pairs agreeing on their order.

4 / WHY THE TWO ANSWERS DIFFER

The same number of reversals can have different rank distances

Keep the first ranking $(1, 2, 3, 4, 5)$ and compare these two second rankings:

second ranking	$\sum d_i^2$	Q	ρ	τ
$(1, 3, 2, 5, 4)$	4	2	$4/5$	$3/5$
$(1, 2, 5, 3, 4)$	6	2	$7/10$	$3/5$

Kendall gives the same score because each list reverses two pairs. Spearman distinguishes their squared displacements.

5 / CHOOSE THE QUESTION, THEN THE COEFFICIENT

Use **Spearman** for agreement of rank positions; use **Kendall** for the balance of pairwise agreements and reversals. Their numerical scales have different interpretations, so a larger coefficient alone does not make its method better.

With ties: give equal measurements their average rank (for example, $2, 2, 7$ gives ranks $1.5, 1.5, 3$) and use the Pearson definition for ρ. For Kendall, use $\tau_b = (P-Q)/\sqrt{(P+Q+T_r)(P+Q+T_s)}$, where T_r, T_s count pairs tied only in the first or only in the second list; pairs tied in both enter neither count. Both coefficients require variation in each variable.

Definitions and ties: SciPy documentation, Spearman correlation and Kendall correlation. All examples calculated from the displayed values.

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THE CUBIC AND ITS DISCRIMINANT ON ONE PAGE

Cardano, Δ, and a near-integer from Apéry

One sign decides how many real roots a cubic has, and two cubic-flavoured numbers lead back to almost-integers.

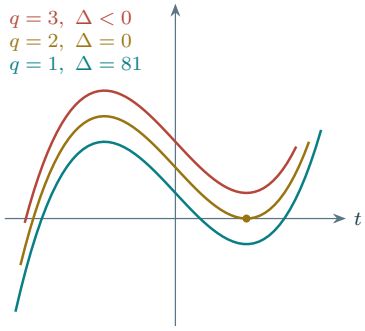
THE RULE

The depressed cubic and Cardano's formula

Every cubic becomes $t^3+pt+q=0$ after a shift of x by $\frac{b}{3a}$. Writing $t=u+v$ with $uv=-\frac{p}{3}$ turns it into a quadratic in u^3, whose own discriminant appears under the square root:

$$t=\sqrt[3]{-\frac{q}{2}+\sqrt{\frac{q^2}{4}+\frac{p^3}{27}}}+\sqrt[3]{-\frac{q}{2}-\sqrt{\frac{q^2}{4}+\frac{p^3}{27}}},\qquad \Delta=-4p^3-27q^2$$

Since $\Delta=-108\left(\frac{q^2}{4}+\frac{p^3}{27}\right)$, the sign of Δ is read straight from the square root: $\Delta>0$ gives three distinct real roots, $\Delta=0$ a repeated root, and $\Delta<0$ one real root with two complex partners.



$y=t^3-3t+q$: three crossings, a touch at $t=1$, then a single crossing.

THREE REAL ROOTS

Cosines rescue Cardano

When $\Delta>0$ the square root in Cardano's formula is imaginary even though all three roots are real, the casus irreducibilis that forced complex numbers into algebra. The identity $4\cos^3\theta-3\cos\theta=\cos 3\theta$ avoids them: for $t^3-3t+1=0$ put $t=2\cos\theta$, so $\cos 3\theta=-\frac{1}{2}$ and

$$t=2\cos 40^\circ,\ 2\cos 160^\circ,\ 2\cos 280^\circ=1.532,\ -1.879,\ 0.347.$$

Here $\Delta=81=9^2$ is a perfect square, the sign that the three roots are rational expressions in one another.

THE GENERAL CASE

The discriminant of ax^3+bx^2+cx+d

Before depressing the cubic,

$$\Delta=b^2c^2-4ac^3-4b^3d-27a^2d^2+18abcd,$$

and with roots r_1,r_2,r_3 it equals $a^4\prod_{i<j}(r_i-r_j)^2$, the squared product of the gaps between roots. That form explains the rule: a repeated root makes a gap zero, and a complex pair makes one squared gap negative. For a quadratic the same recipe gives the familiar b^2-4ac.

A CUBIC OF DISCRIMINANT -23

The plastic number and a Ramanujan-like identity

The cubic $x^3=x+1$ has $p=q=-1$, so $\Delta=4-27=-23$ and one real root, the plastic number:

$$\rho=\sqrt[3]{\tfrac{1}{2}+\sqrt{\tfrac{23}{108}}}+\sqrt[3]{\tfrac{1}{2}-\sqrt{\tfrac{23}{108}}}=1.324\,717\,957\ldots$$

Because $\mathbb{Q}(\sqrt{-23})$ has class number 3 and ρ generates the cubic extension that repairs its factorisation, Weber's modular function satisfies $\mathfrak{f}(\sqrt{-23})=\sqrt{2}\,\rho$ exactly, and expanding $\mathfrak{f}^{24}=e^{\pi\sqrt{23}}+24+\ldots$ gives a relation of the same kind as $e^{\pi\sqrt{163}}\approx 640320^3+744$:

$$e^{\pi\sqrt{23}}\quad\approx\quad 2^{12}\rho^{24}\quad-\quad 24,\qquad \text{error } 7.9\ \times\ 10^{-5}$$

A SUGGESTED BUILD FROM APÉRY

Apéry's growth rate makes near-integers

Apéry's numbers $1,5,73,1445,33001,\ldots$ satisfy $n^3u_n=(34n^3-51n^2+27n-5)u_{n-1}-(n-1)^3u_{n-2}$, whose leading terms give $x^2-34x+1=0$ and the growth rate

$$\alpha=17+12\sqrt{2}=(1+\sqrt{2})^4\approx 33.97.$$

The irrationality proof turns on $\alpha>e^3\approx 20.09$: Apéry's approximations make $q_n\zeta(3)-p_n$ shrink like α^{-n}, clearing denominators costs about e^{3n}, and so non-zero integer combinations of 1 and $\zeta(3)$ become arbitrarily small, which a rational $\zeta(3)$ would forbid. The conjugate $17-12\sqrt{2}$ equals $1/\alpha$, so $\alpha^n+\alpha^{-n}$ is always a whole number and α^n falls short of one by exactly α^{-n}, a Ramanujan-like family with the shortfall known in advance. A modular version would evaluate Beukers' level-6 parametrisation of Apéry's numbers at a CM point, the natural next investigation.

n	$\alpha^n=(17+12\sqrt{2})^n$
3	39 201.999 974 5
4	1 331 713.999 999 25
5	45 239 073.999 999 978
6	1 536 796 801.999 999 999 35

Each shortfall is α^{-n}; at $n=6$ it is 6.5×10^{-10}.

Conventions. $\mathfrak{f}(\tau)=q^{-1/48}\prod_{n\geq 1}(1+q^{n-1/2})$ with $q=e^{2\pi i\tau}$ is Weber's function, and Apéry's numbers are $u_n=\sum_k\binom{n}{k}^2\binom{n+k}{k}^2$.

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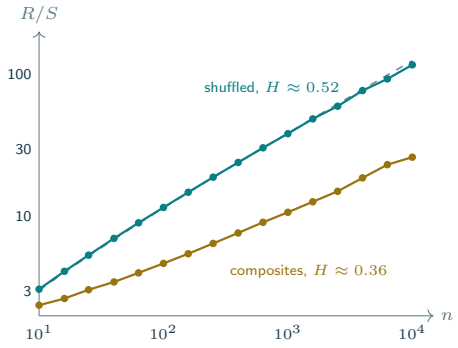
DATA ON ONE PAGE

The Hurst exponent of the composites

Rescaled range analysis asks whether a series wanders further than chance, and the residues of the composites mod 4 stay close to home.

1 / THE RULE AND THE PICTURE

Split the series into blocks of n, and the average rescaled range grows like n^H



R/S against n, log scales, dashed for random data

For a block of n values subtract the block mean, form the running totals Z_t of the deviations, take the range $R = \max Z - \min Z$ and divide by the block's standard deviation S; the average of R/S over blocks then behaves like a power of n,

$$\overline{R/S} \approx c n^H$$

and the slope of the log-log plot is the Hurst exponent. A value near 0.5 is a random series, above 0.5 the series persists in its direction, and below 0.5 it keeps turning back, as the composites do in the picture.

2 / A WORKED BLOCK: THE FIRST TWELVE COMPOSITES

Residues mod 4 with their running totals give $R/S = 2.73$

composite	4	6	8	9	10	12	14	15	16	18	20	21
residue	0	2	0	1	2	0	2	3	0	2	0	1
deviation	−1.08	0.92	−1.08	−0.08	0.92	−1.08	0.92	1.92	−1.08	0.92	−1.08	−0.08
running total Z	−1.08	−0.17	−1.25	−1.33	−0.42	−1.50	−0.58	1.33	0.25	1.17	0.08	0.00

The mean residue is 1.083, the running totals range from -1.50 to 1.33, so $R = 2.83$, and the standard deviation is $S = 1.04$, giving $R/S = 2.83/1.04 = 2.73$ for $n = 12$. The totals never wander far because the residues keep pulling each other back.

3 / WHAT THE COMPOSITES MOD 4 ACTUALLY ARE

The counting cycle $0, 1, 2, 3$ with the primes cut out

from	to 0	to 1	to 2	to 3
0	0	0.851	0.149	0
1	0	0	1	0
2	0.149	0	0	0.851
3	1	0	0	0

Among the first 1.85 million composites the four classes hold 27.0%, 23.0%, 27.0%, 23.0%, an entropy of 1.995 bits against a maximum of 2, so an order-blind measure calls the series random. The transition table shows otherwise: consecutive composites differ by 1, except when a prime sits between them, so the residue steps $+1$ with a jump of 2 at a skipped prime. After an odd composite the next step is certain, and after an even one it jumps with probability 0.149, close to the density $2/\ln x$ of primes among odd numbers; the entropy rate is only 0.328 bits per step.

4 / THE EXPONENT FOR SEVERAL SERIES

Only the composite residues sit far below the random baseline

series	H raw	H corrected	DFA α	reading
composite residues mod 4	0.36	0.32	0.28	strongly anti-persistent, a rhythm
the same residues shuffled	0.52	0.49	0.50	the random baseline
residues mod 4 of odd primes, ± 1	0.46	0.42	0.42	mildly anti-persistent
composite indicator, 0 or 1	0.46	0.43		mildly anti-persistent
prime gaps	0.51	0.47		close to random

Block sizes run from 10 to 10^4 over the first 2×10^5 composites or integers, or 10^5 primes. The corrected column divides each block average by the Anis and Lloyd expectation for random data, and DFA is detrended fluctuation analysis as a second check.

5 / READING THE NUMBERS WITH CARE

A low H here records a rhythm and does not claim a memory process

Rescaled range analysis is biased upward for short blocks, so a random series gives a fitted slope of 0.53 over this range, which is why the shuffled and corrected figures matter. The composites score 0.36 because their residues repeat a cycle, so the running totals stay bounded and R grows slowly, and the cycle is gradually disturbed by the skips at primes, so the local slope climbs from 0.29 below $n = 100$ to 0.36 and then 0.39 per decade above. The prime residues give 0.46, a faint tendency of the race between the two classes to return to level. The window rule of your app is not visible to me, so its values may differ from these by a few hundredths.

ASIDE / ORDER MATTERS

Entropy counts how often each class occurs, whereas the Hurst exponent and the Markov table use the order, and the composites show how far apart the two can be.

LINKS TO OTHER SHEETS

The app is *Composite Numbers mod 4: Information-Theoretic Analysis*; the prime races are in *Prime types: the mod 4 and mod 6 races*; the Poisson view of arrivals is in *The Poisson distribution: counting arrivals*.

Convention. H Hurst exponent; R/S rescaled range; Z_t running total of deviations from the block mean; corrected values use the Anis and Lloyd expectation; counts are over the integers up to 2×10^6.

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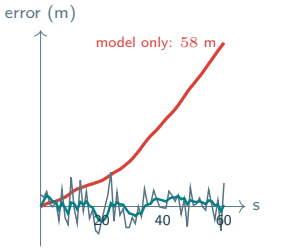
CONTROL THEORY ON ONE PAGE

Why the Kalman filter: guessing well from noisy data

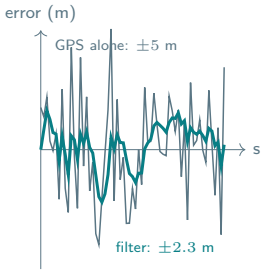
Every sensor is noisy and every model of motion drifts, and the Kalman filter says exactly how to combine the two.

1 / THE RULE AND THE PICTURE

A model alone drifts, a sensor alone jitters, and the filter that blends them beats both



full scale: model only (red), GPS (grey), filter (teal)



zoom on GPS and filter errors

A car moves at about $20\ \mathrm{m\,s^{-1}}$. GPS gives its position each second with an error near 5 m, while a model that integrates a speed guess 2% too high, ignoring the small changes in the true speed, drifts without limit and is 58 m out after a minute. The filter keeps a best guess and a measure of its own doubt, and updates it with each reading:

$$\text{new estimate} = \text{prediction} + K \times (\text{reading} - \text{prediction})$$

It holds the error to about ± 2.3 m, half the GPS error, without ever being told the speed. The same idea guides spacecraft since the 1960s, aircraft, robots, phones and battery meters.

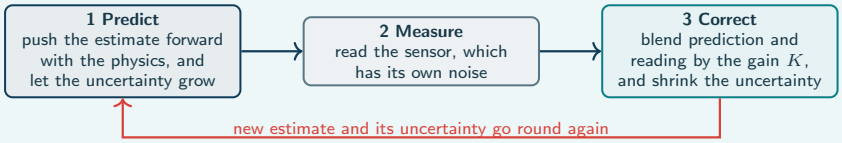
2 / TWO IMPERFECT WITNESSES

The gain K weighs each source by how reliable it is, and the blend is surer than either

Suppose the current guess is 18 with variance $P = 4$ (uncertain by ± 2) and a sensor reads 20.81 with variance $R = 1$ (uncertain by ± 1). The gain is $K = \dfrac{P}{P+R} = \dfrac{4}{5} = 0.8$, so the new estimate is $18 + 0.8\,(20.81 - 18) = 20.25$, close to the more reliable reading, and its variance is $(1-K)P = 0.8$, smaller than both 4 and 1. Combining two uncertain sources always leaves less doubt than either had, and the gain moves towards 1 when the sensor is trusted and towards 0 when the model is.

3 / THE CYCLE

Predict, measure, correct, and repeat, remembering only the estimate and its doubt



In the prediction step the doubt grows by the process noise Q, the amount the truth may wander in one step, and in the correction step it shrinks. Nothing but the latest estimate and its variance needs storing, which made the method cheap enough for the computers of the 1960s.

4 / THE TRUST DIAL

Once settled, the gain is set by the ratio $q = Q/R$ of model doubt to sensor doubt

For a slowly drifting value the settled gain is $K = \frac{1}{2}\big(-q + \sqrt{q^2 + 4q}\big)$: $q = 1$ gives $K = 0.62$ (each reading counts for about 1.6 steps of memory), $q = 0.01$ gives 0.095 (about 10 steps) and $q = 0.0001$ gives 0.010 (about 100).
A settled Kalman filter is therefore the exponential moving average, the RC low-pass, with its averaging time chosen by the signal-to-noise ratio, and a filter that trusts its model more averages for longer.

5 / WHERE IT STOPS

It is optimal only for the world it assumes

Kalman's proof of optimality assumes straight-line dynamics, Gaussian noise and known values of Q and R. When these fail the guarantee goes, and engineers turn to extended, unscented or particle filters, or simply tune Q and R until the filter behaves.

6 / A PERVERSE AFTERTHOUGHT: THE PRIMES

Put the primes in the wrong world and see what the filter makes of them

Take the density of primes as the hidden state and each integer as a reading of 0 or 1. The readings are not Gaussian, the density drifts at only about 5×10^{-9} per integer, and the primes are not Poisson: a low-pass filter of the prime indicator has less than the random variance, falling to 0.23 of the Poisson level at long scales, so the assumed noise is wrong.
The gain collapses towards 0, the estimate becomes a long average equal to $1/\ln n$, and one-step predictions score 0.345 bits per integer, the same as a constant. Adding a seasonal state per residue class mod 6 gains 0.109 bits, which is the sieve rediscovered. The filter finds only the structure its model contains.

ASIDE / ORIGIN

Rudolf Kálmán published the method in 1960 as a recursive solution to the linear filtering problem, and the recursion is what let it run in real time.

LINKS TO OTHER SHEETS

The settled filter is the low-pass of *Primes as a filtered signal*; the matrices for position and velocity are those of *Matrices as transformations*; the noise averaging is in *The normal distribution and the central limit theorem*.

Convention. P variance of the estimate; Q process noise; R sensor noise; K gain; $q = Q/R$; the car example has acceleration noise $0.2\ \mathrm{m\,s^{-2}}$ and GPS error 5 m, simulated with a fixed seed.

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Chapter 6

Change, waves and physical models

What this chapter gathers.

Foothills

- Circular functions
- Hyperbolic functions
- Small-angle approximations
- Exponential functions and the number e
- Euler's formula and the complex exponential
- One equation, $y'' = \pm y$
- Graphs: vertices, edges and degrees
- Paths, cycles, connectivity and trees
- Adjacency matrices

Foothold

- Adding velocities: Galileo, Lorentz and Euclid
- Waves, superposition and the square wave
- Damped oscillations and the three kinds of number
- Taylor series
- cos and cosh: circle and hyperbola, wave and chain
- Doppler shift to the twin paradox
- Reductionism: parts and patterns
- sec and sech as harmonic means
- tanh = sinh sech, tan = sin sec: addition, doubling and Einstein's rule
- RMS current from an oscillating magnet
- RMS in a wire and in a gas: effective value and temperature
- Euler trails and circuits
- Hamilton cycles
- Planar graphs and Euler's formula
- Colouring and bipartite graphs
- A photon's journey: energy through c, h and k
- How equations gain sophistication
- Mass defect and binding energy
- The momentum hinge: one rate equation, three mechanisms
- The potential-energy hinge: configuration, interaction and the chosen zero
- The reciprocal-power hinge: from $1/r$ to $1/r^2$
- The torque hinge: one rotational balance, three origins of torque
- The Wheatstone bridge and random walks
- Three attenuators: ballistic, diffusive and the sieve
- Velocity to displacement, by parts
- What an equation represents: from Coulomb to Maxwell

Handhold

- The gamma function and the beta function
- Hypergeometric functions
- The Laplacian and counting spanning trees
- Simplex numbers in Pascal's triangle and the trees built on their edges
- Alkanes: molecules as trees
- Weighted networks: shortest paths and spanning trees

Follow a point around a circle and record its horizontal and vertical positions. Sine and cosine describe the resulting motion. The hyperbola supplies a related pair of functions, with a different identity and different behaviour. Small-angle approximations begin to show how a curved relationship can become nearly linear within a limited range.

Exponential functions connect these pictures. Euler's formula expresses circular motion through a complex exponential, while equations such as $y'' = \pm y$ separate oscillation from growth and decay. Adding damping gives one physical system several possible behaviours. After an exponential factor is removed, its solutions involve circular, linear or hyperbolic functions. The mathematical distinctions describe observable changes in a spring's motion.

A graph provides another starting language: vertices identify parts and edges record connections. Paths, cycles and trees describe possible routes, while an adjacency matrix makes the connections available for calculation. Questions about tracing every edge, visiting every vertex or colouring neighbours introduce different constraints on the same underlying structure.

A vibrating string brings functions and connections together. Each position responds to its neighbours, and certain shapes repeat as standing waves. Under a linear model, these modes can be superposed to describe more complicated motion. Fourier analysis develops this decomposition; Taylor series provide another way to build functions from simpler pieces. In both cases, we must establish where the representation works.

Physical interpretation gives these calculations their purpose. RMS current matters because heating in a fixed resistance depends on squared current; mean squared speed connects molecular motion with temperature. The momentum, energy and torque examples ask us to identify the system and interaction behind each equation. Units, reference levels and modelling assumptions help determine what a calculated answer can mean.

Networks reveal further connections between mechanisms. Voltages in a resistor network and boundary-hitting probabilities for a suitably weighted random walk obey the same local averaging rule. The graph Laplacian organises these neighbouring differences. It enters models of diffusion and, through a determinant, counts spanning trees. Matching the quantities and boundary conditions allows a calculation in one setting to serve another.

The later pages extend this habit of construction. Gamma and beta functions develop familiar factorials and integrals; hypergeometric functions organise families of solutions. Molecular trees and weighted networks turn connections into counting and optimisation problems. The final ascent brings the book's viewpoints together: choose the parts, state their rules, calculate their combined behaviour, and return to what can be observed.

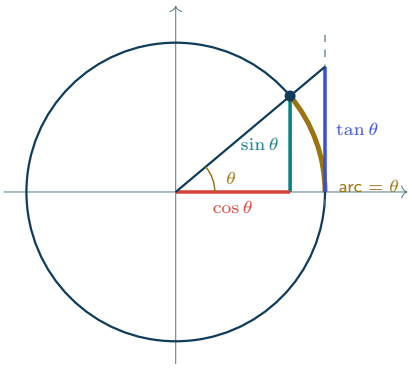
FUNCTIONS LADDER 1 OF 10

Circular functions

Sine and cosine are the coordinates of a point going round a unit circle, and tangent is the slope of the line to that point.

1 / THE RULE AND THE PICTURE

On the unit circle the point at angle θ is $(\cos\theta, \sin\theta)$, and θ is the arc length in radians



A full turn is 2π radians, so $180^\circ = \pi$ and one radian is 57.3°.
The point lies on the circle $x^2 + y^2 = 1$, which gives the Pythagorean identity, and tangent is the ratio of the two coordinates:

$$\cos^2\theta + \sin^2\theta = 1, \qquad \tan\theta = \frac{\sin\theta}{\cos\theta}$$

$\theta = 40^\circ = 0.698$ rad: $\cos = 0.766$, $\sin = 0.643$, $\tan = 0.839$

2 / THE EXACT VALUES / ANGLES WITH SIMPLE SURDS

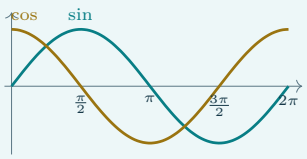
Three right triangles give every entry in the table

angle	sin	cos	tan
0	0	1	0
$\frac{\pi}{6} = 30^\circ$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$\frac{\pi}{4} = 45^\circ$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
$\frac{\pi}{3} = 60^\circ$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$\frac{\pi}{2} = 90^\circ$	1	0	undefined

The 45° values come from the isosceles right triangle with sides $1, 1, \sqrt{2}$, and the 30° and 60° values from half an equilateral triangle with sides $1, \sqrt{3}, 2$.

3 / THE GRAPHS / PERIOD AND PHASE

Sine and cosine repeat every 2π, and cosine is sine shifted by a quarter turn



Both oscillate between -1 and 1, so $A\sin(\omega x + \phi)$ has amplitude A, period $2\pi/\omega$ and a shift ϕ. Since $\cos x = \sin(x + \frac{\pi}{2})$, the cosine curve is the sine curve moved left by $\frac{\pi}{2}$, and tangent repeats with the shorter period π, with vertical asymptotes where $\cos = 0$.

4 / IDENTITIES / COMBINING ANGLES

The addition formulas generate the double-angle formulas

$$\sin(A+B) = \sin A\cos B + \cos A\sin B, \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B.$$

Setting $B = A$ gives $\sin 2A = 2\sin A\cos A$ and $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$. For example $\sin 75^\circ = \sin(45^\circ + 30^\circ) = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \frac{\sqrt{6}+\sqrt{2}}{4} = 0.9659$. Also $\sin(\pi - \theta) = \sin\theta$, $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$.

5 / INVERSES AND EQUATIONS

$\sin x = \frac{1}{2}$ has infinitely many solutions, and the arcsine picks one

The inverse $\arcsin$ returns values in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so $\arcsin\frac{1}{2} = \frac{\pi}{6}$, and the full solution set of $\sin x = \frac{1}{2}$ is $x = \frac{\pi}{6} + 2\pi n$ or $x = \frac{5\pi}{6} + 2\pi n$, the second coming from $\sin(\pi - \theta) = \sin\theta$. Similarly $\arccos$ returns $[0, \pi]$ and $\arctan$ returns $(-\frac{\pi}{2}, \frac{\pi}{2})$.

ASIDE / DEGREES OR RADIANS

Calculus formulas such as $\frac{d}{dx}\sin x = \cos x$ hold only when x is in radians, which is why radians are the natural unit.

LINKS TO OTHER SHEETS

The surds are developed in *The unit circle, the Ailles rectangle and the sine of* 15°; the next sheet is *Hyperbolic functions*.

Convention. θ is in radians unless marked in degrees; $\sin, \cos, \tan$ are the circular functions; $\arcsin$ etc. are principal inverses.

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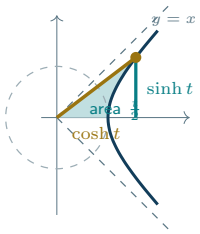
FUNCTIONS LADDER 2 OF 10

Hyperbolic functions

The point $(\cosh t, \sinh t)$ runs along a hyperbola as $(\cos t, \sin t)$ runs round a circle.

1 / THE RULE AND THE PICTURE

The point $(\cosh t, \sinh t)$ lies on $x^2 - y^2 = 1$, and t is twice the shaded area



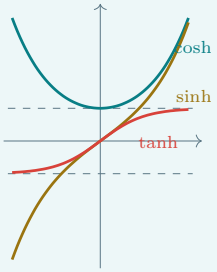
Hyperbolic cosine and sine are the even and odd parts of the exponential:

$$\cosh t = \frac{e^t + e^{-t}}{2}, \qquad \sinh t = \frac{e^t - e^{-t}}{2}, \qquad \cosh^2 t - \sinh^2 t = 1$$

The identity follows from $\frac{1}{4}\big((e^t + e^{-t})^2 - (e^t - e^{-t})^2\big) = \frac{1}{4} \cdot 4 = 1$, and the sector from the origin to the point has area $t/2$, just as the circular sector has area $\theta/2$.

2 / THE GRAPHS / THREE CURVES

Cosh is a symmetric valley, sinh an odd curve, and tanh squashes into $(-1, 1)$



$\cosh t \geq 1$ with minimum 1 at $t = 0$, $\sinh t$ passes through the origin with slope 1, and $\tanh t = \sinh t / \cosh t$ rises from -1 to 1.
For large t both $\cosh t$ and $\sinh t$ are close to $\frac{1}{2}e^t$. Unlike the circular functions none of them is periodic.

3 / VALUES AND IDENTITIES

The identities are those of trigonometry with a sign change

t	$\sinh t$	$\cosh t$	$\tanh t$
0.5	0.5211	1.1276	0.4621
1	1.1752	1.5431	0.7616
2	3.6269	3.7622	0.9640

Osborn's rule says to take a trigonometric identity, write each function as its hyperbolic counterpart, and flip the sign of any product of two sines: $\cosh(A + B) = \cosh A \cosh B + \sinh A \sinh B$, $\sinh 2t = 2 \sinh t \cosh t$, $\cosh 2t = \cosh^2 t + \sinh^2 t = 2\cosh^2 t - 1$ and $1 - \tanh^2 t = \operatorname{sech}^2 t$.
Check with $A = 1$, $B = 2$: $1.5431 \cdot 3.7622 + 1.1752 \cdot 3.6269 = 10.0677 = \cosh 3$.

4 / CALCULUS AND INVERSES

The derivatives are $\frac{d}{dt}\sinh t = \cosh t$ and $\frac{d}{dt}\cosh t = \sinh t$, with no minus sign

The inverses are logarithms, found by solving $\sinh y = x$ as a quadratic in e^y: $\operatorname{arsinh} x = \ln\big(x + \sqrt{x^2 + 1}\big)$, $\operatorname{arcosh} x = \ln\big(x + \sqrt{x^2 - 1}\big)$ for $x \geq 1$, and $\operatorname{artanh} x = \frac{1}{2}\ln\frac{1+x}{1-x}$. For example $\operatorname{arsinh} 1 = \ln(1 + \sqrt{2}) = 0.8814$ and $\operatorname{artanh}\frac{1}{2} = \frac{1}{2}\ln 3 = 0.5493$.

5 / THE HANGING CHAIN / THE CATENARY

A uniform chain hangs in the shape $y = a\cosh(x/a)$

For $a = 1$ m a chain hung from two points 2 m apart, at $x = \pm 1$, sags by $a(\cosh 1 - 1) = 0.543$ m below its ends, and its length is $2a \sinh 1 = 2.350$ m, since the arc length from 0 to x is $a\sinh(x/a)$.
A parabola looks similar near the bottom but is not the same curve, and the inverted catenary is the ideal shape for an arch.

ASIDE / THE CONNECTION

With complex arguments the two families merge: $\cos(ix) = \cosh x$ and $\sin(ix) = i\sinh x$.

LINKS TO OTHER SHEETS

The previous sheet is *Circular functions*; the next is *Small-angle approximations*; Euler's formula explains the link in *Euler's formula and the complex exponential*.

Convention. t is a real number; $\operatorname{sech} = 1/\cosh$; arsinh etc. are inverse hyperbolic functions; a is a length.

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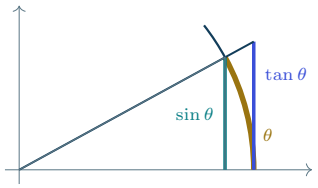
FUNCTIONS LADDER 3 OF 10

Small-angle approximations

For small angles in radians the sine and tangent are close to the angle itself, and the cosine is close to one.

1 / THE RULE AND THE PICTURE

For a small angle in radians $\sin\theta < \theta < \tan\theta$, and all three are nearly equal



$\theta = 0.5$ rad on a unit circle drawn large

In the picture the arc is longer than the vertical side of the small triangle and shorter than the vertical side of the large one, which squeezes

$$\sin\theta \approx \theta, \qquad \tan\theta \approx \theta, \qquad \cos\theta \approx 1 - \frac{\theta^2}{2}$$

and the approximations hold only when θ is measured in radians.

2 / HOW GOOD ARE THEY

The errors grow like θ^2 for the ratios, and stay below 1% for $\sin$ up to about 10°

angle	θ (rad)	$\sin\theta$	error of $\sin \approx \theta$	error of $\tan \approx \theta$	error of $\cos \approx 1 - \frac{\theta^2}{2}$
1°	0.0175	0.0175	0.01%	0.01%	3.9×10^{-9}
5°	0.0873	0.0872	0.13%	0.25%	2.4×10^{-6}
10°	0.1745	0.1736	0.51%	1.02%	3.9×10^{-5}
15°	0.2618	0.2588	1.15%	2.30%	2.0×10^{-4}
30°	0.5236	0.5000	4.72%	9.31%	3.1×10^{-3}

At 10° the error in $\sin\theta \approx \theta$ is 0.51% and in $\tan\theta \approx \theta$ twice that, 1.02%; at 30° the sine is still within 4.7% but the tangent is 9.3% out.

3 / WHY / THE NEXT TERM

The next term of each series gives the size of the error

The series are $\sin\theta = \theta - \frac{\theta^3}{6} + \cdots$, $\tan\theta = \theta + \frac{\theta^3}{3} + \cdots$ and $\cos\theta = 1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} - \cdots$. The relative error of $\sin\theta \approx \theta$ is about $\theta^2/6$, which at $10^\circ = 0.1745$ rad gives 0.0051, matching the table, and the relative error of $\tan\theta \approx \theta$ is about $\theta^2/3$, twice as large.
These are the first terms of the Taylor series, and the squeeze $\sin\theta < \theta < \tan\theta$ gives the limit $\sin\theta/\theta \to 1$ as $\theta \to 0$, the fact that makes $\frac{d}{d\theta}\sin\theta = \cos\theta$ true.

4 / THE PENDULUM

The period of a pendulum is nearly independent of amplitude only for small swings

amplitude θ_0	exact T/T_0	$1 + \theta_0^2/16$
5°	1.00048	1.00048
10°	1.00191	1.00190
30°	1.01741	1.01713
60°	1.07318	1.06854

Replacing $\sin\theta$ by θ in the equation of motion gives simple harmonic motion with $T_0 = 2\pi\sqrt{L/g} = 2.006$ s for $L = 1$ m; the correction factor $1 + \theta_0^2/16$ is accurate to a few parts in a thousand up to 30°.

5 / USES

Angular size, Snell's law and apparent depth

The Moon is 3474 km across at $384\,400$ km, so its angle is $\theta \approx \dfrac{3474}{384\,400} = 0.00904$ rad $= 0.518^\circ$, and one arcsecond is 4.85×10^{-6} rad.
For light crossing a boundary Snell's law $n_1 \sin\theta_1 = n_2 \sin\theta_2$ becomes $n_1\theta_1 \approx n_2\theta_2$ near the normal, so a pool of real depth 3 m and $n = 1.33$ looks about $3/1.33 = 2.26$ m deep when viewed from above.

ASIDE / DEGREES BREAK IT

Using degrees in $\sin\theta \approx \theta$ is the classic mistake, since $\sin 1^\circ = 0.01745$ and not 1.

LINKS TO OTHER SHEETS

The exact pendulum period uses *Hypergeometric functions*; the previous sheet is *Hyperbolic functions* and the next is *Exponential functions and the number e*.

Convention. θ in radians; relative error is (approximation − exact)/exact in absolute value; $g = 9.81$ m/s^2.

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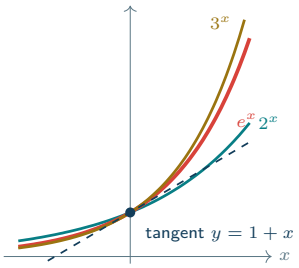
FUNCTIONS LADDER 4 OF 10

Exponential functions and the number e

Quantities that grow in proportion to themselves follow e^x, the exponential whose slope equals its height.

1 / THE RULE AND THE PICTURE

Among all exponentials a^x the one with slope 1 at $x = 0$ is $a = e = 2.71828\ldots$



All the curves pass through $(0, 1)$, and the slope there is $\ln a$: $\ln 2 = 0.693$ for 2^x and $\ln 3 = 1.099$ for 3^x, with e the base that makes the slope exactly 1.
Since the slope of a^x is proportional to its height, e^x is the one whose slope equals its height everywhere:

$$\frac{d}{dx}e^x = e^x, \qquad e^0 = 1, \qquad e^{a+b} = e^a e^b$$

2 / TWO WAYS TO COMPUTE e

A limit of compound interest and a sum of reciprocal factorials

n	$(1 + 1/n)^n$
1	2.00000
2	2.25000
5	2.48832
10	2.59374
100	2.70481
1 000	2.71692
1 000 000	2.71828

Splitting a year into n parts at interest $100\%/n$ gives $(1 + \frac{1}{n})^n$.

terms up to $1/m!$	sum
0	1.00000
1	2.00000
2	2.50000
3	2.66667
4	2.70833
5	2.71667
6	2.71806
7	2.71825

$e = \sum_{k \geq 0} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \cdots$, which converges much faster.

3 / GROWTH AND DECAY

A constant percentage rate gives e^{rt}, with a doubling time of $\ln 2/r$

At 7% per year the doubling time is $\ln 2 / \ln 1.07 = 10.24$ years, close to the rule of 70, $70/7 = 10$, and with continuous compounding it is $\ln 2/0.07 = 9.90$ years.
Investing $£1000$ at 5% for ten years gives $£1628.89$ yearly, $£1647.01$ monthly and $£1648.72 = 1000e^{0.5}$ continuously.
Radioactive decay $N = N_0 e^{-\lambda t}$ with half-life 5730 years for carbon-14 has $\lambda = \ln 2/5730 = 1.21 \times 10^{-4}$ per year.

4 / ANY BASE / $a^x = e^{x \ln a}$

Every exponential is e^x with the axis rescaled

Writing $a = e^{\ln a}$ gives $a^x = e^{x \ln a}$, so $2^x = e^{0.693x}$ and $\frac{d}{dx}a^x = (\ln a)a^x$.
The laws $a^x a^y = a^{x+y}$ and $(a^x)^y = a^{xy}$ follow from $e^{a+b} = e^a e^b$, and the exponential eventually beats every power: at $x = 20$, $e^{20} = 4.85 \times 10^8$ while $20^5 = 3.2 \times 10^6$, a ratio of 0.0066, and the ratio $x^n/e^x \to 0$ for every n.

5 / WHY e / THE DIFFERENTIAL EQUATION

The equation $y' = ky$ has the solution $y = y_0 e^{kt}$

A quantity whose rate of change is proportional to its size, such as a population with unlimited food or a cooling temperature difference, satisfies $y' = ky$, and the exponential is its only solution with $y(0) = y_0$.
Trying $y = e^{rt}$ gives $r = k$, the simplest case of the characteristic equation used for second-order equations in a later sheet.

ASIDE / ARITHMETIC SHORTCUT

For small x, $e^x \approx 1 + x$, so $e^{0.05} \approx 1.05$ against the exact 1.0513.

LINKS TO OTHER SHEETS

The inverse of e^x is in *Logarithms*; the series is developed in *Taylor series*; the previous sheet is *Small-angle approximations*.

Convention. $e = 2.718281828\ldots$; $\ln$ is the natural logarithm; r is a growth rate per unit time; λ a decay rate.

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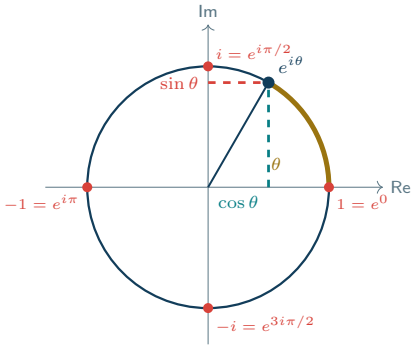
FUNCTIONS LADDER 6 OF 10

Euler's formula and the complex exponential

The exponential of an imaginary number goes round the unit circle, so exponentials and trigonometry are one family.

1 / THE RULE AND THE PICTURE

$e^{i\theta} = \cos\theta + i\sin\theta$ is the point at angle θ on the unit circle



$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

Taking $\theta = \pi$ gives Euler's identity,

$$\boxed{e^{i\pi} + 1 = 0}$$

which links the five constants e, i, π, 1, 0, and a quarter turn $\theta = \frac{\pi}{2}$ gives $e^{i\pi/2} = i$.

2 / WHY IT IS TRUE / TWO DERIVATIONS

Substituting into the series, or following the rotation of the derivative

Substituting ix into $e^x = \sum x^n/n!$ and separating even and odd powers, using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$,

$$e^{i\theta} = \left(1 - \tfrac{\theta^2}{2!} + \tfrac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \tfrac{\theta^3}{3!} + \tfrac{\theta^5}{5!} - \cdots\right) = \cos\theta + i\sin\theta.$$

Geometrically $f(\theta) = e^{i\theta}$ satisfies $f' = if$, and multiplying by i turns a vector by a right angle, so the velocity is always perpendicular to the position and the point moves at unit speed round the circle, covering arc length θ.

3 / SINE AND COSINE AS EXPONENTIALS

Adding and subtracting $e^{i\theta}$ and $e^{-i\theta}$ isolates each

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i},$$

which are the hyperbolic formulas with t replaced by $i\theta$, so $\cos(ix) = \cosh x$ and $\sin(ix) = i\sinh x$; for instance $\cos i = \cosh 1 = 1.5431$ and $\sin i = 1.1752\,i$.
The circular and hyperbolic functions are therefore the same functions seen along the real and imaginary directions.

4 / MULTIPLICATION AND DE MOIVRE / ANGLES ADD

$e^{iA}e^{iB} = e^{i(A+B)}$ contains the addition formulas

Multiplying $(\cos A + i\sin A)(\cos B + i\sin B)$ and comparing real and imaginary parts with $e^{i(A+B)}$ gives $\cos(A+B)$ and $\sin(A+B)$. Repeated multiplication gives de Moivre's theorem $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, whose real part for $n = 3$ is $\cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$; at $\theta = 0.7$ both sides equal -0.5048.
The solutions of $z^6 = 1$ are $e^{2\pi ik/6}$, the six vertices of a regular hexagon.

5 / COMPLEX POWERS AND LOGARITHMS

Exponentiation is a growth by e^x and a turn by y

For $z = x + iy$ we have $e^z = e^x(\cos y + i\sin y)$, so $e^{1+i\pi/3} = e(\frac{1}{2} + \frac{\sqrt{3}}{2}i) = 1.359 + 2.354i$.
Logarithms of complex numbers add multiples of $2\pi i$: $\ln(-1) = i\pi$ and $\ln i = \frac{i\pi}{2}$ on the principal branch.
Then $i^i = e^{i\ln i} = e^{-\pi/2} = 0.2079$, a real number, and every other branch gives $e^{-\pi/2 - 2\pi k}$.

ASIDE / ROTATION

Multiplying by $e^{i\theta}$ rotates the plane by θ, which is why alternating current and waves are written as $e^{i\omega t}$.

LINKS TO OTHER SHEETS

The real parts of roots of unity give the chord lengths in *Cyclotomic fields and the arcs of inscribed polygons*; the previous sheet is *Logarithms* and the next is *One equation, $y'' = \pm y$*.

Convention. $i^2 = -1$; angles in radians; $\ln$ on the complex plane is the principal branch with imaginary part in $(-\pi, \pi]$.

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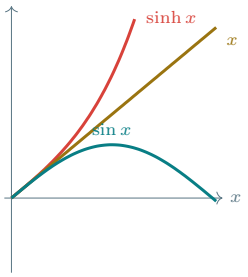
FUNCTIONS LADDER 7 OF 10

One equation, $y'' = \pm y$

The sign of the constant decides whether solutions grow, oscillate or run straight, and the characteristic equation finds them all.

1 / THE RULE AND THE PICTURE

Trying $y = e^{rx}$ in $y'' = ky$ gives the characteristic equation $r^2 = k$



solutions with $y(0) = 0$, $y'(0) = 1$: $\sinh x$, x and $\sin x$

Substituting $y = e^{rx}$ gives $r^2 e^{rx} = k e^{rx}$, and cancelling e^{rx},

$$\boxed{r^2 = k}$$

Two real roots give growth and decay, two imaginary roots give oscillation, and a double root gives a straight line. Every second-order linear equation with constant coefficients is solved by the same trick.

2 / THE THREE CASES

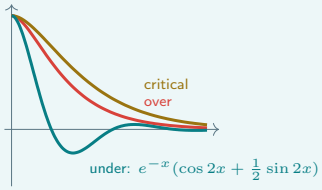
Six familiar functions are the solutions of three equations

equation	characteristic equation	roots	exponential solutions	real forms
$y'' = y$	$r^2 = 1$	$r = 1,\ -1$	$e^{x},\ e^{-x}$	$\cosh x,\ \sinh x$
$y'' = -y$	$r^2 = -1$	$r = i,\ -i$	$e^{ix},\ e^{-ix}$	$\cos x,\ \sin x$
$y'' = 0$	$r^2 = 0$	$r = 0$ twice	$1,\ x$	$1,\ x$

The general solution is a combination of the two basic ones, for example $y = A\cos x + B\sin x$ for $y'' = -y$, and the two constants are fixed by $y(0)$ and $y'(0)$: choosing $y(0) = 1$, $y'(0) = 0$ picks $\cosh x$, $\cos x$ and 1, and $y(0) = 0$, $y'(0) = 1$ picks $\sinh x$, $\sin x$ and x.

3 / ADDING FRICTION / THE THREE NUMBER TYPES

For $y'' + by' + cy = 0$ the characteristic equation $r^2 + br + c = 0$ has three possible kinds of root



The discriminant $b^2 - 4c$ decides: positive gives two real roots and overdamped decay, such as $y'' + 3y' + 2y = 0$ with $r = -1, -2$; zero gives the critical case $(A + Bx)e^{-x}$ for $y'' + 2y' + y = 0$; negative gives complex roots $-\frac{b}{2} \pm i\omega$ and damped oscillation, as for $y'' + 2y' + 5y = 0$ with $r = -1 \pm 2i$ and $y = e^{-x}(A\cos 2x + B\sin 2x)$.

4 / PHYSICAL EXAMPLES

A mass on a spring oscillates, and an inverted pendulum falls exponentially

For $my'' = -ky$ with $m = 0.5$ kg and $k = 20$ N/m, $r^2 = -40$ so $\omega = \sqrt{40} = 6.32$ rad/s and the period is $2\pi/\omega = 0.993$ s.
A pendulum balanced upside down obeys $\theta'' = +\frac{g}{L}\theta$ for small angles, so $r = \pm\sqrt{g/L} = \pm 3.13$ per second for $L = 1$ m, and an error doubles every $\ln 2/3.13 = 0.22$ s, which is why balancing needs fast feedback.

5 / THE SAME IDEA FOR SEQUENCES

The Fibonacci recurrence has characteristic equation $r^2 = r + 1$

Trying $F_n = r^n$ in $F_{n+1} = F_n + F_{n-1}$ gives $r^2 = r + 1$, with roots $\varphi = 1.618$ and $-1/\varphi = -0.618$, so $F_n = A\varphi^n + B(-1/\varphi)^n$, which is the Binet formula.
Differential equations use the trial solution e^{rx} and recurrences the trial solution r^n, and in both cases a quadratic in r determines the behaviour.

ASIDE / THE DIFFERENCE

Real roots give monotone behaviour and complex roots give rotation; for equations the real part of the root sets the growth rate, and for recurrences its modulus does.

LINKS TO OTHER SHEETS

The recurrence version is in *The characteristic equation beneath five presentations*; the previous sheet is *Euler's formula and the complex exponential* and the next is *Taylor series*.

Convention. r is a root of the characteristic equation; ω an angular frequency; b, c constants; y' and y'' derivatives with respect to x.

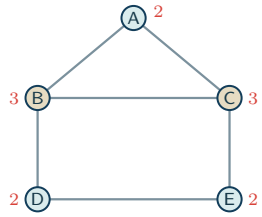
GRAPH THEORY LADDER 1 OF 9

Graphs: vertices, edges and degrees

A graph records which things are linked, and counting the links at each vertex leads to the handshake rule.

1 / THE RULE AND THE PICTURE

Adding up the degrees counts every edge twice



the house graph, degrees in red

A graph is a set of vertices joined by edges, and the degree of a vertex is the number of edges that meet it. Every edge has two ends, so adding the degrees counts each edge twice:

$$\boxed{\sum_v \deg v = 2E}$$

For the house graph the degrees $2, 3, 3, 2, 2$ add to 12, twice its 6 edges. This graph is the running example of the ladder: it has 5 vertices, 6 edges and one triangle.

2 / SOME STANDARD GRAPHS

A few families appear everywhere

graph	vertices	edges	description
path P_n	n	$n-1$	a line of n vertices
cycle C_n	n	n	a ring of n vertices, every degree 2
complete graph K_n	n	$\frac{n(n-1)}{2}$	every pair joined, K_5 has 10 edges
complete bipartite $K_{m,n}$	$m+n$	mn	every vertex of one group joined to every vertex of the other, $K_{3,3}$ has 9
star $K_{1,n}$	$n+1$	n	one centre joined to n leaves

3 / PARTY PUZZLES

The sum of the degrees is always even

Five people cannot each be friends with exactly three others, since the degrees would add to 15, which is odd and so cannot equal $2E$.
For the same reason the number of vertices of odd degree is always even, and in any gathering an even number of people have shaken an odd number of hands.
The degrees do not settle everything, though: a ring of six vertices and two separate triangles have six vertices of degree 2 each, yet the first is connected and the second is not.

4 / WHAT GRAPHS MODEL

Choose the vertices, then decide what an edge means

The vertices can be towns with roads as edges, people with friendships, web pages with links, or atoms with bonds: the carbon skeleton of butane is a path P_4 and the benzene ring is a cycle C_6 with every degree 2.
The picture throws away the lengths and the angles and keeps only the pattern of links, which is why one graph can stand for many different physical situations.

5 / TRY THESE

Count with the handshake rule

(a) How many edges has K_6? Each of 6 vertices has degree 5, so $6 \times 5/2 = 15$.
(b) A graph has degrees $3, 3, 2, 2, 2$: how many edges? $12/2 = 6$.
(c) Can a graph have degrees $3, 3, 3$? Their sum is 9, which is odd, so no.

ASIDE / SIMPLE GRAPHS

A simple graph has no loops and at most one edge between two vertices; the bridges of Königsberg later need repeated edges.

LINKS TO OTHER SHEETS

The next sheet follows paths through the house graph; the carbon chain sheet in the physics collection uses the same graphs for molecules.

Convention. V vertices, E edges, $\deg v$ the degree of v; P_n, C_n, K_n, $K_{m,n}$ are the path, cycle, complete and complete bipartite graphs.

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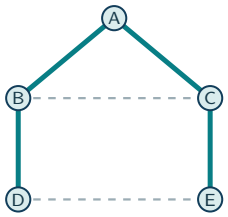
GRAPH THEORY LADDER 2 OF 9

Paths, cycles, connectivity and trees

Walks in a graph lead to cycles, and a connected graph with no cycles is a tree with one edge fewer than vertices.

1 / THE RULE AND THE PICTURE

A tree on n vertices has exactly $n-1$ edges



a spanning tree of the house graph in teal

A tree is a connected graph with no cycle, and removing the two dashed edges from the house leaves a tree through all 5 vertices, a spanning tree:

$V - E = 1$ for a tree, $\quad E - V + 1 =$ number of independent cycles

The house graph has $E - V + 1 = 6 - 5 + 1 = 2$ independent cycles, the triangle and the square, and removing one edge from each leaves a tree with 4 edges.

2 / THE WORDS FOR MOVING ABOUT

Each word allows fewer repeats than the one before

word	meaning	in the house graph
walk	vertices and edges in sequence, repeats allowed	A, B, A, C
trail	a walk with no repeated edge	B, A, C, B, D
path	no repeated vertex	A, B, D, E of length 3
cycle	a closed path of at least 3 vertices	A, B, C (a triangle) or B, D, E, C (a square)
connected	a path joins every pair of vertices	the house graph is connected

3 / WHY A TREE HAS $n-1$ EDGES

Peel off a leaf and count

Every tree with at least two vertices has a leaf, a vertex of degree 1, since a longest path must end at one; removing the leaf and its edge leaves a smaller tree, so by repetition the edges number one fewer than the vertices.
Two more facts follow: any two vertices of a tree are joined by exactly one path, and adding any new edge to a tree creates exactly one cycle.

4 / HOW MANY SPANNING TREES

The house graph has 11 and K_n has n^{n-2}

A spanning tree chooses which edges to keep, and for the house graph there are 11 ways, which the Laplacian sheet later counts with a determinant. Cayley's formula says the complete graph K_n has n^{n-2} spanning trees, so K_4 has 16 and K_5 has 125, the same numbers as the labelled trees on four and five vertices, since every labelled tree is a spanning tree of K_n.

5 / TRY THESE

Find the cycles and the tree

(a) List the cycles of the house graph: the triangle ABC, the square $BDEC$ and the pentagon $ABDEC$.
(b) How many edges must be removed from K_4 to leave a tree? $6 - 3 = 3$ of them.
(c) A graph with 10 vertices and 9 edges that is connected must be a tree, because it has $V - 1$ edges.

ASIDE / FORESTS

A graph with no cycles that is not connected is a forest, and a forest with c components on n vertices has $n - c$ edges.

LINKS TO OTHER SHEETS

The first sheet gives the degree rule used here; the Euler formula sheet turns the cycle count $E - V + 1$ into a count of faces.

Convention. A spanning tree uses all vertices; $E - V + 1$ is the cycle rank of a connected graph; Cayley counted labelled trees.

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GRAPH THEORY LADDER 3 OF 9

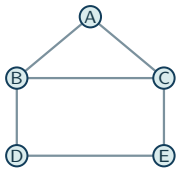
Adjacency matrices

A graph becomes a table of zeros and ones, and the powers of that table count walks.

1 / THE RULE AND THE PICTURE

The entry in row u and column v is 1 when u and v are joined

	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	1	0
C	1	1	0	0	1
D	0	1	0	0	1
E	0	0	1	1	0



Listing the vertices in the order A, B, C, D, E, the matrix M has $M_{uv} = 1$ for an edge and 0 otherwise, and the key property is

$$(M^k)_{uv} = \text{number of walks of length } k \text{ from } u \text{ to } v$$

because a walk of length $k+1$ is a walk of length k followed by one more edge.

2 / READING THE MATRIX

Symmetry, row sums and the diagonal

The matrix is symmetric, since an edge joins u to v exactly when it joins v to u, and the diagonal is zero, since there are no loops. Each row sum is a degree, so the row sums of the house matrix are $2, 3, 3, 2, 2$, and the sum of all entries is $12 = 2E$, the handshake rule. For a directed graph, such as web links, the matrix is no longer symmetric and the rows count outgoing links.

3 / POWERS COUNT WALKS

The squares and cubes of the house matrix

$$M^2 = \begin{pmatrix} 2 & 1 & 1 & 1 & 1 \\ 1 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 2 & 0 \\ 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 0 & 2 \end{pmatrix}, \qquad M^3 = \begin{pmatrix} 2 & 4 & 4 & 2 & 2 \\ 4 & 2 & 6 & 5 & 1 \\ 4 & 6 & 2 & 1 & 5 \\ 2 & 5 & 1 & 0 & 4 \\ 2 & 1 & 5 & 4 & 0 \end{pmatrix}.$$

In M^2 the diagonal gives the degrees $2, 3, 3, 2, 2$, because a walk of length 2 from u back to u goes out along an edge and returns along it, and the entry $(B, E) = 2$ counts the walks B, C, E and B, D, E. In M^3 the entry $(A, B) = 4$ counts four walks of length 3 from A to B, and the sum of all entries of M^2 is $30 = \sum(\deg v)^2$.

4 / TRIANGLES AND EIGENVALUES

Closed walks of length 3 are triangles

The trace of M^3 is 6, which is 6 times the number of triangles, because each triangle gives closed walks of length 3 from three starting points in two directions, and the house graph has one triangle.
The eigenvalues of M are $2.481,\ 0.689,\ 0,\ -1.170,\ -2$, and the largest, 2.481, exceeds the average degree $12/5 = 2.4$, as it always does, while the eigenvalues are what let us study large graphs without drawing them.

5 / TRY THESE

Use the table, not the picture

(a) Which vertex has the largest row sum in the house matrix? B and C with degree 3.
(b) How many walks of length 2 join A to D? Row A, column D of M^2 is 1, the walk A, B, D.
(c) How many walks of length 2 go from A to A? 2, which is the degree of A.

ASIDE / ADJACENCY LISTS

For a large sparse graph a list of neighbours uses far less space than a matrix with mostly zeros, though the matrix makes the algebra possible.

LINKS TO OTHER SHEETS

The linear algebra ladder explains matrix powers and eigenvalues, and the Laplacian sheet later adjusts this matrix by the degrees.

Convention. M adjacency matrix with vertices ordered A, B, C, D, E; $(M^k)_{uv}$ counts walks of length k; the trace is the sum of the diagonal.

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ADDING VELOCITIES ON ONE PAGE

Galileo, Lorentz and Euclid

One addition formula with a single sign switch covers rotations, Galilean shears and Lorentz boosts.

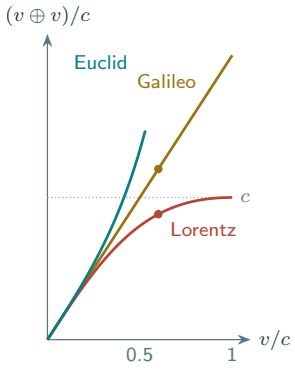
THE RULE

One formula, three geometries

Composing two slopes u and v, whether directions in a plane or velocities in spacetime, follows

$$u \oplus v = \frac{u+v}{1-\kappa uv}, \qquad \kappa = \begin{cases} +1 & \text{Euclidean rotations, } \tan(\alpha+\beta) \\ 0 & \text{Galilean shears, } u+v \\ -1/c^2 & \text{Lorentz boosts, } \dfrac{u+v}{1+uv/c^2} \end{cases}$$

Galileo's rule is the limit $c \to \infty$ of Einstein's, and the two differ only when uv is a noticeable fraction of c^2: two aircraft at $300\,\mathrm{m/s}$ differ from simple addition by one part in 10^{12}, while $0.6c \oplus 0.6c = \frac{15}{17}c \approx 0.882c$ falls well short of $1.2c$.



A slope added to itself: Lorentz never reaches c, while tangents run away.

RAPIDITY

The quantity that does add

Writing $v = c\tanh\varphi$ turns Einstein's rule into the addition formula for $\tanh$, so the rapidities simply add,

$$\tanh(\varphi_1+\varphi_2) = \frac{\tanh\varphi_1 + \tanh\varphi_2}{1+\tanh\varphi_1\tanh\varphi_2}, \qquad \varphi = \operatorname{artanh}\frac{v}{c},$$

just as angles add where tangents do not. For small speeds $\varphi \approx v/c$, which is why everyday velocities seem to add. Rapidity also governs the Doppler factor e^{φ} and the energy and momentum, $E = mc^2\cosh\varphi$ and $p = mc\sinh\varphi$, and a rocket with steady proper acceleration a gains rapidity at the steady rate a/c.

v/c	φ	e^{φ}
0.6	$\ln 2 = 0.693$	2
$0.6 \oplus 0.6 = \frac{15}{17}$	$\ln 4 = 1.386$	4
0.9	1.472	4.36
$0.9 \oplus 0.9 = 0.9945$	2.944	19

Rapidities add and Doppler factors multiply.

EUCLIDEAN, GALILEAN, MINKOWSKIAN

Rotations, shears and boosts as three kinds of complex number

	Euclidean plane	Galilean spacetime	Minkowski spacetime
unit and its square	$\mathrm{i}^2 = -1$	$\varepsilon^2 = 0$	$j^2 = +1$
what is kept	$x^2 + y^2$	time t	$c^2t^2 - x^2$
unit curve	circle	pair of lines $t = \pm 1$	hyperbola
transformation	$\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 0 \\ v & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{smallmatrix}\right)$
parameter that adds	angle θ	velocity v	rapidity φ
slope it gives	$\tan\theta$	v	$\tanh\varphi = v/c$
exponential form	$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$	$\mathrm{e}^{\varepsilon v} = 1 + \varepsilon v$	$\mathrm{e}^{j\varphi} = \cosh\varphi + j\sinh\varphi$

Multiplying two such numbers composes the transformations, so the angle, the velocity or the rapidity adds in the exponent; the Galilean column is the degenerate middle case where the second-order term vanishes, the parabolic member of the elliptic, parabolic and hyperbolic trichotomy.

WHY NOTHING OVERTAKES LIGHT

A product that stays positive

In units with $c = 1$, Einstein's rule satisfies the identity

$$1 - w^2 = \frac{(1-u^2)(1-v^2)}{(1+uv)^2}, \qquad w = u \oplus v,$$

so two speeds below light always combine to a speed below light, and $u \oplus 1 = 1$ for every u. The Euclidean twin, $1 + w^2 = \frac{(1+u^2)(1+v^2)}{(1-uv)^2}$, is $\sec^2$ of a sum, which blows up when the angles reach a right angle.

THE FIRST EXPERIMENTAL HINT

Fizeau's moving water, 1851

Light travels through still water at c/n. Fizeau sent it along water flowing at speed v and found it dragged only partly, by the fraction $1 - 1/n^2 \approx 0.44$ that Fresnel had predicted. Expanding Einstein's rule to first order in v gives exactly

$$\frac{c}{n} \oplus v \approx \frac{c}{n} + v\Big(1 - \frac{1}{n^2}\Big),$$

fifty-four years before special relativity explained it.

Conventions. Velocities lie along one line, ε and j are the dual and split-complex units, and κ selects the geometry.

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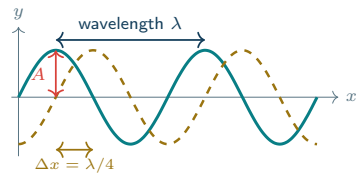
WAVES ON ONE PAGE

Waves, superposition and the square wave

A sine wave has a wavelength, a wave number and a phase, linear waves add, and a square wave is a sum of odd harmonics.

1 / THE RULE AND THE PICTURE

A travelling sine wave $y = A\sin(kx - \omega t + \varphi)$ has wavelength λ, wave number $k = 2\pi/\lambda$ and phase φ



a snapshot at one instant, and the same wave with $\varphi = \pi/2$

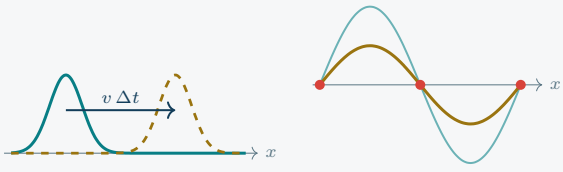
At a fixed time the pattern repeats every wavelength λ, and at a fixed place every period $T = 1/f$, so the wave number k counts radians per metre as $\omega = 2\pi f$ counts radians per second:

$$k = \tfrac{2\pi}{\lambda}, \qquad \omega = 2\pi f, \qquad v = \tfrac{\omega}{k} = f\lambda$$

A phase φ slides the whole pattern by $\Delta x = \varphi/k$. For a 1 kHz sound in air ($343\ \mathrm{m\,s^{-1}}$) $\lambda = 0.343$ m, $k = 18.3\ \mathrm{rad\,m^{-1}}$ and $\omega = 6283\ \mathrm{rad\,s^{-1}}$, while a 1 MHz signal on a cable with velocity factor 0.66 has $\lambda = 198$ m, far larger than the circuit, which is why amplifier tests treat a signal as a function of time alone.

2 / THE WAVE EQUATION / WHY SINES AND WHY SUMS

The equation $\partial_t^2 y = v^2 \partial_x^2 y$ is linear, and its solutions are travelling shapes $f(x - vt) + g(x + vt)$

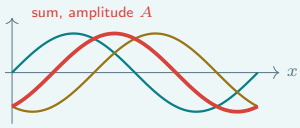


a pulse travelling at speed v, and a standing wave with nodes in red

Any fixed shape moving right, $f(x - vt)$, or left, $g(x + vt)$, satisfies the equation, and a sine gives $\omega^2 = v^2k^2$, the dispersion relation $\omega = vk$, so every frequency travels at the same speed and a sum of harmonics keeps its shape. Linearity means that if y_1 and y_2 are solutions so is $y_1 + y_2$, the principle of superposition. A string of length L fixed at both ends allows only $k_n = n\pi/L$, with shapes $\sum b_n \sin(n\pi x/L)\cos\omega_n t$, a Fourier series, and if ω is not proportional to k the harmonics slip out of step and the shape changes.

3 / ADDING WAVES OUT OF PHASE

Two equal waves a phase δ apart add to a wave of amplitude $2A\cos(\delta/2)$



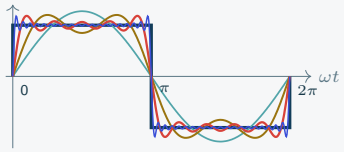
two waves 120° apart add to a wave of amplitude A

$$\sin\theta + \sin(\theta - \delta) = 2\cos\tfrac{\delta}{2}\ \sin\big(\theta - \tfrac{\delta}{2}\big)$$

In phase ($\delta = 0$) the amplitude doubles, at $\delta = 90^\circ$ it is $\sqrt{2}A$, at 120° it is A, and at $\delta = 180^\circ$ the waves cancel. Waves of different wave numbers change the shape and not only the size: $\sin x + \frac{1}{3}\sin 3x$ flattens the crest into two shoulders, while $\sin x - \frac{1}{3}\sin 3x$ sharpens it, so the phases of the harmonics matter as much as their amplitudes.

4 / BUILDING THE SQUARE WAVE

Odd harmonics of amplitude $4/(\pi k)$, all starting in phase, sum to a square wave



sums up to harmonic 1 (teal), 3 (gold), 9 (red), 39 (blue)

$$f(t) = \tfrac{4}{\pi}\sum_{k=1,3,5,\ldots} \tfrac{\sin k\omega t}{k}$$

Multiplying by $\sin kt$ and integrating over a period keeps only the kth term, giving $b_k = \frac{2(1-\cos k\pi)}{k\pi}$, which is $\frac{4}{k\pi}$ for odd k and 0 for even. The sum overshoots each edge by 9% of the jump, settling at $\frac{2}{\pi}\mathrm{Si}(\pi) = 1.179$, and the ringing narrows towards the edge as harmonics are added without ever losing height.

5 / THE AMPLIFIER AS A FILTER

A linear amplifier scales and delays each harmonic, and superposition gives the output

defect	on the scope	measure
limited bandwidth f_H	slow rounded edges	rise time $t_r \approx 0.35/f_H$
poor low-frequency response f_L	sag on the flat tops	tilt $\approx \pi f_L/f$
peaking or low phase margin	overshoot and ringing	a damped sine after each edge
slew-rate limit	straight sloped edges	a sine of amplitude A needs $2\pi fA$
non-linear phase	lopsided tops	harmonics delayed by different times

The output is $\sum |H(k\omega)| \sin\big(k\omega t - \arg H\big)/k$ for the amplifier's response H, so a flat gain with non-linear phase still distorts the square wave.

ASIDE / THE SAME MATHS

A string, a cable and an amplifier are all linear systems, so each responds to a sum of sines by summing its responses to each sine.

LINKS TO OTHER SHEETS

The sines are exponentials in *Euler's formula and the complex exponential*; they solve *One equation*, $y'' = \pm y$; the series idea is in *Taylor series*.

Convention. λ wavelength; k wave number; ω angular frequency; v wave speed; f_H, f_L upper and lower corner frequencies; H, amplifier response; Si sine integral.

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DAMPED OSCILLATIONS AND THE THREE KINDS OF NUMBER

One spring, three kinds of number

Remove the decay: circular motion, a straight line and hyperbolic motion emerge from the same spring.

1 Start with A-level SHM

Measure x from equilibrium; $m, k > 0$ are constant.

$$m\ddot{x} = -kx, \qquad \omega_0 = \sqrt{k/m}.$$

$$x(t) = R\cos(\omega_0 t + \phi).$$

Add viscous drag from the treacle

Model the fluid by $F_d = -b\dot{x}$, with constant $b \geq 0$.

$$m\ddot{x} + b\dot{x} + kx = 0, \qquad \gamma = \frac{b}{2m}.$$

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0.$$

2 Remove the common exponential

$$x = \mathrm{e}^{-\gamma t}y \quad \Longrightarrow \quad \ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = \mathrm{e}^{-\gamma t}\big[\ddot{y} + (\omega_0^2 - \gamma^2)y\big].$$

$$\boxed{\ddot{y} = \Delta y, \qquad \Delta = \gamma^2 - \omega_0^2.}$$

The sign of Δ selects the remaining motion.

3 Three reduced motions: $y(t) = A\,C(t) + B\,S(t)$

	Underdamped complex / elliptic	**Critical** dual / parabolic	**Overdamped** split-complex / hyperbolic
condition	$\gamma < \omega_0$	$\gamma = \omega_0$	$\gamma > \omega_0$
generator	$\mathrm{i}^2 = -1$	$\varepsilon^2 = 0,\ \varepsilon \neq 0$	$\mathrm{j}^2 = +1$
$C(t)$	$\cos(\Omega t)$	1	$\cosh(\kappa t)$
$S(t)$	$\sin(\Omega t)/\Omega$	t	$\sinh(\kappa t)/\kappa$
rates r	$-\gamma \pm \mathrm{i}\Omega$	$-\gamma$ (double)	$-\gamma \pm \kappa$

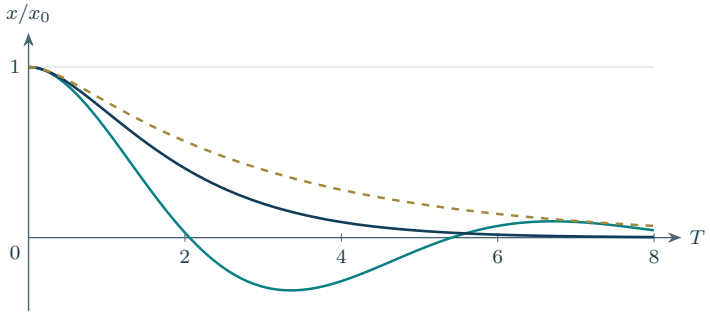
$$\Omega = \sqrt{\omega_0^2 - \gamma^2}, \qquad \kappa = \sqrt{\gamma^2 - \omega_0^2}; \qquad \gamma = \omega_0 \iff b = 2\sqrt{mk}.$$

For $x(0) = x_0$ and $\dot{x}(0) = v_0$, every regime uses the same initial-data formula:

$$\boxed{x(t) = \mathrm{e}^{-\gamma t}\big[x_0 C(t) + (v_0 + \gamma x_0)S(t)\big].}$$

Same release, different drag

Same m, k; $x(0) = x_0 > 0$, $\dot{x}(0) = 0$. Write $T = \omega_0 t$.



Damping ratio $\zeta = \gamma/\omega_0$: under 0.35, critical 1, over 1.6.

Critical: an affine factor

$$x = \mathrm{e}^{-\gamma t}(A + Bt).$$

$$\frac{\sin(\Omega t)}{\Omega} \longrightarrow t \longleftarrow \frac{\sinh(\kappa t)}{\kappa}.$$

As $\Omega, \kappa \to 0$, also $C(t) \to 1$. The reduced motion is the straight line $A + Bt$.

Overdamped still decays

Since $0 < \kappa < \gamma$, both rates below are negative:

$$\mathrm{e}^{-\gamma t}\cosh(\kappa t) = \tfrac{1}{2}\big[\mathrm{e}^{-(\gamma-\kappa)t} + \mathrm{e}^{-(\gamma+\kappa)t}\big].$$

Rotation, shear, boost: the same algebra as velocity addition

Rescale time so that $y_{\theta\theta} = \sigma y$, with $\sigma = -1, 0, +1$. Its state (y, y_θ) has generator

$$Q_\sigma = \begin{pmatrix} 0 & 1 \\ \sigma & 0 \end{pmatrix}, \qquad Q_\sigma^2 = \sigma I.$$

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta, \qquad \mathrm{e}^{\varepsilon\theta} = 1 + \varepsilon\theta, \qquad \mathrm{e}^{\mathrm{j}\theta} = \cosh\theta + \mathrm{j}\sinh\theta.$$

At criticality $Q_0 \neq 0$ but $Q_0^2 = 0$, so $\mathrm{e}^{\theta Q_0} = I + \theta Q_0$: the dual-number case is a shear, and its time dependence is linear.

Physics: OpenStax, *University Physics* 1, §15.5. Geometry: V. V. Kisil (2011), *Ladder Operators in Hypercomplex Mechanics*, arXiv:1103.1120.

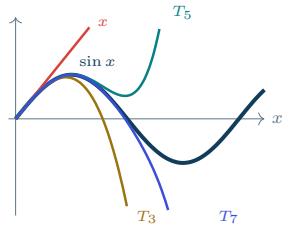
Networked–lattice

Taylor series

A smooth function is the sum of a power series whose coefficients are its derivatives at one point.

1 / THE RULE AND THE PICTURE

Matching derivatives at 0 gives $f(x) = \sum f^{(n)}(0)\,x^n/n!$



$\sin x$ and its polynomials x, T_3, T_5, T_7

A polynomial with the same value and the first n derivatives as f at 0 must have coefficients $f^{(k)}(0)/k!$:

$$f(x) = f(0) + f^{(1)}(0)x + \tfrac{f^{(2)}(0)}{2!}x^2 + \tfrac{f^{(3)}(0)}{3!}x^3 + \cdots$$

For $\sin$ the derivatives at 0 run $0, 1, 0, -1, 0, 1, \ldots$, so $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$, and each extra term hugs the curve for longer.

2 / THE STANDARD SERIES

Seven series that cover most uses

function	series about 0	converges for
e^x	$\sum_{n\ge0} \frac{x^n}{n!}$	all x
$\sin x$	$\sum_{n\ge0} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$	all x
$\cos x$	$\sum_{n\ge0} \frac{(-1)^n x^{2n}}{(2n)!}$	all x
$\frac{1}{1-x}$	$\sum_{n\ge0} x^n$	$\lvert x\rvert < 1$
$\ln(1+x)$	$\sum_{n\ge1} \frac{(-1)^{n+1} x^n}{n}$	$-1 < x \le 1$
$\arctan x$	$\sum_{n\ge0} \frac{(-1)^n x^{2n+1}}{2n+1}$	$\lvert x\rvert \le 1$
$(1+x)^a$	$\sum_{n\ge0} \binom{a}{n} x^n$	$\lvert x\rvert < 1$

The binomial series generalises $(1+x)^n$ to any exponent a, with $\binom{a}{n} = \frac{a(a-1)\cdots(a-n+1)}{n!}$; for $a = -\frac{1}{2}$ it gives $1/\sqrt{1+x} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \cdots$.

3 / HOW ACCURATE

The first omitted term gives the size of the error, for series that alternate

At $x = 1$ the sums of $\sin$ are $T_3 = 0.83333$, $T_5 = 0.84167$ and $T_7 = 0.84147$ against $\sin 1 = 0.841471$, with errors 0.0081, 0.00020 and 0.0000027, each smaller than the first omitted term, $\frac{1}{120}$, $\frac{1}{5040}$ and $\frac{1}{362\,880}$.
Also $e^{0.5} = 1.64872$ against the sum of five terms 1.64844, and $\ln 1.2 = 0.18232$ against $0.2 - 0.02 + 0.00267 - 0.0004 = 0.18227$.

4 / CONVERGENCE / THE RADIUS

A series converges inside a circle in the complex plane that reaches the nearest singularity

The series for $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots$ fails for $\lvert x\rvert \ge 1$ although the function is smooth on the whole real line, because it has poles at $x = \pm i$, at distance 1 from the origin, and the same series for $\arctan x$ converges only for $\lvert x\rvert \le 1$. The Leibniz series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ is the case $x = 1$ and converges slowly: ten terms give 3.0418 and a thousand give 3.1406.

5 / WHAT SERIES ARE FOR

Limits, approximations and integrals that have no closed form

Dividing the series gives limits at once: $\frac{e^x - 1}{x} \to 1$ and $\frac{\ln(1+x)}{x} \to 1$ as $x \to 0$. Keeping two terms gives the small-angle and growth approximations, and integrating term by term handles $\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{x^5}{10} - \cdots$, which no elementary function expresses.
Substituting $i\theta$ into the exponential series proves Euler's formula, and the ratio of successive coefficients is the first hint of the hypergeometric pattern.

ASIDE / ANALYTIC

A function equal to its Taylor series is called analytic, and then its values anywhere are fixed by its behaviour near one point.

LINKS TO OTHER SHEETS

The first terms are used in *Small-angle approximations*; the series for e^x is in *Exponential functions and the number e*; the next sheet is *The gamma function and the beta function*.

Convention. $f^{(n)}(0)$ is the nth derivative at 0; T_n is the Taylor polynomial of degree n; $\binom{a}{n}$ is the generalised binomial coefficient.

COS AND COSH ON ONE PAGE

Circle and hyperbola, wave and chain

One exponential, averaged with its mirror image along the imaginary axis or along the real one.

THE RULE

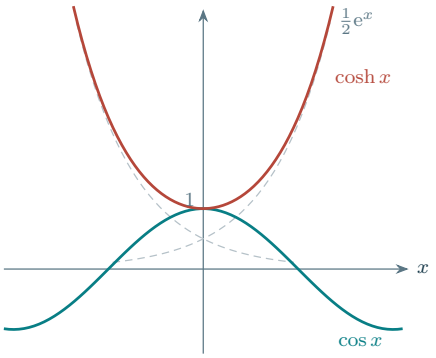
Average e^{z} and e^{-z}

$$\cos x = \frac{\mathrm{e}^{\mathrm{i}x} + \mathrm{e}^{-\mathrm{i}x}}{2}, \qquad \cosh x = \frac{\mathrm{e}^{x} + \mathrm{e}^{-x}}{2}, \qquad \cos(\mathrm{i}x) = \cosh x$$

Euler's formula $\mathrm{e}^{\mathrm{i}x} = \cos x + \mathrm{i}\sin x$ makes $\cos x$ the real part of a point going round the unit circle, and $\cosh x$ is the even part of e^{x}, the same average taken along the real axis. Their series differ only in sign,

$$\cos x = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \dots, \qquad \cosh x = 1 + \tfrac{x^2}{2!} + \tfrac{x^4}{4!} + \dots,$$

so $\cos$ solves $y'' = -y$ and oscillates, while $\cosh$ solves $y'' = y$ and grows; at a general complex point the two mix, $\cos(x + \mathrm{i}y) = \cos x \cosh y - \mathrm{i}\sin x \sinh y$.



Both start at 1 with zero slope; $\cosh$ is the average of the dashed curves.

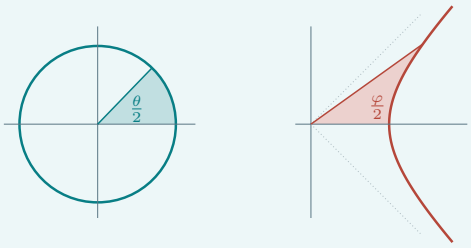
GEOMETRY

Twice the sector area, on two curves

On $x^2 + y^2 = 1$ the point $(\cos\theta, \sin\theta)$ cuts off a sector of area $\frac{\theta}{2}$, and on $x^2 - y^2 = 1$ the point $(\cosh\varphi, \sinh\varphi)$ cuts off a hyperbolic sector of area $\frac{\varphi}{2}$; the same definition of angle gives

$$\cos^2\theta + \sin^2\theta = 1, \qquad \cosh^2\varphi - \sinh^2\varphi = 1.$$

The circular angle returns after 2π, while the hyperbolic angle runs on without limit along a branch that never closes.



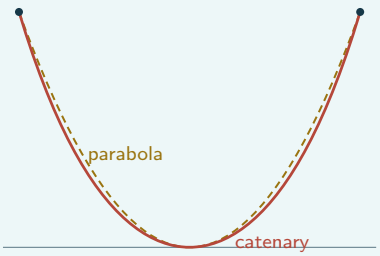
PHYSICAL SHAPES

A wave on a string, a chain hanging from it

A taut string obeys the wave equation, and its simplest solutions are travelling cosines $y = A\cos(kx - \omega t)$ with $\omega = ck$. The same string at rest under its own weight hangs as a catenary,

$$y = a\cosh\frac{x}{a}, \qquad y'' = \frac{1}{a}\sqrt{1 + y'^2},$$

where a is the ratio of horizontal tension to weight per unit length. Inverted, the catenary is the arch that carries its own weight in pure compression, the shape behind Gaudí's hanging models and, weighted, the St Louis Gateway Arch; a parabola through the same ends sags differently because it carries load spread evenly across the span.



Same ends and lowest point: the chain is flatter at the bottom.

FOURIER ANALYSIS

cos builds signals, cosh controls growth

Because $\cos$ repeats, the functions $\cos nx$ are orthogonal on $[-\pi, \pi]$, with $\int \cos mx \cos nx\,\mathrm{d}x = 0$ for $m \neq n$, so any reasonable periodic signal is a sum $\frac{a_0}{2} + \sum(a_n \cos nx + b_n \sin nx)$. $\cosh$ never repeats and cannot serve as such a basis; it enters instead when the exponent turns real, as the Laplace transform e^{-st} becomes the Fourier transform at $s = \mathrm{i}\omega$. Separating Laplace's equation for a potential in a rectangle uses both at once, $u = \sin\frac{n\pi x}{L}\sinh\frac{n\pi y}{L}$, oscillating across and growing up.

BOOSTS

Rotations turn angles, boosts turn rapidities

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad \begin{pmatrix} \cosh\varphi & \sinh\varphi \\ \sinh\varphi & \cosh\varphi \end{pmatrix}$$

A rotation keeps $x^2 + y^2$ and a Lorentz boost keeps $c^2t^2 - x^2$, with $\gamma = \cosh\varphi$ and $v/c = \tanh\varphi$. Rapidities add where speeds do not: $v = \frac{3}{5}c$ has $\varphi = \ln 2$ and $\gamma = \frac{5}{4}$, two such boosts give $\varphi = \ln 4$ and $v = \frac{15}{17}c$, and the Doppler factor is simply $\mathrm{e}^{\varphi} = 2$.

Conventions. $\sinh x = \frac{1}{2}(\mathrm{e}^{x} - \mathrm{e}^{-x})$ and $\sin x = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}x} - \mathrm{e}^{-\mathrm{i}x})$; angles are in radians, φ is the hyperbolic angle or rapidity, and $\tanh\varphi = \sinh\varphi/\cosh\varphi$.

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CLOCK SIGNALS • RAPIDITY • MEDIANTS

Doppler shift to the twin paradox

Two reciprocal Doppler factors; one sum of elapsed times.

1 Start with a receding siren

A siren moves away at speed v through still air. The listener rests in the air. Let u be the sound speed and $P_0 = 1/f_0$ the emitted period.

$$f_{\text{rec}} = f_0 \frac{u}{u+v}, \qquad \frac{P_{\text{rec}}}{P_0} = 1 + \frac{v}{u}.$$

During each emitted period, the siren moves vP_0. That adds flight time vP_0/u:

$$P_{\text{rec}} = P_0 + \frac{vP_0}{u}.$$

For approach, replace $u+v$ by $u-v$, with $v < u$.

2 Light and Bondi's k

For light, periods are measured on the emitter's and receiver's own clocks. Write $\beta = v/c = \tanh\eta$, where η is rapidity. For recession, define

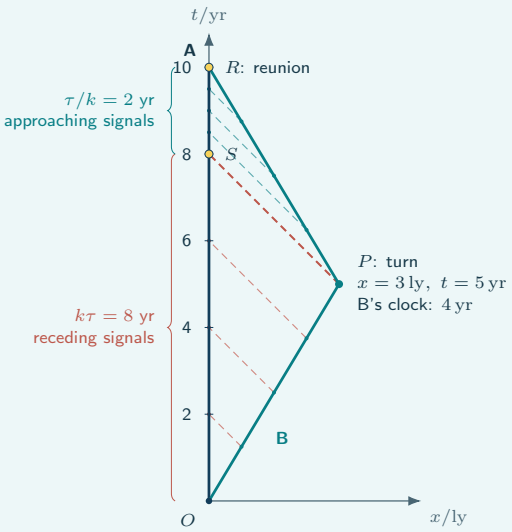
$$k = \frac{P_{\text{rec}}}{P_0} = \sqrt{\frac{1+\beta}{1-\beta}} = e^{\eta} > 1.$$

Motion	P_{rec}/P_0	f_{rec}/f_0
Receding	k	k^{-1}
Approaching	k^{-1}	k

Reversing rapidity exchanges k and $1/k$.

3 One diagram: read the traveller's clock by light

A stays at $x = 0$; B coasts out and back, taking proper time τ per leg. Both send ticks at proper frequency f_0. Work in flat spacetime with an idealised instantaneous turn.



Dashed $45°$ lines carry B's yearly ticks to A. The turnaround message reaches A at S, year 8.

Add the two receiving intervals.

B's leg	B's elapsed time	A receives over
Outbound	τ	$k\tau$
Return	τ	τ/k

$$T_A = k\tau + \frac{\tau}{k}, \qquad T_B = 2\tau.$$

Thus the ratio of their elapsed ages is

$$\boxed{\frac{T_A}{T_B} = \frac{k+k^{-1}}{2} = \cosh\eta = \frac{1}{\sqrt{1-\beta^2}} = \gamma}.$$

The age difference is

$$T_A - T_B = \tau\,\frac{(k-1)^2}{k} \geq 0.$$

The example in the diagram

$$\beta = \frac{3}{5}, \qquad k = 2, \qquad \tau = 4\,\text{yr}.$$

$$T_A = 2(4) + \frac{4}{2} = 10\,\text{yr}, \qquad T_B = 2(4) = 8\,\text{yr}.$$

Earth ages 10 years; the traveller ages 8.

4 Why their views differ

A stays in one inertial frame; B changes to the inbound frame at P. B's received Doppler rate switches at P; A receives the news at S.

	A receives B	B receives A
Receding rate/f_0	$1/k$	$1/k$
For duration	$k\tau$	τ
Approaching rate/f_0	k	k
For duration	τ/k	τ

Durations are measured on the receiver's own clock.

For one tick per year in the example: B receives $2+8=10$ Earth ticks; A receives $4+4=8$ traveller ticks. At reunion both have the complete clock record.

5 The mediant of tick records

Combine actual tick counts N_i and receiving durations D_i by their mediant:

$$\frac{N_1}{D_1}, \frac{N_2}{D_2} \quad\longmapsto\quad \frac{N_1+N_2}{D_1+D_2}.$$

Earth's average received rate is therefore

$$\frac{\overline{f}_{B\to A}}{f_0} = \frac{\tau+\tau}{k\tau+\tau/k} = \frac{2}{k+k^{-1}} = \gamma^{-1}.$$

For the traveller,

$$\frac{\overline{f}_{A\to B}}{f_0} = \frac{\tau/k + k\tau}{2\tau} = \gamma.$$

A: harmonic mean, from equal tick counts. **B: arithmetic mean**, from equal receiving durations.

Reading: OpenStax: sound Doppler; light Doppler; M. Moriconi: Doppler approach, Sections 2 and 6; Einstein Online: travelling twins.

SCALES, SYMMETRY AND WAVES ON ONE PAGE

Reductionism: parts and patterns

From cells and atoms to Fourier series, vibrating drums and the microwave sky.

1 / ASK WHAT COUNTS AS AN ELEMENT

Useful building blocks depend on the question and scale

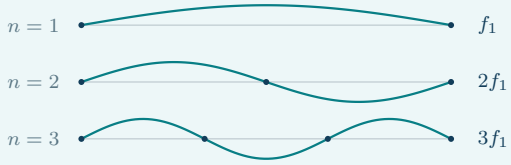
Reductionism studies parts and their interactions; a **normal mode** is a coordinated pattern of the whole.

Perspective	A useful decomposition	Relations still matter
Biology	Cell → organelles → molecules	Transport, reactions, feedback
Chemistry	Molecule → atoms → nuclei and electrons	Bonding and quantum interactions
Cosmology	Cosmic web → galaxies and diffuse matter	Gravity and expansion
Mathematics	Integers → primes; functions → basis functions	Factorisation or linear combination

2 / REDUCE A COMPLICATED SHAPE TO SIMPLE PATTERNS

Fourier series: the string's vocabulary

A string fixed at $x = 0, L$ has nodes at both ends. Its standing-wave shapes are $\sin(n\pi x/L)$.



Each mode oscillates at its own frequency:

$$f_n = \frac{nc}{2L}, \qquad u(x,t) = \sum_{n=1}^{\infty} q_n(t) \sin\frac{n\pi x}{L}.$$

For a fixed initial shape $g(x)$, its Fourier coefficients are

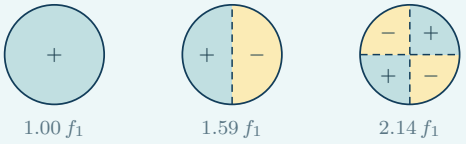
$$a_n = \frac{2}{L}\int_0^L g(x) \sin\frac{n\pi x}{L}\,\mathrm{d}x.$$

Orthogonality isolates each contribution. Initial shape and velocity fix the time coefficients $q_n(t)$.

3 / GIVE THE STRING A SECOND SPATIAL DIMENSION

A drum: geometry chooses the modes

A uniform membrane with a fixed rim has two-dimensional modes; internal nodal lines also stay still.



Signs show displacement. These first three distinct frequencies are not integer harmonics.

$$-\nabla^2\phi = k^2\phi, \qquad f = \frac{ck}{2\pi}.$$

Isoperimetry and Faber–Krahn. For fixed area A, the disk D_A minimizes perimeter P and fundamental frequency:

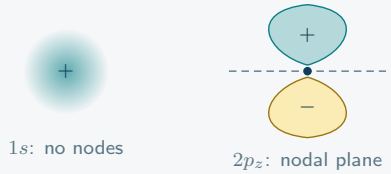
$$P^2 \geq 4\pi A, \qquad f_1(\Omega) \geq f_1(D_A)$$

4 / THE ATOM HAS ITS OWN ALLOWED SPATIAL PATTERNS

Atomic orbitals are quantum modes

Hydrogen's electron is bound by the nucleus's electric potential V. Its allowed states solve the Schrödinger equation:

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2 + V(r)\right]\psi = E\psi.$$



Schematic shapes; colour marks the sign of a real ψ. The probability density is $|\psi|^2$.

Hydrogen separates into radial and angular modes:

$$\psi_{n\ell m}(r,\theta,\varphi) = R_{n\ell}(r)Y_{\ell m}(\theta,\varphi).$$

An energy eigenstate has a time-independent probability density. Transitions give spectral lines $hf = |E_b - E_a|$; many-electron atoms add electron interactions.

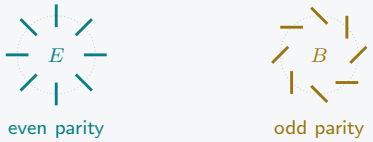
5 / FROM CATALOGUES OF OBJECTS TO MODES OF A FIELD

The microwave sky has a spectrum

Galaxy catalogues describe objects. Cosmic microwave background radiation (CMBR) supplies a field on the sky. Its temperature fluctuations have a spherical Fourier expansion:

$$\Delta T(\theta,\varphi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} a_{\ell m}Y_{\ell m}(\theta,\varphi).$$

Hydrogen uses the same $Y_{\ell m}$. CMB acoustic peaks record oscillations in the early plasma of light and matter.



Polarization separates into **E and B modes** with even and odd parity under reflection (schematic above).

Lensing converts E into B; primordial gravitational waves would also generate B. Separating lensing and foregrounds is essential when testing early-Universe models.

The shared idea. Choose variables and constraints, find the allowed patterns, and determine their weights. Mode decomposition is a change of description within a linear model; an explanation also needs interactions, organisation and evidence.

Conventions. c is the classical wave speed; drums have the same uniform tension and mass per area, with fixed rims. Atomic formulas use the nonrelativistic fixed-nucleus hydrogen model.

Sources. MIT: Fourier and standing waves; MIT: orbitals; Daners: Faber–Krahn; WMAP: angular spectrum; Hanson et al.: lensing B modes.

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MEANS OF AN EXPONENTIAL PAIR ON ONE PAGE

sec and sech are harmonic means

When two numbers multiply to 1, their harmonic mean is the reciprocal of their arithmetic mean.

THE RULE

Four means of two numbers

The harmonic mean is the reciprocal of the average of the reciprocals, the right mean when reciprocals are what add, as times do over equal distances. For two numbers a and b,

$$\mathrm{AM} = \frac{a+b}{2}, \quad \mathrm{GM} = \sqrt{ab}, \quad \mathrm{HM} = \frac{2}{\frac{1}{a}+\frac{1}{b}} = \frac{2ab}{a+b}, \quad \mathrm{RMS} = \sqrt{\frac{a^2+b^2}{2}},$$

and the third is tied to the first two by

$$\mathrm{AM} \times \mathrm{HM} = \mathrm{GM}^2, \qquad \text{so} \qquad ab = 1 \implies \mathrm{HM} = \frac{1}{\mathrm{AM}}$$

Both exponential pairs, $\mathrm{e}^{x}, \mathrm{e}^{-x}$ and $\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$, have product 1, so their geometric mean is 1 and their harmonic mean is the reciprocal of $\cosh x$ or of $\cos x$.

speeds 30 and 60 over equal distances

RMS	47.4
AM	45
GM	42.4
HM	40

The true average speed is the harmonic mean, 40; always HM ≤ GM ≤ AM ≤ RMS for positive numbers.

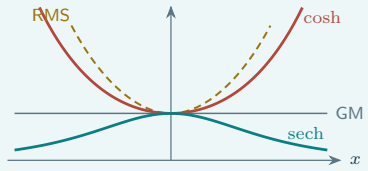
THE REAL PAIR $\mathrm{e}^{x}, \mathrm{e}^{-x}$

Hyperbolic: the order holds

$$\mathrm{AM} = \cosh x, \quad \mathrm{GM} = 1, \quad \mathrm{HM} = \operatorname{sech} x, \quad \mathrm{RMS} = \sqrt{\cosh 2x}$$

Both numbers are positive, so the chain holds, with equality only at $x = 0$:

$$\operatorname{sech} x \le 1 \le \cosh x \le \sqrt{\cosh 2x}.$$



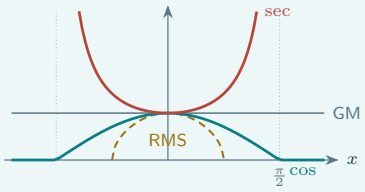
THE IMAGINARY PAIR $\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$

Circular: the same formulas, a new order

$$\mathrm{AM} = \cos x, \quad \mathrm{GM} = 1, \quad \mathrm{HM} = \sec x, \quad \mathrm{RMS} = \sqrt{\cos 2x}$$

The pair is not positive, so on $|x| < \frac{\pi}{2}$ the harmonic and arithmetic means swap places, and the RMS is real only for $|x| \le \frac{\pi}{4}$:

$$\sqrt{\cos 2x} \le \cos x \le 1 \le \sec x.$$



THE IDENTITIES BEHIND THE MEANS

The Pythagorean identities are statements about means

Two exact relations hold for any pair: $\mathrm{AM}^2 - \mathrm{GM}^2 = \left(\frac{a-b}{2}\right)^2$ and $\mathrm{RMS}^2 = \mathrm{AM}^2 + \left(\frac{a-b}{2}\right)^2$. The half-difference of the real pair is $\sinh x$ and of the imaginary pair is $\mathrm{i}\sin x$, so the two relations become the familiar identities.

	$\mathrm{e}^{x}, \mathrm{e}^{-x}$	$\mathrm{e}^{\mathrm{i}x}, \mathrm{e}^{-\mathrm{i}x}$
half-difference $\frac{a-b}{2}$	$\sinh x$	$\mathrm{i}\sin x$
$\mathrm{AM}^2 - \mathrm{GM}^2 = \left(\frac{a-b}{2}\right)^2$	$\cosh^2 x - 1 = \sinh^2 x$	$\cos^2 x - 1 = -\sin^2 x$
$\mathrm{RMS}^2 = \mathrm{AM}^2 + \left(\frac{a-b}{2}\right)^2$	$\cosh 2x = \cosh^2 x + \sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x$
$\mathrm{AM} \times \mathrm{HM} = \mathrm{GM}^2$	$\cosh x \operatorname{sech} x = 1$	$\cos x \sec x = 1$

A BRIDGE BETWEEN THE PAIRS

The Gudermannian swaps the roles

The angle $\operatorname{gd} x = \arctan(\sinh x)$ gives $\sec(\operatorname{gd} x) = \cosh x$ and $\cos(\operatorname{gd} x) = \operatorname{sech} x$, so on this angle the circular harmonic mean equals the hyperbolic arithmetic mean. Its inverse, $\int_0^{\varphi} \sec t \, \mathrm{d}t = \ln(\sec\varphi + \tan\varphi)$, is Mercator's spacing of latitudes.

WHERE THE HARMONIC MEAN APPEARS

The shape of a solitary wave

A shallow-water soliton has the profile $A \operatorname{sech}^2\left(\frac{x-ct}{w}\right)$ and an optical-fibre pulse keeps the shape sech, both built from the harmonic mean of e^{x} and e^{-x}; and $\frac{\mathrm{d}}{\mathrm{d}x}\tanh x = \operatorname{sech}^2 x$ gives the logistic curve its bell-shaped growth rate.

Conventions. $\sec = 1/\cos$, $\operatorname{sech} = 1/\cosh$ and gd is the Gudermannian; the means extend to complex numbers as formulas, but HM ≤ GM ≤ AM ≤ RMS is guaranteed only for positive numbers.

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TANH AND TAN AS RATIOS OF MEANS ON ONE PAGE

$\tanh = \sinh \cdot \operatorname{sech}$

A half-difference times a harmonic mean, which is why speeds compose by Einstein's rule and slopes by the tangent rule.

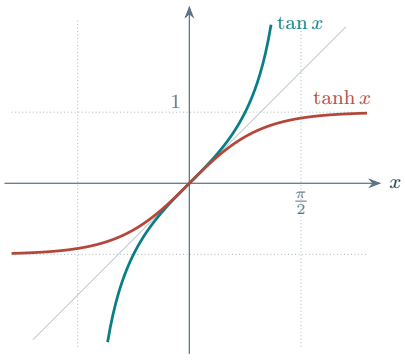
THE RULE

A relative difference of an exponential pair

For the pair $a = \mathrm{e}^{x}$, $b = \mathrm{e}^{-x}$, with product 1, the arithmetic mean is $\cosh x$, the harmonic mean is $\operatorname{sech} x = 1/\cosh x$, and the half-difference $\frac{a-b}{2}$, which is the arithmetic mean of a and $-b$, is $\sinh x$. Their product is

$$\tanh x \;=\; \underbrace{\frac{a-b}{2}}_{\sinh x} \times \underbrace{\frac{2ab}{a+b}}_{\operatorname{sech} x} \;=\; \frac{a-b}{a+b} \;=\; \frac{\mathrm{e}^{2x}-1}{\mathrm{e}^{2x}+1}$$

the relative difference of the pair, which can never exceed 1 in size. The imaginary pair $\mathrm{e}^{\pm \mathrm{i}x}$ gives the circular twin, $\tan x = \sin x \sec x = -\mathrm{i}\,\frac{a-b}{a+b}$, and here the denominator $a+b = 2\cos x$ can vanish, so $\tan$ runs off to infinity at $\pm\frac{\pi}{2}$.



Both leave the origin along $y = x$; $\tanh$ levels off at ± 1, $\tan$ escapes.

ADDITION AND DOUBLING

One sign separates the two addition laws

Writing $A = \mathrm{e}^{2x}$ and $B = \mathrm{e}^{2y}$ makes $\tanh x = \frac{A-1}{A+1}$, and since exponents add, $\tanh(x+y) = \frac{AB-1}{AB+1}$; clearing fractions turns this into the addition law, and the circular case follows from $\tan x = -\mathrm{i}\tanh(\mathrm{i}x)$:

$$\tanh(x + y) \;=\; \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}, \qquad \tan(x + y) \;=\; \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

Setting $y = x$ gives the double-angle forms, where the same sign decides whether the result is bounded:

$$\tanh 2x = \frac{2t}{1+t^2} \;\text{ with } t = \tanh x, \qquad \tan 2x = \frac{2t}{1-t^2} \;\text{ with } t = \tan x.$$

The first never exceeds 1 in size for any t; the second blows up as $t \to 1$, the moment the doubled angle reaches a right angle.

EINSTEIN'S RULE MOTIVATED

Doppler factors multiply, so speeds add as $\tanh$

Bondi's k-calculus starts from the one thing two moving observers can compare directly: light pulses sent every second arrive every k seconds. Chaining two observers multiplies their k factors, so $\varphi = \ln k$ adds, and the speed between them works out as the relative difference of the pair $k, 1/k$:

$$\frac{v}{c} = \frac{k - k^{-1}}{k + k^{-1}} = \tanh \varphi.$$

Multiplying k's is therefore adding rapidities, and the $\tanh$ addition law is exactly Einstein's rule $u \oplus v = \frac{u+v}{1+uv/c^2}$. The double-angle form $\frac{2\beta}{1+\beta^2}$ is the speed reached by two equal boosts, and it is also the Pythagorean ratio $\frac{2pq}{p^2+q^2}$ when $\beta = \frac{p}{q}$.

k	$\varphi = \ln k$	v/c
2	0.693	$\frac{3}{5}$
$2 \times 2 = 4$	1.386	$\frac{15}{17}$
3	1.099	$\frac{4}{5}$
$2 \times 3 = 6$	1.792	$\frac{35}{37}$

$\frac{3}{5} \oplus \frac{3}{5} = \frac{15}{17}$ and $\frac{3}{5} \oplus \frac{4}{5} = \frac{35}{37}$, both Pythagorean speeds.

WHERE THE TAN DOUBLING IS USED

Reflections and half-angle substitution

A mirror tilted by θ turns a reflected ray by 2θ, so slopes double by $\frac{2t}{1-t^2}$. The substitution $t = \tan\frac{\theta}{2}$ gives $\sin\theta = \frac{2t}{1+t^2}$, $\cos\theta = \frac{1-t^2}{1+t^2}$ and $\tan\theta = \frac{2t}{1-t^2}$, turning any rational function of $\sin$ and $\cos$ into an integrable rational function of t, and with $t = \frac{p}{q}$ it produces the triple $(q^2 - p^2,\ 2pq,\ p^2 + q^2)$.

WHERE TANH SATURATES USEFULLY

Bounded responses

Beyond relativity, $\tanh$ describes anything that responds in proportion at first and then saturates: the mean-field magnetisation $m = \tanh\left(\frac{Jm+h}{kT}\right)$, the dispersion of water waves $\omega^2 = gk\tanh(kh)$ between deep and shallow water, and the activation of artificial neurons, where $\tanh x = 2\sigma(2x) - 1$ is a rescaled logistic curve.

Conventions. $\operatorname{sech} = 1/\cosh$ and $\sec = 1/\cos$; k is the relativistic Doppler factor, φ the rapidity and $\beta = v/c$; σ in the last box is the logistic function $1/(1+\mathrm{e}^{-x})$.

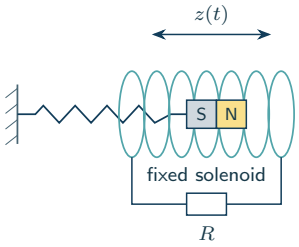
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MOTION, INDUCTION AND THE QUADRATIC MEAN

RMS current from an oscillating magnet

Motion changes flux linkage; the mean square of current measures heating.

A magnet on a spring



Motion ⟶ changing linkage ⟶ current

The magnet's flux linkage is $\Lambda_m(z) = \sum_n \Phi_{m,n}(z)$; it equals $N\Phi_m$ if all turns link the same flux.

$$z(t) = z_0 + A\cos\omega t, \qquad f = \frac{\omega}{2\pi},$$

$$\mathcal{E}_m(t) = -\frac{d\Lambda_m}{dt} = -\Lambda_m'(z(t))\,\dot z(t).$$

Close the loop through total resistance R (coil plus load). Neglecting inductance, $i \simeq \mathcal{E}_m/R$ [1]. Use steady driven motion, or one cycle with weak damping.

A. Nearly constant flux gradient

Over a small range about z_0, approximate

$$\Lambda_m(z) \simeq \Lambda_0 + g(z - z_0), \qquad g = \Lambda_m'(z_0).$$

Since $\dot z = -A\omega\sin\omega t$,

$$i(t) \simeq \frac{gA\omega}{R}\sin\omega t, \qquad I_p = \frac{|g|A\omega}{R}.$$

This current is sinusoidal at electrical angular frequency $\Omega = \omega$. Hence

$$\boxed{I_{\rm rms} = \frac{|g|A\omega}{\sqrt{2}\,R}.}$$

B. Centred in a symmetric coil

With $z_0 = 0$, symmetry gives $\Lambda_m'(0) = 0$. The leading position-dependent term is quadratic:

$$\Lambda_m(z) \simeq \Lambda_0 + \tfrac{1}{2}\kappa z^2, \qquad \kappa = \Lambda_m''(0).$$

Therefore, to leading order,

$$i(t) \simeq \frac{\kappa A^2\omega}{2R}\sin 2\omega t, \qquad \boxed{I_{\rm rms} = \frac{|\kappa|A^2\omega}{2\sqrt{2}\,R}.}$$

Here $\Omega = 2\omega$: the electrical frequency is $2f$. SHM alone does not fix the current waveform.

RMS: square, average in time, then take the root

RMS is the **quadratic mean**. With electrical period T_e, the weights are fractions of time:

$$I_{\rm rms}^2 = \frac{1}{T_e}\int_0^{T_e} i^2(t)\,dt \simeq \sum_j w_j i_j^2, \quad w_j = \frac{\Delta t_j}{T_e}, \quad \sum_j w_j = 1.$$

For $i(t) = I_p\sin\Omega t$, the complementary shares $p = \sin^2\Omega t$, $q = \cos^2\Omega t$ obey $p + q = 1$. A quarter-cycle shift swaps them, so their cycle averages are equal:

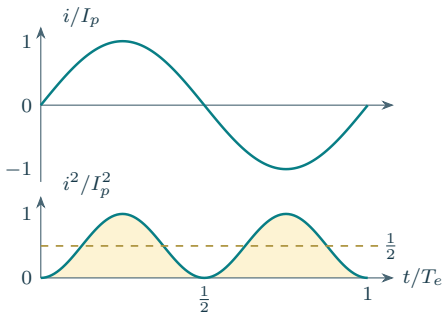
$$\langle p\rangle = \langle q\rangle = \tfrac{1}{2}, \qquad \boxed{I_{\rm rms} = \frac{I_p}{\sqrt{2}}.}$$

The signed mean current is zero. Heating depends on the mean square:

$$\boxed{\overline{P} = R\,\langle i^2\rangle = RI_{\rm rms}^2.}$$

Thus $I_{\rm rms}$ is the equivalent DC heating current [2].

One electrical cycle



Top: $\langle i\rangle = 0$.
Bottom: $\langle i^2\rangle = I_p^2/2$.

Example: the constant-gradient case

Take $g = 0.50\ \mathrm{Wb\,m^{-1}}$, $A = 0.020\ \mathrm{m}$, $f = 2.0\ \mathrm{Hz}$ and $R = 10\ \Omega$. Then

$$\omega = 2\pi f, \qquad \mathcal{E}_p = |g|A\omega = 0.126\ \mathrm{V},$$

$$I_p = 12.6\ \mathrm{mA}, \qquad \boxed{I_{\rm rms} = 8.9\ \mathrm{mA}.}$$

The average heating is $\overline{P} = 0.79\ \mathrm{mW}$. The gradient g already includes all coil turns.

Use measured samples for a distorted trace

With sufficiently dense, equally spaced samples over whole electrical cycles,

$$\boxed{I_{\rm rms} \simeq \sqrt{\frac{i_1^2 + i_2^2 + \cdots + i_n^2}{n}}.}$$

Measure V_j across a series sense resistor R_s and use $i_j = V_j/R_s$. Moving-magnet coils can produce pulses [3]; the $I_p/\sqrt{2}$ shortcut requires a sinusoidal **current** trace.

With inductance [2]. For a steady sinusoidal emf of peak $\mathcal{E}_p$ and angular frequency Ω, with constant R, L, $I_{\rm rms} = \mathcal{E}_p/\sqrt{2[R^2 + (\Omega L)^2]}$.

Sources. [1] OpenStax, *University Physics 2*, §13.1. [2] §15.4 (RMS/power) and §15.3 (impedance). [3] A. Singh et al., *Electromagnetic induction and damping*, physics/0111016 (2001).

ROOT MEAN SQUARE IN A WIRE AND IN A GAS ON ONE PAGE

Effective value and temperature

Wherever energy goes as a square, the root mean square is the honest average, and it earns a name of its own.

THE RULE

Square, average, then take the root

An alternating current and the velocity of a gas molecule both average to zero, yet both carry energy, because heating in a resistor goes as I^2 and kinetic energy goes as v^2. Averaging the square keeps the energy and the square root restores the units:

$$x_{\mathrm{rms}} = \sqrt{\langle x^2 \rangle}, \qquad P = I_{\mathrm{rms}}^2 R \ \text{in the wire}, \qquad \tfrac{1}{2} m v_{\mathrm{rms}}^2 = \tfrac{3}{2} kT \ \text{in the gas}$$

In the wire the root mean square current is called the **effective value**, the steady direct current that would heat the same resistor at the same rate; older British texts call it the *virtual* value and German ones the *Effektivwert*. In the gas the same quantity, rescaled by $m/3k$, is called the **temperature**.

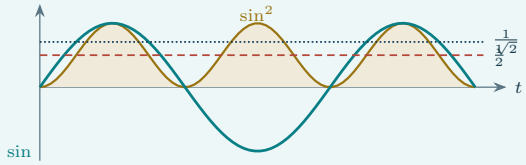
IN THE WIRE

Half the peak squared

For $I = I_0 \sin \omega t$, the square $\sin^2 \omega t = \tfrac{1}{2}(1 - \cos 2\omega t)$ swings evenly about $\tfrac{1}{2}$, so

$$I_{\mathrm{rms}} = \frac{I_0}{\sqrt{2}}, \qquad P = \tfrac{1}{2} I_0^2 R.$$

UK mains is quoted as $230\,\mathrm{V}$ effective, which means peaks of $230\sqrt{2} \approx 325\,\mathrm{V}$.



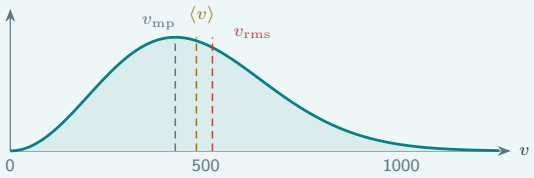
IN THE GAS

Three speeds, one temperature

The Maxwell distribution gives three typical speeds,

$$v_{\mathrm{mp}} = \sqrt{\frac{2kT}{m}}, \quad \langle v \rangle = \sqrt{\frac{8kT}{\pi m}}, \quad v_{\mathrm{rms}} = \sqrt{\frac{3kT}{m}},$$

and only v_{rms} fixes the mean kinetic energy. Nitrogen at $300\,\mathrm{K}$ has 422, 476 and $517\,\mathrm{m/s}$.



SIDE BY SIDE

The same arithmetic in two settings

	alternating current	ideal gas
quantity averaging to zero	current $I(t)$ over a cycle	velocity $\vec{v}$ over the molecules
energy that goes as a square	heating $I^2 R$	kinetic energy $\tfrac{1}{2} m v^2$
the root mean square	$I_{\mathrm{rms}} = I_0/\sqrt{2}$	$v_{\mathrm{rms}} = \sqrt{3kT/m}$
its name	effective (heating, virtual) value	temperature, through $\tfrac{3}{2} kT$
the plain average of the size	rectified mean $\tfrac{2}{\pi} I_0$	mean speed $\langle v \rangle$
ratio of RMS to that average	form factor $\tfrac{\pi}{2\sqrt{2}} = 1.111$	$\sqrt{3\pi/8} = 1.085$

The form factor is why a cheap meter that averages the rectified signal and multiplies by 1.111 reads true only for a pure sine wave, whereas a *true RMS* meter squares, averages and roots, just as the gas does for its temperature.

WHERE THE TWO MEET

A resistor's noise is its temperature

The electrons in any resistor jostle thermally, producing a fluctuating voltage whose effective value is set by temperature alone, the Johnson–Nyquist result

$$V_{\mathrm{rms}} = \sqrt{4kTR\,\Delta f}\,,$$

about $0.41\,\mu\mathrm{V}$ for $1\,\mathrm{k\Omega}$ at $300\,\mathrm{K}$ over a $10\,\mathrm{kHz}$ bandwidth, so an RMS voltmeter can serve as a thermometer.

HOW FAR THE ANALOGY GOES

Temperature is more than an RMS

The link $\tfrac{1}{2} m v_{\mathrm{rms}}^2 = \tfrac{3}{2} kT$ holds for translational motion in an ideal gas, where equipartition gives $\tfrac{1}{2} kT$ to each quadratic term in the energy; temperature itself is defined thermodynamically, through how entropy changes with energy, and stays meaningful for solids, radiation and spins, where no single RMS speed exists.

Conventions. $\langle\ \rangle$ is an average over a cycle or over the molecules, $k = 1.381 \times 10^{-23}\,\mathrm{J/K}$ is Boltzmann's constant, m the molecular mass and Δf the bandwidth of the measurement.

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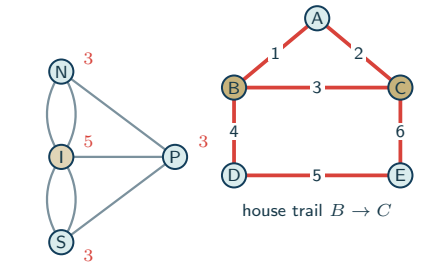
GRAPH THEORY LADDER 4 OF 9

Euler trails and circuits

A walk using every edge once exists when at most two vertices have odd degree.

1 / THE RULE AND THE PICTURE

A connected graph has an Euler circuit exactly when every degree is even



Königsberg, degrees $3, 3, 5, 3$

house trail $B \to C$

Euler asked in 1736 whether one could cross each of the seven bridges of Königsberg exactly once, and his answer is

> Euler circuit $\iff$ connected, all degrees even

with an Euler trail from one vertex to another when exactly two degrees are odd.

2 / THE THREE CASES

Count the vertices of odd degree

degrees	what happens	example
every degree even	an Euler circuit exists, starting anywhere and returning	K_5, every degree 4
exactly two odd degrees	an Euler trail exists, from one odd vertex to the other	the house, from B to C
more than two odd degrees	no Euler trail at all	Königsberg, four odd vertices; K_4, every degree 3

3 / WHY THE RULE WORKS

Each pass through a vertex uses two edges

Whenever a trail passes through a vertex it arrives along one edge and leaves along another, so the edges at that vertex are used in pairs, and an inner vertex needs an even degree.
Only the first and last vertices of an open trail can have odd degree, and a closed circuit has none.
Königsberg has four odd vertices, with degrees $3, 3, 5, 3$, so no route exists, and one needs to add at least two bridges to pair the odd vertices before a circuit appears.

4 / FINDING A TRAIL

Walk until stuck and splice the loops in

For the house graph start at the odd vertex B and walk B, A, C, B, D, E, C, which uses all six edges and ends at the other odd vertex C, as the numbered edges show.
The general method of Hierholzer starts anywhere, walks without repeating an edge until it is stuck, which in an even graph can only happen at the start, and then splices a further loop into the first unused vertex until every edge is used.
The method is fast, so Euler questions are easy.

5 / TRY THESE

Count the odd vertices first

(a) Does K_5 have an Euler circuit? Every degree is 4, so yes.
(b) Does the cube graph, where every vertex has degree 3, have an Euler trail? No, since it has eight odd vertices.
(c) A postman must walk every street of a graph and return: if it has $2k$ odd vertices he repeats at least k streets, which is the Chinese postman problem.

ASIDE / EDGES AGAINST VERTICES

An Euler trail visits every edge once and a Hamilton cycle visits every vertex once, and the first has a simple test while the second has none.

LINKS TO OTHER SHEETS

The degree rule is on the first sheet, and the next sheet turns to Hamilton cycles on the same house graph.

Convention. A trail repeats no edge; an Euler circuit is a closed trail through every edge; the degrees $3, 3, 5, 3$ are those of the four land masses of Königsberg.

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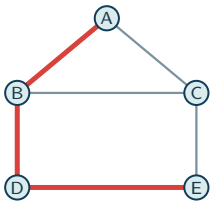
GRAPH THEORY LADDER 5 OF 9

Hamilton cycles

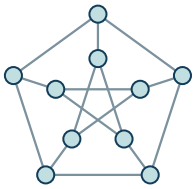
A cycle through every vertex once has no simple test, and the candidate tours explode in number.

1 / THE RULE AND THE PICTURE

A Hamilton cycle visits every vertex exactly once and returns to the start



one Hamilton cycle



Petersen graph, none

Hamilton devised a puzzle in 1857 asking for such a cycle on the dodecahedron, and there is no degree test that decides the question in general; two graphs of the same size can look alike and differ in having one:

Hamilton cycle: n vertices, n edges, no repeats

The house graph has exactly one, and the Petersen graph, drawn on the right, has none.

2 / EULER AND HAMILTON COMPARED

The two questions sound alike and behave very differently

	Euler	Hamilton
visits	every edge once (Euler)	every vertex once (Hamilton)
test	degrees all even: easy	no simple test known
house graph	a trail from B to C	the cycle A, B, D, E, C
hard version	none	the travelling salesman problem

3 / SUFFICIENT CONDITIONS

Dirac and Ore guarantee a cycle once degrees are large

Dirac's theorem says that a simple graph on $n \geq 3$ vertices in which every degree is at least $n/2$ has a Hamilton cycle, and Ore's theorem weakens this to $\deg u + \deg v \geq n$ for every non-adjacent pair.
For $K_{3,3}$ every degree is $3 \geq 3$, so a cycle exists, and indeed there are 6. The house graph has minimum degree $2 < 2.5$, so Dirac says nothing, yet it has a Hamilton cycle, which shows the conditions are sufficient and not necessary.

4 / A GRAPH WITHOUT ONE

The Petersen graph is the classic obstruction

The Petersen graph has 10 vertices, every degree 3, which is large enough to look promising, and an exhaustive search over all orderings shows that it has no Hamilton cycle, although it has Hamilton paths.
On the cube graph Q_3 there are 6 cycles, one of which is the Gray code $000, 001, 011, 010, 110, 111, 101, 100$, in which consecutive codes differ in one bit.

5 / COUNTING TOURS

The complete graph has $(n-1)!/2$ Hamilton cycles

Fixing a start and ignoring direction, K_5 has $4!/2 = 12$ cycles, K_{10} has $9!/2 = 181\,440$, and K_{20} has $19!/2 \approx 6.1 \times 10^{16}$, so a salesman visiting 20 towns cannot simply try every route.
The travelling salesman problem asks for the shortest cycle in a weighted complete graph, and finding a Hamilton cycle at all is among the problems with no known fast method.

6 / TRY THESE

Decide and check

(a) Find a Hamilton cycle of K_4: $1, 2, 3, 4$ returning to 1, and there are 3 in all.
(b) Does the star $K_{1,3}$ have one? No, since a leaf has degree 1 and a cycle needs degree 2 at every vertex.
(c) Is a graph with a vertex whose removal disconnects it Hamiltonian? No, a cycle through all vertices would survive the removal of any one vertex.

ASIDE / A HARD PROBLEM

Deciding whether a graph has a Hamilton cycle is NP-complete, which was shown by Karp in 1972.

LINKS TO OTHER SHEETS

The previous sheet gives the Euler test; the weighted networks sheet meets the travelling salesman problem again beside the easy shortest path problem.

Convention. A Hamilton cycle has n edges on n vertices; K_n is the complete graph; Q_3 the cube graph; Dirac and Ore give sufficient conditions.

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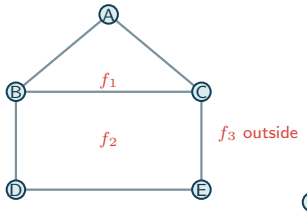
GRAPH THEORY LADDER 6 OF 9

Planar graphs and Euler's formula

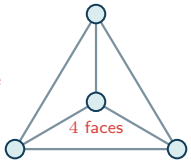
A graph drawn in the plane without crossings satisfies $V - E + F = 2$, which limits how many edges it can have.

1 / THE RULE AND THE PICTURE

Vertices minus edges plus faces equals two for a connected planar graph



$5 - 6 + 3 = 2$



K_4: $4 - 6 + 4 = 2$

A face is a region of the plane cut out by the drawing, counting the outside, and for a connected graph drawn without crossings

$$V - E + F = 2$$

The house has 5 vertices, 6 edges and 3 faces (the roof, the room and the outside), and $5 - 6 + 3 = 2$.

2 / WHY IT HOLDS

Build the graph one edge at a time

A tree has $E = V - 1$ and a single face, the outside, so $V - E + F = V - (V - 1) + 1 = 2$. Adding an edge that closes a cycle splits one face into two, raising E and F by 1 each, and adding an edge to a new vertex raises V and E by 1 each, so $V - E + F$ never changes from 2.

3 / HOW MANY EDGES A PLANAR GRAPH CAN HAVE

At most $3V - 6$, and at most $2V - 4$ without triangles

Each face is bounded by at least 3 edges and each edge borders two faces, so $3F \le 2E$, and combining with $V - E + F = 2$ gives $E \le 3V - 6$. The graph K_5 has $10 > 3 \cdot 5 - 6 = 9$ edges, so it cannot be drawn without crossings, and $K_{3,3}$ has no triangles, so each face has at least 4 edges, giving $E \le 2V - 4 = 8 < 9$, which rules it out too.
Kuratowski showed that a graph is planar exactly when it contains no subdivision of K_5 or $K_{3,3}$.

4 / THE FIVE REGULAR SOLIDS

Every one of them gives 2

solid	V	E	F	$V - E + F$
tetrahedron	4	6	4	2
cube	8	12	6	2
octahedron	6	12	8	2
dodecahedron	20	30	12	2
icosahedron	12	30	20	2

A polyhedron is a planar graph, drawn by flattening it, so the formula holds for each of them.

5 / ON OTHER SURFACES

A doughnut changes the answer to 0

Drawn on a torus without crossings, with faces that are discs, a graph satisfies $V - E + F = 0$, and both K_5 and $K_{3,3}$ can be drawn that way. The complete graph K_7 fits on the torus with 14 triangular faces, giving $7 - 21 + 14 = 0$.

6 / TRY THESE

Use the inequalities

(a) A connected planar graph has $V = 6$ and $E = 9$: how many faces? $F = 2 - 6 + 9 = 5$.
(b) Show that the Petersen graph is not planar: it has $V = 10$, $E = 15$ and girth 5, so $F = 7$ but $5F \le 2E = 30$ gives $F \le 6$, a contradiction.
(c) What is the largest E for a planar graph on 8 vertices? $3 \cdot 8 - 6 = 18$.

ASIDE / FACES AND CYCLES

The count $E - V + 1$ of independent cycles from the tree sheet equals $F - 1$, the number of inner faces, which is the same formula read twice.

LINKS TO OTHER SHEETS

The next sheet colours the faces and vertices of planar graphs, where the four colour theorem lives.

Convention. V vertices, E edges, F faces including the outside; the bound $E \le 3V - 6$ needs $V \ge 3$; $K_{3,3}$ has no triangles.

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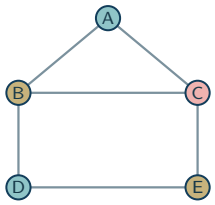
GRAPH THEORY LADDER 7 OF 9

Colouring and bipartite graphs

Neighbours get different colours, and two colours suffice exactly when there is no odd cycle.

1 / THE RULE AND THE PICTURE

The chromatic number is the fewest colours that keep every edge two-coloured



house: 3 colours



$K_{3,3}$: 2 colours

A proper colouring gives adjacent vertices different colours, and the smallest number needed is the chromatic number χ. A graph can be coloured with two colours exactly when it is bipartite:

$$\chi \le 2 \iff \text{no odd cycle}$$

The house needs 3 because of its triangle, and $K_{3,3}$ needs only 2.

2 / KNOWN VALUES

The chromatic number of familiar graphs

graph	χ	reason
path P_n, $n \ge 2$	2	alternate the colours
even cycle C_{2m}	2	alternate round the ring
odd cycle C_{2m+1}	3	two colours fail at the closing edge
complete graph K_n	n	every pair adjacent
complete bipartite $K_{m,n}$	2	one colour for each side
house graph	3	the triangle needs three
Petersen graph	3	two colours fail, three succeed
any planar graph	at most 4	the four colour theorem

3 / COUNTING COLOURINGS

The chromatic polynomial counts them for every number of colours

Colouring the triangle ABC first, then D and E, shows that the house has $k(k-1)(k-2)\big[(k-1)+(k-2)^2\big]$ proper colourings with k colours, which is 0 for $k=2$, 18 for $k=3$ and 168 for $k=4$, and an exhaustive count confirms both numbers.
The 5-cycle has $(k-1)^5-(k-1)$, giving 0 for $k=2$ and 30 for $k=3$, in agreement with the rule that an odd cycle needs three colours.

4 / A SIMPLE METHOD AND ITS LIMIT

Greedy colouring uses at most $\Delta+1$ colours

Colour the vertices in any order, giving each the first colour not already used by a neighbour, and a vertex with Δ neighbours can never be forced beyond $\Delta+1$ colours, so the house with $\Delta=3$ needs at most 4.
The bound is not always tight, since $\chi=3$ here, and Brooks' theorem sharpens it to Δ for every connected graph except complete graphs and odd cycles.
Deciding whether a graph can be coloured with three colours is NP-complete, so no fast method is known.

5 / MAPS AND TIMETABLES

Colouring is scheduling in disguise

To colour a map, make each country a vertex and join countries that share a border, which gives a planar graph, and the four colour theorem of Appel and Haken (1976) says four colours always suffice.
For a timetable, make each exam a vertex, join two exams that share a student, and read the colours as time slots: exams with the house-graph pattern of clashes need three slots.

6 / TRY THESE

Decide the chromatic number

(a) What is $\chi(C_7)$? It is 3, since 7 is odd. **(b)** Can the cube graph be 2-coloured? Yes, by the parity of the number of 1s in the code, so it is bipartite. **(c)** Why does K_4 need four colours? Every vertex is adjacent to every other.

ASIDE / BIPARTITE MEANS TWO SIDES

Matching problems, such as pairing people with jobs, live on bipartite graphs, where Hall's marriage theorem tells when a perfect matching exists.

LINKS TO OTHER SHEETS

Planar graphs, where four colours suffice, are on the Euler formula sheet; the next sheet counts spanning trees with a matrix.

Convention. χ chromatic number; Δ the largest degree; a proper colouring gives adjacent vertices different colours; the house colouring in the picture uses 3 colours.

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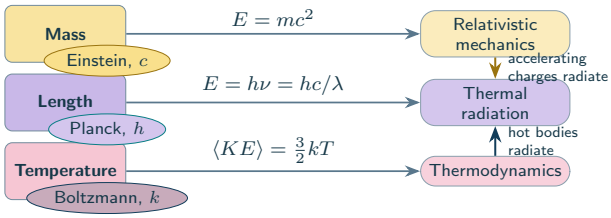
ENERGY ON ONE PAGE

A photon's journey: from $E = mc^2$ to warm air

Three constants, c, h and k, turn mass, frequency and temperature into energy along the way.

1 / THE RULE AND THE PICTURE

Each constant converts one kind of quantity into energy

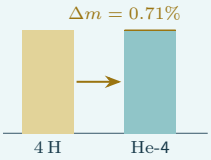


what 1 eV is, through each constant	
mass, $m = E/c^2$	1.78×10^{-36} kg
frequency, $\nu = E/h$	2.42×10^{14} Hz
wavelength, $\lambda = hc/E$	1239.8 nm
temperature, $T = E/k$	11 605 K

Energy is the common currency, and the photon is the carrier that passes it from the Sun's nuclei to the air.

2 / NUCLEAR / BIRTH IN THE CORE, c

Four protons become helium and 0.7% of the mass becomes energy

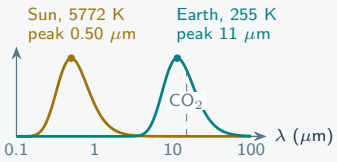


Four hydrogen atoms weigh 4.031300 u and one helium-4 atom 4.002603 u, so $\Delta m = 0.02870$ u and $E = \Delta m\, c^2 = 26.7$ MeV; the Sun's $L = 3.83 \times 10^{26}$ W costs $L/c^2 = 4.3 \times 10^9$ kg every second. The energy appears as MeV gamma rays, but at 1.5×10^7 K the thermal scale is $kT = 1.3$ keV, and repeated absorption and re-emission shares each MeV among about 10^6 photons of roughly 1 eV over 10^4 to 10^5 years.

$$E = \Delta m\, c^2 \approx 26.7 \text{ MeV per helium-4}, \qquad L/c^2 \approx 4.3 \times 10^9 \text{ kg s}^{-1}$$

3 / WAVE AND BLACKBODY / EMISSION AND THE JOURNEY, h WITH c AND k

The photosphere radiates a Planck spectrum, and the light takes 8.3 minutes

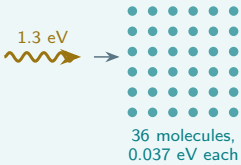


At $T = 5772$ K the spectrum depends on $hc/(\lambda kT)$ and peaks at $\lambda_{\max} = hc/(4.965\,kT) = 502$ nm, where a photon carries 2.5 eV; the mean photon energy is $2.70\,kT = 1.34$ eV. The power $L = 4\pi R^2 \sigma T^4 = 3.83 \times 10^{26}$ W uses $\sigma = 2\pi^5 k^4/(15h^3c^2)$, a law holding all three constants. Light covers $d/c = 499$ s to Earth and thins over $4\pi d^2$ to $S = 1361$ W m^{-2}, about 6×10^{21} photons per square metre per second.

$$S = \frac{L}{4\pi d^2} = 1361 \text{ W m}^{-2}, \qquad t = \frac{d}{c} = 8.3 \text{ min}$$

4 / ATOMIC AND THERMAL / ARRIVAL AND HEATING, k

An absorbed photon becomes the thermal motion of about 36 molecules



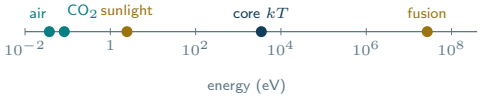
Sunlight is 30% reflected, and the average absorbed flux $S(1 - 0.30)/4 = 238$ W m^{-2} balances σT_e^4 at $T_e = 255$ K, whose spectrum peaks near 11 μm. Ozone takes the ultraviolet and water vapour and CO_2 the infrared, the 15 μm bend of CO_2 carrying 0.083 eV, and collisions convert that into motion with $\langle KE \rangle = \frac{3}{2}kT = 0.037$ eV per molecule at 288 K. One 1.3 eV photon thus warms about 36 molecules, each N_2 moving at $v_{\text{rms}} \approx 507$ m s^{-1}.

$$\langle KE \rangle = \tfrac{3}{2}kT = 0.037 \text{ eV at } 288 \text{ K}; \quad \text{one } 1.3 \text{ eV photon} \approx 36 \text{ molecules}$$

5 / SYNTHESIS

The same energy, from MeV to hundredths of an eV, through three conversion factors

stage	const.	relation	energy
fusion in the core	c	$\Delta m\, c^2$	26.7 MeV
core thermal photon	k	$2.7\,kT$	3.5 keV
sunlight at its peak	h	hc/λ	2.5 eV
CO_2 infrared photon	h	hc/λ	0.083 eV
air molecule	k	$\frac{3}{2}kT$	0.037 eV



ASIDE / THE TWO RADIATION LINKS

Accelerating charges and hot bodies both radiate, which is why the Sun's light is where the constants meet.

ASIDE / CHECK IN PYTHON

`2*pi**5*k**4/(15*h**3*c**2)` gives 5.670×10^{-8}, and `(1361*0.7/4/sigma)**0.25` gives 254.6, with h, c, k, σ from `scipy.constants`.

Convention. Figures are order-of-magnitude solar values ($T = 5772$ K, albedo 0.30); the diagram follows the constants map of mass, length and temperature.

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STRUCTURE, RESPONSE AND THE RETURN OF THE OSCILLATOR

How equations gain sophistication

The unknown grows from a value to a trajectory, a field and a spectrum of modes.

1 / RELATE QUANTITIES AT A SPECIFIED STATE

Algebra: find an unknown value

A relation lets us calculate one quantity from others: for example, $V = RI$ gives $I = V/R$.

Model	Relation	Role
Spring	$F_\mathrm{s} = -kx$	Response law
Resistor	$V = RI$	Response law
Ideal gas	$pV = nR_\mathrm{g}T$	Equation of state

The first two are **constitutive laws**: they specify a response within a model's range of validity. Here k, R describe the particular spring or resistor; R_g is the universal gas constant.

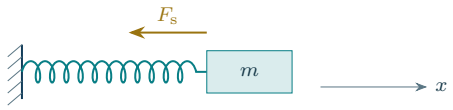
2 / SEPARATE THE FRAMEWORK FROM THE RESPONSE

A structural law supplies the balance

For one coordinate and constant mass, Newton's law is

$$\sum_i F_i \quad = \quad m\ddot{x}.$$

The sum collects the forces; response laws supply them. Dots are time derivatives: $\dot{x} = v$, $\ddot{x} = a$.



3 / COMBINE STRUCTURE WITH CONSTITUTION

An ODE: find an entire trajectory

For a free, frictionless spring, measure x from equilibrium. Insert $F_\mathrm{s} = -kx$ into Newton's law:

$$m\ddot{x} + kx \quad = \quad 0, \qquad \omega_0^2 \quad = \quad \frac{k}{m}.$$

This is an **ordinary differential equation**: the unknown is a function $x(t)$, rather than a single value. For $m, k > 0$, its solutions are

$$x(t) = x_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t).$$

The two initial conditions $x(0) = x_0$, $\dot{x}(0) = v_0$ choose one motion from the family.

4 / GIVE EVERY POSITION ITS OWN VARIABLE

A PDE: find a whole field

A uniform, taut string has displacement $u(z,t)$: one value at each position and time. For small transverse waves,

$$\frac{\partial^2 u}{\partial t^2} \quad = \quad c^2 \frac{\partial^2 u}{\partial z^2}, \qquad c^2 \quad = \quad \frac{\mathcal{T}}{\mu}.$$

Here $\mathcal{T}$ is tension, μ is mass per unit length. A **partial differential equation** relates changes in time and space. For fixed ends,

$$u(0,t) = u(L,t) = 0.$$

Supply two initial profiles: the shape $u(z,0)$ and the velocity $u_t(z,0)$, as well as these boundary conditions.

5 / THE SAME ORGANISING PATTERN IN SPACE

Field equations also need a response law

In electrostatics, use potential ϕ, electric field $\mathbf{E}$, and electric displacement $\mathbf{D}$. For a linear, isotropic dielectric:

$$\mathbf{E} = -\nabla\phi \quad \text{potential relation,}$$

$$\nabla \cdot \mathbf{D} = \rho_\mathrm{f} \quad \text{Gauss's law,}$$

$$\mathbf{D} = \varepsilon \mathbf{E} \quad \text{constitutive law.}$$

The free-charge density ρ_f supplies the source; the permittivity ε specifies the electric response. Combining the three equations gives

$$\nabla \cdot \left(\varepsilon \nabla\phi\right) \quad = \quad -\rho_\mathrm{f}.$$

For constant ε, this is **Poisson's equation**:

$$\nabla^2 \phi = -\frac{\rho_\mathrm{f}}{\varepsilon}.$$

The gradient $\nabla\phi$ measures spatial change; the Laplacian ∇^2 sums second spatial derivatives.

Specify ρ_f, ε and the boundary conditions.

6 / ASK WHICH SHAPES AND FREQUENCIES ARE ALLOWED

An eigenvalue problem: find the modes

For the fixed string, the spatial shapes solve

$$-\psi_n'' = \lambda_n \psi_n, \qquad \psi_n(0) = \psi_n(L) = 0.$$

Non-zero solutions occur at discrete eigenvalues:

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad \psi_n(z) = \sin\frac{n\pi z}{L}, \quad n \geq 1.$$



Write the field as $u(z,t) = \sum_{n\geq 1} q_n(t)\psi_n(z)$. Each mode amplitude then obeys

$$\ddot{q}_n + \omega_n^2 q_n \quad = \quad 0, \qquad \omega_n \quad = \quad \frac{n\pi c}{L}.$$

The field has become a collection of harmonic oscillators. The boundary conditions select the spectrum; the initial profiles fix the mode amplitudes and phases.

7 / QUANTISE THE MODES

For a free bosonic field, each independent mode has discrete oscillator energies. N counts the quanta in mode n.

$$E_{n,N} = \hbar\omega_n\left(N + \tfrac{1}{2}\right)$$
$$N = 0, 1, 2, \ldots$$

Reading an equation: identify the unknown, the structural law, the response model, and the data needed to select a solution.
Explore: *On the Structure of Equations*. **Sources:** Feynman II.12; MIT 8.03, Lecture 9; Tong, *Quantum Field Theory*, §2.

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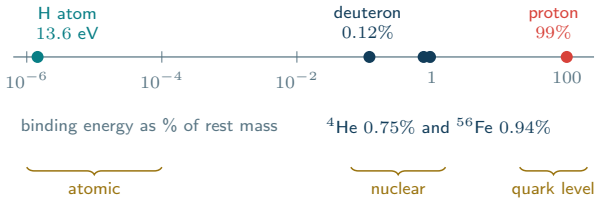
BINDING ENERGY ON ONE PAGE

Mass defect and binding energy

Why a bound system weighs less than its parts, from 13.6 eV in hydrogen to 28.3 MeV in helium-4.

1 / THE RULE AND THE PICTURE

The energy released on binding is the missing mass times c^2



Atoms and nuclei weigh less than their separate parts, and the difference is the energy released when they formed:

$$B = \Delta m\, c^2, \quad \Delta m = \sum m_{\text{parts}} - m_{\text{bound}}$$

Inside a proton the sign reverses, since the three quark masses sum to only 8.8 MeV of its 938.3 MeV, so 99% of the proton's mass is strong-interaction energy.

2 / ATOMIC SCALE / HYDROGEN

$m_H c^2 = (m_p c^2 + m_e c^2) - 13.6$ eV, and a photon is emitted

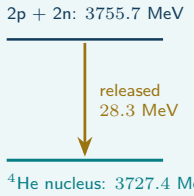
$m_p c^2 + m_e c^2$	$938.272 + 0.511 = 938.783$ MeV
mass defect 13.6 eV/c^2	2.42×10^{-35} kg
fraction of the mass	1.4×10^{-8}
photon emitted, $\lambda = hc/E$	91.2 nm (ultraviolet)

Bringing a proton and electron of opposite charge together does negative work, so energy leaves as a photon and the atom weighs less than its parts: $m_{\text{atom}} = m_{\text{nucleus}} + Zm_e - B_e/c^2$, with B_e counted positive.

The defect is far too small to weigh, which is why atomic binding is read from spectra and nuclear binding from masses: all 79 eV of helium's electrons is 2×10^{-6} of the 28.3 MeV of its nucleus.

3 / NUCLEAR SCALE / THE FIVE-STEP PROCEDURE

Atomic mass, then nuclear mass, then mass defect, then MeV, then per nucleon

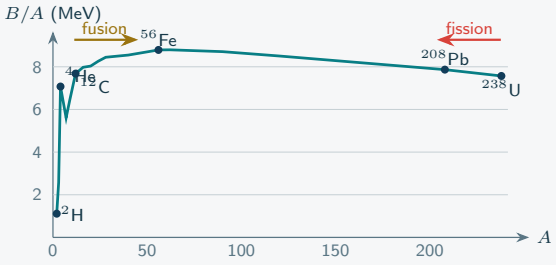


	^{4}He	^{63}Cu
m_{atom} (u)	4.002602	62.929597
$m_{\text{nucleus}} = m_{\text{atom}} - Zm_e$	4.001504	62.913676
$\Delta m = Zm_p + Nm_n - m_{\text{nucleus}}$	0.030378	0.591938
$B = \Delta m \times 931.5$ MeV/u	28.30 MeV	551.4 MeV
B/A	7.07 MeV	8.75 MeV

With 1 u = 931.5 MeV/c^2 no kilograms are needed, and using hydrogen atoms, $\Delta m = Zm_H + Nm_n - m_{\text{atom}}$, cancels the electrons and gives the same 0.030378 u for helium.

4 / THE CURVE / WHICH REACTIONS RELEASE ENERGY

Binding energy per nucleon peaks near iron and nickel



Light nuclei gain binding per nucleon when they fuse, and heavy ones gain it when they split, so both reactions move towards the peak and release energy. Four hydrogen atoms fusing to one helium atom lose 0.028697 u = 26.7 MeV, which is 0.71% of their mass, and iron sits at the top with nothing left to release. Total energy released is $\sum B_{\text{products}} - \sum B_{\text{reactants}}$, using total B and not B/A.

5 / DECAYS / ENERGY RELEASED AS KINETIC ENERGY

Alpha decay from binding energies, and positron decay from atomic masses

decay	working	energy	alpha share
^{223}Ra $\to$ ^{219}Rn $+^4$He	$(219\times7.7238 + 4\times7.074) - 223\times7.6853$	6.0 MeV	5.9 MeV
^{238}U $\to$ ^{234}Th $+^4$He	$(234\times7.6042 + 4\times7.074) - 238\times7.5901$	4.27 MeV	4.20 MeV
^{18}F $\to$ ^{18}O $+e^+ + \nu_e$	$\Delta m\, c^2 - 2m_e c^2 = 1.656 - 1.022$	0.634 MeV	shared with ν_e

The parent is at rest, so momentum conservation gives the alpha the fraction $M_{\text{daughter}}/(M_{\text{daughter}} + M_\alpha)$ of the energy; less energy means a longer half-life, 11.4 days for radium-223 against 4.5×10^9 years for uranium-238.

ASIDE / HEAVIER BY BINDING

In the quark table the proton, neutron and helium are 99.06%, 98.77% and 98.91% strong-interaction energy.

ASIDE / CHECK IN PYTHON

`(2*1.007276+2*1.008665-(4.002602-2*0.000549))*931.5` gives 28.30.

Convention. Binding energies are positive; 1 u = 931.5 MeV/c^2. After the slides E29 to E39c and the binding-energy learner.

Networked-lattice

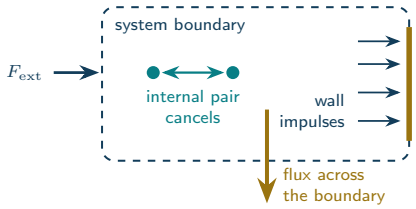
MOMENTUM ON ONE PAGE

The momentum hinge

One rate equation, $F_{\text{ext}} = \mathrm{d}p_{\text{sys}}/\mathrm{d}t$, and three mechanisms of transfer: a collision, a rocket and a gas.

1 / THE RULE AND THE PICTURE

Force is the rate at which momentum changes or crosses a chosen boundary



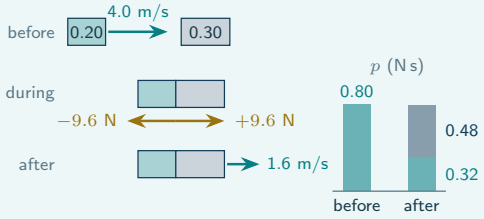
Begin with the rate equation, state the system, then add the closure that makes the force calculable.

$$F_{\text{ext}} = \frac{\mathrm{d}p_{\text{sys}}}{\mathrm{d}t}$$

Collision: an internal pair. **Rocket**: a flux across the boundary. **Gas**: wall impulses per unit area.

2 / COLLISION / IMPULSE AND INTERNAL FORCE

Equal and opposite impulses move momentum between the trolleys

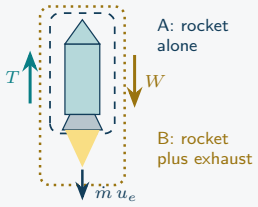


A 0.20 kg trolley at $4.0\ \mathrm{m\,s^{-1}}$ locks onto a stationary 0.30 kg trolley over $\Delta t = 0.050$ s, with negligible external impulse. Conservation, $0.20(4.0) = 0.50\,v_f$, is one equation, and locking closes it: $v_f = 1.6\ \mathrm{m\,s^{-1}}$. Each trolley has $\Delta p = \mp 0.48$ N s, so $F_{\text{av}} = \mp 9.6$ N, an equal and opposite pair, while the pair has $\Delta p = 0$ and $F_{\text{ext}} \approx 0$. A shorter Δt raises F_{av} at fixed impulse, and kinetic energy falls from 1.6 J to 0.64 J.

$$F_{\text{av}} = \Delta p/\Delta t = \pm 9.6\ \text{N on each trolley}, \quad F_{\text{ext}} \approx 0\ \text{on the pair}$$

3 / ROCKET / MOMENTUM CROSSING A BOUNDARY

Exhaust carries momentum away, and the rocket gains the balance

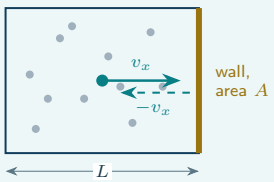


A 1200 kg rocket expels gas at $30\ \mathrm{kg\,s^{-1}}$ with relative speed $u_e = 600\ \mathrm{m\,s^{-1}}$ and $p_e = p_a$. In Δt the expelled mass $\dot{m}\,\Delta t$ carries momentum $u_e\,\dot{m}\,\Delta t$ away, so $T = \dot{m}\,u_e = 1.8 \times 10^4$ N (3600 N s in 0.20 s); with $W = 1.18 \times 10^4$ N the net force is 6240 N and $a = 5.2\ \mathrm{m\,s^{-2}}$ upward. Boundary B sees an internal redistribution, while boundary A sees mass and momentum crossing at the nozzle, so a fixed-mass equation lacks the flux term; if $p_e \neq p_a$ then $T = \dot{m}\,u_e + (p_e - p_a)A_e$.

$$T = \dot{m}\,u_e = 1.8 \times 10^4\ \text{N}, \qquad a = (T - mg)/m = 5.2\ \mathrm{m\,s^{-2}}$$

4 / GAS PRESSURE / MOLECULAR MOMENTUM FLUX

Many small impulses average into a steady pressure



A molecule bouncing elastically off a wall changes momentum by $2mv_x$ and returns after $\Delta t = 2L/|v_x|$, so $F_i = mv_x^2/L$. Summing over N molecules with $V = AL$ gives $P = Nm\langle v_x^2\rangle/V$, and isotropy, $\langle v_x^2\rangle = \frac{1}{3}\langle v^2\rangle$, gives $P = Nmc_{\text{rms}}^2/(3V) \approx 9.7 \times 10^4$ Pa for $L = 0.20$ m, $N = 2.0 \times 10^{23}$, $m = 4.65 \times 10^{-26}$ kg, $c_{\text{rms}} = 500\ \mathrm{m\,s^{-1}}$. Then $PV = \frac{2}{3}E_k$, and with $E_k = \frac{3}{2}NkT$ this is $PV = NkT$ ($T \approx 281$ K): pressure is momentum flux per unit area.

$$P = Nmc_{\text{rms}}^2/(3V) \approx 9.7 \times 10^4\ \text{Pa}, \qquad PV = \tfrac{2}{3}E_k$$

5 / SYNTHESIS

Force is a rate of momentum transfer, but the boundary decides how the transfer is seen

case	momentum system	closure or model	what produces the force
collision	both trolleys, contact pair internal	lock together, no external impulse	equal and opposite contact forces over Δt
rocket	rocket plus exhaust, or rocket alone	exhaust flux, $p_e = p_a$	momentum carried off by expelled mass, $T = \dot{m}\,u_e$
gas	molecules in the box, wall as boundary	elastic collisions, isotropy	repeated molecular impulses, P per unit area

One equation, $F_{\text{ext}} = \mathrm{d}p/\mathrm{d}t$, therefore covers contact force, thrust and pressure; the system and the closure say which.

ASIDE / THREE MISCONCEPTIONS

"Momentum is conserved for each trolley" holds only for the isolated pair. "The exhaust pushes on the air" fails because thrust persists in a vacuum. "Pressure is a stored substance" fails because it is a momentum flux per unit area.

ASIDE / CHECK THE NUMBERS IN PYTHON

`vf = 0.2*4/0.5` gives 1.6, `0.2*(vf-4)/0.05` gives -9.6, `(30*600-1200*9.8)/1200` gives 5.2, and `2e23*4.65e-26*500**2/(3*0.2**3)` gives 9.69×10^4.

Convention. After the worksheet *The Momentum Hinge*, a companion to *The Placeholder Force*; $m = 4.65 \times 10^{-26}$ kg is an N_2 molecule.

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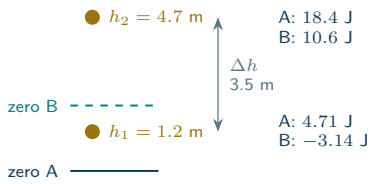
POTENTIAL ENERGY ON ONE PAGE

The potential-energy hinge

Configuration, interaction and the chosen zero, with potential kept distinct from potential energy.

1 / THE RULE AND THE PICTURE

Potential energy belongs to a configuration, and its zero is a choice



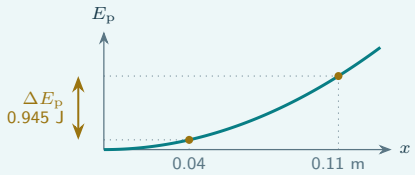
Identify the objects, the variable, the zero and the sign:

$\boxed{\Delta E_\mathrm{p} = -W_\mathrm{cons}}$ interaction law + geometry + reference → E_p.

Lifting 0.40 kg from 1.2 m to 4.7 m gives $\Delta E_\mathrm{p} = mg\Delta h \approx 13.7$ J and $W_g = -13.7$ J. Moving the zero up by 2.0 m shifts every E_p by -7.85 J and leaves differences, work and predictions unchanged; mgh assumes a uniform local field.

2 / ELASTIC ENERGY / THE CLOSURE IS HOOKE'S LAW

$E_\mathrm{p} = \frac{1}{2}kx^2$ because the force kx grows linearly with extension

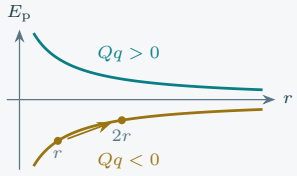


For an ideal spring with $x = 0$ the unstretched position, stretching $k = 180$ N m^{-1} from 0.04 m to 0.11 m gives $\Delta E_\mathrm{p} = \frac{1}{2}k(x_2^2 - x_1^2) = 0.945$ J, so the spring does $W_s = -0.945$ J. The work is the area under the linear force kx, hence the square, valid while Hooke's law holds. Hooke's law supplies the interaction model, energy conservation links transfers to stored energy, and $x = 0$ fixes the reference.

$$\Delta E_\mathrm{p} = \tfrac{1}{2}k(x_2^2 - x_1^2) = 0.945 \text{ J}, \qquad W_s = -0.945 \text{ J}$$

3 / RADIAL FIELDS / THE SAME $1/r$ WITH DIFFERENT SIGNS

$E_\mathrm{p} = -GMm/r$ and $E_\mathrm{p} = kQq/r$, each with zero at infinity



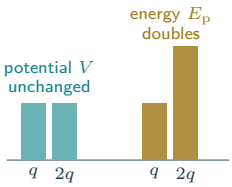
Two separated masses have zero energy and bringing them together releases energy, so the bound configuration is negative; like charges ($Qq > 0$) have positive E_p and unlike charges negative. A satellite moving from r to $2r$ rises from $-GMm/r$ to $-GMm/2r$, less negative, and two positive charges moving apart have $\Delta E_\mathrm{p} < 0$ while the electric force does positive work.

$$r \to 2r: \ \Delta E_\mathrm{p} > 0 \text{ for gravity}; \qquad \Delta E_\mathrm{p} < 0 \text{ for like charges}$$

4 / POTENTIAL AGAINST POTENTIAL ENERGY / THE MASS AND CHARGE ANALOGY

Potential belongs to the source and the place, potential energy to the pair

	gravity $(M,\ m)$	**electric** $(Q,\ q)$
field: force per unit test property	$g = F/m = GM/r^2$	$E = F/q = kQ/r^2$
potential: energy per unit test property	$V_g = E_\mathrm{p}/m = -GM/r$ in J kg^{-1}	$V_e = E_\mathrm{p}/q = kQ/r$ in J C^{-1} = V
potential energy: of the pair	$E_\mathrm{p} = mV_g = -GMm/r$	$E_\mathrm{p} = qV_e = kQq/r$
sign	$V_g < 0$, since mass attracts	V_e follows Q, E_p follows Qq



At 0.10 m from a 1.0 nC charge, $V = 90$ V for every test charge, whereas $E_\mathrm{p} = 1.8 \times 10^{-7}$ J for 2.0 nC and 3.6×10^{-7} J for 4.0 nC. At Earth's surface $V_g = -62.6$ MJ kg^{-1} for any mass, so 1 kg has $E_\mathrm{p} = -62.6$ MJ and 2 kg has -125 MJ.

E_p = (test property) × V: V comes from the source and the place, E_p from the pair

5 / SYNTHESIS

Potential energy describes the energy of a configuration; its expression comes from interaction law

case	objects	variable	reference	closure
near-Earth gravity	Earth and object	height h	chosen level	approximately uniform field
spring	spring and object	extension x	unstretched	Hooke's law
radial gravity	two masses	separation r	zero at infinity	Newtonian gravitation
electric	two charges	separation r	zero at infinity	Coulomb's law and charge signs

ASIDE / mgh AS THE LOCAL LIMIT

$-\frac{GMm}{R+h} \approx -\frac{GMm}{R} + mgh$ with $g = GM/R^2$: mgh is the first-order term of $-GMm/r$, one interaction at two levels of approximation.

ASIDE / MISCONCEPTIONS

Potential energy belongs to the system, not an isolated object; a negative value is meaningful relative to a chosen zero; the zero can move and measurable differences cannot.

Convention. After the worksheet *The Potential-Energy Hinge*; in Python, `0.40*9.81*3.5` gives 13.73 and `0.5*180*(0.11**2-0.04**2)` gives 0.945.

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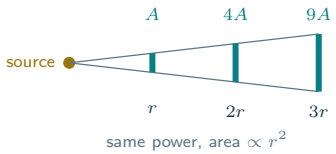
RECIPROCALS ON ONE PAGE

The reciprocal-power hinge

From potential as $1/r$ to field strength and flux density as $1/r^2$, with geometry setting the power.

1 / THE RULE AND THE PICTURE

A point source spreads over a sphere whose area grows as r^2



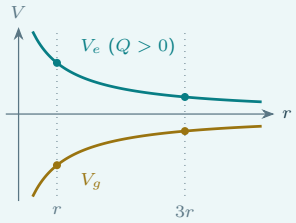
A point source gives two related radial patterns, with the constants and signs set by the source and the power of r set by radial geometry (point source or spherical symmetry, no absorption):

$$\text{potential} \propto \frac{1}{r}, \qquad \text{field strength, flux density} \propto \frac{1}{r^2}$$

Potential: energy per test property. **Field**: force per test property. **Irradiance**: power per unit area.

2 / POTENTIAL / ONE RECIPROCAL POWER

$V_g = -GM/r$ and $V_e = kQ/r$ share the structure $1/r$

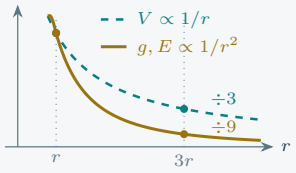


For a point mass and a point charge, $V_g = -GM/r$ and $V_e = kQ/r$ are energies per unit test property, E_p/m and E_p/q, with zero at infinity. Moving from r to $3r$ divides $|V|$ by 3: because V_g is negative this is a rise towards zero, whereas the positive charge's V_e falls towards zero. The numerator carries source strength and coupling, and the sign records gravity's attraction or the source charge. For Earth, V_g rises from $-62.6\ \mathrm{MJ\,kg^{-1}}$ at the surface to $-20.9\ \mathrm{MJ\,kg^{-1}}$ at $3R_E$.

$$|V(3r)| = \tfrac{1}{3}|V(r)|, \qquad V_g \text{ rises towards } 0 \text{ from below}$$

3 / FIELD STRENGTH / TWO RECIPROCAL POWERS

$g = GM/r^2$ and $E = k|Q|/r^2$ fall by nine when the distance triples

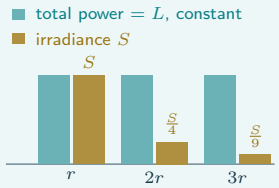


For the same sources $g = GM/r^2$ and $E = k|Q|/r^2$, with $F_g = mg$ and $F_e = qE$, so tripling the distance divides both by 9 while the potential falls only by 3: for Earth, $9.8\ \mathrm{m\,s^{-2}}$ at the surface becomes $1.1\ \mathrm{m\,s^{-2}}$ at $3R_E$. Potential is scalar energy per unit test property and field strength vector force per unit test property, so "both decrease with distance" misses the distinction. $\mathbf{E}$ points away from positive and towards negative charge, whereas $\mathbf{g}$ points towards the source.

$$g(3r) = \tfrac{1}{9}g(r), \quad E(3r) = \tfrac{1}{9}E(r), \quad V(3r) = \tfrac{1}{3}V(r)$$

4 / SPHERICAL SPREADING / FLUX DENSITY

Irradiance falls as $1/r^2$ while the power through each sphere stays at L



A source of power L radiating equally in all directions spreads it over a sphere of area $4\pi r^2$, so the irradiance, or power per unit area, is $S = L/(4\pi r^2)$. Doubling the distance quarters S, and for $L = 3.8 \times 10^{26}\ \mathrm{W}$ at $r = 2.0 \times 10^{11}\ \mathrm{m}$, $S \approx 7.6 \times 10^2\ \mathrm{W\,m^{-2}}$. Because $S(4\pi r^2) = L$ on every sphere, the total power through each complete sphere is unchanged while the irradiance falls, since the same power meets an area growing as r^2.

$$S = \frac{L}{4\pi r^2} \approx 7.6 \times 10^2\ \mathrm{W\,m^{-2}}, \qquad S\,(4\pi r^2) = L$$

5 / SYNTHESIS

The $1/r^2$ expresses spherical spreading; the numerator supplies the source's physics

quantity	dependence	$r \to 2r$	what it measures
potential	$1/r$	$\times\frac{1}{2}$ in magnitude	energy per unit test property
field strength	$1/r^2$	$\times\frac{1}{4}$	force per unit test property
irradiance	$1/r^2$	$\times\frac{1}{4}$	power per unit area
total power through a sphere	constant	unchanged	the source power L

ASIDE / WHY POTENTIAL HAS ONE POWER FEWER

Energy is force times distance, so $V = \int_r^\infty g\,\mathrm{d}r' = GM/r$ loses one power of r on integrating, and conversely $g = -\mathrm{d}V/\mathrm{d}r$.

ASIDE / FOUR MISCONCEPTIONS

Potential and field are not interchangeable; negative potential rises towards zero; flux density and total flux differ; inverse-square needs a point source and no absorption.

Convention. After worksheet *The Reciprocal-Power Hinge*; check in Python: `3.8e26/(4*pi*(2e11)**2)` gives 755.99. Companion to momentum and torque hinges.

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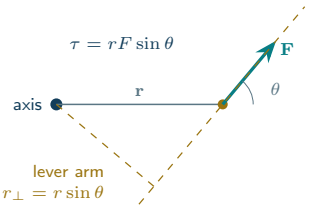
TORQUE ON ONE PAGE

The torque hinge

One rotational balance, $\sum \tau_{\text{ext}} = \mathrm{d}L/\mathrm{d}t$, and three origins of torque: gravity, friction and a magnetic couple.

1 / THE RULE AND THE PICTURE

Choose the axis first, and the lever arm decides how much of a force turns the body



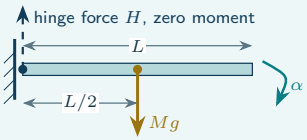
Begin with the chosen axis and the rotational balance, which for a rigid body on a fixed axis with constant I_{rot} becomes $\sum \tau = I_{\text{rot}}\alpha$; it links torque to angular acceleration and leaves the interaction unnamed.

$$\sum \tau_{\text{ext}} = \frac{\mathrm{d}L}{\mathrm{d}t}$$

Gravity: $\mathbf{r} \times M\mathbf{g}$. **Friction**: $\mathbf{r} \times \mathbf{f}$. **Magnetic**: a force couple.

2 / GRAVITY / TORQUE FROM A FIELD INTERACTION

A rod released from horizontal falls with $\alpha = 3g/2L$

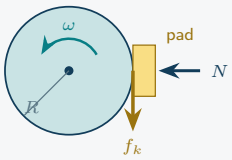


A uniform rod of $1.20\ \mathrm{m}$ and $3.0\ \mathrm{kg}$, hinged at one end, is released from rest while horizontal. The weight acts at the centre of mass, $L/2$ from the hinge, so $\tau_g = MgL/2 = 17.6\ \mathrm{N\,m}$ in the direction of fall, while the hinge force passes through the axis and has zero moment. With $I_{\text{rot}} = \frac{1}{3}ML^2 = 1.44\ \mathrm{kg\,m^2}$, $\alpha = 12.25\ \mathrm{rad\,s^{-2}}$, and the mass cancels because torque and inertia both scale with M, which does not make weight and inertia the same property. The dependence $\tau_g = Mg(L/2)\sin\theta$ belongs to the torque law and the geometry.

$$\tau_g = MgL/2 = 17.64\ \mathrm{N\,m}, \qquad \alpha = 3g/(2L) = 12.25\ \mathrm{rad\,s^{-2}}$$

3 / FRICTION / TORQUE FROM A CONTACT LAW

An empirical law and a geometric lever arm stop the flywheel

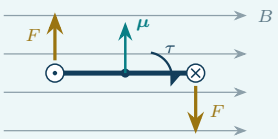


A brake pad presses on the rim of a $4.0\ \mathrm{kg}$, $0.25\ \mathrm{m}$ solid disc turning at $18\ \mathrm{rad\,s^{-1}}$, with $N = 10\ \mathrm{N}$ and $\mu_k = 0.30$. The empirical law $f_k = \mu_k N = 3.0\ \mathrm{N}$ and the geometry $\tau_f = f_k R = 0.75\ \mathrm{N\,m}$ give a torque opposing ω; with $I_{\text{rot}} = \frac{1}{2}MR^2 = 0.125\ \mathrm{kg\,m^2}$, $\alpha = -6.0\ \mathrm{rad\,s^{-2}}$ and $t_{\text{stop}} = 3.0\ \mathrm{s}$, confirmed by angular impulse ($-2.25 = I_{\text{rot}}\Delta\omega$ in $\mathrm{kg\,m^2\,s^{-1}}$). The $20.25\ \mathrm{J}$ of rotational energy becomes thermal, and the disc's L is not conserved because the brake exerts an external torque, though a system with brake, mounting and Earth can conserve it.

$$\tau_f = \mu_k NR = 0.75\ \mathrm{N\,m}, \qquad t_{\text{stop}} = I_{\text{rot}}\omega/\tau_f = 3.0\ \mathrm{s}$$

4 / MAGNETIC INTERACTION / A FORCE COUPLE

Equal and opposite forces sum to zero and still turn the coil



A 50-turn coil of area $8.0\times10^{-3}\ \mathrm{m^2}$ carries $0.60\ \mathrm{A}$ in $0.25\ \mathrm{T}$ with its moment perpendicular to the field. The opposite side forces sum to zero but have separated lines of action, a couple, giving $\tau_B = NIAB\sin\theta = 6.0\times10^{-2}\ \mathrm{N\,m}$, where N, I, A, B set the strength and $\sin\theta$ the orientation (zero at $\theta = 0$). With $I_{\text{rot}} = 2.0\times10^{-2}\ \mathrm{kg\,m^2}$, $\alpha = 3.0\ \mathrm{rad\,s^{-2}}$; reversing the current reverses τ, doubling B doubles α and doubling I_{rot} halves it. The value is instantaneous, since θ changes and a real motor adds commutation, back emf and friction.

$$\tau_B = NIAB = 6.0\times10^{-2}\ \mathrm{N\,m}, \qquad \alpha = \tau_B/I_{\text{rot}} = 3.0\ \mathrm{rad\,s^{-2}}$$

5 / SYNTHESIS

The symbol τ records the moment of an interaction without identifying it

case	axis	physical torque law	coupling or parameter	what I_{rot} represents
gravity	hinge	$\mathbf{r}\times M\mathbf{g} = Mg\frac{L}{2}\sin\theta$	g, centre-of-mass geometry	rod about the hinge, $\frac{1}{3}ML^2$
friction	axle	$\mathbf{r}\times\mathbf{f}$, with $f_k = \mu_k N$	μ_k, an empirical closure	disc about the axle, $\frac{1}{2}MR^2$
magnetic	coil axis	couple, $NIAB\sin\theta$	N, I, A, B	coil and rotor about the axis

$\sum\tau = I_{\text{rot}}\alpha$ needs a rigid body, a fixed axis and constant I_{rot}, so it is a specialised bridge from $\sum\tau_{\text{ext}} = \mathrm{d}L/\mathrm{d}t$.

ASIDE / NOTATIONAL COMPRESSION

The letter I is current in $NIAB$ and moment of inertia in $I_{\text{rot}}\alpha$. Torque and I_{rot} are not properties of a body alone: changing the axis changes the lever arm and I_{rot} together.

ASIDE / CHECK THE NUMBERS IN PYTHON

`3*9.8*0.6/(3*1.2**2/3)` gives 12.25, `-0.3*10*0.25/(0.5*4*0.25**2)` gives -6.0, and `50*0.6*8e-3*0.25/2e-2` gives 3.0.

Convention. After the worksheet *The Torque Hinge*, a companion to *The Placeholder Force*; see also *The Momentum Hinge*.

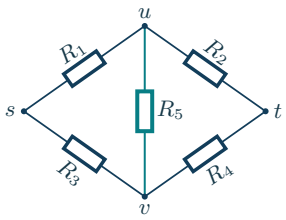
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THE WHEATSTONE BRIDGE AND RANDOM WALKS

A bridge between circuits and chance

Kirchhoff's averaging rule links voltages, probabilities, expected journeys and squared rectangles.

1 The A-level bridge



$$V_s = V, \qquad V_t = 0.$$

$R_5 > 0$ is the finite-resistance detector branch. The random walk will use the five resistor edges.

2 Balance means equal potentials

At balance, $V_u = V_v$, so the middle branch carries zero current. The two arms are voltage dividers:

$$V_u = V\frac{R_2}{R_1+R_2}, \qquad V_v = V\frac{R_4}{R_3+R_4}.$$

With the labels shown,

$$\boxed{V_u = V_v \iff \frac{R_1}{R_2} = \frac{R_3}{R_4}}.$$

The effective resistance is V/I_{total}. At balance,

$$R_{\text{eff}}(s,t) = \left[\frac{1}{R_1+R_2} + \frac{1}{R_3+R_4}\right]^{-1}.$$

3 A voltage becomes a hitting probability

On a finite, connected, undirected resistor network, choose the next edge with probability proportional to its conductance:

$$c_{xy} = \frac{1}{R_{xy}}, \qquad p_{xy} = \frac{c_{xy}}{\sum_z c_{xz}}, \qquad \underbrace{V_x = \sum_y p_{xy}V_y}_{\text{Kirchhoff's law at an interior node}}.$$

Let q_x be the probability, starting at x, of reaching s before t. Looking at the first step gives

$$q_x = \sum_y p_{xy}q_y, \qquad q_s = 1, \quad q_t = 0 \quad \Longrightarrow \quad \boxed{q_x = \frac{V_x}{V}}.$$

The same averaging rule and boundary values give the same solution. Equal resistors make all incident edges equally likely.

4 Resistance measures the expected round trip

Let h_{st} be the expected number of steps to first reach t starting at s. The **commute time** is

$$\boxed{K_{st} = h_{st} + h_{ts} = 2\left(\sum_{e\in E} c_e\right) R_{\text{eff}}(s,t)}.$$

Count each undirected resistor edge once in the sum. For m equal resistors of resistance R_0, $\boxed{K_{st} = 2m\,\dfrac{R_{\text{eff}}(s,t)}{R_0}}$. Time is measured in steps along edges.

5 Five equal resistors

Set $R_1 = \cdots = R_5 = R_0$. Symmetry gives

$$V_u = V_v = \frac{V}{2}, \qquad I_5 = 0.$$

The two outer paths each have resistance $2R_0$:

$$R_{\text{eff}}(s,t) = (2R_0) \parallel (2R_0) = R_0.$$

There are still five edges in the walk:

$$\boxed{K_{st} = 2\cdot 5\cdot \frac{R_0}{R_0} = 10 \text{ steps}}.$$

Check by taking the next step

Let h_x be the expected number of steps from x to t. Then $h_t = 0$. By symmetry, write $h_u = h_v = h$.

$$h_s = 1 + h, \qquad h = 1 + \frac{h_s + h + 0}{3}.$$

The initial 1 counts the step just taken. Solving gives

$$h = 4, \qquad h_{st} = h_s = 5.$$

The return journey has the same mean by symmetry:

$$\boxed{K_{st} = 5 + 5 = 10}.$$

A balanced edge still carries walkers

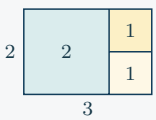
Middle edge	m	R_{eff}/R_0	K_{st}
Present	5	1	10
Removed	4	1	8

At u, the walker chooses v with probability $1/3$, even though the balanced electrical current through R_5 is zero.

The extra edge introduces detours. It also changes the total-conductance factor in the commute-time formula.

6 The rectangle becomes a walk

Three square tiles give three resistors R_0, with $R_{\text{eff}}/R_0 = H/W$.



$$\frac{R_{\text{eff}}}{R_0} = \frac{2}{3}, \qquad m = 3.$$

$$\boxed{K = 2\cdot 3\cdot \frac{2}{3} = 4 \text{ steps}}.$$

Tile height encodes voltage drop; tile width current in scaled units. The same Kirchhoff network now describes chance.

Sources: Doyle–Snell; Chandra et al. (1996); Aldous–Fill, §3.3; Cannon–Floyd–Parry, §1. Companion: *Kirchhoff Circuits & Squared Rectangles*.

Networked–lattice

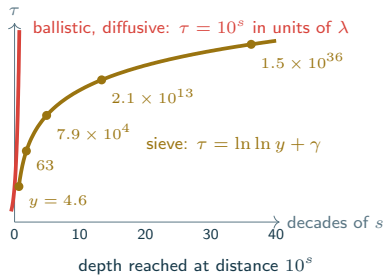
FUNCTIONS ON ONE PAGE

Three attenuators: ballistic, diffusive and the sieve

The same law, transmission equal to $e^{-\tau}$, describes light in dust and the primes, with different opacities.

1 / THE RULE AND THE PICTURE

Every model is a geometric dilution G times $e^{-\tau}$, and only the optical depth $\tau(s)$ differs



depth reached at distance 10^s

Cutting the path into thin independent layers, each transmitting $1-\kappa\,ds$, gives $\prod(1-\kappa\,ds)\to e^{-\tau}$ with $\tau=\int\kappa\,ds$, so

$$f(s)=G(s)\,e^{-\tau(s)},\qquad n=\frac{d\ln G}{d\ln s}-s\,\kappa(s)$$

The local exponent is the geometric slope minus $s\kappa$. For a constant κ the depth grows linearly with distance, so it reaches $\tau=5$ at 5λ, while for the sieve it needs $y=10^{36}$.

2 / THE THREE MODELS SIDE BY SIDE

Constant opacity gives an exponential, and the sieve's opacity $1/(y\ln y)$ thins as it goes

model	geometry G	opacity κ	optical depth τ	exponent n	scale
ballistic beam	r^{-2}	$1/\lambda$	r/λ	$-2-r/\lambda$	λ
diffusive halo	r^{-1}	$1/L$	r/L	$-1-r/L$	L
prime sieve	1	$1/(y\ln y)$	$\ln\ln y+\gamma$	$-1/\ln y$	none

The sieve's opacity follows from the density of primes: the fraction removed by primes between t and $t+dt$ is $\sum 1/p\approx dt/(t\ln t)$, and integrating gives $\ln\ln y$.
Its exponent is $s\kappa=1/\ln y$ with no geometric part, which is exactly the local exponent of $1/\ln y$ found for the density of primes.

3 / HOW FAR TO REACH THE SAME DEPTH

Equal transmission needs a linear distance in light and a doubly exponential one in the sieve

τ	transmission	ballistic distance	sieve y	$\log_{10} y$
1	$e^{-1}=0.3679$	$r=1\lambda$	4.6	0.7
2	$e^{-2}=0.1353$	$r=2\lambda$	63	1.8
3	$e^{-3}=0.0498$	$r=3\lambda$	7.9×10^4	4.9
4	$e^{-4}=0.0183$	$r=4\lambda$	2.1×10^{13}	13.3
5	$e^{-5}=0.0067$	$r=5\lambda$	1.5×10^{36}	36.2

At $y=10^9$ the transmission is $e^{-\gamma}/\ln y=0.027$ and at $y=10^{100}$ it is still 0.0024, whereas a ballistic beam has fallen to 4.5×10^{-5} at only 10λ. The sieve is the thinnest imaginable medium, with τ diverging so slowly that the primes never become rare in any practical range.

4 / BALLISTIC AGAINST DIFFUSIVE / THE SCREENED FORM

In a medium that scatters strongly the halo follows $e^{-r/L}/r$, one power less steep than the ballistic law

For a steady point source in a scattering and absorbing medium the density obeys $D\nabla^2\phi-\mu_a\phi=-S\,\delta$, with solution $\phi=\dfrac{S}{4\pi D}\dfrac{e^{-r/L}}{r}$ and diffusion length $L=\sqrt{\lambda_a\lambda_{tr}/3}$, the same function as the Yukawa potential and Debye screening. With $\lambda_a=100$ m and $\lambda_{tr}=10$ m, $L=18.3$ m. At $r=3L$ the ballistic form $e^{-3}/9$ is 0.0055 while the diffusive $e^{-3}/3$ is 0.0166, a factor of r/L higher, and the exponents are -5 and -4, so the diffusive falloff is slower by exactly one power.

5 / WHAT THE ANALOGY CAPTURES

A true statement about independent filters, and its limits

The sieve fits Beer and Lambert in having multiplicative, memoryless layers, but it has no spreading, its layers are discrete, and the independence is only approximate, which shows as the factor $2e^{-\gamma}=1.123$ between the filter estimate $1.123/\ln n$ and the true density $1/\ln n$.
The remaining ripples, of size about $\sqrt{x}$ around $\mathrm{Li}(x)$ if the Riemann hypothesis holds, are not part of this picture and come from the zeros of the zeta function.

ASIDE / WHICH LIGHT

Ballistic describes a lamp seen through thin haze, and diffusive describes the glow inside a fog bank, where the observer is many scattering lengths from the lamp.

LINKS TO OTHER SHEETS

The primes as a falloff are in *How the primes thin out*; the dust law is in *Combining falloffs*; the decay $e^{-\tau}$ itself is in *Exponential functions and the number e*.

Convention. τ optical depth; κ opacity; λ, L attenuation and diffusion lengths; λ_a, λ_{tr} absorption and transport lengths; y sieve limit.

Networked-lattice

CALCULUS AND KINEMATICS ON ONE PAGE

Velocity to displacement, by parts

Repeat the product-rule identity, solve the recurring integral, and see where the half comes from.

Constant acceleration a; initial velocity u.

$v(\tau) = u + a\tau, \qquad s(t) = \displaystyle\int_0^t v(\tau)\,\mathrm{d}\tau.$

Here $s(t) = x(t) - x(0)$, and $t \geq 0$ is elapsed time.

The product rule, rearranged and integrated

$$\boxed{\int_0^t fg'\,\mathrm{d}\tau = \big[fg\big]_0^t - \int_0^t f'g\,\mathrm{d}\tau.}$$

Primes mean differentiation with respect to τ.

1 First pass: extract a boundary term

Take $f = v(\tau) = u + a\tau$ and $g' = 1$. Then $f' = a$ and $g = \tau$:

$$\begin{aligned} s(t) &= \big[\tau v(\tau)\big]_0^t - \int_0^t \tau a\,\mathrm{d}\tau \\ &= t(u + at) - aJ, \end{aligned}$$

where $J = \displaystyle\int_0^t \tau\,\mathrm{d}\tau$ is the remaining integral.

2 Repeat parts on the remaining integral

For J, choose $f = \tau$ and $g' = 1$. Now $f' = 1$ and $g = \tau$:

$$\begin{aligned} J &= \big[\tau \cdot \tau\big]_0^t - \int_0^t 1 \cdot \tau\,\mathrm{d}\tau \\ &= t^2 - J. \end{aligned}$$

The right-hand integral is exactly the same J. Collect the two copies:

$$2J = t^2 \qquad \Longrightarrow \qquad \boxed{J = \frac{t^2}{2}.}$$

3 Put the correction back

$$\begin{aligned} s(t) &= t(u + at) - \frac{1}{2}at^2 \\ &= \boxed{ut + \frac{1}{2}at^2.} \end{aligned}$$

Equivalently, $s = \dfrac{u + v(t)}{2}\,t$: **mean velocity** $\times$ **elapsed time**.

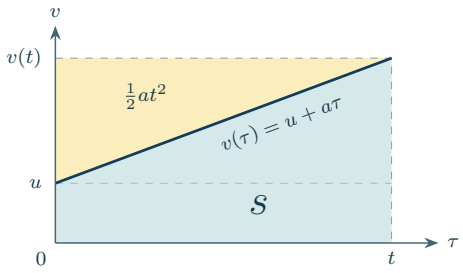
The repeated-parts table

Differentiate v on the left; repeatedly integrate 1 on the right.

sign	differentiate v	integrate 1
$+$	$u + a\tau$	τ
$-$	a	$\tau^2/2$
stop	0	$\tau^3/6$

Add the signed row products; evaluate between 0 and t.

A rectangle, then a correction



Rectangle $t\,v(t)$ minus the gold triangle:

$$s = t\,v(t) - \frac{1}{2}at^2.$$

Sketch: $u, a, t > 0$. In general, s is signed.

The lower endpoint

At $\tau = 0$, both $\tau v(\tau)$ and τ^2 vanish.

A numerical check

$u = 4\,\mathrm{m\,s^{-1}}$, $a = 2\,\mathrm{m\,s^{-2}}$, $t = 3\,\mathrm{s}$.

Then $v(t) = 4 + 2(3) = 10\,\mathrm{m\,s^{-1}}$.

$$s = \underbrace{30\,\mathrm{m}}_{t\,v(t)} - \underbrace{9\,\mathrm{m}}_{at^2/2} = \boxed{21\,\mathrm{m}}.$$

One more term means one more row

Let $v(\tau) = u + a_0\tau + b\tau^2$, with u, a_0, b constant. Its derivatives are v, $a_0 + 2b\tau$, $2b$, 0:

$$s = t(u + a_0 t + bt^2) - \frac{t^2}{2}(a_0 + 2bt) + \frac{t^3}{6}(2b) = \boxed{ut + \frac{1}{2}a_0t^2 + \frac{1}{3}bt^3.}$$

Each by-parts step differentiates one factor and integrates the other. For polynomial velocity, a zero derivative makes the process finite.

Networked–lattice

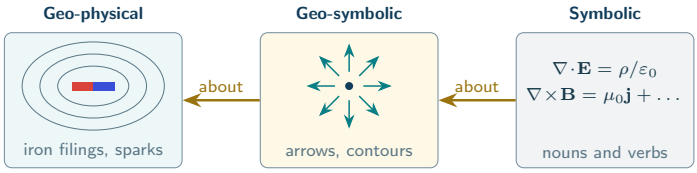
PHYSICS ON ONE PAGE

What does an equation represent?

Maxwell's equations as sentences about patterns, built up from Coulomb's law and the magnetic force law.

1 / THE RULE AND THE PICTURE

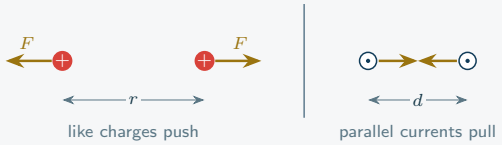
An equation is about a pattern, in the sense of Brentano's aboutness



The filings, sparks and compass needle are a physical pattern; the arrows and contours are a picture of it; the equation says in symbols what the picture does. Its nouns ($\mathbf{E}$, $\mathbf{B}$, ρ, $\mathbf{j}$) name the things and its verbs (spreading, swirling, sloping, changing) name what is done to them.

2 / FORCE LAWS / STATIC CHARGES AND STEADY CURRENTS

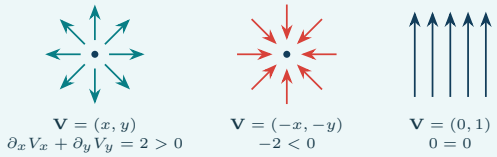
Coulomb's law and the wire-to-wire force, then the fields they imply



Coulomb's law is $F = kQq/r^2$ with $k = 1/(4\pi\varepsilon_0) = 8.99\times10^9$, so $1.0\ \mu$C and $2.0\ \mu$C at 0.30 m push with 0.20 N. Two long parallel wires obey $F/L = \mu_0 I_1 I_2/(2\pi d)$, so 10 A in each at 5.0 cm pull with 4.0×10^{-4} N m^{-1}, using $\mu_0/2\pi = 2\times10^{-7}$. Dividing out the test object leaves fields, $E = F/q = kQ/r^2$ and $B = \mu_0 I/(2\pi r)$, which exist whether or not anything is there to feel them.

3 / SPREADING / DIVERGENCE, FROM FIELD LINES THROUGH A SPHERE

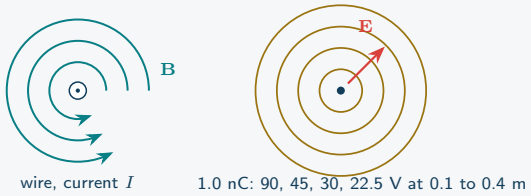
The same lines spread over $4\pi r^2$, which is why Coulomb falls as $1/r^2$



Count the field lines leaving a sphere of radius r round charge Q: the field is the same all over it, so the count is $E\cdot 4\pi r^2$, and if that count is Q/ε_0 for every r then $E = Q/(4\pi\varepsilon_0 r^2) = kQ/r^2$. Locally this is divergence, $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$: field spreads out of a small region in proportion to the charge in it. Spreading arrows read positive, converging arrows negative and parallel arrows zero. Magnets give zero every time, $\nabla\cdot\mathbf{B} = 0$, since each line entering a region also leaves it.

4 / SWIRLING AND SLOPING / CIRCULATION AND THE GRADIENT OF POTENTIAL

Currents make fields swirl, and static fields roll downhill on a potential map

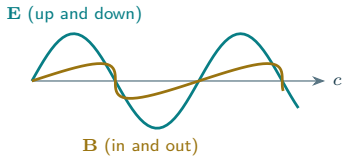


Walking a loop of radius r round a wire and adding the field along the path gives $B\cdot 2\pi r = \mu_0 I$, the field of the force law; this circulation is the swirl in $\nabla\times\mathbf{B} = \mu_0\mathbf{j}$. A changing magnetic flux makes an electric swirl too, $\mathcal{E} = -\mathrm{d}\Phi/\mathrm{d}t$, as when a magnet pushed into a coil lights a bulb. Static electric fields do not swirl, so they come from a landscape of potential with $E = -\mathrm{d}V/\mathrm{d}x$: the field points downhill and is strongest where the contours crowd, and for 1.0 nC the potential $V = 9.0/r$ volts gives 899 V m^{-1} at 0.10 m.

5 / MAXWELL / FOUR SENTENCES ABOUT PATTERNS

Each equation pairs nouns with a verb, and the last two make light

equation	verb	about	in words
$\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$	spreading	static cling, sparks	electric field lines begin and end on charge
$\nabla\cdot\mathbf{B} = 0$	spreading	iron filings, cut magnets	magnetic field lines never begin or end
$\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$	swirl, rate	induction in a coil	a changing $\mathbf{B}$ makes $\mathbf{E}$ swirl
$\nabla\times\mathbf{B} = \mu_0\mathbf{j} + \mu_0\varepsilon_0\,\partial\mathbf{E}/\partial t$	swirl, rate	compass round a wire, light	current and a changing $\mathbf{E}$ make $\mathbf{B}$ swirl



The last two couple the fields, a changing $\mathbf{E}$ making $\mathbf{B}$ and a changing $\mathbf{B}$ making $\mathbf{E}$, so a disturbance sustains itself and travels at

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = 2.998\times10^8\ \mathrm{m\,s^{-1}},\ \text{the speed of light}$$

ASIDE / WHAT THE FILINGS ARE

The iron filings are just there, in themselves; the arrows are our map of them, and the equations are about both.

ASIDE / CHECK IN PYTHON

`1/sqrt(mu_0*epsilon_0)` gives 2.998×10^8 and `k*1e-6*2e-6/0.3**2`, with $k = 8.99\times10^9$, gives 0.1997.

Convention. Differential forms shown, with the integral forms used in the working; vacuum with $\mu_0\varepsilon_0 = 1/c^2$. After the slide on geo-physical, geo-symbolic and symbolic levels.

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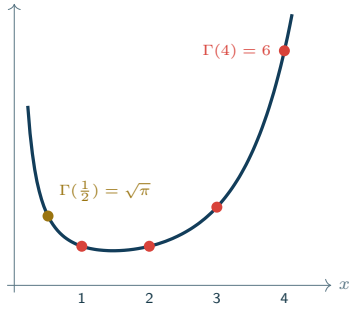
FUNCTIONS LADDER 9 OF 10

The gamma function and the beta function

The gamma function continues the factorial to every complex number, and the beta function is a ratio of gamma values.

1 / THE RULE AND THE PICTURE

The gamma function interpolates the factorials, with $\Gamma(n) = (n-1)!$



The curve passes through the factorials $\Gamma(1) = 1$, $\Gamma(2) = 1$, $\Gamma(3) = 2$, $\Gamma(4) = 6$, and between them it is smooth, with a dip to a minimum 0.886 near $x = 1.46$.
The defining integral and the functional equation obtained by integration by parts are

$$\boxed{\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \qquad \Gamma(z+1) = z\,\Gamma(z)}$$

2 / THE RECURRENCE AND HALF-INTEGERS

Integrating by parts shifts z by one, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ is the Gaussian integral

Writing $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt$ and integrating by parts gives $\big[-t^z e^{-t}\big]_0^\infty + z\int_0^\infty t^{z-1} e^{-t} dt = z\Gamma(z)$, so $\Gamma(n) = (n-1)!$ by induction.
The substitution $t = u^2$ turns $\Gamma(\frac{1}{2})$ into $2\int_0^\infty e^{-u^2} du = \sqrt{\pi}$, and the recurrence then gives every half-integer: $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$, $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$.
Running the recurrence backwards, $\Gamma(z) = \Gamma(z+1)/z$ extends the function to negative arguments, with poles at $0, -1, -2, \ldots$.

3 / SOME VALUES

The reflection formula $\Gamma(z)\Gamma(1-z) = \pi / \sin \pi z$ ties z to $1-z$

argument	exact	value	argument	value
$\Gamma(\frac{1}{2})$	$\sqrt{\pi}$	1.7725	$\Gamma(\frac{7}{2})$	$\frac{15}{8}\sqrt{\pi} = 3.3234$
$\Gamma(\frac{3}{2})$	$\frac{1}{2}\sqrt{\pi}$	0.8862	$\Gamma(-\frac{1}{2})$	$-2\sqrt{\pi} = -3.5449$
$\Gamma(\frac{5}{2})$	$\frac{3}{4}\sqrt{\pi}$	1.3293	$\Gamma(\frac{1}{4})\Gamma(\frac{3}{4})$	$\pi\sqrt{2} = 4.4429$

At $z = \frac{1}{4}$ the reflection formula gives $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi/\sin\frac{\pi}{4} = \pi\sqrt{2} = 4.4429$, matching 3.6256×1.2254, and the unit ball in n dimensions has volume $\pi^{n/2}/\Gamma(\frac{n}{2}+1)$: π for $n = 2$, 4.189 for $n = 3$, 5.264 for $n = 5$ and 2.550 for $n = 10$.

4 / THE BETA FUNCTION / THE POCHHAMMER SYMBOL

The beta function is a ratio of gamma values

$$B(a,b) = \int_0^1 t^{a-1}(1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

For integers $B(2,3) = \dfrac{1!\,2!}{4!} = \frac{1}{12}$, and for $a = b = \frac{1}{2}$ the integral is $\int_0^1 \frac{dt}{\sqrt{t(1-t)}} = \pi = \dfrac{\Gamma(\frac{1}{2})^2}{\Gamma(1)}$. The ratio $(a)_n = \dfrac{\Gamma(a+n)}{\Gamma(a)} = a(a+1)\cdots(a+n-1)$, the rising factorial or Pochhammer symbol, satisfies $(1)_n = n!$ and $(\frac{1}{2})_3 = \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2} = \frac{15}{8}$, and it is the building block of hypergeometric series.

5 / STIRLING'S APPROXIMATION

Large factorials are $n! \approx \sqrt{2\pi n}\,(n/e)^n$

For $n = 10$ the formula gives $3\,598\,696$ against $10! = 3\,628\,800$, an error of 0.83%, and for $n = 20$ it gives 2.4228×10^{18} against 2.4329×10^{18}, an error of 0.42%; the relative error is about $\frac{1}{12n}$.
The same formula holds for $\Gamma(z+1)$ at real z, and it is the reason entropy per particle is $\ln N! \approx N \ln N - N$.

ASIDE / A DEFINITION BY PROPERTIES

The only log-convex function with $f(1) = 1$ and $f(x+1) = xf(x)$ is Γ, which is what makes it the right continuation of the factorial.

LINKS TO OTHER SHEETS

The Gaussian integral also gives the normal distribution in *The normal distribution and the central limit theorem*; the previous sheet is *Taylor series* and the last is *Hypergeometric functions*.

Convention. Γ is the gamma function; B the beta function; $(a)_n$ the rising factorial; π appears through $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

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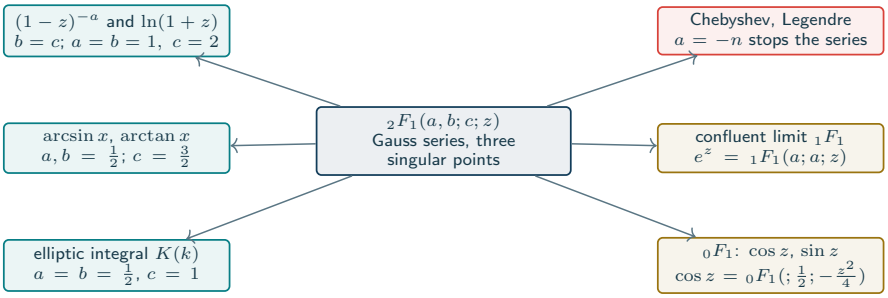
FUNCTIONS LADDER 10 OF 10

Hypergeometric functions

One series with three parameters contains the binomial, logarithm, inverse trigonometric, elliptic and orthogonal polynomial families.

1 / THE RULE AND THE PICTURE

The Gauss series ${}_2F_1(a,b;c;z)$ has many familiar functions as special cases



$$ {}_2F_1(a,b;c;z)=\sum_{n=0}^{\infty}\frac{(a)_n(b)_n}{(c)_n\,n!}\,z^n, \qquad (a)_n=a(a+1)\cdots(a+n-1)=\frac{\Gamma(a+n)}{\Gamma(a)} $$

2 / WHY THIS SERIES / THE RATIO OF TERMS

A series is hypergeometric exactly when the ratio of successive terms is a rational function of n

Here $\dfrac{t_{n+1}}{t_n}=\dfrac{(n+a)(n+b)}{(n+c)(n+1)}\,z$, and for the geometric series the ratio is constant, for e^z it is $z/(n+1)$, and for the binomial series $(n+a)z/(n+1)$, which gives the family ${}_pF_q$ with p factors on top and q below. The series converges for $|z|<1$, and at $z=1$ Gauss's theorem gives the sum when $c>a+b$:

$$ {}_2F_1(a,b;c;1)=\frac{\Gamma(c)\,\Gamma(c-a-b)}{\Gamma(c-a)\,\Gamma(c-b)}, $$

so ${}_2F_1(\frac12,\frac12;\frac32;1)=\dfrac{\Gamma(\frac32)\Gamma(\frac12)}{\Gamma(1)\Gamma(1)}=\dfrac{\sqrt{\pi}}{2}\sqrt{\pi}=\dfrac{\pi}{2}$, which is $\arcsin 1$ from the arcsine row below.

3 / THE IDENTITIES / CHECKED NUMERICALLY

Both sides agree to six figures at the test values

function	as a hypergeometric series	test point	function	series
$(1-z)^{-a}$	${}_2F_1(a,b;b;z)$	$a=2, b=1, z=0.3$	2.04082	2.04082
$\ln(1+z)$	$z\,{}_2F_1(1,1;2;-z)$	$z=0.5$	0.405465	0.405465
$\arcsin x$	$x\,{}_2F_1(\frac12,\frac12;\frac32;x^2)$	$x=0.5$	0.523599	0.523599
$\arctan x$	$x\,{}_2F_1(\frac12,1;\frac32;-x^2)$	$x=0.5$	0.463648	0.463648
$K(k)$	$\frac{\pi}{2}\,{}_2F_1(\frac12,\frac12;1;k^2)$	$k=0.5$	1.685750	1.685750
$T_3(x)=4x^3-3x$	${}_2F_1(-3,3;\frac12;\frac{1-x}{2})$	$x=0.3$	−0.792	−0.792
$P_2(x)=\frac{3x^2-1}{2}$	${}_2F_1(-2,3;1;\frac{1-x}{2})$	$x=0.3$	−0.365	−0.365
$\cos z$	${}_0F_1(;\frac12;-\frac{z^2}{4})$	$z=1$	0.540302	0.540302

For example ${}_2F_1(1,1;2;-z)$ has coefficients $\frac{n!\,n!}{(n+1)!\,n!}=\frac{1}{n+1}$, so $z\,{}_2F_1(1,1;2;-z)=\sum(-1)^n z^{n+1}/(n+1)=\ln(1+z)$, and the first arcsine terms $x+\frac{x^3}{6}+\frac{3x^5}{40}$ come from $(\frac12)_n^2/((\frac32)_n n!)$.

4 / THE DIFFERENTIAL EQUATION / POLYNOMIALS

${}_2F_1$ solves a second-order equation with singular points only at 0, 1 and ∞

The equation is $z(1-z)w''+\big[c-(a+b+1)z\big]w'-abw=0$, and any second-order linear equation with exactly three regular singular points can be brought to this form, which is why so many equations of mathematical physics reduce to it. When $a=-n$ the factor $(a)_k$ vanishes for $k>n$ and the series stops, giving polynomials: Chebyshev $T_n(x)=\cos n\theta$ with $x=\cos\theta$, Legendre $P_n(x)={}_2F_1(-n,n+1;1;\frac{1-x}{2})$ and the Jacobi family.

5 / THE PENDULUM AGAIN / ELLIPTIC INTEGRALS

The exact period is $T=4\sqrt{L/g}\,K(k)$ with $k=\sin\frac{\theta_0}{2}$, and $K=\frac{\pi}{2}\,{}_2F_1(\frac12,\frac12;1;k^2)$

The correction factor is therefore ${}_2F_1(\frac12,\frac12;1;k^2)=1+\frac{k^2}{4}+\frac{9k^4}{64}+\frac{25k^6}{256}+\cdots$, which for a swing of $60°$ ($k^2=\frac14$) gives 1.0732, the exact ratio of 2.1529 s to 2.0061 s for a 1 m pendulum. For small swings $k\approx\theta_0/2$ and the first term is $1+\theta_0^2/16$, the small-angle correction, so the hypergeometric series organises all the improvements.

ASIDE / THE LADDER IN ONE SERIES

The exponential, the circular and hyperbolic functions, the logarithm and the binomial series are all hypergeometric, differing only in which parameters are set.

LINKS TO OTHER SHEETS

The rising factorial comes from *The gamma function and the beta function*; the elliptic integral appears in *The Second Curve* on the lemniscate; the previous sheet is the gamma function.

Convention. ${}_pF_q$ has p upper and q lower parameters; $(a)_n$ is the rising factorial; K the complete elliptic integral; T_n, P_n Chebyshev and Legendre.

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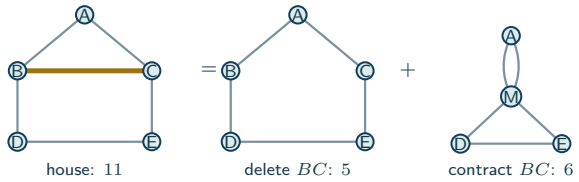
GRAPH THEORY LADDER 8 OF 9

The Laplacian and counting spanning trees

Degrees minus the adjacency matrix give a matrix whose determinant counts spanning trees.

1 / THE RULE AND THE PICTURE

Delete and contract an edge, and the counts add



house: 11 | delete BC: 5 | contract BC: 6

Spanning trees either avoid an edge e, in which case they live in $G - e$, or use it, in which case they correspond to trees of G/e with the ends of e merged, so

$$\tau(G) = \tau(G - e) + \tau(G/e)$$

The house gives $11 = 5 + 6$, the 5-cycle having 5 and the merged graph 6.

2 / THE LAPLACIAN MATRIX

Degrees on the diagonal and minus the adjacency matrix elsewhere

$$L = D - M = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ -1 & 3 & -1 & -1 & 0 \\ -1 & -1 & 3 & 0 & -1 \\ 0 & -1 & 0 & 2 & -1 \\ 0 & 0 & -1 & -1 & 2 \end{pmatrix}.$$

Every row sums to 0, the matrix is symmetric, and for any assignment of numbers x_v to the vertices $x^T L x = \sum_{uv \in E} (x_u - x_v)^2$, the energy of springs along the edges.
The number of zero eigenvalues equals the number of connected components, and for the house the eigenvalues are $0,\ 1.382,\ 2.382,\ 3.618,\ 4.618$.

3 / KIRCHHOFF'S MATRIX TREE THEOREM

Delete any row and the same column, and take the determinant

Removing row and column A leaves a 4×4 matrix whose determinant is 11, and removing any other vertex gives 11 again, so the house has 11 spanning trees.
Equivalently the product of the non-zero eigenvalues divided by n is $\frac{1.382 \times 2.382 \times 3.618 \times 4.618}{5} = 11$. The same recipe gives 16 for K_4, 125 for K_5 and 81 for $K_{3,3}$, in agreement with Cayley's formula and its bipartite version.

4 / KNOWN COUNTS

Standard graphs

graph	spanning trees	value
path P_n	1	nothing to choose
cycle C_n	n	leave out any one edge
complete graph K_n	n^{n-2}	K_4: 16, K_5: 125
complete bipartite $K_{m,n}$	$m^{n-1}n^{m-1}$	$K_{3,3}$: 81
house graph	11	the determinant below

5 / TRY THESE

Count by deleting and contracting

(a) A triangle has 3 spanning trees: delete one edge from three choices.
(b) The second smallest eigenvalue, 1.382 for the house, measures how well connected the graph is: it is 0 exactly when the graph is disconnected.
(c) Check $\tau(K_4) = 16$ by deleting an edge: $K_4 - e$ has 8 trees and K_4/e has 8.

ASIDE / ELECTRICAL NETWORKS

Kirchhoff introduced these matrices in 1847 to analyse electrical circuits, and the same determinant counts the trees that decide currents.

LINKS TO OTHER SHEETS

The adjacency matrix sheet defines M, and the earlier squared rectangles paper uses the same Kirchhoff flows.

Convention. $L = D - M$ with D the diagonal of degrees; $\tau(G)$ the number of spanning trees; $G - e$ deletes and G/e contracts the edge e.

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COMBINATORICS ON ONE PAGE

Simplexes in Pascal, and the trees on their edges

Diagonals of the triangle number the faces, and Cayley's n^{n-2} counts the trees grown leg by leg.

1 / THE RULE AND THE PICTURE

Row n of Pascal's triangle is the face count of the simplex on n vertices

$n=0$				1					
$n=1$			1		1				
$n=2$		1		2		1			
$n=3$		1		3		3		1	
$n=4$		1		4		6		4	1
$n=5$	1		5		10		10		5	1
$n=6$	1		6		15		20		15		6	1
$n=7$	1		7		21		35		35		21		7	1
$n=8$	1		8		28		56		70		56		28		8		1

ones
integers
triangular
tetrahedral
pentatope

vertices n	V	E	triangles	tetra.	5-cells
3: triangle	3	3	1		
4: tetrahedron	4	6	4	1	
5: 4-simplex	5	10	10	5	1
6: 5-simplex	6	15	20	15	6
7: 6-simplex	7	21	35	35	21

A face is a choice of vertices, so the j-faces number $\binom{n}{j+1}$:

$$\#\{j\text{-faces}\} = \binom{n}{j+1}$$

2 / FACES AND EULER'S FORMULA / ALTERNATING ALONG A ROW

$V - E + F - \cdots$ is 1 for the solid simplex and $1 + (-1)^n$ for its surface

Row n alternates to zero, $1 - n + \binom{n}{2} - \cdots = 0$, so the faces of the filled simplex alternate to 1, and removing the single top cell leaves the boundary with $\chi = 1 + (-1)^n$. The tetrahedron's surface gives $4 - 6 + 4 = 2$, your $V - E + F = 2$, and the 5-simplex's gives $6 - 15 + 20 - 15 + 6 = 2$.

$$\chi(\partial\Delta) = 1 + (-1)^n : \quad n = 3, 4, 5, 6, 7 \longmapsto 0,\ 2,\ 0,\ 2,\ 0$$

3 / TREES ON n LABELLED POINTS / CAYLEY'S COUNT

$1, 3, 16, 125, 1296, \ldots = n^{n-2}$, found by counting each shape's labellings



$n = 4$: $12 + 4 = 16$ $\qquad$ $n = 5$: $60 + 60 + 5 = 125$

n	2	3	4	5	6
trees n^{n-2}	1	3	16	125	1296

A shape contributes $n!/|\mathrm{Aut}|$ labelled trees: the 4-path $4!/2 = 12$, the star 4; the 5-path 60, the fork 60, the star 5.

4 / WHERE THE TWO MEET / A TREE IS $n-1$ OF THE SIMPLEX'S EDGES

The $\binom{n}{2}$ edges of the simplex offer $\binom{\binom{n}{2}}{n-1}$ choices, and n^{n-2} of them are trees



n	3	4	5	6	7
edges $\binom{n}{2}$	3	6	10	15	21
choices $\binom{\binom{n}{2}}{n-1}$	3	20	210	3003	54 264
trees n^{n-2}	3	16	125	1296	16 807
not trees (a cycle)	0	4	85	1707	37 457

For $n = 4$ the 20 in the middle of Pascal's row 6 is 16 trees plus 4 triangles with an isolated vertex; for $n = 5$ the 85 non-trees are 15 four-cycles and 70 triangles plus an edge.

5 / BUILDING UP LEGS / PASCAL'S ROW INSIDE CAYLEY'S COUNT

Fix the vertex n and count the trees in which it has d legs

$n \backslash d$	1	2	3	4	5
4	9	6	1		
5	64	48	12	1	
6	625	500	150	20	1

A new vertex joins a tree as a leg on an existing vertex; in the Prüfer code of length $n-2$ the label n appears $d-1$ times when it has d legs, giving $\binom{n-2}{d-1}(n-1)^{n-1-d}$ trees, and summing over d is the binomial theorem on row $n-2$:

$$\sum_{d=1}^{n-1} \binom{n-2}{d-1} (n-1)^{n-1-d} = \big((n-1)+1\big)^{n-2} = n^{n-2}$$

ASIDE / THE MATRIX-TREE THEOREM

The Laplacian of K_n has eigenvalues 0 and n ($n-1$ times), and the product of the non-zero ones over n is n^{n-2}.

ASIDE / CHECK IN PYTHON

`round(np.linalg.det((n*np.eye(n)-1)[1:,1:]))` gives 16 for $n = 4$; `comb(comb(n,2),n-1)` gives 20.

Convention. n counts vertices, so the simplex is $(n-1)$-dimensional and its edge graph is K_n; trees are labelled, as in the problem of joining n points by $n-1$ intervals.

Networked-lattice

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CARBON, CONNECTIVITY AND CAYLEY ON ONE PAGE

Alkanes: molecules as trees

Arthur Cayley's chemical question: how many carbon skeletons can the same formula have?

1 / TURN THE MOLECULE INTO A NETWORK

Four bonds per carbon; count each carbon–carbon bond twice

An alkane contains only carbon and hydrogen, with single bonds and no rings. Its **carbon skeleton** is a tree: a connected graph with no cycles. For n carbon vertices it has $n-1$ edges. If $d(v)$ counts a carbon's carbon neighbours, that carbon needs $4-d(v)$ hydrogens.

$$\underbrace{\sum_v d(v) = 2(n-1)}_{\text{handshaking lemma}} \quad \Longrightarrow \quad H = \sum_v \big(4-d(v)\big) = 4n - 2(n-1) = 2n+2.$$

$$\mathrm{C}_n\mathrm{H}_{2n+2} \qquad \text{with } d(v) \quad \leq \quad 4 \text{ at every carbon vertex.}$$

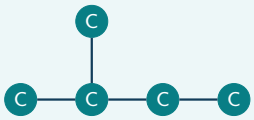
2 / SAME FORMULA, DIFFERENT CONNECTIVITY

Five carbons give three pentanes: $\mathrm{C_5H_{12}}$

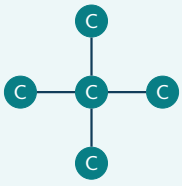
Each circle below is a carbon; hydrogens fill its unused bonds. Bending or rotating a drawing does not change which vertices are joined.



Pentane
$\mathrm{CH_3(CH_2)_3CH_3}$



2-Methylbutane
$\mathrm{(CH_3)_2CHCH_2CH_3}$



2,2-Dimethylpropane
$\mathrm{C(CH_3)_4}$

These are connectivity diagrams; real carbon bonds have three-dimensional geometry.

3 / WHICH OBJECTS ARE BEING COUNTED?

Cayley's formula counts labelled trees

Give the vertices distinct labels $1,\dots,n$. For $n \geq 2$, Cayley's theorem gives

$$T_n \;=\; n^{n-2}, \qquad T_5 \;=\; 5^3 \;=\; 125.$$

Numbering the same carbon skeleton in different ways does not make new isomers. If a skeleton has s graph symmetries (permutations preserving every edge), its distinct labellings number

$$\frac{5!}{s} = \frac{120}{s}.$$

Skeleton	s	Labellings
Pentane	2	60
2-Methylbutane	2	60
2,2-Dimethylpropane	$4! = 24$	5
Three skeletons		125

4 / WHY THE EXPONENT IS $n-2$

Prüfer's code turns a tree into a word

A **leaf** has one neighbour. Repeatedly remove the leaf with the smallest label and record its neighbour's label. Stop when two vertices remain: the word has length $n-2$.

For the path 1–2–3–4–5, the code is $(2,3,4)$. The five-vertex star with centre 1 gives $(1,1,1)$.

Every word of length $n-2$ on n labels reconstructs exactly one labelled tree. There are n choices at every position:

$$\underbrace{n \times n \times \cdots \times n}_{n-2 \text{ positions}} = n^{n-2}.$$

The degree of vertex v is one more than the number of times its label occurs in the word:

$$d(v) = 1 + \#\{\text{occurrences of } v\}.$$

Chemistry adds a restriction: each label may occur at most three times. At $n=6$, the code $(1,1,1,1)$ makes a degree-5 carbon, so Cayley's unrestricted count includes forbidden skeletons.

5 / FORGET THE TEMPORARY LABELS, KEEP THE BONDS

Count unlabelled trees with every degree at most four

Carbons n	1	2	3	4	5	6	7	8	9	10
Alkane isomers	1	1	1	2	3	5	9	18	35	75

These are **constitutional isomers**: different connectivities. Stereoisomers and conformations are not counted separately.

Beyond a tree. With e single C–C bonds, $H = 4n - 2e$. One ring gives $e = n$ and $\mathrm{C}_n\mathrm{H}_{2n}$: a cycloalkane.

Explore: Mathematical Chemistry learner. **Sources:** Rains & Sloane, *On Cayley's Enumeration of Alkanes* (1999); OEIS A000602; MIT notes on Cayley's formula.

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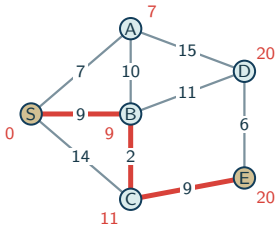
GRAPH THEORY LADDER 9 OF 9

Weighted networks: shortest paths and spanning trees

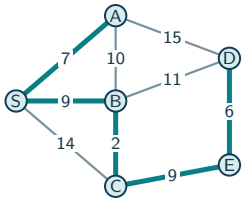
Costs on the edges give practical questions, some with fast answers and some with none.

1 / THE RULE AND THE PICTURE

The shortest route and the cheapest network are different trees



shortest route $S \to E$: 20



minimum spanning tree: 33

Each edge carries a weight such as a length or a cost, and two questions arise: the shortest route from S, solved by Dijkstra, and the cheapest set of edges joining every vertex, solved by Kruskal. The distances from S are in red.

2 / DIJKSTRA'S METHOD

Settle the nearest unsettled vertex and update its neighbours

settled	distance	reached
S	0	start
A	7	from S
B	9	from S
C	11	from B $(9+2)$
D	20	from B $(9+11)$
E	20	from C $(11+9)$

Starting from S the vertices are settled in order of distance, and each distance can only be improved through vertices already settled: C is reached at $9+2=11$ through B, beating the direct edge of 14, and E costs $11+9=20$ through C, beating $20+6=26$ through D.

3 / KRUSKAL'S METHOD

Take the cheapest edge that does not close a cycle

Sorting the edges by weight, $BC=2$, $DE=6$, $SA=7$, $SB=9$ and $CE=9$ are accepted, and $AB=10$, $BD=11$, $SC=14$, $AD=15$ are rejected because each would close a cycle, which leaves 5 edges for 6 vertices of total weight 33.
The shortest-route tree uses SA, SB, BC, BD, CE with total 38, so the cheapest network and the shortest routes differ.

4 / WHICH QUESTIONS ARE EASY

Fast methods exist for some problems and none is known for others

question	problem	status
route through every edge once	Euler circuit	degree test, fast
route through every vertex once	Hamilton cycle	NP-complete (Karp, 1972)
shortest route between two vertices	Dijkstra	fast
shortest tour through every vertex	travelling salesman	no fast method known
cheapest way to link every vertex	minimum spanning tree	Kruskal or Prim, fast
draw without crossings	planarity	linear time (Hopcroft and Tarjan, 1974)
two colours	bipartite test	linear time
three colours	3-colouring	NP-complete

5 / TRY THESE

Compute by hand

(a) What is the shortest route from S to D? S, B, D with length 20, since S, A, D costs 22.
(b) Why can the cheapest network never contain a cycle? Removing the dearest edge of a cycle leaves everything connected and cheaper.
(c) How many edges has a spanning tree on 6 vertices? 5.

ASIDE / THE TRAVELLING SALESMAN

The salesman wants a Hamilton cycle of least total weight, which is a tour and not a tree and has no fast method known, while the spanning tree gives a useful lower bound.

LINKS TO OTHER SHEETS

The Euler and Hamilton sheets give the two route problems; the Laplacian sheet counts the trees that Kruskal chooses among.

Convention. S the start vertex; weights are costs on edges; NP-complete problems have no known fast general method.

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